

GCBASIC documentation

The GCBASIC development team. Copyright 2005 - 2026

Introducing GCBASIC

Hello, and welcome to GCBASIC help. This help file is intended to provide you insights and knowledge to use GCBASIC.

For information on installing GCBASIC and several other programs that may be helpful, please see **Getting Started with GCBASIC**

<http://gcbasic.sourceforge.net/starting.html>

If you are new to programming, you should try the GCBASIC demonstration programs these explains everything in a step-by-step manner, and assumes no prior knowledge.

If you have programmed in another language, then the demonstration files on GitHub (or within your installation) and this command reference may be the best place to start.

If there is anything else that you need help on, please visit the **GCBASIC forum**

http://sourceforge.net/forum/?group_id=169286

Using GCBASIC

Need to compile a program with GCBASIC, but don't know where to begin? Try these simple instructions:

- Complete the installation using the default values - select all the programmers but not the portable mode.
- The installer will automatically start the IDE.
- When a GCBASIC source file is opened, check out the "GCB tools" menu (IDE Tools / GCB tools) - through this menu you can access the oneclick commands. Or try the right mouse button - this will access the same options.
- The IDE Tools... commands (function keys F5 - F8) starts a GCBASIC utility which calls the batchfiles for compiling sourcecode and programming ("flashing")⁽¹⁾ the target microcontroller. You have to select the appropriate programmer in "Edit Programmer Preferences" (IDE Tools / GCB tools / Edit Programmer Preferences or by pressing Ctrl+Alt+E). Find your programmer in the list and drag it to the top beneath the heading "Programmers to use (in order)". GCBASIC will now attempt to flash the microcontroller with that programmer first when you click on "Make HEX and FLASH" (F5) or "FLASH previous made hexfile" (F8).
- In the unlikely event that your programmer is not listed you can add it by pressing "Add..." in "Edit Programmer Preferences". You would have to know the working directory and command line options

etc. for the programmer. See the help tips at the bottom by clicking on the fields.

- For project-specific flashing you can edit the current programmers in “Edit Programmer Preferences” to suit your needs by clicking on “Edit...”. Use the “Use If:” parameter to choose programmer preferences. See the help tips. The chip model is autodetected by the IDE for use in “Use IF:” or in command line options etc.

- Some programmers use a .hex file to "flash" the microcontroller. By selecting "Make HEX" (F5), GCBASIC will compile the program and make a .hex file in the same directory as the GCBASIC file. This method can also be used to check for errors in the GCBASIC program before flashing.

- Included programmer software is:

- Avrdude for AVR,
- PICPgm for PIC,
- PicKit2 and PicKit3
- TinyBootLoader+
- Arduino
- Northern Software Programmer
- Microchip Xpress Board and many, many more.

⁽¹⁾ You need a suitable programmer to do this, and instructions should be included with the programmer on how to download and connect the hardware to the microcontroller.

Programmer Preferences

The “Programmer Preferences” is a software tool to control and set-up the different programmers. See below:

When using GC Code at the IDE

Select Terminal/Run Task or press function <F4> to see the menu

[graphic]

Or, when using GC Code at the IDE

Select the drop down menu to see the menu

[graphic]

When using SynWrite at the IDE

[graphic]

PIC users and Beginners: Start Here

Welcome to GCBASIC. This document is especially important for experienced PIC users moving from MPASM or C so please spend a few seconds here before you start. It could save you hours of frustration.

As a PIC user most of us are conditioned, regardless of the Assembler or Compiler, to reach for the devices data sheet first and try to work out how to setup the Oscillator, interrupt vectors and Configuration bits.

Do not DO IT. read this document first as it will give you some great insights. For the basic operation the only setup and configuration required for a GCBASIC program is the name of the target Device i.e. **#Chip 16f1619**. That is it, honestly, GCBASIC will do the rest and will determine the optimal Oscillator settings, interrupt vectors, Configuration bits etc

Next we would start deciding on and including the Device files and libraries that we intend to use. **STOP.** Let GCBASIC decide. GCBASIC is creating Portable Code, it doesn't care if you use a PIC12, PIC18 or an ATmega328. You write in BASIC and at compile time GCBASIC will decide which core libraries to include based on the instructions you have used and the target device you specified in the #chip statement.

Finally we would decide on the pins to use, their port names, which register bits are needed to make them inputs or outputs and override any Analog function if a digital function is desired.

Again, I say let GCBASIC DO IT..... **Dir PortC.0 In** - Will set Pin RC0 to a Digital Input. There is no need to manually set the TRIS register or see if there is an associated ADCON bit to set or clear.

Putting it all together: An example GCBASIC program.

```
#Chip 16f1619

#Define LED PortC.0

Dir LED Out

Do
    LED = !LED
    Wait 500 ms
Loop
```

That is it. If you have an LED attached to PortC.0 (LED DS1 on the Low Pin Count Board that shipped with the PICKit 2 or PICKit 3 programmer). It will start to Blink confirming that you have a working microcontroller and hardware.

To change target device or family just change the #Chip Entry along with the Pin you have the LED on

and recompile. it Really IS as Simple as that to get started in GCBASIC.

You can manually override GCBASIC and set every register, every flag, every BIT, every Configuration 'Fuse' and every vector if you wish, but why bother doing it upfront? Rather get your code working with the default settings and then adjust from there, if needed, as your confidence grows.

One final bit of advice, the IDE tool bar has a "View Demos" button, use it, there are examples of all of the most common programming challenges and many different devices which, along with the Help files, will answer most of your questions. The Forum is a friendly place too, so do not be shy to introduce yourself and ask for help.

Changes

Formal Release of GCBASIC Compiler v1.xx.xx

Reference	Time Stamp
ASCIIDOCs rendered	2026-01-30 08:21:44 GMT Standard Time
Master ToC information	2026-01-30 08:04:45 GMT Standard Time

Command Line Parameters

About the Command Line Parameters

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/NP] [/M:[Y/1 | N/0]] filename
```

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/WX] [/M:[Y/1 | N/0]] [/NP] filename
```

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/WX] [/M:[Y/1 | N/0]] [/S:Use.ini] [/NP] filename
```

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/WX] [/M:[Y/1 | N/0]] [/S:Use.ini] [/F[0]] [/NP] filename
```

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/WX] [/M:[Y/1 | N/0]] [/S:Use.ini] [/F[0]] [/NP] filename /DO
```

```
GCBASIC [/O:output.asm] [/A:assembler] [/P:programmer] [/K:{C|A}] [/H:[Y/1 | N/0]]  
[/V] [/L] [/WX] [/M:[Y/1 | N/0]] [/S:Use.ini] [/F[0]] [/NP] filename /AO
```

```
GCBASIC /version
```

Switch	Description	Default
<code>/A:assembler</code>	Batch file used to call assembler ⁽¹⁾ . If <code>/A:GCASM</code> is given, GCBASIC will use its internal assembler.	The program will not be assembled
<code>/AO`</code>	Use .build folder for assembly operations.	This is the Assembly Only switch.
<code>/CP</code>	Exports the config bits automatically selected by the compiler to an output file called <code>source_filename.config</code> . The output file is the source filename with the extension of config.	None
<code>/DO`</code>	Use .build folder for debugger operations.	This is the DirectoryOutput switch.
<code>/F:[0 F][1 T]</code>	Used to bypass compilation when not needed, compiler will verify that config settings in the already compiled file match those required for the programmer. If not, a recompilation will be forced. Skip compilation if the hex file is up to date and has correct config. <code>/F:x</code> (F or 0) to force a fresh compile regardless of what ini specifies.	
<code>/FO</code>	Used to bypass compilation and program only. Compiler will verify that config settings in the already compiled file match those required for the programmer. If not, a recompilation will be forced.	
<code>/H:[Y/1 N/0]</code>	Set the production, or not, of the hex output file. <code>/H:1</code> is the default. To prevent production of the hex output file - use <code>/H:0</code>	The default is to produce the hex output file
<code>/K:[C A]</code>	Keep original code in assembly output. <code>/K:C</code> will save comments, <code>/K:A</code> will preserve all input code.	No original code left in output.
<code>/L</code>	Show license and exit.	-
<code>/M:[Y/1 N/0]</code>	Mute the banner messages, or not. <code>/M:1</code> is the default. To prevent banner messages use <code>/M:0</code> .	The default is to output banner messages
<code>/NP</code>	Do not pause on errors. Use with IDEs.	Pause when an error occurs, and wait for the user to press a key.
<code>/O:filename</code>	Sets the name of the assembly file generated to <code>filename</code> .	Same name as the input file, but with a <code>.asm</code> extension.
<code>/P:programmer</code>	Batch file used to call programmer ⁽¹⁾ . This parameter is ignored if the program is not assembled.	The program will not be downloaded.

Switch	Description	Default
<code>/S:fsp</code>	Load the settings from a specified file, rather than use the defaults.	<code>/S:use.ini</code>
<code>/V:[0 F][1 T]</code>	Verbose mode — compiler gives more detailed information about its activities. <code>/Vx</code> will override any configuration in the user ini file.	-
<code>/WX</code>	Force compiler to ensure all include files are valid.	
<code>/version</code>	Show build date and version of the compiler.	
<code>filename</code>	The file to compile.	-

⁽¹⁾ For the `/A:` and `/P:` switches, there are special options available. If `%FILENAME%` is present, it will be replaced by the name of the `.asm` file. `%FN_NOEXT%` will be replaced by the name of the `.asm` file but without an extension, and `%CHIPMODEL%` will be replaced with the name of the chip. The name of the chip will be the same as that on the chip data file.

A batch file to load the `ASM` from GCBASIC into `MPASM`. Command line should be like this:

```
C:\progra~1\microc~1\mpasms~1\MPASMWIN /c- /o- /q+ /l- /x- /w1 %code%.asm
```

A batch file to compile in GCBASIC then load the `ASM` from GCBASIC into `GPASM`. Command line should be like this:

```
gcbasic.exe %1 /NP /K:A /A:"..\gputils\bin\gpasm.exe %~d1%~p1%~n1.asm"
```

To instruct MAKEHEX.BAT to use `GPASM`. You have GPUTILS installed. The batch file should be edited as follows:

```
REM Create the ASM
gcbasic.exe /NP /K:A %1
REM Use GPASM piping to the GCB error log
gpasm.exe "%~d1%~p1%~n1.asm" -k -i -w1 >> errors.txt
```

To summarise, you can use any of the following:

```
gcbasic.exe filetocompile.gcb /A:GCASM /P:"icprog -L%FILENAME%" /V /O:compiled.asm
```

GCBASIC will compile the file, then assemble the program, and run this command:

```
`icprog -lcompiled.hex`
```

You can also create/edit the gcbasic.ini file :

```
Assembler settings
  Assembler = C:\Program Files\Microchip\MPASM Suite\mpasmwin
  AssemblerParams = /c- /o- /q+ /l+ /x- /w1 "%FileName%"

Programmer settings
  Programmer = C:\Program Files\WinPic\Winpic.exe
  ProgrammerParams = /device=PIC%ChipModel% /p "%FileName%"
```

This example will use MPASM to assemble the program. It will run the program specified in the assembler = line, and give it these parameters:

```
`/c- /o- /q+ /l+ /x- /w1 "compiled.asm"`
```

Then, it will run the programmer, and give it these parameters when it calls it:

```
`/device=PIC16F88 /p "compiled.hex"`
```

%ChipModel% will get replaced with the chip you are using, so this the chip GCBASIC will pass to WinPIC.

Errors.txt

The compiler only produces the file errors.txt if there is an error. The creation of the errors.txt file makes it easier for IDEs to detect if the program compiled successfully - if the file was not produced then the IDE would be unable to present the error message to the user.

The file error.txt is always produced in the same folder as the compiler. Typically:
C:\GCStudio\GCBASIC\Errors.txt

USE.INI

USE.INI is the provided setup file for the compiler. The name of **use.ini** is historic and of no relevance.

USE.INI is generally updated by using the **PREFERENCES EDITOR**.

USE.INI is self documenting and open **use.ini** in an editor will shown the full capabilities of setting file.

The details below show the self documentation in a typical **use.ini**

Preferences file for GC BASIC

Location: GCB install (or custom) dir

Documentation for the [gcbasic] section of the use.ini file

```
programmer = arduinouno - the currently selected available programmers
showprogresscounters = n - show percent values as compiler runs. requires Verbose =
y
verbose = y - show verbose compiler information
preserve = n - preserve source program in ASM
warningsaserrors = n - treat Warnings from scripts as errors.
pauseaftercompile = n - pause after compiler. Do not do this with IDEs
flashonly = n - Flash the chip is source older than hex file
assembler = GCASM - currently selected Assembler
hexappendgcbmessage = n - appends a message in the HEX file
laxsyntax = n - use lax syntax when Y, the compiler will not check that
reserved words
mutebanners = n - mutes the post compilation messages
evbs = n - show extra verbose compiler information, requires Verbose
= y
nosummary = n - mutes almost all messages post compilation
extendedverbosemessages = n - show even more verbose compiler information, requires
Verbose = y
conditionaldebugfile = - creates CDF file
columnwidth = 180 - ASM width before wrapping
picasdebug = n - adds PIC-AS preprocessor message to .S file
datfileinspection = y - inspects DAT for memory validation
methodstructureddebug = n - show method structure start & end for validation
floatcapability = 1 - 1 = singles
- 2 = doubles
- 4 = longint
- 8 = uLongINT
compilerdebug = 0 - 1 = COMPILECALCADD
- 2 = VAR SET
- 4 = CALCOPS
- 8 = COMPILECALCMULT
- 16 = AUTOPINDIR
- 32 = ADDRDX
- 64 = GCASM
```

Frequent errors

Frequent errors that may happen, from the initial creation of a program and onwards.

Strange timings: You declared an oscillator frequency, different from the oscillator actually attached to the microcontroller.

No oscillator: Keep in mind that, besides the frequency, you must also set the type of oscillator, internal or external.

No GCBASIC frequency stated: If not declared in your source program - GCBASIC uses a preset frequency suitable for operating the microcontroller as the fastest practical.

External oscillators: It must be explicitly stated, if not stated GCBASIC will attempt to setup the internal oscillator.

Ports: GCBASIC will set the ports automatically but you may need to set the ports outputs or inputs when needed.

Analog levels: When applied on the ports defined as digital inputs. can cause current consumption in the input buffer, which is outside the device specifications. Beware.

Current drawn: Current taken from the microcontroller outputs, exceeding the maximum allowed (not all pins supply the same current). Beware of drawing too much current.

Watchdog Timer (WDT): The WDT is a useful timer. Enable to reset the microcontroller when processing can get stuck in a loop.

Interrupts: A badly controlled interrupt (in some cases) will prevent the execution of the entire program.

No action: The circuit is not powered.

Still no action: The microcontroller is not present or different from the device you expected.

Still no action: The microcontroller inserted incorrectly in the appropriate socket.

Cannot program: Incorrect programmer, Incorrect programmer parameters or circuit connections are incorrect.

Still Cannot program: Values of excessively incorrect circuit resistances.

Serial Communications: The TX and RX pins of the serial port are exchanged, and/or the connections with the level converter, ttl / rs232 or ttl / usb.

Still no Serial Communications: Serial speed, different from the one set in the circuit with which it is intended to communicate or vice versa.

No I2C/TWI: SDA and/or SCL pin exchanged on the I2C/TWI bus connection, and/or no pull-up resistors, and/or no common 0 voltage.

Incorrect timing: Calculation of any timings related to the frequency of the external oscillator, without taking into account the division by 4.

Strange Numeric Values: The variables declared are insufficient to contain the values to be processed.

A Glossary

ADC: analogue digital converter.

Negative power supply: reference to the common point of the circuit power supply, called circuit ground.

Alias: alternative name assigned to a pre-existing entity.

Array: variable able to handle numbers from 0 to 255.

ASCII: acronym for the American Standard Code for information interchange. ASCII is a code for the representation of English characters as numbers.

Assembler: PC software application that converts assembly language into machine language.

Binary: numeric system with base 2, in which there are only two possible values for each digit#0 and 1.

Bit: the smallest element of computer memory. It is a single digit in a binary number (0 or 1). Bit is also a type of variable in GCBASIC.

Bitwise: dealing with bits and binary states instead of numbers or logic.

Byte: 8-bit variable, value from 0 to 255 (2^8-1). Is also a type of variable in GCBASIC.

Boolean: related to a combinatorial system designed by George Boole, which combines propositions with the logical operators AND, OR and IF THEN, except NOT.

CC: direct current.

Machine cycle: oscillator frequency / 4, for PIC (do not forget the PLL where present).

Code: the memory area in a PIC MCU or AVR that contains the program code.

Comment: reminder notes placed in the program.

Compiler: PC software application, which converts a high level language like BASIC into assembly language. In this guide "Compiler" refers to GCBASIC.

Compile-Time: acts during compilation, and is not executed as a command when the program is running on the microcontroller.

Constant: a name that stands for a value defined in the program. The value is replaced instead of the name when the program is compiled and assembled. It is not stored in RAM and cannot be changed during program execution.

D: Digital.

Data Space: is a memory space in a PIC or AVR that is intended for the storage of values (EEPROM memory on chip). Data is accessible in GCBASIC using the EpRead and EpWrite commands for reading and writing.

Dw: referring to a button or actions for the variation of any value, is intended as "decrease".

Debug: used to locate errors, to solve problems encountered when the program is run.

Decimal: numerical system with base 10, composed of 10 numbers from 0 to 9 inclusive. The "point" in a number with base 10 separates the whole part from the bottom to 1.

Device programmer: it is a tool that "writes" the code in machine language in the PIC or AVR microcontroller.

Directive: instruction intended for the compiler or assembler. It is not a command or a compiler statement.

Emdedded System: device controlled by a program, able to independently perform even complex functions, communicate with other similar devices and different architecture, with the personal computer, with a local network and directly via the web.

EPROM: erasable programmable read only memory.

EEPROM: a type of memory that stores data even in the absence of voltage, can be deleted and rewritten about 100,000 times.

Expression: a variable, constant, or combination that represents a stored or calculated value.

Firmware: program compiled and assembled, suitable to be loaded into the program memory, of a programmable device.

Fosc: oscillator frequency.

f.s.: full scale.

Hex: extension of the assembled file.

IDE: integrated development environment, software environment that acts as a code editor, and controls the various programming tools to implement software development.

Set: write in a register or variable, the condition required by the function to be performed.

I / O: input / output.

Integer: 32-bit variable, whose value varies from -32768 to 32767. Is also a type of variable in GCBASIC.

Interrupt: the use of a predefined signal or condition that interrupts normal execution, in favor of a special procedure with high priority.

Kbit / s: one thousand bits per second.

Keywords: keywords for GCBASIC.

Label: word that marks a position in a program.

Least-significant: in reference to binary numbers, a bit or groups of bits that include the "proper" bit. The rightmost bit or bit group, when a number is written in binary.

Assembly language: the programming language that corresponds more closely with machine language codes.

Voltage levels: in this guide we refer to TTL levels, so about 0 Volts for the low level and about 5 Volts or the Vcc of the microcontroller for the high level.

Level 0: equivalent to the low level.

Level 1: equivalent to the high level.

High level: presence of voltage, referring to the particular one is talking about.

Low level: no voltage, voltage close to zero.

Long: numeric entity composed of 32 binary bits, value from 0 to 4294967295 ($2^{32}-1$). Is also a type of variable in GCBASIC.

FLASH MEMORY: non-volatile memory, electrically rewritable numerous times, also called flash / rom.

Microchip: company that produces PIC microcontrollers, now also AVR

Mips: Mega instructions per second.

ms: milliseconds.

Modifier: keyword that somehow changes the interpretation or behavior associated with a command or variable that is written before or after the modifier.

Most-significant: in reference to binary numbers, the bit or group of bits that include the bit that indicates the maximum power of two. The leftmost bit or group of bits when a number is written in binary.

Nibble: a 4-bit binary quantity, can often be used to refer to the 4 most significant or least significant bits of 8-bit bytes. A single hexadecimal digit represents a binary nibble. It is not a variable type in GCBASIC.

ns: nanoseconds.

NC: not connected or, normally closed (depending on the context).

Overflow: the event that occurs when a value in a variable is increased beyond the capacity of the variable type, resulting in an incorrect result.

PC or pc: program counter.

Port: microcontroller port

Porta: Port a.

Portb: Port b.

Portc: Port c.

Portd: Port d.

Porte: Port e.

Pos or pos: postscaler.

Ps or ps: Prescaler

Programmer: you. The person who writes the program.

RAM: the memory area in a PIC MCU that is used to contain the variables. Access to RAM is faster than other memory areas, RAM values are lost when the power is turned off.

Register: an 8-bit memory location that performs a special function in a microcontroller. Registers that (Microchip calls SFR) are integrated in the microcontroller and their functions are described in the technical data sheet published for the device.

ROM: Read Only Memory (read-only memory, can only be written once).

Run-time: executed by the microcontroller when the program is executed (when it is running).

Save to context: save and restore in the context of the interrupt, important variables in the SFR registers.

SFR: registers with special function. Able to represent or process negative and positive numbers.

String: able to deal with number, letters and symbols. Is also a type of variable in GCBASIC.

TMR or tmr: timer.

TWI: I²C Bus.

Two's complement: (complement of 2) a system that allows negative numbers to be represented in

binary.

Typecasting: specify a type of variable for the compiler.

Tp: test point.

Up: referred to a button or actions to change any value, it is intended as "increase".

Underflow: the event that occurs when a value in an unsigned variable decreases below zero (negative number), or when a variable is decreased below the limit value in a negative sense, resulting in an incorrect result.

Unsigned: only able to represent or transform positive numbers. Negative numbers are not valid in integer variables.

Variable: a name that is a synonym of a value that is stored in RAM and can be read and modified during program execution.

Word: a numeric entity composed of 16 binary bits. Value from 0 to 65535 ($2^{16}-1$)

V / I: voltage / current.

μ s or us: microseconds.

Frequently Asked Questions

Why doesn't anything come up when I run GCBASIC.exe?

GCBASIC IS a command line compiler. To compile a file, you can drag and drop it onto the GCBASIC.exe icon.

If you use an Integrated Development Environment (IDE) you can edit your program and press an ICON to send the program to the chip. Several are listed on the GCBASIC website.

The recommended IDE for Windows is GCCODE.

What Microchip PIC, Atmel AVR or LGT microcontrollers does GCBASIC support?

Hopefully, all 8 bit Microchip PIC, Atmel AVR and LGT microcontrollers and (those in the PIC10, PIC12, PIC16 and PIC18 families). If you find one that GCBASIC does not work with properly, please post about it in the Compiler Problems section of the GCBASIC forum.

Is GCBASIC case sensitive?

No! For example, **Set**, **SET**, **set**, **SeT**, etc are all treated exactly the same way by GCBASIC.

Can I specify the bit of a variable to alter using another variable?

GCBASIC support bitwise assignments. As follows:

```
portc.0 = !porta.1
```

You can also use a shift function. As in other languages, by using the Shift Function FnLSL. AN example is:

```
MyVar = FnLSL( 1, BitNum)` is Equivalent to `MyVar = 1<<BitNum`
```

To set a bit of a port and to prevent glitches during operations, use **#option volatile** as follows.

```
'add this option for a specific port.  
#option volatile portc.0  
  
'then in your code  
portc.0 = !porta.1
```

To set a bit of a port or variable. Encapsulate it in the SetWith method, this also eliminates any glitches during the update, use this method.

```
SetWith(MyPORT, MyPORT OR FnLSL( 1, BitNum))
```

To clear a bit of a port, use this method.

```
MyPORT = MyPORT AND NOT FnLSL( 1, BitNum))
```

To set a bit within an array, use this method.

```
video_buffer_A1(video_adress) = video_buffer_A1(video_adress) OR FnLSL( 1, BitNum)
```

To set a bit within a variable, use this method.

```
Dim my_variable as byte
Dim my_bit_address_variable as byte

'example
my_variable = 0
my_bit_address_variable = 7

my_variable.my_bit_address_variable = 1    ' where 1 or 0 or any bit address is valid

'Sets bit 7 of my_variable therefore 128
```

See also [Set](#), [FnLSL](#), [FnLSR](#) and [Rotate](#)

Why is x feature not implemented?

Because it has not been thought of, or no-one has been able to implement it!

If there are any features that you would like to see in GCBASIC, please post them in the "Open Discussion" section of the GCBASIC forum. Or, if you can, have a go at adding the feature yourself!

When using an include file does this use lots of memory?

When using include files, for instance the <ds3231.h> include, if you are not using all the functions of the include file, GCBASIC knows not to include the unused functions within the user program when compiling.

If I am using the hardware I2C, does all the code related to hardware I2C still get compiled in the

code?

GCBASIC only compiles functions and subroutines if they are called. GCBASIC starts by compiling the main routine, then anything called from there. Each time it finds a new subroutine that is called, it compiles it and anything that it calls. If a subroutine is not needed, it does not get compiled.

My LCD will not operate as expected?

Try adding. `#define LCD_SPEED SLOW`

This will slow the writing to the LCD.

Atmel AVR memory usage displayed is incorrect?

Atmel AVR memory values are specified in WORDS in GCBASIC. The GCBASIC compiler uses words, not bytes, for consistency between Microchip PIC and Atmel AVR microcontrollers. This keeps parts of the compiler simpler.

I cannot open the Window Help File?

See <http://digital.ni.com/public.nsf/allkb/B59D2B24D624B823862575FC0056F3D0>

How do I revert the FOR-NEXT loop to the Legacy FOR-NEXT method ?

Some background. In 2021 the GCBASIC compiler was updated to improve the operation of the FOR-NEXT loop. The improvement did increase the size of the ASM generated. The legacy FOR-NEXT loop had some major issues that included never ending loops, incorrect end loop and unexpected operations. This was all caused by the compiler, not the user, and in 2021 the compiler was updated to resolve these issues.

However, there is a risk that the new FOR-NEXT method causes 1) larger ASM that will not fit in small microcontrollers or 2) the new code does not operate as expected. In either case you can disable the new FOR-NEXT method by adding a constant as shown below. Adding this constant will revert the FOR-NEXT loop asm generated to the legacy method.

```
#DEFINE USELEGACYFORNEXT
```

Troubleshooting

Problem	Common Causes	More Assistance
Cannot compile a program	There is an error in the program. Is GCBASIC complaining about a particular line of code?	GCBASIC Forums
	GCBASIC has not been installed correctly - reinstall it.	GCBASIC Forums
	There is a bug in GCBASIC	Post on the GCBASIC Forums. Ensure you state the version of your compiler and attach your code as a ZIP.
A program compiles and downloads fine, but will not run	Oscillator not selected.	Configuration

GCBASIC Compiler Insights

This is the GCBASIC compiler insights section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Compiler Insights

This section will provide some insights into what the compiler does

How does the compiler cope with read only registers in the Chip Family 12 range?

Within this chip range the Option register is a write only register. Reading the register is not permitted.

GCBASIC needs to update this when the user wants to change the configuration - the Sleep process is an example of a user change.

The compiler handles this by the creation of the Option_reg byte variable. This byte is created by the compiler to manage the required write process.

The Option_reg variable is a cache that compiler will create if any bits of option_reg have been set manually.

If the user changes any of the bits in a program, then the compiler will find any uses of the option instruction and insert a "movwf OPTION_REG" immediately before the option instruction to cache the value in the buffer.

If Option_reg bits aren't set individually anywhere, then option_reg doesn't get created, and nothing special is done with the option instruction.

Essentially the compiler maintains a special variable and manages the whole process without the user being aware.

How does the compiler cope with the TRIS register in the 10f products?

The compiler ensures that a TRIS cache matches the actual TRIS register. The TRIS cache is a byte variable called TRISIO. The TRISIO cache is required as TRIS is a write-only register.

All ports default to input (where all TRIS bits to 1) on reset. Therefore, this is assumed to be the value 255.

TRISIO is updated when required by the user code and then used in the writing to the correct register.

The example user code and the associated assembly shows TRISIO cache in use. This method complies with datasheet.

User Code

```
'set as input
dir gpio.0 in
gpio0State = gpio.0
'set as output this will require TRIS GPIO to be set using the TRISIO cache.
dir gpio.0 out
gpio.0 = 1
```

ASM

```
;dir gpio.0 in
    bsf TRISIO,0
    movf TRISIO,W
    tris GPIO
;gpio0State = gpio.0
    clrf GPIO0STATE
    btfsc GPIO,0
    incf GPIO0STATE,F
;dir gpio.0 out
    bcf TRISIO,0
    movf TRISIO,W
    tris GPIO
;gpio.0 = 1
    bsf GPIO,0
```

Anywhere that an individual TRIS bit is set/cleared by change the port direction, the bit in the cache is changed and then that gets written to the TRIS register.

Forcing the ASM to contain comments

It may be useful to force comments into the ASM file. The verbose mode of creating the ASM will include ALL the source program as comments but it may be useful to have specific comments in the ASM to aid the understanding of code or to support debugging.

To force an assembly comment use the following:

```
asm showdebug `comment`
```

Where the **comment** will be placed into the ASM file.

Example.

The source file contains the following, where the comment text is **OSCCON type is 100**

```
asm showdebug OSCCON type is 100
OSCCON1 = 0x60
```

The generated assembly will be as following - this assumes verbose mode is not selected.

```
INITSYS
;oscon type is 100
    movlw 96
    banksel OSCCON1
```

Constants, variables, subs and function and labels

GCBASIC uses a single namespace. A namespace is the set of names used to identify and refer to objects of various kinds. In GCBASIC these can be constants, variables, methods, and labels. Where a label is a true label like the start of sub, function or macro. A namespace ensures that all of a given set of objects have unique names so that they can be identified. This organises constants, variables, methods, labels etc into a single list - the single namespace.

The namespace includes all libraries and source GCBASIC source files. If using MPASM this expands to chip specific INF file. If using PICAS then all of the PICAS toolchain including non-chip specific files.

There are changes already in place to resolve this issue for PICAS as HEX and LINE are reserved with PICAS toolchain and these conflict with GCBASIC methods. These are automatically resolved by the GCBASIC compiler.

So, given that a constants, variables, methods, labels etc are number, the compiler does not know if that is a constant, a variable, a method, or a call to a label. Some are use cases using a constant called **NORMAL** follow. **NORMAL** is defined as a constant with 0.

#1. Code segment

```
#DEFINE NORMAL 0  
CALL Normal
```

The compiler will issue no error. The compiler will assume the following and will do as instructed.
Call normal - this calls normal which has a value of 0

Resulting ASM

```
;CALL Normal  
call 0
```

#2. Code segment

```
#DEFINE NORMAL 0  
CALL Normal()
```

The compiler will issue no error. The compiler will assume the following and will do as instructed.
Call normal() - this calls normal which has a value of 0

Resulting ASM

```
;CALL Normal()  
call 0
```

#3. Code segment

```
#DEFINE NORMAL 0  
Normal
```

The compiler will issue an error message. The compiler will try to resolve the constant normal to a sub but it cannot as it is a value of 0.

Resulting ASM

```
;Normal  
0 ;?F1L8S0I8?
```

#4. Code segment

```
#DEFINE NORMAL 0
Normal()
```

The compiler will issue an error message. The compiler will try to resolve the constant normal to a sub but it cannot as it is a value of 0.

Resulting ASM

```
;Normal()
0() ;?F1L8S0I8?
```

#5. Code segment

```
#DEFINE NORMAL 0
Normal = 1
```

The compiler will issue an error message. This tries to assign a value to the object.

Resulting ASM

```
;Normal = 1
0 = 1
```

#6. Code segment

```
#DEFINE NORMAL 0
Goto normal
```

The compiler will not issue an error message. The compiler will goto (same for jmp) to the value of the object.

Resulting ASM

```
;goto Normal
goto 0
```

Compiler Control

The compiler can be controlled, in terms of the default startup library routines. This may be required to implement a specific control function, or, to disable a default startup behaviour.

Scenario #1:

You have a new LCD. The GCBASIC LCD routines fail to initialise. You want to write your own LCD initialise routine, but you want to ensure the GCBASIC standard INITLCD() does not operate before your own LCD initialise routine. How to do this?

Scenerio #2:

You want to write your own INITSYS routine. You can add your own routine to initialise the microcontroller but the default INITSYS would always be called in the ASM.

In the first sceneria the approach would be to redirect the GCBASIC standard INITLCD() to myInitLCD using #define INITLCD myINITLCD. However, prior to the latest build, this would fail to work. The reason for the failure to redirect to your new routine is the #startup INITLCD directive. The #startup directive was essentially hard coded and all the #startup(s) could not be changed.

In the second scenerio the ASM call to INITSYS is also hard coded. And, you could trick the compiler to call your own initialisation routine but this was not easy and not intuitive.

The new build now supports the updating of the #startup(s) with your own routines, or even to cancel #startup(s).

Examples:

The compiler will search for all #startup(s) and update across all sources (libraries and includes). LCD.H is just an example.

```
#DEFINE INITLCD myINITLCD // This will change any reference in the LCD.h #startup INITLCD to #startup myINITLCD.
```

```
#DEFINE INITLCD // With no second parameter would cancel any #startup in LCD.h.
```

```
#DEFINE INITSYS myINITSYS // This will change the default INITSYS to myINITSYS.
```

```
#DEFINE INITSYS // This will remove the INITSYS from the initialisation of the microcontroller.
```

Example to change LCD initialisation

```
#DEFINE INITLCD myINITLCD

Sub myInitLCD
    // do stuff
End Sub
```

Example to replace INITSYS

```

#DEFINE INITSYS                // Cancel call
#STARTUP myInitSYS, 1          // New init routines, and set as highest priority

Sub myInitSYS
    // do stuff
End Sub

```

Scripts can now change the #startup. You can add a script to change the behaviour dependent on a specific condition (the existant of another constant).

In user program

```
#DEFINE LCD_OCULAR_OM1614
```

Supported within LCD.H

```

#SCRIPT
    If Def( LCD_OCULAR_OM1614 ) Then
        'Change INITLCD to specific Initialisation sub
        INITLCD = INIT_OCULAR_OM1614_LCD
    End if
#ENDSCRIPT

.....

Sub INIT_OCULAR_OM1614_LCD
    .. lots of code
End Sub

```

will generate ASM like this...

```

;Program_memory_page: 0
    ORG 5
BASPROGRAMSTART
;Call initialisation routines
    call    INITSYS
    call    INIT_OCULAR_OM1614_LCD

;Start_of_the_main_program

```

This new capability to give you more control of the compiler.

For more help, see: `#define`, `#startup`

Libraries Overview

About Libraries

GCBASIC (as with most other microcontroller programming languages) supports libraries.

You can create you own device specific library, you are not limited to those shown below. If you create a new device specific library - please submit for inclusion in the next release via the GCBASIC forum.

Maintenance of these libraries is completed by the GCBASIC development team. If you wish to adapt these libraries you should create a local copy, edit and save within your development file structure. The development team may update these libraries as part of a release and we do not want you to lose your local changes.

To use a library, simple include the following in your user code

```
#include <3PI.H>      'this will include the 3PI capabilities within your program
```

To use a local copy of a library, simple include the following in your user code

```
#include "C:\mydev\library\3pi.h"      'this will include a local copy of the the 3PI capabilities within your program
```

GCBASIC supports the following device libraries.

Library	Class	Usage
3PI	Polulu 3pi robot	A library that interfaces the switch and the motors.
47XXX_EEPROM.H	I2C EEPROM memory	A device specific library for the Microchip EEPROM device classes
ALPS-EC11	Rotary Encoder	A device specific library for a rotary encoder.
ADS7843	Touch Shield	A library that interfaces with the ADS7843 touch screen.
BME280	Temp, Humidity and Pressure sensor	A library that interfaces with the BME280 and the BMP280 sensor.
CHIPINO	Shield	A library that interfaces the Chipino board with Arduino like port addresses.
DHT	Temperature and Humidity	A library that supports the DHT22 and the DHT11 Temperature and Humidity sensors.
DS1307	Clock	A library that supports the timer clock and NVRAM functions.

Library	Class	Usage
DS1672	Clock	A library that supports the timer clock and NVRAM functions.
DS18B20	Temperature	A library that supports the temperature functions.
DS18SB0MultiPort	Temperature	A library that supports the temperature functions with devices attached to multiple ports.
DS18S20	Temperature	A library that supports the temperature functions.
DS2482	Clock	A library that supports the I2C to Dallas OneWire functions.
DS3231	Clock	A library that supports the timer clock and NVRAM functions.
DUEMILANOVE	Shield	A library that interfaces the Duemilanove board with Arduino like port addresses.
EMC1001	Temperature	A library that supports the temperature functions and the other device capabilities.
FRAM	I2C Eeprom	A library that supports memory functions.
GETUSERID	Microchip read ID	A library that supports the identification of Microchip microcontrollers.
EPD_EPD2In13	Graphical e-Paper display	A core library for Graphical LCD support.
EPD_EPD7in5	Graphical e-Paper display	A core library for Graphical LCD support.
GLCD_	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_HX8347	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ILI9340	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ILI9341	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ILI9481	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ILI9486L	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_NT7108C	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_IMAGESANDB ONTS_ADDIN3	Graphical LCD	A library to increase the capabilities of the Graphical LCDs.
GLCD_KS0108	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_NEXTION	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_PCD8544	Graphical LCD	A device specific library for an Graphical LCD.

Library	Class	Usage
GLCD_SH1106	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_SSD1289	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_SSD1306	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_SSD1331	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ST7735	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_ST7920	Graphical LCD	A device specific library for an Graphical LCD.
GLCD_T6963_64	Graphical T6963 LCD with 240 x 64 pixels	A device specific library for an Graphical LCD.
GLCD_T6963_128	Graphical T6963 LCD with 240 x 64 pixels	A device specific library for an Graphical LCD.
HEFLASH	HEF Memory Driver	A library that supports the HEF memory functions.
HMC5883L	Triple-axis Magnetometer	A library that supports the magnetometer functions.
HWI2C_ISR_HANDLER	I2C Slave Driver	A library that supports the use of a Microchip microcontroller as an I2C slave.
HWI2C_MESSAGEINTERFACE	I2C Slave	A support library that supports the use of a Microchip microcontroller as an I2C slave.
HWI2C_ISR_HANDLER_ERKMODE	I2C Slave Driver	A library that supports the use of a Microchip microcontroller as an I2C slave.
HWI2C_MESSAGEINTERFACE_ERKMODE	I2C Slave	A support library that supports the use of a Microchip microcontroller as an I2C slave.
I2CEEPROM	I2C EEPROM memory	A library that supports memory functions.
LCD2SERIALREDIRECT	LCD to Serial Handler	A library that supports the use of a serial and PC terminal as a psuedo LCD.
LEGO-PF	Lego Mindstorms shield	A library that supports the Lego Mindstorms robot
LEGO	Lego Mindstorms shield	A library that supports the Lego Mindstorms robot
MATHS	Maths routines	A library that supports maths functions such as logs, power and atan.
MAX6675	Temperature	A library that supports the temperature functions.
MAX7219_ledmatrix_driver	LED 8*8 Matrix driver	A library that supports the MAX7219 8*8 LED matrixes
MCP23008	i2C to serial	A library that supports the I2C to serial functions.

Library	Class	Usage
MCP23017	i2C to serial	A library that supports the I2C to serial functions.
MCP4XXXDIGITALPOT	Digital Pot	A library that supports the MCPxxxx range of digital potentiometers.
MCP7940N	Clock	A library that supports the timer clock and NVRAM functions.
MILLIS	Clock	A library that supports the 1000ms timer event cycle.
NUNCHUCK	Game controller	A library that supports the NunChuck game controller.
PCA9685	PWM	A device specific library for the 16channel PWM driver. See the demonstrations for example on usage. Support up to four devices via the I2C bus.
PCF8574	GLCD	A device specific library for an Graphical LCD.
PCF85X3	Clock	A library that supports the timer clock and alarms.
SD	SD Card	A device specific library for an SD Card.
SMT_Timers	Signal Measurment Timer	A library for Signal Measurment Timer for specific Microchip microcontrollers.
SOFTSERIAL	Serial	A library for software serial.
SOFTSERIALCH1	Serial	A library for software serial.
SOFTSERIALCH2	Serial	A library for software serial.
SOFTSERIALCH3	Serial	A library for software serial.
SONGLAY	Music	A library for play music. Supports QBASIC and RTTTL format.
SONYREMOTE	Infrared	A library that supports the functions of a Sony remote control.
SRF02	Distance Sensor	A library that supports the SRF02 ultrasonic sensor.
SRAM	Memory devices	A library that supports 23LC1024, 23LCV1024, 23LC1024, 23A1024, 23LCV512, 23LC512, 23A512, 23K256, 23A256, 23A640 or 23K640 devices
SRF04	Distance Sensor	A library that supports the SRF04 ultrasonic sensor. See GitHub demos here
TEA5767	I2C Radio	A library that supports the TEA5767 radio.
TM1637	7 Segment LED display	A library that supports the TM1637 7-Segment LED displays
TRIG2PLACES	Maths functions	A maths library that supports trigonometry to two places.

Library	Class	Usage
TRIG3PLACES	Maths functions	A maths library that supports trigonometry to three places
TRIG4PLACES	Maths functions	A maths library that supports trigonometry to four places
UNO_MEGA328P	Shield	A library that interfaces the shield with Arduino like port addresses.
USB	USB Supoport	A library that interfaces the USB for 16f and 18f microcontrollers.

GCBASIC supports the following core libraries. These libraries are automatically included in your user program therefore you do not need to use '#include' to access the libraries capabilities.

Library	Class	Usage
7SEGMENT	7 Segment LED display	A library that interfaces the device. See also TM1637a library.
A-D	Analog to Digital	A library that supports the ADC functionality.
EEPROM	EEProm	A library that supports I2C eeprom devices.
HWI2C	I2C	A library that supports the MSSP and TWI hardware modules of I2C
HWI2C2	I2C	A library that supports the MSSP and TWI hardware modules of I2C on channel two
HWSPI	SPI	A library that supports the MSSP and TWI hardware modules of SPI
I2C	I2C	A library that supports software I2C
KEYPAD	KeyPad	A library that supports a keypad.
PS2	I2C	A library that supports keyboard functionality
LCD	LCD	A library that supports LCD functionality, library supports many different communications methods.
PWM	Pulse Width Modulation	A library supports PWM functionality.
RANDOM	Random Numbers	A library supports random number functionality.
REMOTE	Infrared	A library that supports the functions of a NEC remote control.

Library	Class	Usage
RS232	Serial	A library for serial communications.
SOUND	Tones	A library for sound and tone generation
STDBASIC	Utility Functions	The library that contains many of the utility methods.
STRING	String	The library that contains the string methods.
SYSTEM	System	The library that contains the system methods.
TIMER	Timers	The library that contains the timer methods.
USART	Serial	The library that contains the hardware serial methods that use the MSSP or AVR equivalent hardware module.
XPT2046	Touch Shield	A library that interfaces with the APT2026 and the ADS7843 touch sensors.

Acknowledgements

Developers and Contributors:

Hugh Considine - Main developer of GCBASIC

Stefano Bonomi - Two-wire LCD subroutines

Geordie Millar - Swap and Swap4 subroutines

Jacques Nilo - HEFM and help file conversion to asciidoc

Finn Stokes - 8-bit multiply routine, program memory access code

Evan Venn - Utilities, revised I2C routines, this help file and generally everything else!

Translation Contributors:

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Murat Inceer - Turkish

Other Contributors:

Russ Hensel - GCBASIC Notes.

Chuck Hellebuyck - His documentation for the GLCD and other pieces, see <http://www.elproducts.com>.

Frank Steinberg - GCode IDE for GCBASIC.

Alexy T. - SynWrite IDE used for GCB IDE, see <http://www.uvviewsoft.com/synwrite>

Thomas Henry for the Select Case and the Sine Table examples.

William Roth for the LCD code and supporting diagrams.

Theo Loermans for the revised LCD sections and the serial library.

Chris Roper for the bitwise methods including the library including FnEquBit, FnNotBit, FnIs1Bit, FnIsrBit, SetWith and 47xxx.

Jberg2024 for the adaption of the Software Serial routines to improve usage.

Angel Mier for the USB driver installation

Conversion of asciidoctor documentation files:

See the [asciidoctor Web site](#) and the [support forum](#).

Tricks and Tips

This is a collation of tricks and tips that may be useful to you.

[RAM, variables and resets](#)

[Reverting the FOR-NEXT loop to the Legacy FOR-NEXT method](#)

[Change the compilers behaviour when the compiler states a capability is not available](#)

[Create a minimal ASM source with no config and/or initsys](#)

[PPS microcontrollers and multiple USARTs](#)

TIP: RAM, variables and resets

When you define a variable it will be mapped to a RAM location. As you develop your solution you should always do the following to ensure the variable are initialised correctly.

- Always initialise variables to a known state

A variable will not show up in the ASM source code unless it is used somewhere in code. Adding `Variable = 0` will assure that the variable is initialised and will show up in the ASM. This is very useful for troubleshooting. This is essential when debugging ASM to look at variables that are defined using "EQU". If you do not initialise or use the variable then the variable will not be shown in the EQU list of variables. So, initialise all your variables.

- Always power cycle the microcontroller after programming

A soft reset when debugging/testing/programming will not reset the RAM to a known state. This is essential when debugging ASM to look at variables that are defined using "EQU". A soft reset does not change the contents of RAM. Where a hard reset reverts ram back to an undefined /random state! So, a power cycle is good practice.

TRICK: Reverting the FOR-NEXT loop to the Legacy FOR-NEXT method ?

Why do this? To reduce the PROGMEM size. But, you must assure yourself that the loop variable cannot overflow as the legacy FOR-NEXT does not prevent an overflow of the loop variable.

Some background. In 2021 the GCBASIC compiler was updated to improve the operation of the FOR-NEXT loop. The improvement did increase the size of the ASM generated. The legacy FOR-NEXT loop had some major issues that included never ending loops, incorrect end loop and unexpected operations. This was all caused by the compiler, not the user, and in 2021 the compiler was updated to

resolve these issues.

However, there is a risk that the new FOR-NEXT method causes 1) larger ASM that will not fit in small microcontrollers or 2) the new code does not operate as expected. In either case you can disable the new FOR-NEXT method by adding a constant as shown below. Adding this constant will revert the FOR-NEXT loop asm generated to the legacy method.

```
#DEFINE USELEGACYFORNEXT
```

TRICK: How to change the compilers behaviour when the compiler states a capability is not available when I know it is ?

The compiler is issuing an error message that a EEPROM, HEF, SAF, PWM16 or hardware USART is not available... but, it is.

This is caused by the microcontroller DAT file. The microcontroller DAT file is missing key information that informs the compiler that a specific capability is available. This information was added to prevent silent failures where you could use a capability when it is not available.

The compiler thinks your microcontroller does not have the selected capability. Simply use the table below to resolve. Adding the constant defined to your source program.

Then, let us know via the Forum so we can correct the source microcontroller DAT file.

EEPROM

```
#DEFINE CHIPEEPROM = 1
```

HEF

```
#DEFINE CHIPHEFWORDS = 128
```

SAF

```
#DEFINE CHIPSAFWORDS = 128
```

PWM16

```
#DEFINE CHIPPWM16TYPE = 1
```

USART hardware

```
#DEFINE CHIPUSART = 1
```

TRICK: How do I create a minimal ASM source with no config and/or initsys?

Very easy. Simple add two **#OPTION** statements.

#OPTION UserCodeOnly ENTERBOOTLOADER: This will instruct the compiler to NOT call the INITSYS() method. And, to jump to a label. The label is mandated. The label specified will be included in the ASM generated.

#OPTION NoConfig This will instruct the compiler to NOT add the microcontroller specific config statements.

Example:

```
#chip 16f877a, 4
#OPTION Explicit

#OPTION UserCodeOnly ENTERBOOTLOADER:
#OPTION NoConfig

ENTERBOOTLOADER:
```

The example above yields the following asm. Comment lines have been removed for clarity.

```
LIST p=16F877A, r=DEC
#include <P16F877A.inc>

;Vectors
ORG 0
pagesel ENTERBOOTLOADER
goto ENTERBOOTLOADER

;ORG 5

ENTERBOOTLOADER

;ORG 2048
;ORG 4096
;ORG 6144

END
```

TIP: PPS and multiple USARTs

You can set up multiple pins to simultaneously operate as a peripheral output on microcontrollers with Peripheral Pin Select (PPS).

PPS microcontrollers can set up to simultaneously output specific modules. The example below shows the method to output two TX ports. Hardware Serial (TX1) data will now be output on both B.6 and C.6

```
Sub InitPPS
  'Module: UART pin directions
  Dir PORTC.6 Out    ' Make TX1 pin an output
  Dir PORTB.6 Out    ' Make TX1 pin an output
  'Module: UART1 to two ports
  RC6PPS = 0x0020    'TX1 > RC6
  RB6PPS = 0x0020    'TX1 > RB6

End Sub
```

UNO as ISP programmer

So, you have brought some ATtiny88 breakout boards online. They are advertised as Nano equivalents but are inferior to the Nano in having low RAM (512 bytes vs 2048) and missing some other features. Specifically the lack of a USB comport for programming.

The ATtiny88 USB interface only works in Arduino IDE with some tweaking, and, you are not in the mood for learning how to write sketches after being in the GCB environment for years.

This is an all-in-one tutorial for programming the ATtiny88 via AVRdude using GCB.

NOTE | The only baud rate that works is 19200. Every other baud rate failed in testing

The process described will create a new programmer entry in the GCB Programmer Options to fully automate the compile & program progress.

NOTE | This refers to an ATtiny88 but you can use this method for many AVR's which used in conjunction with AvrDude.

The Process

1. Obtain an Arduino UNO or mega. Upload this [hex file](#) to convert the UNO into an ISP programmer or follow steps 2 -5 below.
2. Download the Arduino IDE software. This is used to upload a sketch to the UNO that converts it into an ISP programmer.
3. Connect the UNO to your PC via USB. In Arduino IDE goto Tools → Set board and select "Arduino UNO". Select the correct com port for the Arduino Uno as show in device manager.
4. Goto file → examples → ArduinoISP to select the sketch that will convert the UNO to an ISP

programmer. I found a better(?) working version at adafruit. Simply copy all the text from this link into a new sketch <https://raw.githubusercontent.com/adafruit/ArduinoISP/master/ArduinoISP.ino> (or download the ino file attached and open it in Arduino IDE) and goto step 5

5. Click upload and confirm the sketch uploaded correctly by checking the status window at the bottom of the Arduino IDE
6. Build a cable to connect the ISP headers on the UNO and target (ATtiny88) board as described below. Search online for the UNO ISP header pinout, the ISP header happens to be labelled underneath the ATtiny88 breakout board.
7. Connect pin 10 of the UNO to the reset pin on target ISP header
8. Connect VCC to VCC, MOSI to MOSI, MISO to MISO, GND to GND, SCK to SCK.
9. Open Synwrite → "IDE tools" → "GCB tools" → "Edit Programmer preferences", or, in GCStudio "Edit Programmer preferences"
10. Click "add" and a program editor window opens
11. Enter name Arduino as ISP or similar
12. In the "Use if" box paste DEF(AVR)
13. In the "File" box paste %instdir%..\avrdude\avrdude.exe
14. In the "command line parameters" paste -c avrisp -p t88 -P %Port% -b 19200 -U flash:w:"%FileName%":a
15. Select the com port that corresponds to the connected UNO port
16. Click ok

Enter the sample code here into GCB IDE

```
#chip tiny88, 12

dir portd.0 out

Do
  set portd.0 on
  wait 500 ms
  set portd.0 off
  wait 500 ms
Loop
```

Now you can select "Hex/Flash" to upload the code to the Attiny88. If all goes well the LED should blink on and off every second

Microcontroller Fundamentals

Inputs/Outputs

About Inputs and Outputs

Most general purpose pins on a microcontroller can function in one of two modes: input mode, or output mode.

When acting as an input, the general purpose input/output pin will be placed in a high impedance state. The microcontroller will then sense the general purpose input/output pin, and the program can read the state of the general purpose input/output pin and make decisions based on it.

When in output mode, the microcontroller will connect the general purpose input/output pin to either Vcc (the positive supply), or Vss (ground, or the negative supply). The program can then set the state of the general purpose input/output pin to either high or low.

GCBASIC will attempt to determine the direction of each general purpose input/output pin, and set it appropriately, when possible. GCBASIC will try to set the direction of the general purpose input/output pin. However, if the general purpose input/output pin is read from and written to in your program, then the general purpose input/output pin must be configured to input or output mode by the program, using the appropriate [Dir](#) commands.

Example of **dir** commands.

```
'The port address is microcontroller specific. Portx.x is a general case for PICs
and AVRs
dir portb.0 in
dir portb.1 out

'The port address is microcontroller specific. gpiox.x is a general case for some
PICs
dir gpio.0 in
dir gpio.1 Out

'Set the whole port as an output
dir portb out
dir gpio out

'Set the whole port as an input
dir portc in
dir gpio in
```

Microchip specifics for read/write operations

For the specific ports and general purpose input/output pins available for a specific microcontroller please refer to the datasheet.

Port	Purpose	Example
PORTx maps to the microcontrollers digital pins 0 to 7. Where x can be a,b,c,d,e,f or g	Read: PORTx the port data register for a read operation.	uservar=PORTA uservar=PORTA.1
PORTx maps to microcontrollers digital pins 0 to 7. Where x can be a,b,c,d,e,f or g	Write: PORTx the port data register for a write operation, and, where LATx is not required as GCBASIC will implement LATx when needed. See Option NoLatch for more information on LAT registers and how to disable this automatic function.	PORTA=255 PORTA.1=1

To read a general purpose input/output pin, you need to ensure the direction is correct **DIR Portx IN** is set (default is IN) or a specific set of port bits. Where **uservar = PORTx.n** can be used.

Examples:

```
uservar = PORTb.0
uservar = PORTb
```

To write to a general purpose input/output pin, you need to ensure the direction is correct **DIR Portx OUT** for port or a specific set of port bits. Where **PORTx.n = uservar** can be used.

Examples:

```
PORTb.0 = uservar
PORTb = uservar
```

ATMEL specifics for read/write operations

Using a Mega328p as a general the following provides insights for the AVR devices. For the specific ports and general purpose input/output pins available for a specific microcontroller please refer to the datasheet.

Port	Write operation	Read operation
PORTD maps to Mega328p (and, the AVR microcontrollers) digital pins 0 to 7	PORTD - The Port D Data Register - write operation (a read operation to a port will provide the pull-up status)	PIND - The Port D Input Pins Register - read only

Port	Write operation	Read operation
PORTB maps to Mega328p (and, the AVR microcontrollers) digital pins 8 to 13. The two high bits (6 & 7) map to the crystal pins and are not usable	PORTB - The Port B Data Register - write operation (a read operation to a port will provide the pull-up status)	PINB - The Port B Input Pins Register - read only
PORTC maps to Mega328p (and, the AVR microcontrollers) analog pins 0 to 5. Pins 6 & 7 are only accessible on the Mega328p (and, the AVR microcontrollers) Mini	PORTC - The Port C Data Register - write operation (a read operation to a port will provide the pull-up status)	PINC - The Port C Input Pins Register - read only

To read a general purpose input/output pin, you need to ensure the direction is correct **DIR Portx IN** is set (default is IN) or a specific set of port bits. Where **uservar = PINx.n** can be used and therefore to read data port use **uservar = PINx**.

Examples:

```
uservar = PINb.0
uservar = PINb
```

To write to a general purpose input/output pin you need to ensure the direction is correct **DIR Portx OUT** for port or a specific set of port bits. Where **PORTx.n = uservar** can be used and therefore to write to a data port use **PORTx = uservar**.

Examples:

```
PORTb.0 = uservar
PORTb = uservar
```

Setting Ports and Port.bit

You can set a port as shown above with a variable, or, you can set with a constant or any combination using the bitwise and logical operators.

```
#define InitStateofPort 0b11110000
PORTb = InitStateofPort           'will unconditionally set bits 4:7

PORTb = 0b11110000                'will unconditionally set bits 4:7

PORTb = uservar OR 0b11110000      'will OR bits 4:7 to ensure bits 4:7 are set
```

The following is also valid - read a port.bit and then set port.bit with a variable or port value. As shown below.

```
dir portb out

portb.0 = NOT portb.0
```

The user code above may cause issues with glitches when the read and write operations occurs. Let us look at the generated assembler.

```
;portb.0 = NOT portb.0
banksel SYSTEMP1
clrf SysTemp1
btfsc PORTB,0
incf SysTemp1,F
comf SysTemp1,F
bcf PORTB,0
btfsc SysTemp1,0
bsf PORTB,0
```

To resolve any glitches add **#option Volatile** to your user code.

```
#option Volatile portb.0

dir portb out

portb.0 = NOT portb.0
```

This option provides the following assembler resolving the glitch issue.

```
;portb.0 = NOT portb.0
banksel SYSTEMP1
clrf SysTemp1
btfsc PORTB,0
incf SysTemp1,F
comf SysTemp1,F
btfsc SysTemp1,0
bsf PORTB,0
btfss SysTemp1,0
bcf PORTB,0
```

Configuration

About Microcontroller Configuration

For PICs

This section applies to Microchip PIC microcontrollers. For AVR and LGT microcontrollers see the sections below.

Every Microchip PIC has a CONFIG word. This is an area of memory on the chip that stores settings which govern the operation of the chip.

The following aspects of the chip are governed by the CONFIG word:

- Oscillator selection - will the chip run from an internal oscillator, or is an external one attached?
- Automatic resets - should the chip reset if the power drops too low? If it detects it is running the same piece of code over and over?
- Code protection - what areas of memory must be kept hidden once written to?
- Pin usage - which pins are available for programming, resetting the chip, or emitting PWM signals?

The exact configuration settings vary amongst chips. To find out a list of valid settings, please consult the datasheet for the microcontrollers that you wish to use.

This can all be rather confusing - hence, GCBASIC will automatically set some config settings, unless told otherwise:

- **Low Voltage Programming (LVP) is turned off.** This enables the PGM pin (usually B3 or B4) to be used as a normal I/O pin.
- **Watchdog Timer (WDT) is turned off.** The WDT resets the chip if it runs the same piece of code over and over - this can cause trouble with some of the longer delay routines in GCBASIC.
- **Master Clear (MCLR) is disabled where possible.** On many newer chips this allows the MCLR pin (often PORTA.5) to be used as a standard input port. It also removes the need for a pull-up resistor on the MCLR pin.
- **An oscillator mode will be selected, based on the following rules:**
 - If the microcontroller has an internal oscillator, and the internal oscillator is capable of generating the speed specified in the #chip line, then the internal oscillator will be used.
 - If the clock speed is over 4 Mhz, the external HS oscillator is selected
 - If the clock speed is 4 MHz or less, then the external XT oscillator mode is selected.

Note that these settings can easily be individually overridden whenever needed. For example, if the Watchdog Timer is needed, adding the line

```
#config WDT = ON
```

This will enable the watchdog timer, without affecting any other configuration settings.

For AVR

This section applies to Atmel AVR microcontrollers. Generally, Atmel AVR microcontrollers do have a similar configuration settings, but they are controlled through "Configuration Fuses". GCBASIC cannot set these - you MUST use the programmer software.

The exception to the general case are the ATTiny4-5-9-10 and ATTiny102-104. These microcontrollers have software selectable frequencies for the following frequencies:

```
ChipMHz 8  
ChipMHz 4  
ChipMHz 2  
ChipMHz 1  
ChipMHz 0.5  
ChipMHz 0.25  
ChipMHz 0.125  
ChipMHz 0.0625  
ChipMHz 0.03125
```

Therefore, you can use (an example)

```
#chip tiny10, 0.25
```

For LGT

This section applies to LGT microcontrollers.

All LGT microcontroller have software selectable frequencies for the following frequencies:

```
ChipMHz 8  
ChipMHz 4  
ChipMHz 2  
ChipMHz 1  
ChipMHz 0.5  
ChipMHz 0.25  
ChipMHz 0.125  
ChipMHz 0.0625  
ChipMHz 0.03125
```

Therefore, you can use (an example)

```
#chip #chip LGT8F328P, 0.25
```

Using Configuration

For PICs only.

Once the necessary CONFIG options have been determined, adding them to the program is easy. On a new line type "#config" and then list the desired options separated by commas, such as in this line:

```
#config OSC = RC, BODEN = OFF
```

GCBASIC also supports this format on 10/12/16 series chips:

```
#config INTOSC_OSC_NOCLKOUT, BODEN_OFF
```

However, for upwards compatibility with 18F chips, you should use the = style config settings.

It is possible to have several #config lines in a program - for instance, one in the main program, and one in each of several #include files. However, care must then be taken to ensure that the settings in one file do not conflict with those in another.

For more help, see [#config Directive](#)

USB Drivers Installer

WARNING

Installing the USB driver is only required when using the GCBASIC USB library.

Description:

The drivers for windows x86 and x64 correspond to the USB LIBKWIN capability of GCBasic for supported PIC microcontrollers.

For security reasons, in Microsoft windows for a driver to be installed, it is necessary that it be digitally signed by Microsoft.

Microsoft did make a special “Test” mode for developers to install MANUALLY unsigned drivers for debug and testing, being a technical advanced and not user-friendly procedure; at the same time the windows developers make efforts to disable the capability of doing this in an automated fashion by the concerns of being used as a vulnerability of the operating system.

This scenario will make installing test drivers difficult and frustrating for the uninitiated, at the same time for a useful Hobby project it will be not practical to make end users to take all this drama.

This driver installer method resolves the constraints imposed by the Windows operating system, and, therefore will allow you to install the drivers in the easiest way possible, almost like any driver of a well-known company.

Usage:

WARNING

The installer will reboot the system without notice. Please close all programs and save any work you have open before begin whit the driver install.

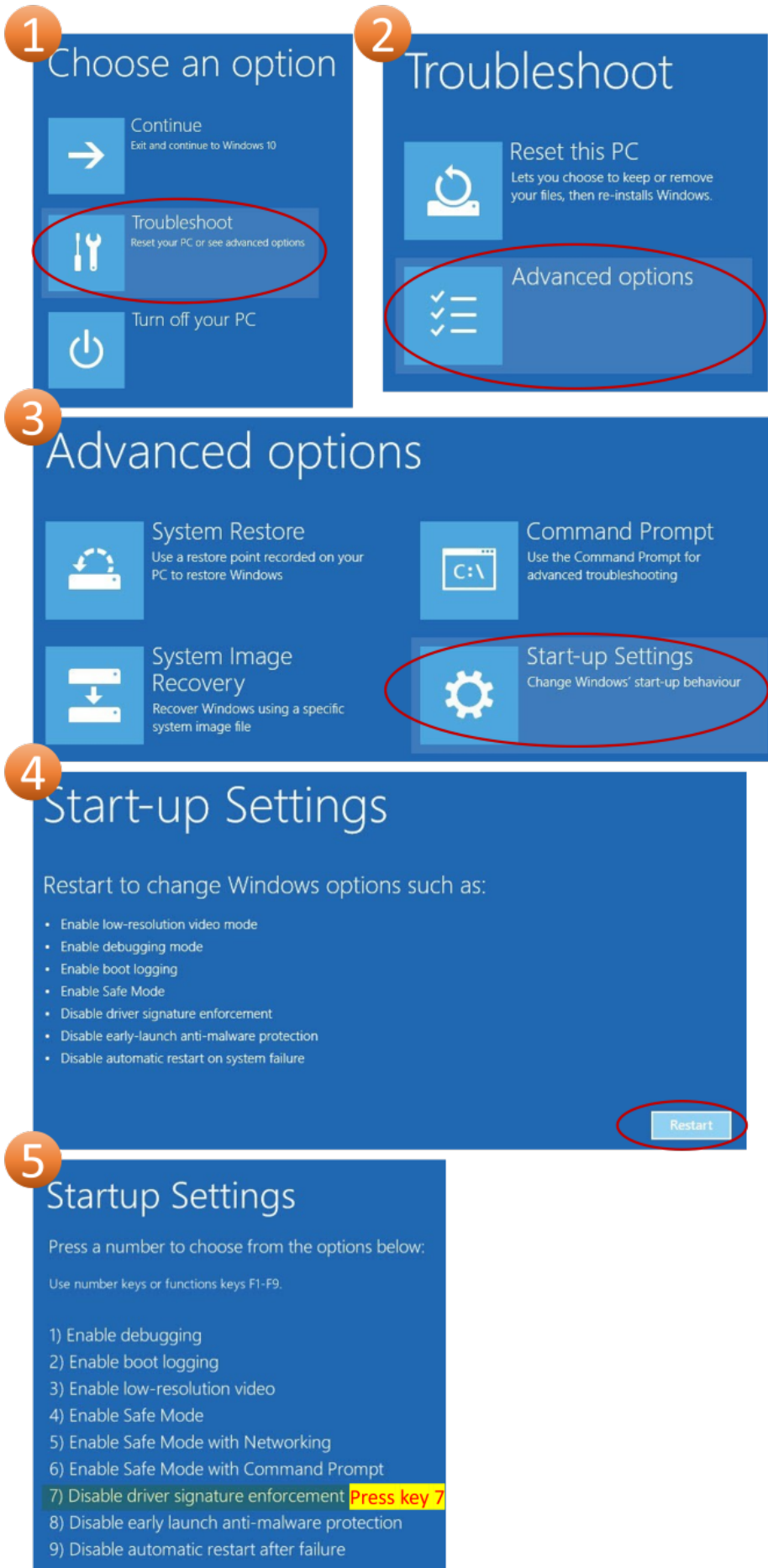
- 1 - Open the installer, it will request admin rights.
- 2 - Navigate thru the wizard to automatically extract the driver files (there aren't any options to select).
- 3 - At the end of the wizard, after you click the exit button, the system will restart automatically

WARNING

In the case where your computer has Secure Boot enabled, the installer will advise you of extra steps needed after reboot, at the end of this page you will find a graphic reflecting those steeps and what elements you need to select.

- 4 - After restart and login in to your user account, a window will inform you that the driver is not signed and you will be asked if you want to install the driver, please allow it.
- 5 - when the driver has been installed, the computer will restart automatically.

Secure Boot Enabled, Boot menu steps



USB Driver details

The driver uses the following USB flags.

```
USB_VID 0x1209  
USB_PID 0x2006  
USB_REV 0x0000
```

For others, need to modify and recompile the USB library.

USB_PRODUCT_NAME and USB_VENDOR_NAME can change without problem (windows device manager will show the name reported by the hardware not the driver)

Tested on (but not limited to)

```
Windows 11 pro x64 secureboot disabled, os build Dev 21H2 22000.194  
Windows 11 pro x64 secureboot enabled, os build Dev rs_prerelease 22458.1000  
Windows 10 pro x64 secureboot disabled, os build stable 20H2 19042.867  
Windows 7 pro x86 secureboot disabled, os service pack 1 build 6.1.7601
```

Variables

Data Types

This section discusses the different types and sizes of data variables used by GCBASIC, and how they are interpreted or handled by GCBASIC methods.

The section also provides an insight of which type of variable to use and when.

What variable sizes are supported by GCBASIC?

GCBASIC implements support for Bit, Byte, Word, Integer and Long Variable Types, all of which are described below.

Supported variables are **Bit** (1 Bit), **Byte** (8 Bit), **Word** (16 Bit), **Long** (32 Bit). GCBASIC does not support decimal numbers.

Bit is used as a Flag or a Port Pin and has two states which may be:

ON or OFF
TRUE or FALSE
HIGH or LOW
1 or 0
SET or RESET

other complementary states depending on how your application interprets and handles the data.

Byte is the most common size in 8 Bit devices and could represent a Number, an ASCII Character, a Port, two Nibbles (as used by Hex or BCD number systems), an Internal Register, an 8 bit Variable or any user defined collection of to eight Bits such as a group of flags.

Word is normally used for its Numeric value. 16 Bits will allow it to store Numbers from Zero to 65535 which is large enough to store the product of any two 8 bit Bytes without overflowing. However, it is not confined to being used as a numeric value. A Word may be used in any manner that your application needs depending on how it interprets the 16 Bits of data. Examples may be a memory address or a data pointer.

- **Note:** The Word size of a device (as opposed to the Word Type above) is a representation of the number of Bits that it can manipulate simultaneously by the chip. The number of Bits for PIC and AVR Microcontrollers supported by GCBASIC are 8 Bits and so they are considered to have an 8 Bit Word.*

Long is for situations where Values exceeding 65535 have to be handled and has a range of zero to 4294967295 ($2^{32}-1$). It is rarely used in 8 Bit devices but is invaluable on the rare occasions that it is needed. The `Millis()` is an example that uses the Long Data Type to handle time periods of up to 50

days.

All of the above can be considered to be Integer Values of varying magnitude as they can hold non Fractional Positive Whole Numbers, but try not to confuse **Integer Values** with the **Integer Variable Type**, they are complementary but separate concepts as you will see below.

An **integer** is a whole number (not a fractional number) that can be Positive, Negative, or Zero.

In your application there may be a need to be able represent Negative Numbers in our variables and that is where the GCBASIC **Integer Variable Type** is useful. An **Integer Variable** is similar to the **Word Variable** as they are both 16 bits. The difference how the GCBASIC compiler interprets the data bits that it contains.

The compiler will treat a **Word Variable Type** as a Variable that can store the values $0 < 65535$ but it will see the **Integer Variable Type** as a Variable that can store values of $-32768 < 0 < 32767$.

Variable size

Each type of variable is defined in various bit lengths, in this case GCBASIC these are:

Byte	8 Bit
Integer	16 Bit
Word	16 Bit
Long	32 Bit

All four of the above are number types are true Integers. In that they are representations of a integer non fractional number as follows:

8 Bit - an 8 Bit number can be in the range of 0 to 255
16 Bit - a 16 Bit number can be in the range of 0 to 65535
32 Bit - a 32 Bit number can be in the range of 0 to 4294967295 ($2^{32}-1$)

But, they can only represent positive numbers. In Mathematics there is a need for an Integer that can be Positive, Negative, or Zero. Note that Zero is always a Positive Whole Number.

Two's Complement

To take the Two's Complement of a number it is inverted then incremented:

$$\text{MyVar} = \text{NOT MyVar} + 1$$

The increment, of adding 1, has two effects, it avoids the possible creation of a negative zero as a value of 1000000 would be seen as -128 and it allows subtraction to be achieved through addition.

If MyVar contained a value of 1 the 8 Bit representation would be:

```
00000001
```

The NOT will make it

```
11111110
```

Note that the Most significant Bit is now 1 so as a signed value it is negative.

The increment will result in a value of:

```
11111111
```

So Minus one using an 8 Bit representation in Two's Complement notation is 11111111

Let's test it by adding -1 to plus 3

```
11111111    -1
00000011 +   3
=====
00000010     2
```

We have successfully subtracted 1 from 3 by adding Minus 1 to 3 and obtaining a result of 2.

Notice that while a Byte is normally used to represent $0 < 255$ by making the MSB (Most Significant Bit) into a sign bit the maximum value is now 127. A signed 8 Bit integer can represent numbers in the range $-128 < 0 < 127$. That is still 256 values including Zero but they can now be Negative or Positive numbers.

The benefit of the two's complement method is that it works for any size of variable:

```
MyByte = NOT MyByte +1
MyWord = NOT MyWord +1
MyLong = NOT MyLong +1
```

All of the above will result in a Negated version of the original contents.

But not all, in fact relatively few, methods of a microcontroller require negative values so in situations where negative values are not required the loss of half of the magnitude of a Byte or Word can be significant. That is why it is necessary to be able to specify if a value is Signed or Unsigned, that is if the MSB is the sign bit or part of the value.

It is obviously important from the above that the user programs need to know what sort of data to expect as a value of 0xFF could be considered to be both 255 and -1 depending on the interpretation of the variable. That is why it is important to have Signed and Unsigned Data Types so that the compiler can decide how to handle or interpret the contents. As shown above in GCBASIC those types are referred to as Integer and Word respectively.

Summary

GCBASIC implements support for Bit, Byte, Word, Integer and Long Variable Types, all of which are described above.

And, that negative numbers are represented as two's complement.

Variable Types

About Variables and Variable Types

A variable is an area of memory on the microcontroller that can be used to store a number or a series of letters. This is useful for many purposes, such as taking a sensor reading and acting on it, or counting the number of times the microcontroller has performed a particular task.

Each variable must be given a name, such as "MyVariable" or "PieCounter". Choosing a name for a variable is easy - just don't include spaces or any symbols (other than _), and make sure that the name is at least 2 characters (letters and/or numbers) long.

Variable Types

There are several different types of variable, and each type can store a different sort of information. These are the variable types that GCBASIC can currently use:

Variable type	Information that this variable can store	Example uses for this type of variable
Bit	A bit (0 or 1)	Flags to track whether or not a piece of code has run
Byte	A whole number between 0 and 255	General purpose storage of data, such as counters
Word	A whole number between 0 and 65535	Storage of extra large numbers
Integer	A whole number between -32768 and 32767	Anything where a negative number will occur
Long	A whole number between 0 and $2^{32}-1$ (4.29 billion)	Storing very, very big numbers
Array	A list of whole numbers, each of which may be a byte, word, integer, or long	Logs of sensor readings

Variable type	Information that this variable can store	Example uses for this type of variable
String	A series of letters, numbers and symbols.	Messages that are to be shown on a screen

Using Variables

Byte variables do not need any special commands to set them up - just put the name of the variable in to the command where the variable is needed. However, it is good practice to "dimension" all byte variables and to use `#OPTION EXPLICIT`. `#OPTION EXPLICIT` mandates the "dimensioning" of all variables in the user program. Using `#OPTION EXPLICIT` will improve the quality of the program.

Other types of variable can be used in a very similar way, except that they must be "dimensioned" first. This involves using the DIM command, to tell GCBASIC that it is dealing with something other than a byte variable.

A key feature of variables is that it is possible to have the microcontroller check a variable, and only run a section of code if it is a given value. This can be done with the IF command.

Number Variables

You can assign values to number variables using ` `= ``.

A simple, but typical example follows. This is the typical for numeric variable assignment.

```
#OPTION EXPLICIT

dim myByteVariable as Byte
myByteVariable = 127      'assign the value of 127
```

GCBASIC support bitwise assignments as follows:

```
portc.0 = !porta.1  'set a single bit to the value of another bit
```

The function `FnLSL` performs the shift operation found in other languages. Here is an example:

```
MyVar = FnLSL( 1, BitNum)`  is Equivalent to `MyVar = 1<<BitNum`
```

To set a bit of a port and to prevent glitches during operations, use `#option volatile` as follows:

```
'add this option for a specific port.  
#option volatile portc.0  
  
'then in your code  
portc.0 = !porta.1
```

To set a bit of a port or variable, encapsulate it in the **SetWith** method. Using this method also eliminates any glitches during the update.

```
SetWith(MyPORT, MyPORT OR FnLSL( 1, BitNum))
```

To clear a bit of a port, use this method:

```
MyPORT = MyPORT AND NOT FnLSL( 1, BitNum))
```

To set a bit within an array, use this method:

```
video_buffer_A1(video_adress) = video_buffer_A1(video_adress) OR FnLSL( 1, BitNum)
```

To set a bit within a variable, use this method:

```
Dim my_variable as byte  
Dim my_bit_address_variable as byte  
  
'example  
my_variable = 0  
my_bit_address_variable = 7  
  
my_variable.my_bit_address_variable = 1    ' where 1 or 0 or any bit address is valid  
  
'Sets bit 7 of my_variable therefore 128
```

String Variables

Strings are defined as follows:

```
'Create buffer variables to store received messages  
  
Dim Buffer As String
```

String variables default to the following rules and the RAM constraints of a specific chip.

- 10 bytes for chips with less than 16 bytes of RAM.
- 20 bytes for chips with 16 to 367 bytes of RAM.
- 40 bytes for devices with more RAM than 367 bytes.
- For chips that have less RAM than the required RAM to support the user define strings the strings (and therefore the RAM) will be NOT be allocated. Please reduce string size.

You cannot store a string 20 characters long in a chip with 16 bytes of RAM.

You can change the default string size handled internally by the GCBASIC compiler by changing the **STRINGSIZE** constant:

```
'set the default string to 24 bytes
#define STRINGSIZE 24
```

Defining a length for the string is the best way to limit memory usage. It is good practice if you need a string of a certain size to set the length of a strings, since the default length for a string variable changes depending on the amount of memory in the microcontroller (see above).

To set the length of a string, see the example below:

```
'Create buffer variables to store received messages as 16 bytes long
Dim OutBuffer As String * 16
```

To place quotation marks (") in a string of text. For example:

```
She said, "You deserve a treat!"
```

To place quotation marks (") in a string of text, use two quotation marks in a row instead of one for each quote mark. The following example shows two ways of printing **She said, "You deserve a treat!"**. This technique works for all output methods (HSerPrint, Print, etc.)

```
HSerPrint "She said, ""You deserve a treat!"" "
```



```
Dim myString As String * 39
myString = "She said, ""You deserve another treat!"" "
```

```
HSerPrint myString
```

Variable Aliases

Some variables are aliases, which are used to refer to memory locations used by other variables. These

are useful for joining predefined byte variables together to form a word variable.

Aliases are not like pointers in many languages - they must always refer to the same variable or variables and cannot be changed.

When setting a register/variable bit (i.e `my_variable.my_bit_address_variable`) and using a alias for the variable then you must ensure the bytes that construct the variable are consecutive.

The coding approach should be to DIMension the variable (word, integer, or long) first, then create the byte aliases:

```
Dim my_variable as LONG
Dim ByteOne   as Byte alias my_variable_E
Dim ByteTwo   as Byte alias my_variable_U
Dim ByteThree as Byte alias my_variable_H
Dim ByteFour  as Byte alias my_variable

Dim my_bit_address_variable as Byte
my_bit_address_variable = 23

'set the bit in the variable
my_variable.my_bit_address_variable = 1

'then, use the four byte variables as you need to.
```

To set a series of registers that are not consecutive, it is recommended to use a mask variable then apply it to the registers:

```
Dim my_variable as LONG
Dim my_bit_address_variable as Byte
my_bit_address_variable = 23

'set the bit in the variable
my_variable.my_bit_address_variable = 1

porta = my_variable_E
portb = my_variable_E
portc = my_variable_E
portd = my_variable_E
```

Casting

Casting changes the type of a variable or value. To tell the compiler to perform a type conversion, put the desired type in square brackets before the variable. The following example will cause two byte variables added together to be treated as a word variable.

```
Dim MyWord As Word
MyWord = [word]ByteVar + AnotherByteVar
```

Why do this? Suppose that **ByteVar** is 150, and **AnotherByteVar** is 231. When added, this will come to 381 - which will overflow, leaving 125 in the result. However, when the cast is added, GCBASIC will treat **ByteVar** as if it were a word, and so will use the word addition code. This will cause the correct result to be calculated.

It is good practice to cast when calculating an average:

```
MyAverage = ([word]Value1 + Value2) / 2
```

It's also possible to cast the second value instead of the first:

```
MyAverage = (Value1 + [word]Value2) / 2
```

The result will be exactly the same.

To apply operations to individual bits of variables see, [Set](#), [Rotate](#)

To check variables and apply logic based on their value, see [If](#), [Do](#), [For](#), [Conditions](#)

For more help, see: [Declaring variables with DIM](#), [Setting Variables](#)

Advanced VariableTypes

About Advanced Variable Types

A variable is an area of memory on the microcontroller that can be used to store a number or other data. This is useful for many purposes, such as taking a sensor reading and acting on it, or counting the number of times the microcontroller has performed a particular task.

Each variable must be given a name, such as "MyVariable" or "PieCounter". Choosing a name for a variable is easy - do not include spaces or any symbols (other than `_`), and make sure that the name is at least 2 characters (letters and/or numbers) long.

Advanced Types

There are a number different types of advanced variable types, and each type can store a different range of numeric information.

With respect to advanced variables GCBASIC supports:

- single floats which can be signed and unsigned.

With respect to used advanced variables - please use Singles in your program as these have been tested. The other types are documented for completeness and should be used by developers in libraries.

- double floats, and the large integers which can be signed & unsigned

Using advanced variable type maths is also much slower than integer maths when performing calculations and loops, therefore should be avoided if. You should convert float calculations to integer maths to increase operation of your solution. The example program (shown below) shows how use a float maths and you should try to do the same with integers and time the overall time for comparison. Typically, floats are 18%-20% slower than similar integer maths operations.

The advanced variable types that GCBASIC supports are:

Advanced Variable type	Supported	Information that this variable can store	Example uses for this type of variable
Single	Yes	A numeric floating point values that range from -3.4×10^{38} and $+3.4 \times 10^{38}$ with up to seven significant digits.	Storing decimal numbers that could be a negative number and positive.
Developers Only	Developers Only	Developers Only	Developers Only
LongINT	Libraries only	A list of whole numbers between $-(2^{63})$ and $2^{63} - 1$	Storing very, very big integer numbers that could be a negative number. The GCBASIC range is -9999999999999990 to 9999999999999990. This range is an implementation constraint with the GCBASIC compiler.
uLongINT	Libraries only	A whole number between 0 and $2^{64} - 1$	Storing very, very, very big integer numbers
Double	Libraries only	A numeric floating point values that range from -1.7×10^{308} and $+1.7 \times 10^{308}$ with up to 15 significant digits.	Storing decimal numbers that could be a negative number and positive.

The format for single and double floats is defined by the IEEE 754 standard. Sign, exponent and mantissa are all in the positions described here: <https://www.geeksforgeeks.org/ieee-standard-754-floating-point-numbers/>

Organisation of advanced variables

GCBASIC stores advanced variables in bytes. The format of these bytes is:

```
_ D, _C, _B, _A, _E, _U, _H, variable_name ( from high to Low)
```

You can access the bytes within advanced variables using the following as a guide using the suffixes _A, _B, _C etc.

Example of accessing the lowest byte, the _H, _U and the _A bytes.

```
Dim workvariable as Single
workvariable = 21845
Dim lowb as byte
Dim highb as byte
Dim upperb as byte
Dim lastb as byte

lowb = workvariable
highb = workvariable_H
upperb = workvariable_U
lastb = workvariable_A
```

Using the Byte components of Advanced Variables

This is strict. Accessing BYTE values of advanced variables requires the use cast. Failure to use cast will cause issue with the low byte (the low byte will tranformed into a Long integer and you will provide the low byte of the Long integer).

Example. Mandated use of cast for single/float

```
Dim sNumC as Single

HserPrint "Hex with [CAST] / "
HserPrint "0x"
HserPrint Hex([BYTE]sNumC_E)
HserPrint Hex([BYTE]sNumC_U)
HserPrint Hex([BYTE]sNumC_H)
HserPrint Hex([BYTE]sNumC)
HserPrintCRLF
```

Example assigning a HEX value to a single/float


```
// Assing 0x3F19999A equates to 0.6

[BYTE]mySingle = 0x9A      // Strict usage of BYTE cast to ensure the correct value
is assinged the low byte of the single variable.
mySingle_H= 0x99           // Assign _H byte
mySingle_U= 0x19           // Assign _U byte
mySingle_E= 0x3f           // Assign _E byte
```

Working example of assigning d0.5 or 0x3F000000 (which is the IEEE574 hex value for d0.5)

```
// Decimal assignement
mySingle = 0.5

// Hex assignment
[BYTE]mySingle = [single]0x00
mySingle_H     = 0x00
mySingle_U     = 0x00
mySingle_E     = 0x3f
```

Using Advanced Variables

Advanced variables must be "DIMensioned" first. This involves using the DIM command, to tell GCBASIC that it is dealing with an advanced variable.

```
Dim mySingle as Single
mySingle= 1.1

// The following types are for Libraries only

Dim myLongInt as LongInt
myLongInt = 9999999999999999          'see the Help for constraints

Dim myuLongInt as uLongInt
myuLongInt = 0xFFFFFFFFFFFFFFFF      'see the Help for constraints

Dim myDouble as Double
myDouble=3.141592
```

Using Advanced Variables

Advanced variables are only supported by a subset of the functions of GCBASIC.

The functional characteristics are:

- Dimensioning of longInt, ulongInt, single and double advanced variable types.
- Assigning advanced variables creation of values from constants.
- Assigning a single to double and double to single.
- Assigning single to long and long to single.
- Assigning double to long and long to double.
- The assignment of a single or a double to a long also deals with byte and word. This is very inefficient.
- Copying between variables of the same type (so double to double, and single to single and other advanced variables).
- Extract of the unit value of a single or double variable to a long variable.
- Setting of advanced variable bits.
- Addition and subtraction of advanced variables.
- Rotate of longInt & ulongInt advanced variables.
- Negate of longInt & ulongInt advanced variables.
- Boolean operators working on advanced variables.
- Use of float variable(s) as global variables. Passing float variable(s) as parameters to methods (sub, function and macro) not supported.
- Support for conditional statements
- Support for overload subs/functions
- Passing float variable(s) as parameters to methods (sub, function and macro)
- Extraction of mantissa value
- Multiplication
- Division
- Modulo
- SingleToString
- StringToSingle
- Advanced variable(s) to string functions
- Math functions for float variable(s) (plus pseudo functions shown below)

Assigning Values to Advanced Variables

You can assign values to advanced variables using ``='``.

A simple, but typical example follows. This is the typical for numeric variable assignment.

```
Dim mySingle as Single  
mySingle = 123.4567      'assign the value
```

Another example is bitwise assignments as follows:

```
mySingle.16 = 1  'set the single bit to 1
```

+

INT() and SINGLRROUND()

Floating point numbers are not exact, and may yield unexpected results when compared using conditions (IF etc). For example $6.0 / 3.0$ may not equal 2.0. Users should instead check that the absolute value of the difference between the numbers is less than some small number.

These techniques can replace the INT() and SINGLEROUND() functions.

Alternative to INT()

Assignment of a Single variable to an Integer variable is supported.

So, use the conversion from floating point to integer as this results in integer truncation.

```
dim mySingleVar as Single  
mySingleVar = 2.9  'A float type variable  
  
dim myLongVar as Long  
myLongVar = mySingleVar ' will set myLongVar to 2
```

Alternative to ROUNDSSINGLE()

As an alternative to using the ROUNDSSINGLE() function.

Create your own round conversion, add 0.5 to return the nearest integer. As follows:

'Add 0.5 to a single or double and then assign to an integer variable

```
dim mySingleVar as Single  
mySingleVar = 2.9
```

```
dim myLongVar as Long  
myLongVar= mySingleVar + [single]0.5
```

Example Program

This program shows the values of calculation of $4.5 \times$ multiplied by a number ($4.5 \times$ a range of 0 to 40,000). The program shows setting up the advanced variables, assigned a value and completing the multiplication of the initial value using a for-next loop.

```

HSerPrintCRLF 2
HSerPrint "Maths test "
HSerPrintCRLF 2

DIM multiplier as Word
DIM ccount as Single
Dim result as Single

HSerPrint "Use floats with multiplier maths"
HSerPrintCRLF

'Assign a value to the variable
ccount = 4.5

'Do some maths... multiplier x ccount
For multiplier = 0 to 40000 step 2500

    HSerPrint SingleToString(ccount)
    HSerPrint " x "
    HSerPrint left(WordToString(multiplier)+", 10 )
    HSerPrint " = "

    'Calculate the result
    result = multiplier * ccount
    HSerPrint left(SingleToString(result)+", 10 )
    HSerPrintCRLF
next

Do Forever
Loop

```

To check variables and apply logic based on their value, see [If](#), [Do](#), [For](#), [Conditions](#)

For more help, see: [Declaring variables with DIM](#), [Setting Variables](#)

Variable Memory Allocation, Addressing & Control

This section discusses the allocation of variables to RAM (GPR, SRAM or other TLA).

Variables in GCBASIC can be bits, bytes, words, integers, longs, arrays or reals. This section will NOT address reals as these are developmental variables only.

Variables can also be defined as Aliases - this is discussed later in this section.

Basic variable allocation

Variables of type *byte*, *word*, *integer*, *longs* are placed in RAM using the following simple rules.

1. A RAM memory location is automatically assigned starting at the first available memory location.
2. The first memory location is first RAM location as defined in the chip datasheet.
3. Once a variable is allocated the RAM location is marked as used and this specific location can be reviewed in the ASM source.
4. Bytes use a single RAM location, words two RAM locations, integer and longs four RAM locations.
5. Subsequent variables of type byte, word, integer, longs are placed in RAM at the next available RAM location.

Variables of *array* and *strings* type are placed in RAM using the following simple rules.

1. A RAM memory location is automatically assigned from the end of RAM less the (size of the array + 1 byte).
2. The last memory location is last RAM location as defined in the chip datasheet.
3. Once an array is allocated the RAM location is marked as used and the start of the array RAM location can be reviewed in the ASM source.
4. Subsequent variables of type array in RAM at the next available RAM location subtracted from the start the of previous RAM location minus the size of this next array.

Variables of *bit* type are placed in RAM using the following simple rules.

1. Bit memory location is automatically assigned to the first bit with the creation of a BYTE variable at a RAM memory location that is automatically assigned starting at the first available memory location. This byte can handle 8 bits.
2. Once a bit is allocated the byte is marked as used and this specific location can be reviewed in the ASM source.
3. Subsequent bits are allocation either to an existing byte variable, or when 8 bits are allocated to an existing byte variable another byte variable will be created.

Addressing Variables

Addressing a variable memory address can be achieved by using the @ prefix. This will return the address of the variable (@ applies to table data and any data block).

The following example shows registers DManSSAU, DManSSAH, DManSSAL being loaded with the address of the array WaveArray.

```
' Source start address
Dim addressdummy as byte
Dim DManSS as long ALIAS addressdummy, DManSSAU, DManSSAH, DManSSAL
DManSS = @WaveArray
```

AT allocation

The Dim variable command can be used to instruct GCBASIC to allocate variables at a specific memory location using the parameter AT.

The compiler will inspect the provided AT memory location and if the memory location is already used (by an existing variable), lower than the minimum memory location or greater than the maximum memory location an error will be issued.

Variable Aliases

Variable can be defined as aliases. Aliases are used to refer to existing memory locations SFR or RAM and aliases can be used to construct other variables. Constructed variables can be a mix (or not) of SFR or RAM. These are useful for joining predefined byte variables together to form a word/long variable.

Aliases are not like pointers in many languages - they must always refer to an existing variable or variables and cannot be changed.

When setting a register/variable bit (i.e my_variable.my_bit_address_variable) and using a alias for the variable then you must ensure the bytes that construct the variable are consecutive.

Aliases are shown in the ASM source in the ;ALIAS VARIABLES section.

The coding approach should be to DIMension the variable (word, integer, or long) first, then create the byte aliases:

```
Dim my_variable as LONG
Dim ByteOne   as Byte alias my_variable_E
Dim ByteTwo   as Byte alias my_variable_U
Dim ByteThree as Byte alias my_variable_H
Dim ByteFour  as Byte alias my_variable

Dim my_bit_address_variable as Byte
my_bit_address_variable = 23

'set the bit in the variable
my_variable.my_bit_address_variable = 1

'then, use the four byte variables as you need to.
```

To set a series of registers that are not consecutive, it is recommended to use a mask variable then apply it to the registers:

```

Dim my_variable as LONG
Dim my_bit_address_variable as Byte
my_bit_address_variable = 23

'set the bit in the variable
my_variable.my_bit_address_variable = 1

porta = my_variable_E
portb = my_variable_E
portc = my_variable_E
portd = my_variable_E

```

Memory Specification

All memory specifics like RAM size, lower and upper RAM addresses are specified in the chip specific dat file.

The dat file details should be reviewed in PICINFO application. See the PICINFO/CHIPDATA tab for RAM and MaxAddress etc.

A simple calculation is $\text{MaxAddress} - \text{RAM} + 1$ = the 'first memory address'. And, 'first memory address' + RAM -1 = 'the last memory address'.

This can be confirmed by review the DAT file. See the section [FreeRAM] for the start and end of RAM.

The dat file also has a [NoBankRAM]. NoBankRAM is somewhat misnamed - it is used for the definition of (any) access bank locations. If a memory location is defined in both NoBankRAM and FreeRAM, then the compiler knows that it is access bank RAM. If an SFR location is in one of the NoBankRAM ranges, then the compiler knows not to do any bank selection when accessing that register.

The [NoBankRAM] section includes two ranges, one for access bank RAM, one for access bank SFRs. The first range MUST be the ACCESS RAM range The first range is the FAST SFR range

If there are no ranges defined in NoBankRAM, the compiler will try to guess them. On 18Fs, it will guess based on where the lowest SFR is, and from what the total RAM on the chip is. If there's only one range defined. in the NoBankRAM locations, the compiler will assume that is the range for the RAM, and then will guess where the range for the access bank SFRs is.


```

'GCBASIC/GCGB Chip Data File
'Chip: 18F27Q43

[ChipData]

.... many other data rows

'This constant is exposed as ChipRAM
RAM=8192          'Dec values

.... many other data rows

'This constant is exposed as ChipMaxAddress
MaxAddress=9471   'Dec values

.... many other data rows

[FreeRAM]
500:24FF          'Hex value

[NoBankRAM]
500:55F           'Hex value
460:4FF           'Hex value

.... many other data rows

```

In the example shown above the following can be extracted.

1. RAM size: RAM = 8192d
2. Minimum RAM address: FREERAM = 0x500
3. Maximum RAM address: FREERAM = 0x24FF
4. Maximum RAM address: MAXADDRESS=9471d or 0x24FF
5. ACCESS RAM: NOBANKRAM = 0x500-0x55F
6. BANKED SFR: NOBANKRAM = 0x460-0x4FF

Reference Data

Efficient Implementation of Lookup Reference Tables in GCBASIC

Introduction

This section explores the efficient implementation of lookup reference tables in embedded systems, specifically GCBASIC, focusing on the use of PROGMEM memory to store fixed data sets. It addresses common misconceptions about data storage and initialization, compares different methods of data handling, and provides advanced techniques for optimizing memory usage.

Lookup reference tables are essential in embedded systems for storing fixed data sets that can be accessed during runtime. This section aims to clarify the correct implementation of these tables, debunking common myths and providing practical solutions for efficient data management.

Conventional Misconceptions

A common misconception is that the data required by a fixed lookup table is defined by its content and declared in a Dim statement, with its data filled at runtime. This implies that an array (in RAM memory) is empty initially and populated during initialization, leading to data duplication and wasted memory resources.

Correct Implementation

A fixed lookup table is a set of data (bytes, words, etc.) stored in the PROGMEM memory. The correct implementation involves using the TABLE and READTABLE commands: * Definition: `TABLE tablename... data... END TABLE` * Reading: `READTABLE` There is no DIM in the definition process, and the data is part of the hex file, not filled at runtime.

Memory Efficiency

Storing data in PROGMEM ensures that there is only one copy of the data, avoiding duplication. Copying data to an array is redundant as reading the table can replace the array.

Practical Solutions

Using `TABLE - END TABLE` is the simplest way to handle data. For data sets smaller than the EEPROM in the chip, load the table directly to EEPROM and use `READTABLE` to read the data.

Advanced Techniques

1. PROGMEM Page Size Constraint: On the 16F, `TABLE-END TABLE` is constrained by PROGMEM page size (2048 items).
2. EEPROM Storage: Use `EEPROM .. END EEPROM` and a direct method like `PROGREAD`. Data is stored in

EEPROM, constrained by its size and typically byte values.

3. Direct PROGMEM Storage: Use `DATA .. END DATA` and `PROGREAD`. Data is stored in PROGMEM, constrained by unused PROGMEM size and typically word values (max 0x3FFF for 16F chips).

Arrays in Embedded Systems

An array is a special type of variable that can store multiple values, addressed individually using an index. Arrays can be bytes, longs, integers, or words, and are held in RAM. Loading an array can be done element by element or all at once.

Comparison of Methods

Using arrays can be costly in terms of RAM and PROGMEM. The following examples illustrate the difference:

Using an Array

64 words Progmem / 13 bytes RAM

```
#CHIP 18F2550
#option Explicit

Dim myResult, myIndex as Byte

// Using an array 64 words Progmem / 13 bytes RAM
Dim myArray(10)
  myArray = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
For myIndex = 1 to 10
  myResult = myArray(myIndex)
Next
```

Using a Table

56 words Progmem / 2 bytes RAM

```
#CHIP 18F2550
#option Explicit

Dim myResult, myIndex as Byte

// Using a table 56 words Progmem / 2 bytes RAM
For myIndex = 1 to 10
    ReadTable myTable, myIndex, myResult
Next

Table myTable
    1, 2, 3, 4, 5, 6, 7, 8, 9, 10
End Table
```

Usage

The **ReadTable** method provides data set capabilities to chips with limited RAM, using fewer resources and offering faster performance. Advanced techniques and proper understanding of memory usage can significantly optimize embedded system performance.

Notes

- A byte array is handled similarly to a string, which can be resource-intensive.

By following these guidelines, developers can efficiently implement lookup reference tables in embedded systems, optimizing memory usage and performance.

Syntax

Arrays

About Arrays

An array is a special type of variable - one which can store several values at once. It is essentially a list of numbers in which each one can be addressed individually through the use of an "index".

The numbers can be bytes (the default), longs, integers, or words. The index is a value in brackets immediately after the name of the array.

All the numbers stored in an array must be of the same type. For instance, you cannot store bytes and words in the same array.

Array are are 1-based. The first element is element zero.

Examples of array names are:

Array/Index	Meaning
<code>Fish(10)</code>	Definition of an array containing bytes with 10 elements called <code>Fish</code>
<code>x(5) As Word</code>	Definition of an array containing words with 5 elements called <code>x</code>
<code>DataLog(2)</code>	The second element in an array named <code>DataLog</code>
<code>ButtonList(Temp)</code>	An element in the array <code>ButtonList</code> that is selected according to the value in the variable <code>Temp</code>

Defining an array

Use the DIM command to define an array.

```
DIM array_title ( number_of_elements ) [As _type_]
```

The number of elements can be number or a constant - not a variable.

The value for the number elements in an array must be a number or constant. The compiler allocates RAM for arrays at compile time, and therefore you cannot use a variable because during compilation the value of a variable cannot be determined.

Assigning values to an array

It is possible to set several elements of a byte array with a single line of code. This short example shows how:

```
Dim TestVar(10)
TestVar = 1, 2, 3, 4, 5, 6, 7, 8, 9
```

When using this method above element 0 of the array TestVar will be set to the number of items in the list, which in this case is 9. Each element of the array will then be loaded with the corresponding value in the list - so in the example, TestVar(1) will be set to 1, TestVar(2) to 2, and so on. Element 0 will only be set to number of items in the array when using this method. For microcontrollers with less than 2048 bytes of RAM the limit is 250 elements or the array cannot exceed the microcontrollers RAM size. For microcontrollers with more than 2048 bytes of RAM the limit is 255 elements.

This only works for **byte** arrays, however. For arrays of type **integer**, **word**, or **long**, each element must be set separately:

```
Dim TestVar(5) As Word
TestVar(1) = 20
TestVar(2) = 50
TestVar(3) = 60
TestVar(4) = 80
TestVar(5) = 100
```

If each element has the same value, this can be shortened using a loop:

```
Dim TestVar(5) As Word
For i = 1 to 5
    TestVar(i) = 0
Next
```

Array Length

Element 0 should not be used to obtain the length of the array. Element 0 will only be a consistent with respect to the length of the array when the array is set as shown above.

The correct method is to use a constant to set the array size and use the constant within your code to obtain the array length.

```
#Define ArraySizeConstant 500
Dim TestVar( ArraySizeConstant )

SerPrint ArraySizeConstant    'or, other usage
```

Using Arrays

To use an array, its name is specified, then the index. Arrays can be used everywhere that a normal variable can be used.

Maximum Array Size

The limit on the array size is dependent on the chip type, the amount of RAM, and the number of other variable you use in your program.

Use the following simple program to determine the maximum array size. Set **CHIP** to your device, **MAXSCOPE** to a value which is less the total RAM, and the data type of **test_array** to the data type to be stored in the array.

The data type of **imaxscope** must be set to match the size of the constant **MAXSCOPE**. If **MAXSCOPE** $\leq$ 255, **imaxscope** should be a byte. If **MAXSCOPE** > 255, **imaxscope** should be a word.

If the array is too large to fit, the compiler will issue an error message. Reduce **MAXSCOPE** until the error message is not issued. The largest **MAXSCOPE** value without an error message is the largest useable array of this type for this chip.

```
#CHIP 12f1571
#OPTION Explicit

#define MAXSCOPE 111
DIM imaxscope As Byte
DIM test_array( MAXSCOPE ) As Byte

For imaxscope = 0 to MAXSCOPE
    test_array( imaxscope ) = imaxscope
Next
```

For the Atmel AVR, LGT 328p or an 18F array sizes are limited to 10,000 elements.

If a memory limit is reached, the compiler will issue an error message.

Get the most from the available memory

Array RAM usage is determined by the architecture of the chip type. Getting most out of the available memory is determined by the allocation of the array within the available banks of memory.

An example is an array of 6 or 7 bytes when there is only 24 bytes of RAM and the 24 bytes is split across multiple memory banks. Assume in this example that 18 bytes have allocated to other variables and there is 29 bytes total available. An array of 6 bytes will fit into the free space in one bank, but the array of 7 will not.

GCBASIC currently cannot split an array over banks, so if there are 6 bytes free in one bank and 5 in another, you cannot have an array of 7 bytes. This would be very hard to do efficiently on 12F/16F as there would be a series of special function registers in the middle of the array when using a 12F or 16F.

This constraint is not the case on 16F1/18F as linear addressing makes it easy to span banks because the SFRs are not making the problem (as with 12F/16F).

Using Tables as an alternative.

If there are many items in the array, it may be better to use a Lookup Table to store the items, and then copy some of the data items into a smaller array as needed.

For more help, see [Declaring arrays with DIM](#), [Declaring memory with ALLOC](#)

Comments

About Comments

Adding comments to your GCBASIC program can be done using a number of methods. Explanatory notes embedded within the code. Comments are used to remind yourself and to inform others about the function of your program. Comments are ignored by the compiler

You can comment out sections of code if you want just by placing an apostrophe at the beginning of each line. The GCBASIC IDE has a feature to do this automatically.

You can also use a REM (for REMark statement), a semi-colon or two forward slashes.

Multiline comments are support for large text descriptions of code or to comment out chunks of code while debugging applications.

Syntax:

```
/*  
    block comment  
*/
```

Warning: Graphical GCBASIC uses semi-colons to mark comments that it has inserted automatically. It does not read these comments when opening a file, so any comments in a GCBASIC program starting with a semi-colon will be deleted if the program is opened using Graphical GCBASIC.

Example:


```

' The number of pins to flash
#define FlashPins 2

REM You can create a header using an apostrophe before each line
REM This is a great way to describe your program
REM You can also use it to describe the hardware connections.

' You can place comments above the command or on the same line
Dir PORTB Out ' Initialise PORTB to all Outputs

; The Main loop
do
PORTB = 0 ' All Pins off
Wait 1 S ' Delay 1 second
PORTB = 0xFF ' All pins on
Wait 1 s ' Delay 1 second
Loop

```

Line Continuation

About Line Continuation

A single `_` (underscore) character at the end of a line of code tells the compiler that the line continues in the next line. This allows a single statement (line of code) to be spread across multiple lines in the input file, which can provide nice formatting.

Be careful when adding the `_` line continuation character right behind an identifier or keyword. It **MUST** be separated with at least **one space** character, otherwise it would be treated as part of the identifier or keyword.

Example 1:

```
#CHIP 18f27k42
```

```
Dim sMyString As String  
  sMyString ="one _  
              two _  
              three _  
              four _  
              five _  
              six _  
              seven _  
              eight _  
              nine _  
              ten _  
              eleven _  
              twelve _  
              thirteen _  
              fourteen _  
              fifteen _  
              sixteen _  
              seventeen _  
              eighteen _  
              nineteen _  
              twenty _  
              twentyOne _  
              twentyTwo _  
              twentyThree _  
              twentyFour _  
              twentyFive"
```

```
HSerPrint sMyString
```

This example will print on the serial terminal the string "one two three four five six seven eight nine ten eleven twelve thirteen fourteen fifteen sixteen seventeen eighteen nineteen twenty twentyOne twentyTwo twentyThree twentyFour twentyFive"

Example 2:

```

Sub Aiguillages (In voie_principale As Byte, _
                In voie_marchandises As Byte, _
                In voie_gravier As Byte)

    ' code segment
    ' code segment
    ' code segment

End Sub

```

This example improves the layout of definition of the sub-routine.

Example 3:

```

#DEFINE Ouvrir_voie_marchandises Aiguillages _
    (0, Marche_avant, Marche_arriere)

```

This example creates a constants over two lines. This improves readability.

Conditions

About Conditions

In GCBASIC (and most other programming languages) a condition is a statement that can be either true or false. Conditions are used when the program must make a decision. A condition is generally given as a value or variable, a relative operator (such as = or >), and another value or variable. Several conditions can be combined to form one condition through the use of logical operators such as AND and OR.

GCBASIC supports these relative operators:

Symbol	Meaning
=	Equal
<>	Not Equal
<	Less Than
>	Greater Than
≤	Less than or equal to
≥	Greater than or equal to

In addition, these logical operators can be used to combine several conditions into one:

Name	Abbreviation	Condition true if
AND	&	both conditions are true
OR		at least one condition is true
XOR	#	one condition is true
NOT	!	the condition is not true

NOT is slightly different to the other logical operators, in that it only needs one other condition. Other arithmetic operators may be combined in conditions, to change values before they are compared, for example.

GCBASIC has two built in conditions - TRUE, which is always true, and FALSE, which is always false. These can be used to create Conditional tests and infinite loops.

The condition `bit_variable = TRUE` is treated as TRUE if the bit is on. Any non-zero value will be treated as equal to a high bit. The condition `bit_variable = other_type_of_variable` generates a warning. If the `byte_variable` is set to TRUE and then compared to the bit, it will always be FALSE because the high bit will be treated as a 1. But the new warning will be generated, "Comparison will fail if %nonbit% is any value other than 0 or 1"

It is also possible to test individual bits in conditions. To do this, specify the bit to test, then 1 or 0 (or ON and OFF respectively). Presently there is no way to combine bit tests with other conditions - NOT, AND, OR and XOR will not work.

Example conditions:

Condition	Comments
Temp = 0	Condition is true if Temp = 0
Sensor <> 0	Condition is true if Sensor is not 0
Reading1 > Reading2	True if Reading1 is more than Reading2
Mode = 1 AND Time > 10	True if Mode is 1 and Time is more than 10
Heat > 5 OR Smoke > 2	True if Heat is more than 5 or Smoke is more than 2
Light >= 10 AND (NOT Time > 7)	True if Light is 10 or more, and Time is 7 or less
Temp.0 ON	True if Temp bit 0 is on

Constraints when using Conditional Test

As GCBASIC is very flexible with the use of variables type this can cause issues when testing constants

and/or functions.

A few simple rules. **Always put the function or constant first, or, always call the function with the addition of the braces.**

The example code below shows the correct method and an example that does compile but will not work as expected.

```
'Example A - works
'Call the function by adding the braces
,
Do
Loop While HSerReceive() <> 62

'Example B - works
'Please the constant first - this is the general rule - put the constant first.
,
Do
Loop While 62 <> HSerReceive
```

This fails as the function will not be called

```
'Example C - compiles but does not operate as expected
Do
Loop While HSerReceive <> 62
```

Constants

About Constants

A constant tells the compiler to find a given string, and replace it with another string. The **#Define** directive create constants.

Constants are useful for situations where a routine needs to be easily altered. For example, a define could be used to specify the amount of time to run an alarm for once triggered.

It is also possible to use defines to specify port.pin(s) - thus defines can be used to aid in the creation of code that can easily be adapted to run on a different microcontroller with different ports.

GCBASIC makes considerable use of defines internally. For instance, the LCD code uses defines to set the ports that it must use to communicate with the LCD.

About Defines

To create a constant is a matter of using the #define directive. Here are some examples of defines:

```
#define LINE 34
#define LIGHT PORTB.0
#define LIGHTON Set PORTB.0 ON
```

LINE is a simple constant - GCBASIC will find **LINE** in the program, and replace it with the number 34. This could be used in a line following program, to make it easier to calibrate the program for different lighting conditions.

LIGHT is a port.pin - it represents a particular pin on the microcontroller. This would be of use if the program had many lines of code that controlled the light, and there was a possibility that the port the light was attached to would need to change in the future.

LIGHTON is a define used to make the program more readable. Rather than typing **Set PORTB.0 ON** over and over, it would then be made possible to type **LIGHTON**, and have the compiler do the hard work.

GCBASIC Defined constants

There are many GCBASIC standard constants, some are shown below.

```
#define ON 1
#define OFF 0
#define TRUE 255
#define FALSE 0

'Names for symbols
#define AND &
#define OR |
#define XOR #
#define NOT !
#define MOD %
```

A GCBASIC special constant

FOREVER is a special constant. For Graphical GCBASIC users think of this as 'false'. For those not using Graphical GCBASIC think of this as a non numeric value that has no value. You can use **FOREVER** in a DO-LOOP but not in a REPEAT-END REPEAT loop, as in the later case the REPEAT will have no value and you will create an error condition.

Precedence of Constants within GCBASIC.

The `#define` command creates constants, and, scripts also creates constants. See the `#script` section of the Help.

The precedence is as follows:

- Constants defined in the main user program is read first,
- then, the constants defined in the include files. Constants defined in the include files are ignored if they conflict or are different to a constant, with the same title, in the main program.
- then, the scripts are processed. Scripts that create constants can therefore override any constant value previously defined.

All constants are listed in the Constant Debug File (CDF). The CDF shows all constants and the end state of the constants.

See `#define`

Functions

```
Function identifier [( arg1 [ as Type ], arg2... argx) ] [As return_type]
    statements
    ...
    identifier = return_value
    ...
End Function
```

About Functions

Functions are a special type of subroutine that can return a value. This means that when the name of the function is used in the place of a variable, GCBASIC will call the function, get a value from it, and then put the value into the line of code in the place of the variable.

Functions are strict. The function **MUST** be assigned to an appropriate variable or passed to another subroutine. Calling a function with no assignment or returned value usage will raise an error condition.

Functions may have parameters - these are treated in exactly the same way as parameters for subroutines. The only exception is that brackets are required around any parameters when calling a function. The argument's type is given by "As type" following the parameter. If a parameter in the declaration is given a default value, the parameter is optional. Array parameters are specified by following an identifier with an empty parenthesis.

Returning values : `return_type` specifies the data type returned by a function upon exit. If no data type is specified, then the function will return the default data type which is a byte. Functions return values by assigning the Function keyword or the function's `identifier` to the desired return value, this method do not cause the function to exit, however.

Exit Function keyword in a same function is unsupported when returning objects with constructors. Since functions return values, function calls evaluate to expressions. Thus, function calls can be made wherever an expression is expected, like in Assignments or If statements. Parentheses surrounding the argument list are required on function calls in expressions and even highly recommended if there are no arguments.

Using Functions

This program uses a function called `AverageAD` to take two analog readings, and then make a decision based on the average:

```
'Select chip
#chip 16F88, 20

'Define ports
#define LED PORTB.0
#define Sensor AN0

'Set port directions
dir LED out
dir PORTA.0 in

'Main code
Do
    Set PORTB.0 Off
    If AverageAD > 128 Then Set PORTB.0 On
    wait 10 ms
Loop

Function AverageAD
    'Get 2 readings, divide by 2, store in AverageAD
    'Note the cast, the result of ReadAD needs to be converted to
    'a word before adding, or the result may overflow.
    AverageAD = ([word]ReadAD(Sensor) + ReadAD(Sensor)) / 2
end function
```

See Also [Subroutines](#), [Exit](#)

Labels

About Labels

Labels are used as markers throughout the program. Labels are used to mark a position in the program to ‘jump to’ from another position using a goto, gosub or other command.

Labels can be any word (that is not already a reserved keyword) and may contain digits and the underscore character. Labels must start with a letter or underscore (not digit), and are followed directly by a colon (:) at the marker position. The colon is not required within the actual commands.

The compiler is not case sensitive. Lower and/or upper case may be used at any time.

Example:

```
'This program will flash the light until the button is pressed
'off. Notice the label named SWITCH_OFF.

#chip 16F628A, 4

#define BUTTON PORTB.0
#define LIGHT PORTB.1
Dir BUTTON In
Dir LIGHT Out

Do
PulseOut LIGHT, 500 ms
If BUTTON = 1 Then Goto SWITCH_OFF
Wait 500 ms
If BUTTON = 1 Then Goto SWITCH_OFF
Loop

SWITCH_OFF:
Set LIGHT Off
'Chip will enter low power mode when program ends
```

For more help, see [Goto](#), [Gosub](#)

Lookup Tables

Overview

A lookup table is a list of values stored in the microcontroller’s memory and accessed using [ReadTable](#).

Key Features

- Efficient memory usage
- Supports [byte](#), [word](#), [long](#), and [integer](#) types
- ASCII string definitions supported

- Constants and inline calculations permitted
- External data sources allowed

Note: Decimal values are not supported

Contents

- [Defining Tables](#)
- [String Tables](#)
- [ASCII Escape Sequences](#)
- [Reading Lookup Tables](#)
- [Importing from External File](#)
- [EEPROM Table Storage](#)
- [Reference](#)

Defining Tables

Single Values

```
Table TestDataSource
  12
  24
  36
  48
  60
  72
End Table
```

Multiple Values

```
Table TestDataSource
  12, 24, 36
  48, 60, 72
End Table
```

Constants & Calculations

```
#define calculation_constant 2

Table TestDataSource
    1 * calculation_constant
    2 * calculation_constant
    3 * calculation_constant
    8 * calculation_constant
    4 * calculation_constant
    5 * calculation_constant
End Table
```

String Tables

Simple Examples

```
Table Test_1
    "ABCDEFGHJIJ"
End Table
```

```
Table MnuTxt_1
    "  Display_1    Display_2    Display_3  "
End Table

Table MnuTxt_2
    "1: Display"
    "2: System Setup"
    "3: Config 1"
    "4: Config 2"
    "5: Data Log"
    "6: Diagnostic"
    "7: Help+"
End Table
```

Equivalent Representations

- "String1", "String2", "String3"
- "String1String2String3"
- "String1" "String2" "String3"

ASCII Escape Sequences

Escape	Meaning
\a	Beep

Escape	Meaning
\b	Backspace
\f	Formfeed
\n or \l	Newline
\r	Carriage Return
\t	Tab
\0	Null value (ASCII 0)
\&nnn	ASCII character (decimal)
\\	Backslash
\"	Double quote
\'	Single quote

Reading Lookup Tables

Basic Reading

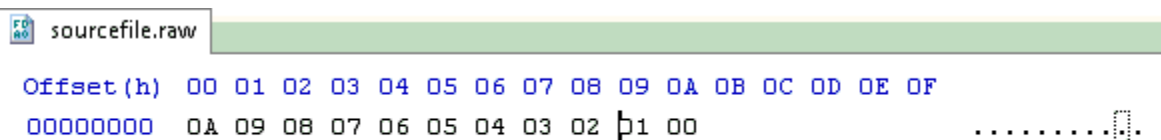
```
Dim TableCounter, Invalue as byte

CLS
For TableCounter = 1 to 6
    ReadTable TestDataSource, TableCounter, Invalue
    Print Invalue
    Print ", "
Next
```

Reading Table Length

```
Dim lengthoftable as word
ReadTable TestDataSource, 0, lengthoftable
Print lengthoftable
```

Importing from External File

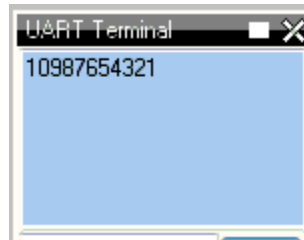


```
sourcefile.raw
```

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00000000	0A	09	08	07	06	05	04	03	02	01	00					

```
#chip 16f877a
Table TestDataSource from "sourcefile.raw"

For nn = 1 to 10
  ReadTable TestDataSource, nn, inc
  Print inc
Next
```



EEPROM Table Storage

```
#chip 16F628

TableLoc = 2
ReadTable TestDataSource, TableLoc, SomeVar

EPWrite 1, 45 'Optional write

Table TestDataSource Store Data
  12
  24
  36
  48
  60
  72
End Table
```

EEPROM Limitations

- Only **BYTE** values are supported
- **WORD**, **INTEGER**, or **LONG** are not compatible with EEPROM tables

Reference

See: [ReadTable](#)

Miscellaneous

About Miscellaneous things....

It is possible to combine multiple instructions on a single line, by separating them with a colon. For example, this code:

```
Set PORTB.0 On
Set PORTB.1 On
Wait 1 sec
Set PORTB.0 Off
Set PORTB.0 Off
```

could also be written as:

```
Set PORTB.0 On: Set PORTB.1 On
Wait 1 sec
Set PORTB.0 Off: Set PORTB.0 Off
```

In most cases, it will make no difference if commands share a line or not. However, special care should be taken with If commands, as this code:

```
Set PORTB.0 Off
Set PORTB.1 Off
If Temp > 10 Then Set PORTB.0 On: Set PORTB.1 On
Wait 1 s
```

Will be equivalent to this:

```
Set PORTB.0 Off
Set PORTB.1 Off
If Temp > 10 Then
Set PORTB.0 On
Set PORTB.1 On
End If
Wait 1 s
```

Also, the commands used to start and end subroutines, data tables and functions must be alone on a line. For example, this is WRONG:

```
Sub Something: Set PORTB.0 Off: End Sub
```

ReadTable

About ReadTable

The **ReadTable** command is used to read information from lookup tables. **TableName** is the name of the table that is to be read, **Item** is the line of the table to read, and **Output** is the variable to write the retrieved value in to.

Syntax:

```
ReadTable TableName, Item, Output
```

Command Availability:

Available on all microcontrollers.

Explanation:

Item is 1 for the first line of the table, 2 for the second, and so on. If the Table is more than 256 elements then **Item** must be WORD variable. Care must be taken to ensure that the program is not instructed to read beyond the end of the table as Zero will be returned.

The type of **Output** should match the type of data stored in the table. For example, if the table contains Word values then **Output** should be a Word variable. If the type does not match, GCBASIC will attempt to convert the value.

Example:

```

'Chip Settings
#chip 16F88, 20

'Hardware Settings
#define LED PORTB.0
Dir LED Out

'Main Routine
ReadTable TimesTwelve, 4, Temp
Set LED Off
If Temp = 48 Then Set LED On

'Lookup table named "TimesTwelve"
Table TimesTwelve
12
24
36
48
60
72
84
96
108
120
132
144
End Table

```

For more help, see [Lookup Tables](#)

Scripts

About Scripts

A script is a small section of code that GCBASIC runs when it starts to compile a program. Uses include performing calculations that are required to adjust the program for different chip frequencies.

Scripts are not compiled or downloaded to the microcontroller - GCBASIC reads them, executes them, then removes them from the program and then the results calculated can be used as *constants* in the user program.

Inside a script, *constants* are treated like variables. Scripts can read the values of *constants*, and set them to contain new values.

Using Scripts

Scripts start with **#script** and end with **#endscript**. Inside, they can consist of the following commands:

```
If  
Assignment (=)  
Error  
Warning  
Int()
```

If is similar to the If command in normal GCBASIC code, except that it does not have an **Else** clause. It is used to compare the values of the script constants.

The **=** sign is identical to that in GCBASIC programs. The *constant* that is to be set goes on the left side of the **=** and the new value goes to the right of the **=**.

Error is used to display an error message. Anything after the **Error** command is displayed at the end of compilation, and is saved in the error log for the program.

Warning is used to display a warning message. Anything after the **Warning** command is displayed at the end of compilation but warning does not halt compilation.

Int() will calculate the integer value of a calculation. Using **Int()** is critical to set the *constant* to the integer component of the calculation.

Notes:

Use **Warning** to display constant values when creating and debugging scripts.

Scripts have a limited syntax and limited error checking when compiling. The compiler may halt if you get something wrong.

Scripts that are incorrectly formatted may also halt the compiler or return unrelated error.

Scripts used for calculations should use the **Int(expression)** where you may have a floating point numbers returned.

Scripts do use floating point for all calculations and a failure to use **Int()** may set the script constant and the resulting *constant* to 0.

Scripts may require that complex math expressions may require definition in multiple steps/line to simplify the calculation.

The returned value could be incorrect if simplification is not implemented.

Scripts can only access existing **constants** both user and system defined.

User defines variables are not accessible within the scope of a script.

Scripts has precedence over **#define**. A **#define** constant statements are read first, then scripts run. So, a script will always overwrite a constant that was set with **#define**.

Use **Warning** to display constants values when creating and debugging scripts.

Example Script

This script is used in the pwm.h file. It takes the values of the user defined *constants* PWM_Freq, PWM_Duty and system *constant* ChipMHz and calculates the results using the equations. These calculation are based on information from a Microchip PIC datasheet to calculate the correct values to setup Pulse Width Modulation (PWM).

```
#script
PR2Temp = INT((1/PWM_Freq)/(4*(1/(ChipMHz*1000))))
T2PR = 1
If PR2Temp > 255 Then
    PR2Temp = INT((1 / PWM_Freq) / (16 * (1 / (ChipMHz * 1000))))
    T2PR = 4
    If PR2Temp > 255 Then
        PR2Temp = INT(( 1 / PWM_Freq) / (64 * (1 / (ChipMHz * 1000))))
        T2PR = 16
        If PR2Temp > 255 Then
            Error Invalid PWM Frequency value
        End If
    End If
End If

DutyCycle = (PWM_Duty * 10.24) * PR2Temp / 1024
DutyCycleH = (DutyCycle AND 1020) / 4
DutyCycleL = DutyCycle AND 3
#endscript
```

During the execution of the script the calculations and assignment uses the constants in the script.

After this script has completed the *constants* **PR2Temp**, **DutyCycleH** and **DutyCycleL** are set using the constants and/or the calculations.

The *constants* assigned in this script, **PR2Temp**, **DutyCycleH** and **DutyCycleL**, are made available as *constants* in the user program.

Subroutines

About Subroutines

A subroutine is a small program inside of the main program. Subroutines are typically used when a task needs to be repeated several times in different parts of the main program.

There are two main uses for subroutines:

- Keeping programs neat and easy to read
- Reducing the size of programs by allowing common sections of code to be reused.

When the microcontroller comes to a subroutine it saves its location in the current program before jumping to the start of, or calling, the subroutine. Once it reaches the end of the subroutine it returns to the main program, and continues to run the code where it left off previously.

Normally, it is possible for subroutines to call other subroutines. There are limits to the number of times that a subroutine can call another sub, which vary from chip to chip:

Microcontroller Family	Instruction Width	Number of subs called
10F*, 12C5*, 12F5*, 16C5*, 16F5*	12	1
12C*, 12F*, 16C*, 16F*, except those above	14	7
18F*, 18C*	16	31

These limits are due to the amount of memory on the microcontroller which saves its location before it jumps to a new subroutine. Some GCBASIC commands are subroutines, so you should always allow for 2 or 3 subroutine calls more than your program has.

On 16F chips, the program memory is divided into pages. Each page holds 2048 instructions. If the program jumps from code on one page to code on another, the compiler has to select the new page. Having to do this makes the program bigger, so try to avoid this. To keep jumps between pages down, GCBASIC imposes a rule that each subroutine must be entirely within one page, so that only jumps to other subroutines require the page selection. As an example, say you have two pages of memory, each 2048 instructions (words) long.

If you have a main sub that is 1500 words, and four other subroutines each 600 words long, your total program size would be 3900 words and you might expect it to fit into the 4096 words available. The problem though is that once the main routine takes 1500 words from page 1, nothing else will fit after it. Three of the 600 word subroutines would fit onto page 2, but that leaves one 600 word subroutine that will not fit into the 500 left on page 1 or the 200 left on page 2. If you want to reduce the chance of this happening, the best option is to keep your subroutines smaller - move anything out of the main routine and into another one - this will resolve memory page constraints.

Atmel AVR microcontrollers have no fixed limit on how many subroutines can be called at a time, but if too many are called then some variables on the chip may be corrupted. To check if there are too many subroutines, work out the most that will be called at once, then multiply that number by 2 and create an array of that size. If an out of memory error message comes up, there are too many calls.

Another feature of subroutines is that they are able to accept parameters. These are values that are passed from the main program to the subroutine when it is called, and then passed back when the subroutine ends.

Using Subroutines

To call a subroutine is very simple - all that is needed is the name of the sub, and then a list of parameters. This code will call a subroutine named "Buzz" that has no parameters:

```
Buzz
```

If the sub has parameters, then they should be listed after the name of the subroutine. This would be the command to call a subroutine named "MoveArm" that has three parameters:

```
MoveArm NewX, NewY, 10
```

Or, you may choose to put brackets around the parameters, like so:

```
MoveArm (NewX, NewY, 10)
```

All that this does is change the appearance of the code - it doesn't make any difference to what the code does. Decide which one meets your own personal preference, and then stick with it.

Creating subroutines

To create a subroutine is almost as simple as using one. There must be a line at the start which has **sub**, and then the name of the subroutine. Also, there needs to be a line at the end of the subroutine which reads **end sub**. To create a subroutine called **Buzz**, this is the required code:

```
sub Buzz  
  
'code for the subroutine goes here  
  
end sub
```

If the subroutine has parameters, then they need to be listed after the name. For example, to define the **MoveArm** sub used above, use this code:

```
sub MoveArm(ArmX, ArmY, ArmZ)  
  
'code for the subroutine goes here  
  
end sub
```

In the above sub, **ArmX**, **ArmY** and **ArmZ** are all variables. If the call from above is used, the variables will have these values at the start of the subroutine:

```
ArmX = NewX
ArmY = NewY
ArmZ = 10
```

When the subroutine has finished running, GCBASIC will copy the values back where possible. **NewX** will be assigned to **ArmX**, and **NewY** will be assigned to **ArmY**. GCBASIC will not attempt to set the number 10 to **ArmZ**.

Controlling the direction data moves in

It is possible to instruct GCBASIC not to copy the value back after the subroutine is called. If a subroutine is defined like this:

```
sub MoveArm(In ArmX, In ArmY, In ArmZ)
'code for the subroutine goes here

end sub
```

Then GCBASIC will copy the values to the subroutine, but will not copy them back.

GCBASIC can also be prevented from copying the values back, by adding **Out** before the parameter name. This is used in the EEPROM reading routines - there is no point copying a data value into the read subroutine, so **Out** has been used to avoid wasting time and memory. The EPRead routine is defined as **Sub EPRead(In Address, Out Data)**.

Many older sections of code use **#NR** at the end of the line where the parameters are specified. The **#NR** means "No Return", and when used has the same effect as adding **In** before every parameter. Use of **#NR** is not recommended, as it does not give the same level of control.

Using different data types for parameters

It is possible to use any type of variable as a parameter for a subroutine. Just add **As** and then the data type to the end of the parameter name. For example, to make all of the parameters for the **MoveArm** subroutine word variables, use this code:

```
sub MoveArm(ArmX As Word, ArmY As Word, ArmZ As Word)
...
end sub
```

Optional parameters

Sometimes, the same value may be used over and over again for a parameter, except in a particular case. If this occurs, a default value may be specified for the parameter, and then a value for that parameter only needs to be given in a call if it is different to the default.

For example, suppose a subroutine to create an error beep is required. Normally it emits ! 440 Hz tone, but sometimes a different tone is required. To create the sub, this code would be use:

```
Sub ErrorBeep(Optional OutTone As Word = 440)
    Tone OutTone, 100
End Sub
```

Note the **Optional** before the parameter, and the **= 440** after it. This tells GCBASIC that if no parameter is supplied, then set the **OutTone** parameter to 440.

If called using this line:

```
ErrorBeep
```

then a 440 Hz beep will be emitted. If called using this line:

```
ErrorBeep 1000
```

then the sub will produce a 1000 Hz tone.

When using several parameters, it is possible to make any number of them optional. If the optional parameter/s are at the end of the call, then no value needs to be specified. If they are at the start or in the middle, then you must insert commas to allow GCBASIC to tell where the optional parameters are.

Overloading

It is possible to have 2 subroutines with the same name, but different parameters. This is known as overloading, and GCBASIC will automatically select the most appropriate subroutine for each call.

An example of this is the Print routine in the LCD routines. There are actually several Print subroutines; for example, one has a byte parameter, one a word parameter, and one a string parameter. If this command is used:

```
Print 100
```

Then the Print (byte) subroutine will be called. However, if this command is used:

```
Print 30112
```

Then the Print (word) subroutine will be called. If there is no exact match for a particular call, GCBASIC will use the option that requires the least conversion of variable types. For example, if this

command is used:

```
Print PORTB.0
```

The byte print will be used. This is because byte is the closest type to the single bit parameter.

See Also [Functions](#), [Exit](#)

Converters

About Converters

Converters allow GCBASIC to read files that have been created by other programs. A converter can convert these files into GCBASIC libraries or any GCBASIC instruction or a GCBASIC dataset.

A typical use case is when you have a data source file from another computer system and you want to consume the data within your GCBASIC program. The data source file could be database, graphic, reference data or music file. The converter will read these source files and convert them into a format that can be processed by GCBASIC. The conversion process is completed by external application which can be written by the developer or you can use one of the converters provided with the GCBASIC release.

The GCBASIC release includes the converter for BMP files and standard Text files.

With an appropriate Converter installed, and an associated `#include` to these non-GCBASIC files, GCBASIC will detect that the file extension and hand the processing to the external converting program. When the external converting program had complete, GCBASIC will then continue with the converted source file as a GCBASIC source file.

An example of a converter is to read an existing picture file, convert the picture file to a GCB table and then refer to the picture file table to display the picture file on a GLCD.

Conversion is achieved by including a command within the source program to transform external data. The command used is the instruction `#include` followed by the data source. An example:

```
'Convert ManLooking.BMP to a GCBASIC usable format.  
  
#include <..\converters\ManLooking.BMP>
```

The inclusion of the `#include` line within a GCBASIC program will enable the commencement of the following process:

1. GCBASIC will examine the `..\converters` folder structure for a configuration file that will handle the file extension specified in the include statement.

2. GCBASIC will examine the configuration file(s) *.INI for command line instructions.
3. GCBASIC will at stage examine the folder structure for the source file and the target transformed file. If the source file is older than the transformed file the next step will not be executed, goto step 6.
4. GCBASIC will execute the command as specified within the configuration file to transform the source file to the target file.

The Conversion program must create the output file extension as specified in the configuration file. If the include statement as an extension of .TXT and the configuration files states the input file extension as .TXT and the output as .GCB the converted file must have the extension of .GCB.

```
#include <..\converters\ManLooking.BMP>
```

Init file is input file as BMP and output as GCB, then the file expected is ..\converters\ManLooking.GCB

5. GCBASIC will attempt to include the transformed target file (with the file extension as specified in the configuration file) within the GCBASIC program.
6. GCBASIC will resume normal processing of the GCBASIC program including the transformed target file, therefore, with normal compiling and errors handling.

For example programs see [here](#).

More about Converters

1. The configuration file

The configuration file MUST have the extension of .INI. No leading spaces are permitted in the configuration file. Specification of the configuration file. The file has four items: desc, in, out and exe. Where:

```
desc      : Is the description shown in GCGB
in        : Is the source file extension to be transformed
out       : Is the target transformed file extension.
exe       : Is the executable to be run for this specific configuration file.
params    : Optional, is the required parameter to be passed from the compiler.
Example:  params = %filename% %chipmodel%
deletetarget : Optional, will always recreate the target transformed file. The default
is to retain the target transformed file unless source has changed. Options are Y or N
```

You can have multiple configuration files within the ..\converters folder structure.

GCBASIC will examine all configuration file to match the extension as specified in the #include command.

Example 1 :

BMP (Black and White) conversion configuration file is called **BMP2GCBasic.ini**. The source extension is **.bmp**, the transformed file extension is **.GCB**, and the executable is called **BMP2GCBASIC.exe**.

```
desc = BMP file (*.bmp)
in = bmp
out = GCB
exe = BMP2GCBASIC .exe
```

An example :

```
#include <..\converters\ManLooking.BMP>
```

Will be converted by the **BMP2GCBASIC .EXE** to **..\converters\ManLooking.GCB**

Example 2 :

Data file conversion configuration file is called **TXT2GCB.ini**. The source extension is **.TXT**, the transformed file extension is **.GCB**, and the command line called **AWKRUN.BAT**.

```
desc = Infrared Patterns (*.txt)
in = txt
out = GCB
exe = awkrun.bat
```

An example :

```
#include <..\converters\InfraRedPatterns.TXT>
```

Will be converted by the **AWKRUN.BAT** to **..\converters\ InfraRedPatterns.GCB**

The example would require a supporting batch file and a script process to complete the transformation.

2. Conversion Executable

The conversion executable may be written in any language (compiled or interpreted).

The conversion executable MUST create the converted file with the correct file extension as specified in the configuration file.

The conversion executable will be passed one parameter - the source file name. Using example #1 the conversion executable would be passed `..\converters\ManLooking.BMP`

The conversion executable MUST create a GCBASIC compatible source file. Any valid commands/instruction are permitted.

3. Installation

The `INI` file, the source file and the conversion executable MUST be located in the `..\converters` folder. The converters folder is relative to the `GCBASIC.EXE` compiler folder.

Example 3 : Converter Program

This program converts the `InfraRedPatterns.TXT` into `InfraRedPatterns.GCB` that will have a GCBASIC table called `DataSource`. This example is located in the converter folder of the GCBASIC installation.

```
#chip16f877a, 16
#include <..\converters\InfraRedPatterns.TXT>

dir portb out

' These must be WORDs as this could be large table.
dim TableReadPosition, TableLen as word

dir portb out

' Read the table length
TableReadPosition = 0
ReadTable DataSource, TableReadPosition, TableLen

Do Forever
    For TableReadPosition = 1 to TableLen step 2
        ReadTable DataSource, TableReadPosition, TransmissionPattern
        ReadTable DataSource, TableReadPosition+1 , PulseDelay
        portb = TransmissionPattern
        wait PulseDelay ms
    next
Loop
```

Example 4 : Dynamic Import

This program converts a chip specific configuration file into **manifest.GCB** that will have a GCBASIC functions called **DataIn** and **DataOut**. This example is located in the converter folder of the GCBASIC installation.

```
#chip 16f18326

#include <..\converters\manifest.mcc>

DataOut ( TX, RA0 ) 'this method is created during the convert process. They do
not exist withiut the converter.
DataIn ( Rx, RC6 ) 'this method is created during the convert process. They do
not exist withiut the converter.
```

This example would use the optional parameters of **params** and **deletetarget** in the converter configuration file as follows:

```
desc = PPS file (*.PPS)
params = %filename% %chipmodel%
in = mcc
out = GCB
exe = DataHandler.exe
deletetarget= y
```

Example 5 : Add build numbers and time/date details to your programs

This converter is used to expose two string variables as follows:

```
GCBBuildStr
GCBBuildTimeStr
```

The user code is simple. Using the **#include** statement specify any filename with an extension must be **.cnt**. As follows:

```
#include "GCBVersionNumber.cnt"
```

Complete code would like this - this not optimised - this shows the use of the exposed strings.

```
#include "GCBVersionNumber.cnt"

dim versionString as string * 40
versionString = "Max7219 build"+GCBBuildStr
versionString = versionString + "@"+GCBBuildTimeStr
Print versionString
```

This outputs the following - where #20 is the current build and the date/time is correct for build time.

```
Max7219 build20@01-06-2021 08:00:21
Commence main program
```

This works as the support INI file instructs the compiler to call a utility that automatically creates a build number tracker file and the supporting string functions. The utility creates a tracker file and the methods files in the same folder as your source program - so, each tracker is specific to each project. The converter requires the following files - these are included within your Installation.

```
GCBVersionStamp.exe - the utility called by the converter capability.
cnt2gcb.ini - the supporting ini file used by the compiler to handle this converter.
```

Command References

Analog/Digital conversion

This is the Analog/Digital conversion section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Analog/Digital Conversion Overview

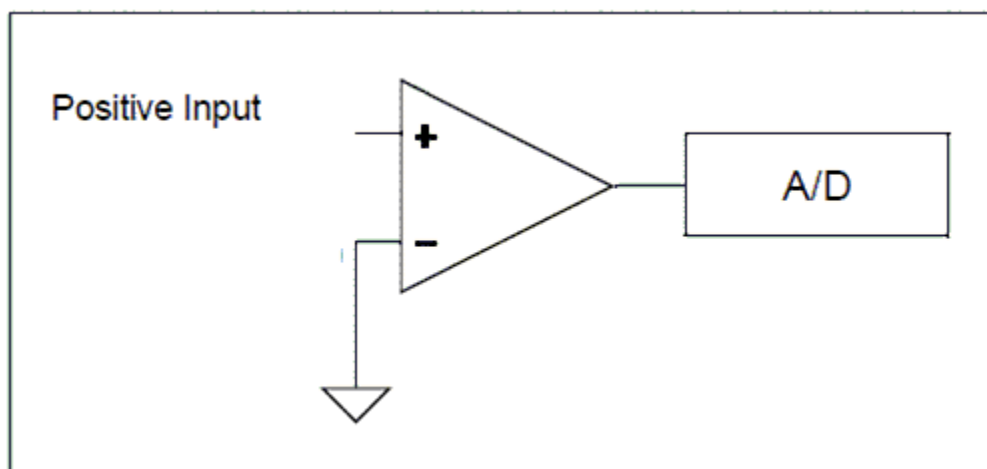
About Analog to Digital Conversion

The analog to digital converter (ADC or A/D) module support is implemented by GCBASIC to provide 8-bit, 10-bit and 12-bit Single channel measurement mode and Differential Channel Measurement mode.

GCBASIC configures the analog to digital converter clock source, the programmed acquisition time and justification of the response byte, word or integer (as defined in the GCBASIC method).

Normal or Single channel measurement mode

The Single channel measurement mode is the default method for reading the ADC port. The positive input is attached to suitable device (a light sensor or adjustable resistor) and the command ReadADC, ReadADC10, ReadADC12 with return a byte, word or word value respectively.



The A/D module on most microcontrollers only supports single-ended mode. Single channel mode uses a single A/D port and the returned Value represents the difference between the voltage on the analog pin and a fixed negative reference which is usually ground or Vss.

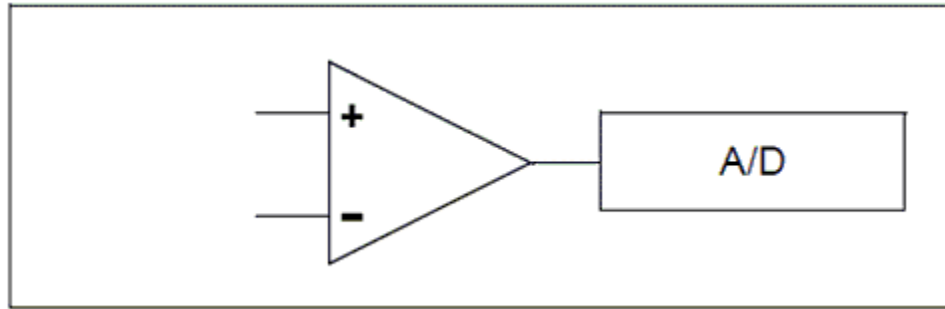
The syntax for single-ended A/D is Returned_Value = ReadAD(Port)

Example

```
Print ReadAD10(AN3)
```

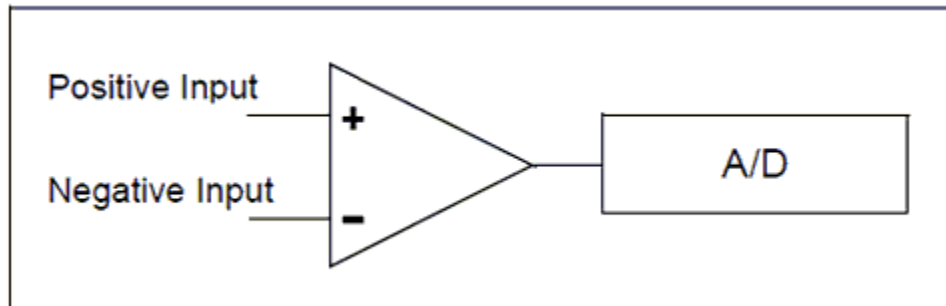
Differential channel measurement mode

Some of the in the Microchip PIC family of devices also support differential analog to digital conversion. With differential conversion, the differential voltage between two channels is measured and converted to a digital value. The returned value can be either positive or negative (therefore an integer value).



When configured to differential channel measurement mode, the positive channel is connected to the defined positive analog pin (ANx), and the negative channel is connected to the defined negative analog pin. These two pins are internally connected (within the microcontroller) to a unity gain differential amplifier and once the amplifier has completed the comparison the result is returned as an integer.

The positive channel Input is selected using the CHSx bits and the negative channel input is selected using the CHSNx bits. These bits are managed by GCBASIC. The programmer only needs to supply the correct analog pin designators in the ReadADx commands.



The 12-bit returned result is available on the ADRESH and ADRESL registers which is returned by the GCBASIC methods as an integer variable.

Some Microchip PIC microcontrollers have differential A/D modules and support differential Mode as well as 12-Bit A/D. With Differential mode the returned value can be either a positive or negative number that represents the voltage differential between the two A/D ports.

The syntax for differential A/D is ReadAD(PositiveANPort , NegativeANPort). Note: if "negative port" is omitted readAd() will perform a single-ended read on the positive AN port.

Example

```
Print ReadAD12( AN3, AN4 )
```

Using Voltage Reference

Voltage references come in many forms and offer different features across the PICs, AVR and LGTs microcontrollers. But, in the end, accuracy and stability are a voltage reference's most important features, as the main purpose of the reference is to provide a known output voltage. Variation from this known value is an error. Therefore, it is useful to use the internal voltage reference provided within the microcontroller.

To use a voltage reference source for ADC operation set the `AD_REF_SOURCE` constant to your chosen source. The default is the VCC pin, and there for the constant is set by default to `AD_REF_AVCC`. The voltage reference is specific to the microcontroller but the options are as follows:

AD_REF_SOURCE Constant	Reference Voltage
AD_REF_AVCC	VCC supplied Voltage
AD_REF_1024	1.024v internal reference source
AD_REF_2048	2.048v internal reference source
AD_REF_4096	4.096v internal reference source
AD_REF_AREF	External voltage reference source
AD_REF_256	AD_REF_256 for ATMegs

Optimising GCBASIC Code

GCBASIC supports a wide range of A/D modules and the supporting library addresses up to 34 channels. To reduce the size of the code produced you can define which channels are specifically supported. See [Optimising ADC code](#) for more details.

See also [ReadAD](#), [ReadAD10](#) and [ReadAD12](#)

For the latest Microchip PIC microcontrollers that support Differential and 12-bit A/D please refer to Microchip MAPS or the microcontrollers datasheet.

ADFormat (Deprecated - Do not use)

Syntax:

```
ADFormat ( Format_Left | Format_Right )
```

Command Availability:

Available only on Microchip PIC microcontrollers.

Explanation:

Left justified means 8 bits in the high byte, 2 in the low. Right justified means 2 in the high byte, and the remaining 8 in the low byte. It's only supported on Microchip PIC microcontrollers.

ADOff

This command is obsolete. There should be no need to call it. GCBASIC will automatically disable the A/D converter and set all pins to digital mode when starting the program, and after every use of the ReadAD function.

It is recommended that this command be removed from all programs.

ReadAD

Syntax:

For a normal (also called a Single Channel) read use.

```
byte_variable = ReadAD( ANX )
```

For a Differential Channel read use the following. Where ANpX is the positive port, and ANnY is the negative port.

```
byte_variable = ReadAD( ANpX , ANnY )
```

To obtain a byte value from an AD Channel use the following to force an 8 bit AD Channel to respond with a byte value [0 to 255].

```
byte_variable = ReadAD( ANX , TRUE )
```

Command Availability:

When using **ReadAD** (ANx) the returned value is an 8 bit number [0- 255]. The byte variable assigned by the function can be a byte, word, integer or long.

When using **ReadAD** (ANpX , ANnY) the returned value is an integer, as negative values can be returned.

When using **ReadAD** (ANpX , TRUE) the returned value is an integer, but you should treat as a byte.

ReadAD is a function that can be used to read the built-in analog to digital converter that many microcontroller chips include. port should be specified as AN0, AN1, AN2, etc., up to the number of analog inputs available on the chip that is in use. Those familiar with Atmel AVR microcontrollers can also refer to the ports as ADC0, ADC1, etc. Refer to the datasheet for the microcontroller chip to find the number of ports available. (Note: it's perfectly acceptable to use ANx on AVR, or ADCx on the

microcontroller)

Other functions that are similar are **ReadAD10** and **ReadAD12**. See the relevant Help page for the specific usage of each function.

The constant **AD_Delay** controls the acquisition delay. The default value is 20 us. This can be changed by adding the following constant.

```
#define AD_Delay 2 10us
```

ADSpeed controls the source of the clock for the ADC module. It varies from one chip to another. InternalClock is a Microchip PIC microcontroller only option that will drive the ADC from an internal RC oscillator. The default value is 128.

Using ADSPEED

```
'default value
#define ADSpeed MediumSpeed

'pre-defined constants
#define HighSpeed 255
#define MediumSpeed 128
#define LowSpeed 0
```

AD_VREF_DELAY controls the charging time for VRef capacitor on Atmel AVR microcontrollers only. This therefore controls the charge from internal VRef. ReadAD will not be accurate for internal reference without this.

AD_Acquisition_Time_Select_bits also controls the Acquisition Time Select bits. Acquisition time is the duration that the AD charge holding capacitor remains connected to AD channel from the instant the read is commenced is set until conversions begins.

The default value of AD_Acquisition_Time_Select_bits is 0b100 or decimal 4, where all three ACQT bits will be set. To change use the following.

```
'change the default value
#define AD_Acquisition_Time_Select_bits 0b001    'this will only set ACQT bit 0, ACQT
bits 1 and 2 will be cleared.
```

Example 1

This example reads the ADC port and writes the output to the EEPROM.

```
#chip 16F819, 8

'Set the input pin direction
Dir PORTA.0 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

    'Take a reading and log it
    EPWrite CurrentAddress, ReadAD(AN0)

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 2

This example reads the ADC port and writes the output to the EEPROM. The output value will be in the range of [0-255].

```
#chip 16F1789, 8

'Set the input pin direction
Dir PORTA.0 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

    'Take a reading and log it
    EPWrite CurrentAddress, ReadAD(AN0, TRUE)

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 3

This example used the differential capabilities of ADC port and writes the output to the EEPROM. The output value will be in the range of [-255 to 255].

AN0 and AN2 are used for the differential ADC reading.

```
#chip 16F1789, 8

'Set the input pin direction
Dir PORTA.0 In
Dir PORTA.2 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

    'Take a reading and log it
    EPWrite CurrentAddress, ReadAD( AN0, AN2 )

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

See Also [ReadAD10](#), [ReadAD12](#)

ReadAD10

Syntax:

For a normal (also called a Single Channel) read use.

```
word_variable = ReadAD10( ANX )
```

For a Differential Channel read use the following. Where ANpX is the positive port, and ANnY is the negative port.

```
integer_variable = ReadAD10( ANpX , ANnY )
```

To obtain a 10-bit value from an AD Channel use the following to force a 10 bit AD Channel to respond with the correct value, in terms of the range [0 to 1023]

```
integer_variable = ReadAD10( ANX , TRUE )
```

Command Availability:

ReadAD10 is a function that can be used to read the built-in analog to digital converter that many microcontroller chips include. The port should be specified as AN0, AN1, AN2, etc., up to the number of analog inputs available on the chip that is in use. Those familiar with Atmel AVR microcontrollers

can also refer to the ports as ADC0, ADC1, etc. Refer to the datasheet for the microcontroller chip to find the number of ports available. (Note: it's perfectly acceptable to use ANx on AVR, or ADCx on the microcontroller.)

When using **ReadAD10** (ANX) the returned value is the **full range** of the ADC module. Therefore, the method will return an 8 bit value [0-255], or an 10 bit value [0-1023] or an 12 bit value [0-4095]. This is dependent on the microcontrollers capabilities. For a 10 bit value [0-1023] always to be returned use `user_variable = ReadAD10( ANX , TRUE )`. The user variable can be a byte, word, integer or long but typically a word is recommended.

When using **ReadAD10** (ANpX , ANnY), for differential ADC reading, the returned value is an integer as negative values will be returned.

When using **ReadAD10** (ANpX , TRUE), to force a 10 bit ADC reading, the returned value is an integer.

Other functions that are similar are **ReadAD** and **ReadAD12**. See the relevant Help page for the specific usage of each function.

AD_Delay controls is the acquisition delay. The default value is 20 us. This can be changed to a longer acquisition delay by adding the following constant.

```
#define AD_Delay 4 10us
```

ADSpeed(controls the source of the clock for the ADC module. It varies from one chip to another. InternalClock is a microcontroller only option that will drive the ADC from an internal RC oscillator. The default value is 128.

```
'default value
#define ADSpeed MediumSpeed

'pre-defined constants
#define HighSpeed 255
#define MediumSpeed 128
#define LowSpeed 0
```

AD_Acquisition_Time_Select_bits also controls the Acquisition Time Select bits. Acquisition time is the duration that the AD charge holding capacitor remains connected to AD channel from the instant the read is commenced is set until conversions begins.

The default value of `AD_Acquisition_Time_Select_bits` is 0b100 or decimal 4, where all three ACQT bits will be set. To change use the following.

```
'change the default value
#define AD_Acquisition_Time_Select_bits 0b001    'this will only set ACQT bit 0, ACQT
bits 1 and 2 will be cleared.
```

Example 1 - Read 10-bit ADC

```
#chip 16F819, 8

'Set the input pin direction
Dir PORTA.0 In

'Print 255 reading
For CurrentAddress = 0 to 255

    'Take a reading and show it
    Print str(ReadAD10(AN0))

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 2 - Reading Reference Voltages:

When selecting the reference source for ADC on ATmega328 GCBASIC will overwrite anything that you put into the ADMUX register - but this option allows you to change the ADC reference source on Atmel AVR microcontrollers. You can set the AD_REF_SOURCE constant to whatever you want to use. It defaults to the VCC pin, as an example you can set the Atmel AVR to use the 1.1V reference with this: **#define AD_REF_SOURCE AD_REF_256** where 256 refers to the 2.56V reference on some older AVRs, but the same code will select the 1.1V reference on an ATmega328p

```
' Dynamically switching reference.
#define AD_REF_SOURCE ADRefSource
#define AD_VREF_DELAY 5 ms
AdRefSource = AD_REF_AVCC
HSerPrint ReadAD10(AN1)
HSerPrint ", "
AdRefSource = AD_REF_256
HSerPrint ReadAD10(AN1)
```

The example above sets the AD_REF_SOURCE to a variable, and then changes the value of the variable to select the source. With this approach, we also need to allow time to charge the reference capacitor to the correct voltage.

Example 3 - Read 10-bit ADC forcing a 10-bit value to be returned

```
#chip 16F1789, 8

'Set the input pin direction
Dir PORTA.0 In

'Print 255 reading
For CurrentAddress = 0 to 255

    'Take a reading and show it
    Print str(ReadAD10(AN0), TRUE)

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 4

This example used the differential capabilities of ADC port and writes the output to the EEPROM. The output value will be in the range of [-1023 to 1023].

AN0 and AN2 are used for the differential ADC reading.

```
#chip 16F1789, 8

'USART settings
#define USART_BAUD_RATE 9600 'Initializes USART port with 9600 baud
#define USART_TX_BLOCKING ' wait for tx register to be empty
wait 100 ms

'Set the input pin direction
Dir PORTA.0 In
Dir PORTA.2 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

    'Take a reading and log it
    HSerPrint ReadAD10( AN0, AN2 )
    HserPrintCRLF
    'Wait 10 minutes before getting another reading
    Wait 10 min

Next
```

See Also [ReadAD](#), [ReadAD12](#)

ReadAD12

Syntax:

For a normal (also called a Single Channel) read use.

```
user_variable = ReadAD12( ANX )
```

For a Differential Channel read use the following. Where ANpX is the positive port, and ANnY is the negative port.

```
user_variable = ReadAD12( ANpX , ANnY )
```

To obtain a 12-bit value from an AD Channel use the following to force a 12 bit AD Channel to respond with the correct value, in terms of the range of [0 to 4095]

```
user_variable = ReadAD12( ANX , TRUE )
```

Command Availability:

When using **ReadAD12** (ANX) the returned value is an 12 bit number [0-4095]. The user variable can be a word, integer or long.

When using **ReadAD12** (ANpX , ANnY) the returned value is an integer as negative values can be returned.

ReadAD12 is a function that can be used to read the built-in analog to digital converter that many microcontroller chips include. Port should be specified as AN0, AN1, AN2, etc., up to the number of analog inputs available on the chip that is in use. Those familiar with Atmel AVR microcontrollers can also refer to the ports as ADC0, ADC1, etc. Refer to the datasheet for the microcontroller chip to find the number of ports available. (Note: it's perfectly acceptable to use ANx on AVR, or ADCx on the microcontroller.)

Other functions that are similar are **ReadAD** and **ReadAD10**. See the relevant Help page for the specific usage of each function.

AD_Delay controls is the acquisition delay. The default value is 20 us. This can be changed to a longer acquisition delay by adding the following constant.

```
#define AD_Delay 4 10us
```

ADSpeed(controls the source of the clock for the ADC module. It varies from one microcontroller to another. InternalClock is a Microchip PIC microcontroller only option that will drive the ADC from an internal RC oscillator. The default value is 128.

```
'default value
#define ADSpeed MediumSpeed

'pre-defined constants
#define HighSpeed 255
#define MediumSpeed 128
#define LowSpeed 0
```

AD_Acquisition_Time_Select_bits also controls the Acquisition Time Select bits. Acquisition time is the duration that the AD charge holding capacitor remains connected to AD channel from the instant the read is commenced is set until conversions begins.

The default value of **AD_Acquisition_Time_Select_bits** is 0b100 or decimal 4, where all three ACQT bits will be set. To change use the following.

```
'change the default value
#define AD_Acquisition_Time_Select_bits 0b001    'this will only set ACQT bit 0, ACQT
bits 1 and 2 will be cleared.
```

Example 1 - Read 12-bit ADC

```
#chip 16F1788, 8

'Set the input pin direction
Dir PORTA.0 In

'Print 255 readings
For CurrentAddress = 0 to 255

    'Take a reading and show it
    Print str(ReadAD12(AN0))

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 2 - Force a 12-bit value to be returned

```
#chip 16F1788, 8

'Set the input pin direction
Dir PORTA.0 In

'Print 255 readings
For CurrentAddress = 0 to 255

    'Take a reading and show it
    Print str(ReadAD12(AN0), TRUE)

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next
```

Example 3

This example used the differential capabilities of ADC port and writes the output to the EEPROM. The

output value will be in the range of [-4095 to 4095].

AN0 and AN2 are used for the differential ADC reading.

```
#chip 16F1789, 8

'USART settings
#define USART_BAUD_RATE 9600 'Initializes USART port with 9600 baud
#define USART_TX_BLOCKING ' wait for tx register to be empty
wait 100 ms

'Set the input pin direction
Dir PORTA.0 In
Dir PORTA.2 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

    'Take a reading and log it
    HSerPrint ReadAD12( AN0, AN2 )
    HserPrintCRLF
    'Wait 10 minutes before getting another reading
    Wait 10 min

Next
```

See Also [ReadAD](#), [ReadAD10](#)

Analog/Digital Conversion Code Optimisation

About Analog/Digital Conversion Code Optimisation

The analog to digital converter (ADC or A/D) module support is implemented by GCBASIC to provide 8-bit, 10-bit and 12-bit Single channel measurement mode and Differential Channel Measurement with support up to 34 channels. For compatibility all channels are supported.

There are two methods to optimise the code.

1. To minimise the code, use the constants to disable support for a specific channels
2. To adapt the ADC configuration by inserting specific commands to set registers or register bits.

1. Minimise the code

The example below would disable support for ADC port 0 (AD0).

```
#define USE_AD0 FALSE
```

The following tables show the #defines that can be used to reduce the code size - these are the defines for the standard microcontrollers. For 16f1688x and similar microcontrollers please see the second table.

Channel	Optimisation Value	Default Value
USE_AD0	FALSE	TRUE
USE_AD1	FALSE	TRUE
USE_AD2	FALSE	TRUE
USE_AD3	FALSE	TRUE
USE_AD4	FALSE	TRUE
USE_AD5	FALSE	TRUE
USE_AD6	FALSE	TRUE
USE_AD7	FALSE	TRUE
USE_AD8	FALSE	TRUE
USE_AD9	FALSE	TRUE
USE_AD10	FALSE	TRUE
USE_AD11	FALSE	TRUE
USE_AD12	FALSE	TRUE
USE_AD13	FALSE	TRUE
USE_AD14	FALSE	TRUE
USE_AD15	FALSE	TRUE
USE_AD16	FALSE	TRUE
USE_AD17	FALSE	TRUE
USE_AD18	FALSE	TRUE
USE_AD19	FALSE	TRUE
USE_AD20	FALSE	TRUE
USE_AD21	FALSE	TRUE
USE_AD22	FALSE	TRUE
USE_AD23	FALSE	TRUE
USE_AD24	FALSE	TRUE
USE_AD25	FALSE	TRUE
USE_AD26	FALSE	TRUE

Channel	Optimisation Value	Default Value
USE_AD27	FALSE	TRUE
USE_AD28	FALSE	TRUE
USE_AD29	FALSE	TRUE
USE_AD30	FALSE	TRUE
USE_AD31	FALSE	TRUE
USE_AD32	FALSE	TRUE
USE_AD33	FALSE	TRUE
USE_AD34	FALSE	TRUE

For 16f1688x devices see the table below.

Channel	Optimisation Value	Default Value
USE_ADA0	FALSE	TRUE
USE_ADA1	FALSE	TRUE
USE_ADA2	FALSE	TRUE
USE_ADA3	FALSE	TRUE
USE_ADA4	FALSE	TRUE
USE_ADA5	FALSE	TRUE
USE_ADA6	FALSE	TRUE
USE_ADA7	FALSE	TRUE
USE_ADC0	FALSE	TRUE
USE_ADC1	FALSE	TRUE
USE_ADC2	FALSE	TRUE
USE_ADC3	FALSE	TRUE
USE_ADC4	FALSE	TRUE
USE_ADC5	FALSE	TRUE
USE_ADC6	FALSE	TRUE
USE_ADC7	FALSE	TRUE
USE_ADD0	FALSE	TRUE
USE_ADD1	FALSE	TRUE
USE_ADD2	FALSE	TRUE
USE_ADD3	FALSE	TRUE

Channel	Optimisation Value	Default Value
USE_ADD4	FALSE	TRUE
USE_ADD5	FALSE	TRUE
USE_ADD6	FALSE	TRUE
USE_ADD7	FALSE	TRUE
USE_ADE0	FALSE	TRUE
USE_ADE1	FALSE	TRUE
USE_ADE2	FALSE	TRUE

This is an example - disables every channel except the specified channel by defining every channel except USE_AD0 as FALSE.

This will save 146 bytes of program memory.

```
#chip 16F1939

'USART settings
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Set the input pin direction
Dir PORTA.0 In

'Print 255 reading
For CurrentAddress = 0 to 255

    'Take a reading and show it
    HSerPrint str(ReadAD10(AN0))

    'Wait 10 minutes before getting another reading
    Wait 10 min
Next

#define USE_AD0 TRUE
#define USE_AD1 FALSE
#define USE_AD2 FALSE
#define USE_AD3 FALSE
#define USE_AD4 FALSE
#define USE_AD5 FALSE
#define USE_AD6 FALSE
#define USE_AD7 FALSE
#define USE_AD8 FALSE
```

```

#define USE_AD9 FALSE
#define USE_AD10 FALSE
#define USE_AD11 FALSE
#define USE_AD12 FALSE
#define USE_AD13 FALSE
#define USE_AD14 FALSE
#define USE_AD15 FALSE
#define USE_AD16 FALSE
#define USE_AD17 FALSE
#define USE_AD18 FALSE
#define USE_AD19 FALSE
#define USE_AD20 FALSE
#define USE_AD21 FALSE
#define USE_AD22 FALSE
#define USE_AD23 FALSE
#define USE_AD24 FALSE
#define USE_AD25 FALSE
#define USE_AD26 FALSE
#define USE_AD27 FALSE
#define USE_AD28 FALSE
#define USE_AD29 FALSE
#define USE_AD30 FALSE
#define USE_AD31 FALSE
#define USE_AD32 FALSE
#define USE_AD33 FALSE
#define USE_AD34 FALSE

```

For 16f18855 family of microcontrollers this is a example. This will save 149 bytes of program memory.

```

''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board
''' Date: 13.3.2021
'''

'Chip Settings.
#CHIP 16F18855,32
#CONFIG MCLRE_ON
#OPTION EXPLICIT

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D1 |-

```

```

''
''

#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT

#define SWITCH_DOWN      0
#define SWITCH_UP        1

#define SWITCH            PORTA.5

'Setup an Interrupt event when porta.5 goes negative.
IOCAN5 = 1
On Interrupt PORTABChange Call InterruptHandler

do

'Read the value from the EEPROM from register Zero in the EEPROM
EPRead ( 0, OutValue )

'Leave the Top Bytes alone and set the lower four bits
PortA = ( PortA & 0XF0 ) OR ( OutValue / 16 )
Sleep

loop

sub InterruptHandler

    if IOCAF5 = 1 then
        IOCAN5 = 0
InterruptHandler routine
        IOCAF5 = 0

        wait 5 ms
still held down
        if ( SWITCH = SWITCH_DOWN ) then
            'Read the ADC
            'S2 was just pressed
            'Prevent the event from reentering the
            'We must clear the flag in software
            'debounce by waiting and seeing if

```



```

        adc_value = readad ( AN4 )
        'Write the value to register Zero in the EEPROM
        EPWrite ( 0, adc_value )
    end if
    IOCAN5 = 1                                'ReEnable the InterruptHandler
routine

    end if

end sub

#define USE_ADA0 FALSE
#define USE_ADA1 FALSE
#define USE_ADA2 FALSE
#define USE_ADA3 FALSE
#define USE_ADA4 TRUE
#define USE_ADA5 FALSE
#define USE_ADA6 FALSE
#define USE_ADA7 FALSE
#define USE_ADB0 FALSE
#define USE_ADB1 FALSE
#define USE_ADB2 FALSE
#define USE_ADB3 FALSE
#define USE_ADB4 FALSE
#define USE_ADB5 FALSE
#define USE_ADB6 FALSE
#define USE_ADB7 FALSE
#define USE_ADC0 FALSE
#define USE_ADC1 FALSE
#define USE_ADC2 FALSE
#define USE_ADC3 FALSE
#define USE_ADC4 FALSE
#define USE_ADC5 FALSE
#define USE_ADC6 FALSE
#define USE_ADC7 FALSE
#define USE_ADD0 FALSE
#define USE_ADD1 FALSE
#define USE_ADD2 FALSE
#define USE_ADD3 FALSE
#define USE_ADD4 FALSE
#define USE_ADD5 FALSE
#define USE_ADD6 FALSE
#define USE_ADD7 FALSE
#define USE_ADE0 FALSE
#define USE_ADE1 FALSE
#define USE_ADE2 FALSE

```

2. Adapt the ADC configuration

Example 1:

The following example will set the specific register bits. The instruction will be added to the compiled code.

```
#define ADReadPreReadCommand  ADCON.2=0:ANSELA.0=1
```

The constant **ADReadPreReadCommand** can be used to adapt the ADC methods. The constant can enable registers or register bit(s) that are required to managed for a specific solution.

In the example above the following ASM will be added to your code. This WILL be added just before the ADC is enabled and the setting of the acquisition delay.

```
;ADReadPreReadCommand  
banksel ADCON  
bcf ADCON,2  
banksel ANSELA  
bsf ANSELA,0
```

Example 2:

The following example can be used to change the ADMUX to support a sensor on ADC4.

This supports reading the internal temperature sensor on the ATTINY85. This method will work on other similar chips. Please refer the chip specific datasheet.

This will call a macro to change the ADMUX to read the ATTINY85 internal temperature sensor, set the reference voltage to 1v1 and then wait 100 ms.

```
#define ADREADPREREADCOMMAND ATTINY85ReadInternalTemperatureSensor
```

```
Macro ATTINY85ReadInternalTemperatureSensor
```

```
/*
```

```
17.12 of the datasheet
```

The temperature measurement is based on an on-chip temperature sensor that is coupled to a single ended ADC4

channel. Selecting the ADC4 channel by writing the MUX[3:0] bits in ADMUX register to 1111 enables the temperature sensor. The internal 1.1V reference must also be selected for the ADC reference source in the

temperature sensor measurement. When the temperature sensor is enabled, the ADC converter can be used in

single conversion mode to measure the voltage over the temperature sensor.

The measured voltage has a linear relationship to the temperature as described in Table 17-2 The sensitivity is

approximately 1 LSB / °C and the accuracy depends on the method of user calibration. Typically, the measurement

accuracy after a single temperature calibration is $\pm 10^\circ\text{C}$, assuming calibration at room temperature. Better

accuracies are achieved by using two temperature points for calibration.

```
*/
```

```
IF ADReadPort=4 then
```

```
    ADMUX = ( ADMUX and 0X20 ) or 0X8F
```

```
    wait 100 ms
```

```
End if
```

```
End Macro
```

This will generate the following ASM.

```
;ADREADPREREADCOMMAND 'adds user code below
```

```
    lds SysCalcTempA,ADREADPORT
```

```
    cpi SysCalcTempA,4
```

```
    brne ENDIF2
```

```
    ldi SysTemp2,32
```

```
    in  SysTemp3,ADMUX
```

```
    and SysTemp3,SysTemp2
```

```
    mov SysTemp1,SysTemp3
```

```
    ldi SysTemp2,143
```

```
    or  SysTemp1,SysTemp2
```

```
    out ADMUX,SysTemp1
```

```
    ldi SysWaitTempMS,100
```

```
    ldi SysWaitTempMS_H,0
```

```
    rcall Delay_MS
```

```
ENDIF2:
```

Bitwise

This is the Bitwise section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Bitwise Operations Overview

About Bitwise Operations

GCBASIC (as with most other microcontroller programming languages) supports bitwise operations.

Bitwise operations are performed on one or more bit patterns at the level of their individual bits.

GCBASIC supports the following methods.

Method	Meaning
Set	Assigns a Bit value of On or Off
SetWith	Evaluates an expression and assigns the result
FnLSL	Performs a Bitwise LEFT shift
FnLSR	Performs a Bitwise RIGHT shift
Rotate	Performs a rotation of a variable of one bit in a specified direction

For more help, see: [Set](#), [SetWith](#), [FnLSL](#), [FnLSR](#) and [Rotate](#)

FnLSL

Syntax:

```
BitsOut = FnLSL(BitsIn, NumBits)
```

Command Availability:

Available on all microcontrollers.

Explanation:

FnLSL (Logical Shift Left) will perform a Bitwise left shift. **FnLSL** will return BitsIn shifted NumBits to the left, it is equivalent to the 'C' operation:

```
BitsOut = BitsIn << NumBits
```

Each left shift is the equivalent of multiplying BitsIn by 2. BitsIn and NumBits may be may be a

variable and of type: Bit, Byte, Word, Long, Constant or another Function. Zeros are shifted in from the right, Bits that are shifted out are lost.

It is useful for mathematical and logical operations, as well as creating serial data streams or manipulating I/O ports.

Example:

```
' This program will shift the LEDs on the Microchip PIC Low Pin
' Count Demo Board from Right to Left, that is DS1(RC0) to
' DS4(RC3) and repeat

#chip    16f690          ' declare the target Device

#define  LEDPORT PORTC ' LEDs on pins 16, 15, 14 and 7

Dim LEDMask as Byte      ' Pattern to be displayed
LEDMask = 0b0001          ' Initialise the Patten
Dir LEDPORT Out          ' Enable the LED Port.

Do
    LEDMask = FnLSL(LEDMask, 1) & 0x0F      ' Mask the lower 4 bits
    if LEDPORT.3 then LEDMask.0 = 1          ' Restart the sequence
    LEDPORT = LEDMask                        ' Display the Pattern
    wait 500 ms
Loop
End
```

See Also [Bitwise Operations Overview](#) and [Conditions](#)

FnLSR

Syntax:

```
BitsOut = FnLSR(BitsIn, NumBits)
```

Command Availability:

Available on all microcontrollers.

Explanation:

FnLSR (Logical Shift Right) will perform a Bitwise right shift. **FnLSR** will return BitsIn shifted NumBits to the right, it is equivalent to the 'C' operation:

```
BitsOut = BitsIn >> NumBits
```

Each right shift is the equivalent of dividing BitsIn by 2.

BitsIn and NumBits may be a variable and of type: Bit, Byte, Word, Long, Constant or another Function.

Zeros are shifted in from the left, Bits that are shifted out are lost.

It is useful for mathematical and logical operations, as well as creating serial data streams or manipulating I/O ports.

Example:

```
' This program will shift the LEDs on the Microchip PIC Low Pin Count Demo Board
' from Right to Left, that is DS4(RC3) to DS1(RC0) and repeat.

#chip    16f690          ' declare the target Device

#define  LEDPORT PORTC ' LEDs on pins 16, 15, 14 and 7

Dim LEDMask as Byte      ' Pattern to be displayed
LEDMask = 0b1000          ' Initialise the Pattern
Dir LEDPORT Out          ' Enable the LED Port.

Do
    LEDPORT = LEDMask      ' Display the Pattern
    wait 500 ms
    LEDMask = FnLSR(LEDMask, 1) & 0x0F ' Mask the lower 4 bits
    if LEDPORT.0 then LEDMask.3 = 1    ' Restart the sequence
Loop
End
```

See Also [Bitwise Operations Overview](#) and [Conditions](#)

SetWith

Syntax:

```
SetWith(TargetBit, Source)
```

Command Availability:

Available on all microcontrollers.

Explanation:

SetWith is an extended version of SET, it allows a Bit Field to be set or cleared by evaluating the content of Source. **SetWith** should always be used when TargetBit is an I/O Bit and Source is a Function, in order to avoid the possibility of I/O jitter.

Source may be a variable and of type: Bit, Byte, Word or Long, a Constant, an expression or a Function.

It will SET TargetBit to 1 if Source evaluates to anything other than zero. TargetBit will always be a 1 or a 0 regardless of the variable type of TargetBit.

Example:

```
' This program will reflect the state of SW1(RA3) on LED DS1(RC0) of the Microchip
' Low Pin Count Demo Board. Notice that because SW1 is normally High the state has to
' be inverted to turn on the LED (DS1) when SW1 is pressed.

#chip    16f690    ' declare the target Device

#Define SW1 PORTA.3
#Define DS1 PORTC.0

DIR DS1 Out
DIR SW1 In

Do
    ' set the Bit DS1 to equal the Bit SW1
    SetWith( DS1, !SW1 )
Loop
END
```

See Also [Bitwise Operations Overview](#) and [Conditions](#)

Memory

This is the Memory section of the Help file. Please refer the sub-sections for details using the contents/folder view.

MCU EEPROM (DFM)

This is the EEPROM (PFM) section of the Help file. Please refer the sub-sections for details using the contents/folder view.

EPRead

Syntax:

```
EPRead location, store
```

Command Availability:

Available on all Microchip PIC and Atmel AVR microcontrollers with EEPROM data memory.

Explanation:

EPRead is used to read information from the EEPROM data storage that many microcontroller chips are equipped with. **location** represents the location to read data from, and varies from one chip to another. **store** is the variable in which to store the data after it has been read from EEPROM.

Note Do not exceed the location (also known as the EEPROM address) of the physical EEPROM. If the EEPROM size is 256 ensure location is in the range of 0 to 255; If the EEPROM size is 512 ensure location is in the range of 0 to 511 and use a Word variable as the location parameter. Example:

Example:


```

'Program to turn a light on and off
'Will remember the last status

#chip tiny2313, 1
#define Button PORTB.0
#define Light PORTB.1

Dir Button In
Dir Light Out

'Load saved status
EPRead 0, LightStatus

If LightStatus = 0 Then
    Set Light Off
Else
    Set Light On
End If

Do
    'Wait for the button to be pressed
    Wait While Button = On
    Wait While Button = Off
    'Toggle value, record
    LightStatus = !LightStatus
    EPWrite 0, LightStatus

    'Update light
    If LightStatus = 0 Then
        Set Light Off
    Else
        Set Light On
    End If
Loop

```

For more help, see [EPWrite](#)

EPWrite

Syntax:

```
EPWrite location, data
```

Command Availability:

Available on all Microchip PIC and Atmel AVR microcontrollers with EEPROM data memory.

Explanation:

EPWrite is used to write information to the EEPROM data storage, so that it can be accessed later by a programmer on the PC, or by the **EPRead** command. **location** represents the location to write data to, and the location varies from one chip to another. **data** is the data that is to be written to the EEPROM, and can be a value or a variable.

NOTE

Do not exceed the location (also known as the EEPROM address) of the physical EEPROM. If the EEPROM size is 256 ensure location is in the range of 0 to 255; If the EEPROM size is 512 ensure location is in the range of 0 to 511 and use a Word variable as the location parameter.

Example:

```
#chip 16F819, 8

'Set the input pin direction
Dir PORTA.0 In

'Loop to take readings until the EEPROM is full
For CurrentAddress = 0 to 255

'Take a reading and log it
EPWrite CurrentAddress, ReadAD(AN0)

'Wait 10 minutes before getting another reading
Wait 10 min

Next
```

For more help, see [EPRead, Creating EEPROM data from a Lookup Table](#)

Dataset for EEPROM

Syntax:

```
EEPROM DataSetName [[,]address]
    // multiples values, strings etc.
    0,1,2,3
END EEPROM
```

Command Availability:

Available on all PIC microcontrollers with EEPROM memory. AVR support required use of AVR-ASM assembler. GCASM does not support AVR EEPROM operations.

Explanation:

The EEPROM construct creates an EEPROM dataset for use with the specific microcontroller. An EEPROM dataset is a list of values that are stored in the EEPROM memory of the microcontroller, which then can be accessed using the EPREAD() command or other EEPROM read operations.

The advantage of an EEPROM dataset is that they are memory efficient being loaded directly into the EEPROM during programming operations.

EEPROM datasets are defined as follows:

1. Byte values,
2. EEPROM addresses and EEPROM datasets CANNOT overlap,
3. EEPROM addresses must not overlap TABLE data,
4. TABLE data has precedence from address 0x00 until the the end of TABLE all data,
5. Strings must be expressed as ASCII byte value(s),
6. Multiple elements on a single line separated by commas,
7. Constants and calculations within the single line dataset entries are permitted,
8. Decimal values are NOT supported,
9. Access is via EPRread(), not supported by READTABLE().
10. 18F devices must use even address for EEPROM location, and, 18F will pad (with 0x00) datasets to even number length. This is MPASM constraint and therefore the compiler and assembler will issue specific error messages for odd EEPROM locations.

Defining EEPROM datasets

Single data values

A single value on each line with in the dataset. The example dataset, shown below, has the data on different line in within the set.

Simple example: This creates an EEPROM dataset at the first EEPROM location, then, the values of 12, 24, ... 72 are the consecutive values.

```
EEPROM EEDataSet
    12
    24
    36
    48
    60
    72
End EEPROM
```

Multiple data values of the same line

The following example creates the EEPROM dataset at EEPROM offset address of 0x10.

Multiple elements on a single line separated by commas. The example dataset, shown below, has the data separated by , and on different line in within the dataset.

```
EEPROM EEDataSource 0x10
    12, 24, 36
    48, 60, 72
End EEPROM
```

Data values as constants, and, with data transformation

Constants and calculations within the single line. The example dataset, shown below, uses a defined constant to multiple the data with the dataset.

```
#define calculation_constant 2

EEPROM EEDataSource 0x20
1 * calculation_constant
2 * calculation_constant
3 * calculation_constant
8 * calculation_constant
4 * calculation_constant
5 * calculation_constant
End EEPROM
```

Data values as Strings

Strings can be defined. Strings are delimited by double quotes. The following examples show the methods.

Any ASCII characters between any two " " (double quotes) will be converted to dataset data. Also see ASCII escape codes.

A source string can be one string per line or comma separated strings, therefore, on the same line.

Example:

```
EEPROM Test_1
"ABCDEFGHJIJ"
End EEPROM
```

ASCII Escape code

Accepted escape strings are shown in the dataset below.

Escape sequence	Meaning
\a	beep
\b	backspace
\f	formfeed
\l or \n	newline
\r	carriage return
\t	tab
\0	Null value, equates to ASCII 0. Same as \&000
\&nnnn	ascii char in decimal
\\	backslash
\"	double quote
\'	single quote

Complete working example program

This example creates several EEPROM datasets. The example also create a lookup table. The EEPROM dataset are addressed with the additional parameter to ensure there is no EEPROM dataset overlap.

```
#chip 16F886
#option explicit

#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

```

Dim EEdataaddress, myvar as Byte
EEdataaddress = 2

Readtable TwoBytes,EEdataaddress,myVar
HSerPrint myVar

// ***** EXAMPLE EE DATA *****
// * THIS IS ONLY ACCESSIBLE VIA EPREAD or other EE read functions.
/*
Usage:      EEPROM EEPROMBlockName [[,] OffSet Address ]
            OffSet address defaults to 0x00 if not stated.

            Addresses and datasets CANNOT overlap.
            Addresses must not overlap TABLE data.
            TABLE data has precedence from address 0x00 until the the end of
TABLE data
*/

EEPROM EEDataset1 0x10    // Locate EE Data at address
3,2,1
End EEPROM

EEPROM VersionData 0x20    // Locate EE Data at address
"   PWM2Laser   "
"   Fabrice ENGEL   "
"   Version 1.4   "
"   November 2023   "
End EEPROM

EEPROM EEDataset2 0x0D    // Locate EE Data at address
1,2,3
End EEPROM

EEPROM EEDataset 0X04      // Locate EE Data at address
1,2,3
End EEPROM

// ***** EXAMPLE TABLE DATA BEING LOADED INTO EE BY THE
COMPIILER
// *                      THIS IS ONLY ACCESSIBLE VIA READTABLE

Table TwoBytes STORE data // EE Data Address Allocated by compiler
    0X55,0XAA,0X55
End Table

```

For more help, see [EPRead, Creating EEPROM data from a Lookup Table](#)

HEFM (PFM)

This is the HEFM (PFM) section of the Help file. Please refer the sub-sections for details using the contents/folder view.

HEFM Overview

Introduction:

Some enhanced mid-range Microchip PIC devices support High-Endurance Flash (HEF) memory. These devices lack the data EEPROM found on other devices. Instead, they implement an equivalent amount of special flash memory, called HEF memory, that can provide an endurance comparable to that of a traditional data EEPROM. HEF memory can be erased and written 100,000 times. HEF memory appears in the regular program memory space and can be used for any purpose, like regular flash program memory.

As with all flash memory, data must be erased before it can be written and writing this memory will stall the device. Methods to read, write and erase the HEF memory are included in GCBASIC and they are described in this introduction. Also see Microchip application note AN1673, Using the PIC16F1XXX High-Endurance Flash (HEF) Block.

The `hefsaf.h` library supports HEF operations for GCBASIC.

Note: By default, GCBASIC will use HEF memory for regular executable code unless it is told otherwise. If you wish to store data here, you should reserve the HEF memory by using the compiler option, as shown below to reserve 128 words of HEF memory:

```
#option ReserveHighProg 128
```

HEF memory is a block of memory locations found at the top of the flash program memory. Each memory location can be used to hold a 8-bit byte value. To further explain, the PIC 16F Enhanced Midrange Services memory architecture is 14-bits wide. Therefore, for a single 14-bit memory location it is only practical to store an 8-bit byte value, and two 14-bit memory locations to hold one 16-bit word value. This is because the memory architecture only allows the use of the the lower 8-bits of each 14-bit flash memory location for HEF usage

The main difference between HEF memory and EEPROM is that EEPROM allows byte-by-byte erase whereas the HEF memory does not. With HEF memory, data must be erased before a write and the erase can only be performed in blocks of memory. The blocks, also called rows, are a fixed size associated with the specific device.

GCBASIC handles the erase operation automatically. When a write operation is used by a user the GCBASIC library reads to a cache, updates the cache, erase the block and finally write the caches. The complexity of using HEF memory is reduced with the automatically handling of these operations.

The `hefsaf.h` library provides a set of methods to support the use of HEF memory.

Method	Parameters	Usage
<code>HEFWrite</code>	a subroutine with the parameters: location, byte value	<code>HEFWrite (location, byte_variable)</code>
<code>HEFWriteWord</code>	a subroutine with the parameters: location, word_value	<code>HEFWriteWord (location, word_variable)</code>
<code>HEFRead</code>	a function with the parameters: location returns a byte value	<code>byte_variable = HEFRead (location)</code>
<code>HEFRead</code>	a subroutine with the parameters: location, byte_value	<code>HEFRead (location , out_byte_variable)</code>
<code>HEFReadWord</code>	a function with the parameters: location returns a word value	<code>word_variable = HEFRead (location)</code>
<code>HEFReadWord</code>	a subroutine with the parameters: location, word_value	<code>HEFRead (location , out_word_variable)</code>

HEFEraseBlock	a subroutine with the parameters: block_number	HEFEraseBlock (0) A value of 0,1,2,3 etc.
HEFWriteBlock	a subroutine with the parameters: block_number, buffer() [, HEF_ROWSIZE_BYTES]	HEFWriteBlock(0, myMemoryBuffer) 'where myMemoryBuffer is an Array or a String The Array or a String will contain the values to be wrttin to the HEFM.
HEFReadBlock	a subroutine with the parameters: block_number, buffer() [, HEF_ROWSIZE_BYTES]	HEFReadBlock(0, myMemoryBuffer) 'where myMemoryBuffer is an Array or a String. The Array or a String will contain the values from the HEFM.

The library also defines a set constants that are specific to the device. These may be useful in the user program. These constants are used by the library. A user may use these public constants.

Constant	Type	Usage
HEF_ROWSIZE_BYTES	Byte	Size of an HEFM block in words
HEF_WORDS and HEF_BYTES	Word or a Byte	ChipHEFMemWords parameter from the device .dat file
HEF_START_ADDR	Word	Starting address of HEFM
HEF_NUM_BLOCKS	Byte	Number of blocks of HEFM
CHIPWORDS	Word	Device specific constant for the total flash size
CHIPHEFMEMWORDS	Word	Device specific constant for the number of HEFM words available

CHIPERASEROWSIZEWORDS	Word	Device specific constant for the number of HEFM in an erase row
-----------------------	------	---

Warning

Whenever you update the hex file of your Microchip PIC micro-controller with your programmer you MAY erase the data that are stored in HEF memory. If you want to avoid that you will have to flash your Microchip PIC micro-controller with software that allows memory exclusion when flashing. This is the case with Microchip PIC MPLAB IPE (Go to [Advanced Mode/Enter password/Select Memory/Tick "Preserve Flash on Program"/ Enter Start and End address](#) of your HEFM). Or, simply use the PICKitPlus suite of software to preserve HEF memory during programming.

See also [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

HEFRead

Syntax:

```
'as a subroutine
HEFRead ( location, data )

'as a function
data = HEFRead ( location )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory

Explanation:

HEFRead is used to read information, byte values, from HEFM, so that it can be accessed for use in a user program.

[location](#) represents the location or relative address to read. The location will range from location 0 to HEF_BYTES - 1, or for all practical purposes 0-127 since all PIC Microcontrollers with HEF support 128 bytes of HEF Memory. HEF_BYTES is a GCBASIC constant that represents the number of bytes of HEF Memory.

data is the data that is to be read from the HEFM data storage area. This can be a byte value or a byte variable.

This method reads data from HEFM given the specific relative location. This method is similar to the EPRead method for EEPROM.

Example 1:

```
'... code preamble to select part
'... code to setup PPS
'... code to setup serial

'The following example reads the HEFM data value into the byte variable "byte_value"
using a subroutine.

Dim data_byte as byte

;Write a byte of data to HEFM Location 34
HEFWrite( 34, 144)

;Read the byte back from HEFM location 34
HEFread( 34, byte_value )

;Display the data on a terminal
HserPrint "byte_value = "
Hserprint byte_value
```

Example 2:

```
'... code preamble to select part '... code preamble to select part  
'... code to setup PPS  
'... code to setup serial
```

'The following example reads the HEFM data value into the byte variable "byte_value" using a function.

```
Dim data_byte as byte  
  
;Write a byte of data to HEF Location 34  
HEFWrite( 34, 144)  
  
;Read the byte back from HEF location 34  
byte_value = HEFread( 34 )  
  
;Display the data on a terminal  
HserPrint "byte_value = "  
Hserprint byte_value
```

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

HEFReadWord

Syntax:

```
'as a subroutine  
HEFReadWord ( location, data_word_variable )  
  
'as a function  
data_word_variable = HEFReadWord ( location )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory

Explanation:

HEFReadWord is used to read information, word values, from HEFM so that it can be accessed for use in a user program.

location represents the location or relative address to read. The location will range from location 0 to HEF_BYTES - 1, or for all practical purposes 0-127 since all PIC Microcontrollers with HEF support 128

bytes of HEF Memory. HEF_BYTES is a GCBASIC constant that represents the number of bytes of HEF Memory.

data is the data that is to be read from the HEFM data storage. This must be a word variable.

This method reads data from HEFM given the specific relative location.

Example 1:

```
'... code preamble to select part
'... code to setup serial

'The following example reads the HEFM value into the word variable
"data_word_variable" by initially writing some word values.

dim data_word_variable as word
HEFWriteWord( 254, 4660 )

HEFReadWord( 254, data_word_variable )

HSerPrint "Value = "
HSerPrint data_word_variable
HSerPrintCRLF
```

If example 1 were displayed on a serial terminal. The result would show:

```
Value = 4660
```

Example 2:

```
'... code preamble to select part  
'... code to setup serial
```

'The following example uses a function to read the HEFM value into the word variable "data_word_variable".

```
dim data_word_variable as word  
HEFWriteWord( 254, 17185 )  
  
data_word_variable = HEFReadWord( 254 )  
  
HSerPrint "Value = "  
HSerPrint data_word_variable  
HSerPrintCRLF
```

If example 2 were displayed on a serial terminal. The result would show:

```
Value = 17185
```

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFeraseBlock](#)

HEFWrite

Syntax:

```
HEFWrite ( location, data )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory

Explanation:

HEFWrite is used to write information, byte values, to HEFM so that it can be accessed later for use in a user program.

location represents the location or relative address to write. The location will range from location 0 to HEF_BYTES - 1, or for all practical purposes 0-127 since all PIC Microcontrollers with HEF support 128 bytes of HEF Memory. HEF_BYTES is a GCBASIC constant that represents the number of bytes of HEF

Memory.

data is the data that is to be written to the HEFM location. This can be a byte value or a byte variable.

This method writes information to the HEFM given the specific location. This method is similar to the EPWrite method for EEPROM.

Example 1:

```
'... code preamble to select part
'... code to setup serial

'The following example writes a byte value of 126 into HEFM location 34

HEFWrite( 34, 126 )
```

Example 2:

```
'... code preamble to select part
'... code to setup serial

'This example will populate all 128 bytes of HEF memory with a value that is same as
the HEFM location

Dim Rel_Address, DataByte as Byte
Dim NVM_Address as Long
Dim DataWord, as Word
Dim HEFaddress as Byte

For Rel_Address = 0 to 127
    HEFWrite ( Rel_Address, Rel_Address )
Next
HEFM_DUMP

End

; This subroutine displays the High Endurance Flash Memory on a terminal.
; Words are in reverse byte order relative to address.
; HEF data resides in the low byte of each 14bit program memory word.
; The high byte is not HEF and should always read "3F".

Sub HEFM_DUMP
```



```

Dim Blocknum as Byte
NVM_Address = HEF_START_ADDR
BlockNum = 0

Repeat HEF_BYTES ;128

    If NVM_Address % HEF_ROWSIZE_BYTES = 0 then
        If BlockNum > 0 then HSERPRINTCRLF
        HSerprintCRLF
        HserPrint "Block"
        HSerprint BlockNum
        HSerprint "    0    1    2    3    4    5    6    7"
        BlockNum++
    End if

    IF NVM_Address % 8 = 0 then
        HSerPrintCRLF
        hserprint hex(NVM_Address_H)
        hserprint hex(NVM_ADDRESS)
        hserprint "    "
    end if

    Rel_Address = (NVM_ADDRESS - HEF_START_ADDR)
    HEFRead(Rel_Address, DataWord)

    hserprint hex(DataWord_H)
    hserprint hex(DataWord)
    hserprint "    "

    NVM_Address++
End Repeat
HserPrintCRLF
End sub

```

If example 2 were displayed on a serial terminal. The result would show:

Block0	0	1	2	3	4	5	6	7
3F80	3F00	3F01	3F02	3F03	3F04	3F05	3F06	3F07
3F88	3F08	3F09	3F0A	3F0B	3F0C	3F0D	3F0E	3F0F
3F90	3F10	3F11	3F12	3F13	3F14	3F15	3F16	3F17
3F98	3F18	3F19	3F1A	3F1B	3F1C	3F1D	3F1E	3F1F
Block1	0	1	2	3	4	5	6	7
3FA0	3F20	3F21	3F22	3F23	3F24	3F25	3F26	3F27
3FA8	3F28	3F29	3F2A	3F2B	3F2C	3F2D	3F2E	3F2F
3FB0	3F30	3F31	3F32	3F33	3F34	3F35	3F36	3F37
3FB8	3F38	3F39	3F3A	3F3B	3F3C	3F3D	3F3E	3F3F
Block2	0	1	2	3	4	5	6	7
3FC0	3F40	3F41	3F42	3F43	3F44	3F45	3F46	3F47
3FC8	3F48	3F49	3F4A	3F4B	3F4C	3F4D	3F4E	3F4F
3FD0	3F50	3F51	3F52	3F53	3F54	3F55	3F56	3F57
3FD8	3F58	3F59	3F5A	3F5B	3F5C	3F5D	3F5E	3F5F
Block3	0	1	2	3	4	5	6	7
3FE0	3F60	3F61	3F62	3F63	3F64	3F65	3F66	3F67
3FE8	3F68	3F69	3F6A	3F6B	3F6C	3F6D	3F6E	3F6F
3FF0	3F70	3F71	3F72	3F73	3F74	3F75	3F76	3F77
3FF8	3F78	3F79	3F7A	3F7B	3F7C	3F7D	3F7E	3F7F

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

HEFWriteWord

Syntax:

```
HEFWriteWord ( location, data_word_value )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory

Explanation:

HEFWriteWord is used to write information, word values, to HEFM, so that it can be accessed in a user program via the HEFReadWord command.

location presents the location or relative address to write write. A data Word requires 2 HEF Locations,

therefore the location will range from 0 to 126 in steps of 2.

data is the data that is to be written to the HEFM location. This can be a word value or a word variable.

This method writes information to the HEFM given the specific location in the HEFM data storage . This method is similar to the methods for EEPROM but this method supports Word values.

Example 1:

```
'... code preamble to select part
'... code to setup serial

'The following example stores a word value in HEFM location 0

HEFWrite( 0, 0x1234)
```

Example 2:

```
'... code preamble to select part
'... code to setup serial

'This example will write two word values to two specific locations.
HEFWriteWord (16, 0xAA01)
HEFWriteWord (18, 0xBB02)
```

If example 2 were displayed on a serial terminal. The result would show, where -- is the existing value.

```
Block0
3F00  -- -- -- -- -- -- -- -- -- -- -- -- -- --
3F10  01 AA 02 BB -- -- -- -- -- -- -- -- -- --
3F20  -- -- -- -- -- -- -- -- -- -- -- -- -- --
3F30  -- -- -- -- -- -- -- -- -- -- -- -- -- --
```

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#),

HEFReadBlock

Syntax:

```
HEFReadBlock ( block_number, buffer(), [, num_bytes] )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory.

Explanation:

HEFReadBlock is used to read information from the HEFM data storage into the buffer. Once the buffer is populated it can be accessed for use within a user program.

The parameters are as follows:

block_number represents the block to be written to. The block_number parameter is used to calculate the physical memory location(s) that are updated.

buffer() represents an array or string. The buffer will be used as the data target for the block read operation. The buffer is handled as a buffer of bytes values. In most cases the buffer should be the same size as a row/block of HEFM. For most PIC Microcontrollers this will be 32 bytes. For PIC microcontrollers with 2KW or less of Flash Program Memory this will be 16 Bytes. Once data is read into the buffer from HEFM, the user program must handle the data as Byte, Word or String values, as appropriate.

num_bytes is an optional parameter, and can be used to specify number of bytes to read from HEFM, starting at the first location in the selected HEFM block. This parameter is not normally required as the default is set to the GCBASIC constant **HEF_ROWSIZE_BYTES**.

Example 1:

```

'... code preamble to select part
'... code to setup serial

Dim My_Buffer(HEF_ROW_SIZE_BYTES)
Dim index as byte

;Write some data to Block 2
My_Buffer =
1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32
HEFWriteBlock(2, My_Buffer())

;Read the data back from HEFM using HEFReadBlock
HEFReadBlock( 2 , My_buffer() )

;Send the data to a terminal in decimal format
index = 1
Repeat HEF_ROW_SIZE_BYTES
    Hserprint(My_Buffer(index))
    HserPrint " "
    index++
End Repeat

```

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

HEFWriteBlock

Syntax:

```
HEFWriteBlock ( block_number, buffer(), [, num_bytes] )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory.

Explanation:

HEFWriteBlock is used to write information from a user buffer to HEFM. Once the block is written it can be accessed for use within a user program.

The parameters are as follows:

block_number represents the block to be written to. The block_number parameter is used to calculate

the physical memory location(s) that are updated.

`buffer()` represents an array or string. The buffer will be used as the data source that is written to the HEFM block. The buffer is handled as a buffer of bytes values. In most cases the buffer should be the same size as a row/block of HEFM. For most PIC Microcontrollers this will be 32 bytes. For PIC microcontrollers with 2KW or less of Flash Program Memory this will be 16 Bytes. Best practice is to size the buffer using the `HEF_ROWSIZE_BYTES` constant. If the size of the buffer exceeds the device specific `HEF_ROWSIZE_BYTES`, the excess data will not be handled and the buffer will be truncated at the `HEF_ROWSIZE_BYTES` limit.

`num_bytes` is an optional parameter, and can be used to specify number of bytes to write to HEFM, starting at the first location in the selected HEFM block. This parameter is not normally required as the default is set to the GCBASIC constant `HEF_ROWSIZE_BYTES`.

Example 1:

```
'... code preamble to select part
'... code to setup serial

Dim My_Buffer(HEF_ROWSIZE_BYTES)

My_Buffer =
1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32

'HEFwriteBlock operation!!
HEFwriteBlock(2, My_Buffer)

'A utility method to show the contents of HEFM.
HEFM_Dump
```

For HEFM_Dump routine, see [HEFRead](#)

If example 1 were displayed on a serial terminal using HEFM_Dump. The result would show. Note the value display at the start of block 2 @ 0x3F80.

Block0	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F00	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F10	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F20	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F30	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
Block1	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F40	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F50	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F60	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7F70	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
Block2	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F80	0201	0403	0605	0807	0A09	0C0B	0E0D	100F
7F90	1211	1413	1615	1817	1A19	1C1B	1E1D	201F
7FA0	2120	2322	2524	2726	2928	2B2A	2D2C	2F2E
7FB0	3130	3332	3534	3736	3938	3B3A	3D3C	3F3E
Block3	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7FC0	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7FD0	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7FE0	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF
7FF0	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

HEFEraseBlock

Syntax:

```
HEFEraseBlock ( block_number )
```

Command Availability:

Available on all PIC micro-controllers with HEFM memory.

Explanation:

HEFEraseBlock is used to erase all data locations within the HEFM block. HEFM data within the HEFM block to the erase state value of the device. This Value is 0xFF and will read 0x3FFF if the entire 14bit program memory word is displayed. Use Caution. Once the HEFM block is erased, the HEFM data is gone forever and cannot be recovered unless it was previpusly saved.

The single parameter is as follows:

block_number represents the block to be erased. The block_number parameter is used to calculate the physical memory location(s) that are updated.

Example 1:

Erase a specific block of HEFM.

```
'... code preamble to select part  
'... code to setup serial, if needed  
  
'Erase block 2 of HEFM  
HEFEraseBlock ( 2)
```

See also [HEFM Overview](#), [HEFRead](#), [HEFReadWord](#), [HEFWrite](#), [HEFWriteWord](#), [HEFReadBlock](#), [HEFWriteBlock](#), [HEFEraseBlock](#)

PROGMEM (PFM)

This is the PROGMEM (PFM) section of the Help file. Please refer the sub-sections for details using the contents/folder view.

PFMRead

Syntax:

```
PFMRead (location, store)
```

Command Availability:

Available on all Microchip PIC microcontrollers with PFM self write capability.

Explanation:

PFMRead reads information from the program memory on chips that support this feature. **location** is a word variable, and **store** can be a byte of word.

The largest value possible for **location** depends on the amount of program memory on the Microchip PIC microcontroller.

This is an advanced command which should only be used by advanced developers.

For more help, see [PFMWrite](#)

PFMWrite

Syntax:

```
PFMWrite (location, value)
```

Command Availability:

Available on all Microchip PIC microcontrollers with PFM self write capability.

Explanation:

PFMWrite writes information to the program memory on chips that support this feature. **location** is a word variable, and **store** can be a byte of word.

The largest value possible for **location** depends on the amount of program memory on the microcontroller.

This is an advanced command which should only be used by advanced developers.

Example:

For more help, see [*PFMRead](#)

DATA**Syntax:**

```
DATA DataSetName [as Byte | Word]
    // multiples values, strings etc.
    0,1,2,3
END DATA
```

Command Availability:

Available on all PIC microcontrollers with DATA memory.

Explanation:

The DATA construct creates an DATA dataset, or DATA block, for use with the specific microcontroller. A DATA dataset, or DATA block, is a list of values that are stored in the PROGMEM memory of the microcontroller, which then can be accessed using the ProgRead() command or other DATA read operations.

The advantage of an DATA dataset, or DATA block, is that they are memory efficient being loaded directly into the DATA during programming operations.

DATA datasets are defined as follows:

1. Byte or Word values,
2. Multiple numeric elements on a single line separated by commas,
3. Constants and calculations within the single line dataset entries are permitted,
4. Decimal values are NOT supported,
5. Access is via ProgRead().

Defining DATA datasets*Single data values*

A single value on each line within the dataset. The example dataset, shown below, has the data on different line within the set.

Simple example: This creates an DATA dataset at the first DATA location, then, the values of 12, 24, ... 72 are the consecutive values.

```
DATA EEDataset as Byte
    12
    24
    36
    48
    60
    72
End DATA
```

Multiple data values of the same line

The following example creates the DATA dataset at DATA offset address of 0x10.

Multiple elements on a single line separated by commas. The example dataset, shown below, has the data separated by , and on different line in within the dataset.

```
DATA EEDataSource as Byte
    12, 24, 36
    48, 60, 72
End DATA
```

Data values as constants, and, with data transformation

Constants and calculations within the single line. The example dataset, shown below, uses a defined constant to multiple the data with the dataset.

```
#define calculation_constant 2

DATA EEDataSource as Word
1 * calculation_constant
2 * calculation_constant
3 * calculation_constant
8 * calculation_constant
4 * calculation_constant
5 * calculation_constant
End DATA
```

Data values as Strings

Strings can be defined. Strings are delimited by double quotes. The following examples show the methods.

Any ASCII characters between any two " " (double quotes) will be converted to dataset data. Also see ASCII escape codes.

A source string can be one string per line or comma separated strings, therefore, on the same line.

Example:

```
DATA Test_1 as Byte
  "ABCDEFGHIJ"
End DATA
```

ASCII Escape code

Accepted escape strings are shown in the dataset below.

Escape sequence	Meaning
\a	beep
\b	backspace
\f	formfeed
\l or \n	newline
\r	carriage return
\t	tab
\0	Null value, equates to ASCII 0. Same as \&000
\&nnnn	ascii char in decimal
\\	backslash
\"	double quote
\'	single quote

Maximum Stored Value

The **maximum value** that can be stored in a single program memory **word** location across the PIC families you asked about (PIC10, PIC12, PIC14, PIC16, PIC18), formatted as clean markdown:

Family	Instruction Word Size	Program Memory Word Width	Maximum Value per Word (unsigned decimal)	Hex Range	Can store full 16-bit value (0–65535) in one word?

PIC10	12-bit	12 bits	4095	0x000 – 0xFFFF	No
PIC12 (baseline)	12-bit	12 bits	4095	0x000 – 0xFFFF	No
PIC12 (enhanced mid-range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC14	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC16 (mid- range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC16 (enhanced mid-range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC18	16-bit	16 bits	65535	0x0000 – 0xFFFF	Yes

Quick Summary Table (most common cases)

Family group	Typical word size	Max value you can store in one program memory word	Equivalent to storing a full 16-bit number?
PIC10 / baseline PIC12	12 bits	4095 (0xFFFF)	No
PIC14 / PIC16 / enhanced PIC12	14 bits	16383 (0x3FFF)	No
PIC18	16 bits	65535 (0xFFFF)	Yes

Important Notes

- **PIC18** is the only 8-bit PIC family where you can directly store any 16-bit value (0–0xFFFF) in a single program memory word location.
- On all earlier families (PIC10, PIC12 baseline, PIC14, PIC16, enhanced mid-range), you must split any value > 0x3FFF (16383) across **two words** if you need the full 16-bit range.
- The values above apply when storing constants, lookup table entries, **retlw** literals, **data/db** directives, etc. — i.e., the largest number that fits in one program memory **word**.

Complete working example program

This example creates several DATA datasets. The example also create a lookup table. The DATA dataset are addressed with the additional parameter to ensure there is no DATA dataset overlap.

```

#chip 16F886
#option explicit

#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF

Dim dataaddress, datavalue as Byte

DATA DataSet1 as Byte
    3,2,1
End DATA

DATA VersionData as Byte
"    PWM2Laser    "
"    Fabrice ENGEL "
"    Version 1.4   "
"    November 2023 "
End DATA

For dataaddress = 0 to 2
    ProgramRead ( @DataSet1 + dataaddress , datavalue )
    HserPrint datavalue
Next

```

For more help, see [ProgramRead](#), [Creating data from a Lookup Table](#)

ProgramErase

Syntax:

```
ProgramErase (location)
```

Command Availability:

Available on all Microchip PIC microcontrollers with self write capability. Not available on Atmel AVR at present.

Explanation:

ProgramErase erases information from the program memory on chips that support this feature. The largest value possible for **location** depends on the amount of program memory on the Microchip PIC microcontroller, which is given on the datasheet.

This command must be called before writing to a block of memory. It is slow in comparison to other

GCBASIC commands. Note that it erases memory in 32-byte blocks - see the relevant Microchip PIC microcontroller datasheet for more information.

This is an advanced command which should only be used by advanced developers. Care must be taken with this command, as it can easily erase the program that is running on the microcontroller.

For more help, see [ProgramRead](#) and [ProgramWrite](#)

ProgramRead

Syntax:

```
ProgramRead (location, store)

or for the 18FxxQ41 family of chips use:
PFMRead (location, store)
```

Command Availability:

Available on all Microchip PIC microcontrollers with self write capability. Not available on Atmel AVR at present.

Explanation:

ProgramRead reads information from the program memory on chips that support this feature. **location** and **store** are both word variables, meaning that they can store values over 255.

The largest value possible for **location** depends on the amount of program memory on the Microchip PIC microcontroller, which is given on the datasheet. **store** range is explained in the next section.

Maximum Stored Value

The **maximum value** that can be stored in a single program memory **word** location across the PIC families you asked about (PIC10, PIC12, PIC14, PIC16, PIC18), formatted as clean markdown:

Family	Instruction Word Size	Program Memory Word Width	Maximum Value per Word (unsigned decimal)	Hex Range	Can store full 16-bit value (0-65535) in one word?
PIC10	12-bit	12 bits	4095	0x000 – 0xFFFF	No
PIC12 (baseline)	12-bit	12 bits	4095	0x000 – 0xFFFF	No

PIC12 (enhanced mid-range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC14	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC16 (mid-range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC16 (enhanced mid-range)	14-bit	14 bits	16383	0x0000 – 0x3FFF	No
PIC18	16-bit	16 bits	65535	0x0000 – 0xFFFF	Yes

Quick Summary Table (most common cases)

Family group	Typical word size	Max value you can store in one program memory word	Equivalent to storing a full 16-bit number?
PIC10 / baseline PIC12	12 bits	4095 (0xFFFF)	No
PIC14 / PIC16 / enhanced PIC12	14 bits	16383 (0x3FFF)	No
PIC18	16 bits	65535 (0xFFFF)	Yes

Important Notes

- **PIC18** is the only 8-bit PIC family where you can directly store any 16-bit value (0–0xFFFF) in a single program memory word location.
- On all earlier families (PIC10, PIC12 baseline, PIC14, PIC16, enhanced mid-range), you must split any value > 0x3FFF (16383) across **two words** if you need the full 16-bit range.
- The values above apply when storing constants, lookup table entries, `retlw` literals, `data/db` directives, etc. — i.e., the largest number that fits in one program memory **word**.

This is an advanced command which should only be used by advanced developers.

For more help, see [ProgramErase](#) and [ProgramWrite](#)

ProgramWrite

Syntax:

```
ProgramWrite (location, value)
```


Command Availability:

Available on all Microchip PIC microcontrollers with self write capability. Not available on Atmel AVR at present.

Explanation:

ProgramWrite writes information to the program memory on chips that support this feature. **location** and **value** are both word variables.

The largest value possible for **location** depends on the amount of program memory on the microcontroller , which is given on the datasheet. **value** is 14 bits wide, and thus can store values up to 16383.

This is an advanced command which should only be used by advanced developers. ProgramErase must be used to clear a block of memory BEFORE **ProgramWrite** is called.

Example:

For more help, see [ProgramErase](#) and [ProgramRead](#)

PROGMEM (MCU Configuration)

This is the PROGMEM (MCU Configuration) section of the Help file. Please refer the sub-sections for details using the contents/folder view.

DeviceConfigurationRead

Syntax:

```
deviceconfigurationRead (location, store)
```

Command Availability:

Available on all Microchip PIC microcontrollers with self read capability. Not available on Atmel AVR at present.

Explanation:

`deviceconfigurationRead` reads information from the configurations area of the memory on chips that support this feature. `location` and `store` are both word variables, meaning that they can be values greater than 255.

The `location` depends on the amount Microchip PIC microcontroller, which is given on the datasheet. `store` is 14 bits wide, and thus can store values up to 16383.

This is an advanced command which should only be used by advanced developers.

SAFM

This is the SAFM section of the Help file. Please refer the sub-sections for details using the contents/folder view.

SAFM Overview

Introduction:

Some Advanced (18F) and some Enhanced Mid-Range (16F) Microchip PIC devices support Storage Area Flash (SAF) memory. These devices also include EEPROM memory. SAF memory is not High Endurance, meaning it does not have an endurance of 100K write cycles. SAF has the same endurance as regular flash memory, usually specified as 10K write cycles.

SAF memory appears at the top of program memory space and can be used for any purpose, like regular flash program memory. Storage Area Flash is intended to be used to store data, such a device calibration data, RF device register settings, and other data. SAFEM can be Read as frequently as necessary. However, it is not intended to be written frequently like EEPROM. If non-volatile memory need to be written frequently, it is best to use the EEPROM on these devices.

As with all flash memory, data must be erased before it can be written and writing this memory will stall the device for a few ms. Methods to read, write and erase the SAF memory are included in GCBASIC and they are described in this introduction.

The `hefsaf.h` library supports SAF operations for GCBASIC.

Note: By default, GCBASIC will use SAF memory for regular executable code unless it is told otherwise. If you wish to store data here, you should reserve the SAF memory by using the compiler option, as shown below to reserve 128 Words of SAF memory: This equates to 256 bytes on PIC 18F microcontrollers and 128 Bytes on PIC 16F microcontrollers

```
#option ReserveHighProg 128
```

SAF memory is a block of memory locations found at the top of the Flash program memory. Each memory location can be used to hold a variable value, either a byte or a word dependent on the specific device. The main difference between SAF memory and EEPROM is that EEPROM allows byte-by-byte erase whereas the SAF memory does not. With SAF memory data must be erased before a write and the erase can only be performed in blocks of memory. The blocks, also called rows, are a fixed size associated with the specific device.

GCBASIC handles the erase operation automatically. When a write operation is used by a user the GCBASIC library reads to a buffer, update the buffer, erase the block and finally write the buffer back to SAFM. The complexity of using SAF memory is reduced with the automatically handling of these operations.

The library provides a set of methods to support use of SAF memory.

Method	Parameters	Usage
SAFWrite	a subroutine with the parameters: location, byte value	SAFWrite (location, byte_variable)
SAFWriteWord	a subroutine with the parameters: location, word_value	SAFWriteWord (location, word_variable)
SAFRead	a function with the parameters: location returns a byte value	byte_variable = SAFRead (location)
SAFRead	a subroutine with the parameters: location, byte_value	SAFRead (location , out_byte_variable)
SAFReadWord	a function with the parameters: location returns a word value	word_variable = SAFRead (location)
SAFReadWord	a subroutine with the parameters: location, word_value	SAFRead (location , word_variable)

SAFEraseBlock	a subroutine with the parameters: block_number	SAFEraseBlock (0) A value of 0,1,2,3 etc.
SAFWriteBlock	a subroutine with the parameters: block_number, buffer() [,num_blocks]	SAFWriteBlock(0, myMemoryBuffer) 'where myMemoryBuffer is an Array or a String The Array or a String will contain the values to be wrttin to the SAFM.
SAFReadBlock	a subroutine with the parameters: block_number, buffer() [, num_blocks]	SAFReadBlock(0, myMemoryBuffer) 'where myMemoryBuffer is an Array or a String. The Array or a String will contain the values from the SAFM.

The library also defines a set constants that are specific to the device. These may be useful in the user program. These constants are used by the library. A user may use these public constants.

Constant	Type	Usage
SAF_ROW_SIZE_BYTES	Byte	Size of an SAFM block in bytes
SAF_WORDS and SAF_BYTES	Word or a Byte	ChipSAFMemWords parameter from the device .dat file
SAF_START_ADDR	Word	Starting address of SAFM
SAF_NUM_BLOCKS	Byte	Number of block of SAFM
CHIPWORDS	Word	Device specific constant for the total flash size
CHIPSAFMEMWORDS	Word	Device specific constant for the number of SAFM words available

CHIPERASEROWSIZEWORDS	Word	Device specific constant for the number of SAFM in an erase row
-----------------------	------	---

Warning

Whenever you update the hex file of your Microchip PIC micro-controller with your programmer you MAY erase the data that are stored in SAF memory. If you want to avoid that you will have to flash your Microchip PIC micro-controller with software that allows memory exclusion when flashing. This is the case with Microchip PIC MPLAB IPE (Go to **Advanced Mode/Enter password/Select Memory/Tick "Preserve Flash on Program"/ Enter Start and End address** of your SAFM). Or, simply use the PICkitPlus suite of software to preserve SAF memory during programming.

See also [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFeraseBlock](#)

SAFRead

Syntax:

```
'as a subroutine
SAFRead ( location, data )

'as a function
data = SAFRead ( location )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory

Explanation:

SAFRead is used to read information, byte values, from SAFM, so that it can be accessed for use in a user program.

location represents the location or relative address to read. The location will range from location 0 to SAF_BYTES - 1. This can be from 0-127 or 0-255m depending upon the specific device. HEF_BYTES is a GCBASIC constant that represents the number of bytes of SAF Memory.

data is the data that is to be read from the SAFM data storage area. This can be a byte value or a byte variable.

This method reads data from SAFM given the specific relative location. This method is similar to the EPRead method for EEPROM.

Example 1:

```
'... code preamble to select part  
'... code to setup serial  
'... code to setup PPS
```

'The following example reads the SAFM data value into the byte variable "byte_value" using a subroutine.

```
Dim data_byte as byte  
  
;Write a byte of data to SAF Location 34  
SAFWrite( 34, 144)  
  
;Read the byte back from SAF location 34  
byte_value = SAFread( 34 )  
  
;Display the data on a terminal  
HserPrint "byte_value = "  
Hserprint byte_value
```

Example 2:

```
'... code preamble to select part  
'... code to setup serial  
'... code to setup PPS
```

'The following example reads the SAFM data value into the byte variable "byte_value" using a function.

```
Dim data_byte as byte  
  
;Write a byte of Data to SAF Location 34  
SAFWrite( 34, 144)  
  
;Read the byte back from SAF location 34  
byte_value = SAFread( 34 )  
  
;Display the data on a terminal  
HserPrint "byte_value = "  
Hserprint byte_value
```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFReadWord

Syntax:

```
'as a subroutine
SAFReadWord ( location, data_word_variable )

'as a function
data_word_variable = SAFReadWord ( location )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory

Explanation:

SAFReadWord is used to read information, word values, from SAFM so that it can be accessed for use in a user program.

location represents the location or relative address to read. The location will range from 0 to SAF_BYTES -1. Each data Word requires 2 SAF Locations, therefore the location will range from either 0 to 254 or 0 to 126 (in steps of 2), depending upon the device.

data is the word data that is to be read from the SAFM location. This must be a word variable.

This method reads word information from SAFM given the relative location in SAFM.

Example 1:


```
'... code preamble to select part  
'... code to setup serial
```

'The following example uses a subroutine to read an SAFM location into a word variable.

```
dim data_word_variable as word  
  
;Write a word to SAF location 64  
SAFWriteWord( 64, 0x1234 )  
  
; Read the Word from SAF location 64  
SAFReadWord ( 64, data_word_variable )  
  
HSerPrint "Value = "  
HSerPrint data_word_variable  
HSerPrintCRLF
```

If example 1 were displayed on a serial terminal. The result would show:

```
Value = 4660
```

Example 2:

```
'... code preamble to select part  
'... code to setup serial
```

'The following example uses a function to read an SAFM location into a word variable.

```
dim data_word_variable as word  
  
;Write a word to SAF location 64  
SAFWriteWord( 64, 0x4321 )  
  
; Read the Word from SAF location 64  
data_word_variable = SAFReadWord ( 64 )  
  
HSerPrint "Value = "  
HSerPrint data_word_variable  
HSerPrintCRLF
```

If example 2 were displayed on a serial terminal. The result would show:

```
Value = 17185
```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFWrite

Syntax:

```
SAFWrite ( location, data )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory

Explanation:

SAFWrite is used to write information, byte values, to SAFM so that it can be accessed later for use in a user program.

location represents the location or relative address to write. The location will range from location 0 to SAF_BYTES - 1, or for all practical purposes 0-255 since all PIC Microcontrollers with SAFM support 256 bytes of SAF Memory. HEF_BYTES is a GCBASIC constant that represents the number of bytes of SAF Memory.

data is the data that is to be written to the SAFM location. This can be a byte value or a byte variable. This method writes information to SAFM given the specific location. This method is similar to the EPWrite method for EEPROM.

Example 1:

```
#chip 18F24K42, 16
'... code to setup PPS
'... code to setup serial

'The following example writes a byte value of 126 into HEFM location 34

SAFWrite( 34,126 )
```

Example 2:

```
#chip 18F24K42, 16
'... code to setup PPS
'... code to setup serial
```

'This example will populate the 256 bytes of SAF memory with a value that is same as the SAFM location

```
Dim Rel_Address, DataByte as Byte
Dim NVM_Address as Long
Dim DataWord, as Word
```

```
For Rel_Address = 0 to 255
    SAFWrite ( Rel_Address, Rel_Address )
Next
```

```
SAFM_Dump
end
```

```
; This subroutine displays the SAF Flash Memory on a terminal
; Words in reverse byte order relative to address
sub SAFM_Dump
```

```
Dim Blocknum as Byte
NVM_Address = SAF_START_ADDR
BlockNum = 0
```

```
Repeat SAF_WORDS ;128
    If NVM_Address % SAF_ROW_SIZE_BYTES = 0 then
        If BlockNum > 0 then HSERPRINTCRLF
        HSerprintCRLF

        HserPrint "Block"
        HSerprint BlockNum
        HSerprint " 1 0 3 2 5 4 7 6 9 8 B A D C F E"
        BlockNum++
    End if

    IF NVM_Address % 16 = 0 then
        HSerPrintCRLF
        hserprint hex(NVM_Address_H)
        hserprint hex(NVM_Address)
        hserprint " "
    end if
```

```

    Rel_Address = NVM_ADDRESS - SAF_START_ADDR
    SAFReadWord(Rel_Address,DataWord)

    hserprint hex(DataWord_H)
    hserprint hex(DataWord)
    hserprint " "

    NVM_Address+=2 ' Next "WORD"
End Repeat
End sub

```

If example 2 were displayed on a serial terminal. The result would show:

Block0	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F00	0100	0302	0504	0706	0908	0B0A	0D0C	0F0E
7F10	1110	1312	1514	1716	1918	1B1A	1D1C	1F1E
7F20	2120	2322	2524	2726	2928	2B2A	2D2C	2F2E
7F30	3130	3332	3534	3736	3938	3B3A	3D3C	3F3E
Block1	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F40	4140	4342	4544	4746	4948	4B4A	4D4C	4F4E
7F50	5150	5352	5554	5756	5958	5B5A	5D5C	5F5E
7F60	6160	6362	6564	6766	6968	6B6A	6D6C	6F6E
7F70	7170	7372	7574	7776	7978	7B7A	7D7C	7F7E
Block2	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7F80	8180	8382	8584	8786	8988	8B8A	8D8C	8F8E
7F90	9190	9392	9594	9796	9998	9B9A	9D9C	9F9E
7FA0	A1A0	A3A2	A5A4	A7A6	A9A8	ABAA	ADAC	AFAE
7FB0	B1B0	B3B2	B5B4	B7B6	B9B8	BBBA	BDBC	BFBE
Block3	1 0	3 2	5 4	7 6	9 8	B A	D C	F E
7FC0	C1C0	C3C2	C5C4	C7C6	C9C8	CBCA	CDCC	CFCE
7FD0	D1D0	D3D2	D5D4	D7D6	D9D8	DBDA	DDDC	DFDE
7FE0	E1E0	E3E2	E5E4	E7E6	E9E8	EBEA	EDEC	EFEE
7FF0	F1F0	F3F2	F5F4	F7F6	F9F8	FBFA	FDFC	FFFE

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFWriteWord

Syntax:

```
SAFWriteWord ( location, data_word_value )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory

Explanation:

SAFWriteWord is used to write information, word values, to the SAFM data storage, so that it can be accessed later by a programmer on a Personal, or by the SAFRead commands.

location presents the location or relative address to write. The location will range from 0 to SAF_BYTES-1. Each data Word requires 2 SAF Locations, therefore the location will range from either 0 to 254 or 0 to 126 (in steps of 2), depending upon the device.

data is the data that is to be written to the SAFM location. This can be a word value or a word variable.

This method writes information to SAFM given the specific location in SAFM. This method is similar to the methods for EEPROM, but supports Word values.

Example 1:

```
'... code preamble to select part
'... code to setup serial

'The following example stores in the word value of 0x1234 as SAFM location 34

SAFWriteWord( 34, 0x1234 )
```

Example 2:

```

#chip 18F24K42, 16
'... code to setup PPS
'... code to setup serial

'This example will write two word values to two specific locations.

dim Word_Variable1 as Word
dim Word_Variable2 as Word

;Write the data
SAFWriteWord (16, 0x1234) 'location 16, in this device, equates to 0x7F10
SAFWriteWord (18, 0x4321) 'location 18, in this device, equates to 0x7F12

;Read the data and send to terminal
SAFReadWord(16, Word_Variable1 )
SAFReadWord(18, Word_Variable2 )

HserPrint "Word_Variable1 = "
Hserprint Word_Variable1
HserPrintCRLF
HserPrint "Word_Variable2 = "
Hserprint Word_Variable2
HserPrintCRLF

```

If example 2 were displayed on a serial terminal. The result would show, where ---- is the existing value.

```

Word_Variable1 = 4660
Word_Variable2 = 17185

```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFReadBlock

Syntax:

```
SAFReadBlock ( block_number, buffer(), [, num_bytes] )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory.

Explanation:

HEFReadBlock is used to read information from the HEFM data storage into the buffer. Once the buffer is populated it can be accessed for use within a user program.

The parameters are as follows:

block_number represents the block to be written to. The block_number parameter is used to calculate the physical memory location(s) that are updated.

buffer() represents an array or string. The buffer will be used as the data target for the block read operation. The buffer is handled as a buffer of bytes values. In most cases the buffer should be the same size as a row/block of SAFM. For most PIC Microcontrollers with SAFM this will be 32 bytes.

num_bytes is an optional parameter, and can be used to specify number of bytes to read from SAFM, starting at the first location in the selected SAFM block. This parameter is not normally required as the default is set to the GCBASIC constant **SAF_ROWSIZE_BYTES**.

Example 1:

```
#chip 18F24K42, 16
'... code preamble to setup PPS
'... code to setup serial

Dim My_Buffer(HEF_ROWSIZE_BYTES)
Dim index as byte

;Write some data to Block 2
My_Buffer =
1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32
SAFWriteBlock(2, My_Buffer())

;Read the data back from SAFM using SAFReadBock
SAFReadBlock( 2 , My_buffer() )

;Send the data to a terminal in decimal format
index = 1
Repeat SAF_ROWSIZE_BYTES
    Hserprint(My_Buffer(index))
    HserPrint " "
    index++
End Repeat
```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFWriteBlock

Syntax:

```
SAFWriteBlock ( block_number, buffer(), [, num_bytes] )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory.

Explanation:

SAFWriteBlock is used to write information from a user buffer to SAFM. Once the block is written it can be accessed for use within a user program.

The parameters are as follows:

block_number represents the block to be written to. The block_number parameter is used to calculate the physical memory location(s) that are updated.

buffer() represents an array or string. The buffer will be used as the data source that is written to the SAFM block. The buffer is handled as a buffer of bytes values. In most cases the buffer should be the same size as a row/block of SAFM. For most PIC Microcontrollers this will be 32 bytes. Best practice is to size the buffer using the SAF_ROW_SIZE_BYTES constant. If the size of the buffer exceeds the device specific SAF_ROW_SIZE_BYTES, the excess data will not be handled and the buffer will be truncated at the SAF_ROW_SIZE_BYTES limit.

num_bytes is an optional parameter, and can be used to specify the number of bytes to write to HEFM, starting at the first location in the selected HEFM block. This parameter is not normally required as the default is set to the GCBASIC constant **HEF_ROW_SIZE_BYTES**.

Example 1:


```

#chip 18F24K42, 16
'... code preamble to setup PPS
'... code to setup serial

Dim My_Buffer(HEF_ROWSIZE_BYTES)
Dim index as byte

;Write some data to Block 2
My_Buffer =
1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32
SAFWriteBlock(2, My_Buffer())

;Read the data back from SAFM using SAFReadBock
SAFReadBlock( 2 , My_buffer() )

;Send the data to a terminal in decimal format
index = 1
Repeat SAF_ROWSIZE_BYTES
    Hserprint(My_Buffer(index))
    HserPrint " "
    index++
End Repeat

```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

SAFEraseBlock

Syntax:

```
SAFEraseBlock ( block_number )
```

Command Availability:

Available on all PIC micro-controllers with SAFM memory. **Explanation:**

SAFEraseBlock is used to erase all data locations within the SAFM block. HEFM data within the HEFM block to the erase state value of the device. This Value is 0xFF for each location and will read 0xFFFF if the program memory word is displayed. Use Caution. Once the SAFM block is erased, the SAFEM data is gone forever and cannot be recovered unless it was previpusly saved.

The single parameter is as follows:

block_number represents the block to be erased. The **block_number** parameter is used to calculate the physical memory location(s) that are updated.

Example 1:

Erase a specific block of SAFEM.

```
'... code preamble to select part  
'... code to setup PPS, if needed  
'... code to setup serial, if needed  
  
'Erase block 2 of HEFM  
HEFEraseBlock ( 2)
```

See also [SAFM Overview](#), [SAFRead](#), [SAFReadWord](#), [SAFWrite](#), [SAFWriteWord](#), [SAFReadBlock](#), [SAFWriteBlock](#), [SAFEraseBlock](#)

EERAM (Device)

This is the EERAM section of the Help file. Please refer the sub-sections for details using the contents/folder view.

47xxx EERam Devices

This section covers the 47xxx EERam devices.

The 47xxx EERam device is a memory device is organized as 512 x 8 bits or 2,048 x 8 bits of memory and utilizes the I2C serial interface.

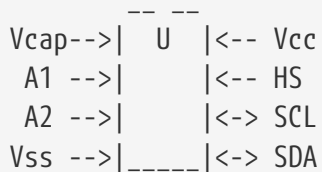
The 47xxx provides infinite read and write cycles to the SRAM while EEPROM cells provide high-endurance nonvolatile storage of data with more than one million store cycles to EEPROM & a Data retention of > 200 years.

With an external capacitor (~10uF), SRAM data is automatically transferred to the EEPROM upon loss of power, giving the advantages of NVRAM whilst eliminating the need for backup batteries.

Data can also be backed up manually by using either the Hardware Store pin (HS) or software control.

On power-up, the EEPROM data is automatically recalled to the SRAM. EEPROM data Recall can also be initiated through software control.

Connectivity is shown below:



Modes of Operation

The SRAM allows for fast reads and writes and unlimited endurance. As long as power is present, the data stored in the SRAM can be updated as often as desired.

To preserve the SRAM image, the AutoStore function copies the entire SRAM image to an EEPROM array whenever it detects that the voltage drops below a predetermined level. The power for the AutoStore process is provided by the externally connected VCAP capacitor. Upon power-up, the entire memory contents are restored by copying the EEPROM image to the SRAM. This automatic restore operation is completed in milliseconds after power-up, at the same time as when other devices would be initializing.

There is no latency in writing to the SRAM. The SRAM can be written to starting at any random address, and can be written continuously throughout the array, wrapping back to the beginning after the end is reached. There is a small delay, specified as TWC in the data sheet, when writing to the

nonvolatile configuration bits of the STATUS Register (SR).

Besides the AutoStore function, there are two other methods to store the SRAM data to EEPROM:

- One method is the Hardware Store, initiated by a rising edge on the HS pin.
- The other method is the Software Store, initiated by writing the correct instruction to the command register via I2C.

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Explanation

The GCBASIC constants and commands shown below control the configuration of the 47xxx EE-RAM device. GCBASIC supports I2C hardware and software connectivity - this is shown in the tables below.

To use the 47xxx driver simply include the following in your user code. This will initialise the driver.

```
#include <47xxx_EERAM.H>

; ----- Define Hardware settings for EERAM Module
#define I2C_Adr_EERAM 0x30      ; EERAM base Address
#define EERAM_HS PortB.1      ; Optional hardware Store Pin

Dir EERAM_HS Out              ; Rising edge initiates Backup

EERAM_AutoStore(ON)           ; Enable Automatic Storage on power loss

'EERAM_AutoStore(OFF)        ; Disable Automatic Storage on power loss
```

The device parameters for the device are shown in the table below.

Part Number	Density (bits)	VCC Range	Max. I2C Frequency	Tstore Delay	Trecall Delay
47L04	4K	2.7-3.6V	1 MHz	8ms	25ms
47C04	4K	4.5-5.5V	1 MHz	8ms	2ms
47L16	16K	2.7-3.6V	1 MHz	25ms	5ms
47C16	16K	4.5-5.5V	1 MHz	25ms	5ms

The GCBASIC constants for control of the device are:

Constant	Context	Example	Default
EERAM_I2C_Adr	8-bit I2C Address of device	#define I2C_Adr_EERAM 0x30	Default is 0x30. This is mandated
EERAM_HS	Optional hardware Store Pin	#define EERAM_HS portb.1	No default - this is not mandated
EERAM_Tstore	Delay period for write to device	#define EERAM_Tstore 25	25 (ms)
EERAM_Trecall	Delay period to read from device	#define EERAM_Trecall 5	5 (ms)

The GCBASIC commands for control of the device are:

Command	Context	Example
EERAM_AutoStore	Enable Automatic Storage on power loss or Disable Automatic Storage on power loss	EERAM_AutoStore(ON), or EERAM_AutoStore(OFF)
EERAM_Status	Read the Status Register	User_byte_variable = EERAM_Status()
EERAM_Backup	Backup / Store Now	EERAM_Backup()
EERAM_Recall	Restore Now	EERAM_Recall()
EERAM_HWStore	Force Backup with HS Pin	EERAM_HWStore()
EERAM_Write	Write a Byte of Data to address at the specified address. The address must be a word value and the data is byte value.	ERAM_Write(EERAM_Address_word, EERAM_Data_byte)
EERAM_Read	Read a Byte of Data from address. The address must be a word value and returned data is byte value.	User_byte_variable = EERAM_Read(EERAM_Address_word)

This example shows how to use the device.

Example:

```
#CHIP 16F18855,32
#OPTION EXPLICIT

#include <47XXX_EERAM.H>

#startup InitPPS, 85
```

Sub InitPPS

'PPS is explicit to a specific chip. Use PPSTool to ensure the PPS settings are correct.

'Module: EUSART

RC0PPS = 0x0010 'TX > RC0

TXPPS = 0x0008 'RC0 > TX (bi-directional)

'Module: MSSP1

SSP1DATPPS = 0x0013 'RC3 > SDA1

RC3PPS = 0x0015 'SDA1 > RC3 (bi-directional)

RC4PPS = 0x0014 'SCL1 > RC4

SSP1CLKPPS = 0x0014 'RC4 > SCL1 (bi-directional)

End Sub

; ----- Define Hardware Serial Print

#DEFINE USART_BAUD_RATE 115200

#DEFINE USART_TX_BLOCKING

; ----- Define Hardware settings for hwi2c

#DEFINE HI2C_BAUD_RATE 400

#DEFINE HI2C_DATA PORTC.3

#DEFINE HI2C_CLOCK PORTC.4

'I2C pins need to be input for legacy I2C modules

DIR HI2C_DATA IN

DIR HI2C_CLOCK IN

'Initialise I2C Master

hi2cMode Master

; ----- Define Hardware settings for EERAM Module

#define EERAM_I2C_Adr 0x30 ; EERAM base Address

#define EERAM_HS PortB.1 ; Optional hardware Store Pin

Dir EERAM_HS Out ; Rising edge initiates Backup

'Library function

EERAM_AutoStore(ON) ; Enable Automatic Storage on power loss

; ----- Main body of program commences here.

dim Idx as Byte

```
HserPrintCRLF 2
```

```
HserPrint "Hardware I2C EERAM Read Test at I2C Adr 0x"
```

```
HserPrint Hex(EERAM_I2C_Adr)
```

```
HserPrint " Reading RAM addresses 0x0 to 0xF" : HserPrintCRLF 2
```

```
for Idx = 0x0 to 0xF
```

```
    HserPrint hex(Idx) + " = " : HserPrint Hex(EERAM_Read(Idx))
```

```
    If Idx = 7 or Idx = 0xf then
```

```
        HserPrintCRLF
```

```
    Else
```

```
        HserPrint " : "
```

```
    End if
```

```
next
```

```
HserPrintCRLF : HserPrint "Control Byte = " Hex(EERAM_Status()) : HserPrintCRLF 2
```

```
wait 100 ms ; time for serial operations to complete
```

```
end
```

For more help, see [Software I2C](#) or [Hardware I2C](#)

SRAM (Device)

This is the SRAM section of the Help file. Please refer the sub-sections for details using the contents/folder view.

SRAM Overview

Introduction:

Serial SRAM is a standalone volatile memory that provides an easy and inexpensive way to add more RAM to application. These are 8-pin low-power devices. They are high-performance devices have unlimited endurance and zero write times, making them ideal for applications involving continuous data transfer, buffering, data logging, audio, video, internet, graphics and other math and data-intensive functions.

These devices are available from 64 Kbit up to 1 Mbit in density and support SPI, SDI and SQI™ bus modes.

The GCBASIC library only supports SPI bus mode. The GCBASIC library supports hardware and software SPI - this is controlled via a constant, see below.

To use the SRAM libray simply include the following in your user code.

This will initialise the driver.

```
#define SPISRAM_CS      Porta.2      'Also known as SS, or Slave Select
#define SPISRAM_SCK     Portc.3      'Also known as CLK
#define SPISRAM_DO      Portc.5      'Also known as MOSI
#define SPISRAM_DI      Portc.4      'Also known as MISO

#define SPISRAM_HARDWARESPI
#define SPISRAM_TYPE     SRAM_23LC1024
```

SRAM memory operations.

The library exposes a set of method to support use of SRAM memory.

Method	Parameters	Usage
SRAM Write	eepAddr as long, eepromVal as byte	A subroutine that required the address and the value to be written to SRAM.

SRAMRead	eeAddr as long, eePromVal as byte	A subroutine that required the address and variable to be updated with the byte value from the specified SRAM address.
SRAMRead	eeAddr as long	A function that required the address. The function returns a byte value from the specified SRAM address.

The library requires a set of constants to support use of SRAM memory.

Constant	Parameters	Usage
SPISRAM_TYPE	Specifies the type of SRAM.	Requires one of the following constants SRAM_23LC1024, SRAM_23LCV1024, SRAM_23LC1024, SRAM_23A1024, SRAM_23LCV512, SRAM_23LC512, SRAM_23A512, SRAM_23K256, SRAM_23A256, SRAM_23A640, or SRAM_23K640
SPISRAM_CS	Specifies the port for the chip select port.	Required
SPISRAM_SCK	Specifies the port for the SPI clock port.	Required
SPISRAM_DO	Specifies the port for the SPI data out, or MOSI, port.	Required
SPISRAM_DI	Specifies the port for the data in, or MISO, port.	Required
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFast for specific AVR's only.

SPISRAM_H ARDWARES PI	Instructs the library to use hardware SPI, remove or comment out if you want to use software SPI.	Optional
-----------------------------	---	----------

The library also exposes a constant that is specific to the device.

These may be useful in the user program.

This constant is used by the library.

A user may use this public constant.

Constant	Type	Usage
SPISRAM_CAPACITY	Long value	Use to determine the size of the SRAM device

Examples

```
#include <uno_mega328p.h>
#option explicit

' USART settings
#define USART_BAUD_RATE 57600
#define USART_DELAY 0 ms
#define USART_BLOCKING
#define USART_TX_BLOCKING

'SD card attached to SPI bus as follows:
,

'UNO:  MOSI - pin 11, MISO - pin 12, CLK - pin 13, CS - pin 4 (CS pin can be
changed) and pin #10 (SS) must be an output
'Mega:  MOSI - pin 51, MISO - pin 50, CLK - pin 52, CS - pin 4 (CS pin can be
changed) and pin #52 (SS) must be an output
'Leonardo: Connect to hardware SPI via the ICSP header

#define SPISRAM_CS      DIGITAL_5      'Also known as SS, or Slave Select
#define SPISRAM_SCK     DIGITAL_13     'Also known as CLK
#define SPISRAM_DO      DIGITAL_11     'Also known as MOSI
#define SPISRAM_DI      DIGITAL_12     'Also known as MISO
```

```

#define SPISRAM_HARDWARESPI
#define SPISRAM_TYPE      SRAM_23LC1024

#define HWSPIMode MASTERULTRAFAST      'MASTERSLOW | MASTER | MASTERFAST |
MASTERULTRAFAST for specific AVRs only. Defaults to MASTERFAST

'*****

'Main program

'Wait 2 seconds to open the serial terminal
wait 2 s

HSerPrintCRLF 2
HSerPrint "Writing..."
HSerPrintCRLF
For SRAM_location=0 to SPISRAM_CAPACITY - 1
  SRAMWrite ( [long]SRAM_location, SRAM_location and 255 )
Next

dim spirambyteread as Byte
spirambyteread = 11
HSerPrintCRLF 2
dim SRAM_location as long
HSerPrint "Reading..."
HSerPrintCRLF
For SRAM_location=0 to SPISRAM_CAPACITY - 1
  'choose one....
  'SRAMread ( SRAM_location, spirambyteread )
'or, as a function
  spirambyteread = SRAMread ( SRAM_location )

  if spirambyteread = ( SRAM_location and 255 ) then
    HSerPrint hex(spirambyteread)
  else
    HSerPrint "***"
  end if
  HSerPrint ":"
Next
HSerPrintCRLF
HSerPrint "Wait..."
HSerPrintCRLF
Wait 2 s

HSerPrint "Rewriting to 0x00 ..."

```

```

HSerPrintCRLF
For SRAM_location=0 to SPIRAM_CAPACITY - 1
  SRAMWrite ( [long]SRAM_location, 0 )
Next

Dim errorcount as long
errorcount = 0
For SRAM_location=0 to SPIRAM_CAPACITY - 1
  SRAMRead ( SRAM_location, spirambyteread )
  if spirambyteread <> 0 then
    errorcount++
  end if
Next
HSerPrint "Error Count (should be 0) = "
HSerPrint errorcount
HSerPrintCRLF
HSerPrint "End..."
HSerPrintCRLF
end

```

or, for a PIC with PPS

```

/*
  An example program for GCGB and GCB.
  -----
  This program blinks the onboard LED of a microcontroller - simple - the demonstration
  should work across many microcontrollers.

  This sample file can be adapted to any of the supported microcontrollers by simply
  changing the #chip definition - do this by changing to your chip. Other common chips are
  shown.

  To make your programmer or your bootloader work, simply do the following.

  From within this IDE, press function_key F4, then select 'Edit Programmer
  Preferences'. This will show you the 'Programmer Preferences', select the 'Programmer'
  tab.

  For PIC: Typically, if you are using a PICKIT2 then select 'Microchip PICKit 2 CLI -
  Default' will work for you - select and drag this to the top of the dialog area.

  For AVR: Typically, for an UNO the programmer 'Arduino UNO/328p@115200' will work,
  select the option from the list of programmers, then, select 'Edit',
  please edit the communication 'Port', select 'OK', and, then select this option again
  and drag this to the top of the dialog area.

```

For further adventures with GCBASIC Select F4 and then "view Demos" for 1000's of sample programs.

Enjoy.

*/

// *****

'Chip Settings.

#chip 18F47k42, 64

#config MCLRE = ON

#option explicit

'PPS Tool version: 0.0.5.27

'PinManager data: v1.78

'Generated for 18F47K42

,

'Template comment at the start of the config file

,

#startup InitPPS, 85

#define PPSToolPart 18F47K42

Sub InitPPS

'This has been added to turn off PPS SPI when in SPI software mode

#ifdef SPISRAM_HARDWARESPI

'Module: SPI1

RC3PPS = 0x001E 'SCK1 > RC3

SPI1SCKPPS = 0x0013 'RC3 > SCK1 (bi-directional)

RC5PPS = 0x001F 'SD01 > RC5

SPI1SDIPPS = 0x0014 'RC4 > SDI1

'Module: UART pin directions

#endif

'Module: UART pin directions

Dir PORTC.6 Out ' Make TX1 pin an output

'Module: UART1

RC6PPS = 0x0013 'TX1 > RC6

End Sub

'Template comment at the end of the config file

' USART settings

#define USART_BAUD_RATE 57600

#define USART_DELAY 0 ms

#define USART_BLOCKING

#define USART_TX_BLOCKING

```

#define SPISRAM_CS      Porta.2      'Also known as SS, or Slave Select
#define SPISRAM_SCK     Portc.3      'Also known as CLK
#define SPISRAM_DO      Portc.5      'Also known as MOSI
#define SPISRAM_DI      Portc.4      'Also known as MISO

#define SPISRAM_HARDWARESPI
#define SPISRAM_TYPE     SRAM_23LC1024

'*****

'Main program

'Wait 2 seconds to open the serial terminal
wait 2 s
dim sizeofSPIRAM as long
sizeofSPIRAM = SPISRAM_CAPACITY
HSerPrintCRLF 2
HSerPrint "Writing...SPIRAM_CAPACITY = 0x"
HSerPrint LongTohex(sizeofSPIRAM)
HSerPrintCRLF
wait 100 ms

dim SRAM_location as long
For SRAM_location=0 to SPISRAM_CAPACITY - 1
    SRAMWrite ( [long]SRAM_location, SRAM_location and 255 )
Next

dim spirambyteread as Byte
spirambyteread = 11 'could be any number....
HSerPrintCRLF 2

HSerPrint "Reading..."
HSerPrintCRLF
For SRAM_location=0 to SPISRAM_CAPACITY - 1
    'choose one....
    'SRAMRead ( SRAM_location, spirambyteread )
    'or, as a function
    spirambyteread = SRAMRead ( SRAM_location )

    if spirambyteread = ( SRAM_location and 255 ) then
        HSerPrint hex(spirambyteread)
    else
        HSerPrint "**"
    end if
    HSerPrint ":"
Next

```

```

HSerPrintCRLF
HSerPrint "Wait..."
HSerPrintCRLF
Wait 2 s

HSerPrint "Rewriting to 0x00 ..."
HSerPrintCRLF
For SRAM_location=0 to SPISRAM_CAPACITY - 1
    SRAMWrite ( [long]SRAM_location, 0 )
Next

Dim errorcount as long
errorcount = 0
For SRAM_location=0 to SPISRAM_CAPACITY - 1
    SRAMRead ( SRAM_location, spirambyteread )
    if spirambyteread <> 0 then
        errorcount++
    end if
Next
HSerPrint "Error Count (should be 0) = "
HSerPrint errorcount
HSerPrintCRLF
HSerPrint "End..."
HSerPrintCRLF

do

loop

```

For more help, see [SRAMRead](#) or [SRAMWrite](#)

SRAMRead

Syntax:

```

SRAMRead location, store

or

store = SRAMRead location

```

Command Availability:

Available on all Microchip PIC and Atmel AVR microcontrollers with SRAM data memory attached.

Explanation:

SRAMRead is the method, a function or a subroutine, used to read information from the SRAM data storage.

location represents the location to read data from.

store is the variable in which to store the data after it has been read from SRAM.

Example:

```
#include <uno_mega328p.h>
#option explicit

'Set up SRAM
#define SPISRAM_CS      DIGITAL_5      'Also known as SS, or Slave Select
#define SPISRAM_SCK     DIGITAL_13     'Also known as CLK
#define SPISRAM_DO      DIGITAL_11     'Also known as MOSI
#define SPISRAM_DI      DIGITAL_12     'Also known as MISO

#define SPISRAM_HARDWARESPI
#define SPISRAM_TYPE     SRAM_23LC1024

'*****

'Main program

dim in_byte as byte

'Using a function: Read from SRAM location 0x10 and place the results in the variable
in_byte
in_byte = SRAMRead ( 0x10 )

'Using a subroutine: Read from SRAM location 0x10 and place the results in the
variable in_byte
SRAMRead ( 0x10, in_byte )
```

For more help, see [SRAM Overview](#) or [SRAMWrite](#)

SRAMWrite

Syntax:

```
SRAMWrite location, data
```

Command Availability:

Available on all Microchip PIC and Atmel AVR microcontrollers with SRAM data memory attached.

Explanation:

SRAMWrite is the method used to write information to the SRAM data storage, so that it can be accessed by the **SRAMRead** command.

location represents the location to read data from, and this location will vary from one application/solution to another.

data is the data that is to be written to the SRAM, a byte value or a byte variable.

Example:

```
#include <uno_mega328p.h>
#option explicit

'Set up SRAM
#define SPISRAM_CS      DIGITAL_5      'Also known as SS, or Slave Select
#define SPISRAM_SCK     DIGITAL_13     'Also known as CLK
#define SPISRAM_DO      DIGITAL_11     'Also known as MOSI
#define SPISRAM_DI      DIGITAL_12     'Also known as MISO

#define SPISRAM_HARDWARESPI
#define SPISRAM_TYPE     SRAM_23LC1024

'*****

'Main program

dim out_byte as byte

'A subroutine: Weite to SRAM location 0x10 and the variable out_byte
SRAMRead ( 0x10, out_byte )
```

For more help, see [SRAMOverview](#) or [SRAMRead](#)

Flow control

This is the Flow control section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Do

Syntax:

```
Do [{While | Until} condition]
...
program code
...
<condition> Exit Do
...
Loop [{While | Until} condition]
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Do** command will cause the code between the **Do** and the **Loop** to run repeatedly while **condition** is true or until **condition** is true, depending on whether **While** or **Until** has been specified.

Note that the **While** or **Until** and the condition can only be specified once, or not at all. If they are not specified, then the code will repeat endlessly.

Optionally, you can specify a condition to **EXIT** the **Do-Loop** immediately.

Example 1:

```
'This code will flash a light until the button is pressed
#chip 12F629, 4

#define BUTTON GPIO.3
#define LIGHT GPIO.5

Dir BUTTON In
Dir LIGHT Out

Do Until BUTTON = 1
    PulseOut LIGHT, 1 s
    Wait 1 s
Loop
```

Example 2:

This code will also flash a light until the button is pressed. This example uses **EXIT DO** within a continuous loop.

```
#chip 12F629, 4

#define BUTTON GPIO.3
#define LIGHT GPIO.5

Dir BUTTON In
Dir LIGHT Out

Do
    PulseOut LIGHT, 1 s
    Wait 1 s
    if BUTTON = 1 then EXIT DO
Loop
```

For more help, see [Conditions](#)

End

Syntax:

```
End
```

Command Availability:

Available on all microcontrollers.

Explanation:

When the **End** command is used, the program will immediately stop running. There are very few cases where this command is needed - generally, the program should be an endless loop.

Example:

```
'This program will turn on the red light, but not the green light
Set RED On
End
Set GREEN On
```

Exit**Syntax options:**

Exit Sub | Exit Function | Exit Do | Exit For | Exit Repeat

Command Availability:

Available on all microcontrollers.

Explanation:

This command will make the program exit the routine it is currently in, as it would if it came to the end of the routine.

Applies to Subroutines, Functions, For-Next loops, Do-Loop loops and Repeat loops.

Example:

```
#chip tiny13, 1

#define SENSOR PORTB.0
#define BUZZER PORTB.1
#define LIGHT PORTB.2
Dir SENSOR In
Dir BUZZER Out
Dir LIGHT Out

Do
  Burglar
Loop

'Burglar Alarm subroutine
Sub Burglar
  If SENSOR = 0 Then
    Set BUZZER Off
    Set LIGHT Off
    Exit Sub
  End If
  Set BUZZER On
  Set LIGHT On
End Sub
```

For more help, see [Do](#), [For](#), [Sub](#), [Functions](#) and [Repeat](#)

For

Syntax:

```
For counter = start To end [Step increment]
...
program code
...
<condition> Exit For
...
Next
```

Command Availability:

Available on all microcontrollers.

Explanation:

The For command is ideal for situations where a piece of code needs to be run a set number of times,

and where it is necessary to keep track of how many times the code has run. When the For command is first executed, `counter` is set to `start`. Then, each successive time the program loops, `increment` is added to `counter`, until `counter` is equal to `end`. Then, the program continues beyond the Next.

`Step` and `increment` are optional. If Step is not specified, GCBASIC will increment `counter` by 1 each time the code is run.

Step is an integer value. Step value can positive or negative. When using advanced variable you must cast the step value as an integer, see the example below.

`increment` can be a positive or negative constant or an integer.

The `Exit For` is optional and can be used to exit the loop upon a specific condition.

WARNING

`#define USELEGACYFORNEXT` to enable legacy FOR-NEXT support. The GCBASIC compiler was revised in 2021 to improve the handling of the FOR-NEXT support. You can revert to the legacy FOR-NEXT support by using `#DEFINE USELEGACYFORNEXT` but using this legacy support will cause your program to operate incorrectly. The use of `#DEFINE USELEGACYFORNEXT` is NOT recommended.

Examples.

Example 1:

```
'This code will flash a green light 6 times.

#chip 16F88, 8

#define LED PORTB.0
Dir LED Out

For LoopCounter = 1 to 6

    PulseOut Led, 1 s
    Wait 1 s

Next
```

Example 2:

'This code will flash alternate LEDS until the switch is pressed.

```
#chip 16F88, 8

#define LED1 PORTB.0
Dir LED1 Out
#define LED2 PORTB.2
Dir LED2 Out

#define SWITCH1 PORTA.0
Dir SWITCH1 In
main:
PulseOut LED1, 1 s
For LoopCounterOut = 1 to 250

    PulseOut LED2, 4 Ms
    if switch = On then Exit For

Next
Set LED2 OFF
goto main
```

Example 3:

This example show casting the step value as an integer. The step value in this example is the integer value of 2.

```
#script
    // Create a constant
    Pi = 22/7
#endscript

Dim myCircumference, myRadius as Single
Dim myDiameter as Single Alias myCircumference_E, myCircumference_U,
myCircumference_H, myCircumference

HserPrintCRLF

For MyRadius = 0.5 to 10.5 step [integer]2
    myCircumference=myRadius * Pi * 2
    HserPrint "myRadius = " + ltrim(SingleToString(myRadius))
    HserPrintStringCRLF " myCircumference = " +
ltrim(SingleToString(myCircumference))
next
```

For more help, see [Repeat](#)

Gosub

Syntax:

```
Gosub label
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Gosub** command is used to jump to a label as a subroutine, in a similar way to **Goto**. The difference is that **Return** can then be used to return to the line of code after the **Goto**.

NOTE

Gosub should NOT be used if it can be avoided. It is not required to call a subroutine that has been defined using **Sub**, just write the name of the subroutine.

Example:

```
'This program will flash an LED on portb bit 0 and play a beep on
'porta bit 4. until the microcontroller is turned off.

#chip 16F628A, 4 'Change this to suit your circuit

#define SOUNDOUT PORTA.4
#define LIGHT PORTB.0
Dir LIGHT Out

Do
    'Flash Light
    PulseOut LIGHT, 1 s
    Wait 1 s
    'Beep
    Gosub PlayBeep
Loop

PlayBeep:
Tone 200, 10
Tone 100, 10
Return
```


For more help, see [Goto](#) and [Labels](#)

Goto

Syntax:

```
Goto label
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Goto** command will make the microcontroller jump to the line specified, and continue running the program from there. The **Goto** command is mainly useful for exiting out of loops - if you need to create an infinite loop, use the **Do** command instead.

Be careful how you use **Goto**. If used too much, it can make programs very hard to read.

To define a label, put the name of the label alone on a line, with just a colon (:) after it.

Example:

```
'This program will flash the light until the button is pressed  
'off. Notice the label named SWITCH_OFF.
```

```
#chip 16F628A, 4 'Change this line to suit your circuit
```

```
#define BUTTON PORTB.0
```

```
#define LIGHT PORTB.1
```

```
Dir BUTTON In
```

```
Dir LIGHT Out
```

```
Do
```

```
    PulseOut LIGHT, 500 ms
```

```
    If BUTTON = 1 Then Goto SWITCH_OFF
```

```
    Wait 500 ms
```

```
    If BUTTON = 1 Then Goto SWITCH_OFF
```

```
Loop
```

```
SWITCH_OFF:
```

```
Set LIGHT Off
```

```
'Chip will enter low power mode when program ends
```

For more help, see [Gosub](#) and [Labels](#)

If

Syntax:

Short form:

```
If condition Then command
```

Long form:

```
If condition Then  
...  
program code  
...  
End If
```

Using Else:

```
If condition Then  
    code to run if true  
Else  
    code to run if false  
End If
```

Using If Else:

```
If condition Then  
    code to run if true  
Else if nextcondition then  
    code to run if nextcondition true  
Else  
    code to run if false  
End If
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **If** command is the most common command used to make decisions. If **condition** is **true**, then **command** (short) or **program code** (long) will be run. If it is **false**, then the microcontroller will skip to the code located on the next line (short) or after the **End If** (long form).

If **Else** is used, then the condition between **If** and **Else** will run if the condition is **true**, and the code between **Else** and **End If** will run if the condition is **false**.

If **Else if** is used, then the condition after the **Else if** will run if the condition is **true**.

Note: **Else** must be on a separate line in the source code.

Supported:

```
<instruction> 'is supported
Else
<instruction>
```

```
<instruction> Else 'Not Supported, but will compile
<instruction>
```

Example:

```
'Turn a light on or off depending on a light sensor

#chip 12F683, 8

#define LIGHT GPIO.1
#define SENSOR AN3
#define SENSOR_PORT GPIO.4

Dir LIGHT Out
Dir SENSOR_PORT In

Do
  If ReadAD(SENSOR) > 128 Then
    Set LIGHT Off
  Else
    Set LIGHT On
  End If
Loop
```

For more help, see [Conditions](#)

IndCall

Syntax:

```
IndCall Address
```

Command Availability:

Available on all microcontrollers.

Explanation:

IndCall provides a basic implementation of function pointers. **Address** is the program memory location of the subroutine that is to be called. There are two ways to specify this - either by providing a direct reference to the subroutine using the @ operator, or by specifying a word variable that contains the address.

This command is useful for callbacks. For example, a particular subroutine might read bytes from a serial connection, but different actions may need to be taken at different times. A different subroutine could be created for each action, and then the subroutine for the appropriate action could be passed to the serial connection reading routine each time it is called.

Note: Calling subroutines that have parameters using **IndCall** is not supported. Errors may occur. If data needs to be passed, use a variable instead.

Example:

```
'Flash an LED using an indirect call
#chip 12F683

'Create a word variable, and set it to the memory location of the
'Blink subroutine.
Dim FlashingSub As Word
FlashingSub = @Blink

'Main loop
Do
'Indirect call to subroutine at location FlashingSub
    IndCall FlashingSub
Loop

'LED flashing subroutine
Sub Blink
    PulseOut GPIO.0, 500 ms
    Wait 500 ms
End Sub
```

Pause

Syntax:

Fixed Length Delay:
Pause time_ms

Command Availability:

Available on all microcontrollers.

Explanation:

The Pause command will cause the program to pause for a specified time in milliseconds. The only unit of time permitted is milliseconds.

Please use the **wait** command to use other units of time.

For more help, see [Wait](#)

Repeat

Syntax:

```
Repeat times
...
program code
...
<condition> Exit Repeat
...
End Repeat
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Repeat** command is ideal for situations where a piece of code needs to be run a set number of times. It uses less memory and runs faster than the **For** command, and should be used wherever it is not necessary to count how many times the code has run.

Optionally, you can specify a condition to **Exit** the Repeat-Loop immediately.

Repeat has a maximum repeat value of 65535.

Example:

```
'This code will flash a green light 6 times.
```

```
#chip 16F88, 20
```

```
#define LED PORTB.0
```

```
dir LED out
```

```
Repeat 6
```

```
PulseOut LED, 1 s
```

```
Wait 1 s
```

```
End Repeat
```

See Also [For](#)

Select

Syntax:

```
Select Case var
```

```
Case value1
```

```
code1
```

```
Case value2
```

```
code2
```

```
Case value_3 To _value4
```

```
code3
```

```
Case Else
```

```
code4
```

```
End Select
```

Command Availability:

Available on all microcontrollers.

Explanation:

The Select Case control structure is used to select and run a particular section of code, based on the value of `var`. If `var` equals `value1` then `code1` will be run. Once `code1` has run, the chip will jump to the `End Select` command and continue running the program. If none of the other conditions are true, then the code under the `Case Else` section will be run.

Case **var TO var** is a range of values. If the value is within the range the code section will be executed.

Case Else is optional, and the program will function correctly without it.

If there is only one line of code after the **Case**, the code may look neater if the line is placed after the **Case**. This is shown below in the example, for cases 3, 4 and 5.

It is important to note that **only one section of code will be run** when using **Select Case**.

There are two examples shown below.

Example 1:

'Program to read a value from a potentiometer, and display a
'different word based on the result

```
#chip 16F877a, 4

'LCD connection settings
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7
#define LCD_RS PORTD.0
#define LCD_NO_RW
#define LCD_Enable PORTD.2

DIR PORTA.0 IN
Do
  Temp = ReadAD(AN0) / 20
  CLS
  Select Case Temp
    Case 0
      Print "None!"
    Case 1
      Print "One"
    Case 2
      Print "Two"
    Case 3: Print "Three"
    Case 4: Print "Four"
    Case 5: Print "Five"
    Case Else
      Print "A lot!"
  End Select
  Wait 250 ms
Loop
```

Example 2:

This code demonstrates how to receive codes from a handheld remote control unit. This has been tested and supports a Sony TV remote and also a universal remote set to Sony TV mode.

The program gets both the device number and the key number, and also translates the key number to English. The received results are displayed on an LCD.

The circuit for the IR receiver and the chip is shown below.


```

'A program to receive IR codes sent by a Sony
'compatible handheld remote control.

#chip 16F88, 8          'PIC16F88 running at 8 MHz
#config mclr=off        'reset handled internally

'----- Constants

#define LCD_IO          4          '4-bit mode
#define LCD_WIDTH 20          ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS          PortB.2    'pin 8 is Register Select
#define LCD_Enable      PortB.3    'pin 9 is Enable
#define LCD_DB4         PortB.4    'DB4 on pin 10
#define LCD_DB5         PortB.5    'DB5 on pin 11
#define LCD_DB6         PortB.6    'DB6 on pin 12
#define LCD_DB7         PortB.7    'DB7 on pin 13
#define LCD_NO_RW       'ground RW line on LCD
#define IR              PortA.0    'sensor on pin 17

'----- Variables

dim device, cmd, count, i as byte
dim pulse(12)          'pulse count array
dim button as string    'ASCII for button label

'----- Program

dir PortA in            'A.0 is IR input
dir PortB out           'B.2 - B.6 for LCD

cls                     'clear the LCD
print "Dev:   Cmd:"     'logo for top line
locate 1,0
print "Button:"         'logo for second line

do
  getIR, cmd            'wait for IR signal
  printCmd              'show device and command
  printKey              'show key label
  wait 10 mS            'ignore any repeats
loop                   'repeat forever

'----- Subroutines

sub getIR
  tarry1:
    count = 0           'wait for start bit

```

```

do while IR = 0          'measure width (active low)
    wait 100 uS          '24 X 100 uS = 2.4 mS
    count += 1
loop
if count < 20 then goto tarry1 'less than this so wait

for i=1 to 12            'read/store the 12 pulses
    tarry2:
        count = 0
        do while IR = 0  'zero = 6 units = 0.6 mS
            wait 100 uS  'one = 12 units = 1.2 mS
            count += 1
        loop
        if count < 4 then goto tarry2 'too small to be legit
        pulse(i) = count             'else store pulse width
    next

cmd = 0                  'command built up here
for i = 1 to 7           '1st seven bits are the cmd
    cmd = cmd / 2        'shift into place
    if pulse(i) > 10 then 'longer than 10 mS
        cmd = cmd + 64   'so call it a one
    end if
next

device = 0               'device number built up here
for i=8 to 12            'next 5 bits are device number
    device = device / 2
    if pulse(i) > 10 then
        device = device + 16
    end if
next
end sub

sub printCmd              'print device number
    locate 0,5
    print "    "
    locate 0,5
    print device

    locate 0,13           'print raw command number
    print "    "
    locate 0,13
    print cmd
end sub

sub PrintKey              'print translated button
    locate 1,9

```

```

print "      "
locate 1,9

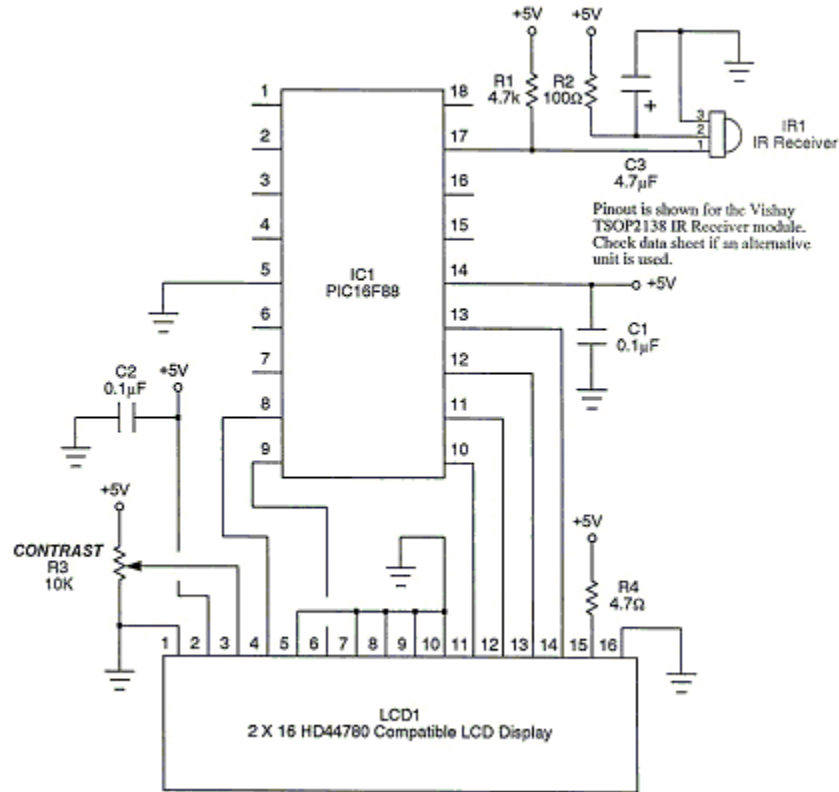
select case cmd      'translate command code
  case 0
    button = "One"
  case 1
    button = "Two"
  case 2
    button = "Three"
  case 3
    button = "Four"
  case 4
    button = "Five"
  case 5
    button = "Six"
  case 6
    button = "Seven"
  case 7
    button = "Eight"
  case 8
    button = "Nine"
  case 9
    button = "Zero"
  case 10
    button = "#####"
  case 11
    button = "Enter"
  case 12
    button = "#####"
  case 13
    button = "#####"
  case 14
    button = "#####"
  case 15
    button = "#####"
  case 16
    button = "Chan+"
  case 17
    button = "Chan-"
  case 18
    button = "Vol+"
  case 19
    button = "Vol-"
  case 20
    button = "Mute"
  case 21
    button = "Power"

```

```

    case else
        button = "    "
    end select
    print button
end sub

```



Wait

Syntax:

Fixed Length Delay:

```
Wait time units
```

Conditional Delay:

```
Wait {While | Until} condition
```

Using a variable to specific US Delay with Warning supression:

```
Wait timevalue US #OVERRIDEWARNING
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Wait** command will cause the program to wait for either a specified amount of time (such as 1 second), or while/until a condition is true.

When using the fixed-length delay, there is a variety of units that are available:

Unit	Length of unit	Delay range
us	1 microsecond	1 us - 65535 us
10us	10 microseconds	10 us - 2.55 ms
ms	1 millisecond	1 ms - 65535 ms
10ms	10 milliseconds	10 ms - 2.55 s
s	1 second	1 s - 255 s
m	1 minute	1 min - 255 min
h	1 hour	1 hour - 255 hours

At one stage, GCBASIC variables could not hold more than 255. The **10us** and **10ms** units were added as a way to work around this limit. There is now no such limit (**Wait 1000 ms** will work for example), so these are not really needed. However, you may see them in some older examples or programs, and the **10us** units are sometimes the shortest delay that will work accurately.

WARNING

PIC Devices Only

MS Delays at Clock frequency's below 28kHz are not supported and will silently fail.

US Delays at Clock frequency's below 250kHz are not supported and will silently fail.

US Delays at lower Clock frequency's is accurate ONLY when nn is divisible by 4. This is caused by the minimum ASM delay loop being a specific number of instructions.

US Delays at lower Clock frequency' when not divisible by 4 will silently accept the nn value and incorrect delays will be produced. + Use **#OVERRIDEWARNING** to suppress warnings. Delays at Clock frequency's below 500kHz may be impacted by previous instructions; testing of actual delays is advised.

Example:

'This code will wait until a button is pressed, then it will flash
'a light every half a second and produce a 440 Hz tone.

```
#chip 16F819, 8

#define BUTTON PORTB.0
#define SPEAKER PORTB.1
#define LIGHT PORTB.2
Dir BUTTON In
Dir SPEAKER Out
Dir LIGHT Out

'Assumes Button switches on when pressed
Wait Until BUTTON = 1
Wait Until BUTTON = 0

Do
  'Flash the light
  Set LIGHT On
  Wait 500 ms
  Set LIGHT Off

  'Produce the tone
  '440 Hz = 880 changes = tone on for 1.14 ms
  Repeat 440
    PulseOut SPEAKER, 1140 us
    Wait 114 10us 'Wait for 114 x 10 us (1.14 ms)
  End Repeat
Loop
```

To suppress warnings when using US.

```
dim timevariable as Word
timevariable = 100 // 100 is an example value that assigns the variable.

// Use #OVERRIDEWARNING to prevent warning messages
wait timevariable US #OVERRIDEWARNING
```

For more help, see [Conditions](#)

Fixed Voltage Reference

This is the Fixed Voltage Reference section of the Help file. Please refer the sub-sections for details using the contents/folder view.

FVRInitialize

Syntax:

```
FVRInitialize ( FVR_OFF | FVR_1x | FVR_2x | FVR_4x )
```

Command Availability:

Available on all Microchip microcontrollers with the Fixed Voltage Reference (FVR) module.

Explanation:

The method is a subroutine that sets the state of the FVR.

FVR_Off = Fixed Voltage Reference is set to OFF

FVR_1x = Fixed Voltage Reference is set to 1.024v

FVR_2x = Fixed Voltage Reference is set to 2.048v

FVR_4x = Fixed Voltage Reference is set to 4.096v

Using the the following device's datasheet, as a general case, <http://ww1.microchip.com/downloads/en/DeviceDoc/40001419F.pdf> that can be downloaded from the device's page, <http://www.microchip.com/wwwproducts/en/pic16f1828> parameter AD06 in table 30-8 at page 359, and the corresponding Note 4, tell us that the Vref voltage (Vref+ minus Vref-) should not be less than 1.8V, regardless of the reference voltage used, in order for the ADC module to work within the datasheet specifications. Also, as Vref- cannot be a negative voltage (voltages below GND) the lowest voltage on it is 0V. Then an FVR of 1.024V cannot be used as VREF+ for the ADC, but only 2.048 and 4.098 values.

The 1.024V FVR value exists for usage with other modules not just the ADC module.

Example:

```
'// use FVR 4096 as Reference
FVRInitialize ( FVR_4x )
wait while FVRIsOutputReady = false
ADVal = ReadAd(AN0)

'// Turn off FVR
FVRInitialize ( FVR_Off )
```

For more help, see [FVRIsOutputReady](#)

FVRIsOutputReady

Syntax:

```
user_var = FVRIsOutputReady()
```

Command Availability:

Available on all Microchip microcontrollers with the Fixed Voltage Reference (FVR) module.

Explanation:

The method is a function that returns the state of the FVR. The returned value can be assigned to a variable to be used as a function.

The method returns 0 or 1. As follows:

0 = Fixed Voltage Reference output is not ready or not enabled

1 = Fixed Voltage Reference output is ready for use

Example:

```
'// use FVR 4096 as Reference
FVRInitialize ( FVR_4x )
wait while FVRIsOutputReady = false
ADVal = ReadAd(AN0)
```

For more help, see [FVRInitialize](#)

Interrupts

This is the Interrupt section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Interrupts overview

Introduction

Interrupts are a feature of many microcontrollers. They allow the microcontroller to temporarily pause (interrupt) the code it is running and then start running another piece of code when some event occurs. Once it has dealt with the event, it will return to where it was and continue running the program.

Many events can trigger an interrupt, such as a timer reaching its limit, a serial message being received, or a special pin on the microcontroller receiving a signal.

Using Interrupts

There are two ways to use interrupts in GCBASIC. The first way is to use the On Interrupt command. This will automatically enable a given interrupt, and run a particular subroutine when the interrupt occurs.

The other way to deal with interrupts is to create a subroutine called Interrupt. GCBASIC will call this subroutine whenever an interrupt occurs, and then your code can check the "flag" bits to determine which interrupt has occurred, and what should be done about it. If you use this approach, then you'll need to enable the desired interrupts manually. It is also essential that your code clears the flag bits, or else the interrupt routine will be called repeatedly.

Some combination of these two methods is also possible - the code generated by On Interrupt with check to see if the interrupt is one it recognises. If the interrupt is recognised, On Interrupt will deal with it - if not, the Interrupt subroutine will be called to deal with the interrupt.

The recommended way is to use On Interrupt, as it is both more efficient and easier to set up.

During some sections of code, it is desirable not to have any interrupts occur. If this is the case, then use the IntOff command to disable interrupts at the start of the section, and IntOn to re-enable them at the end. If any interrupt events occur while interrupts are disabled, then they will be processed as soon as interrupts are re-enabled. If the program does not use interrupts, IntOn and IntOff will be removed automatically by GCBASIC.

See Also [IntOff](#), [IntOn](#), [On Interrupt](#)

IntOff

Syntax:

IntOff

Command Availability:

Available on Microchip PIC and Atmel AVR microcontrollers with interrupt support. Will be automatically removed on chips without interrupts.

Explanation:

IntOff is used to disable interrupts on the microcontroller. It should be used at the start of code which is timing-sensitive, and which would not function correctly if paused and restarted.

It is essential that **IntOn** is used to turn interrupts on again after the timing-sensitive code has finished running. If not, no interrupts will be handled.

It is recommended that IntOff be placed before all code that is timing sensitive, in case interrupts are implemented later.

IntOff will be removed from the assembler if no interrupts are used.

See also [IntOn](#), [Interrupts](#)

IntOn

Syntax:

IntOn

Command Availability:

Available on Microchip PIC and Atmel AVR microcontrollers with interrupt support. Will be automatically removed on chips without interrupts.

Explanation:

IntOn is used to enable interrupts on the microcontroller after **IntOff** has disabled them. It should be used at the end of code which is timing-sensitive.

IntOn will be removed from the assembler if no interrupts are used.

See also [IntOff](#), [Interrupts](#)

On Interrupt

Syntax:

On Interrupt event Call handler
On Interrupt event Ignore

Command Availability:

Available on Microchip PIC and Atmel AVR microcontrollers with interrupt support.

Explanation:

On Interrupt will add code to call the subroutine *handler* whenever the interrupt *event* occurs. When Call is specified, GCBASIC will enable the interrupt, and call the interrupt handler when it occurs. When Ignore is specified, GCBASIC will disable the interrupt handler and prevent it from being called when the event occurs. If the event occurs while the handler is disabled, then the handler will be called as soon as it is re-enabled. The only way to prevent this from happening is to manually clear the flag bit for the interrupt.

There are many possible interrupt events that can occur, and the events vary greatly from chip to chip. GCBASIC will display an error if a given chip cannot support the specified event.

On Interrupt may require the setting or clearing of the interrupt register bit(s), and, On Interrupt may require setting of explicit enable register bits. You should always consult the device datasheet for these On Interrupt additional specific settings of register bits. Typically, you will need define the 1) source event register bit(s) in the main program, and, 2) clear or set the register bit at the start of the of the interrupt handler subroutine.

GCBASIC has many demonstrations showing how to set and enable appropriate interrupt register bits to support the On Interrupt method.

If On Interrupt is used to handle an event, then the Interrupt() subroutine will not be called for that event. However, it will still be called for any events not dealt with by On Interrupt.

Events:

GCBASIC supports the events shown on the table below. Some events are only implemented on a few specialised chips. Events in **grey** are supported by Microchip PIC and Atmel AVR microcontrollers, events in **blue** are only supported by some Microchip PIC microcontrollers, and events in **red** are only supported by Atmel AVR microcontrollers.

Note that GCBASIC doesn't fully support all of the hardware which can generate interrupts - some work may be required with various system variables to control the unsupported peripherals.

Event Name	Description	Supported
ADCReady	The analog/digital converter has finished a conversion	Microchip& AVR

Event Name	Description	Supported
BatteryFail	The battery has failed in some way. This is only implemented on the ATmega406	AVR
CANActivity	CAN bus activity is taking place	Microchip
CANBadMessage	A bad CAN message has been received	Microchip
CANError	Some CAN error has occurred	Microchip&AVR
CANHighWatermark	CAN high watermark reached	Microchip
CANRx0Ready	New message present in buffer 0	Microchip
CANRx1Ready	New message present in buffer 1	Microchip
CANRx2Ready	New message present in buffer 2	Microchip
CANRxReady	New message present	Microchip
CANTransferComplete	Transfer of data has been completed	AVR
CANTx0Ready	Buffer 0 has been sent	Microchip
CANTx1Ready	Buffer 1 has been sent	Microchip
CANTx2Ready	Buffer 2 has been sent	Microchip
CANTxReady	Sending has completed	Microchip
CCADCAccReady	CC ADC accumulate conversion finished (ATmega406 only)	AVR
CCADCReady	CC ADC instantaneous conversion finished (ATmega406 only)	AVR
CCADCRegular	CC ADC regular conversion finished (ATmega406 only)	AVR
CCP1	The CCP1 module has captured an event	Microchip
CCP2	The CCP2 module has captured an event	Microchip
CCP3	The CCP3 module has captured an event	Microchip
CCP4	The CCP4 module has captured an event	Microchip
CCP5	The CCP5 module has captured an event	Microchip
Comp0Change	The output of comparator 0 has changed	Microchip&AVR
Comp1Change	The output of comparator 1 has changed	Microchip&AVR

Event Name	Description	Supported
Comp2Change	The output of comparator 2 has changed	Microchip&AVR
Crypto	The KEELOQ module has generated an interrupt	Microchip
EEPROMReady	An EEPROM write has finished	Microchip&AVR
Ethernet	The Ethernet module has generated an interrupt. This must be dealt within the handler.	Microchip
ExtInt0	External Interrupt pin 0 has been detected	Microchip&AVR
ExtInt1	External Interrupt pin 1 has been detected	Microchip&AVR
ExtInt2	External Interrupt pin 2 has been detected	Microchip&AVR
ExtInt3	External Interrupt pin 3 has been detected	Microchip&AVR
ExtInt4	External Interrupt pin 4 has been detected	AVR
ExtInt5	External Interrupt pin 5 has been detected	AVR
ExtInt6	External Interrupt pin 6 has been detected	AVR
ExtInt7	External Interrupt pin 7 has been detected	AVR
GPIOChange	The pins on port GPIO have changed	Microchip
LCDReady	The LCD is about to draw a segment	Microchip&AVR
LPWU	The Low Power Wake Up has been detected	Microchip
OscillatorFail	The external oscillator has failed, and the microcontroller is running from an internal oscillator.	Microchip
PinChange	Logic level of PCINT pin has changed	AVR
PinChange0	Logic level of PCINT0 pin has changed	AVR
PinChange1	Logic level of PCINT1 pin has changed	AVR
PinChange2	Logic level of PCINT2 pin has changed	AVR
PinChange3	Logic level of PCINT3 pin has changed	AVR
PinChange4	Logic level of PCINT4 pin has changed	AVR
PinChange5	Logic level of PCINT5 pin has changed	AVR

Event Name	Description	Supported
PinChange6	Logic level of PCINT6 pin has changed	AVR
PinChange7	Logic level of PCINT7 pin has changed	AVR
PMPReady	A Parallel Master Port read or write has finished	Microchip
PORTChange	The pins on ports ABCDEF have changed. This is generic port change interrupt. You must inspect the source to ensure you are handling the correct interrupt.	Microchip
PORTAChange	The pins on port A have changed	Microchip
PORTABChange	The pins on port A and/or B have changed	Microchip
PORTBChange	The pins on port B have changed	Microchip&AVR
PSC0Capture	The counter for Power Stage Controller 0 matches the value in a compare register, the value of the counter has been captured, or a synchronisation error has occurred	AVR
PSC0EndCycle	Power Stage Controller 0 has reached the end of its cycle	AVR
PSC1Capture	The counter for Power Stage Controller 1 matches the value in a compare register, the value of the counter has been captured, or a synchronisation error has occurred	AVR
PSC1EndCycle	Power Stage Controller 1 has reached the end of its cycle	AVR
PSC2Capture	The counter for Power Stage Controller 2 matches the value in a compare register, the value of the counter has been captured, or a synchronisation error has occurred	AVR
PSC2EndCycle	Power Stage Controller 2 has reached the end of its cycle	AVR
PSPReady	A Parallel Slave Port read or write has finished	Microchip
PWMTimeBase	The PWM time base matches the PWM Time Base Period register (PTPER)	Microchip
SPIReady	The SPI module has finished the previous transfer	AVR
SPMReady	A write to program memory by the spm instruction has finished	AVR
SPPReady	A SPP read or write has finished	Microchip
SSP1Collision	SSP1 has detected a bus collision	Microchip
SSP1Ready	The SSP/SSP1/MSSP1 module has finished sending or receiving	Microchip
SSP2Collision	SSP2 has detected a bus collision	Microchip
SSP2Ready	The SSP2/MSSP2 module has finished sending or receiving	Microchip

Event Name	Description	Supported
Timer0Capture	An input event on the pin ICP0 has caused the value of Timer 0 to be captured in the ICR0 register	AVR
Timer0Match1	Timer 0 matches the Timer 0 output compare register A (OCR0A)	AVR
Timer0Match2	Timer 0 matches the Timer 0 output compare register B (OCR0B)	AVR
Timer0Overflow	Timer 0 has overflowed	Microchip&AVR
Timer1Capture	An input event on the pin ICP1 has caused the value of Timer 1 to be captured in the ICR1 register	AVR
Timer1Error	The Timer 1 Fault Protection unit has been detected by an input on the INT0 pin	AVR
Timer1Match1	Timer 1 matches the Timer 1 output compare register A (OCR1A) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer1Match2	Timer 1 matches the Timer 1 output compare register B (OCR1B) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer1Match3	Timer 1 matches the Timer 1 output compare register C (OCR1C) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer1Match4	Timer 1 matches the Timer 1 output compare register D (OCR1D) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer1Overflow	Timer 1 has overflowed	Microchip&AVR
Timer2Match	Timer 2 matches the Timer 2 output compare register (PR2) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	Microchip
Timer2Match1	Timer 2 matches the Timer 2 output compare register A (OCR2A) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer2Match2	Timer 2 matches the Timer 2 output compare register B (OCR2B) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer2Overflow	Timer 2 has overflowed	AVR
Timer3Capture	An input event on the pin ICP3 has caused the value of Timer 3 to be captured in the ICR3 register	AVR

Event Name	Description	Supported
Timer3Match1	Timer 3 matches the Timer 3 output compare register A (OCR3A) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer3Match2	Timer 3 matches the Timer 3 output compare register B (OCR3B) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer3Match3	Timer 3 matches the Timer 3 output compare register C (OCR3C) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer3Overflow	Timer 3 has overflowed	Microchip & AVR
Timer4Capture	An input event on the pin ICP4 has caused the value of Timer 4 to be captured in the ICR4 register	AVR
Timer4Match	Timer 4 matches the Timer 4 output compare register (PR4) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	Microchip
Timer4Match1	Timer 4 matches the Timer 4 output compare register A (OCR4A) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer4Match2	Timer 4 matches the Timer 4 output compare register B (OCR4B) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer4Match3	Timer 4 matches the Timer 4 output compare register C (OCR4C) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer4Overflow	Timer 4 has overflowed	AVR
Timer5CAP1	An input on the CAP1 pin has caused the value of Timer 5 to be captured in CAP1BUF	Microchip
Timer5CAP2	An input on the CAP2 pin has caused the value of Timer 5 to be captured in CAP2BUF	Microchip
Timer5CAP3	An input on the CAP3 pin has caused the value of Timer 5 to be captured in CAP3BUF	Microchip
Timer5Capture	An input event on the pin ICP5 has caused the value of Timer 5 to be captured in the ICR5 register	AVR
Timer5Match1	Timer 5 matches the Timer 5 output compare register A (OCR5A) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR

Event Name	Description	Supported
Timer5Match2	Timer 5 matches the Timer 5 output compare register B (OCR5B) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer5Match3	Timer 5 matches the Timer 5 output compare register C (OCR5C) Within the Interrupt handling sub routine ensure the timer reset and clear timer is set appropriately.	AVR
Timer5Overflow	Timer 5 has overflowed	Microchip & AVR
Timer6Match	Timer 6 matches the Timer 6 output compare register (PR6)	Microchip
Timer7Overflow	Timer 7 has overflowed	Microchip
Timer8Match	Timer 8 matches the Timer 8 output compare register (PR8)	Microchip
Timer10Match	Timer 10 matches the Timer 10 output compare register (PR10)	Microchip
Timer12Match	Timer 12 matches the Timer 12 output compare register (PR12)	Microchip
TWICConnect	The Atmel AVR has been connected to or disconnected from the TWI (I2C) bus	Microchip & AVR
TWIReady	The TWI has finished the previous transmission and is ready to send or receive more data	Microchip & AVR
UsartRX1Ready	UART/USART 1 has received data	Microchip & AVR
UsartRX2Ready	UART/USART 2 has received data	Microchip & AVR
UsartRX3Ready	UART/USART 3 has received data	AVR
UsartRX4Ready	UART/USART 4 has received data	AVR
UsartTX1Ready	UART/USART 1 is ready to send data	Microchip & AVR
UsartTX1Sent	UART/USART 1 has finished sending data	AVR
UsartTX2Ready	UART/USART 2 is ready to send data	Microchip & AVR
UsartTX2Sent	UART/USART 2 has finished sending data	AVR
UsartTX3Ready	UART/USART 3 is ready to send data	AVR
UsartTX3Sent	UART/USART 3 has finished sending data	AVR
UsartTX4Ready	UART/USART 4 is ready to send data	AVR
UsartTX4Sent	UART/USART 4 has finished sending data	AVR

Event Name	Description	Supported
USBEndpoint	A USB endpoint has generated an interrupt	AVR
USB	The USB module has generated an interrupt. This must be dealt with in the handler.	Microchip&AVR
USIOverflow	The USI counter has overflowed from 15 to 0	AVR
USIStart	The USI module has detected a start condition	AVR
VoltageFail	The input voltage has dropped too low	Microchip
VoltageRegulator	An interrupt has been generated by the voltage regulator (ATmega16HVA only)	AVR
WakeUp	The Wake-Up timer has overflowed	AVR
WDT	An interrupt has been generated by the Watchdog Timer	AVR

Example 1:

```

' This program increments a counter every time Timer1 overflows
#chip 16F877A, 20

' LCD connection settings
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

InitTimer1 Osc, PS1_1/8
StartTimer 1
CounterValue = 0

Wait 100 ms
Print "Int Test"

On Interrupt Timer1Overflow Call IncCounter

Do
    CLS
    Print CounterValue
    Wait 100 ms
Loop

Sub IncCounter
    CounterValue ++
End Sub

```

Example 2:

```

'This example reflects the input signal on the output port.
#chip mega328p, 16
#option explicit

'set out SOURCE interrupt port as an output
dir portb.0 in

'set/enable the mask for the specific input port
'this is crucial - for a lot of the On Interrupt methods you will need to specify the
interrupt source via a mask.bit.
PCINT0 = 1

'set out signal port as an output
dir portB.5 out

'setup the On Interrupt method
On Interrupt PinChange0 Call TogglePin

'maintain a loop
do

loop

'handle the output signal
'Note. The AVR automatically clears the Interrupt. Please study the datasheet for
each specific microcontroller

sub togglePin
    portb.5 = !pinb.5
end sub

```

Example 3:

```

    'This example reflects the input signal on the output port from the external
interrupt port.
    #Chip mega328p, 16
    #option explicit

    'Set external interrupt INT0 input pin as an input
    dir portd.2 in

    'set out signal port as an output
    dir portB.5 out

    'hardware interrupt on Port D2
    INT0 = 1

    'set interrupt to a falling or rising edge
    'interrupt on falling edge
    EICRA = b'00000010'
    'or, alternatively you can set to a rising edge
    'EICRA = b'00000011'

    'set out signal port as an output
    dir portB.5 out

    'setup the On Interrupt method on external interrupt 0
    On Interrupt EXTINT0 Call togglePin

    'maintain a loop
    do

    loop

    'handle the output signal
    'Note. The AVR automatically clears the Interrupt. Please study the datasheet for
each specific microcontroller

    sub togglePin
        portb.5 = !pinb.5
    end sub

```

For more help, see [InitTimer0](#) article contains an example of using Timer 0 and On Interrupt to generate a Pulse Width Modulation signal to control a motor.

See also [IntOff](#), [IntOn](#)

On Interrupt: The default handler

Introduction

GCBASIC supports a default interrupt handler in two modes:

1. You can define the interrupt flags and the default handler (a sub routine) will executed
2. You can define an On Interrupt event Call handler where the handler is executed that matches the event and where all other define/valid events are handled by the default handler (a sub routine), The easiest way to write an interrupt handler is to write it in GCBASIC in conjunction with the On Interrupt statement. On Interrupt tells microcontroller to activate its internal interrupt handling and to jump to a predetermined interrupt handler (a sub routine that has been defined) when the interrupt handler (the sub routine) has completed processing returns to correct address in the program. See [On Interrupt](#).

This method of supports the handling interrupts by enabling a default interrupt subroutine.

Example 1

This example shows if an event occurs the microcontroller will be program to jump to the interrupt vector and the program will not know the event type, it will simple execute the Interrupt subroutine. This code is not intended as a meaningful solution and intended to show the functionality only. An LED is attached to PORTB.1 via a suitable resistor. It will light up when the Interrupt event has occurred.

```
#chip 16f877a, 4
dir PORTB.1 out
Set PORTB.1 Off

'Note: there is NO On Interrupt handler
InitTimer1 Osc, PS1_8
SetTimer 1, 1
StartTimer 1
'Manually set Timer1Overflow to the overflow event
'this will event will be handled by the Interrupt sub routine
TMR1IE = 1
end

Sub Interrupt
  Set PORTB.1 On
  TMR1IF = 0
End Sub
```

Example 2

Any events that are not dealt with by On Interrupt will result in the code in the Interrupt subroutine executing. This example shows the operation of two interrupt handlers - is not intended as a

meaningful solution.

LEDs are attached to PORTB.1 and PORTB.2 via suitable resistors. They will light up when the Interrupt events occur.

```
#chip 16f877a, 4
On Interrupt Timer10verflow call Overflowed

dir PORTB.1 out
Set PORTB.1 Off

dir PORTB.2 out
Set PORTB.2 Off

InitTimer1 Osc, PS1_8
SetTimer 1, 1
StartTimer 1

InitTimer2 PS2_16, PS2_16
SetTimer 2, 255
StartTimer 2

'Manually set Timer20verflow to create a second event
'this will event will be handled by the Interrupt sub routine
TMR2IE = 1
end

Sub Interrupt
    Set PORTB.2 On
    TMR2IF = 0
End Sub

Sub Overflowed
    Set PORTB.1 On
    TMR1IF = 0
End Sub
```

Keypad

This is the Keypad section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Keypad Overview

Introduction

The keypad routines allow for a program to read from a 4 x 4 matrix keypad.

There are two ways that the keypad routines can be set up. One option is to connect the wires from the keypad in a particular order, and then to set the KeypadPort constant. The other option is to connect the keypad in whatever way is easiest, and then set the `KEYPAD_ROW_x` and `KEYPAD_COL_x` constants. The option (setting `KeypadPort`) will generate slightly more efficient code.

Configuration using `KEYPAD_ROW_x` and `KEYPAD_COL_x`:

These constants must be set:

Constant Name	Controls	Default Value
<code>KEYPAD_ROW_1</code>	The pin on the microcontroller that connects to the Row 1 pin on the keypad	N/A
<code>KEYPAD_ROW_2</code>	The pin on the microcontroller that connects to the Row 2 pin on the keypad	N/A
<code>KEYPAD_ROW_3</code>	The pin on the microcontroller that connects to the Row 3 pin on the keypad	N/A
<code>KEYPAD_ROW_4</code>	The pin on the microcontroller that connects to the Row 4 pin on the keypad	N/A
<code>KEYPAD_COL_1</code>	The pin on the microcontroller that connects to the Col 1 pin on the keypad	N/A
<code>KEYPAD_COL_2</code>	The pin on the microcontroller that connects to the Col 2 pin on the keypad	N/A
<code>KEYPAD_COL_3</code>	The pin on the microcontroller that connects to the Col 3 pin on the keypad	N/A
<code>KEYPAD_COL_4</code>	The pin on the microcontroller that connects to the Col 4 pin on the keypad	N/A

If using a 3 x 3 keypad, do not set the `KEYPAD_ROW_4` or `KEYPAD_COL_4` constants.

Configuration using `KeypadPort`:

When setting up the keypad code using the `KeypadPort` constant, only `KeypadPort` needs to be set.

Pull-ups or pull-downs go on the columns only, and are typically 4.7k to 10k in value.

Constant Name	Controls	Default Value
<code>KeypadPort</code>	The port on the microcontroller chip that the keypad is connected to.	N/A

Configuration when using Pull down resistors

The keypad routine has a feature when using pull-down resistors, simply add the constant to your program and the scan logic will be inverted appropriately.

Constant Name	Controls	Default Value
<code>KEYPAD_PULLDOWN</code>	Support pull down resistors.	N/A

For this to work, the keypad must be connected as follows:

Microcontroller port pin	Keypad connector
0	Row 1
1	Row 2
2	Row 3
3	Row 4
4	Column 1
5	Column 2
6	Column 3
7	Column 4

Note: To use a 3 x 3 keypad in this mode, the pins on the microcontroller for any unused columns must be pulled up.

KeypadData

Syntax:

```
var = KeypadData
```

Command Availability:

Available on all microcontrollers.

Explanation:

This function will return a value corresponding to the key that is pressed on the keypad. Note that if two or more keys are pressed, then only one value will be returned. `var` can have one of the following values:

Value	Constant Name	Key Pressed
0		0
1		1
2		2
3		3
4		4
5		5
6		6
7		7
8		8
9		9
10	KEY_A	A
11	KEY_B	B
12	KEY_C	C
13	KEY_D	D
14	KEY_STAR	Asterisk/Star (*)
15	KEY_HASH	Hash (#)
255	KEY_NONE	None

Example:

```

'Program to show the value of the last pressed key on the LCD
#chip 18F4550, 20

'LCD connection settings
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

'Keypad connection settings
#define KeypadPort PORTB

'Main loop
Do
    'Get key
    Temp = KeypadData

    'If a key is pressed, then display it
    If Temp <> KEY_NONE Then
        CLS
        Print Temp
        Wait 100 ms
    End If
Loop

```

For more help, see [Keypad Overview](#)

KeypadRaw

Syntax:

```
largevar = KeypadRaw
```

Command Availability:

Available on all microcontrollers.

Explanation:

This function will return a 16 bit value, in which each bit corresponds to a key on the keypad. If the key is pressed its bit will hold 1, and if it is released its bit will contain a 0.

This table shows the key that each bit corresponds to:

Bit	Key Position (row, col)	Common Key Symbol
15	1,1	1
14	1,2	2
13	1,3	3
12	1,4	A
11	2,1	4
10	2,2	5
9	2,3	6
8	2,4	B
7	3,1	7
6	3,2	8
5	3,3	9
4	3,4	C
3	4,1	*
2	4,2	0
1	4,3	#
0	4,4	D

Example:

```

'Program to show the keypad status using LEDs
#chip 16F877A, 20

'Keypad connection settings
#define KeypadPort PORTB

'LEDs
#define LED1 PORTC
#define LED2 PORTD
Dir LED1 Out
Dir LED2 Out

'Declare a 16 bit variable for the key value
Dim KeyStatus As Word

'Main loop
Do
    'Get key
    KeyStatus = KeypadRaw

    'Display
    LED1 = KeyStatus_H 'High Byte
    LED2 = KeyStatus 'Low Byte
Loop

```

For more help, see [Keypad Overview](#)

Graphical LCD

This is the GLCD section of the Help file. Please refer the sub-sections for details using the contents/folder view.

GLCD Overview

The GLCD commands are used to control a Graphical Liquid Crystal Display (GLCD) based on the a number of GLCD chipsets. These are often 128x64 pixel displays but the size can vary. GLCD devices draw graphical elements by enabling or disabling pixels.

A GLCD is an upgrade from the popular 16x2 LCDs (see [Liquid Crystal Display Overview](#)) but the GLCD allows full graphical control of the display.

Typical displays are

- Color or mono displays
- Low power white LED, OLED with or without back-light
- e-Paper with low power consumption
- Driven by on-board interface chipsets and/or interface controllers
- The GLCDs are very common and well documented
- Small to large view areas
- Typically requires from 3-pin to 36-pin header connections and 10K contrast pot
- Typically have back-lit pixels
- Require memory in the microcontroller to support graphical operations or can be used in text and picture mode

GCBASIC makes this type of device easier to control with the commands for the GLCD.

Microcontroller Requirements: Specific GLCDs require different configurations of a microcontroller. Parameters include

- Communications protocol: These include 8 wire bus, I2C, SPI etc
- Operating voltage: These are typically 3.3v or 5.v
- Memory required: For full GLCD capabilities you will require 1k or more, for text only and JPG mode low memory devices are supported

Review your choice of microcontroller and GLCD carefully before commencing your project.

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
1	KS0108	2.9 inch and less. various sizes	128 * 64	Large	Monochrome	LCD typically with backlight	8-bit parallel PIC and AVR: Software device specific protocol	Typically operates at VCC 5. Always check voltage specifications 8-bit bus required.	Bit 7 of the bus is read/write – this could cause potential lockup – this is low risk. Uses the KS0107B (or NT7107C) a 64-channel common driver which generates the timing signal to control the two KS0108B segment drivers.	Requires 12 ports/connections.	These are low cost mono devices.
2	ILI9481	3.2inch	320 * 240	Large	Color	TFT LCD 8-bit parallel	PIC: Set per bit. AVR: via Shield set via AND PORT command	+VCC from 2.7 to 5. Always check voltage specifications	UNO shield is excellent. Very fast display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good GLCD with very good GLCD performance.

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
3	PCD8544	1.77inch	Nokia 3310 or 5110	160*128	Small Mono LCD with LED	SPI	PIC and AVR: Device specific SPI command, all in software.	Display can operate in text mode only for low RAM microcontrollers as full GLCD capabilities requires 512bytes of RAM. +VCC 3.3. Always check voltage specifications Nice display. Sensitive to operating voltages.	Minimum RAM required is 512 bytes then add user variables for graphics mode, this display can operate in text mode only for low RAM microcontrollers.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance
4	ILI9341	2.8 Inch or 3.2 Inch	320*240	Medium	Color TFT	SPI PIC and AVR: Hardware and software SPI	Typically operates at VCC 5. Always check voltage specifications	+VCC from 2.7 to 5. Always check voltage specifications	Very nice display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance
5	SSD1289	3.2inch	240*320	Large	Color	TFT LCD	16-bit parallel AVR: Software device specific protocol.	Typically operates at VCC 5. Always check voltage specifications	Mega2560 shield required.	Connectivity requires 20 ports.	Good for Mega2560 type shields

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
6	ST7735	1.8 Inch	128 * 64	Large	Color	TFT LCD	SPI	PIC and AVR: Hardware and software SPI	Typically operates at VCC 5. Always check voltage specifications Very nice display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance
7	ST7735R	1.8 Inch	128 * 160	Large	Color	TFT LCD	SPI	PIC and AVR: Hardware and software SPI	Typically operates at VCC 5. Always check voltage specifications Very nice display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance
8	ST7735R_160_80	1.8 Inch	160 * 80	Large	Color	TFT LCD	SPI	PIC and AVR: Hardware and software SPI	Typically operates at VCC 5. Always check voltage specifications Very nice display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance
9	ILI9340	2.2 Inch	240 * 320	Medium	Monochrome	TFT LCD	SPI	PIC and AVR: Hardware and software SPI	Typically operates at VCC 5. Always check voltage specifications	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
10	ILI9486 L or ILI9486	4inch	RPI 240*320	Large	Color	TFT LCD	SPI and 8Bit Bus	PIC and AVR: Hardware and software SPI AVR: 8Bit Bus using an UNO Shield. PIC: 8bit port supported.	Typically operates at VCC 5. Always check voltage specifications Great pixel display.	SPI requires 4 ports plus 2 ports. Typically 6 in total. 8Bit Bus requires 8 ports plus 4 control ports. Typically 13 in total using an 8bit bus solution.	An expensive option
11	Nextion	ITEAD Nexion	240*320 to 800*480	Large - 2.4 to 7inches	Color	TFT LCD	Serial - hardware or software serial is supported.	Nextion specific and GLCD command set	Typically operates at VCC 5 with external power supply. Always check voltage specifications Great command set, you need to learn the GUI and then interface to GCBASIC.	2 ports for the read/write serial operations.	A very nice option but if you need flexibility then the best!

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
1 2	SH106	1.3 inch or 0.96inch	128 * 64	Small	Monochrome OLED	I2C	PIC and AVR: Hardware and software I2C	Always at 3.3v. Always check voltage specifications	RAM for Full Mode GLCD is 1024 bytes or Low Memory GLCD is 128 bytes or 0 bytes for Text GLCD Mode then add user variables for graphics mode.	I2C requires 2 ports.	Good OLED display, excellent value for money

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
13	SDD1306	0.96inch	128 * 64	Small	Mono	OLED	I2C and SPI	PIC and AVR: Hardware and software I2C, and software SPI	RAM for Full Mode GLCD is 1024 bytes or Low Memory GLCD is 128 bytes or 0 bytes for Text GLCD Mode then add user variables for graphics mode. Typically operates at VCC 5. Always check voltage specifications Very good OLED display. Driver supports gaming. Minimum RAM required is 1024 bytes then add user variables for graphics mode. Display can operate in text mode only for low RAM microcontrollers	SPI requires 4 ports plus 2 ports. Typically 6 in total. I2C requires 2 ports.	Good OLED display, excellent value for money

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
14	SDD1306 Twin Screen	0.96inch * 2	128 * 128	Small	Monochrome	OLED	I2C and SPI	PIC and AVR: Hardware and software I2C, and software SPI	RAM for Full Mode GLCD is 2048 bytes or Low Memory GLCD is 128 bytes or 0 bytes for Text GLCD Mode then add user variables for graphics mode. Typically operates at VCC 5. Always check voltage specifications Very good OLED display. Driver supports gaming. Minimum RAM required is 1024 bytes then add user variables for graphics mode. Display can operate in text mode only for low RAM microcontrollers	SPI requires 4 ports plus 3 ports. Typically 7 in total. I2C requires 2 ports.	Good OLED display, excellent value for money

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
15	SDD1306_32	0.96inch	128*32	Very small	Monochrome	OLED	I2C and SPI	PIC and AVR: Hardware and software I2C, and software SPI	RAM for Full Mode GLCD is 512 bytes or Low Memory GLCD is 128 bytes or 0 bytes for Text GLCD Mode then add user variables for graphics mode. Typically operates at VCC 5. Always check voltage specifications Best small OLED display. Driver supports gaming. Minimum RAM required is 512 bytes then add user variables for graphics mode, this display can operate in text mode only for low RAM microcontrollers	SPI requires 4 ports plus 2 ports. Typically 6 in total. I2C requires 2 ports.	Good OLED display, excellent value for money

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
16	ST7920	2.9inch	128 * 64	Large	Monochrome	LCD typically with backlight 8-bit parallel	PIC and AVR: Software device specific protocol .	Typically operates at VCC 5. Always check voltage specifications	8-bit bus required. Bit 7 of the bus is read/write – this could cause potential lockup – this is low risk. This looks like a KS0108 but it is NOT! Supports Chinese font set.	Requires 12 ports.	A very slow device.
17	HX8347G	2.2inch	240 * 320	Large	Color	TFT LCD	SPI	AVR 8 bit bus	Typically operates at VCC 5. Always check voltage specifications Great pixel display.	Controller requires 8 ports plus 5 control ports. Typically 13 in total with an UNO shield.	An very nice display
18	SDD1331	0.96inch	96 * 48	Small	Color	OLED	SPI	PIC and AVR: Hardware and software I2C, and software SPI	Typically operates at VCC 5. Always check voltage specifications	SPI requires typically 6 in total.	Very good color OLED display, excellent value for money

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
19	ILI9326	3.00inch	400*320	Large	Color	OLED	8 bit bus	PIC and AVR: 8 bit bus	Typically operates at VCC 3.3. Always check voltage specifications	Requires typically 13 in total plus 0v, VCC and LED.	Good color OLED display, good value for money as it is fast. But, the rotate is all executed in software and this does slow down processing. The LED connected is typically to ground. You can solder the GND connect to make the backlite permanently on.

[illegible]

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
21	T6963_64	4inches by 2inches	240 * 64	Large	Monochrome	LCD typically with backlight	8-bit parallel PIC and AVR: Software device specific protocol	Typically operates at VCC 5. Always check voltage specifications 8-bit bus required.	Operating similar to KS0108 and an LCD. segment drivers.	Requires 12 ports/connections.	These are medium cost mono devices.
22	T6963_128	4inches by 2inches	240 * 128	Large	Monochrome	LCD typically with backlight	8-bit parallel PIC and AVR: Software device specific protocol	Typically operates at VCC 5. Always check voltage specifications 8-bit bus required.	Operating similar to KS0108 and an LCD. segment drivers.	Requires 12 ports/connections.	These are medium cost mono devices.

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
23	UC1601	4.00inch	132 * 22	Medium	Monochrome	OLED	I2C and SPI	PIC and AVR: Hardware and software I2C, and software SPI	RAM for Full Mode GLCD is 396 bytes or Low Memory GLCD is 128 bytes or 0 bytes for Text GLCD Mode then add user variables for graphics mode. Typically operates at VCC 2.8v. Always check voltage specifications Very good display. Driver supports gaming. Minimum RAM required is 396 bytes then add user variables for graphics mode.	Requires up to 5 ports/connections.	Low cost device

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
24	SDD1351	1.50inch	128*128	Small	Color	OLED	SPI	PIC and AVR: Hardware and software I2C, and software SPI	Typically operates at VCC 3.3 or 5. Always check voltage specifications	SPI requires typically 6 in total.	Very good color OLED display, excellent value for money
25	Waveshare e-Paper	Various Size from 2.13 to 7.5 inches	104*112 to 640*384	Small to very large	Black and White	Micro encapsulated Electrophoretic Display	SPI	PIC and AVR: Hardware and software I2C, and software SPI	Typically operates at VCC 3.3. Always check voltage specifications	SPI requires typically 6 in total.	Very good color e-Paper displays, excellent value for money Display can operate in text mode only for low RAM microcontrollers using SRAM solution.
26	ST7789	2.0 Inch	240*240 or 320*240	Medium	Color TFT	SPI PIC and AVR: Hardware and software SPI	Typically operates at 3v3. Always check voltage specifications	+VCC from 3v3. Always check voltage specifications	Very nice display.	SPI requires 3 ports (data, clock & command/data) plus 1 port (reset). Typically 4 in total.	Good for cost and performance
27	ST7735R_160_80	1.8 Inch	160*80	Large	Color	TFT LCD	SPI	PIC and AVR: Hardware and software SPI	Typically operates only at VCC 3.3. Always check voltage specifications Very nice display.	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Good for cost and performance

#	ChipSet	Size	Pixels	Depth	Type	I/O	Support	Operating	Comments	Requirements	Assessment
28	ILI9488	3.2inch	320*240	Large	Color	TFT LCD SPI	PIC/AVR: SPI Only	+VCC from 3v3 to 5. GLCD I/O is ONLY 3v3. Always check voltage specifications.	Display is good, however, slower than comparable (size) GLCDs as the color definitions are four bytes (typical color definitions are two bytes)	SPI requires 4 ports plus 2 ports. Typically 6 in total.	Acceptable GLCD performance.
29	ST7567	1.9inch	128*64	Medium	Monochrome	LCD	I2C and SPI	PIC and AVR: Software I2C, and, hardware software SPI. Hardware I2C fails as the ST7567 does not comply with the I2C standard.	+VCC from 3v3 to 5. GLCD I/O is ONLY 3v3. Always check voltage specifications.	SPI requires 4 ports plus 2 ports. Typically 6 in total. I2C requires 2 ports.	Typically operates at VCC 3v3 but may support 5v0. Always check voltage specifications Very good LCD display. Driver supports gaming. Minimum RAM required is 1024 bytes then add user variables for graphics mode.

Setup:

You **must** include the `glcd.h` file at the top of your program. The file needs to be in brackets as shown below.

```
#include <GLCD.h>
```

Defines:

There are several connections that must be defined to use the GLCD commands with a GLCD display. The *I/O pin* is the pin on the Microchip PIC or the Atmel AVR microcontroller that is connected to that specific pin on the graphical LCD.

Example: KS0108 connectivity

```
#define GLCD_RW    _I/O pin_ 'Read/Write pin connection
#define GLCD_RESET _I/O pin_ 'Reset pin connection
#define GLCD_CS1   _I/O pin_ 'CS1 pin connection
#define GLCD_CS2   _I/O pin_ 'CS2 pin connection
#define GLCD_RS    _I/O pin_ 'RS pin connection
#define GLCD_ENABLE _I/O pin_ 'Enable pin Connection
#define GLCD_DB0    _I/O pin_ 'Data pin 0 Connection
#define GLCD_DB1    _I/O pin_ 'Data pin 1 Connection
#define GLCD_DB2    _I/O pin_ 'Data pin 2 Connection
#define GLCD_DB3    _I/O pin_ 'Data pin 3 Connection
#define GLCD_DB4    _I/O pin_ 'Data pin 4 Connection
#define GLCD_DB5    _I/O pin_ 'Data pin 5 Connection
#define GLCD_DB6    _I/O pin_ 'Data pin 6 Connection
#define GLCD_DB7    _I/O pin_ 'Data pin 7 Connection
#define GLCD_PROTECTOVERRUN 'prevent screen overdrawing      'SSD1306 GLCD only
#define GLCDDirection      'Invert GLCD Y axis                'KS0108 GCD only
```

Common commands supported across the range of supported GLCDs are:

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])

Command	Purpose	Example
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)

Public variable supported across the range of supported GLCDs are shown in the table below. These variables control the user definable parameters of a specific GLCD.

Variable	Purpose	Type
GLCDBackground	Color of GLCD background.	Can be monochrome or color. For mono GLCDs the default is White or 0x0001. For color GLCDs the default is White or 0xFFFF.
GLCDForeground	Color of GLCD foreground.	Can be monochrome or color. For mono GLCDs the default is non-white or 0x0000. For color GLCDs the default is Black or 0x0000.
GLCDFontWidth	Width of the current GLCD font.	Default is 6 pixels.
GLCDfntDefault	Size of the current GLCD font.	Default is 0.+ This equates to the standard GCB font set.
GLCDfntDefaultsize	Size of the current GLCD font.	Default is 1.+ This equates to the 8 pixel high.

For color TFT displays any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF., see <http://www.barth-dev.de/online/rgb565-color-picker/> for a wider range of color parameters.

The following color constants are pre-defined.

TFT_BLACK	0x0000
TFT_NAVY	0x000F
TFT_DARKGREEN	0x03E0
TFT_DARKCYAN	0x03EF
TFT_MAROON	0x7800
TFT_PURPLE	0x780F
TFT_OLIVE	0x7BE0
TFT_LIGHTGREY	0xC618
TFT_DARKGREY	0x7BEF
TFT_BLUE	0x001F
TFT_GREEN	0x07E0
TFT_CYAN	0x07FF
TFT_RED	0xF800
TFT_MAGENTA	0xF81F
TFT_YELLOW	0xFFE0
TFT_WHITE	0xFFFF
TFT_ORANGE	0xFD20
TFT_GREENYELLOW	0xAFE5
TFT_PINK	0xF81F

This example shows how to drive a KS0108 based Graphic LCD module with the built in commands of GCBASIC. See [Graphic LCD](#) for details, this is an external web site.

Example:


```

;Chip Settings
#chip 16F886,16
'#config MCLRE = on 'enable reset switch on CHIPINO
#include <GLCD.h>

;Defines (Constants)
#define GLCD_RW PORTB.1 'D9 to pin 5 of LCD
#define GLCD_RESET PORTB.5 'D13 to pin 17 of LCD
#define GLCD_CS1 PORTB.3 'D12 to actually since CS1, CS2 can be inverted
#define GLCD_CS2 PORTB.4 'D11 to actually since CS1, CS2 can be inverted
#define GLCD_RS PORTB.0 'D8 to pin 4 D/I pin on LCD
#define GLCD_ENABLE PORTB.2 'D10 to Pin 6 on LCD
#define GLCD_DB0 PORTC.7 'D0 to pin 7 on LCD
#define GLCD_DB1 PORTC.6 'D1 to pin 8 on LCD
#define GLCD_DB2 PORTC.5 'D2 to pin 9 on LCD
#define GLCD_DB3 PORTC.4 'D3 to pin 10 on LCD
#define GLCD_DB4 PORTC.3 'D4 to pin 11 on LCD
#define GLCD_DB5 PORTC.2 'D5 to pin 12 on LCD
#define GLCD_DB6 PORTC.1 'D6 to pin 13 on LCD
#define GLCD_DB7 PORTC.0 'D7 to pin 14 on LCD

Start:
GLCDCLS
GLCDPrint 0,10,"Hello"          'Print Hello
wait 5 s
GLCDPrint 0,10, "ASCII #:"      'Print ASCII #:
Box 18,30,28,40                'Draw Box Around ASCII Character
for char = 15 to 129            'Print 0 through 9
    GLCDPrint 17, 20 , Str(char)+" "
    GLCDdrawCHAR 20,30, char
    wait 125 ms
next
line 0,50,127,50                'Draw Line using line command
for xvar = 0 to 80              'Draw line using Pset command
    pset xvar,63,on
next
Wait 1 s
GLCDPrint 0,10,"End  "          'Print Hello
wait 1 s
Goto Start

```

For more help, see [Graphical LCD Demonstration](#), [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#), [Pset](#)

Fonts and Characters

This section covers GLCD fonts and characters.

GLCD Support for Fonts

lorem ipsum.

GLCD Support for Characters

lorem ipsum.

GLCD Character Table

lorem ipsum.

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00		32	20		64	40	Q	96	60	'	128	80	Ç	160	A0	á	192	C0	À	224	E0	Ó
1	01		33	21	!	65	41	Q	97	61	a	129	81	ü	161	A1	í	193	C1	Á	225	E1	Ô
2	02		34	22	"	66	42	B	98	62	b	130	82	é	162	A2	ó	194	C2	Â	226	E2	Õ
3	03		35	23	#	67	43	C	99	63	c	131	83	â	163	A3	ú	195	C3	Ë	227	E3	Ö
4	04		36	24	\$	68	44	D	100	64	d	132	84	ä	164	A4	ü	196	C4	Ì	228	E4	Ç
5	05		37	25	%	69	45	E	101	65	e	133	85	å	165	A5	ñ	197	C5	É	229	E5	È
6	06		38	26	&	70	46	F	102	66	f	134	86	ç	166	A6	ª	198	C6	Ê	230	E6	É
7	07		39	27	'	71	47	G	103	67	g	135	87	ç	167	A7	»	199	C7	Ë	231	E7	Ê
8	08		40	28	(	72	48	H	104	68	h	136	88	ê	168	A8	¿	200	C8	Ë	232	E8	Ë
9	09		41	29	)	73	49	I	105	69	i	137	89	ë	169	A9	®	201	C9	Ì	233	E9	Ì
10	0A		42	2A	*	74	4A	J	106	6A	j	138	8A	è	170	AA	¬	202	CA	Í	234	EA	Í
11	0B		43	2B	+	75	4B	K	107	6B	k	139	8B	ÿ	171	AB	½	203	CB	Î	235	EB	Î
12	0C		44	2C	,	76	4C	L	108	6C	l	140	8C	î	172	AC	¾	204	CC	Ï	236	EC	Ï
13	0D		45	2D	-	77	4D	M	109	6D	m	141	8D	ï	173	AD	¿	205	CD	Ñ	237	ED	Ñ
14	0E		46	2E	.	78	4E	N	110	6E	n	142	8E	ï	174	AE	«	206	CE	Ò	238	EE	Ò
15	0F	—	47	2F	/	79	4F	O	111	6F	o	143	8F	ä	175	AF	»	207	CF	Ó	239	EF	Ó
16	10	►	48	30	0	80	50	P	112	70	p	144	90	é	176	B0		208	D0	Ô	240	FO	Ô
17	11	◄	49	31	1	81	51	Q	113	71	q	145	91	æ	177	B1		209	D1	Õ	241	F1	Õ
18	12	"	50	32	2	82	52	R	114	72	r	146	92	æ	178	B2		210	D2	Ö	242	F2	Ö
19	13	"	51	33	3	83	53	S	115	73	s	147	93	ö	179	B3		211	D3	Σ	243	F3	Σ
20	14	▲	52	34	4	84	54	T	116	74	t	148	94	ö	180	B4		212	D4	Τ	244	F4	Τ
21	15	▼	53	35	5	85	55	U	117	75	u	149	95	ö	181	B5	Á	213	D5	Υ	245	F5	Υ
22	16	◆	54	36	6	86	56	V	118	76	v	150	96	û	182	B6	Â	214	D6	Φ	246	F6	Φ
23	17	◀	55	37	7	87	57	W	119	77	w	151	97	ù	183	B7	À	215	D7	Χ	247	F7	Χ
24	18	↑	56	38	8	88	58	X	120	78	x	152	98	ÿ	184	B8	©	216	D8	Υ	248	F8	Υ
25	19	↓	57	39	9	89	59	Y	121	79	y	153	99	ÿ	185	B9	©	217	D9	Ω	249	F9	Ω
26	1A	→	58	3A	:	90	5A	Z	122	7A	z	154	9A	ÿ	186	BA	©	218	DA	Γ	250	FA	Γ
27	1B	←	59	3B	;	91	5B	[	123	7B	{	155	9B	ø	187	BB	©	219	DB	Γ	251	FB	Γ
28	1C	≤	60	3C	<	92	5C	\	124	7C		156	9C	£	188	BC	©	220	DC	à	252	FC	à
29	1D	≥	61	3D	=	93	5D	]	125	7D	}	157	9D	£	189	BD	©	221	DD	é	253	FD	é
30	1E	▲	62	3E	>	94	5E	^	126	7E	~	158	9E	×	190	BE	¥	222	DE	ñ	254	FE	ñ
31	1F	▼	63	3F	?	95	5F	_	127	7F	␣	159	9F	f	191	BF	ı	223	DF	ı	255	FF	ı

GLCD Controlling Constants

The GCBASIC constants for control of fonts and characters are shown in the table below.

Constants	Controls	Options
CONSTANT	Words	Explanation
ANOTHER CONSTANT	Words	Explanation

For more help, see [GLCDCLS](#), [GLCDDrawChar](#)

e-Paper Controllers

This section covers GLCD devices known as e-Papers.

An e-paper device is a Microencapsulated Electrophoretic Display, MED.

A MED display uses tiny spheres, in which the charged color pigments are suspending in the transparent oil and would move depending on the electronic charge. The e-paper screen display patterns by reflecting the ambient light, so it has no background light requirement. Under sunshine, the e-paper screen still has high visibility with a wide viewing angle of 180 degree. It is the ideal choice for e-reading or providing information that can be refreshed at a slow rate of change.

GLCD Support for e-Papers

GCBASIC supports covers the full range of GLCD capabilities like line, circle, print.

GCBASIC supports SPI communications for the e-Papers - both hardware and software. And, GCBASIC supports low memory configurations and SRAM for the display buffer.

See the demonstration programs to show you how to use these GLCD capabilities.

Memory Usage

The GCBASIC library uses RAM to buffer the e-paper display. The amount of RAM used is specific to the total pixel of the specific e-paper display. You can control the amount of RAM used as the buffer using the device specific constants, see below. Each device specific library has four memory options. Each of the memory options uses different amount RAM. The greater the amount of RAM used the faster the process of updating the e-paper display. Conversely, the smaller the amount of RAM used the slower the process of updating the e-paper display.

GLCD Page Transactions

To make the operation of the library seamless - the library supports GLCDTransaction. GLCDTransaction automatically manages the methods to update the e-paper via the buffer, where the buffer can be small. The process of transaction sends GLCD commands to the e-paper display on a page and page basis. Each page is the size of the buffer and for a large e-paper display the number of pages may be equivalent to the numbers of pixels high (height).

GLCDTransaction simplifies the operation by ensuring the buffer is setup correctly, handles the GLCD appropriately, handles the sending of the buffer and then closes out the process to update the display.

To use GLCDTransaction use the following two methods.

```
GLCD_Open_PageTransaction
....
glcd commands
.....

GLCD_Close_PageTransaction
```

It is recommended to use GLCDTransactions at all times. These methods remove the complexity of the e-paper update process.

When using GLCDTransaction you must start the transaction with `GLCD_Open_PageTransaction` then include a series of GLCD commands and then terminate the transaction with `GLCD_Close_PageTransaction`.

GLCDTransaction Insight: When using GLCDtransactions the number of buffer pages is probably be greater than 1 (unless using the SRAM option), so the process of incrementing variables and calls to non-GLCD methods must be considered carefully. The transaction process **will** increment variables and call non-GLCD methods the same number of times as the number of pages. Therefore, design GLCDTransaction operations with this in mind.

SRAM as the e-paper buffer

To improve memory usage the e-paper the e-Paper libraries support the use of SRAM. SRAM can be used as an alternative to the microcontrollers RAM. Using SRAM does have a small performance impact but does free up the critical resource of the microcontroller RAM. The use of SRAM within the e-paper library is transparent to the user. To use SRAM as the e-paper buffer you will need to set-up the SRAM library. See the SRAM library for more details on SRAM usage.

When using SRAM for the e-paper buffer it is still recommended to use GLCDTransaction as this ensures the SRAM buffer is correctly initialised.

Refresh mode

This library uses Full refresh: The e-Paper will flicker when full refreshing. This flicker removes the ghost image from the display. You could use Partial refresh as this does not flicker. Note that you cannot use Partial refresh all the time, you should full refresh e-paper regularly, otherwise, the ghost problem will get worse and even damage the display.

Refresh rate

When using the e-Paper library, you should set the update interval at least 180 seconds, except when using Partial mode.

Please set the e-Paper to sleep mode in software or remove the power directly, otherwise, the e-Paper will be damaged because of working in high voltage for extended time periods. You need to update the content of the e-Paper at least once every 24 hours to avoid from burn-in problem.

Operating Voltages

The e-Paper should be driven with 3V3 operating voltages and signals.

If your Microcontroller (PIC, AVR and therefore an Arduino) cannot drive the e-Paper successfully. You must convert the level to 3.3V. The I/O level of Arduino is 5V. **HEALTH WARNING:** You can also try to connect the Vcc pin to the 5V of Arduino to see whether the e-Paper works, but we recommend you not to use 5V for a long time.

The e-Paper looks a little black or grey

You can try to change the value of Vcom the library by setting the VCOM_AND_DATA_INTERVAL constant. See the Vcom and data interval in the data sheet. VCOM_AND_DATA_INTERVAL can be 0x00 to 0x0F

GCBASIC library supports Black/White NOT Black/White/Red

The default is Black/White. To support Black/White/Red add ``#define PANEL_SETTING_KWR 0x00`` to your user program.

The constants are the TFT_BLACK and TFT_WHITE constants.

The e-paper has ghosting problem after working for some days

Please set the e-paper to sleep mode or disconnect it if you do not refresh the e-paper but need to power on your solution.

Do NOT leave power on for extended periods, otherwise, the voltage of panel remains high and it will damage the e-paper display.

e-Paper Guidelines

Remove power if practical.

ALWAYS use `GLCDDisplay Off` or sleep mode.

When in storage CLEAR the screen.... avoid burn it - use

```
GLCDCLD TFT_WHITE
GLCDDisplay Off
```

The recommended method is:

```
GLCDCLS TFT_WHITE
GLCDDisplay Off
do
loop
```

Using the e-Paper Library

To use the e-Paper driver for a specific simply include the following in your user code.

This will initialise the driver.

```
'Setup for the e-Paper
#include <glcd.h>

#define GLCD_TYPE GLCD_TYPE_EPD_EPD7in5
#define GLCD_EXTENDEDFONTSET1
#define GLCD_OLED_FONT
#define GLCD_TYPE_EPD7in5_LOWMEMORY4_GLCD_MODE fastest but uses a lot of RAM
'#define GLCD_TYPE_EPD7in5_LOWMEMORY3_GLCD_MODE
'#define GLCD_TYPE_EPD7in5_LOWMEMORY2_GLCD_MODE
'#define GLCD_TYPE_EPD7in5_LOWMEMORY1_GLCD_MODEslowest uses the least amount of RAM

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC portA.0 ' Data(hight)/ command(low) line
#define GLCD_CS portC.1 ' Chip select line (negate)
#define GLCD_RESETportD.2 ' Reset line (negate)
#define GLCD_DO portC.5 ' GLCD MOSI connect to MCU SDO
#define GLCD_SCKportC.3 ' Clock Line
#define GLCD_Busy portC.0 ' Busy Line

'The following should be used for hardware SPI remove or comment out if you want to
use software SPI.
#define EPD_HardwareSPI
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_EPD_EPD7in5	GLCD_TYPE_EPD_EPD7in5 and GLCD_TYPE_EPD_EPD2in13D supported

Constants	Controls	Options
GLCD_TYPE_<device_memory_mode>	Memory usage for the display buffer. Memory management is crucial when using the e-paper displays.	GLCD_TYPE_EPD7in5_LOWMEMORY4_GLCD_MODE ... GLCD_TYPE_EPD7in5_LOWMEMORY1_GLCD_MODE, or, GLCD_TYPE_EPD2in13D_LOWMEMORY4_GLCD_MODE ... GLCD_TYPE_EPD2in13D_LOWMEMORY1_GLCD_MODE
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required
GLCD_BUSY	Specifies the output pin that is connected to Busy pin on the GLCD.	Required
EPD_HardwareSPI	Instructs the library to use hardware SPI, remove or comment out if you want to use software SPI.	#define EPD_HardwareSPI
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFast for specific AVR only.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	Specific to the e-Paper selected This cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	Specific to the e-Paper selected This cannot be changed
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6 or 5 for the OLED font set.

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDDisplay	Enables sleep mode, or, enables operations	GLCDDisplay Off, or, GLCDDisplay On
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCD_Open_PageTransaction	Commence a series of GLCD commands with memory buffer management. Must be followed a GLCD_Close_PageTransaction command.	GLCD_Open_PageTransaction. Parameters may be passed where the two parameters are the range of pages to be updated
GLCD_Close_PageTransaction	Terminate a series of GLCD commands. Must follow a GLCD_Open_PageTransaction command.	GLCD_Close_PageTransaction. Terminates the GLCDTransaction.

Example Usage:

```
#chip mega328p, 16
#include <uno_mega328p.h>
#option explicit

'*****
*****

'Setup the E-Paper
#include <glcd.h>

#define HWSPIMode ULTRAFAST
```



```

#define GLCD_TYPE GLCD_TYPE_EPD_EPD2in13D
#define GLCD_EXTENDEDFONTSET1
#define GLCD_TYPE_EPD2in13D_LOWMEMORY4_GLCD_MODE
#define GLCD_OLED_FONT
#define GLCD_PROTECTOVERRUN

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC DIGITAL_9
#define GLCD_CS DIGITAL_10
#define GLCD_RESETDIGITAL_8
#define GLCD_DO DIGITAL_11
#define GLCD_SCKDIGITAL_13
#define GLCD_Busy DIGITAL_7

#define EPD_HARDWARESPI

' *****
*****

'Main program

GLCDForeground=TFT_BLACK
GLCDBackground=TFT_WHITE

GLCD_Open_PageTransaction
  GLCDPrintStringLN ("GCBASIC")
  GLCDPrintStringLN ("")
  GLCDPrintStringLN ("Test of the e-Paper")
  GLCDPrintStringLN ("")
  GLCDPrintStringLN ("December 2021")
GLCD_Close_PageTransaction
GLCDDisplay Off

wait 2 s
GLCDDisplay On
GLCDCLS
GLCDDisplay off

do

loop

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#), [Pset](#) or

Supported in <GLCD.H>

HX8347G Controllers

This section covers GLCD devices that use the HX8347G graphics controller.

HX8347G is a 262k-color single-chip SoC driver for a-TFT liquid crystal display with resolution of 240 RGB x 320 dots.

The HX8347-G is designed to provide a single-chip solution that combines a gate driver, a source driver, power supply circuit for 262k colors to drive a TFT panel with 240RGBx320 dots at maximum.

GCBASIC supports 65K-color mode operations.

The HX8347-G can be operated in low-voltage (1.4V) condition for the interface and integrated internal boosters that produce the liquid crystal voltage, breeder resistance and the voltage follower circuit for liquid crystal driver. In addition, The HX8347-G also supports various functions to reduce the power consumption of a LCD system via software control.

The GCBASIC constants shown below control the configuration of the HX8347G controller. The GCBASIC constants for control and data line connections are shown in the table below.

Connectivity is via an 8-bit bus. Where 8 pins are connected between the microcontroller and the GLCD to control the data bus plus 5 control pins. This is simple when using an Arduino GLCD Shield.

To use the HX8347G driver simply include the following in your user code. This will initialise the driver.

8-bit mode

'This GLCD driver supports 8 bit only. UNO ports can be replaced with `porta.b` constants.

```
#include <glcd.h>
#include <UNO_mega328p.h >
#define GLCD_TYPE GLCD_TYPE_HX8347

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_RD      ANALOG_0      ' read command line
#define GLCD_WR      ANALOG_1      ' write command line
#define GLCD_RS      ANALOG_2      ' Command/Data line
#define GLCD_CS      ANALOG_3      ' Chip select line
#define GLCD_RST      ANALOG_4      ' Reset line

#define GLCD_DB0      DIGITAL_8
#define GLCD_DB1      DIGITAL_9
#define GLCD_DB2      DIGITAL_2
#define GLCD_DB3      DIGITAL_3
#define GLCD_DB4      DIGITAL_4
#define GLCD_DB5      DIGITAL_5
#define GLCD_DB6      DIGITAL_6
#define GLCD_DB7      DIGITAL_7
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_HX8347	
GLCD_DB0..7	Specifies the pin that is connected to DB0..7 IO pin on the GLCD (8 bit DBI).	Required
GLCD_RST	Specifies the output pin that is connected to Reset IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to Data/Command pin on the GLCD.	Required
GLCD_WR	Specifies the output pin that is connected to Data In (RW or WDR) pin on the GLCD.	Required

Constants	Controls	Options
GLCD_RD	Specifies the output pin that is connected to Data Out (RD or RDR) pin on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	480
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6 for GCB fonts, and 5 for OLED fonts.
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDfontDefaultsize can be set to 1 or 2 only. GLCDfontDefaultsize= 1. A small 8 height pixel font with variable width. GLCDfontDefaultsize= 2. A larger 10 width * 16 height pixel font.	Optional

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]

Command	Purpose	Example
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
HX8347G_ [color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below. Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

HX8347G_BLACK	'hexidecimal value 0x0000
HX8347G_RED	'hexidecimal value 0xF800
HX8347G_GREEN	'hexidecimal value 0x0400
HX8347G_BLUE	'hexidecimal value 0x001F
HX8347G_WHITE	'hexidecimal value 0xFFFF
HX8347G_PURPLE	'hexidecimal value 0xF11F
HX8347G_YELLOW	'hexidecimal value 0xFFE0
HX8347G_CYAN	'hexidecimal value 0x07FF
HX8347G_D_GRAY	'hexidecimal value 0x528A
HX8347G_L_GRAY	'hexidecimal value 0x7997
HX8347G_SILVER	'hexidecimal value 0xC618
HX8347G_MAROON	'hexidecimal value 0x8000
HX8347G_OLIVE	'hexidecimal value 0x8400
HX8347G_LIME	'hexidecimal value 0x07E0
HX8347G_AQUA	'hexidecimal value 0x07FF
HX8347G_TEAL	'hexidecimal value 0x0410
HX8347G_NAVY	'hexidecimal value 0x0010
HX8347G_FUCHSIA	'hexidecimal value 0xF81F

These examples show how to drive a HX8347G based Graphic LCD module with the built in commands of GCBASIC. The 8 bit DBI example uses a UNO shield, this can easily adapted to Microchip architecture. The 16 bit DBI example uses a Mega2560 board.

Example:

```

#chip mega328p, 16
#option explicit

#include <glcd.h>
#include <UNO_mega328p.h >

#define GLCD_TYPE GLCD_TYPE_HX8347
#define GLCD_OLED_FONT

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_RD      ANALOG_0      ' read command line
#define GLCD_WR      ANALOG_1      ' write command line
#define GLCD_RS      ANALOG_2      ' Command/Data line
#define GLCD_CS      ANALOG_3      ' Chip select line
#define GLCD_RST      ANALOG_4      ' Reset line


#define GLCD_DB0      DIGITAL_8
#define GLCD_DB1      DIGITAL_9
#define GLCD_DB2      DIGITAL_2
#define GLCD_DB3      DIGITAL_3
#define GLCD_DB4      DIGITAL_4
#define GLCD_DB5      DIGITAL_5
#define GLCD_DB6      DIGITAL_6
#define GLCD_DB7      DIGITAL_7


GLCDRotate ( Portrait )
GLCDCLS HX8347_RED
GLCDPrint(0, 0, "Test of the HX8347G Device")
end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9326 Controllers

This section covers GLCD devices that use the ILI9326 graphics controller. The ILI9326 is a TFT LCD Single Chip Driver with 400RGBx320 Resolution and 262K colors.

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ILI9326 controller. GCBASIC supports 8 bit bus connectivity - this is shown in the tables below.

To use the ILI9326 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9326

'Pin mappings for ILI9326 - these MUST be specified
#define GLCD_RD      porta.3      ' read command line
#define GLCD_WR      porta.2      ' write command line
#define GLCD_RS      porta.1      ' Command/Data line
#define GLCD_CS      porta.0      ' Chip select line
#define GLCD_RST      porta.5      ' Reset line
#define GLCD_DataPort portD
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ILI9326	
GLCD_RD	Specifies the output pin that is connected to RD IO pin on the GLCD.	Required
GLCD_WR	Specifies the output pin that is connected to WR on the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to RS pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to CS pin on the GLCD.	Required
GLCD_RST	Specifies the output pin that is connected to RST pin on the GLCD.	Required
GLCD_DataPort	Specifies the output port that is connected to DB0 to DB7 pins on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	240
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6 for GCB fonts, and 5 for OLED fonts.

Constants	Controls	Default
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDfntDefaultsize can be set to 1 or 2 only. GLCDfntDefaultsize= 1. A small 8 height pixel font with variable width. GLCDfntDefaultsize= 2. A larger 10 width * 16 height pixel font.	Optional

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
ILI9326_[color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below.

ILI9326_BLACK	'hexidecimal value 0x0000
ILI9326_RED	'hexidecimal value 0xF800
ILI9326_GREEN	'hexidecimal value 0x07E0
ILI9326_BLUE	'hexidecimal value 0x001F
ILI9326_WHITE	'hexidecimal value 0xFFFF
ILI9326_PURPLE	'hexidecimal value 0xF11F
ILI9326_YELLOW	'hexidecimal value 0xFFE0
ILI9326_CYAN	'hexidecimal value 0x07FF
ILI9326_D_GRAY	'hexidecimal value 0x528A
ILI9326_L_GRAY	'hexidecimal value 0x7997
ILI9326_SILVER	'hexidecimal value 0xC618
ILI9326_MAROON	'hexidecimal value 0x8000
ILI9326_OLIVE	'hexidecimal value 0x8400
ILI9326_LIME	'hexidecimal value 0x07E0
ILI9326_AQUA	'hexidecimal value 0x07FF
ILI9326_TEAL	'hexidecimal value 0x0410
ILI9326_NAVY	'hexidecimal value 0x0010
ILI9326_FUCHSIA	'hexidecimal value 0xF81F

For a ILI9326 datasheet, please refer to Google.

This example shows how to drive a ILI9326 based Graphic LCD module with the built in commands of GCBASIC.

Example #1

```
;Chip Settings
#chip 16F1789,32

#config MCLRE=on
#option explicit
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9326

#define GLCD_RD      porta.3      ' read command line
#define GLCD_WR      porta.2      ' write command line
#define GLCD_RS      porta.1      ' Command/Data line
#define GLCD_CS      porta.0      ' Chip select line
#define GLCD_RST      porta.5      ' Reset line
#define GLCD_DataPort portD

GLCDPrint(0, 0, "Test of the ILI9326 Device")
end
```

Example #2 This example shows how to drive a ILI9341 with the OLED fonts. Note the use of the `GLCDfntDefaultSize` to select the size of the OLED font in use.

```
'Chip Settings
#chip 16F1789,32

#config MCLRE=on
#option explicit
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9326

#define GLCD_RD      porta.3      ' read command line
#define GLCD_WR      porta.2      ' write command line
#define GLCD_RS      porta.1      ' Command/Data line
#define GLCD_CS      porta.0      ' Chip select line
#define GLCD_RST      porta.5      ' Reset line
#define GLCD_DataPort portD

#define GLCD_OLED_FONT              'The constant is required to support OLED fonts

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ILI9326" )
GLCDPrint ( 0, 34, "Size: 400 x 240" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9340 Controllers

This section covers GLCD devices that use the ILI9340 graphics controller. The ILI9340 is a TFT LCD Single Chip Driver with 240RGBx320 Resolution and 262K colors.

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ILI9340 controller. GCBASIC supports SPI hardware and software connectivity - this is shown in the tables below.

To use the ILI9340 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9340

'Pin mappings for ILI9340 - these MUST be specified
#define GLCD_DC      porta.0      'example port setting
#define GLCD_CS      porta.1      'example port setting
#define GLCD_RESET   porta.2      'example port setting
#define GLCD_DI      porta.3      'example port setting
#define GLCD_DO      porta.4      'example port setting
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Const ants	Controls	Options
GLCD_T YPE	GLCD_TYPE_ILI9340	
GLCD_D C	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_C S	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_R eset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_D I	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD.	Required
GLCD_D O	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_S CK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD. #define GLCD_SCK porta.5 'example port setting	Required
HWSPIM ode	User can specify the hardware SPI mode. Must be one of MasterSlow, Master, Masterfast	Optional. Defaults to Masterfast when chipMhz is less than 64mhz

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320+ Cannot be changed.
GLCD_HEIGHT	The height parameter of the GLCD	240+ Cannot be changed.
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
ILI9340_[color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below. Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

ILI9340_BLACK	'hexidecimal value 0x0000
ILI9340_RED	'hexidecimal value 0xF800
ILI9340_GREEN	'hexidecimal value 0x07E0
ILI9340_BLUE	'hexidecimal value 0x001F
ILI9340_WHITE	'hexidecimal value 0xFFFF
ILI9340_PURPLE	'hexidecimal value 0xF11F
ILI9340_YELLOW	'hexidecimal value 0xFFE0
ILI9340_CYAN	'hexidecimal value 0x07FF
ILI9340_D_GRAY	'hexidecimal value 0x528A
ILI9340_L_GRAY	'hexidecimal value 0x7997
ILI9340_SILVER	'hexidecimal value 0xC618
ILI9340_MAROON	'hexidecimal value 0x8000
ILI9340_OLIVE	'hexidecimal value 0x8400
ILI9340_LIME	'hexidecimal value 0x07E0
ILI9340_AQUA	'hexidecimal value 0x07FF
ILI9340_TEAL	'hexidecimal value 0x0410
ILI9340_NAVY	'hexidecimal value 0x0010
ILI9340_FUCHSIA	'hexidecimal value 0xF81F

For a ILI9340 datasheet, please refer [here](#).

This example shows how to drive a ILI9340 based Graphic LCD module with the built in commands of GCBASIC.

Example:

```
;Chip Settings
#chip 16F1937,32
#config MCLRE_ON      'microcontroller specific configuration

#include <glcd.h>

'Defines for ILI9340
#define GLCD_TYPE GLCD_TYPE_ILI9340

'Pin mappings for ILI9340
#define GLCD_DC porta.0
#define GLCD_CS porta.1
#define GLCD_RESET porta.2
#define GLCD_DI porta.3
#define GLCD_D0 porta.4
#define GLCD_SCK porta.5

GLCDPrint(0, 0, "Test of the ILI9340 Device")
end
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9341 Controllers

This section covers GLCD devices that use the ILI9341 graphics controller. The ILI9341 is a TFT LCD Single Chip Driver with 240RGBx320 Resolution and 262K colors.

GCBASIC supports 65K-color mode operations.

The GCBASIC supports different methods to controller the GLCD. These methods are shown below control the configuration of the ILI9341 controller. GCBASIC supports SPI hardware and software connectivity - this is shown in the tables below.

To use the ILI9341 driver simply include the following in your user code. This will initialise the driver of a SPI method of connection.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9341

'Pin mappings for ILI9341 - these MUST be specified
#define GLCD_DC      porta.0      'example port setting
#define GLCD_CS      porta.1      'example port setting
#define GLCD_RESET   porta.2      'example port setting
#define GLCD_DI      porta.3      'example port setting
#define GLCD_DO      porta.4      'example port setting
#define GLCD_SCK     porta.5      'example port setting
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Const ants	Controls	Options
GLCD_ TYPE	GLCD_TYPE_ILI9341	Define the constant only
SPI Method The method can use hardware or software SPI		
GLCD_ DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_ CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_ Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required

Const ants	Controls	Options
GLCD_ DI	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD. If using hardware SPI this must be the hardware SPI port.	Required
GLCD_ DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD. If using hardware SPI this must be the hardware SPI port.	Required
GLCD_ SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD. If using hardware SPI this must be the hardware SPI port.	Required
UNO Shield Method The method uses the ILI9341 attached via the UNO shield		
GLCD_ CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_ Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_ RD	Specifies the output pin that is connected to Read (RD) pin on the GLCD.	Required
GLCD_ WR	Specifies the output pin that is connected to Write (WR) pin on the GLCD.	Required
GLCD_ RS	Specifies the output pin that is connected to Data/Command (RS) pin on the GLCD.	Required
GLCD_ DB0	DIGITAL_8	Mandated to this port
GLCD_ DB1	DIGITAL_9	Mandated to this port
GLCD_ DB2	DIGITAL_2	Mandated to this port
GLCD_ DB3	DIGITAL_3	Mandated to this port
GLCD_ DB4	DIGITAL_4	Mandated to this port
GLCD_ DB5	DIGITAL_5	Mandated to this port
GLCD_ DB6	DIGITAL_6	Mandated to this port

Constants	Controls	Options
GLCD_DB7	DIGITAL_7	Mandated to this port
8Bit Port Method The method uses a contiguous 8bit port for the data port.		
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_RD	Specifies the output pin that is connected to Read (RD) pin on the GLCD.	Required
GLCD_WR	Specifies the output pin that is connected to Write (WR) pin on the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to Data/Command (RS) pin on the GLCD.	Required
GLCD_PORT	Any valid 8 bit port, like PORTC	Required
SPI Controls		
HWSPI Mode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFast for specific AVR's only.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	240
GLCDfontWidth	Specifies the font width of the GCBASIC font set.	6 for GCB fonts, and 5 for OLED fonts.
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDfontDefaultsize can be set to 1 or 2 only. GLCDfontDefaultsize= 1. A small 8 height pixel font with variable width. GLCDfontDefaultsize= 2. A larger 10 width * 16 height pixel font.	Optional

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the

appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
ILI9341_color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below.
ReadPixel	Read the pixel color at the specified XY coordination. Returns long variable with Red, Green and Blue encoded in the lower 24 bits.	ReadPixel(Xosition , Yposition) or ReadPixel_ILI9341(Xosition , Yposition) Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

ILI9341_BLACK	'hexidecimal value 0x0000
ILI9341_RED	'hexidecimal value 0xF800
ILI9341_GREEN	'hexidecimal value 0x07E0
ILI9341_BLUE	'hexidecimal value 0x001F
ILI9341_WHITE	'hexidecimal value 0xFFFF
ILI9341_PURPLE	'hexidecimal value 0xF11F
ILI9341_YELLOW	'hexidecimal value 0xFFE0
ILI9341_CYAN	'hexidecimal value 0x07FF
ILI9341_D_GRAY	'hexidecimal value 0x528A
ILI9341_L_GRAY	'hexidecimal value 0x7997
ILI9341_SILVER	'hexidecimal value 0xC618
ILI9341_MAROON	'hexidecimal value 0x8000
ILI9341_OLIVE	'hexidecimal value 0x8400
ILI9341_LIME	'hexidecimal value 0x07E0
ILI9341_AQUA	'hexidecimal value 0x07FF
ILI9341_TEAL	'hexidecimal value 0x0410
ILI9341_NAVY	'hexidecimal value 0x0010
ILI9341_FUCHSIA	'hexidecimal value 0xF81F

For a ILI9341 datasheet, please refer [here](#).

This example shows how to drive a ILI9341 based Graphic LCD module with the built in commands of GCBASIC.

Example #1

```
;Chip Settings
#chip 16F1937,32
#config MCLRE_ON      'microcontroller specific configuration

#include <glcd.h>

'Defines for ILI9341
#define GLCD_TYPE GLCD_TYPE_ILI9341

'Pin mappings for ILI9341
#define GLCD_DC porta.0
#define GLCD_CS porta.1
#define GLCD_RESET porta.2
#define GLCD_DI porta.3
#define GLCD_D0 porta.4
#define GLCD_SCK porta.5

GLCDPrint(0, 0, "Test of the ILI9341 Device")
end
```

Example #2 This example shows how to drive a ILI9341 with the OLED fonts. Note the use of the `GLCDfntDefaultSize` to select the size of the OLED font in use.

```
#define GLCD_OLED_FONT                'The constant is required to support OLED fonts

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ILI9341" )
GLCDPrint ( 0, 34, "Size: 320 x 240" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

Example #2 This example shows how to disable the large OLED Fontset. This disables the font to reduce memory usage.

When the extended OLED fontset is disabled every character will be shown as a block character.

```
#define GLCD_OLED_FONT                'The constant is required to support OLED fonts
#define GLCD_Disable_OLED_FONT2      'The constant to disable the extended OLED
fontset.

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ILI9341" )
GLCDPrint ( 0, 34, "Size: 320 x 240" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9481 Controllers

This section covers GLCD devices that use the ILI9481 graphics controller.

ILI9481 is a 262k-color single-chip SoC driver for a-TFT liquid crystal display with resolution of 320 RGB x 480 dots, comprising a 960-channel source driver, a 480-channel gate driver, 345,600 bytes GRAM for graphic data.

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ILI9481 controller. The GCBASIC constants for control and data line connections are shown in the table below. Two options are available for connectivity:

- 1) The 8-bit mode where 8 pins are connected between the microcontroller and the GLCD to control the data bus.
- 2) The 16-bit mode where two data ports (8 pins each) are connected between the microcontroller and the GLCD to control the data bus.

To use the ILI9481 driver simply include the following in your user code. This will initialise the driver.

8-bit mode

```
'Pin mappings for Data Bus Interface (DBI)
'this GLCD driver supports 8 bit and 16 bit parallel data lines

'8 bit DBI
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9481

'8 bit control and parallel data lines (UNO Board)
#define GLCD_RD      ANALOG_0      ' read command line
#define GLCD_WR      ANALOG_1      ' write command line
#define GLCD_RS      ANALOG_2      ' Command/Data line
#define GLCD_CS      ANALOG_3      ' Chip select line
#define GLCD_RST      ANALOG_4      ' Reset line

#define GLCD_DB0      DIGITAL_8      'Data port'
#define GLCD_DB1      DIGITAL_9      'Data port'
#define GLCD_DB2      DIGITAL_2      'Data port'
#define GLCD_DB3      DIGITAL_3      'Data port'
#define GLCD_DB4      DIGITAL_4      'Data port'
#define GLCD_DB5      DIGITAL_5      'Data port'
#define GLCD_DB6      DIGITAL_6      'Data port'
#define GLCD_DB7      DIGITAL_7      'Data port'
```

16-bit mode

```
'16 bit DBI
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9481
#define GLCD_ILI9481_16bit

'16 bit control and dual data port lines (Mega2560 Board)
#define ILI9481_GLCD_CS PortG.1    'Chip Select line
#define ILI9481_GLCD_RS PortD.7    'DC data command line
#define ILI9481_GLCD_WR PortG.2    'Write command line
#define ILI9481_GLCD_RST PortG.0    'Reset line

#define ILI9481_DataPortH PortA    'DB[15:8]
#define ILI9481_DataPortL PortC    'DB[7:0]
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ILI9481	
GLCD_ILI9481_16bit	Specifies 16 bit DBI mode	
GLCD_DB0..7	Specifies the pin that is connected to DB0..7 IO pin on the GLCD (8 bit DBI).	Required
ILI9481_DataPortH	Specifies the port DB[15:8] pins on the GLCD (16 bit DBI).	Required
ILI9481_DataPortL	Specifies the port DB[7:0] pins on the GLCD (16 bit DBI).	Required
GLCD_RST	Specifies the output pin that is connected to Reset IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to Data/Command pin on the GLCD.	Required
GLCD_WR	Specifies the output pin that is connected to Data In (RW or WDR) pin on the GLCD.	Required

Constants	Controls	Options
GLCD_RD	Specifies the output pin that is connected to Data Out (RD or RDR) pin on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	480
GLCDDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])

Command	Purpose	Example
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
ILI9481_[color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below. Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

```
ILI9481_BLACK    'hexidecimal value 0x0000
ILI9481_RED      'hexidecimal value 0xF800
ILI9481_GREEN    'hexidecimal value 0x0400
ILI9481_BLUE     'hexidecimal value 0x001F
ILI9481_WHITE    'hexidecimal value 0xFFFF
ILI9481_PURPLE   'hexidecimal value 0xF11F
ILI9481_YELLOW   'hexidecimal value 0xFFE0
ILI9481_CYAN     'hexidecimal value 0x07FF
ILI9481_D_GRAY   'hexidecimal value 0x528A
ILI9481_L_GRAY   'hexidecimal value 0x7997
ILI9481_SILVER   'hexidecimal value 0xC618
ILI9481_MAROON   'hexidecimal value 0x8000
ILI9481_OLIVE    'hexidecimal value 0x8400
ILI9481_LIME     'hexidecimal value 0x07E0
ILI9481_AQUA     'hexidecimal value 0x07FF
ILI9481_TEAL     'hexidecimal value 0x0410
ILI9481_NAVY     'hexidecimal value 0x0010
ILI9481_FUCHSIA  'hexidecimal value 0xF81F
```

These examples show how to drive a ILI9481 based Graphic LCD module with the built in commands of

GCBASIC. The 8 bit DBI example uses a UNO shield, this can easily adapted to Microchip architecture. The 16 bit DBI example uses a Mega2560 board.

Examples:

```
'8 bit DBI
#include <glcd.h>
#include <UNO_mega328p.h >

#define GLCD_TYPE GLCD_TYPE_ILI9481

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_RD      ANALOG_0      ' read command line
#define GLCD_WR      ANALOG_1      ' write command line
#define GLCD_RS      ANALOG_2      ' Command/Data line
#define GLCD_CS      ANALOG_3      ' Chip select line
#define GLCD_RST      ANALOG_4      ' Reset line

#define GLCD_DB0      DIGITAL_8
#define GLCD_DB1      DIGITAL_9
#define GLCD_DB2      DIGITAL_2
#define GLCD_DB3      DIGITAL_3
#define GLCD_DB4      DIGITAL_4
#define GLCD_DB5      DIGITAL_5
#define GLCD_DB6      DIGITAL_6
#define GLCD_DB7      DIGITAL_7

GLCDPrint(0, 0, "Test of the ILI9481 Device")
end
```

```
'16 bit DBI
#chip mega2560, 16
#include <glcd.h>

#define GLCD_TYPE GLCD_TYPE_ILI9481
#define GLCD_ILI9481_16bit

#define ILI9481_GLCD_CS PortG.1
#define ILI9481_GLCD_RS PortD.7
```

```

#define ILI9481_GLCD_WR PortG.2
#define ILI9481_GLCD_RST PortG.0
#define ILI9481_DataPortH PortA
#define ILI9481_DataPortL PortC

#define ILI9481_YELLOW1 0xFFC1
#define ILI9481_BlueViolet 0x895C

GLCDCLS_ILI9481 ILI9481_Black
wait 1 s
GLCDCLS_ILI9481 ILI9481_White
wait 1 s

GLCDfntDefaultsize = 3
GLCDBackground = ILI9481_BlueViolet
GLCDForeground = ILI9481_Yellow1
GLCDCLS
wait 1 s

Start:

'demonstrate screen rotation
GLCDRotate (Portrait)
GLCDCLS
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 24, ILI9481_GLCD_HEIGHT/2 - 62, "GCB")
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 120, ILI9481_GLCD_HEIGHT/2 - 24, "ILI9481
Driver")
wait 5 s

GLCDRotate (Landscape)
GLCDCLS
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 24, ILI9481_GLCD_HEIGHT/2 - 62, "GCB")
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 120, ILI9481_GLCD_HEIGHT/2 - 24, "ILI9481
Driver")
wait 5 s

GLCDRotate (Portrait_REV)
GLCDCLS
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 24, ILI9481_GLCD_HEIGHT/2 - 62, "GCB")
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 120, ILI9481_GLCD_HEIGHT/2 - 24, "ILI9481
Driver")
wait 5 s

GLCDRotate (Landscape_REV)
GLCDCLS
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 24, ILI9481_GLCD_HEIGHT/2 - 62, "GCB")
GLCDDrawString ( ILI9481_GLCD_WIDTH/2 - 120, ILI9481_GLCD_HEIGHT/2 - 24, "ILI9481
Driver")

```

```
wait 5 s  
  
goto Start
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9486(L) Controllers

This section covers GLCD devices that use the ILI9486(L) graphics controller.

The ILI9486(L) is a 262kcolor single-chip SoC driver for a-Si TFT liquid crystal display with resolution of 320RGBx480 dots, comprising a 960-channel source driver, a 480-channel gate driver, 345,600bytes GRAM for graphic data of 320RGBx480 dots.

The GCBASIC constants shown below control the configuration of the ILI9486(L) controller. GCBASIC supports 1) SPI using the SPI hardware module, 2) software SPI, 3) UNO shields and 4) an 8bit port bus - this is detailed in the tables below.

GCBASIC supports 65K-color mode operations.

To use the ILI9486(L) driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>  
#DEFINE GLCD_TYPE GLCD_TYPE_ILI9486L
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ILI9486L or GLCD_TYPE_ILI9486	
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DI	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required

Constants	Controls	Options
GLCD_SLK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required

The GCBASIC constants for the communication protocol for the controller are shown in the table below.

Communications Constants	Use	Comments
ILI9486L_HardwareSPI	Specifies that hardware SPI will be used	SPI ports MUST be defined that match the SPI module for each specific microcontroller #define ILI9486L_HardwareSPI
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFast for specific AVR's only.
UNO_8bit_Shield	Specifies that a UNO shield will be used	The shield will use 13 ports. These ports are pre-defined by the shield. These ports must be specified. #define UNO_8bit_Shield
GLCD_DataPort	Specifies that a full 8 port will be used	The microcontroller will use 13 ports. These port is defined as 8 contiguous bits. These control port and the data port must be specified. #define GLCD_DataPort portb

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	480
GLCDDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])

Command	Purpose	Example
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional In LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported
ILI9486L_[color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below. Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

TFT_BLACK	'hexidecimal value 0x0000
TFT_RED	'hexidecimal value 0xF800
TFT_GREEN	'hexidecimal value 0x07E0
TFT_BLUE	'hexidecimal value 0x001F
TFT_WHITE	'hexidecimal value 0xFFFF
TFT_PURPLE	'hexidecimal value 0xF11F
TFT_YELLOW	'hexidecimal value 0xFFE0
TFT_CYAN	'hexidecimal value 0x07FF
TFT_D_GRAY	'hexidecimal value 0x528A
TFT_L_GRAY	'hexidecimal value 0x7997
TFT_SILVER	'hexidecimal value 0xC618
TFT_MAROON	'hexidecimal value 0x8000
TFT_OLIVE	'hexidecimal value 0x8400
TFT_LIME	'hexidecimal value 0x07E0
TFT_AQUA	'hexidecimal value 0x07FF
TFT_TEAL	'hexidecimal value 0x0410
TFT_NAVY	'hexidecimal value 0x0010
TFT_FUCHSIA	'hexidecimal value 0xF81F

For a ILI9486L datasheet, please refer to [Google](#).

This example shows how to drive a ILI9486L based Graphic LCD module with the built in commands of GCBASIC.

Example:

```

#chip mega328p, 16
#option explicit

#include <glcd.h>
#include <UNO_mega328p.h >

#define GLCD_TYPE GLCD_TYPE_ILI9486L

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      DIGITAL_8      ' Data command line
#define GLCD_CS      DIGITAL_10     ' Chip select line
#define GLCD_RST     DIGITAL_9      ' Reset line

#define GLCD_DI      DIGITAL_13     ' Data in | MISO
#define GLCD_DO      DIGITAL_11     ' Data out | MOSI
#define GLCD_SCK     DIGITAL_13     ' Clock Line

#define ILI9486L_HardwareSPI          ' Remove/comment out if you want to use
software SPI.

GLCDPrint(0, 0, "Test of the ILI9486L Device")
end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ILI9488 Controllers

This section covers GLCD devices that use the ILI9488 graphics controller.

ILI9488 is a 262k-color single-chip SoC driver for a-TFT liquid crystal display with resolution of 320 x 240 resolution, 16.7M-color and with internal GRAM .

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ILI9488 controller. The GCBASIC constants for control and data line connections are shown in the table below. Only SPI is available for connectivity:

To use the ILI9488 driver simply include the following in your user code. This will initialise the driver.

SPI mode

'Pin mappings for SPI

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9488

#define GLCD_DC      PORTB.3      ' Data command line
#define GLCD_CS      PORTB.5      ' Chip select line
#define GLCD_RST     PORTB.4      ' Reset line

#define GLCD_DI      PORTB.2      ' Data in | MISO
#define GLCD_DO      PORTB.0      ' Data out | MOSI
#define GLCD_SCK     PORTB.1      ' Clock Line
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ILI9488	
ILI9488_HARDWARE_SPI	Specifies to use the microcontrollers SPI module. For PPS microcontrollers the library assumes PPS for SPI has been configured.	Optional
HWSPIMODE_MASTERFAST	Specifies the speed of the SPI communications.	Optional
GLCD_RST	Specifies the output pin that is connected to Reset IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to Data/Command pin on the GLCD.	Required
GLCD_DI	Specifies the output pin that is connected to Data In (RW or WDR) pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (RD or RDR) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock pin on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320
GLCD_HEIGHT	The height parameter of the GLCD	480
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported

Command	Purpose	Example
ILI9488_TFT_[color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below, butm you can use the generic TFT color scheme. Any color can be defined using a valid hexidecimal word value between 0x0000 to 0xFFFF.

```
ILI9488_TFT_BLACK      //0x000000
ILI9488_TFT_RED        //0xFC0000
ILI9488_TFT_GREEN      //0x00FC00
ILI9488_TFT_BLUE       //0x0000FC
ILI9488_TFT_WHITE      //0xFFFFFF

ILI9488_TFT_CYAN       //0x003F3F
ILI9488_TFT_DARKCYAN   //0x00AFAF
ILI9488_TFT_DARKGREEN  //0x002100
ILI9488_TFT_DARKGREY   //0xAAAAAA
ILI9488_TFT_GREENYELLOW //0x93FC33
ILI9488_TFT_LIGHTGREY  //0xC9C9C9
ILI9488_TFT_MAGENTA    //0xCC00CC
ILI9488_TFT_MAROON     //0x7E007E
ILI9488_TFT_NAVY       //0x00003E
ILI9488_TFT_OLIVE      //0x783E00
ILI9488_TFT_ORANGE     //0xFC2900
ILI9488_TFT_PINK       //0xFC000F
ILI9488_TFT_PURPLE     //0xF01F9E
ILI9488_TFT_YELLOW     //0xFC7E00
```

These examples show how to drive a ILI9488 based Graphic LCD module with the built in commands of GCBASIC.

Examples - PPS Enabled

```
#chip 18F26K83, 64
#option Explicit
```

```

'Generated by PIC PPS Tool for GCBASIC
#startup InitPPS, 85
#DEFINE PPSToolPart 18f26k83

Sub InitPPS
  'Module: UART pin directions
  Dir PORTC.7 Out      ' Make TX1 pin an output
  'Module: UART1
  RC7PPS = 0x0013      'TX1 > RC7

  #IFDEF ILI9488_HardwareSPI
    UNLOCKPPS
    'Module: SPI1
    RB0PPS = 0x001F      'SD01 > RB0
    RB1PPS = 0x001E      'SCK1 > RB1
    SPI1SCKPPS = 0x0009    'RB1 > SCK1 (bi-directional)
    SPI1SDIPPS = 0x000A    'RB2 > SDI1
  #ELSE
    RB0PPS = 0
    RB1PPS = 0
  #ENDIF
End Sub
// Template comment at the end of the config file

#include <glcd.h>
#DEFINE GLCD_TYPE GLCD_TYPE_ILI9488
#DEFINE ILI9488_HARDWARESPI
#DEFINE HWSPIMODE MASTERFAST
#DEFINE GLCD_DC      PORTB.3      ' Data command line
#DEFINE GLCD_CS      PORTB.5      ' Chip select line
#DEFINE GLCD_RST      PORTB.4      ' Reset line

#DEFINE GLCD_DI      PORTB.2      ' Data in | MISO
#DEFINE GLCD_DO      PORTB.0      ' Data out | MOSI
#DEFINE GLCD_SCK      PORTB.1      ' Clock Line

'''*****

'main program start here

// Set the background
#DEFINE DEFAULT_GLCDBACKGROUND TFT_WHITE

GLCDPrint 0, 0, "Test of the ILI9488 Device", TFT_BLACK
end

```

Examples - Legacy non PPS microcontroller

```
#chip 16F1939
#option Explicit

#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ILI9488
#define GLCD_DC      PORTB.3      ' Data command line
#define GLCD_CS      PORTB.5      ' Chip select line
#define GLCD_RST     PORTB.4      ' Reset line

#define GLCD_DI      PORTB.2      ' Data in | MISO
#define GLCD_DO      PORTB.0      ' Data out | MOSI
#define GLCD_SCK     PORTB.1      ' Clock Line

'''*****

'main program start here

// Set the background
#define DEFAULT_GLCDBACKGROUND TFT_WHITE

GLCDPrint 0, 0, "Test of the ILI9488 Device", TFT_BLACK
end
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

Developer Notes

The ILI9488 library implemented uses BRG color scheme which is different from other GLCD libraries.

The ILI9488 library implemented also uses 18bits for color definition where the color scheme is defined as shown below:

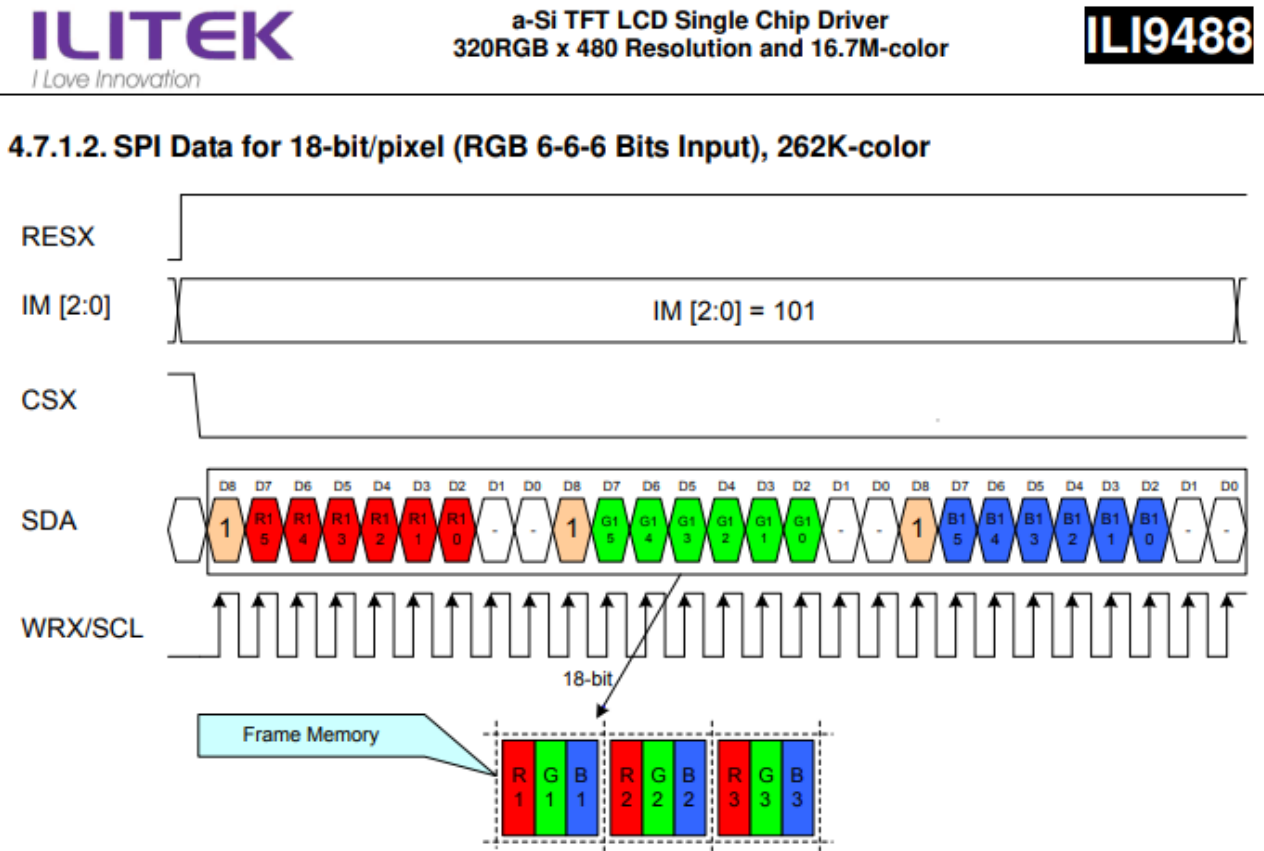


Figure 103: SPI Data for 18-bit/pixel (RGB 6-6-6 Bits Input), 262K-color

Notes:

1. One pixel data contains 18-bit color depth information.
2. The most significant bits are: R x 5, G x 5, and B x 5.
3. The least significant bits are: R x 0, G x 0, and B x 0.
4. '-' = void

The ILI9488 library implemented there has the following differences from a typical GLCD library.

1. The colors are defined as RGB left justified 6 bits.
2. The colors are defined as Longs (not Words other GLCDs are Words).
3. The color information uses a 18bit macro for SPI communications. Color information is sent to the GLCD in three bytes.
4. The color constraints are based on the SPI constraints specified in the ILI9488 datasheet.

KS0108 Controllers

This section covers GLCD devices that use the KS0108 graphics controller.

The KS0108 is an LCD is driven by on-board 5V parallel interface chipset KS0108 and KS0107. They are extremely common and well documented

The GCBASIC constants shown below control the configuration of the KS0108 controller. The only connectivity option is the 8-bit mode where 8 connections (for the data) are required between the microcontroller and the GLCD to control the data bus.

The KS0108 is a monochrome device.

To use the KS0108 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_KS0108

#define GLCD_RW      PORTB.1      'chip specific configuration
#define GLCD_RESET  PORTB.5      'chip specific configuration
#define GLCD_CS1     PORTB.3      'chip specific configuration
#define GLCD_CS2     PORTB.4      'chip specific configuration
#define GLCD_RS      PORTB.0      'chip specific configuration
#define GLCD_ENABLE  PORTB.2      'chip specific configuration
#define GLCD_DB0     PORTC.7      'chip specific configuration
#define GLCD_DB1     PORTC.6      'chip specific configuration
#define GLCD_DB2     PORTC.5      'chip specific configuration
#define GLCD_DB3     PORTC.4      'chip specific configuration
#define GLCD_DB4     PORTC.3      'chip specific configuration
#define GLCD_DB5     PORTC.2      'chip specific configuration
#define GLCD_DB6     PORTC.1      'chip specific configuration
#define GLCD_DB7     PORTC.0      'chip specific configuration
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_KS0108	
GLCD_RS	Specifies the output pin that is connected to Register Select on the GLCD.	Required
GLCD_RW	Specifies the output pin that is connected to Read/Write on the GLCD. The R/W pin can be disabled.	Must be defined (unless R/W is disabled) see GLCD_NO_RW

Constants	Controls	Options
GLCD_CS1	Specifies the output pin that is connected to CS1 on the GLCD.	Required
GLCD_CS2	Specifies the output pin that is connected to CS2 on the GLCD.	Required
GLCD_ENABLE	Specifies the output pin that is connected to Enable on the GLCD.	Required
GLCD_DB0	Specifies the output pin that is connected to DB0 on the GLCD.	Required
GLCD_DB1	Specifies the output pin that is connected to DB1 on the GLCD.	Required
GLCD_DB2	Specifies the output pin that is connected to DB2 on the GLCD.	Required
GLCD_DB3	Specifies the output pin that is connected to DB3 on the GLCD.	Required
GLCD_DB4	Specifies the output pin that is connected to DB4 on the GLCD.	Required
GLCD_DB5	Specifies the output pin that is connected to DB5 on the GLCD.	Required
GLCD_DB6	Specifies the output pin that is connected to DB6 on the GLCD.	Required
GLCD_DB7	Specifies the output pin that is connected to DB7 on the GLCD.	Required
GLCD_NO_RW	Disables read/write inspection of the device during read/write operations	Optional, but recommend NOT to set. The R/W pin can be disabled by setting the GLCD_NO_RW constant. If this is done, there is no need for the R/W to be connected to the chip, and no need for the LCD_RW constant to be set. Ensure that the R/W line on the LCD is connected to ground if not used.
GLCD_DATA_PORT	Not Available for this controller.	Not applicable.

The GCBASIC constants defined for the controller type are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128 This constant cannot be changed

Constants	Controls	Default
GLCD_HEIGHT	The height parameter of the GLCD	64 This constant cannot be changed
GLCDDirection	Defining this will invert the Y Axis	Not defined
KS0108ReadDelay	Read delay	Default is 9 Can be set to improve overall performance.
KS0108WriteDelay	Write delay	Default is 1 Can be set to improve performance.
KS0108ClockDelay	Clock Delay	Default is 1 Can be set to improve performance.

The GCBASIC constants for control display characteristics are shown in the table below.

Variables	Controls	Default
GLCDFontWidth	Width of the current GLCD font.	Default is 6 pixels.
GLCDFntDefault	Size of the current GLCD font.	Default is 0. This equates to the standard GCB font set.
GLCDFntDefaultsize	Size of the current GLCD font.	Default is 1. This equates to the 8 pixel high.

The GCBASIC commands supported for this GLCD are shown in the table below.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])

Command	Purpose	Example
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
GLCDOn	Turn the KS0108 On. The GLCD is On by default	GLCDOn
GLCDOff	Turn the KS0108 Off. The GLCD is On by default	GLCDOff
Private methods		
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte

For a KS0108 datasheet, please refer [here](#).

This example shows how to drive a KS0108 based Graphic LCD module with the built in commands of GCBASIC. See [Graphic LCD](#) for details, this is an external web site.

```

;Chip Settings
#chip 16F886,16
'#config MCLRE = on 'enable reset switch on CHIPINO
#include <GLCD.h>

;Defines (Constants)
#define GLCD_RW PORTB.1 'D9 to pin 5 of LCD
#define GLCD_RESET PORTB.5 'D13 to pin 17 of LCD
#define GLCD_CS1 PORTB.3 'D12 to actually since CS1, CS2 can be reversed on some
devices
#define GLCD_CS2 PORTB.4 'D11 to actually since CS1, CS2 can be reversed on some
devices
#define GLCD_RS PORTB.0 'D8 to pin 4 D/I pin on LCD
#define GLCD_ENABLE PORTB.2 'D10 to Pin 6 on LCD
#define GLCD_DB0 PORTC.7 'D0 to pin 7 on LCD
#define GLCD_DB1 PORTC.6 'D1 to pin 8 on LCD
#define GLCD_DB2 PORTC.5 'D2 to pin 9 on LCD
#define GLCD_DB3 PORTC.4 'D3 to pin 10 on LCD
#define GLCD_DB4 PORTC.3 'D4 to pin 11 on LCD
#define GLCD_DB5 PORTC.2 'D5 to pin 12 on LCD
#define GLCD_DB6 PORTC.1 'D6 to pin 13 on LCD
#define GLCD_DB7 PORTC.0 'D7 to pin 14 on LCD

Do forever
  GLCDCLS
  GLCDPrint 0,10,"Hello" 'Print Hello
  wait 5 s
  GLCDPrint 0,10, "ASCII #:" 'Print ASCII #:
  Box 18,30,28,40 'Draw Box Around ASCII Character
  for char = 15 to 129 'Print 0 through 9
    GLCDPrint 17, 20 , Str(char)+" "
    GLCDDrawCHAR 20,30, char
    wait 125 ms
  next
  line 0,50,127,50 'Draw Line using line command
  for xvar = 0 to 80 'draw line using Pset command
    pset xvar,63,on
  next
  Wait 1 s
  GLCDPrint 0,10,"End " 'Print Hello
  wait 1 s
Loop

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

NEXTION Controllers

This section covers GLCD devices that use the serially attached Nextion graphics displays.

Nextion includes hardware part (a series of TFT boards) and software part (the Nextion editor (<http://nextion.itead.cc/>)).

The Nextion TFT board uses only one serial port to communicate. It lets you avoid the hassle of wiring. Nextion editor has mass components such as button, text, progress bar, slider, instrument panel etc. to enrich your interface design. And, the drag-and-drop function ensures that you spend less time in programming

The Nextion displays are 2.4 to 7.0 inches and range from 320*240 to 800*480 pixels. The connections are 5v, 0v, SerialIn and SerialOut. GCBASIC supports hardware and software serial connectivity.

See GITHUB for the set of GCBASIC demonstrations from the Nextion displays. See [Nextion demonstrations on GITHUB](#).

To use the Nextion driver simply include the following in your user code. This will initialise the driver.

Setup for Hardware Serial

```
' ----- Configuration
  'Chip Settings.
  #chip 16f18855,32
  #option explicit

' ----- Set up the Nextion GLCD
  #include <glcd.h>
  #define GLCD_TYPE GLCD_TYPE_Nextion

  ;VERY IMPORTANT!!
  ;Change the width and height to match the rotation in the Nextion Editor
  #define GLCD_WIDTH 320 'could be 320 | 400 | 272 | 480 but any valid dimension
will work.
  #define GLCD_HEIGHT 240 'could be 240 | 480 | 800 but any valid dimension will work.

  ;VERY IMPORTANT!!
  ;Fonts installed in the Nextion MUST match the fonts parameters loading to the GLCD.
  ;Obtain parameters from Nextion Editor/Font dialog.
```

```

#define NextionFont0      0, 8, 16    'Arial 8x16
#define NextionFont1      1, 12, 24   '24point 12x24 charset
#define NextionFont2      2, 16, 32   '32point 16x32 charset

' ----- End of set up for Nextion GLCD

' ----- Set up for Hardware Serial
;VERY IMPORTANT!!
;The Nextion MUST be setup for 9600 bps.
#define USART_BAUD_RATE 9600
#define USART_BLOCKING

;VERY IMPORTANT!!
;These two are optional. These constants are set in the library.
#define GLCD_NextionSerialPrint HSerPrint
#define GLCD_NextionSerialSend  HSerSend

' ----- End of set up for Serial

'Generated by PIC PPS Tool for GCBASIC
'PPS Tool version: 0.0.5.11
'PinManager data: v1.55
,
'Template comment at the start of the config file
,
#startup InitPPS, 85

Sub InitPPS

    'Module: EUSART
    RXPPS = 0x0016    'RC6 > RX

    'Module: EUSART
    RC0PPS = 0x0010    'TX > RC0
    TXPPS = 0x0010    'RC0 > TX (bi-directional)
    RC5PPS = 0x0010    'TX > RC5
    TXPPS = 0x0015    'RC5 > TX (bi-directional)

End Sub
'Template comment at the end of the config file

' ----- Main program starts
....

```

Setup for Software Serial

```

' ----- Configuration
  'Chip Settings.
  #chip 16f18855,32
  #option explicit

' ----- Set up the Nextion GLCD
  #include <glcd.h>
  #define GLCD_TYPE GLCD_TYPE_Nextion

  ;VERY IMPORTANT!!
  ;Change the width and height to match the rotation in the Nextion Editor
  #define GLCD_WIDTH 320  'could be 320 | 400 | 272 | 480 but any valid dimension
will work.
  #define GLCD_HEIGHT 240 'could be 240 | 480 | 800 but any valid dimension will work.

  ;VERY IMPORTANT!!
  ;Fonts installed in the Nextion MUST match the fonts parameters loading to the GLCD.
  ;Obtain parameters from Nextion Editor/Font dialog.
  #define NextionFont0      0, 8, 16  'Arial 8x16
  #define NextionFont1      1, 12, 24  '24point 12x24 charset
  #define NextionFont2      2, 16, 32  '32point 16x32 charset

' ----- End of set up for Nextion GLCD

' ----- Set up for Software Serial - this is optional - shown to explain the method.
  ;Remove Hardware Serial before using Software serial
  ;You MUST also remove PPS setup, for hardware serial, when using Software serial
  #include <SoftSerial.h>

  ; ----- Config Serial UART for sending:
  #define SER1_BAUD 9600      ; baudrate must be defined
  #define SER1_TXPORT PORTC ; I/O port (without .bit) must be defined
  #define SER1_TXPIN 5        ; portbit must be defined

  ;VERY IMPORTANT!!
  ;These two constants are required to support the the library.
  #define GLCD_NextionSerialPrint  Ser1Print
  #define GLCD_NextionSerialSend   Ser1Send
,
' ----- End of set up for Serial

' ----- Main program starts

```

The GCBASIC constants shown below control the configuration of the Nextion controller. The GCBASIC constants for control and data line connections are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_Nextion	
GLCD_NextionSerialPrint	Default is HSerPrint for hardware serial can be SernPrint when using software serial.	Required
GLCD_NextionSerialSend	Default is HSerSend for hardware serial can be SernSend when using software serial.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	Mandated. The width parameter of the GLCD	320
GLCD_HEIGHT	Mandated. The height parameter of the GLCD	480

The GCBASIC Nextion specific commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDPrint_Nextion	Print string of characters on GLCD using Nextion font set	GLCDPrint(Xposition, Yposition, Stringvariable [,NextionFont])
GLCDLocateString_Nextion	Locate the screen coordinates at a specific location.	GLCDLocateString_Nextion(Xposition, Yposition)
GLCDPrintString_Nextion	Print string of characters on GLCD using Nextion font set	GLCDPrintString_Nextion(Stringvariable)

Command	Purpose	Example
<code>GLCDPrintStringLn_Nextion</code>	Print string of characters on GLCD using Nextion font set adding a newline and carriage return to move cursort to start of next line.	<code>GLCDPrintStringLn_Nextion(Stringvariable)</code>
<code>GLCDSendOpInstruction_Nextion</code>	Send the Nextion display a specific command and a specific value	<code>GLCDSendOpInstruction_Nextion(Nextion_command, command_value)</code>
<code>GLCDUpdateObject_Nextion</code>	Update a Nextion display object with a specific value	<code>GLCDUpdateObject_Nextion(Nextion_object, object_value)</code>
<code>myReturnedWordValue = GLCDGetTouch_Nextion("nextion_command_string")</code>	A function that returns a long, that can be treated as word variable, value of the Touch event.. As follows: "tch0" for current x co-ordinate touched "tch1" for current y co-ordinate touched "tch2" for last x co-ordinate touched "tch3" for last y co-ordinate touched The function is non-blocking. 1. Checks for three bytes of 0xFF. If Four 0xff are received then exit = non-block. 2. If at any time a 0x71 is recieved then we have data for the event. 3. If seven bytes arrive, but the method did not receive a 0x71 then exit = non-block. 4. The method supports software and hardware serial. As does all the other methods. 5. The method uses a function to receive the data not a sub-routine. 6. The method returns 0xBEEF if there is an invalid read, and, functional value for GLCDGetTouch_Nextion will also be set to 0xDEADBEEF	<code>myReturnedWordValue = GLCDGetTouch_Nextion("tch2")</code> or, <code>myReturnedWordValue = GLCDGetTouch_Nextion("tch3")</code>

The GCBASIC commonn commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])

```

TFT_BLACK    'hexidecimal value 0x0000
TFT_RED      'hexidecimal value 0xF800
TFT_GREEN    'hexidecimal value 0x0400
TFT_BLUE     'hexidecimal value 0x001F
TFT_WHITE    'hexidecimal value 0xFFFF
TFT_PURPLE   'hexidecimal value 0xF11F
TFT_YELLOW   'hexidecimal value 0xFFE0
TFT_CYAN     'hexidecimal value 0x07FF
TFT_D_GRAY   'hexidecimal value 0x528A
TFT_L_GRAY   'hexidecimal value 0x7997
TFT_SILVER   'hexidecimal value 0xC618
TFT_MAROON   'hexidecimal value 0x8000
TFT_OLIVE    'hexidecimal value 0x8400
TFT_LIME     'hexidecimal value 0x07E0
TFT_AQUA     'hexidecimal value 0x07FF
TFT_TEAL     'hexidecimal value 0x0410
TFT_NAVY     'hexidecimal value 0x0010
TFT_FUCHSIA  'hexidecimal value 0xF81F

```

For more help, see [GLCDCLS](#)

Supported in <GLCD.H>

NT7108C Controllers

This section covers GLCD devices that use the NT7108C graphics controller.

The NT7108C is an GLCD is driven by on-board 5V parallel interface chipset NT7108C. They are similar to the KS0108.

The GLCD controller is the Winstar WDG0151-TMI module, which is a 128×64 pixel monochromatic display. It uses two Neotic display controller chips: NT7108C and NT7107C, which are similar with Samsung KS0108B and KS0107B controllers. The controller uses a dot matrix LCD segment driver with 64 channel output, and therefore, the WDG0151 module contains two sets of it to drive 128 segments.

The GCBASIC constants shown below control the configuration of the NT7108C controller. The connectivity options are as follows, This is required between the microcontroller and the GLCD to control the data bus.:

- A full port mode. Where a full data port therefore eight contiguous port.bits. The port is used the data communications.
 - Eight port.bits mode. This option allows for greater flexibility with the configuration but will operate slower then the full port mode. These port.bits are used the data communications.
- To use the NT7108C driver simply include the following in your user code. This will initialise the driver.

```
;Full port mode
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_NT7108C

#define GLCD_DATA_PORT PORTD      'Data Port

#define GLCD_CS1 PORTC.1          'CS1 control line
#define GLCD_CS2 PORTC.0          'CS2 control line
#define GLCD_RS PORTe.0           'RS control line
#define GLCD_Enable PORTe.2       'Enable control line
#define GLCD_RW PORTc.3           'RW control line
#define GLCD_RESET PORTC.2        'Reset control line
```

or

```

;Eight port.bits mode
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_NT7108C

;Defines (Constants)
;Define port as 8 port,bit(s)
#define GLCD_DB0 PORTA.2    'Data Port.bit 0
#define GLCD_DB1 PORTC.0    'Data Port.bit 1
#define GLCD_DB2 PORTC.1    'Data Port.bit 2
#define GLCD_DB3 PORTC.2    'Data Port.bit 3
#define GLCD_DB4 PORTB.4    'Data Port.bit 4
#define GLCD_DB5 PORTB.5    'Data Port.bit 5
#define GLCD_DB6 PORTB.6    'Data Port.bit 6
#define GLCD_DB7 PORTB.7    'Data Port.bit 7
;End of define as 8 port,bit(s)

#define GLCD_CS1 PORTC.7    'CS1 control line
#define GLCD_CS2 PORTC.6    'CS2 control line
#define GLCD_RS PORTC.5     'RS control line
#define GLCD_ENABLE PORTA.4 'Enable control line
#define GLCD_RW PORTC.4     'RW control line
#define GLCD_RESET PORTC.3  'Reset control line

```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_NT7108C	
GLCD_RS	Specifies the output pin that is connected to Register Select on the GLCD.	Required
GLCD_RW	Specifies the output pin that is connected to Read/Write on the GLCD.	Required
GLCD_CS1	Specifies the output pin that is connected to CS1 on the GLCD.	Required
GLCD_CS2	Specifies the output pin that is connected to CS2 on the GLCD.	Required
GLCD_ENABLE	Specifies the output pin that is connected to Enable on the GLCD.	Required
Full port mode		
GLCD_DATA_PORT	Specifies the port that is connected to 8 connections on the GLCD.	Required when using full port mode

Constants	Controls	Options
Eight port.bits mode		
GLCD_DB0 GLCD_DB1 .. GLCD_DB7	Specifies the port.bit that is connected to a single connection on the GLCD.	Required when using eight port.bits mode

The GCBASIC constants defined for the controller type are shown in the table below. The NT7108C is very sensitive to clock timings. You may to adjust the clock timing to ensure the display operates correctly.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128 This constant cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	64 This constant cannot be changed
GLCDDirection	Defining this will invert the Y Axis	Not defined
NT7108CReadDelay	Read delay	Default is 7 Can be set to improve overall performance.
NT7108CWriteDelay	Write delay	Default is 7 Can be set to improve performance.
NT7108CClockDelay	Clock Delay	Default is 7 Can be set to improve performance.

The GCBASIC constants for control display characteristics are shown in the table below.

Variables	Controls	Default
GLCDFontWidth	Width of the current GLCD font.	Default is 6 pixels.
GLCDfntDefault	Size of the current GLCD font.	Default is 0. This equates to the standard GCB font set.
GLCDfntDefaultsize	Size of the current GLCD font.	Default is 1. This equates to the 8 pixel high.

The GCBASIC commands supported for this GLCD are shown in the table below.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS

Command	Purpose	Example
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte

For a NT7108C datasheet, please refer [here](#).

This example shows how to drive a NT7108C based Graphic LCD module with the built in commands of GCBASIC. See [Graphic LCD](#) for details, this is an external web site.

```

;Chip Settings
#chip 16F1939,32
#option explicit
#config MCLRE_On

#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_NT7108C           ' Specify the GLCD type
#define GLCDDirection 0                       ' Flip the GLCD  0 do not flip, 1
flip

'Setup the device
#define GLCD_CS1 PORTC.1      'D12 to actually since CS1, CS2 can be reversed on some
devices
#define GLCD_CS2 PORTC.0
#define GLCD_DATA_PORT PORTD
#define GLCD_RS PORTE.0
#define GLCD_Enable PORTE.2
#define GLCD_RW PORTC.3
#define GLCD_RESET PORTC.2

GLCDPrint ( 4, 1, "GCBASIC 2021")                ; Print some text

Box 0, 0, 127, 10
Line 63, 10, 63, 63
Line 0, 37, 127, 37
Circle 63, 37, 15

End

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

PCD8544 Controllers

This section covers GLCD devices that use the PCD844 graphics controller.

The PCD8544 is a low power CMOS LCD controller/driver, designed to drive a graphic display of 48 rows and 84 columns. All necessary functions for the display are provided in a single chip, including on-chip generation of LCD supply and bias voltages, resulting in a minimum of external components and low power consumption. The PCD8544 interfaces to microcontrollers through a serial bus interface.

The GCBASIC constants shown below control the configuration of the PCD844 controller. GCBASIC supports SPI software connectivity only - this is shown in the tables below.

The PCD8544 is a monochrome device.

The PCD844 can operate in two modes. Full GLCD mode and Text/JPG mode the full GLCD mode requires a minimum of 512 bytes. For microcontrollers with limited memory the text only can be selected by setting the correct constant.

To use the PCD844 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_PCD8544

' Pin mappings for software SPI for Nokia 3310 Device
#define GLCD_D0      portc.5      'example port setting
#define GLCD_SCK     portc.3      'example port setting
#define GLCD_DC      portc.2      'example port setting
#define GLCD_CS      portc.1      'example port setting
#define GLCD_RESET   portc.0      'example port setting
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_PCD8544	
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_D0	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_TYPE_PCD8544_CHARACTER_MODE_ONLY	Specifies that the display controller will operate in text mode and BMP draw mode only. For microcontrollers with less than 1kb of RAM this will be set to default.	Optional
PCD8544ClockDelay	Specifies the clock delay, if required for slower microcontroller,	Optional. Set to 0 as the default value
PCD8544WriteDelay	Specifies the write delay, if required for slower microcontroller,	Optional. Set to 0 as the default value

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	160 This cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	128 This cannot be changed
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte

*For a PCD8544 datasheet, please refer [here](#)

This example shows how to drive a PCD8544 based Graphic LCD module with the built in commands of

Example:

```
#chip 16lf1939,32
#option Explicit
#config MCLRE_ON

#include <glcd.h>

#define GLCD_TYPE GLCD_TYPE_PCD8544

' Pin mappings for software SPI for Nokia 3310 Device
#define GLCD_D0 portc.5
#define GLCD_SCK portc.3
#define GLCD_DC portc.2
#define GLCD_CS portc.1
#define GLCD_RESET portc.0

Dim outString as string
Dim ccount, byteNumber as Byte
Dim longNumber as Long
Dim wordNumber as Word
GLCDCLS

DO forever
  for CCount = 31 to 127
    GLCDPrint (0, 0, "PrintStr")
    GLCDDrawString (0, 9, "DrawStr")
    GLCDPrint ( 44 , 21, "      ")
    GLCDPrint ( 44 , 29, "      ") ' word value
    GLCDPrint ( 44 , 37, "      ") ' Byte value

    outstring = hex( longNumber_U)
    GLCDPrint ( 44 , 21,outstring )
    outstring = hex( longNumber_H)
    GLCDPrint ( 55 , 21, outstring)
    outstring = hex( longNumber)
    GLCDPrint ( 67 , 21,  outstring )
    GLCDPrint ( 44 , 29, mid( str(wordNumber),1, 6))
    GLCDPrint ( 44 , 37, byteNumber)

    box 46,9,57,19
    GLCDDrawChar(48, 9, CCount )
    outString = str( CCount )
    ' draw a box to overwrite existing strings
    FilledBox(58,9,GLCD_WIDTH-1,17,GLCDBackground )
```

```

        GLCDDrawString(58, 9, outString )

        box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1
        box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH- 1, GLCD_HEIGHT-1
        filledbox 2,30,6,38, wordNumber
        Circle( 25,30,8,1)           ;center
        FilledCircle( 25,30,4,longNumber xor 1) ;center

        line 0, GLCD_HEIGHT-1 , GLCD_WIDTH/2, (GLCD_HEIGHT /2) +1
        line GLCD_WIDTH/2, (GLCD_HEIGHT /2) +1 ,0, (GLCD_HEIGHT /2) +1

        longNumber = longNumber + 7
        wordNumber = wordNumber + 3
        byteNumber++
    NEXT
LOOP

end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H> and <glcd_PCD8544.h>

SDD1289 Controllers

This section covers GLCD devices that use the SDD1289 graphics controller. The SDD1289 is a 240 x 320 single chip controller driver IC for 262k color (RGB) amorphous TFT LCD.

The GCBASIC constants shown below control the configuration of the SDD1289 controller. GCBASIC supports SPI hardware and software connectivity - this is shown in the tables below.

GCBASIC supports 65K-color mode operations.

To use the SDD1289 driver simply include the following in your user code. This will initialise the driver.

```

#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_SDD1289
'Pin mappings for SDD1289
#define GLCD_DC      porta.0      'example port setting
#define GLCD_CS      porta.1      'example port setting
#define GLCD_RESET   porta.2      'example port setting
#define GLCD_DI      porta.3      'example port setting
#define GLCD_DO      porta.4      'example port setting
#define GLCD_SCK     porta.5      'example port setting

```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_TYPE	GLCD_TYPE_SDD1289	
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DI	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required

The GCBASIC constants for control display characteristics are shown in the table below.

Constant	Purpose	Default
GLCD_WIDTH	The width parameter of the GLCD	Set automatically
GLCD_HEIGHT	The height parameter of the GLCD	Set automatically
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])

Command	Purpose	Example
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
SDD1289_ [color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below. Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

```

SSD1289_BLACK    'hexidecimal value 0x0000
SSD1289_RED      'hexidecimal value 0xF800
SSD1289_GREEN    'hexidecimal value 0x07E0
SSD1289_BLUE     'hexidecimal value 0x001F
SSD1289_WHITE    'hexidecimal value 0xFFFF
SSD1289_PURPLE   'hexidecimal value 0xF11F
SSD1289_YELLOW   'hexidecimal value 0xFFE0
SSD1289_CYAN     'hexidecimal value 0x07FF
SSD1289_D_GRAY   'hexidecimal value 0x528A
SSD1289_L_GRAY   'hexidecimal value 0x7997
SSD1289_SILVER   'hexidecimal value 0xC618
SSD1289_MAROON   'hexidecimal value 0x8000
SSD1289_OLIVE    'hexidecimal value 0x8400
SSD1289_LIME     'hexidecimal value 0x07E0
SSD1289_AQUA     'hexidecimal value 0x07FF
SSD1289_TEAL     'hexidecimal value 0x0410
SSD1289_NAVY     'hexidecimal value 0x0010
SSD1289_FUCHSIA  'hexidecimal value 0xF81F

```

For a SDD1289 datasheet, please refer [here](#).

This example shows how to drive a SDD1289 based Graphic LCD module with the built in commands of GCBASIC.

Example:

```

;Chip Settings
#chip 16F1937,32
#config MCLRE_ON

#include <glcd.h>

'Defines for SDD1289
#define GLCD_TYPE GLCD_TYPE_SDD1289
'Pin mappings for SDD1289
#define GLCD_DC porta.0
#define GLCD_CS porta.1
#define GLCD_RESET porta.2
#define GLCD_DI porta.3
#define GLCD_D0 porta.4
#define GLCD_SCK porta.5

GLCDPrint(0, 0, "Test of the SDD1289 Device")
end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) OR [Pset](#)

Supported in <GLCD.H>

SH1106 Controllers

This section covers GLCD devices that use the SH1106 graphics controller. The SH1106 is a single-chip CMOS OLED/PLED driver with controller for organic/polymer light emitting diode dot-matrix graphic display system. SH1106 consists of 132 segments, 64 commons that can support a maximum display resolution of 132 X 64. It is designed for Common Cathode type OLED panel.

The GCBASIC constants shown below control the configuration of the SH1106 controller. GCBASIC supports i2C hardware and software connectivity - this is shown in the tables below.

The SH1106 is a monochrome device.

To use the SH1106 driver simply include the following in your user code. This will initialise the driver. You can select Full Mode GLCD, Low Memory Mode GLCD or Text mode these require 1024, 128 or 0 byte GLCD buffer respectively - your microcontroller requires sufficient RAM to support the selected mode of GLCD operation.

Specific to the Rajguru Electronics 13002-Series display you can specify a GLCD_SubType 13002 to enable support.

```

#include <glcd.h>

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SH1106
#define GLCD_I2C_Address 0x78
'#define GLCD_TYPE_SH1106_LOWMEMORY_GLCD_MODE      'select Low Memory mode of
operation
'#define GLCD_TYPE_SH1106_CHARACTER_MODE_ONLY      'select Text mode of operation

'#define GLCD_SubType 13002                          'add to support Rajguru
Electronics 13002-Series display

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_SH1106	Required
GLCD_I2C_Address	I2C address of the GLCD.	Required
HI2C_BAUD_RATE	HI2C_BAUD_RATE	400 or 100
HI2C_DATA	HI2C_DATA	Mandated, plus HI2CMode Master is required.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128
GLCD_HEIGHT	The height parameter of the GLCD	64
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6
GLCD_TYPE_SH1106_CHARACTER_MODE_ONLY	Specifies that the display controller will operate in text mode and BMP draw mode only. For microcontrollers with low RAM this will be set be default. When selected ONLY text related commands are supported. For graphical commands you must have sufficient memory to use Full GLCD mode or use GLCD_TYPE_SH1106_LOWMEMORY_GLCD_MODE	Optional

Constants	Controls	Default
GLCD_TYPE_SH1106_LOWMEMORY_GLCD_MODE	Specifies that the display controller will operate in Low Memory mode.	Optional

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCD_Open_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must be followed a GLCD_Close_PageTransaction command.	GLCD_Close_PageTransaction 0, 7 where 0 and 7 are the range of pages to be updated
GLCD_Close_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must follow a GLCD_Open_PageTransaction command.	

The additional GCBASIC commands for this GLCD are shown in the table below.

Command	Purpose
GLCDSetDisplayInvertMode	Inverts the display
GLCDSetDisplayNormalMode	Set the display to normal mode
GLCDSetContrast (dim_state)	Sets the constrast between 0 and 255. The contrast increases as the value increases. Parameter is dim value

For a SH1106 datasheet, please refer [here](#).

This example shows how to drive a SH1106 based Graphic LCD module with the built in commands of GCBASIC.

```
; ----- Configuration
#chip mega328p,16
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SH1106
#define GLCD_I2C_Address 0x78

GLCDCLS
GLCDPrint 0, 0, "GCBASIC"
GLCDPrint (0, 16, "Anobium 2021")

wait 3 s
GLCDCLS

' Prepare the static components of the screen
GLCDPrint ( 0, 0, "PrintStr") ; Print some text
GLCDPrint ( 64, 0, "@")
; Print some more text
GLCDPrint ( 72, 0, ChipMhz) ; Print chip speed
GLCDPrint ( 86, 0, "Mhz") ; Print some text
GLCDDrawString( 0,8,"DrawStr") ; Draw some text
box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
Circle( 44,41,15) ; Draw a circle
line 64,31,0,31 ; Draw a line
```



```

DO forever
  for CCount = 31 to 127
    GLCDPrint ( 64 , 36, hex(longNumber_E ) ) ; Print a HEX string
    GLCDPrint ( 76 , 36, hex(longNumber_U ) ) ; Print a HEX string
    GLCDPrint ( 88 , 36, hex(longNumber_H ) ) ; Print a HEX string
    GLCDPrint ( 100 , 36, hex(longNumber ) ) ; Print a HEX string
    GLCDPrint ( 112 , 36, "h" ) ; Print a HEX string

    GLCDPrint ( 64 , 44, pad(str(wordNumber), 5 ) ) ; Print a padded string
    GLCDPrint ( 64 , 52, pad(str(byteNumber), 3 ) ) ; Print a padded string

    box (46,9,56,19) ; Draw a Box
    GLCDDrawChar(48, 9, CCount ) ; Draw a character
    outString = str( CCount ) ; Prepare a string
    GLCDDrawString(64, 9, pad(outString,3) ) ; Draw a string

    filledbox 3,43,11,51, wordNumber ; Draw a filled box

    FilledCircle( 44,41,9, longNumber xor 1) ; Draw a filled box
    line 0,63,64,31 ; Draw a line

    ; Do some simple maths
    longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
  NEXT
LOOP
end

```

This example shows how to drive a SH1106 based Graphic I2C LCD module with the built in commands of GCBASIC using Low Memory Mode GLCD.

Note the use of `GLCD_Open_PageTransaction` and `GLCD_Close_PageTransaction` to support the Low Memory Mode of operation and the contraining of all GLCD commands with the transaction commands. The use Low Memory Mode GLCD the two defines `GLCD_TYPE_SH1106_LOWMEMORY_GLCD_MODE` and `GLCD_TYPE_SH1106_CHARACTER_MODE_ONLY` are included in the user program.

```

#chip mega328p,16
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings

```

```

#define GLCD_TYPE GLCD_TYPE_SH1106 'for 128 * 64 pixels support
#define GLCD_I2C_Address 0x78
#define GLCD_TYPE_SH1106_LOWMEMORY_GLCD_MODE
#define GLCD_TYPE_SH1106_CHARACTER_MODE_ONLY

dim outString as string * 21

GLCDCLS
GLCD_Open_PageTransaction 0,7
    GLCDPrint 0, 0, "GCBASIC"
    GLCDPrint (0, 16, "Anobium 2021")
GLCD_Close_PageTransaction
wait 3 s
GLCDCLS

DO forever

    for CCount = 31 to 127

        outString = str( CCount ) ; Prepare a string

        GLCD_Open_PageTransaction 0,7

            ' Prepare the static components of the screen
            GLCDPrint ( 0, 0, "PrintStr" ) ; Print some text
            GLCDPrint ( 64, 0, "@" )
            ; Print some more text
            GLCDPrint ( 72, 0, ChipMhz ) ; Print chip speed
            GLCDPrint ( 86, 0, "Mhz" ) ; Print some text
            GLCDDrawString( 0,8,"DrawStr" ) ; Draw some text
            box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            Circle( 44,41,15 ) ; Draw a circle
            line 64,31,0,31 ; Draw a line

            GLCDPrint ( 64 , 36, hex(longNumber_E ) ) ; Print a HEX string
            GLCDPrint ( 76 , 36, hex(longNumber_U ) ) ; Print a HEX string
            GLCDPrint ( 88 , 36, hex(longNumber_H ) ) ; Print a HEX string
            GLCDPrint ( 100 , 36, hex(longNumber ) ) ; Print a HEX string
            GLCDPrint ( 112 , 36, "h" ) ; Print a HEX string

            GLCDPrint ( 64 , 44, pad(str(wordNumber), 5 ) ) ; Print a padded string
            GLCDPrint ( 64 , 52, pad(str(byteNumber), 3 ) ) ; Print a padded string

            box (46,8,56,19) ; Draw a Box
            GLCDDrawChar(48, 9, CCount ) ; Draw a character

            GLCDDrawString(64, 9, pad(outString,3) ) ; Draw a string

```

```

        filledbox 3,43,11,51, wordNumber ; Draw a filled box

        FilledCircle( 44,41,9, longNumber xor 1) ; Draw a filled box
        line 0,63,64,31 ; Draw a line

    GLCD_Close_PageTransaction

; Do some simple maths
    longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
NEXT
LOOP
end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

SSD1306 Controllers

This section covers GLCD devices that use the SSD1306 graphics controller.

The SSD1306 is a single-chip CMOS OLED/PLED driver with controller for organic / polymer light emitting diode dot-matrix graphic display system. It consists of 128 segments and 64 commons. This IC is designed for Common Cathode type OLED panel.

The SSD1306 embeds with contrast control, display RAM and oscillator, which reduces the number of external components and power consumption. It has 256-step brightness control. Data/Commands are sent from general MCU through the hardware selectable 6800/8000 series compatible Parallel Interface, I2C interface or Serial Peripheral Interface. It is suitable for many compact portable applications, such as mobile phone sub-display, MP3 player and calculator, etc.

The GCBASIC constants shown below control the configuration of the SSD1306 controller. GCBASIC supports SPI and I2C hardware & software connectivity - this is shown in the tables below.

To use the SSD1306 driver simply include the following in your user code. This will initialise the driver.

The SSD1306 library supports 128 * 64 pixels or 128 * 32 pixels. The default is 128 * 64 pixels.

The SSD1306 is a monochrome device.

The SSD1306 can operate in three modes. Full GLCD mode, Low Memory GLCD mode or Text/JPG mode the full GLCD mode requires a minimum of 1k bytes or 512 bytes for the 128x64 and the 128x32 devices respectively in Full GLCD mode. For microcontrollers with limited memory the third mode of operation - Text mode. These can be selected by setting the correct constant.

To use the SSD1306 drivers simply include one of the following configuration. You can select Full Mode

GLCD, Low Memory Mode GLCD or Text mode these require 1024, 128 or 0 byte GLCD buffer respectively - you microcontroller requires sufficient RAM to support the selected mode of GLCD operation.

Performance of the SSD1306 has been validated at 16Mhz and 400Hz I2C baud. Using other frequencies should be fully tested.

```
'An I2C configuration
#include <glcd.h>

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306
#define GLCD_I2C_Address 0x78
#define GLCD_TYPE_SH1306_LOWMEMORY_GLCD_MODE      'select Low Memory mode of
operation
#define GLCD_TYPE_SH1306_CHARACTER_MODE_ONLY      'select Text mode of operation

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master
```

or,

```
'An SPI configuration'
#include <glcd.h>

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306

; ----- Define Hardware settings
#define S4Wire_DATA

#define MOSI_SSD1306 PortB.1
#define SCK_SSD1306  PortB.2
#define DC_SSD1306   PortB.3
#define CS_SSD1306   PortB.4
#define RES_SSD1306  PortB.5
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_SSD1306	Required
GLCD_I2C_Address	I2C address of the GLCD.	Required

The GCBASIC constants for SPI/S4Wire control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_SSD1306	Required to support 128 * 64 pixels. Mutually exclusive to GLCD_TYPE_SSD1306_32
GLCD_TYPE	GLCD_TYPE_SSD1306_32	Required to support 128 * 32 pixels. Mutually exclusive to GLCD_TYPE_SSD1306
S4Wire_Data	4 wire SPI Mode	Required
MOSI_SSD1306	Specifies output pin connected to serial data in D1 pin	Must be defined
SCK_SSD1306	Specifies output pin connected to serial clock D0 pin	Must be defined
DC_SSD1306	Specifies output pin connected to data control DC pin	Must be defined
CS_SSD1306	Specifies output pin connected to chip select CS pin	Must be defined
RES_SSD1306	Specifies output pin connected to reset RES pin	Must be defined

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128
GLCD_HEIGHT	The height parameter of the GLCD	64 or 32
GLCD_PROTECTOVERRUN	Define this constant to restrict pixel operations with the pixel limits	Not defined
Rotate GCLD Constants		
GLCDYFLIP	Define this constant to rotate the GLCD display	Not defined
GLCDXFLIP	Define this constant to rotate the GLCD display	Not defined

Constants	Controls	Default
GLCDXYFLIP	Define this constant to rotate the GLCD display	Not defined
Memory Management Constants		
GLCD_TYPE_SSD1306_CHARACTER_MODE_ONLY	Specifies that the display controller will operate in text mode and BMP draw mode only. For microcontrollers with low RAM this will be set be default. When selected ONLY text related commands are supported. For graphical commands you must have sufficient memory to use Full GLCD mode or use GLCD_TYPE_SSD1306_LOWMEMORY_GLCD_MODE	Optional
GLCD_TYPE_SSD1306_LOWMEMORY_GLCD_MODE	Specifies that the display controller will operate in Low Memory mode.	Optional

The GCBASIC variables for control display characteristics are shown in the table below. These variables control the user definable parameters of a specific GLCD.

Variable	Purpose	Type
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDFNTDEFAULTSIZE can be set to 1 or 2 only. GLCDFNTDEFAULTSIZE= 1. A small 8 height pixel font with variable width. GLCDFNTDEFAULTSIZE= 2. A larger 10 width * 16 height pixel font.	Optional
GLCDBACKGROUND	GLCD background state.	A monochrome value. For mono GLCDs the default is White or 0x0001.
GLCDFOREGROUND	Color of GLCD foreground.	A monochrome value. For mono GLCDs the default is non-white or 0x0000.
GLCDFONTWIDTH	Width of the current GLCD font.	Default is 6 pixels.
GLCDFNTDEFAULT	Size of the current GLCD font.	Default is 0. This equates to the standard GCB font set.
GLCDFNTDEFAULTSIZE	Size of the current GLCD font.	Default is 1. This equates to the 8 pixel high.

The GCBASIC commands supported for this GLCD are shown in the table below.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS

Command	Purpose	Example
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCD_Open_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must be followed a GLCD_Close_PageTransaction command.	GLCD_Close_PageTransaction 0, 7 where 0 and 7 are the range of pages to be updated
GLCD_Close_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must follow a GLCD_Open_PageTransaction command.	

The GCBASIC specific commands for this GLCD are shown in the table below.

Command	Purpose
Stopscroll_SSD1306	Stops all scrolling
Startscrollright_SSD1306 (start , stop [,scrollspeed])	Activate a right handed scroll for rows start through stop Hint, the display is 16 rows tall. To scroll the whole display, execute: startscrollright_SSD1306(0x00, 0x0F) Parameters are Start row, End row, optionally Scrollspeed
Startscrollleft_SSD1306 (start , stop [,scrollspeed])	Activate a left handed scroll for rows start through stop Hint, the display is 16 rows tall. To scroll the whole display, execute: startscrollleft_SSD1306(0x00, 0x0F) Parameters are Start row, End row, optionally Scrollspeed

Command	Purpose
<code>Startscrolldiagright_SSD1306 (start , stop [,scrollspeed])</code>	Activate a diagright handed scroll for rows start through stop Hint, the display is 16 rows tall. To scroll the whole display, execute: <code>startscrolldiagright_SSD1306(0x00, 0x0F)</code> Parameters are <code>Start row</code> , <code>End row</code> , optionally <code>Scrollspeed</code>
<code>Startscrolldiagleft_SSD1306 (start , stop [,scrollspeed])</code>	Activate a diagleft handed scroll for rows start through stop Hint, the display is 16 rows tall. To scroll the whole display, execute: <code>startscrolldiagleft_SSD1306(0x00, 0x0F)</code> Parameters are <code>Start row</code> , <code>End row</code> , optionally <code>Scrollspeed</code>
<code>GLCDSetContrast (dim_state)</code>	Sets the constrast between 0 and 255. The contrast increases as the value increases. Parameter is dim value

For a SSD1306 datasheet, please refer [here](#).

This example shows how to drive a SSD1306 based Graphic I2C LCD module with the built in commands of GCBASIC using Full Mode GLCD

```
#chip mega328p,16
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306 'for 128 * 64 pixels support
#define GLCD_I2C_Address 0x78

dim outString as string * 21

GLCDCLS
GLCDPrint 0, 0, "GCBASIC"
GLCDPrint (0, 16, "Anobium 2021")

wait 3 s
GLCDCLS

' Prepare the static components of the screen
GLCDPrint ( 0, 0, "PrintStr" ) ; Print some text
GLCDPrint ( 64, 0, "@" )
```



```

; Print some more text
GLCDPrint ( 72, 0, ChipMhz) ; Print chip speed
GLCDPrint ( 86, 0, "Mhz") ; Print some text
GLCDDrawString( 0,8,"DrawStr") ; Draw some text
box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
Circle( 44,41,15) ; Draw a circle
line 64,31,0,31 ; Draw a line

DO forever
  for CCount = 31 to 127
    GLCDPrint ( 64 , 36, hex(longNumber_E ) ) ; Print a HEX string
    GLCDPrint ( 76 , 36, hex(longNumber_U ) ) ; Print a HEX string
    GLCDPrint ( 88 , 36, hex(longNumber_H ) ) ; Print a HEX string
    GLCDPrint ( 100 , 36, hex(longNumber ) ) ; Print a HEX string
    GLCDPrint ( 112 , 36, "h" ) ; Print a HEX string

    GLCDPrint ( 64 , 44, pad(str(wordNumber), 5 ) ) ; Print a padded string
    GLCDPrint ( 64 , 52, pad(str(byteNumber), 3 ) ) ; Print a padded string

    box (46,9,56,19) ; Draw a Box
    GLCDDrawChar(48, 9, CCount ) ; Draw a character
    outString = str( CCount ) ; Prepare a string
    GLCDDrawString(64, 9, pad(outString,3) ) ; Draw a string

    filledbox 3,43,11,51, wordNumber ; Draw a filled box

    FilledCircle( 44,41,9, longNumber xor 1) ; Draw a filled box
    line 0,63,64,31 ; Draw a line

    ; Do some simple maths
    longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
  NEXT
LOOP
end

```

This example shows how to drive a SSD1306 based Graphic I2C LCD module with the built in commands of GCBASIC using Low Memory Mode GLCD.

Note the use of `GLCD_Open_PageTransaction` and `GLCD_Close_PageTransaction` to support the Low Memory Mode of operation and the contraining of all GLCD commands with the transaction commands. The use Low Memory Mode GLCD the two defines `GLCD_TYPE_SSD1306_LOWMEMORY_GLCD_MODE` and `GLCD_TYPE_SSD1306_CHARACTER_MODE_ONLY` are included in the user program.

```
#chip mega328p,16
```

```

#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306 'for 128 * 64 pixels support
#define GLCD_I2C_Address 0x78
#define GLCD_TYPE_SSD1306_LOWMEMORY_GLCD_MODE
#define GLCD_TYPE_SSD1306_CHARACTER_MODE_ONLY

dim outString as string * 21

GLCDCLS

'To clarify - page updates
'0,7 correspond with the Text Lines from 0 to 7 on a 64 Pixel Display
'In this example Code would be GLCD_Open_PageTransaction 0,1 been enough
'But it is allowed to use GLCD_Open_PageTransaction 0,7 to show the full screen
update
GLCD_Open_PageTransaction 0,7
    GLCDPrint 0, 0, "GCBASIC"
    GLCDPrint (0, 16, "Anobium 2021")
GLCD_Close_PageTransaction
wait 3 s
DO forever

    for CCount = 31 to 127

        outString = str( CCount ) ; Prepare a string

        GLCD_Open_PageTransaction 0,7

            ' Prepare the static components of the screen
            GLCDPrint ( 0, 0, "PrintStr" ) ; Print some text
            GLCDPrint ( 64, 0, "@" )
            ; Print some more text
            GLCDPrint ( 72, 0, ChipMhz ) ; Print chip speed
            GLCDPrint ( 86, 0, "Mhz" ) ; Print some text
            GLCDDrawString( 0,8,"DrawStr" ) ; Draw some text
            box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            Circle( 44,41,15 ) ; Draw a circle
            line 64,31,0,31 ; Draw a line

```

```

GLCDPrint ( 64 , 36, hex(longNumber_E ) ) ; Print a HEX string
GLCDPrint ( 76 , 36, hex(longNumber_U ) ) ; Print a HEX string
GLCDPrint ( 88 , 36, hex(longNumber_H ) ) ; Print a HEX string
GLCDPrint ( 100 , 36, hex(longNumber  ) ) ; Print a HEX string
GLCDPrint ( 112 , 36, "h" ) ; Print a HEX string

GLCDPrint ( 64 , 44, pad(str(wordNumber), 5 ) ) ; Print a padded string
GLCDPrint ( 64 , 52, pad(str(byteNumber), 3 ) ) ; Print a padded string

box (46,8,56,19) ; Draw a Box
GLCDDrawChar(48, 9, CCount ) ; Draw a character

GLCDDrawString(64, 9, pad(outString,3) ) ; Draw a string

filledbox 3,43,11,51, wordNumber ; Draw a filled box

FilledCircle( 44,41,9, longNumber xor 1) ; Draw a filled box
line 0,63,64,31 ; Draw a line

GLCD_Close_PageTransaction

; Do some simple maths
longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
NEXT
LOOP
end

```

This example shows how to drive a SSD1306 based Graphic SPI LCD module with the built in commands of GCBASIC.

```

'Chip model
#chip mega328p, 16
#include <glcd.h>

'Defines for a 7 pin SPI module
'RES pin is pulsed low in glcd_SSD1306.h for proper startup
#define MOSI_SSD1306 PortB.1
#define SCK_SSD1306 PortB.2
#define DC_SSD1306 PortB.3
#define CS_SSD1306 PortB.4
#define RES_SSD1306 PortB.5
; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306 'for 128 * 64 pixels support
#define S4Wire_DATA

dim longnumber as Long
longnumber = 123456
dim wordnumber as word
wordnumber = 62535
dim bytenumber as Byte
bytenumber =255

#define led PortB.0
dir led out

Do
    SET led ON
    wait 1 s
    SET led OFF

    GLCDCLS
    GLCDPrint (30, 0, "Hello World!")
    Circle (18,24,10)
    FilledCircle (48,24,10)
    Box (70,14,90,34)
    FilledBox (106,14,126,34)
    GLCDDrawString (32,35,"Draw String")
    GLCDPrint (0,46,longnumber)
    GLCDPrint (94,46,wordnumber)
    GLCDPrint (52,55,bytenumber)
    Line (0,40,127,63)
    Line (0,63,127,40)
    wait 3 s

Loop

```

This example shows how to drive a SSD1306 based Graphic I2C LCD module with 128 * 32 pixel support.

```
#chip mega328p,16
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_SSD1306_32 'for 128 * 32 pixels support
#define GLCD_I2C_Address 0x78

GLCDCLS
GLCDPrint 0, 0, "GCBASIC"
GLCDPrint (0, 16, "Anobium 2021")
```

This example shows how to drive a SSD1306 with the OLED fonts. Note the use of the `GLCDfntDefaultSize` to select the size of the OLED font in use.

```
#define GLCD_OLED_FONT

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: SSD1306" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

This example shows how to set the SSD1306 OLED the lowest contrast level by using a OLED chip specific command.

```
'Use the GCB command to set the lowest constrast
GLCDSetContrast ( 0 )
'Then, use the Write command to set the output between 0 and 255
Write_Command_SSD1306(SSD1306_SETVCOMDETECT)
Write_Command_SSD1306(15)      ' 0x40 default, to lower the contrast, put 0 for
lowest and 255 for highest.
```

```
GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: SSD1306" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

This example shows how to disable the large OLED Fontset. This disables the font to reduce memory usage.

When the large OLED fontset is disabled every character will be shown as a block character.

```
#define GLCD_OLED_FONT           'The constant is required to support OLED fonts
#define GLCD_Disable_OLED_FONT2 'The constant to disable the large fontset.

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: SSD1306" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

SSD1331 Controllers

This section covers GLCD devices that use the SSD1331 graphics controller. The SSD1331 is a single-chip controller/driver for 262K-color, graphic type OLED-LCD.

The GCBASIC constants shown below control the configuration of the SSD1331 controller. GCBASIC supports SPI, hardware and software SPI, connectivity. This is shown in the tables below.

GCBASIC supports 65K-color mode operations.

To use the SSD1331 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#include <UNO_mega328p.h >

#define GLCD_TYPE GLCD_TYPE_SSD1331

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      portb.0      ' Data command line
#define GLCD_CS      portb.2      ' Chip select line
#define GLCD_RESET   portb.1      ' Reset line
#define GLCD_DO      portb.3      ' Data out | MOSI
#define GLCD_SCK     portb.5      ' Clock Line

#define SSD1331_HardwareSPI      ' remove/comment out if you want to use software
SPI.
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_SSD1331	
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required

Constants	Controls	Options
SSD1331_HardwareSPI	Specifies that hardware SPI will be used	SPI ports MUST be defined that match the SPI module for each specific microcontroller #define SSD1331_HardwareSPI
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFAST for specific AVRs only.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	96 This cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	48 This cannot be changed
GLCDDFontWidth	Specifies the font width of the GCBASIC font set.	6 or 5 for the OLED font set.

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])

Command	Purpose	Example
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1) Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

Colour	RGB
TFT_BLACK	0x0000
TFT_NAVY	0x000F
TFT_DARKGREEN	0x03E0
TFT_DARKCYAN	0x03EF
TFT_MAROON	0x7800
TFT_PURPLE	0x780F
TFT_OLIVE	0x7BE0
TFT_LIGHTGREY	0xC618
TFT_DARKGREY	0x7BEF
TFT_BLUE	0x001F
TFT_GREEN	0x07E0
TFT_CYAN	0x77FF
TFT_RED	0xF800
TFT_MAGENTA	0xF81F
TFT_YELLOW	0xFFE0
TFT_WHITE	0xFFFF
TFT_ORANGE	0xFD20
TFT_GREENYELLOW	0xAFE5
TFT_PINK	0xF81F

Example:

```

#chip mega328p, 16
#option explicit

#include <glcd.h>
#include <UNO_mega328p.h >

#define GLCD_TYPE GLCD_TYPE_SSD1331

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      portb.0      ' Data command line
#define GLCD_CS      portb.2      ' Chip select line
#define GLCD_RESET   portb.1      ' Reset line
#define GLCD_DO      portb.3      ' Data out | MOSI
#define GLCD_SCK      portb.5      ' Clock Line

#define SSD1331_HardwareSPI      ' remove/comment out if you want to use software SPI.

'GLCD selected OLED font set.
#define GLCD_OLED_FONT
GLCDfntDefaultsize = 1

GLCDCLS
GLCDPrintStringLN ("GCBASIC")
GLCDPrintStringLN ("")
GLCDPrintStringLN ("Test of the SSD1331")
GLCDPrintStringLN ("")
GLCDPrintStringLN ("Anobium 2021")
end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

SSD1351 Controllers

This section covers GLCD devices that use the SSD1351 graphics controller. The SSD1351 is a single-chip controller/driver for 262K-color, graphic type OLED-LCD.

The GCBASIC constants shown below control the configuration of the SSD1351 controller. GCBASIC supports SPI, hardware and software SPI, connectivity. This is shown in the tables below.

GCBASIC supports 65K-color mode operations.

To use the SSD1351 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_SSD1351

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      portb.0      ' Data command line
#define GLCD_CS      portb.2      ' Chip select line
#define GLCD_RESET   portb.1      ' Reset line
#define GLCD_DO      portb.3      ' Data out | MOSI
#define GLCD_SCK      portb.5      ' Clock Line

#define SSD1351_HardwareSPI      ' remove/comment out if you want to use software
SPI. If you are using PPS to setup the SPI - ensure that PPS SPI is disabled to use
software SPI.
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_SSD1351	
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DO	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required
SSD1351_HardwareSPI	Specifies that hardware SPI will be used	SPI ports MUST be defined that match the SPI module for each specific microcontroller #define SSD1351_HardwareSPI
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFAST for specific AVRs only.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128 This cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	128 This cannot be changed
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6 or 5 for the OLED font set.

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1) Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

```
SSD1351_BLACK    'hexidecimal value 0x0000
SSD1351_BLUE     'hexidecimal value 0xF800
SSD1351_RED      'hexidecimal value 0x001F
SSD1351_GREEN    'hexidecimal value 0x07E0
SSD1351_CYAN     'hexidecimal value 0xFFE0
SSD1351_MAGENTA  'hexidecimal value 0xF81F
SSD1351_YELLOW   'hexidecimal value 0x07FF
SSD1351_WHITE    'hexidecimal value 0xFFFF
```

Example:

```
#chip mega328p, 16
#option explicit

#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_SSD1351

'Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      portb.0      ' Data command line
#define GLCD_CS      portb.2      ' Chip select line
#define GLCD_RESET   portb.1      ' Reset line
#define GLCD_DO      portb.3      ' Data out | MOSI
#define GLCD_SCK     portb.5      ' Clock Line

#define SSD1351_HardwareSPI      ' remove/comment out if you want to use software SPI.

'GLCD selected OLED font set.
#define GLCD_OLED_FONT
GLCDfntDefaultsize = 1

GLCDCLS
GLCDPrintStringLN ("GCBASIC")
GLCDPrintStringLN ("")
GLCDPrintStringLN ("Test of the SSD1351")
GLCDPrintStringLN ("")
GLCDPrintStringLN ("October 2021")
end
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ST7567 Controllers

This section covers GLCD devices that use the ST7567 graphics controller.

The ST7567 is a single-chip CMOS LCD driver with controller for organic / polymer light emitting diode dot-matrix graphic display system. It consists of 128 segments and 64 commons. This IC is designed for Common Cathode type LCD panel.

ST7567 is a single-chip dot matrix LCD driver which incorporates LCD controller and common/segment drivers. A ST7567 can be connected directly to a microprocessor with I2C or 4-line serial interface (SPI-4). Display data sent from microprocessor is stored in the internal Display Data RAM (DDRAM) of 65x132 bits. The display data bits which are stored in DDRAM are directly related to the pixels of LCD panel. The ST7567 contains 132 segment-outputs, 64 common-outputs and 1 icon-common-output, however the address pixels are 128 * 64. The ST7567 has built-in oscillation circuit and low power consumption power circuit, ST7567 generates LCD driving signal without external clock or power, so that it is possible to make a display system with the fewest components and minimal power consumption.

There are different types of ST75xx GLCDs. The table below shows the different types and the GCBASIC support.

Index	GLCD MPU	Interfaces	Datasheet Ref	Support
1	ST7565	Parallel 8080&6080	Ver 1.0a;Page 12	Not supported
2	ST7565S	Parallel 8080&6080	Ver 0.6b;Page 23	Not supported
3	ST7567	4 Pin SPI;Parallel 8080&6080	Ver1.4b;Page 12	3&4 Pin SPI
4	ST7567S	3&4 Pin SPI;I2C;Parallel 8080&6080	Ver1.4;Page 17	3&4 Pin SPI & I2C
5	ST7576	3&4 Pin SPI;I2C;Parallel 8080&6080	Ver1;Page 18	3&4 Pin SPI & I2C

The ST7567 embeds with contrast control, display RAM and it is suitable for many compact portable applications, such as mobile phone sub-display, MP3 player and calculator, etc.

The GCBASIC constants shown below control the configuration of the ST7567 controller. GCBASIC supports SPI and I2C software connectivity - this is shown in the tables below.

The ST7567 library supports 128 * 64 pixels.

The ST7567 is a monochrome device. The library supports difference bias settings for the different types of LCD. See the constant **ST7567_BIAS** for the options.

The ST7567 can operate in three modes. Full GLCD mode, Low Memory GLCD mode or Text/JPG mode the full GLCD mode requires a minimum of 1k bytes or 512 bytes for the 128x64 respectively in Full GLCD mode. For microcontrollers with limited memory the third mode of operation - Text mode. These can be selected by setting the correct constant.

To use the ST7567 drivers simply include one of the following configuration. You can select Full Mode GLCD, Low Memory Mode GLCD or Text mode these require 1024, 128 or 0 bytes GLCD buffer respectively - you microcontroller requires sufficient RAM to support the selected mode of GLCD

operation.

To use the ST7567 driver simply include the following in your user code. This will initialise the driver.

```
'An I2C configuration
#include <glcd.h>

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_ST7567
#define GLCD_I2C_Address 0x7E
#define ST7567_BIAS      ST7567_SET_BIAS_7      ' ST7567_SET_BIAS_7 or ST7567_SET_BIAS_9

; ----- Define software IC2 settings
#define I2C_MODE MASTER
#define I2C_DATA PORTB.4
#define I2C_CLOCK PORTB.6
#define I2C_DISABLE_INTERRUPTS ON
```

or,

```
'An SPI configuration'
#include <glcd.h>

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_ST7567

; ----- Define Hardware settings
#define S4Wire_DATA

#define MOSI_ST7567 PortB.1
#define SCK_ST7567  PortB.2
#define DC_ST7567   PortB.3
#define CS_ST7567   PortB.4
#define RES_ST7567  PortB.5
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ST7567	Required
GLCD_I2C_Address	I2C address of the GLCD.	Required defaults to 0x7E

The GCBASIC constants for SPI/S4Wire control display characteristics are shown in the table below.

Const ants	Controls	Options
GLCD_T YPE	GLCD_TYPE_ST7567	Required to support 128 * 64 pixels. Mutually exclusive to GLCD_TYPE_ST7567_32
ST7567 _BIAS	Bias ratio of the voltage required to driving the LCD at a fixed duty of 1/65 (see the datasheet)	Defaults to ST7567_SET_BIAS_7. Can be either ST7567_SET_BIAS_7 or ST7567_SET_BIAS_9
S4Wire _Data	4 wire SPI Mode	Required
MOSI_S T7567	Specifies output pin connected to serial data in D1 pin	Must be defined
SCK_ST 7567	Specifies output pin connected to serial clock D0 pin	Must be defined
DC_ST7 567	Specifies output pin connected to data control DC pin	Must be defined
CS_ST7 567	Specifies output pin connected to chip select CS pin	Must be defined
RES_ST 7567	Specifies output pin connected to reset RES pin	Must be defined

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128
GLCD_HEIGHT	The height parameter of the GLCD	64
GLCD_PROTECTOVERRUN	Define this constant to restrict pixel operations with the pixel limits	Not defined
GLCD_TYPE_ST7567_CHARACTER_MODE_ONLY	Specifies that the display controller will operate in text mode and BMP draw mode only. For microcontrollers with low RAM this will be set by default. When selected ONLY text related commands are supported. For graphical commands you must have sufficient memory to use Full GLCD mode or use GLCD_TYPE_ST7567_LOWMEMORY_GLCD_MODE	Optional
GLCD_TYPE_ST7567_LOWMEMORY_GLCD_MODE	Specifies that the display controller will operate in Low Memory mode.	Optional
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDfntDefaultsize can be set to 1 or 2 only. GLCDfntDefaultsize= 1. A small 8 height pixel font with variable width. GLCDfntDefaultsize= 2. A larger 10 width * 16 height pixel font.	Optional

The GCBASIC variables for control display characteristics are shown in the table below. These variables control the user definable parameters of a specific GLCD.

Variable	Purpose	Type
GLCDBackground	GLCD background state.	A monochrome value. For mono GLCDs the default is White or 0x0001.
GLCDForeground	Color of GLCD foreground.	A monochrome value. For mono GLCDs the default is non-white or 0x0000.
GLCDFontWidth	Width of the current GLCD font.	Default is 6 pixels.
GLCDfntDefault	Size of the current GLCD font.	Default is 0. This equates to the standard GCB font set.
GLCDfntDefault size	Size of the current GLCD font.	Default is 1. This equates to the 8 pixel high.

The GCBASIC commands supported for this GLCD are shown in the table below.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte

Command	Purpose	Example
GLCD_Open_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must be followed a GLCD_Close_PageTransaction command.	GLCD_Close_PageTransaction 0, 7 where 0 and 7 are the range of pages to be updated
GLCD_Close_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must follow a GLCD_Open_PageTransaction command.	

The GCBASIC specific commands for this GLCD are shown in the table below.

Command	Purpose
GLCDSetContrast (dim_state)	Sets the constrast between 0 and 255. The contrast increases as the value increases. Parameter is dim value

This example shows how to drive a ST7567 based Graphic I2C LCD module with the built in commands of GCBASIC using Full Mode GLCD

```
#CHIP 18F26Q71
#OPTION Explicit

#startup InitPPS, 85
  #define PPSToolPart 18F26Q71

  Sub InitPPS
    // Ensure PPS is NOT set for Software I2C
    UNLOCKPPS
    RB6PPS = 0
    RB4PPS = 0
  End Sub
  'Template comment at the end of the config file

'' -----PORTA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' IO:    -----
'' -----

'' -----PORTB-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' IO:    ----SCL----SDA-----
'' -----
''
```

```

'' -----PORTC-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' IO:    -----
'' -----

' Define Software I2C settings
  #DEFINE I2C_MODE MASTER
  #DEFINE I2C_DATA PORTB.4
  #DEFINE I2C_CLOCK PORTB.6
  #DEFINE I2C_DISABLE_INTERRUPTS ON

'*****
*****
'Main program commences here.. everything before this is setup for the chip.

  Dim DeviceID As Byte
  Dim DISPLAYNEWLINE As Byte

  #include <glcd.h>
  #DEFINE GLCD_TYPE GLCD_TYPE_ST7567
  #DEFINE GLCDDIRECTION INVERTED

; ----- Define variables
  Dim BYTENUMBER, CCOUNT as Byte

  CCount = 0
  dim longNumber as long
  longNumber = 123456 ' max value = 4294967290
  dim wordNumber as Word
  dim outstring as string
  wordNumber = 0
  byteNumber = 0

; ----- Main program

  GLCDPrint 0, 0, "GCBASIC"
  GLCDPrint (0, 16, "Anobium 2024")
  GLCDPrint (0, 32, "Portability Demo")
  GLCDPrint (0, 48, ChipNameStr )

  wait 3 s
  GLCDCLS

  ' Prepare the static components of the screen
  GLCDPrint ( 2, 2, "PrintStr")
                                     ; Print some
text

```

	GLCDPrint (64, 2, "@")	; Print some
more text	GLCDPrint (72, 2, ChipMhz)	; Print chip
speed	GLCDPrint (86, 2, "Mhz")	; Print some
text	GLCDDrawString(2,10,"DrawStr")	; Draw some
text	box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1	; Draw a box
	box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1	; Draw a box
	Circle(44,41,15)	; Draw a circle
	line 64,31,0,31	; Draw a line
	DO forever	
	for CCount = 32 to 127	
	GLCDPrint (64 , 36, hex(longNumber_E))	; Print a HEX
string	GLCDPrint (76 , 36, hex(longNumber_U))	; Print a HEX
string	GLCDPrint (88 , 36, hex(longNumber_H))	; Print a HEX
string	GLCDPrint (100 , 36, hex(longNumber))	; Print a HEX
string	GLCDPrint (112 , 36, "h")	; Print a HEX
string		
	GLCDPrint (64 , 44, pad(str(wordNumber), 5))	; Print a
padded string	GLCDPrint (64 , 52, pad(str(byteNumber), 3))	; Print a
padded string		
	box (46,9,56,19)	; Draw a Box
	GLCDDrawChar(48, 10, CCount)	; Draw a
character		
	outString = str(CCount)	; Prepare a
string		
	GLCDDrawString(64, 10, pad(outString,3))	; Draw a
string		
	filledbox 3,43,11,51, wordNumber	; Draw a
filled box		
	FilledCircle(44,41,9, longNumber xor 1)	; Draw a
filled box		

```

                                line 0,63,64,31                                ; Draw a line

                                ; Do some
simple maths
                                longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
                                NEXT
                                LOOP
                                end

```

This example shows how to drive a ST7567 based Graphic I2C LCD module with the built in commands of GCBASIC using Low Memory Mode GLCD.

Note the use of `GLCD_Open_PageTransaction` and `GLCD_Close_PageTransaction` to support the Low Memory Mode of operation and the contraining of all GLCD commands with the transaction commands. The use Low Memory Mode GLCD the two defines `GLCD_TYPE_ST7567_LOWMEMORY_GLCD_MODE` and `GLCD_TYPE_ST7567_CHARACTER_MODE_ONLY` are included in the user program.

```

#chip {any valid chip}
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define I2C_MODE MASTER
#define I2C_DATA PORTB.4
#define I2C_CLOCK PORTB.6
#define I2C_DISABLE_INTERRUPTS ON

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_ST7567 'for 128 * 64 pixels support
#define GLCD_I2C_Address 0x7E
#define GLCD_TYPE_ST7567_LOWMEMORY_GLCD_MODE
#define GLCD_TYPE_ST7567_CHARACTER_MODE_ONLY

dim outString as string * 21

GLCDCLS

'To clarify - page udpates
'0,7 correspond with the Text Lines from 0 to 7 on a 64 Pixel Display
'In this example Code would be GLCD_Open_PageTransaction 0,1 been enough
'But it is allowed to use GLCD_Open_PageTransaction 0,7 to show the full screen
update
GLCD_Open_PageTransaction 0,7
GLCDPrint 0, 0, "GCBASIC"
GLCDPrint (0, 16, "Anobium 2024")

```

```

GLCD_Close_PageTransaction
wait 3 s
DO forever

    for CCount = 31 to 127

        outString = str( CCount ) ; Prepare a string

        GLCD_Open_PageTransaction 0,7

            ' Prepare the static components of the screen
            GLCDPrint ( 0, 0, "PrintStr" ) ; Print some text
            GLCDPrint ( 64, 0, "@" )
            ; Print some more text
            GLCDPrint ( 72, 0, ChipMhz ) ; Print chip speed
            GLCDPrint ( 86, 0, "Mhz" ) ; Print some text
            GLCDDrawString( 0,8,"DrawStr" ) ; Draw some text
            box 0,0,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            box GLCD_WIDTH-5, GLCD_HEIGHT-5,GLCD_WIDTH-1, GLCD_HEIGHT-1 ; Draw a box
            Circle( 44,41,15 ) ; Draw a circle
            line 64,31,0,31 ; Draw a line

            GLCDPrint ( 64 , 36, hex(longNumber_E ) ) ; Print a HEX string
            GLCDPrint ( 76 , 36, hex(longNumber_U ) ) ; Print a HEX string
            GLCDPrint ( 88 , 36, hex(longNumber_H ) ) ; Print a HEX string
            GLCDPrint ( 100 , 36, hex(longNumber ) ) ; Print a HEX string
            GLCDPrint ( 112 , 36, "h" ) ; Print a HEX string

            GLCDPrint ( 64 , 44, pad(str(wordNumber), 5 ) ) ; Print a padded string
            GLCDPrint ( 64 , 52, pad(str(byteNumber), 3 ) ) ; Print a padded string

            box (46,8,56,19) ; Draw a Box
            GLCDDrawChar(48, 9, CCount ) ; Draw a character

            GLCDDrawString(64, 9, pad(outString,3) ) ; Draw a string

            filledbox 3,43,11,51, wordNumber ; Draw a filled box

            FilledCircle( 44,41,9, longNumber xor 1 ) ; Draw a filled box
            line 0,63,64,31 ; Draw a line

        GLCD_Close_PageTransaction

        ; Do some simple maths
        longNumber = longNumber + 7 : wordNumber = wordNumber + 3 : byteNumber++
    NEXT
LOOP
end

```

This example shows how to drive a ST7567 based Graphic SPI LCD module with the built in commands of GCBASIC.

```

#chip {any valid chip}
#include <glcd.h>

'Defines for a 7 pin SPI module
'RES pin is pulsed low in glcd_ST7567.h for proper startup
#define MOSI_ST7567 PortB.1
#define SCK_ST7567 PortB.2
#define DC_ST7567 PortB.3
#define CS_ST7567 PortB.4
#define RES_ST7567 PortB.5
; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_ST7567 'for 128 * 64 pixels support
#define S4Wire_DATA

dim longnumber as Long
longnumber = 123456
dim wordnumber as word
wordnumber = 62535
dim bytenumber as Byte
bytenumber =255

#define led PortB.0
dir led out

Do
    SET led ON
    wait 1 s
    SET led OFF

    GLCDCLS
    GLCDPrint (30, 0, "Hello World!")
    Circle (18,24,10)
    FilledCircle (48,24,10)
    Box (70,14,90,34)
    FilledBox (106,14,126,34)
    GLCDDrawString (32,35,"Draw String")
    GLCDPrint (0,46,longnumber)
    GLCDPrint (94,46,wordnumber)
    GLCDPrint (52,55,bytenumber)
    Line (0,40,127,63)
    Line (0,63,127,40)
    wait 3 s

Loop

```


This example shows how to drive a ST7567 with the OLED fonts. Note the use of the `GLCDfntDefaultSize` to select the size of the OLED font in use.

```
#define GLCD_OLED_FONT

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ST7567" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

This example shows how to set the ST7567 OLED the lowest contrast level by using a OLED chip specific command.

```
'Use the GCB command to set the lowest contrast
GLCDSetContrast ( 0 )

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ST7567" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

This example shows how to disable the large OLED Fontset. This disables the font to reduce memory usage.

When the large OLED fontset is disabled every character will be shown as a block character.

```

#define GLCD_OLED_FONT           'The constant is required to support OLED fonts
#define GLCD_Disable_OLED_FONT2 'The constant to disable the large fontset.

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ST7567" )
GLCDPrint ( 0, 34, "Size: 128x64" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ST7735 Controllers

This section covers GLCD devices that use the ST7735 graphics controller. The ST7735 or ST7735R is a single-chip controller/driver for 262K-color, graphic type TFT-LCD.

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ST7735 or ST7735R controller.

GCBASIC supports an 8 bit bus connectivity. The 8 bit must be a single port of consecutive bits - this is shown in the tables below.

To use the ST7735 driver simply include the following in your user code. This will initialise the driver.

```

#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ST7735R
#define ST7735TABCOLOR ST7735_BLACKTAB ; can also be ST7735_GREENTAB or
ST7735_REDTAB or GLCD_TYPE_ST7735R_160_80

'Pin mappings for ST7735
#define GLCD_DC      porta.0      'example port setting
#define GLCD_CS      porta.1      'example port setting
#define GLCD_RESET   porta.2      'example port setting
#define GLCD_DI      porta.3      'example port setting
#define GLCD_DO      porta.4      'example port setting
#define GLCD_SCK     porta.5      'example port setting

```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ST7735 or GLCD_TYPE_ST7735R or GLCD_TYPE_ST7735R_160_80	
ST7735TABCOLOR	Specifies the type of ST7735 chipset. The default is ST7735_BLACKTAB	Options are ST7735_BLACKTAB, ST7735_GREENTAB or ST7735_REDTAB. Each tab is a different ST7735 configuration. If you do not know your type try each constant and test.
GLCD_DATA_PORT	Not Available for this controller.	Not applicable.
GLCD_DC	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_CS	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Required
GLCD_Reset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_DI	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD.	Required
GLCD_D0	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_SCLK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required
ST7735_HardwareSPI	Specifies that hardware SPI will be used	SPI ports MUST be defined that match the SPI module for each specific microcontroller #define ST7735_HardwareSPI
HWSPIMode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFAST for specific AVRs only.
ST7735_XSTART	Specifies the adjustment made to the X axis when writing to the GLCD. This is used to correct any geometry correction required for specific GLCDs.	Optional. Defaults are set for each specific GLCD.
ST7735_YSTART	Specifies the adjustment made to the Y axis when writing to the GLCD. This is used to correct any geometry correction required for specific GLCDs.	Optional. Defaults are set for each specific GLCD.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	160 This cannot be changed
GLCD_HEIGHT	The height parameter of the GLCD	128 This cannot be changed
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
ST7735 [color]	Specify color as a parameter for many GLCD commands	Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF., see http://www.barth-dev.de/online/rgb565-color-picker/ for a wider range of color parameters.

For a ST7735 datasheet, please refer [here](#).

For a ST7735R datasheet, please refer [here](#).

Example:

```
;Chip Settings
#chip 16F1937,32
#config MCLRE_ON

#include <glcd.h>

'Defines for ST7735
GLCD_TYPE GLCD_TYPE_ST7735R
'Pin mappings for ST7735
GLCD_DC porta.0
GLCD_CS porta.1
GLCD_RESET porta.2
GLCD_DI porta.3
GLCD_D0 porta.4
GLCD_SCK porta.5

GLCDPrint(0, 0, "Test of the ST7735 Device")
end
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ST7789 Controllers

This section covers GLCD devices that use the ST7789 graphics controller. The ST7789 is a TFT LCD Single Chip Driver with 240x240 or 320x240 Resolution and 65K colors.

GCBASIC supports 65K-color mode operations.

The GCBASIC constants shown below control the configuration of the ST7789 controller. GCBASIC supports SPI hardware and software connectivity - this is shown in the tables below.

To use the ST7789 driver simply include the following in your user code. This will initialise the driver.

```

#include <glcd.h>
#define GLCD_TYPE      GLCD_TYPE_ST7789_240_240
// #define GLCD_TYPE    GLCD_TYPE_ST7789_320_240

'Pin mappings for ST7789 - these MUST be specified
#define GLCD_DC        porta.0          'example port setting
#define GLCD_RESET     porta.2          'example port setting
#define GLCD_DO        porta.4          'example port setting
#define GLCD_SCK       porta.5          'example port setting

'Optional to use the following - please check the datasheet for the specific GLCD.
#define GLCD_CS        porta.1          'example port setting
#define GLCD_DI        porta.3          'example port setting

```

The GCBASIC constants for the interface to the controller are shown in the table below.

Const ants	Controls	Options
GLCD_T YPE	GLCD_TYPE_ST7789_240_240 or GLCD_TYPE_ST7789_320_240	Select one option to set geometry
GLCD_D C	Specifies the output pin that is connected to Data/Command IO pin on the GLCD.	Required
GLCD_R eset	Specifies the output pin that is connected to Reset pin on the GLCD.	Required
GLCD_D O	Specifies the output pin that is connected to Data Out (GLCD in) pin on the GLCD.	Required
GLCD_S CK	Specifies the output pin that is connected to Clock (CLK) pin on the GLCD.	Required
GLCD_D I	Specifies the output pin that is connected to Data In (GLCD out) pin on the GLCD.	Optional
GLCD_C S	Specifies the output pin that is connected to Chip Select (CS) on the GLCD.	Optional
HWSPIM ode	Specifies the speed of the SPI communications for Hardware SPI only.	Optional defaults to MASTERFAST. Options are MASTERSLOW, MASTER, MASTERFAST, or MASTERULTRAFAST for specific AVRs only.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	320 or 240
GLCD_HEIGHT	The height parameter of the GLCD	240
GLCDFontWidth	Specifies the font width of the GCBASIC font set.	6 for GCB fonts, and 5 for OLED fonts.
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The GLCDfntDefaultsize can be set to 1 or 2 only. GLCDfntDefaultsize= 1. A small 8 height pixel font with variable width. GLCDfntDefaultsize= 2. A larger 10 width * 16 height pixel font.	Optional

The GCBASIC commands supported for this GLCD are shown in the table below. Always review the appropriate library for the latest full set of supported commands.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS [,Optional LineColour]
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode [,Optional LineColour])
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable [,Optional LineColour])
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour]
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2 [,Optional LineColour])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour)
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte
GLCDRotate	Rotate the display	LANDSCAPE, PORTRAIT_REV, LANDSCAPE_REV and PORTRAIT are supported

Comm and	Purpose	Example
ST7789_ [color]	Specify color as a parameter for many GLCD commands	Color constants for this device are shown in the list below.
ReadPixel	Read the pixel color at the specified XY coordination. Returns long variable with Red, Green and Blue encoded in the lower 24 bits.	ReadPixel(Xosition , Yposition) or ReadPixel_ST7789(Xosition , Yposition) Any color can be defined using a valid hexadecimal word value between 0x0000 to 0xFFFF.

```

TFT_BLACK      0x0000
TFT_NAVY       0x000F
TFT_DARKGREEN  0x03E0
TFT_DARKCYAN   0x03EF
TFT_MAROON     0x7800
TFT_PURPLE     0x780F
TFT_OLIVE      0x7BE0
TFT_LIGHTGREY  0xC618
TFT_DARKGREY   0x7BEF
TFT_BLUE       0x001F
TFT_GREEN      0x07E0
TFT_CYAN       0x07FF
TFT_RED        0xF800
TFT_MAGENTA    0xF81F
TFT_YELLOW     0xFFE0
TFT_WHITE      0xFFFF
TFT_ORANGE     0xFD20
TFT_GREENYELLOW 0xAFE5
TFT_PINK       0xF81F

```

This example shows how to drive a ST7789 based Graphic LCD module with the built in commands of GCBASIC.

The library support PIC, AVR and LGT - change to suit your configuration.

Example #1


```

#chip LGT8F328P
#include <LGT8F328P.h>
#option explicit

#include <glcd.h>
#include <glcd_st7789.h>
#define ST7789_HardwareSPI
#define HWSPIMode MASTERULTRAFAST

// Can be either pixels geometry
#define GLCD_TYPE GLCD_TYPE_ST7789_240_240
//#define GLCD_TYPE GLCD_TYPE_ST7789_320_240

//Pin mappings for SPI - this GLCD driver supports Hardware SPI and Software SPI
#define GLCD_DC      DIGITAL_8      ' Data command line
#define GLCD_CS      DIGITAL_10     ' Chip select line
#define GLCD_RESET   DIGITAL_9      ' Reset line
#define GLCD_DI      DIGITAL_12     ' Data in | MISO    - Not used therefore
not really required
#define GLCD_DO      DIGITAL_11     ' Data out | MOSI
#define GLCD_SCK      DIGITAL_13    ' Clock Line

#define GLCD_EXTENDEDFONTSET1
GLCDBackground = TFT_BLACK
GLCDCLS TFT_BLACK

GLCDfntDefaultsize = 2

GLCDRotate Portrait_Rev
GLCDPrint (0,0,"Hello World",TFT_GREEN)

GLCDRotate Portrait
GLCDPrint (0,0,"Hello World",TFT_GREEN)

GLCDROTATE Landscape
GLCDPrint (0,0,"Hello World",TFT_GREEN)

GLCDROTATE Landscape_Rev
GLCDPrint (0,0,"Hello World",TFT_GREEN)

```

Example #2

This example shows how to drive a ST7789 using a PIC with PPS.

```

#chip 16F15376
#option Explicit

#startup InitPPS, 85

Sub InitPPS
    #ifdef ST7789_HardwareSPI

        'This #ifdef is added to enable easy change from hardware SPI (using PPS)
to software PPS that just uses the port assignments shown below.

        SSP1CLKPPS = 0x1    //RC3->MSSP1:SCK1
        RC3PPS = 0x15       //RC3->MSSP1:SCK1
        RC5PPS = 0x16       //RC5->MSSP1:SD01
        SSP1DATPPS = 0x14   //RC4->MSSP1:SDI1

    #endif
End Sub

' ***** Setup the GLCD
*****

#include <glcd.h>
#define GLCD_TYPE          GLCD__TYPE_ST7789_240_240
// #define GLCD_TYPE      GLCD__TYPE_ST7789_320_240

'This is a PPS chip, so, need to make the DO/SDO & SCK match the PPS assignments
#define GLCD_DO          portC.5
#define GLCD_SCK          portC.3

'Additinal pin assignments for GLCD
#define GLCD_DC          portA.4
#define GLCD_RESET      portA.1
'It is optional on the ST7789 to set the GLCD_CS... therefore, here but commented
out
'#define GLCD_CS          porte.0

'Uncomment out the next line... enable or disable the PPS!!!
#define ST7789_HardwareSPI  ' remove/comment out if you want to use software
SPI.0

' ***** DEMO REALLY STARTS HERE
*****

GLCDPrint(0, 0, "Test of the ST7789 Device")
end

```

Example #3

This example shows how to drive a ILI3941 with the OLED fonts. Note the use of the `GLCDfntDefaultSize` to select the size of the OLED font in use.

```
#define GLCD_OLED_FONT           'The constant is required to support OLED fonts

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ST7789" )
GLCDPrint ( 0, 34, "Size: "+ Str(GLCD_WIDTH) +" x 240" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

Example #4

This example shows how to disable the large OLED Fontset. This disables the font to reduce memory usage.

When the extended OLED fontset is disabled every character will be shown as a block character.

```
#define GLCD_OLED_FONT           'The constant is required to support OLED fonts
#define GLCD_Disable_OLED_FONT2 'The constant to disable the extended OLED
fontset.

GLCDfntDefaultSize = 2
GLCDFontWidth = 5
GLCDPrint ( 40, 0, "OLED" )
GLCDPrint ( 0, 18, "Typ: ST7789" )
GLCDPrint ( 0, 34, "Size: "+ Str(GLCD_WIDTH) +" x 240" )

GLCDfntDefaultSize = 1
GLCDPrint(20, 56,"https://goo.gl/gjrxkp")
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ST7920 Controllers

This section covers GLCD devices that use the ST7920 graphics controller.

The GCBASIC constants for control of the connectivity are shown in the table below. The only connectivity option is the 8-bit mode where 8 pins are connected between the microcontroller and the GLCD to control the data bus.

The ST7920 GLCD is graphics and character mixed mode display.

ST7920 LCD controller/driver IC can display alphabets, numbers, Chinese fonts and self-defined characters. It supports 3 kinds of bus interface, namely 8-bit, 4-bit and serial. GCBASIC currently supports 8-bit only. For LCD only operations (text characters only) you can use the GCBASIC LCD routines.

All functions, including display RAM, Character Generation ROM, LCD display drivers and control circuits are all in a one-chip solution. With a minimum system configuration, a Chinese character display system can be easily achieved.

The ST7920 includes character ROM with 8192 16x16 dots Chinese fonts and 126 16x8 dots half-width alphanumerical fonts. It supports 64x256 dots graphic display area for graphic display (GDRAM). Mix-mode display with both character and graphic data is possible. ST7920 has built-in CGRAM and provide 4 sets software programmable 16x16 fonts.

To use the ST7920 driver simply include the following in your user code. This will initialise the driver.

```
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_ST7920

#define GLCD_Enable      PORTA.1      'example port setting
#define GLCD_RS          PORTa.0      'example port setting
#define GLCD_RW          PORTA.2      'example port setting
#define GLCD_RESET       PORTA.3      'example port setting
#define GLCD_DATA_PORT   PORTD        'example port setting
```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_ST7920	Required
GLCD_DATA_PORT	Specifies the output port that is connected between the microcontroller and the GLCD.	Required
GLCD_RS	Specifies the output pin that is connected to Register Select on the GLCD.	Required

Constants	Controls	Options
GLCD_RW	Specifies the output pin that is connected to Read/Write on the GLCD. The R/W pin can be disabled*.	Must be defined unless R/W is disabled), see GLCD_NO_RW
GLCD_RESET	Specifies the output pin that is connected to Reset on the GLCD.	Required
GLCD_ENABLE	Specifies the output pin that is connected to Enable on the GLCD.	Required
GLCD_NO_RW	Disables read/write inspection of the device during read/write operations	Optional, but recommend NOT to set. The R/W pin can be disabled by setting the GLCD_NO_RW constant. If this is done, there is no need for the R/W to be connected to the chip, and no need for the LCD_RW constant to be set. Ensure that the R/W line on the LCD is connected to ground if not used.
Constants that control the timing of the library		
ST7920_READDELAY	Set the time delay between read data transmissions.	Optional, set to 20 us as the default value.
ST7920_WRITEDELAY	Set the time delay between write data transmissions.	Optional, set to 2 us as the default value. ' read delay of 25 is required at 32mhz, this can be reduced to 0 for slower clock speeds #DEFINE ST7920_READDELAY 25 ' write delay of 2 is required at 32mhz. this can be reduced to 1 for slower clock speeds #DEFINE ST7920_WRITEDELAY 2

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	128 cannot be changed.
GLCD_HEIGHT	The height parameter of the GLCD	64 cannot be changed.

The GCBASIC commands supported for this GLCD are shown in the table below. For device specific see the commands with the prefix of ST7920*.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)

Command	Purpose	Example
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
The following methods (calls) are available for expert use.		
GLCDWriteByte	Set a byte value to the controller, see the datasheet for usage.	GLCDWriteByte (LCDByte)
GLCDReadByte	Read a byte value from the controller, see the datasheet for usage.	bytevariable = GLCDReadByte

For a TS7920 datasheet, please refer [here](#).

This example shows how to drive a ST7920 based Graphic LCD module with the built in commands of GCBASIC. See [Graphic LCD](#) for details, this is an external web site.

Example 1:

```

;Chip Settings
#CHIP 16F1937,32
#CONFIG MCLRE_ON

#include <GLCD.H>
#define GLCD_TYPE GLCD_TYPE_ST7920
#define GLCD_IO 8
#define GLCD_WIDTH 128
#define GLCD_HEIGHT 160
' read delay of 25 is required at 32mhz, this can be reduced to 0 for slower clock speeds
#define ST7920READDELAY 25

```

```

' write delay of 2 is required at 32mhz. this can be reduced to 1 for slower clock
speeds
#DEFINE ST7920WRITEDELAY 2

#DEFINE GLCD_RS PORTA.0
#DEFINE GLCD_ENABLE PORTA.1
#DEFINE GLCD_RW PORTA.2
#DEFINE GLCD_RESET PORTA.3
#DEFINE GLCD_DATA_PORT PORTD

ST7920GLCDEnableGraphics
GLCDClearGraphics_ST7920
GLCDPrint 0, 1, "GCBASIC "
wait 1 s

GLCDCLS
GLCDClearGraphics_ST7920

rrun = 0
dim msg1 as string * 16

dim xradius, yordinate , radiusErr, incrementalxradius, orginalxradius,
orginalyordinate as Integer

Do forever
  GLCDCLS
  GLCDClearGraphics_ST7920 ;clear screen
  GLCDDrawString 30,0,"ChipMhz@" ;print string
  GLCDDrawString 78,0, str(ChipMhz) ;print string
  Circle(10,10,10,0) ;upper left
  Circle(117,10,10,0) ;upper right
  Circle(63,31,10,0) ;center
  Circle(63,31,20,0) ;center
  Circle(10,53,10,0) ;lower left
  Circle(117,53,10,0) ;lower right
  GLCDDrawString 30,54,"PIC16F1937" ;print string
  wait 1 s ;wait
  FilledBox( 0,0,128,63) ;create box
  for ypos = 0 to 63 ;draw row by row
    Line 0,ypos,128, 0 ;draw line
  next
  wait 1 s ;wait
  GLCDClearGraphics_ST7920 ;clear
loop

```

Example 2:

```

;Chip Settings
#CHIP 16F1937,32
#CONFIG MCLRE_ON

#include <GLCD.H>
#define GLCD_TYPE GLCD_TYPE_ST7920
#define GLCD_IO 8
#define GLCD_WIDTH 128
#define GLCD_HEIGHT 160

' read delay of 25 is required at 32mhz, this can be reduced to 0 for slower clock
speeds
#define ST7920READDELAY 25
' write delay of 2 is required at 32mhz. this can be reduced to 1 for slower clock
speeds
#define ST7920WRITEDELAY 2

#define GLCD_RS PORTA.0
#define GLCD_ENABLE PORTA.1
#define GLCD_RW PORTA.2
#define GLCD_RESET PORTA.3
#define GLCD_DATA_PORT PORTD

WAIT 1 S
GLCDEnableGraphics_ST7920
GLCDClearGraphics_ST7920
Tile_ST7920 "A"
GLCDPrint 0, 1, "GCBASIC "

GLCDCLS

rrun = 0
dim msg1 as string * 16

do forever

GLCDEnableGraphics_ST7920
GLCDClearGraphics_ST7920
GTile_ST7920 0x55, 0x55
wait 1 s

GLCDClearGraphics_ST7920
Lineh_ST7920(0, 0, GLCD_WIDTH)
Lineh_ST7920(0, GLCD_HEIGHT - 1, GLCD_WIDTH)
Linev_ST7920(0, 0, GLCD_HEIGHT)
Linev_ST7920(GLCD_WIDTH - 1, 0, GLCD_HEIGHT)

```



```
Box 18,30,28,40
```

```
WAIT 2 S
```

```
FilledBox 18,30,28,40
```

```
GLCDClearGraphics_ST7920
```

```
Start:
```

```
GLCDDrawString 0,10,"Hello" 'Print Hello
```

```
wait 1 s
```

```
GLCDDrawString 0,10, "ASCII #:" 'Print ASCII #:
```

```
Box 18,30,28,40 'Draw Box Around ASCII Character
```

```
for char = 0x30 to 0x39 'Print 0 through 9
```

```
    GLCDDrawString 16, 20 , Str(char)+" "
```

```
    GLCDDrawCHAR 20, 30, char
```

```
    wait 250 ms
```

```
next
```

```
line 0,50,127,50 'Draw Line using line command
```

```
for xvar = 0 to 80 'draw line using Pset command
```

```
    pset xvar,63,on
```

```
next
```

```
FilledBox 18,30,28,40 'Draw Box Around ASCII Character
```

```
Wait 1 s
```

```
GLCDClearGraphics_ST7920
```

```
GLCDDrawString 0,10,"End "
```

```
wait 1 s
```

```
GLCDClearGraphics_ST7920
```

```
workingGLCDDrawChar:
```

```
GLCDEnableGraphics_ST7920
```

```
dim gtext as string
```

```
gtext = "ST7920 @QC12864B"
```

```
for xchar = 1 to gtext(0) 'Print 0 through 9
```

```
    xxpos = (1+(xchar*6)-6)
```

```
    GLCDDrawChar xxpos , 0 , gtext(xchar)
```

```
next
```

```
GLCDDrawString 1, 9, "GCBASIC"
```

```
GLCDDrawString 1, 18,"GLCD 128*64"
```

```
GLCDDrawString 1, 27,"Using GLCD.H from GCB"
```

```
GLCDDrawString 1, 37,"Using GLCD.H GCB"
```

```
GLCDDrawString 1, 45,"GLCDDrawChar method"
```

```
GLCDDrawString 1, 54,"Test Routines"
```

```
wait 1 s
```

```
GLCDClearGraphics_ST7920
ST7920GLCDDisableGraphics
GLCDCLS
```

```
msg1 = "Run = " +str(rrun)
rrun++
GLCDPrint 0, 0, "ST7920 @QC12864B"
GLCDPrint 0, 1, "GCBASIC "
GLCDPrint 0, 2, "GLCD 128*64"
GLCDPrint 0, 3, msg1
wait 5 s
GLCDCLS
```

```
' show all chars... takes some time!
ST7920CallBuiltinChar
```

```
wait 1 s
GLCDCLS
```

' See <http://www.khngai.com/chinese/charmap/tblbig.php?page=0>
' and see <https://sourceforge.net/projects/vietunicode/files/hannom/hannom%20v2005/>
for the FONTS!!

```
dim BIG5code as word
```

'ST7920 can display half-width HCGROM fonts, user- defined CGRAM fonts and full 16x16 CGROM fonts. The

'character codes in 0000H~0006H will use user- defined fonts in CGRAM. The character codes in 02H~7FH will use

'half-width alpha numeric fonts. The character code larger than A1H will be treated as 16x16 fonts and will be

'combined with the next byte automatically. The 16x16 BIG5 fonts are stored in A140H~D75FH while the 16x16 GB

'fonts are stored in A1A0H~F7FFH. In short:

'1. To display HCGROM fonts:

'Write 2 bytes of data into DDRAM to display two 8x16 fonts. Each byte represents 1 character.

'The data is among 02H~7FH.

'2. To display CGRAM fonts:

'Write 2 bytes of data into DDRAM to display one 16x16 font.

'Only 0000H, 0002H, 0004H and 0006H are acceptable.

'3. To display CGROM fonts:

'Write 2 bytes of data into DDRAM to display one 16x16 font.

'A140H~D75FH are BIG5 code, A1A0H~F7FFH are GB code.

'To display HCGROM fonts

' Write 2 bytes of data into DDRAM to display two 8x16 fonts. Each byte represents 1 character.

```

' The data is among 02H~7FH.
' The english characters set...

linetest1:

    GLCDEnableGraphics_ST7920

    wait 1 s
    GLCDClearGraphics_ST7920

'lineh test
LineH_ST7920(0, 0, GLCD_WIDTH)
LineH_ST7920(0, GLCD_HEIGHT - 1, GLCD_WIDTH)
LineV_ST7920(0, 0, GLCD_HEIGHT)
LineV_ST7920(GLCD_WIDTH - 1, 0, GLCD_HEIGHT)

' box test
LineH_ST7920(10 ,0 , 118 )
LineH_ST7920(0 ,8 , 128)
LineH_ST7920(16 ,16 , 96 )
LineH_ST7920(10 ,32 , 108 )
LineH_ST7920(0, 16, GLCD_WIDTH)
LineH_ST7920(0, 24, GLCD_WIDTH)
LineH_ST7920(0, 32, GLCD_WIDTH)
LineH_ST7920(0, 40, GLCD_WIDTH)
LineH_ST7920(0, 48, GLCD_WIDTH)
LineH_ST7920(0, 56, GLCD_WIDTH)
LineH_ST7920(0, 63, GLCD_WIDTH)
LineV_ST7920(16, 0, GLCD_HEIGHT)
LineV_ST7920(17, 0, GLCD_HEIGHT)
LineV_ST7920(15, 0, GLCD_HEIGHT)

LineV_ST7920(46, 0, GLCD_HEIGHT)
LineV_ST7920(47, 0, GLCD_HEIGHT)
LineV_ST7920(48, 0, GLCD_HEIGHT)

LineV_ST7920(46, 0, GLCD_HEIGHT)
LineV_ST7920(47, 0, GLCD_HEIGHT)
LineV_ST7920(48, 0, GLCD_HEIGHT)

LineV_ST7920(96, 0, GLCD_HEIGHT)
LineV_ST7920(97, 0, GLCD_HEIGHT)
LineV_ST7920(98, 0, GLCD_HEIGHT)

for HCGROM = 0 to GLCD_WIDTH step 8
    LineV_ST7920(HCGROM, 0, GLCD_HEIGHT)
next

```

GraphicTestPlace:

```
GLCDClearGraphics_ST7920
GraphicTest_ST7920
GLCDClearGraphics_ST7920
```

```
' Test draw a line
for yrowpos = 0 to 63 step 4
  LineH_ST7920(0, yrowpos, GLCD_WIDTH)
next
```

```
GLCDClearGraphics_ST7920
ST7920GLCDDisableGraphics
GLCDCLS
```

```
SetIcon_ST7920( 1, 0x55 )
```

loop

```
sub ST7920CallBuiltinChar
```

```
' 0xA140 ~ 0xA15F
for ii = 0 to 31
```

```
  WriteData_ST7920( 0xA1)
  WriteData_ST7920( 0x40 + ii)
```

```
next
```

```
wait 1 s
```

```
GLCDCLS
```

```
' 0xA140 ~ 0xA15F
for ii = 0 to 31
```

```
  WriteData_ST7920( 0xA1)
  WriteData_ST7920( 0xb0 + ii)
```

```
next
```

```
wait 1 s
```

```
GLCDCLS
```

```
' 0xA140 ~ 0xA15F
for ii = 0 to 31
```

```
  WriteData_ST7920( 0xA4)
  WriteData_ST7920( 0x40 + ii)
```

```
next
wait 1 s
GLCDCLS
end sub
```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

ST7920GLCDClearGraphics

Syntax:

```
ST7920GLCDClearGraphics
```

Explanation:

This command clears the GCLD display.

Example usage:

```
ST7920GLCDClearGraphics 'clear the screen
```

ST7920GLCDDisableGraphics

Syntax:

```
ST7920GLCDDisableGraphics
```

Explanation:

This command sets the GCLD display controller to text mode.

Example usage:

```
ST7920GLCDDisableGraphics 'Set to text mode
```

ST7920GLCDEnableGraphics

Syntax:

```
ST7920GLCDEnableGraphics
```

Explanation:

This command sets the GCLD display controller to text mode.

Example usage:

```
ST7920GLCDEnableGraphics 'Set to text mode
```

ST7920GraphicTest**Syntax:**

```
ST7920GraphicTest
```

Explanation:

This command tests the graphics functionality of the GLCD display.

Example usage:

```
ST7920GraphicTest 'Test the display
```

ST7920LineHs**Syntax:**

```
ST7920LineHs ( Xpos, Ypos, XLength, Style)
```

Explanation:

This command draws a line with a specific style. The style is based on the bits value of the byte passed to the routine.

Example usage:

```
ST7920LineHs ( 0, 31,128 , 0x55) 'will draw a dashed line
```

ST7920Locate**Syntax:**

```
ST7920Locate ( Xpos, Ypos)
```

Explanation:

This command locates the pixel at the specific X and Y location of the text screen. Subsequent printing to the GLCD will place a character to the GLCD controller on the specified row and column. Due to the design of the ST7920 controller (to accomodate Mandarin and Cyrillic), you must place the text on the column according to the numbers above the diagram below. The addressing is handle by the command.

```
|--0--|--1--|--2--|...      ...|--7--|  
+---+---+---+---+---+---+---+  
|H |e |l |l |o |   ...      | <- row 0 (address 0x80)  
+---+---+---+---+---+---+---+  
|T |h |i |s |   |i ...      | <- row 1 (address 0x90)  
+---+---+---+---+---+---+---+  
|' |' |' |' |' |' ...      | <- row 2 (address 0x88)  
+---+---+---+---+---+---+---+  
|- |- |- |- |- |- ...      | <- row 3 (address 0x98)  
+---+---+---+---+---+---+---+
```

Writing 'a' onto the 1st column, and 1st row:

```

|--0--|--1--|--2--|...      ...|--7--|

+--+--+--+--+--+--+-----+

|  |  |  |  |  |  ...      | <- row 0 (address 0x80)

+--+--+--+--+--+--+-----+

|  |  |a|  |  |  ...      | <- row 1 (address 0x90)

+--+--+--+--+--+--+-----+

|  |  |  |  |  |  ...      | <- row 2 (address 0x88)

+--+--+--+--+--+--+-----+

|  |  |  |  |  |  ...      | <- row 3 (address 0x98)

+--+--+--+--+--+--+-----+

```

Example usage:

```
ST7920Locate ( 64, 31) 'the pixel at the mid screen point
```

ST7920Tile

Syntax:

```
ST7920Tile ( word variable )
```

Explanation:

This command tiles the screen with the word value provided.

Example usage:

```
Dim tileValue as word
tileValue = (0x55 * 256 ) + 0x55
ST7920Tile (tileValue) 'tile the screen with a nice cross hatch
```

ST7920cTile

Syntax:


```
ST7920cTile ( word variable )
```

Explanation:

Tiles screen with a Chinese Symbol.

This required 2 bytes of data into DDRAM to display one 16x16 font from memory location A140H~D75FH are BIG5 code, A1A0H~F7FFH are GB code.

Example usage:

```
Dim CTileValue as word  
cTileValue = (0xA140H * 256 ) + 0xA140H  
ST7920Tile (CTileValue) 'tile the screen with a nice cross hatch
```

ST7920gLocate**Syntax:**

```
ST7920gLocate ( Xpos, Ypos)
```

Explanation:

This command locates the pixel at the specific X and Y location of the graphical screen.

Example usage:

```
ST7920gLocate ( 64, 31) 'the pixel at the mid screen point
```

ST7920gTile**Syntax:**

```
ST7920gTile ( byte variable , byte variable)
```

Explanation:

Tile LCD screen with two bytes in Graphic Mode.

Example usage:

```
ST7920gTile (0x55, 0x85) 'tile the screen with an odd cross hatch
```

ST7920lineh

Syntax:

```
ST7920lineh ( Xpos, Ypos, xUnitsStyle, )
```

Explanation:

This command draws a horizontal line with the specific style. The style can be ON or OFF. Default is ON.

This is called by the GLCD common routines.

Example usage:

```
ST7920lineh ( 0, 31,128 , ON) 'will draw a line
```

ST7920linev

Syntax:

```
ST7920lineh ( Xpos, Ypos, xUnitsStyle, )
```

Explanation:

This command draws a vertical line with the specific style. The style can be ON or OFF. Default is ON

This is called by the GLCD common routines.

Example usage:

```
ST7920lineh ( 0, 31,128 , ON) 'will draw a line
```

ST7920GLCDReadByte

Syntax:

```
byte_variable = ST7920GLCDReadByte
```

Explanation:

This function return the word value (16 bits) of the GLCD display for the current XY position.

This is called by the GLCD common routines.

See the data sheet for more information.

Example usage:

```
SET GLCD_RS OFF

ST7920WriteByte( SysCalcPositionY )
ST7920WriteByte( SysCalcPositionX )
' read data
GLCDDataTempWord = ST7920GLCDReadByte
GLCDDataTempWord = ST7920GLCDReadByte
GLCDDataTempWord = (GLCDDataTempWord*256) + ST7920GLCDReadByte
```

ST7920WriteByte

Syntax:

```
ST7920GLCDWriteByte
```

Explanation:

This command write to the appropriate location as specified by the current XY position.

This is called by the GLCD common routines.

See the data sheet for more information.

Example usage:

```
...

SET GLCD_RS OFF

ST7920WriteByte( SysCalcPositionY )
ST7920WriteByte( SysCalcPositionX )
' read data
GLCDDataTempWord = ST7920GLCDReadByte
GLCDDataTempWord = ST7920GLCDReadByte
GLCDDataTempWord = (GLCDDataTempWord*256) + ST7920GLCDReadByte
...
```

ST7920WriteCommand

Syntax:

```
ST7920GWriteCommand ( byte_variable)
```

Explanation:

This command writes a command to the controller.

See the data sheet for more information.

Example usage:

```
...
ST7920WriteCommand(0x36) ' set the graphics mode on
GLCD_TYPE_ST7920_GRAPHICS_MODE = true
...
```

ST7920WriteData

Syntax:

```
ST7920GWriteData ( byte_variable)
```

Explanation:

This command writes data to the controller.

See the data sheet for more information.

Example usage:

```
...
for yy = 0 to ( GLCD_HEIGHT - 1 )
  ST7920gLocate(0, yy)
  for xx = 0 to ( GLCD_COLS -1 )
    ST7920WriteData( 0x55 )
    T7920WriteData( 0x55 )
  next
next
...
```

ST7920gReaddata

Syntax:

```
byte_variable = ST7920gReaddata
```

Explanation:

This function return the word value (16 bits) of the GLCD display for the current XY position.

See the data sheet for more information.

Example usage:

```
...  
' Read a word from the display device.  
word_variable = ST7920GLCDReadData
```

T6963 Controllers

This section covers Graphical Liquid Crystal Display (GLCD) devices that use the Toshiba T6963 graphics controller. The T6963 is a monochrome device that typically is blue or white. The GLCD can be provided in a number of pixels sizes - 240 * 64 or 240 * 128.

The Toshiba T6963 is a very popular LCD controller for use in small graphics modules. It is capable of controlling displays with a resolution up to 240x128. Because of its low power and small outline it is most suitable for mobile applications such as PDAs, MP3 players or mobile measurement equipment.

A number of GLCD modules have this controller built-in these include the SP12N002 & SP14N001. Although this controller is small, it has the capability of displaying and merging text and graphics and it manages all the interfacing signals to the displays Row and Column drivers. The GCBASIC library supports the complex capabilities of the T6963.

The T6963 is an LCD is driven by on-board 5V parallel interface chipset T6963. For the specific operating voltage always verify the operating voltages in the device specific datasheet.

The GCBASIC connectivity option is the 8-bit mode - where 8 connections (for the data) are required between the microcontroller and the GLCD to control the data bus.

To use the T6963 driver simply include the following in your user code. This will initialise the driver.

```
#chip 16f1939,32  
#option explicit
```

```

'*****
*****
'Specify this GLCD - a 240 x 64 pixels display
#include <glcd.h>
#define GLCD_TYPE GLCD_TYPE_T6963_64

'*****
*****
'define the connectivity - the 8bit port
#define GLCD_DATA_PORT PORTD          'Library support contiguous 8-bit port

' or
' #define GLCD_DB0      PORTD.0      'chip specific configuration where the
library supports 8-bit port defined via 8 constants
' #define GLCD_DB1      PORTD.1      'chip specific configuration
' #define GLCD_DB2      PORTD.2      'chip specific configuration
' #define GLCD_DB3      PORTD.3      'chip specific configuration
' #define GLCD_DB4      PORTD.4      'chip specific configuration
' #define GLCD_DB5      PORTD.5      'chip specific configuration
' #define GLCD_DB6      PORTD.6      'chip specific configuration
' #define GLCD_DB7      PORTD.7      'chip specific configuration

#define GLCD_CS          PORTA.7      'Chip Enable (Active Low)
#define GLCD_CD          PORTA.0      'Command or Data control line port
#define GLCD_RD          PORTA.1      'Read control line port
#define GLCD_WR          PORTA.2      'Write control line port
#define GLCD_RESET       PORTA.3      'Reset port
#define GLCD_FS          PORTA.5      'FS port
#define GLCD_FS_SELECT 1              'FS1 Font Select port. 6x8 font: FS1="High
"=1 8x8 font FS1="Low"=0 for GLCD_FS_SELECT

'*****
*****
' *
' * Note    :   The T6963 controller's RAM address space from $0000 - $7FFF, total
32kbyte RAM, or it could be 64kbyte RAM best check!!
' *

'*****
*****
#define TEXT_HOME_ADDR    0x0000
'This is specific to the GLCD display
#define GRH_HOME_ADDR     0x3FFF
'This is specific to the GLCD display

```

```

#define CG_HOME_ADDR      0x77FF
'This is specific to the GLCD display
#define COLUMN            40    'Set column number to be 40 , 32, 30 etc.
This is specific to the GLCD display
#define MAX_ROW_PIXEL     64    'MAX_ROW_PIXEL the physical matrix length (y
direction) This is specific to the GLCD display
#define MAX_COL_PIXEL     240   'MAX_COL_PIXEL the physical matrix width (x
direction) This is specific to the GLCD display

'*****
*****
' * End of configuration

'*****
*****

```

The GCBASIC constants for the interface to the controller are shown in the table below.

Constant	Controls	Options
GLCD_TYPE	GLCD_TYPE_TYPE_T6963_64 or GLCD_TYPE_T6963_128	Required
GLCD_DATA_PORT	A full 8-bit port. 8 contiguous input/outputs.	or use GLCD_DB0..GLCD_DB7
GLCD_DB0..7	A 8-bit port using 8 input/outputs.	or use GLCD_DATA_PORT
GLCD_CS	Specifies the output pin that is connected to Chip Select on the GLCD.	Required
GLCD_CD	Specifies the output pin that is connected to Command/Data on the GLCD.	Required
GLCD_RD	Specifies the output pin that is connected to Read on the GLCD.	Required
GLCD_WR	Specifies the output pin that is connected to Write on the GLCD.	Required
GLCD_RESET	Specifies the output pin that is connected to Reset on the GLCD.	Required
GLCD_FS	Specifies the output pin that is connected to FS on the GLCD. The FS specifies the font size. Please set to 6 setting GLCD_FS_SELECT = 1	Required

Constants	Controls	Options
GLCD_FS_SELECT	Specifies the output pin that is connected to FS on the GLCD. The FS specifies the font size. Please set to 6 setting GLCD_FS_SELECT = 1	Required. Can be 1 or 0. Default setting is 1.

The T6963 differs from most other GLCD controllers in its use of the display RAM. Where a fixed area of memory is normally allocated for text, graphics and the external character generator, but with the T6963 the size for each area **MUST** be set by software commands. This means that the area for text, graphics and external character generator can be freely allocated within the external memory, up to 64 kByte. Check the specific device for the amount of memory available. This can range from 4 kbyte to 64 kbyte.

For more information on memory management refer the device specific datasheet.

The GCBASIC constants control the memory configuration of the T6963 controller.

Constants	Value	Comments
TEXT_HOME_ADDR	0x0000	This is specific to the GLCD display
GRH_HOME_ADDR	0x3FFF	This is specific to the GLCD display
CG_HOME_ADDR	0x77FF	This is specific to the GLCD display
COLUMN	40	Set column number to be 40 , 32, 30 etc. This is specific to the GLCD display
MAX_ROW_PIXEL	64	MAX_ROW_PIXEL the physical matrix length (y direction) This is specific to the GLCD display
MAX_COL_PIXEL	240	MAX_COL_PIXEL the physical matrix width (x direction) This is specific to the GLCD display

The GCBASIC library supports the following capabilities. Please refer to the relevant Help section and the device specific demonstrations.

Commands Supported for the LCD and GLCD

The GLCD command set covers the standard GLCDCLS, Line, Circle and all the GLCD methods and the LCD command set: CLS, Locate, Print, LCDHEX etc. The demonstrations show how to load BMP loading via external data sources and GLCD and LCD page swapping.

The table below shows the specific implementations of the command set for this device. Refer to the GLCD and LCD in the Help for the generic GLCD and LCD commands.

Commands	Usage
CLS	Clear the screen of the current LCD page
LOCATE	Locate the cursor at a specific screen position
PRINT	Print numbers or strings
PUT	Put a specific ASCII code at a specific screen position
LCDHOME	Set output position of 0, 0
LCDcmd	Send specific command to the device to control the device.
LCDdata	Send specific data to the device to control the device.
LCDHex	Print Hex value of a number to the LCD screen
LCDSpace	Print a number of space to the LCD screen
LCDCursor	Send specific commands to the device to control the cursor
GLCDCLS	Clear the screen of the current GLCD page
GLCDRotate	Rotate the GLCD screen. Only Landscape rotation is supported.
SelectGLCDPage_T6963	Select a specific GLCD page.
SelectLCDPage_T6963	Select a specific LCD page.

GLCD and LCD page swapping

To support GLCD and LCD page swapping - this can be used to support fixed pages of information, BMPs or scrolling the following constants have are available to the user.

For GLCD memory addressing

```
GLCDPage0_T6963
GLCDPage1_T6963
GLCDPage2_T6963
... etc
GLCDPage10_T6963
```

Ten pages are automatically created but the number of pages available is constrained by the memory configuration.

For LCD memory addressing

```
LCDPage0_T6963
LCDPage1_T6963
LCDPage2_T6963
...etc
LCDPage10_T6963
```

Ten pages are automatically created but the number of pages available is constrained by the memory configuration.

To use add the following to you user program. See the demonstration programs for more detailed usage. After calling the `SelectGLCDPage` or `SelectLCDPage` methods all GLCD or LCD commands will be applied to the current GLCD or LCD page.

```
'Select the GLCD page 1 memory
SelectGLCDPage ( GLCDPage1_T6963 )

'Select the LCD page 2 memory
SelectLCDPage ( LCDPage2_T6963 )
```

The `SelectLCDPage` and `SelectLCDPage` and "Set Text Home Address" methods change the screen being viewed on the device.

The key is to establish what you want your memory map to look like. Below is a map for one of my 240 x 64 pixel device. The default is for 10 screen pages (some newer LCD's may have more RAM for more screens). If you write the appropriate value (0x1000, or 0x11b0, or 0x1360, etc) to the text home address, the display will instantly change to that screen - using `SelectLCDPage` and `SelectLCDPage` method with the appropriate constant as parameter.

You can write your screens "ahead of time", in my case during the "splash screen" delay interval, and

instantly change to them later as desired. You can do this by setting **current_grh_home_addr** to the appropriate page. And, then execute the GLCD commands you would normal use.

The graphic and text screens are independant but can be overlaid for a variety of useful effects.

Although, not tested, the LCD text screens can be scrolled 1 full text line at a time, while the GLCD screens can be scrolled 1 pixel row at a time, provided you've set up your memory map accordingly with adequate RAM for the graphic area.

Default Memory Map

```

'
' *****
'
' LCD MEMORY MAP
'
' *****
'
'
'
' -----
' |                                | 0x0000
' | TEXT RAM AREA                | Each page has the numnbers of bytes + extra
' | ( 10 SCREENs )              | few bytes need to attributes. This is
' |                              | mentioned in the datasheet but imperical
' |                              | testing shows... you need the extra bytes
' |                              |
' |-----|
' | xx bytes unused              |
' |                              |
' |-----|
' |                                | 0x3fff
' |                                |
' | GCLD RAM AREA                |
' | ( 10 SCREENS )              |
' |                              |
' |                              |
' |                              |
' |-----|
' |                                | 0x77ff
' | CG RAM AREA                  | (Sacrosanct)
' |                              |
' |-----| 0x7ffff
'
'

```

Other methods and constants

There are many other methods and constants that support this device. Reviewing the library will assist in understanding how these private methods and constants support the overall solution for this library.

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#) or [Pset](#)

Supported in <GLCD.H>

UC1601 Controllers

This section covers GLCD devices that use the UC1601 graphics controller.

The UC1601 is an advanced high-voltage mixed signal CMOS IC, especially designed for the display needs of ultra-low power hand-held devices.

The UC1601 embeds with contrast control, display RAM and oscillator, which reduces the number of external components and power consumption. It has 256-step brightness control. Data/Commands are sent from general MCU through the hardware selectable 6800/8000 series compatible Parallel Interface, I2C interface or Serial Peripheral Interface. It is suitable for many compact portable applications, such as mobile phone sub-display, MP3 player and calculator, etc.

The UC1601 library supports 132 * 22 pixels. The UC1601 library supports monochrome devices.

[graphic]

The UC1601 can operate in three modes. Full GLCD mode, Low Memory GLCD mode or Text/JPG mode the full GLCD mode requires a minimum of 396 bytes or 128 bytes for the respective modes. For microcontrollers with limited memory the third mode of operation - Text mode. These can be selected by setting the correct constant.

To use the UC1601 driver simply include the following in your user code. This will initialise the driver.

The GCBASIC constants shown below control the configuration of the UC1601 controller. GCBASIC supports hardware I2C & software I2C connectivity - this is shown in the tables below.

To use the UC1601 drivers simply include one of the following configuration.

```
'An I2C configuration
#include <glcd.h>

#define GLCD_TYPE GLCD_TYPE_UC1601
#define GLCD_I2C_Address      0x70           'I2C address
#define GLCD_RESET            portc.0        'Hard Reset pin connection
#define GLCD_PROTECTOVERRUN
#define GLCD_OLED_FONT

; ----- Define Hardware settings for I2C
' Define I2C settings - CHANGE PORTS
#define I2C_MODE Master
#define I2C_DATA PORTb.5
#define I2C_CLOCK PORTb.7
#define I2C_DISABLE_INTERRUPTS ON
```

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Options
GLCD_TYPE	GLCD_TYPE_UC1601	Required
GLCD_I2C_Address	I2C address of the GLCD.	Fixed at 0x70.

The GCBASIC constants for control display characteristics are shown in the table below.

Constants	Controls	Default
GLCD_WIDTH	The width parameter of the GLCD	132
GLCD_HEIGHT	The height parameter of the GLCD	22
GLCD_PROTECTOVERRUN	Define this constant to restrict pixel operations with the pixel limits	Recommended
GLCD_TYPE_UC1601_CHARACTER_MODE_ONLY	Specifies that the display controller will operate in text mode and BMP draw mode only. For microcontrollers with low RAM this will be set by default. When selected ONLY text related commands are supported. For graphical commands you must have sufficient memory to use Full GLCD mode or use <code>GLCD_TYPE_UC1601_LOWMEMORY_GLCD_MODE</code>	Optional
GLCD_TYPE_UC1601_LOWMEMORY_GLCD_MODE	Specifies that the display controller will operate in Low Memory mode.	Optional
GLCD_OLED_FONT	Specifies the use of the optional OLED font set. The <code>GLCDfontDefaultsize</code> can be set to 1 or 2 only. <code>GLCDfontDefaultsize= 1</code> . A small 8 height pixel font with variable width. <code>GLCDfontDefaultsize= 2</code> . A larger 10 width * 16 height pixel font.	Optional

The GCBASIC variables for control display characteristics are shown in the table below. These variables control the user definable parameters of a specific GLCD.

Variable	Purpose	Type
GLCDBackground	GLCD background state.	A monochrome value. For mono GLCDs the default is White or 0x0001.
GLCDForeground	Color of GLCD foreground.	A monochrome value. For mono GLCDs the default is non-white or 0x0000.
GLCDFontWidth	Width of the current GLCD font.	Default is 6 pixels.
GLCDfontDefault	Size of the current GLCD font.	Default is 0. This equates to the standard GCB font set.

Variable	Purpose	Type
GLCDfntDefault size	Size of the current GLCD font.	Default is 1. This equates to the 8 pixel high.

The GCBASIC commands supported for this GLCD are shown in the table below.

Command	Purpose	Example
GLCDCLS	Clear screen of GLCD	GLCDCLS
GLCDPrint	Print string of characters on GLCD using GCB font set	GLCDPrint(Xposition, Yposition, Stringvariable)
GLCDDrawChar	Print character on GLCD using GCB font set	GLCDDrawChar(Xposition, Yposition, CharCode)
GLCDDrawString	Print characters on GLCD using GCB font set	GLCDDrawString(Xposition, Yposition, Stringvariable)
Box	Draw a box on the GLCD to a specific size	Box (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour as 0 or 1])
FilledBox	Draw a box on the GLCD to a specific size that is filled with the foreground colour.	FilledBox (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
Line	Draw a line on the GLCD to a specific length that is filled with the specific attribute.	Line (Xposition1, Yposition1, Xposition2, Yposition2, [Optional In LineColour 0 or 1])
PSet	Set a pixel on the GLCD at a specific position that is set with the specific attribute.	PSet(Xposition, Yposition, Pixel Colour 0 or 1)
GLCD_Open_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must be followed a GLCD_Close_PageTransaction command.	GLCD_Close_PageTransaction 0, 3 where 0 and 3 are the range of pages to be updated
GLCD_Close_PageTransaction	Commence a series of GLCD commands when in low memory mode. Must follow a GLCD_Open_PageTransaction command.	
Open_Transaction_UC1601	Send command instruction to GLCD. Handles I2C and SPI protocols.	Transaction must be closed by using Close_Transaction_UC1601
Open_Transaction_Data_UC1601	Send data instruction to GLCD. Handles I2C and SPI protocols.	Transaction must be closed by using Close_Transaction_UC1601
Write_Transaction_Data_UC1601	Send transactional, a stream of, data to GLCD.	Transaction must be opened and closed by using transaction commands.
Close_Transaction_UC1601	Close the communications to the GLCD.	Transaction must be opened by using Open_Transaction_UC1601 or Open_Transaction_Data_UC1601

The GCBASIC specific commands for this GLCD are shown in the table below.

Command	Purpose
Stopscroll_UC1601	Stops all scrolling
Startscroll_UC1601 (start)	Activates a vertical scroll for rows start.
GLCDSetContrast (contrast_state)	Sets the constrast between 0 and 255. The contrast increases as the value increases. Parameter is contrast value

For a UC1601 datasheet, please refer [here](#).

This example shows how to drive a UC1601 based Graphic I2C LCD module with the built in commands of GCBASIC using Full Mode GLCD


```

; ----- Configuration
#chip 16f18446, 32
#option explicit

; ----- Define GLCD Hardware settings
#include <glcd.h>

#define GLCD_TYPE GLCD_TYPE_UC1601
#define GLCD_I2C_Address      0x70           'I2C address
#define GLCD_RESET           portc.0        'Hard Reset pin connection
#define GLCD_PROTECTOVERRUN
#define GLCD_OLED_FONT

; ----- Define Hardware settings

' Define I2C settings - CHANGE PORTS
#define I2C_MODE Master
#define I2C_DATA PORTb.5
#define I2C_CLOCK PORTb.7
#define I2C_DISABLE_INTERRUPTS ON

; ----- Define variables

; ----- Main program

'You can treat the GLCD like an LCD....
GLCDPrintStringLN "User the GLCD like an LCD...."
GLCDPrintStringLN "The GLCDPrintString commands...."
GLCDPrintString "Enjoy....."
wait 4 s

end

```

This example shows how to drive a UC1601 based Graphic I2C LCD module with the built in commands of GCBASIC using Low Memory Mode GLCD.

Note the use of **GLCD_Open_PageTransaction** and **GLCD_Close_PageTransaction** to support the Low Memory Mode of operation and the contraining of all GLCD commands with the transaction commands. The use Low Memory Mode GLCD the two defines **GLCD_TYPE_UC1601_LOWMEMORY_GLCD_MODE** and **GLCD_TYPE_UC1601_CHARACTER_MODE_ONLY** are included in the user program.

```

#chip mega328p,16
#include <glcd.h>

; ----- Define Hardware settings
' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA
HI2CMode Master

; ----- Define GLCD Hardware settings
#define GLCD_TYPE GLCD_TYPE_UC1601
#define GLCD_TYPE_UC1601_LOWMEMORY_GLCD_MODE
#define GLCD_TYPE_UC1601_CHARACTER_MODE_ONLY

dim outString as string * 21

GLCDCLS

'To clarify - page updates
'0,7 correspond with the Text Lines from 0 to 3 on a 22 Pixel Display
'In this example Code would be GLCD_Open_PageTransaction 0,1 been enough
'But it is allowed to use GLCD_Open_PageTransaction 0,3 to show the full screen
update
  GLCD_Open_PageTransaction 0,3
  GLCDPrint 0, 0, "GCBASIC"
  GLCDPrint (0, 16, "Anobium 2021")
GLCD_Close_PageTransaction

end

```

For more help, see [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#) or [Pset](#)

Supported in <GLCD.H>

Box

Syntax:

```
Box(LineX1,LineY1, LineX2, LineY2 [, LineColour ] )
```

Explanation:

Draws a box on a graphic LCD from the upper corner of pixel position X1, Y1 location to pixel position

X2,Y2 location.

LineColour can be specified. Typical the value is 0 or 1 for GLCDForeGround and GLCDBackGround respectively.

See also [FilledBox](#)

Circle

Circle:

```
Circle(XPixelPosition, YPixelPosition, Radius [ [,Optional LineColour] [,Optional Rounding] ] )
```

Explanation:

Draws a circle on a GLCD at **XPixelPosition**, **YPixelPosition** with a specific **Radius**.

The constant **GLCD_PROTECTOVERRUN** can be added to prevent circles from re-drawing at the screen edges. Ensure the **GLCD_Width** and **GLCD_HEIGHT** constants are set correctly when using this additional constant.

Example:

```
#include <glcd.h>

circle(10,10,10) ;upper left
circle(117,10,10) ;upper right
circle(63,31,10) ;center
circle(63,31,20) ;center
circle(10,53,10) ;lower left
circle(117,53,10) ;lower right
```



Ellipse

Ellipse:

```
Ellipse(XPixelPosition, YPixelPosition, XRadius, YRadius [,Optional LineColour] )
```

Explanation:

Draws a Ellipse on a GLCD at **XPixelPosition**, **YPixelPosition** with a specific vertex of **XRadius** and **YRadius**.

The constant **GLCD_PROTECTOVERRUN** can be added to prevent Ellipses from re-drawing at the screen edges. Ensure the **GLCD_Width** and **GLCD_HEIGHT** constants are set correctly when using this additional constant.

Example:

```
#include <glcd.h>

Ellipse(63, 31, 20, 10)
```

FilledBox

Syntax:

```
FilledBox(LineX1,LineY1, LineX2, LineY2, Optional LineColour = 1)
```

Explanation:

Draws a filled box on a graphic LCD from the upper corner of pixel X1, Y1 location to pixel X2,Y2 location.

See also [Box](#)

FilledCircle

Circle:

```
FilledCircle(XPixelPosition, YPixelPosition, Radius [,Optional LineColour] )
```

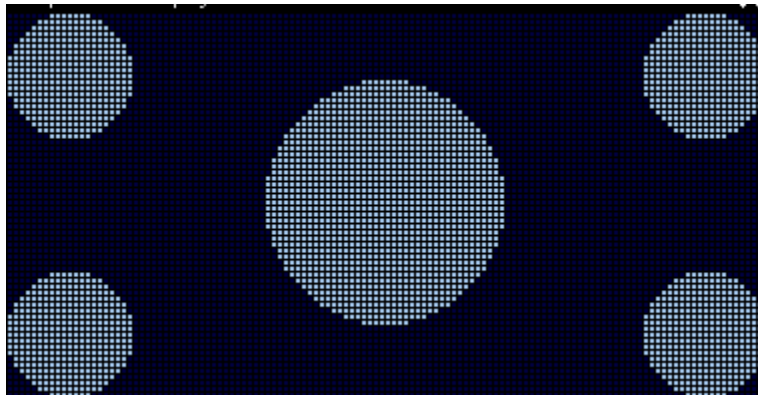
Explanation:

Draws a circle on a GLCD at **XPixelPosition**, **YPixelPosition** with a specific **Radius**.

Example:

```
#include <glcd.h>

filledcircle(10,10,10) ;upper left
filledcircle(117,10,10) ;upper right
filledcircle(63,31,10) ;center
filledcircle(63,31,20) ;center
filledcircle(10,53,10) ;lower left
filledcircle(117,53,10) ;lower right
```



FilledEllipse

FilledEllipse:

```
FilledEllipse(XPixelPosition, YPixelPosition, XRadius, YRadius [,Optional LineColour]
)
```

Explanation:

Draws a FilledEllipse on a GLCD at **XPixelPosition**, **YPixelPosition** with a specific vertex of **XRadius** and **YRadius**.

The constant **GLCD_PROTECTOVERRUN** can be added to prevent FilledEllipses from re-drawing at the screen edges. Ensure the **GLCD_Width** and **GLCD_HEIGHT** constants are set correctly when using this additional constant.

Example:

```
#include <glcd.h>

FilledEllipse(63, 31, 20, 10)
```

FilledTriangle

FilledTriangle:

```
FilledTriangle( XPixelPosition1, YPixelPosition1, XPixelPosition2, YPixelPosition2,  
XPixelPosition3, YPixelPosition3 [,Optional LineColour] )
```

Explanation:

Draws a FilledTriangle on a GLCD at **XPixelPositionN**, **YPixelPositionN**.

The constant **GLCD_PROTECTOVERRUN** can be added to prevent FilledTriangles from re-drawing at the screen edges. Ensure the **GLCD_Width** and **GLCD_HEIGHT** constants are set correctly when using this additional constant.

Example:

```
#include <glcd.h>  
  
FilledTriangle(0, 0, 31, 63, 127, 0 )
```

GLCDCLS

Syntax:

```
GLCDCLS [GLCDBackground]
```

Explanation:

Clears the screen of a Graphic LCD. This command is supported by all GLCD displays.

For colour GLCD displays only. The optional parameter can be used to clear the screen to a specific colour. Using this additional parameter will also change the GLCDBackground colour to this same colour.

Specific to the ST7920 GLCD devices. This command supports the clearing the GLCD to either text mode or graphics mode.

GLCDDisplay

Syntax:

```
GLCDDisplay Off | On
```

Explanation:

Places the GLCD in sleep mode or enables the GLCD for normal operations.

The options are:

OFF
ON

GLCDDrawChar**Syntax:**

```
GLCDDrawChar(CharLocX, CharLocY, CharCode [, Optional Colour] )
```

CharLocX is the **X** coordinate location for the character

CharLocY is the **Y** coordinate location for the character

CharCode is the ASCII number of the character to display. Can be decimal hex or binary.

Colour can be **ON** or **OFF**. For the ST7735 devices this can be any word value that represents the color palette.

Explanation:

Displays an ASCII character at a specified X and Y location. On a 128x64 Graphic LCD:

X = 1 to 128

Y = 1 to 64

GLCDDrawString**Syntax:**

```
GLCDDrawString(CharLocX, CharLocY, String [, Optional Colour] )
```

CharLocX is the X coordinate location for the character

CharLocY is the Y coordinate location for the character

String is the string of characters to display

Colour can be ON or OFF. For the ST7735 devices this can be any word value that represents the color palette

Explanation:

Displays an ASCII character at a specified X and Y location.

On a 128x64 Graphic LCD :

X = 1 to 128

Y = 1 to 64

GLCDPrint

Syntax:

```
GLCDPrint(PrintLocX, PrintLocY, PrintData_Byte [, Optional Colour] )    ',or  
GLCDPrint(PrintLocX, PrintLocY, PrintData_Word [, Optional Colour] )    ',or  
GLCDPrint(PrintLocX, PrintLocY, PrintData_Long [, Optional Colour] )    ',or  
  
GLCDPrint(PrintLocX, PrintLocY, PrintData_String [, Optional Colour] )
```

PrintLocX is the X corrdinate location for the data

PrintLocY is the Y coordinate location for the data

PrintData_[type] is a variable or constant to be displayed

Explanation:

Prints data values (byte, word, long or string) at a specified location on the GLCD screen.

To display an integer use:

```
GLCDPrint(PrintLocX, PrintLocY, strinteger(integer_value) )
```

GLCDPrintLargeFont

Syntax:

```
GLCDPrintLargeFont( PrintLocX, PrintLocY, PrintData_String [, Optional Colour] )
```

GLCD supports for a larger fixed font of 13 pixels. GLCDPrintLargeFont supports strings only.

PrintLocX is the X corrdinate location for the data

PrintLocY is the Y coordinate location for the data
PrintData_[type] is a variable or constant to be displayed

Explanation:

Prints data values (byte, word, long or string) at a specified location on the GLCD screen.

As an example, to display a string use:

```
GLCDPrintLargeFont( 0, 0, "13 Pixels Fixed Font" )
```

GLCDPrintWithSize

Syntax:

```
GLCDPrintWithSize(PrintLocX, PrintLocY, PrintData_Byte , FontSize [, Color ] )  
,or  
GLCDPrintWithSize(PrintLocX, PrintLocY, PrintData_Word , FontSize [, Color ] )  
,or  
GLCDPrintWithSize(PrintLocX, PrintLocY, PrintData_Long , FontSize [, Color ] )  
,or  
GLCDPrintWithSize(PrintLocX, PrintLocY, PrintData_String , FontSize [, Color ] )
```

PrintLocX is the X coordinate location for the data
PrintLocY is the Y coordinate location for the data
PrintData_[type] is a variable or constant to be displayed
FontSize is the GLCD fontsize. Typical values are 1, 2 or 3
Color is an optional parameter to change the color the GLCD printed text.

Explanation:

Prints data values (byte, word, long or string) at a specified location on the GLCD screen with a specific font size.

To display a string using font size two use:

```
GLCDPrintWithSize(PrintLocX, PrintLocY, "Using font size #2", 2 )
```

GLCDLocateString

Syntax:

```
GLCDLocateString(PrintLocX, PrintLocY )
```

Explanation:

Moves the GLCD string pointer to the specified location on the GLCD screen.

PrintLocX is the X coordinate location for the data

PrintLocY is the Y coordinate location for the data

For the purpose of this command. The screen addressing is the first line equates to the parameter 1, the second line equates to the parameter 2 etc.

An example:

```
GLCDLocateString( 0, 1 )    'The first line of the display  
GLCDLocateString( 0, 6 )    'The sixth line of the display
```

Example:

```
GLCDPrintStringLn ( "1.First Ln" )  
GLCDPrintStringLn ( "2.Second Ln" )  
GLCDPrintStringLn ( "" )  
GLCDPrintStringLn ( "4.Forth Ln" )  
GLCDLocateString( 0, 5 )  
GLCDPrintString ( "5." )  
GLCDPrintStringLn ( "Fifth Ln" )  
  
GLCDPrintStringLn ( "6.Sixth Ln" )  
GLCDLocateString( 0, 3 )  
dim val3 as Byte  
val3 = 3  
GLCDPrintStringLn ( str( val3 ) + ".Third Ln" )
```

GLCDPrintString

Syntax:

```
GLCDPrintString( String )
```

Explanation:

Prints string character(s) at a current XY location on the GLCD screen.

Where **String** is a String or String variable of the data to display

This command will **NOT** move the to start of the next line after the string has been displayed

Example:

```
GLCDPrintStringLn ( "1.First Ln" )
GLCDPrintStringLn ( "2.Second Ln" )
GLCDPrintStringLn ( "" )
GLCDPrintStringLn ( "4.Forth Ln" )
GLCDLocateString( 0, 5 )
GLCDPrintString ( "5." )
GLCDPrintStringLn ( "Fifth Ln" )

GLCDPrintStringLn ( "6.Sixth Ln" )
GLCDLocateString( 0, 3 )
dim val3 as Byte
val3 = 3
GLCDPrintStringLn ( str( val3 ) + ".Third Ln" )
```

GLCDPrintStringLn

Syntax:

```
GLCDPrintStringLn( String )
```

Explanation:

Prints string character(s) at a current XY location on the GLCD screen.

Where **String** is a String or String variable of the data to display

This command will move to the start of the next line after the string has been displayed

Example:

```

GLCDPrintStringLn ( "1.First Ln" )
GLCDPrintStringLn ( "2.Second Ln" )
GLCDPrintStringLn ( "" )
GLCDPrintStringLn ( "4.Forth Ln" )
GLCDLocateString( 0, 5 )
GLCDPrintString ( "5." )
GLCDPrintStringLn ( "Fifth Ln" )

GLCDPrintStringLn ( "6.Sixth Ln" )
GLCDLocateString( 0, 3 )
dim val3 as Byte
val3 = 3
GLCDPrintStringLn ( str( val3 ) + ".Third Ln" )

```

GLCDRotate

Syntax:

```
GLCDROTATE LANDSCAPE | PORTRAIT_REV | LANDSCAPE_REV | PORTRAIT
```

Explanation:

Rotate the GLCD display to a relative position.

GLCD rotation needs to be supported by the GLCD chipset. **NOT** all GLCD chipset support these commands.

The options are:

```

LANDSCAPE
PORTRAIT_REV
LANDSCAPE_REV
PORTRAIT

```

The command will rotate the screen and set the following variables using the global variables shown below.

```

GLCD_WIDTH
GLCD_HEIGHT

```

The command is supported by the following global constants.

```
#define LANDSCAPE      1
#define PORTRAIT_REV   2
#define LANDSCAPE_REV  3
#define PORTRAIT       4
```

GLCDReadByte

Syntax:

```
byte_variable = GLCDReadByte
```

Explanation:

Reads a byte of data from the Graphic LCD memory

GLCDTimeDelay

Syntax:

```
GLCDTime
```

Explanation:

This will call the delay routine that delays data transmissions. By default this is set to **20**, which equate to **20 us**. **GLCDTimeDelay** default of **20us** is for 32Mhz support. The can be reduced for slower chip speeds by change the constant **ST7920WriteDelay**.

Example usage:

```
GLCDTime           'call the delay routine
#define ST7920WriteDelay 1 'set the delay to 1 us
```

GLCDTransaction

Syntax:

```
GLCD_Open_PageTransaction
....
    additional number of other GLCD methods
....
GLCD_Close_PageTransaction
```

Explanation:

To make the operation of GLCD seamless - specific library supports GLCDTransaction. GLCDTransaction automatically manages the methods to update the GLCD display via a RAM memory buffer, where this buffer can be small relative to the size of the total number of GLCD pixels.

The process of GLCDtransaction sends GLCD commands to the GLCD display on a page and page basis. Each page is the size of the buffer and for a large GLCD display the number of pages may be equivalent to the numbers of pixels high (height).

GLCDTransaction simplifies the operation by ensure the buffer is setup correctly, handles the GLCD appropriately, handles the sending of the buffer and then close out the update to the display.

To use GLCDTransaction use the following methods.

```
GLCD_Open_PageTransaction
....
additional number of other GLCD methods
....
GLCD_Close_PageTransaction
```

It recommended to use GLCDTransactions at all times when using the e-Paper libraries. Other GLCD libraries support GLCDTransaction to reduce the memory requirement.

Thes GLCDTransactions methods remove the complexity of the GLCD display update process when RAM within the microcontroller is limited.

When using GLCDTransaction you must commence with GLCD_Open_PageTransaction then a series of GLCD commands and then terminate with GLCD_Close_PageTransaction.

GLCDTransaction Insight: When using GLCDtransactions the number of buffer pages is probably be greater then 1 (unless using the SRAM option), so the process of incrementing variables and calls to non-GLCD methods must be considered carefully. The transaction process will increment variables and call non-GLCD methods the same number of times as the number of pages. Therefore, design GLCDTransaction operations with this is mind.

SRAM as the display buffer

To improve memory usage the e-paper the e-Paper libraries support the use of SRAM. SRAM can be used as an alternative to the microcontrollers RAM. Using SRAM does have a small performance impact but does free up the critical resource of the microcontroller RAM. The use of SRAM within the e-paper library is transparent to the user. To use SRAM as the e-paper buffer you will need to set-up the SRAM library. See the SRAM library for more details on SRAM usage.

When using SRAM for the e-paper buffer it is still remcommend to use GLCDTransaction as this ensure the SRAM buffer is correctly initialised.

Optional GLCD_Open_Transaction parameters

Syntax:

```
GLCD_Open_PageTransaction ( low_page, high_page )
```

Explanation:

You can optionally pass `GLCD_Open_PageTransaction` two parameters. The parameters will constrain the GLCD display update process to the specific pages.

This can be used when only updating a portion of the screen to improve performance.

GLCDWriteByte

Syntax:

```
GLCDWriteByte (LCDByte)
```

Explanation:

Writes a byte of data to the Graphic LCD memory

Line

Syntax:

```
Line(LineX1,LineY1, LineX2, LineY2, Optional LineColour = 1)
```

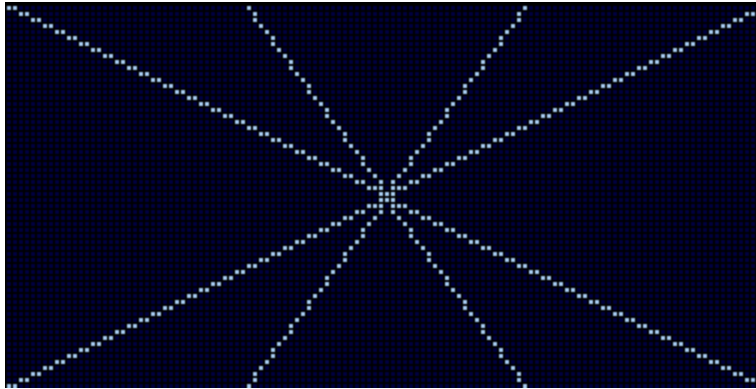
Explanation:

Draws a line on a GLCD from pixel X1, Y1 location to pixel X2,Y2 location.

Example:

```
#include <glcd.h>

line 0,0,127,63
line 0,63,127,0
line 40,0,87,63
line 40,63,87,0
```



Hyperbole

Syntax:

```
Hyperbole (x, y, a_axis, b_axis, type, ModeStop, optional
LineColour=GLCDForeground)
```

Explanation:

Draws on a GLCD an hyperbole with equation $(x/a)^2 - (y/b)^2 = 1$, centered at pixel positions (x, y) with axis a and b.

The hyperbole can be aligned either along the x axis or along the y axis.

Both cases $a_axis \geq b_axis$ and $a_axis < b_axis$ are accepted.

The hyperbole is an unbounded curve made by four branches

Drawing hyperbole on the screen can be stopped by following two different criteria: - a branch has reached a border of the display - all branches have reached the display border

For an hyperbole centered on the display these criteria are equivalent.

Input parameters:

Parameter	Controls
x	X coordinates of hyperbole center (in pixel positions)
y	Y coordinates of hyperbole center (in pixel positions) The x or y coordinates are Word value.
a_axis	The a axis of the hyperbole
b_axis	The b axis of the hyperbole

Parameter	Controls
type	type=1 the hyperbole is aligned along x axis type=2 the hyperbole is aligned along y axis
modestop	modestop=1 drawing stops when a display border is encountered by a hyperbole branch. modestop=2 drawing stops when all the reachable display borders are encountered by all the hyperbole branches
LineColour	Color of the hyperbole

Example:

```
'Example for a 240x320 pixels GLCD

#include <glcd.h>

Hyperbole(120, 160, 40,20, 1, 2, GLCDForeground) ; centered, a=40, b=20, x_axis
alined, stops when all branches have reached a a border
Hyperbole(120, 160, 40,20, 1, 1, GLCDForeground) ; centered, a=40, b=20, x_axis
alined, stops when a border is reached
Hyperbole(120, 160, 40,20, 2, 1, GLCDForeground) ; centered, a=40, b=20, y_axis
alined, stops when a border is reached,

Hyperbole(180, 80, 40,20, 1, 1, GLCDForeground) ; upper right, a=40, b=20, x_axis
alined, stops when a border is reached,
Hyperbole(60, 240, 40,20, 1, 2, GLCDForeground) ; lower left, a=40, b=20, x_axis
alined, stops when all branches have reached a border

Hyperbole(180, 80, 40,20, 2, 1, GLCDForeground) ; upper right, a=40, b=20, y_axis
alined, stops when a border is reached,
Hyperbole(60, 240, 40,20, 2, 2, GLCDForeground) ; lower left, a=40, b=20, y_axis
alined, stops when all branches have reached a border
```

Parabola

Syntax:

```
Parabola (x, y, p_factor, type, modestop, optional LineColour=GLCDForeground)
```

Explanation:

Draws on a GLCD a parabola with equation $y^2=2*p_factor*x$, centered at pixel positions (x, y) .

The parabola is an unbounded curve.

The parabola can be aligned either along the x axis or along the y axis.

Drawing parabola on the screen can be constrained by following two different criteria: - a branch has reached a border of the display. - both branches have reached the display border.

For a parabola centered on the display these criteria are equivalent.

Input parameters:

Parameter	Controls
x, y	X, Y coordinates of the parabola vertex. X is the minimum x value of the parabola when aligned along X. Y is the minimum y value of the parabola when aligned along y in pixel positions The x or y coordinates are Word value, p_factor is word value, type and ModeStop are byte values .
p_factor	The factor such that $y^2=2*p_factor*x$ is the equation of the parabola
type	type=1 the parabola is aligned along x axis type=2 the parabola is aligned along y axis
modestop	modestop=1 drawing stops when a display border is encountered by a parabola branch. modestop=2 drawing stops when all the parabola branches encountered a border
LineColor	Color of the parabola

Example:

'Example for a 240x320 pixels GLCD

```
#include <glcd.h>
```

```
Parabola(120, 160, 20, 1, 2, GLCDForeground) ; centered, p_factor=20, x_axis  
alined, stops when all branches have reached a a border
```

```
Parabola(120, 160, 20, 1, 1, GLCDForeground) ; centered, p_factor=20, x_axis  
alined, stops when a border is reached
```

```
Parabola(120, 160, 20, 2, 1, GLCDForeground) ; centered, p_factor=20, y_axis  
alined, stops when a border is reached,
```

```
Parabola(180, 80, 20, 1, 1, GLCDForeground) ; upper right, p_factor=20, x_axis  
alined, stops when a border is touched,
```

```
Parabola(60, 240, 20, 1, 2, GLCDForeground) ; lower left, p_factor=20, x_axis  
alined, stops when all branches have reached a border
```

```
Parabola(180, 80, 20, 2, 1, GLCDForeground) ; upper right, p_factor=20, y_axis  
alined, stops when a border is touched,
```

```
Parabola(60, 240, 20, 2, 2, GLCDForeground) ; lower left, p_factor=20, y_axis  
alined, stops when all branches have reached a border
```

Pset

Syntax:

```
PSet(XPosition, YPosition, GLCDState)
```

Explanation:

Sets or Clears a Pixel at the specified XPosition, YPosition. Use GLCDState set to 1 to set the pixel and a 0 clears the pixel.

Triangle

Triangle:

```
Triangle(XPixelPosition1, YPixelPosition1, XPixelPosition2, YPixelPosition2,  
XPixelPosition3, YPixelPosition3 [,Optional LineColour] )
```

Explanation:

Draws a Triangle on a GLCD at XPixelPositionN, YPixelPositionN.

The constant `GLCD_PROTECTOVERRUN` can be added to prevent Triangles from re-drawing at the screen

edges. Ensure the `GLCD_Width` and `GLCD_HEIGHT` constants are set correctly when using this additional constant.

Example:

```
#include <glcd.h>

Triangle(0, 0, 31, 63, 127, 0 )
```

Touch Screen

This is the Touch Screen section of the Help file. Please refer the sub-sections for details using the contents/folder view.

ADS 7843 Serial Driver

Syntax:

```
ADS7843_Init
```

```
ADS7843_GetXY
```

```
ADS7843_SetPrecision
```

Command Availability:

Available on all microcontrollers. Requires the inclusion of the following:

```
#include <ADS7843.h>
```

Explanation:

The ADS7843 device is a 12-bit sampling Analog-to-Digital Converter (ADC) with a synchronous serial interface and low on resistance switches for driving touch screens.

The GCBASIC driver is integrated with the SDD1289 GLCD driver. To use the ADS7843 driver the following is required to be added to the GCBASIC source file.

ADS7843_Init is required to initialise the touch screen. This is mandated.

ADS7843_GetXY this sub-routine returns the X and Y coordinates of touched point.

ADS7843_SetPrecision this sub-routine sets the level of precision of the touch screen.

Required Constants:

Constants	Controls/Direction	Default Value
ADS7843_DOUT (IN)	The chip output pin	Mandated
ADS7843_IRQ (IN)	The interrupt pin	Mandated
ADS7843_CS (OUT)	The chip select pin	Mandated
ADS7843_CLK (OUT)	The clock pin	Mandated

Constants	Controls/Direction	Default Value
ADS7843_DIN (OUT)	The chip input pin	Mandated

The GCBASIC commands supported for this chip are:

Command	Purpose	Example
ADS7843_Init	Initialise the device.	ADS7843_Init [Optional precision = PREC_EXTREME]
ADS7843_Get XY	Returns the X and Y coordinates of touched point.	ADS7843_GetXY (TP_X, TP_Y)
ADS7843_Set Precision	Set the precision of the conversion result.	ADS7843_SetPrecision(precision) (with PREC_EXTREME the conversion error is less than 3%)

Precision can be set to four values as shown in the table below. Passing a parameter of ADS7843_SetPrecision changes the precision controls.

Constants	Defined Value	Default Value
#define PREC_LOW	1	
#define PREC_MEDIUM	2	
#define PREC_HI	3	
#define PREC_EXTREME	4	Default Value

Example:

For more information see <http://www.ti.com/product/ads7843>.

This example shows how to drive a SDD1289 based Graphic LCD module with ADS7843 touch controller.

```
'Chip Settings
#chip mega2560, 16

'Include for GLCD
#include <glcd.h>

'Include for ADS7843
#include <ADS7843.h>

'GLCD Device Selection
#define GLCD_TYPE GLCD_TYPE_SSD1289
'Define ports for the SSD1289 display - ALL are required
#define GLCD_WR PORTG.2
```

```

#define GLCD_CS    PORTG.1
#define GLCD_RS    PORTD.7
#define GLCD_RST   PORTG.0

#define GLCD_DB0   PORTC.0
#define GLCD_DB1   PORTC.1
#define GLCD_DB2   PORTC.2
#define GLCD_DB3   PORTC.3
#define GLCD_DB4   PORTC.4
#define GLCD_DB5   PORTC.5
#define GLCD_DB6   PORTC.6
#define GLCD_DB7   PORTC.7
#define GLCD_DB8   PORTA.0
#define GLCD_DB9   PORTA.1
#define GLCD_DB10  PORTA.2
#define GLCD_DB11  PORTA.3
#define GLCD_DB12  PORTA.4
#define GLCD_DB13  PORTA.5
#define GLCD_DB14  PORTA.6
#define GLCD_DB15  PORTA.7

'GLCD font control
#define GLCD_EXTENDED_FONTSET1

'Define ports for ADS7843
#define ADS7843_DOUT  PORTE.5 ' Arduino Mega D3
#define ADS7843_IRQ   PORTE.4 ' Arduino Mega D2
#define ADS7843_CS    PORTE.3 ' Arduino Mega D5
#define ADS7843_CLK   PORTH.3 ' Arduino Mega D6
#define ADS7843_DIN   PORTG.5 ' Arduino Mega D4
#define ADS7843_BUSY  PORTH.4 ' Arduino Mega D7

Wait 100 ms
num=0
Do Forever

'Library function
if ADS7843_IRQ=0 then

    num++
    GLCDPrint 10, 15, str(num), SSD1289_YELLOW, 2

'Library sub routine - returns two variables
ADS7843_GetXY ( TP_X , TP_Y )

if TP_X>=100 then GLCDPrint 100, 50, Str(TP_X), SSD1289_YELLOW, 2
if TP_X>=10 and TP_X<100 then GLCDPrint 100, 50, Str(TP_X)+" ", SSD1289_YELLOW,

```

```

2   if TP_X<10 then GLCDPrint 100, 50, Str(TP_X)+" ",SSD1289_YELLOW, 2
    if TP_Y>=100 then GLCDPrint 100, 70, Str(TP_Y),SSD1289_YELLOW, 2
    if TP_Y>=10 and TP_Y<100 then GLCDPrint 100, 70, Str(TP_Y)+" ", SSD1289_YELLOW,

    if TP_Y<10 then GLCDPrint 100, 70, Str(TP_Y)+" ",SSD1289_YELLOW, 2

    'Set the pixel to yellow using the GLCD PSET sub routine
    Pset TP_X, TP_Y, SSD1289_YELLOW

    end if
    Wait 1 ms

Loop

```


Liquid Crystal Display

This is the LCD section of the Help file. Please refer the sub-sections for details using the contents/folder view.

LCD Overview

Introduction:

The LCD routines in this section allow GCBASIC programs to control an alphanumeric Liquid Crystal Displays based on the **HD44780** IC. This covers most 16 x 1, 16 x 2, 20 x 4 and 40 x 4 LCD displays.

The GCBASIC methods allow the displays to be connected to the microcontroller

Connection Modes:

The table below shows the connection modes. These modes support the connection to the LCD using differing methods.

Connection Mode	Required Connections
0	No configuration is required directly by this method. The LCD routines must be provided with other subroutines which will handle the communication. This is useful for communicating with LCDs connected through RS232 or I2C. This is an advanced method of driving an LCD.
1	Uses a combined data and clock line. This mode is used when the LCD is connected through a shift register 74HC595, as detailed at here . This method of driving an LCD requires an additional integrated circuit and other passive components. This is not recommended for the beginner.
2	Uses separated Data and Clock lines. This mode is used when the LCD is connected through a 74LS174 shift register IC, as detailed at here . This method of driving an LCD requires additional integrated circuits and other passive components. This is not recommended for the beginner.
3	DB , CB , EB are connected to the microcontroller as the Data, Clock and Enable Bits. This a common method to connect a microcontroller to an LCD. This requires 3 data ports on the microcontroller.
4	R/W , RS , Enable and the highest 4 data lines (DB4 through DB7) are connected to the microcontroller. The use of the R/W line is optional. This a common method to connect a microcontroller to an LCD. This requires 7(6) data ports on the microcontroller.

Connection Mode	Required Connections
8	R/W, RS, Enable and all 8 data lines. The data lines must all be connected to the same I/O port, in sequential order. For example, DB0 to PORTB.0, DB1 to PORTB.1 and so on, with `DB7` going to PORTB.7. This is a common method to connect a microcontroller to a LCD. This requires 11(10) data ports on the microcontroller.
10	The LCD is controlled via I2C. A type 10 LCD 12C adapter. Set LCD_IO to 10 for the YwRobot LCD1602 IIC V1 or the Sainsmart LCD_PIC I2C adapter This is a common method and requires two data ports on the microcontroller.
12	The LCD is controlled via I2C. A type 12 LCD 12C adapter. Set LCD_IO to `12` for the Ywmjkdz I2C adapter with a potentiometer (variable resistance) bent over top of chip. This is a common method and requires two data ports on the microcontroller.
107	The LCD is controlled via serial. Set LCD_IO to 107 or K107. The K107 requires one serial data ports on the microcontroller.

Supported LCDs mapped to Connection Mode

The support of various types of LCD displays are shown in the following table.

NOTE

Supported LCD Type number of characters x number of lines	Connection Mode
16 x 1, 16 x 2, 20 x 2, 20 x 4 type LCD displays, also known as 1601, 1602, 2002, 2004 type LCD displays.	0,1,2,4,8,10 and 12
40 x 4 LCD displays, also known as 4004 type LCD displays.	4
16 x 1 LCD displays, with a non-standard/non-consecutive memory map. This LCD sub type is supported using a specific constant. Use #define LCD_VARIANT 1601a to use this sub variant. Also known as 1601 type LCD displays.	Supports any LCD_IO mode.

Communication Performance

There may be a need to change the communication performance for a specific LCD as some LCD's are slower to operate. GCBASIC supports change the communications speed.

To change the performance (communications speed) of the LCD use `#DEFINE LCD_SPEED`. This method allows the timing to be optimised.

Example

NOTE

```
#DEFINE LCD_SPEED FAST
```

Define	Performance Characteristics
<code>LCD_SPEED</code>	<code>FAST</code> - The speed is approximately 20,000 CPS. <code>MEDIUM</code> - The speed is approximately 15,000 CPS. <code>SLOW</code> - The speed is approximately 10,000 CPS. <code>OPTIMAL</code> - The speed is approximately 30,000 CPS.

If `LCD_SPEED` is not defined, the speed defaults to `SLOW`

Using `LCD_Speed OPTIMAL`

OPTIMAL disables fixed delays and allows the LCD operate as fast as it can. In this mode, The the busy flag is polled before each byte is sent to the HD44780 controller. This not only optimizes speed, but also assures that data is not sent to the diplay controler until it is ready to receive the data.

With most displays this equates to a speed of about 30,000 characters per second. For comparision about 10 times faster than I2C using a PC8574 Expander (See LCD_IO 10 or See LCD_IO 112)

OPTIMAL is only supported in LCD_IO 4,8 and only when LCD_NO_RW is not defined (RW Mode). When **#DEFINE LCD_NO_RW** is defined, reading data from the HD44780 is not possible since this disables Read Mode on the controller. In this case busy flag checking is not available and the GET subroutine is not avaiable.

In order to enable busy flag checking, and, therefore to use the **GET** command the following criteria must be true.

NOTE

1. LCD I/O Mode must be either 4-wire or 8-wire
2. **#DEFINE LCD_NO_RW** is not defined
3. An I\O pin is connected between the microcontroller and the RW connection on the LCD Display
4. **'DEFINE LCD_RW port.pin** is defined in the GCBASIC source code

Example:

```
#DEFINE LCD_IO 4
#DEFINE LCD_SPEED OPTIMAL

#DEFINE LCD_DB7 PORTB.5
#DEFINE LCD_DB6 PORTB.4
#DEFINE LCD_DB5 PORTB.3
#DEFINE LCD_DB4 PORTB.2

#DEFINE LCD_RW PORTA.3      'Must be defined for RW Mode
#DEFINE LCD_RS PORTA.2
#DEFINE LCD_ENABLE PORTA.1
```

NOTE

Changing the LCD Width

To change the LCD width characteristics use **#define LCD_WIDTH**

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 3](#), [LCD_IO_2 74xx164](#), [LCD_IO_2 74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

and,

[LCD_Width](#), [LCD_Speed](#)

LCD_IO 0

Using connection mode 0:

To use connection mode 0, a subroutine to write a byte to the LCD **must** be provided.

Optionally, another subroutine to read a byte from the LCD can also be defined. If the LCD was to be read, the function `LCDReadByte` would be set to the name of a function that reads the LCD and returns the data byte from the LCD. If there is no way (or no requirement) to read from the LCD, then the `LCD_NO_RW` constant must be set.

In connection mode 0, the `LCD_RS` constant will be set automatically to an unused bit variable. The higher level LCD commands (such as `Print` and `Locate`) will set it, and the subroutine is responsible for writing to the LCD. The subroutine should handle the process and then set the RS pin on the LCD appropriately.

Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specific constants to support the connection mode using `#define`.

When using connection mode 0 only one constant must be set - all others are optional or can be ignored.

Constant Name	Controls	Value
<code>LCD_IO</code>	The I/O mode.	<code>0</code>

For a code example of connection mode 0 program, download [here](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#),[LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 1

Using connection mode 1:

This approach uses a single connectivity line that supports a combined data and clock signal between the microcontroller and the LCD display. This approach is used when the LCD is connected through a shift register 74HC595, as detailed at [here](#). This connection method is also called a 1-wire connection.

This solution approach recognises the original work provided in the Elektor Magazine.

Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specify constants to support the connection mode using #define.

When using connection mode 1, only two constants must be set - all others are optional or can be ignored.

How to connect and control the LCD background led: see [LCDBacklight](#).

Constant Name	Controls	Default Value
LCD_IO	The I/O mode.	1
LCD_CD	The clock/data pin used in 1-bit mode.	Mandated

LCD.h supports in 1-wire mode the control of pin 4 of the 74HC595 for the background led.

For a code example download [One Wire LCD Example](#).

See for further code examples see [0,1 and 2 Wire LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#) [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 2_74xx164

Using connection mode 2_74XX164:

Use a Data and a Clock line. This manner is used when the LCD is connected through a shift register IC either using a 74HC164 or a 74LS164, as detailed at [here](#). This connection method is also called a 2-wire connection.

This is the preferred two wire method to connect via a shift register to an LCD display.

Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specify constants to support the connection mode using #define.

When using connection mode 2_74XX164 only three constants must be set - all others are optional or can be ignored.

Constant Name	Controls	Default Value
LCD_IO	The I/O mode.	2
LCD_DB	The data pin used in 2-bit mode.	Mandated
LCD_CB	The clock pin used in 2- bit mode.	Mandated

LCD.h supports in connection mode 2_74XX164 via the control of pin 11 of the 74HC164 / 74LS164 the background led/backlight.

How to connect and control the LCD background led: see http://gcbasic.sourceforge.net/help/_lcdbacklight.html

For a code example download [Two Wire LCD Example](#).

See for further code examples see [Two Wire LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx74](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 2

Using connection mode 2:

This method uses a Data and a Clock line via a shift register to control the LCD display. This method is used when the LCD is connected through a shift register IC either using a 74HC164 or a 74LS174, as detailed at [here](#). This connection method is also called a 2-wire connection.

This is a **deprecated** method mode to connect an LCD display to a microcontroller via a shift registry either a 74LS174 (or a 74LS164 with diode connected to pin 11). This method does not support backlight control and has no additional input/output pin.

If you have used the 2-wire mode prior to August 2015, please choose this method for your existing code.

See [LCD_IO 2 74xx164](#) for the preferred method to connect an LCD display to a microcomputer via a shift register.

Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specific constants to support the connection mode using #define. When using 2-bit mode only three constants must be set - all others are optional

or can be ignored.

Constant Name	Controls	Default Value
LCD_IO	The I/O mode.	2
LCD_DB	The data pin used in 2-bit mode.	Mandated
LCD_CB	The clock pin used in 2- bit mode.	Mandated

For a code example download [Two Wire LCD Example](#).

See for further code examples see [Two Wire LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

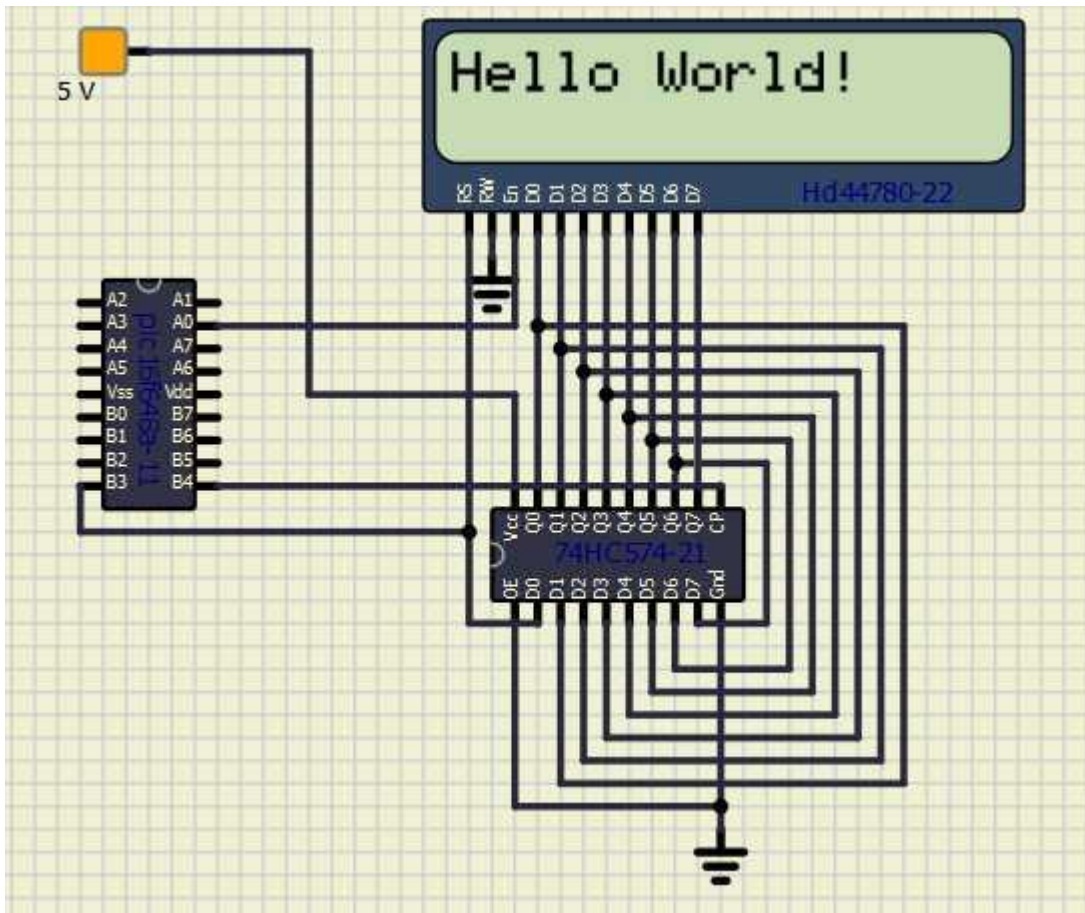
LCD_IO 3

Using connection mode 3:

This method uses a Data and a Clock line via a shift register to control the LCD display plus an Enable line. This method is used when the LCD is connected through a shift register IC using a LS74574.

This connection method is also called a 3-wire connection.

The diagram below shows a method to connect the LCD to a microcontroller.



Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specific constants to support the connection mode using `#define`. When using 3-bit mode only three constants must be set.

Constant Name	Controls	Default Value
<code>LCD_IO</code>	The I/O mode.	3
<code>LCD_DB</code>	The data pin used in 3-bit mode.	Mandated
<code>LCD_CB</code>	The clock pin used in 3- bit mode.	Mandated
<code>LCD_EB</code>	The enable pin used in 3- bit mode.	Mandated

Example:

```

#chip 16f628a, 4
#option explicit

;Setup LCD Parameters
#define LCD_IO 3

'Change ports as necessary
#define LCD_DB      PORTb.3      ; databit
#define LCD_CB      PORTb.4      ; clockbit
#define LCD_EB      PORTa.0      ; enable bit

    Dim BV as Byte

'Program Start

PRINT "GCBASIC"
Locate 1,0
PRINT "@2021"
Wait 4 s

DO Forever
    CLS
    WAIT 2 s
    PRINT "Test LCDHex "
    wait 3 s
    CLS
    wait 1 s

    for bv = 0 to 16
        locate 0,0
        Print "DEC " : Print BV
        locate 1,0
        Print "HEX "
        LCDHex BV
        Locate 1, 8
        LCDHEX BV, LeadingZeroActive

        wait 500 ms
    next
    CLS
    wait 1 s
    Print "END TEST"
LOOP

```

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 2_74xx174

LCD_IO 2_74xx174 has been deprecated as preferred method mode to connect an LCD display to a microcontroller via a shift register either a 74LS174 (or a 74LS164 with diode connected to pin 11). This method does not support backlight control and has no additional input/output pin.

See [LCD_IO 2_74xx164](#) for the preferred method to connect an LCD display to a microcontroller via a shift register.

For more help, see [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx164](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 4

Using connection mode 4:

To use connection mode 4 the R/W, RS, Enable control lines and the highest 4 data lines (DB4 through DB7) must be connected to the microcontroller.

Relevant Constants:

Specific constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set these constants the main program should specify constants to support the connection mode using #define. Constants required for connection mode 4.

Constants are required for 4-bit mode as follows.

Constant Name	Controls	Default Value
LCD_SPEED	FAST, MEDIUM or SLOW.	MEDIUM
LCD_IO	Must be 4	4
LCD_RS	Specifies the output pin that is connected to Register Select on the LCD.	Must be defined as <code>port.bit</code>
LCD_RW	Specifies the output pin that is connected to Read/Write on the LCD. The R/W pin can be disabled*.	Must be defined as <code>port.bit</code> (unless R/W is disabled)
LCD_Enable	Specifies the output pin that is connected to Read/Write on the LCD.	Must be defined as <code>port.bit</code>

Constant Name	Controls	Default Value
LCD_DB4	Output pin used to interface with bit 4 of the LCD data bus	Must be defined as <code>port.bit</code>
LCD_DB5	Output pin used to interface with bit 5 of the LCD data bus	Must be defined as <code>port.bit</code>
LCD_DB6	Output pin used to interface with bit 6 of the LCD data bus	Must be defined as <code>port.bit</code>
LCD_DB7	Output pin used to interface with bit 7 of the LCD data bus	Must be defined as <code>port.bit</code>
LCD_VFD_DELAY	Specifies a delay between transmission of data nibbles to LCD or VFD. Usage must include number value and unit of time. <code>#DEFINE LCD_VFD_DELAY 1 ms</code> Only applicable when using LCD_IO 4	None.
LCD_OCULAR_OM1614	Specifies OCULAR OM1614 support. This changes the initialisation routine to a specific routine for the OCULAR devices.	To specify explicit OCULAR_OM1614 support <code>#DEFINE LCD_OCULAR_OM1614</code> The OCULAR devices requires LCD_RW

The R/W pin can be disabled by setting the `LCD_NO_RW` constant. If this is done, there is no need for the R/W to be connected to the chip, and no need for the `LCD_RW` constant to be set. Ensure that the R/W line on the LCD is connected to ground if not used.

For a code example download [Four Wire LCD Example](#).

Also see for further code examples see [Four Wire LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 8](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 8

Using connection mode 8:

Using connection mode will require R/W, RS, Enable and all 8 data lines.

The data lines must all be connected to the same I/O port, in sequential order. For example, DB0 to PORTB.0, DB1 to PORTB.1 and so on, with DB7 going to PORTB.7.

Relevant Constants:

These constants are used to control settings for the Liquid Crystal Display routines included with

GCBASIC. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

Constants are required for 8-bit mode as follows.

Constant Name	Controls	Default Value
<code>LCD_SPEED</code>	<code>FAST</code> , <code>MEDIUM</code> or <code>SLOW</code> .	<code>MEDIUM</code>
<code>LCD_IO</code>	The I/O mode. Can be 2, 4 or 8.	<code>8</code>
<code>LCD_RS</code>	Specifies the output pin that is connected to Register Select on the LCD.	<i>Must be defined</i>
<code>LCD_RW</code>	Specifies the output pin that is connected to Read/Write on the LCD. The R/W pin can be disabled*.	<i>Must be defined (unless R/W is disabled)</i>
<code>LCD_Enable</code>	Specifies the output pin that is connected to Read/Write on the LCD.	<i>Must be defined</i>
<code>LCD_DATA_PORT</code>	Output port used to interface with LCD data bus	<i>Must be defined</i>
<code>LCD_LAT</code>	Drives the port with <code>LATx</code> support. Resolves issues with faster Mhz and the Microchip PIC read/write/modify feature. See example below.	Optional

The `R/W` pin can be disabled by setting the `LCD_NO_RW` constant. If this is done, there is no need for the `R/W` to be connected to the chip, and no need for the `LCD_RW` constant to be set. Ensure that the `R/W` line on the LCD is connected to ground if not used.

For a code example download [Eight Wire LCD example](#).

For code examples see [Eight Wire Examples](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 10](#) or [LCD_IO 12](#)

LCD_IO 10

LCD_IO 10 provides support for character LCD modules that use an I2C/TWI interface based on the **PCF8574 or PCF8574A I/O expander**. These I/O expanders are commonly found on low-cost LCD “I2C backpacks” sold under many brand names (YwRobot, Sainsmart, Joy-It, generic blue/black I2C adapters, etc.).

The PCF8574 device converts an I2C bus signal into 8 digital output lines, which are then wired to the LCD’s RS, EN, D4–D7, and backlight control pins.

LCD_IO 10 implements the correct signalling for this type of **I/O expander**.

Use **LCD_IO 10** whenever your LCD module uses a **PCF8574-based I2C adapter**, regardless of brand or PCB layout. You can configure the specific LCD port layout, if required.

To use mode 10:

1. Configure your I2C/TWI pins normally in your GCBASIC program.
2. Define the LCD type by using **LCD_IO 10**
3. Set the I2C address of the PCF8574 adapter, if required. Typical PCF8574 I2C addresses range from **0x40 to 0x4E**, depending on the A0–A2 jumper settings on the adapter board.
4. Optionally set LCD speed and backlight behaviour.
5. Optionally set the LCD port layout.

Relevant Constants

These constants configure the Liquid Crystal Display routines in GCBASIC. Add them to your main program using **#define**.

Constant Name	Controls	Value
LCD_IO	Selects the I/O mode. Must be 10 for PCF8574 I2C LCDs.	10
Constant Name	Controls	Default
I2C_LCD_E	LCD Enable - Expander Port Bit	I2C_LCD_BYTE.2
I2C_LCD_RW	LCD RW - Expander Port Bit	I2C_LCD_BYTE.1
I2C_LCD_RS	LCD RS - Expander Port Bit	I2C_LCD_BYTE.0
I2C_LCD_BL	LCD Back Light - Expander Port Bit	I2C_LCD_BYTE.3
I2C_LCD_D4	LCD Data Bit4 - Expander Port Bit	I2C_LCD_BYTE.4
I2C_LCD_D5	LCD Data Bit5 - Expander Port Bit	I2C_LCD_BYTE.5
I2C_LCD_D6	LCD Data Bit6 - Expander Port Bit	I2C_LCD_BYTE.6
I2C_LCD_D7	LCD Data Bit7 - Expander Port Bit	I2C_LCD_BYTE.7
LCD_I2C_ADDRESS_1	PCF8574 I2C address (A0–A2 = 111)	0x4E
LCD_I2C_ADDRESS_2	PCF8574 I2C address (A0–A2 = 110)	0x4C
LCD_I2C_ADDRESS_3	PCF8574 I2C address (A0–A2 = 101)	0x4A
LCD_I2C_ADDRESS_4	PCF8574 I2C address (A0–A2 = 100)	0x48
LCD_I2C_ADDRESS_5	PCF8574 I2C address (A0–A2 = 011)	0x46
LCD_I2C_ADDRESS_6	PCF8574 I2C address (A0–A2 = 010)	0x44

Constant Name	Controls	Value
LCD_I2C_ADDRESS_7	PCF8574 I2C address (A0–A2 = 001)	0x42
LCD_I2C_ADDRESS_8	PCF8574 I2C address (A0–A2 = 000)	0x40

Example Usage

This example demonstrates using two PCF8574-based I2C LCDs on the same I2C bus. Up to eight LCDs may be connected simultaneously, each with a unique PCF8574 address (0x40–0x4E).

Set up I2C-LCD

```
#DEFINE LCD_IO 10
```

```
/* LCD_IO 10 is for PCF8574-based I2C LCD adapters.
   Examples include YwRobot LCD1602 IIC V1, Sainsmart LCD_PIC,
   and most generic blue/black I2C LCD backpacks.
```

```
   LCD_IO 12 is used for adapters based on the Ywmjkdz layout
   with the potentiometer bent over the IC.
```

```
*/
```

```
#DEFINE LCD_I2C_ADDRESS_1 0x4E    ' First LCD at address 0x4E
```

```
PRINT "Hello World"
```

This example demonstrates using two PCF8574-based I2C LCDs on the same I2C bus. Up to eight LCDs may be connected simultaneously, each with a unique PCF8574 address (0x40–0x4E).

Set up I2C-LCD

```
#DEFINE LCD_IO 10

/* LCD_IO 10 is for PCF8574-based I2C LCD adapters.
   Examples include YwRobot LCD1602 IIC V1, Sainsmart LCD_PIC,
   and most generic blue/black I2C LCD backpacks.

   LCD_IO 12 is used for adapters based on the Ywmjkdz layout
   with the potentiometer bent over the IC.
*/

#DEFINE LCD_I2C_ADDRESS_1 0x4E    ' First LCD at address 0x4E
#DEFINE LCD_I2C_ADDRESS_6 0x44    ' Second LCD at address 0x44

' Switch between LCDs:
LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_6    ' Activate second LCD
PRINT "Hello World"

LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_1    ' Activate first LCD
PRINT "Hello World"
```

See other sections of the Help file for details on alternative connection modes.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#), [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 12](#)

LCD_IO 10 Port Configuration

LCD_IO 10 provides support for character LCD modules that use an I2C/TWI interface based on the **PCF8574 or PCF8574A I/O expander**. These expanders are commonly found on low-cost LCD “I2C backpacks” sold under many brand names (YwRobot, Sainsmart, Joy-It, generic blue/black adapters, and others).

Using mode 10

When using LCD_IO 10, the target **PCF8574 or PCF8574A I/O expander** port is configured by default as shown in the table below. Each bit of the byte variable `i2c_lcd_byte` is mapped to the appropriate LCD control or data line.

Constant Name	Controls	Default
I2C_LCD_E	LCD Enable – Expander Port Bit	<code>I2C_LCD_BYTE.2</code>
I2C_LCD_RW	LCD RW – Expander Port Bit	<code>I2C_LCD_BYTE.1</code>
I2C_LCD_RS	LCD RS – Expander Port Bit	<code>I2C_LCD_BYTE.0</code>

Constant Name	Controls	Default
I2C_LCD_BL	LCD Backlight – Expander Port Bit	I2C_LCD_BYTE.3
I2C_LCD_D4	LCD Data Bit 4 – Expander Port Bit	I2C_LCD_BYTE.4
I2C_LCD_D5	LCD Data Bit 5 – Expander Port Bit	I2C_LCD_BYTE.5
I2C_LCD_D6	LCD Data Bit 6 – Expander Port Bit	I2C_LCD_BYTE.6
I2C_LCD_D7	LCD Data Bit 7 – Expander Port Bit	I2C_LCD_BYTE.7

If your I2C LCD adapter or PCB uses a different wiring arrangement, you can override the default configuration to match your hardware.

These overrides should be placed in your main program.

To apply an alternative configuration, define the following eight constants. Doing so replaces the default mappings shown in the table above.

```
#DEFINE I2C_LCD_E    I2C_LCD_BYTE.1
#DEFINE I2C_LCD_RW   I2C_LCD_BYTE.2
#DEFINE I2C_LCD_RS   I2C_LCD_BYTE.0
#DEFINE I2C_LCD_BL   I2C_LCD_BYTE.3
#DEFINE I2C_LCD_D4   I2C_LCD_BYTE.7
#DEFINE I2C_LCD_D5   I2C_LCD_BYTE.6
#DEFINE I2C_LCD_D6   I2C_LCD_BYTE.5
#DEFINE I2C_LCD_D7   I2C_LCD_BYTE.4
```

LCD_IO 12

Using connection mode 12:

The LCD is controlled via I2C. A type 12 is the Ywmjkdz I2C adapter with potentiometer variable resistor) bent over top of chip. To use mode 12 you must define the I2C ports as normal in your GCB code. Then, define the LCD type, set the I2C_address of the LCD adapter and the LCD speed, if required.

Relevant Constants:

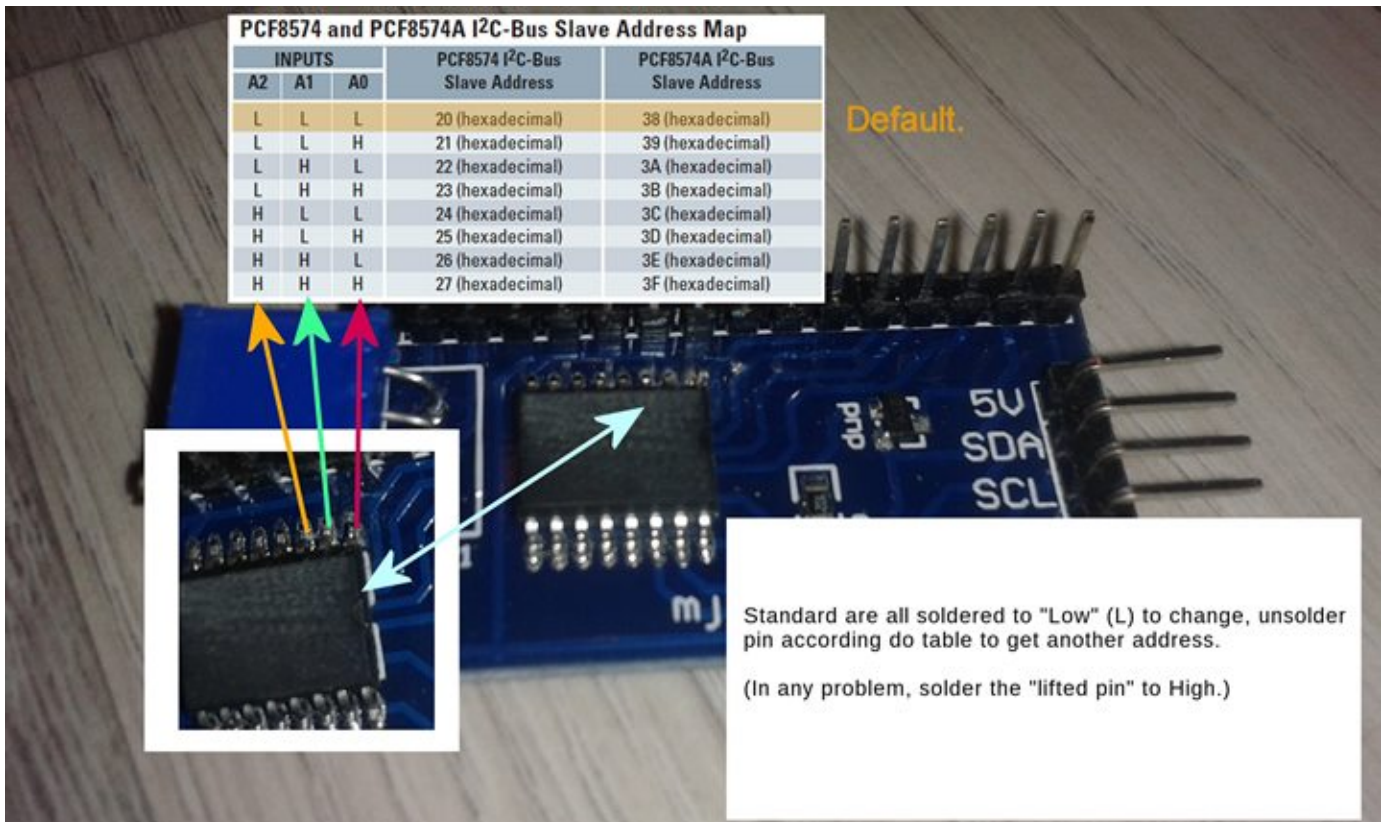
These constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set them, place a line in the main program file that uses **#define** to assign a value to the particular constant.

When using 2-bit mode only three constants must be set - all others can be ignored.

Constant Name	Controls	Value
LCD_IO	I/O mode	12

Constant Name	Controls	Value
LCD_I2C_Address_1	Address of I2C adapter	Default 0x4E could also be 0x27
LCD_I2C_Address_2	Address of I2C adapter	Not set
LCD_I2C_Address_2	Address of I2C adapter	Not set
LCD_I2C_Address_2	Address of I2C adapter	Not set

To set the correct address see the picture below:



For code examples see [I2C LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#)

LCD_IO 12 Port Configuration

Using mode 12:

When using I2C LCD mode 12 the target I2C device address is setup as shown below. Each bit of the the variable `i2c_lcd_byte` is defined to address the correct LCD display port.

```
i2c_lcd_e = i2c_lcd_byte.4
i2c_lcd_rw = i2c_lcd_byte.5
i2c_lcd_rs = i2c_lcd_byte.6
i2c_lcd_bl = i2c_lcd_byte.7
i2c_lcd_d4 = i2c_lcd_byte.0
i2c_lcd_d5 = i2c_lcd_byte.1
i2c_lcd_d6 = i2c_lcd_byte.2
i2c_lcd_d7 = i2c_lcd_byte.3
```

If you have an I2C LCD display adapter with a different set of connection of the adapter then change this configuration to suit the specific of the adapter as follows. This should be done in the your main program code.

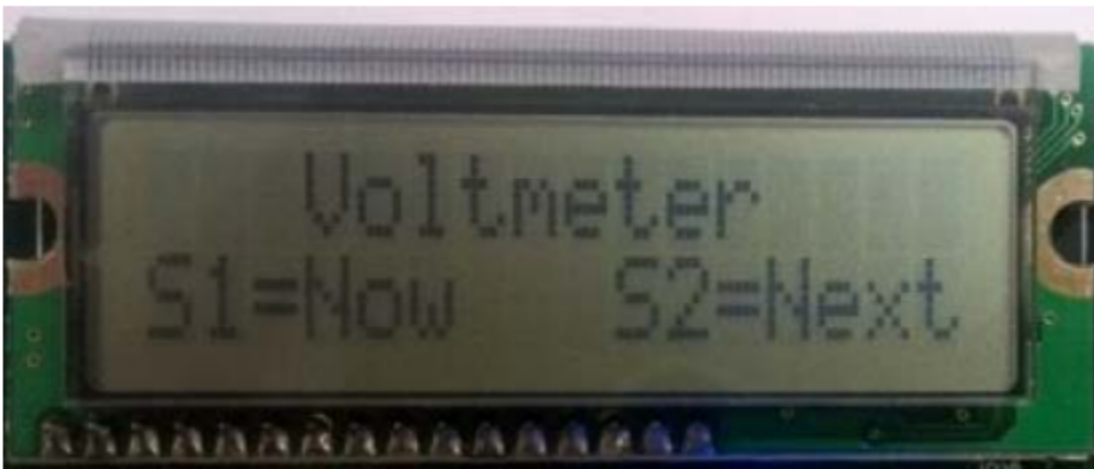
```
#define i2c_lcd_e i2c_lcd_byte.4
#define i2c_lcd_rw i2c_lcd_byte.5
#define i2c_lcd_rs i2c_lcd_byte.6
#define i2c_lcd_bl i2c_lcd_byte.7
#define i2c_lcd_d4 i2c_lcd_byte.3
#define i2c_lcd_d5 i2c_lcd_byte.2
#define i2c_lcd_d6 i2c_lcd_byte.1
#define i2c_lcd_d7 i2c_lcd_byte.0
```

LCD_IO 14

Using connection mode 14:

Using this LCD IO method the LCD is controlled via an SPI expander.

To use mode 14 you must define the SPI ports as normal in your GCB code. Then, define the LCD type, set the SPI address of the SPI expander, and, the LCD speed, if required.



Relevant Constants:

These constants are used to control settings for the LCD routines included with GCBASIC. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

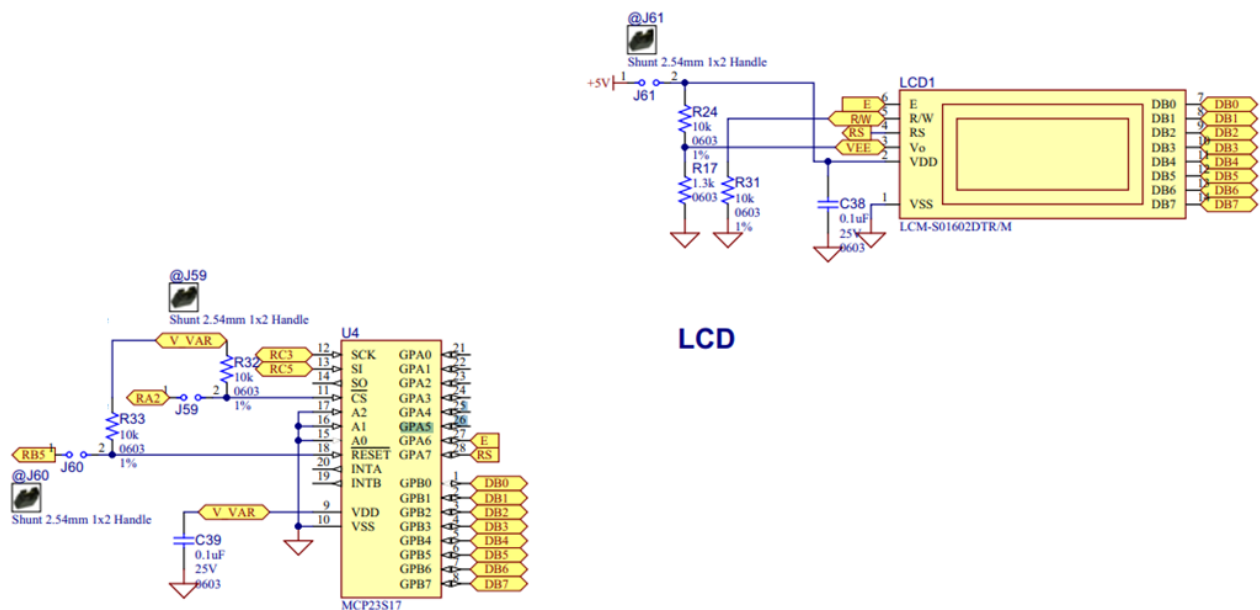
When using this mode only three constants are mandated - all others can be ignored.

Constant Name	Controls	Value
LCD_IO	I/O mode	14
LCD_SPI_D0	Microcontroller SPI data out port	Required
LCD_SPI_SCK	Microcontroller SPI clock out port	Required
LCD_SPI_CS	Microcontroller SPI chip select port	Required

Connectivity

The connectivity is shown below. The microcontroller connections are as shown below. This is an example using the Microchip Explorer 8 board.

RC3 > Expander SPI SCK (clock)
 RC5 > Expander SPI SI (slave in)
 RA2 > Expander SPI CS (chip select) - could be set to 0v0
 RB5 > Expander Reset (optional)



Optional configuration

There are some options you can tweak. See the example setup below. You can play with the use of

hardware or software SPI, SPI frequency (HWSPIMODE MASTERFAST). LED speed, the connectivity between the expander and the LCD and other options.

```
//Constants - LCD connectivity type; controls whether to use HW SPI; The inter
character delay
#define LCD_IO 14
#define LCD_HARDWARESPI
#define LCD_SPEED FAST
#define HWSPIMODE MASTERFAST

//These are the physical connections from the expander to the LCD. These are
automatically set in the library and are shown here purely for clarity.
#define LCD_SPI_EXPD_ADDRESS 0x40 // address of the expander
#define LCD_SPI_EXPANDER_E_ADDRESS 0x40 // GPA6 on the expander
#define LCD_SPI_EXPANDER_RS_ADDRESS 0x80 // GPA7 on the expander

//Pin mappings for LCD IO SPI Expander
#define LCD_SPI_DO portc.5 // constant is mandated
#define LCD_SPI_SCK portc.3 // constant is mandated
#define LCD_SPI_CS porta.2 // constant is required.
// Optional(s) reset Port.Pin connection to expander, select one.
// #define LCD_SPI_RESET_IN portb.5
#define LCD_SPI_RESET_OUT portb.5
```

For code examples see [LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#), [LCD_IO 12](#)

LCD_IO 14 Port Configuration

Using mode 14:

When using LCD mode 14 this is an example program to show a working solution,

```
#chip 18F67K40, 8
#option explicit

'PPS Tool version: 0.0.6.3
// Generated for 18f67k40

#startup InitPPS, 85
#define PPSToolPart 18f67k40
```

```

Sub InitPPS

    #ifdef LCD_HARDWARESPI
        SSP1CLKPPS = 0x13;    //RC3->MSSP1:SCK1;
        RC3PPS = 0x19;    //RC3->MSSP1:SCK1;
        RC5PPS = 0x1A;    //RC5->MSSP1:SD01;
        SSP1DATPPS = 0x14;    //RC4->MSSP1:SDI1;
    #endif

End Sub

//Constants - LCD connectivity type
#define LCD_IO 14

//Comment out to use software SPI
#define LCD_HARDWARESPI

#define LCD_SPEED FAST

//Optional. Can also select MASTERSLOW or MASTER. The compiler will set
automatically.
#define HWSPIMODE MASTERFAST

//These are physical connections from the expander to the LCD. These are
automatically set in the library and are shown here purely for clarity.
#define LCD_SPI_EXPD_ADDRESS    0x40
#define LCD_SPI_EXPANDER_E_ADDRESS    0x40    // GPA6 on the expander
#define LCD_SPI_EXPANDER_RS_ADDRESS    0x80    // GPA7 on the expander

//Mandated Pin mappings for LCD IO SPI Expander
#define LCD_SPI_DO    portc.5
#define LCD_SPI_SCK    portc.3
#define LCD_SPI_CS    porta.2
// Optional(s) reset Port.Pin connection to expander, select one.
// #define LCD_SPI_RESET_IN    portb.5
#define LCD_SPI_RESET_OUT    portb.5

; ----- Main body of program commences here.

CLS
Print "Hello World"

```

LCD_IO 16

Using connection mode 16:

Using this LCD IO method the LCD is controlled via the Microchip PIC16LF72 SPI expander.

To use mode 16 you must define the SPI ports as shown below.

Relevant Constants:

These constants are used to control settings for the LCD routines included with GCBASIC. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

When using this mode only three constants are mandated - all others can be ignored.

Constant Name	Controls	Value
LCD_IO	I/O mode	16
LCD_SPI_DO	Microcontroller SPI data out port	Required
LCD_SPI_SCK	Microcontroller SPI clock out port	Required

Connectivity

The connectivity is shown below. The microcontroller connections are as shown below. This is an example using the Microchip PICDEM 4 2003 board.

```
//Constants - LCD connectivity type;
    #DEFINE LCD_IO 16

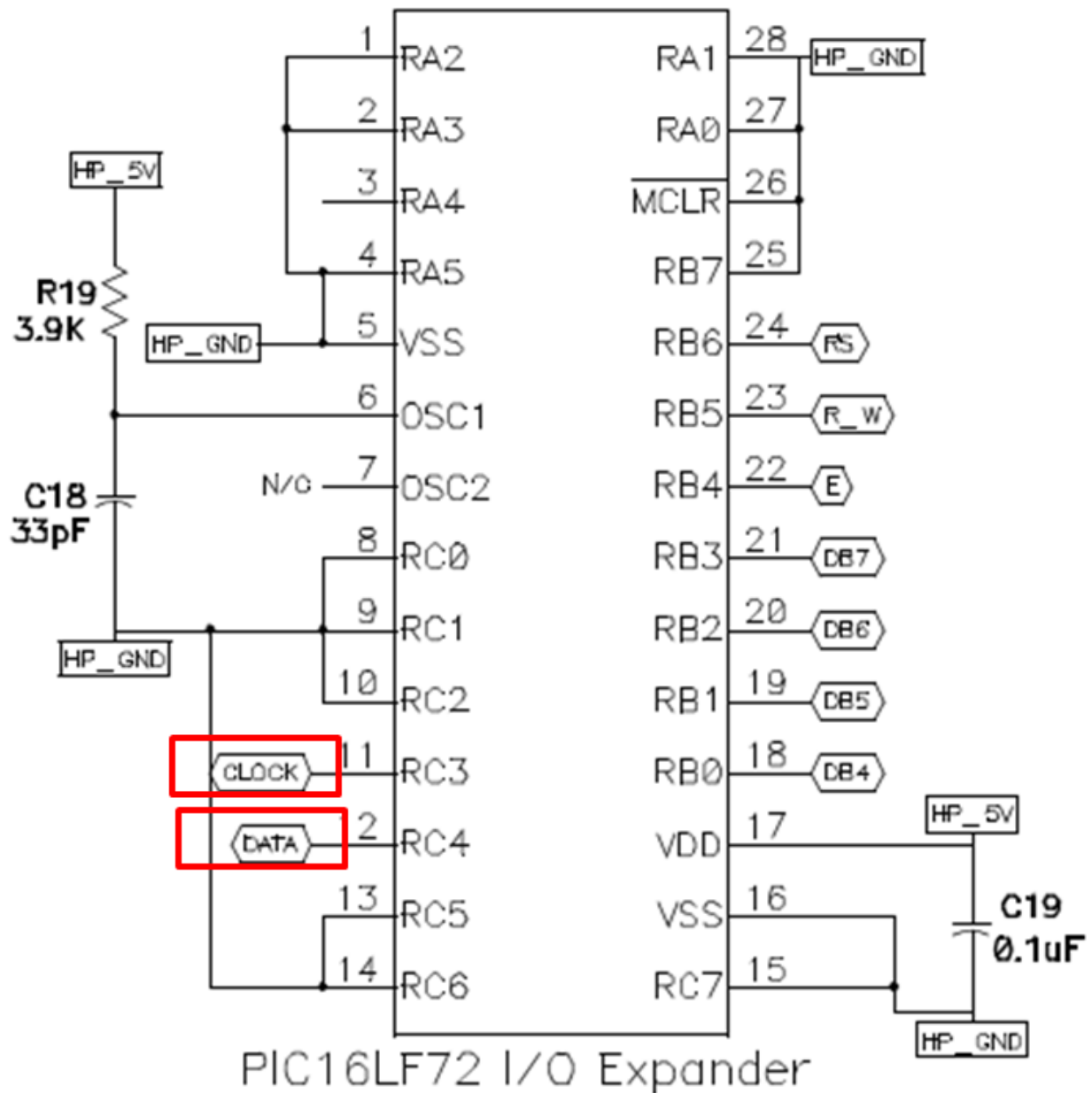
    //PIN MAPPINGS FOR PIC16LF72 LCD IO SPI EXPANDER

    // CONSTANT IS MANDATED - DATA LINE
    #DEFINE LCD_SPI_DO      PORTB.2

    // CONSTANT IS MANDATED - CLOCK LINE
    #DEFINE LCD_SPI_SCK     PORTB.5

//! Main program

Print "GCBASIC Rocks"
End
```



For code examples see [LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#), [LCD_IO 12](#)

LCD_IO 107

Using connection mode 107:

The LCD is controlled via the serial port. A type 107 is a K107 serial adapter. To use mode 107 you must define the serial port as normal in your GCB code. Then, serial speed to match the K107 adapter.

Relevant Constants:

These constants are used to control settings for the Liquid Crystal Display routines included with GCBASIC. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

When using 107 mode only one constants must be set - all others can be ignored.

Constant Name	Controls	Value
LCD_IO	I/O mode	107 or K107

Example Code:

```

#chip 16f18313
#option Explicit

'Generated by PIC PPS Tool for GCBASIC
'Generated for 16f18313
,

#startup InitPPS, 85
#define PPSToolPart 16f18313

Sub InitPPS

    'Module: EUSART
    RA5PPS = 0x0014    'TX > RA5

End Sub
'Template comment at the end of the config file

'USART settings for USART1
#define USART_BAUD_RATE 115200
#define USART_TX_BLOCKING
#define USART_DELAY OFF

#define LCD_IO 107    'K107

do Forever
    CLS
    Print "GCBASIC 2021"
    Locate 1, 0
    Print "Reading ADC ANA0"

    Locate 3, 0
    Print "Scaled = "
    Print Scale( ReadAD( ANA0 ), 0, 236, 0, 100 )
    wait 100 ms
loop

```

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#)

LCD_VARIANT

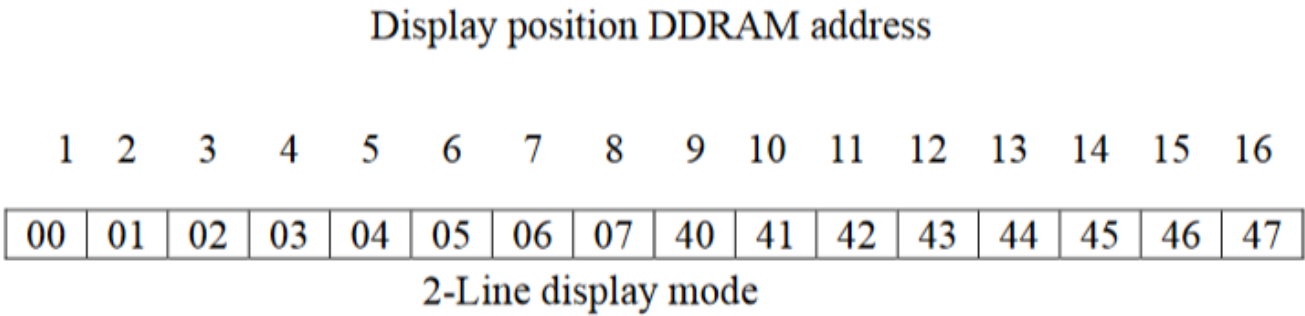
Using LCD_VARIANT:

Some LCDs are non-standard. The non-standard LCDs may have a different memory architecture

where the memory is non-consecutive or different delay timing is required for the LCD IC. Use `LCD_VARIANT` to change the operating behaviour of GCBASIC with respect to LCD operations. If a `LCD_VARIANT` adaption has been created in the library then the non-standard LCD can be supported.

```
#DEFINE LCD_VARIANT 1601a
```

Use `#define LCD_VARIANT 1601a` to use this sub variant. Requires a `LCD_IO` then this sub type modifier. This variant has a non consecutive memory as shown in the diagram below.



Example:

This example shows how to use the `LCD_VARIANT` constant. This example shows the use of software I2C - any LCD mode can be used not just software I2C.

```

#chip tiny84,1

'Set up LCD
#define LCD_IO 10
#define LCD_VARIANT 1601a
#define LCD_WIDTH 16

'You may need to use SLOW or MEDIUM if your LCD is a slower device.
#define LCD_SPEED FAST

' ----- Define Hardware settings
' Define I2C settings - CHANGE PORTS FOR YOUR NEEDS
#define LCD_I2C_Address 0x0E
#define I2C_MODE Master
#define I2C_DATA PORTA.4
#define I2C_CLOCK PORTA.5
#define I2C_DISABLE_INTERRUPTS ON

'You may need to invert these states. Dependent of LCD I2C adapter.
#define LCD_Backlight_On_State 1
#define LCD_Backlight_Off_State 0

; ----- Main body of program commences here.
Locate 0,0
PRINT "GCBASIC"

Do
Loop

```

For code examples see [I2C Variants LCD Solutions](#).

See the separate sections of the Help file for the specifics of each Connection Mode.

For more help, see [LCD_IO 0](#), [LCD_IO 1](#), [LCD_IO 2](#) [LCD_IO 2_74xx164](#), [LCD_IO 2_74xx174](#), [LCD_IO 4](#), [LCD_IO 8](#), [LCD_IO 10](#)

LCD_SPEED

Using LCD_SPEED:

The communication performance of a LCD display can be controlled via a **#DEFINE**. This method allows the timing to be optimised.

Example

```
#DEFINE LCD_SPEED FAST
```

Define	Required Connections
LCD_SPEED	Options are: FAST - The speed is approximately 20,000 CPS. MEDIUM - The speed is approximately 15,000 CPS. SLOW - The speed is approximately 10,000 CPS. OPTIMAL - The speed is approximately 30,000 CPS.

If **LCD_SPEED** is not defined, the speed defaults to **SLOW**

To change the performance (communications speed) of the LCD use **#DEFINE LCD_SPEED**. This method allows the timing to be optimised.

Example

```
#DEFINE LCD_SPEED FAST
```

If **LCD_SPEED** is not defined, the speed defaults to **SLOW**

Speed Parameter **OPTIMAL**

WHEN **LCD_NO_RW** is not defined, **OPTIMAL** disables fixed delays and allows the LCD operate as fast as it can.

In this mode, The the busy flag is polled before each byte is sent to the HD44780 controller. This not only optimizes speed, but also assures that data is not sent to the diplay controler until it is ready to receive the data.

With most displays this equates to a speed of about 30,000 characters per second. For comparision about 10 times faster than I2C using a PC8574 Expander (See LCD_IO 10 or See LCD_IO 112)

OPTIMAL is only supported in LCD_IO 4,8 and only when **LCD_NO_RW** is not defined (RW Mode)

When **#DEFINE LCD_NO_RW** is defined, reading data from the HD44780 is not possible since this disables Read Mode on the controller. In this case busy flag checking is not available and the GET subroutine is not available.

In order to enable busy flag checking, and, therefore to use the **GET** command the following criteria must be true.

1. LCD I/O Mode must be either 4-wire or 8-wire
2. **#DEFINE LCD_NO_RW** is not defined
3. An I/O pin is connected between the microcontroller and the RW connection on the LCD Display

4. 'DEFINE LCD_RW port.pin is defined in the GCBASIC source code

Example:

```
#DEFINE LCD_IO 4
#DEFINE LCD_SPEED OPTIMAL
#DEFINE LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#DEFINE LCD_DB7 PORTB.5
#DEFINE LCD_DB6 PORTB.4
#DEFINE LCD_DB7 PORTB.3
#DEFINE LCD_DB6 PORTB.2

#DEFINE LCD_RW PORTA.3 'Must be defined for RW Mode
#DEFINE LCD_RS PORTA.2
#DEFINE LCD_ENABLE PORTA.1
```

LCD_WIDTH

Using LCD_WIDTH:

This constant changes the width characteristics of a LCD display. The standard width is assumed to be 20 characters.

This constant allows the width to be optimised for specific LCD chipsets.

Example

```
#DEFINE LCD_WIDTH 16
```

Define	Required Connections
LCD_WIDTH	Default is 20 16 - Set the WIDTH 16 characters

If LCD_WIDTH is not defined, the WIDTH defaults to 20

CLS

Syntax:

```
CLS
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **CLS** command clears the contents of the LCD, and returns the cursor to the top left corner of the screen

Example :

```
'A Flashing text "Hello World" program for GCBASIC

'General hardware configuration
#chip 16F877A, 20

'LCD connection settings
#define LCD_IO 8
#define LCD_DATA_PORT PORTC
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is
the default width

'Main routine
Do
    Print "Hello World"
    Wait 1 sec
    CLS
    Wait 1 sec
Loop
```

For more help, see [LCD Overview](#)

Supported in <LCD.H>

Get

Syntax:

```
var = Get(Line, Column)
```

Command Availability:

Available on all microcontrollers with the LCD R/W line (pin 5) connected and if the following constant definition is used; **#define LCD_RW**. Only available when the LCD is connected using the 4 or 8 bit mode

and when the constant definition `#define LCD_NO_RW` is NOT used.

Explanation:

The `Get` function reads the ASCII character code at a given location on the LCD.

For more help, see [Put](#), [LCD Overview](#)

Supported in <LCD.H>

LCDBacklight

Syntax:

```
LCDBacklight ( On | Off )
```

Command Availability:

Available on all microcontrollers

Explanation:

Sets the LCD backlight on or off

Do not connect the LCD backlight directly to the microcontroller! Always refer to the datasheet for the correct method to drive the LCD backlight.

For 0, 4, 8, 404 LCD types you **must** define the controlling port.pin for the LCD backlight.

```
'this port.pin is connected to the LCD backlight via a suitable circuit
#define LCD_Backlight porta.4
...
...
...
...
LCDBacklight ( On )

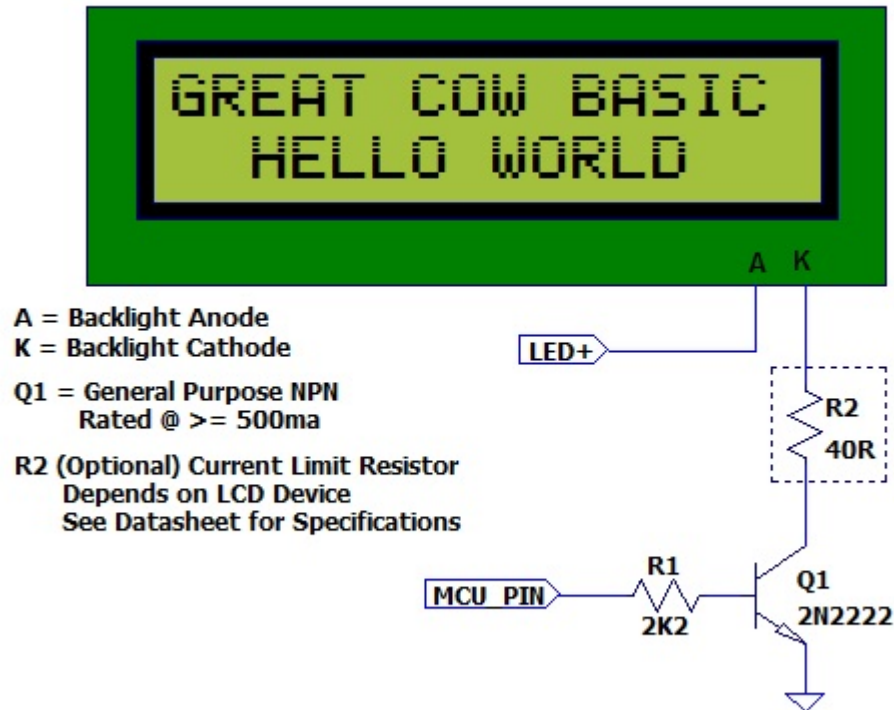
.... more user code...
LCDBacklight ( Off )
```

Inverting the State of the LCD

You may need to invert the state of the LCD backlight control port. This can be achieved by setting the following constants.


```
'Invert the LCD Backlight States to suit the circuit board
#define LCD_Backlight_On_State 0    'the default constant value is 1
#define LCD_Backlight_Off_State 1   'the default constant value is 0
```

The diagram below shows a method to connect the LCD backlight to a microcontroller.



The

diagram above was provided by William Roth, January 2015.

Supported in <LCD.H>

LCDCreateChar

Syntax:

```
LCDCreateChar char, chardata()
```

Command Availability:

Available on all microcontrollers.

Explanation:

The LCDCreateChar command is used to send a custom character to the LCD.

Each character on the LCD is made up from an 8 row by 5 column (5x8) matrix of pixels. The data to be sent to the LCD is composed of an 8 element array, where each element corresponds to a row. Inside each element, the 5 lowest bits make up the data for the corresponding row. When a bit is set a dot will be drawn at the matching location; when it is cleared, no dot will appear.

An array of more than 8 elements may be used, but only the first 8 will be read.

`char` is the ASCII value of the character to create. ASCII codes 0 through 7 are usually used to store custom characters.

`chardata()` is an array containing the data for the character.

Example:

```

'This program draws a smiling face character

'General hardware configuration
#chip 16F877A, 20

'LCD connection settings
#define LCD_IO 8
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DATA_PORT PORTC
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

'Create an array to store the character until it is copied
Dim CharArray(8)

'Set the array to hold the character
'Binary has been used to improve the readability of the code, but is not essential
CharArray(1) = b'00011011'
CharArray(2) = b'00011011'
CharArray(3) = b'00000000'
CharArray(4) = b'00000100'
CharArray(5) = b'00000000'
CharArray(6) = b'00010001'
CharArray(7) = b'00010001'
CharArray(8) = b'00001110'

'Copy the character from the array to the LCD
LCDCreateChar 0, CharArray()

'Draw the custom character
LCDWriteChar 0

```

For more help, see [LCDWriteChar](#), [LCD Overview](#)

Supported in <LCD.H>

LCDCreateGraph

Syntax:

```
LCDCreateGraph value
```

Command Availability:

Available on all microcontrollers.

Explanation:

The LCDCreateGraph command will create a graph like character which can then be displayed on the LCD

Example :

```
;Chip Settings
#chip 16F88,8

;Defines (Constants)
#define LCD_IO 4
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PORTA.6
#define LCD_NO_RW
#define LCD_Enable PORTA.7
#define LCD_DB4 PORTB.4
#define LCD_DB5 PORTB.5
#define LCD_DB6 PORTB.6
#define LCD_DB7 PORTB.7

Locate 0,0
Print "Reset"
wait 1 s
cls

Graph_Tests:

cls
'Draw the custom character - fill the LCD
repeat 64
    LCDWriteChar 0
end Repeat

' Update the characters at high speed without re-printing on LCD
for graphvalue = 0 to 8
    LCDCreateGraph ( 0 , graphvalue )
    wait 100 ms
next
```

Supported in <LCD.H>

LCDCmd

Syntax:

```
LCDCMD value
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command set LCD specific instructions to the LCD display. As shown in the table below.

INSTRUCTION	Decimal	Hexadecimal
Scroll display one character right (all lines)	28	1E
Scroll display one character left (all lines)	24	18
Home (move cursor to top/left character position)	2	2
Move cursor one character left	16	10
Move cursor one character right	20	14
Turn on visible underline cursor	14	0E
Turn on visible blinking-block cursor	15	0F
Make cursor invisible	12	0C
Blank the display (without clearing)	8	08
Restore the display (with cursor hidden)	12	0C
Clear Screen	1	01
Set cursor position (DDRAM address)	128 + addr	80+ addr
Set pointer in character-generator RAM (CG RAM address)	64 + addr	40+ addr

Example 1:

```
;Chip Settings
#chip 16F88,8

;Defines (Constants)
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
```

```

#define LCD_RS PORTA.6
#define LCD_NO_RW
#define LCD_Enable PORTA.7
#define LCD_DB4 PORTB.4
#define LCD_DB5 PORTB.5
#define LCD_DB6 PORTB.6
#define LCD_DB7 PORTB.7

Locate 0,0
Print "Reset"
wait 1 s
cls

LCD_Command_Tests:

locate 0,8
print "123456"
'Scroll display one character right (all lines)      28
,

lcdcmd 28
wait 1 s
lcdcmd 28
wait 1 s
lcdcmd 28
wait 1 s
lcdcmd 28
wait 1 s

'Scroll display one character left (all lines)      24
,

lcdcmd 24
wait 1 s
lcdcmd 24
wait 1 s
lcdcmd 24
wait 1 s
lcdcmd 24
wait 1 s

'Home (move cursor to top/left character position)  2
,

lcdcursor flash
lcdcmd 2
wait 1 s

'Move cursor one character left                      16
,

```

```

lcdcursor flash
locate 0,8

lcdcmd 16
wait 1 s
lcdcmd 16
wait 1 s
lcdcmd 16
wait 1 s
lcdcmd 16
wait 1 s

'Move cursor one character right          20
,

lcdcmd 20
wait 1 s
lcdcmd 20
wait 1 s
lcdcmd 20
wait 1 s
lcdcmd 20
wait 1 s

```

Example 2:

```

#chip 16F877A,20
#option Explicit

'Use LCD in 4 pin mode and define LCD pins
#define LCD_IO 4
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7

;Here are various LCD commands which can be used.
;These are the LCD commands for the HD44780 controller
#define clrHome = 1           ;clear the display, home the cursor
#define home    = 2           ;home the cursor only
#define RtoL    = 4           ;print characters right to left
#define insR    = 5           ;insert characters to right
#define LtoR    = 6           ;print characters left to right

```

```

#define insL    = 7      ;insert characters to left
#define lcdOff  = 8      ;LCD screen off
#define lcdOn   = 12     ;LCD screen on, no cursor
#define curOff  = 12     ;an alias for the above
#define block   = 13     ;LCD screen on, block cursor
#define under   = 14     ;LCD screen on, underline cursor
#define undblk  = 15     ;LCD screen on, blinking and underline cursor
#define CLeft   = 16     ;cursor left
#define CRight  = 20     ;cursor right
#define panR    = 24     ;pan viewing window right
#define panL    = 28     ;pan viewing window left
#define bus4    = 32     ;4-bit data bus mode
#define bus8    = 48     ;8-bit data bus mode
#define mode1   = 32     ;one-line mode (alias)
#define mode2   = 40     ;two-line mode
#define line1   = 128    ;go to start of line 1
#define line2   = 192    ;go to start of line 2
;----- Variables
dim char, msn, lsn, index, ii as byte
;----- Main Program
LoadEeprom                ;load the EEprom with strings

do forever
    printMsg(0)            ;print first message
    wait 3 S               ;pause 3 seconds
    printMsg(2)            ;print next message
    wait 3 S               ;pause 3 seconds
    repeat 5               ;blink it five times
        LCDCmd(lcdOff)    ;display off
        wait 500 mS       ;pause
        LCDCmd(lcdOn)     ;display on
        wait 500 mS       ;pause
    end repeat
    wait 1 S               ;pause before next demo
    ;demonstrate panning
    printMsg(4)            ;print next message
    wait 3 S               ;pause 3 seconds
    repeat 16
        LCDCmd(panL)      ;pan left a step at a time
        wait 300 mS       ;slow down to avoid blur
    end repeat
    repeat 16
        LCDCmd(panR)      ;then pan right
        wait 300 mS
    end repeat
    wait 1 S               ;pause before next demo
    ;demonstrate moving the cursor
    printMsg(6)            ;print next message

```



```

wait 3 S                ;pause 3 seconds
LCDHome
LCDCmd(under)           ;choose underline cursor
for ii = 0 to 15         ;move cursor across first line
    LCDCmd(line1+ii)
    wait 200 mS
next i
for ii = 0 to 15         ;move cursor across second line
    LCDCmd(line2+ii)
    wait 200 mS
next i
for ii = 15 to 0 step -1 ;move cursor back over second line
    LCDCmd(line2+ii)
    wait 200 mS
next i
for ii = 15 to 0 step -1 ;move cursor back over first line
    LCDCmd(line1+ii)
    wait 200 mS
next i
wait 3 S
;demonstrate blinking block cursor
printMsg(8)             ;print next message
LCDHome                 ;home the cursor
LCDCmd(block)           ;choose blinking block cursor
wait 4 S                ;pause 4 seconds
LCDCmd(mode1)           ;change to one long line mode
LCDHome                 ;home the cursor again
LCDCmd(curOff)          ;and disable it

;demonstrate scrolling a lengthy one-line marquee
for ii = 0xd0 to 0xff    ;print next message - the remaining EEPROM
    ERead ii, char       ;fetch directly from eeprom
    print chr(char)
next i
wait 1 S
LCDHome                 ;home cursor once more
repeat 141              ;scroll message twice
    LCDCmd(panR)
    wait 250 mS
end repeat
wait 2 S
LCDCmd(mode2)           ;change back to two line mode
CLS                     ;clear the screen
;demonstrate all of the characters
printMsg(11)            ;print next message
for ii = 33 to 127      ;print first batch of ASCII characters
    LCDCmd(line1+12)    ;overwrite each character displayed

```

```

        print chr(ii)           ;this is the ASCII code
        wait 500 mS
    next i
    for ii = 161 to 255         ;print next batch of ASCII characters
        LCDCmd(line1+12)
        print chr(ii)
        wait 500 mS
    next i
    ;say good-bye
    LCDCmd(line2)
    printMsg(11)               ;print next message
    LCDHome                   ;home the cursor
loop
end

```

```

;----- Print a message to the LCD
;The parameter 'row' points to the start of the string.
sub printMsg(in row as byte, in Optional StringLength As Byte = 15)
    Locate 0, 0                ;get set for first line

    for ii = 0 to StringLength
        index = row*16+ii
        EPread index, char      ;fetch next character and
        print chr(char)         ;transmit to the LCD
    next

    Locate 1,0                 ;get set for second line
    for ii = 0 to StringLength
        index = (row+1)*16+ii
        EPread index, char      ;fetch next character and
        print chr(char)         ;transmit to the LCD
    next
end sub

```

```

sub loadEeprom

```

' Strings for EEPROM, Strings should be limited to 16 characters for the first 13 sstrings, then a long string to fill eeprom

```

WriteEeprom "First we'll show"
WriteEeprom "this message.  "
WriteEeprom "Then we'll blink"
WriteEeprom "five times.    "
WriteEeprom "Now lets pan   "
WriteEeprom "left and right. "
WriteEeprom "Watch the line  "
WriteEeprom "cursor move.    "
WriteEeprom "A block cursor  "
WriteEeprom "is available.   "

```

```

WriteEeprom "Characters:  "
WriteEeprom "Bye!        "
WriteEeprom "in one line mode"
WriteEeprom "Next well scroll this long message as a marquee"
end sub

```

```

; Write to the device eeprom
sub WriteEeprom ( in Estring() )

```

Dim eeLocation as Byte 'if the EEPROM size was larger than 256 bytes then this would need to be a WORD

```

    for eeLocation = 1 to len ( Estring )
        HSersend Estring( eeLocation )
        epwrite eeLocation, Estring( eeLocation )
    next
end sub

```

Supported in <LCD.H>

LCDCursor

Syntax:

```
LCDCursor value
```

Command Availability:

Available on all microcontrollers.

Explanation:

The LCDCursor command will accept the following parameters:

LCDON, **LCDOFF**, **CURSORON**, **CURSOROFF**, **FLASHON**, **FLASHOFF**

FLASH, and **ON/OFF** have been retained for backward compatibility with older releases of GCB.

LCDON will turn on (restore) the LCD display.

LCDOFF will turn off (hide) the LCD display.

CURSORON will turn on the cursor.

CURSOROFF will turn off the cursor.

FLASHON will flash the cursor.

FLASHOFF will stop flashing the cursor.

Example

:

#config osc = intrc

```
#chip 16f877a, 8

;Defines (Constants)
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PORTA.6
#define LCD_NO_RW
#define LCD_Enable PORTA.7
#define LCD_DB4 PORTB.4
#define LCD_DB5 PORTB.5
#define LCD_DB6 PORTB.6
#define LCD_DB7 PORTB.7

Start:
CLS
WAIT 3 s
PRINT "START DEMO"
locate 1,0
PRINT "DISPLAY ON"

wait 3 s

CLS
Locate 0,0
Print "Cursor ON"
Locate 1,0
LCDcursor CursorOn
wait 3 S

CLS
LCDcursor CursorOFF
locate 0,0
Print "Cursor OFF"
wait 3 s

CLS
Locate 0,0
Print "FLASH ON"
Locate 1,0
LCDcursor FLASHON
wait 3 s

CLS
```

```

Locate 0,0
Print "FLASH OFF"
LCDCURSOR FLASHOFF
wait 3 sec

Locate 0,0
Print "CURSOR&FLASH ON" 'Both are on at the same time
locate 1,0
LCDCURSOR CURSORON
LCDCURSOR FLASHON
Wait 3 sec

Locate 0,0
Print "CURSOR FLASH OFF"
locate 1,0
LCDCURSOR CursorOFF
LCDCURSOR FLASHOFF
Wait 3 sec

CLS
Locate 0,4
PRINT "Flashing"
Locate 1,4
Print "Display"
wait 500 ms

repeat 5
    LCDCURSOR LCDOFF
    wait 500 ms
    LCDCURSOR LCDON
    wait 500 ms
end repeat

CLS
Locate 0,0
Print "DISPLAY OFF"
Locate 1,0
Print "FOR 5 SEC"
Wait 2 SEC
LCDCURSOR LCDOFF
WAIT 5 s

CLS
Locate 0,0
LCDCURSOR LCDON
Print "END DEMO"
wait 3 s
goto start

```

LCDHex

Syntax:

LCDHex value

LCDHex value, `LeadingZeroActive`

Command Availability:

Available on all microcontrollers.

Explanation:

The LCDHex will display the byte value as a 1 or 2 character HEX string.

`value` is a byte value from 0 to 255.

`LeadingZeroActive` is a constant or byte value of 2.

Example :

```
;Set chip model required:
#chip mega328p, 16
;Setup LCD Parameters
#define LCD_IO 4
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_NO_RW
#define LCD_Speed MEDIUM 'FAST IS OK ON ARDUINO UNO R3

'Change as necessary
#define LCD_RS PortC.0
#define LCD_Enable PortC.1
#define LCD_DB4 PortC.2
#define LCD_DB5 PortC.3
#define LCD_DB6 PortC.4
#define LCD_DB7 PortC.5

' #chip 16f877a, 8
' ;Setup LCD Parameters
' #define LCD_IO 4
' #define LCD_NO_RW
' #define LCD_Speed fast 'FAST IS OK ON 16f877a
```

```

' ;Change as necessary
' #define LCD_RS PortB.2
' #define LCD_Enable PortB.3
' #define LCD_DB4 PortB.4
' #define LCD_DB5 PortB.5
' #define LCD_DB6 PortB.6
' #define LCD_DB7 PortB.7

'Program Start
DO Forever
  CLS
  WAIT 2 s
  PRINT "Test LCDHex "
  wait 3 s
  CLS
  wait 1 s

  for bv = 0 to 255
    locate 0,0
    Print "DEC " : Print BV
    locate 1,0
    Print "HEX "
    LCDHex BV, LeadingZeroActive ; display leading Zero
    ' LCDHex BV ; do not display leading zero
    wait 1 s
  next
  CLS
  wait 1 s
  Print "END TEST"
LOOP

```

Supported in <LCD.H>

LCDHome

Syntax:

```
LCDHome
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **LCDHome** command will return the cursor to home position.

The current contents of the LCD screen will be retained.

Example:

```
;Chip Settings
#chip 16F88,8

;Defines (Constants)
#define LCD_IO 4
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PORTA.6
#define LCD_NO_RW
#define LCD_Enable PORTA.7
#define LCD_DB4 PORTB.4
#define LCD_DB5 PORTB.5
#define LCD_DB6 PORTB.6
#define LCD_DB7 PORTB.7

Locate 0,0
Print "Reset"
wait 1 s
CLS

Cursor_Home_Tests:

cls
lcdcursor flash
print "Test Home Cmd"
LCDHome
wait 3 s
```

Supported in <LCD.H>

LCDDisplayOn

Syntax:

```
LCDDisplayOn
```

Explanation:

Will turn on (restore) the LCD display

See also [LCDCursor](#)

Supported in <LCD.H>

LCDDisplayOff

Syntax:

```
LCDDisplayOff
```

Explanation:

Will turn off (hide) the LCD display.

See also [LCDCursor](#)

Supported in <LCD.H>

LCDSpace

Syntax:

```
LCDSpace value
```

Command Availability:

Available on all microcontrollers.

Explanation:

The LCDSpace command will print the required number of spaces on the LCD display

value is a byte value from 1 to 255. Where the **value** is the number of spaces required.

Example :

```
Locate 0,0
Print "Reset"
wait 1 s
cls

LCD_Space_Tests:

lcdcursor flash

lcdspace 12

print "**"
```

Supported in <LCD.H>

LCDWriteChar

Syntax:

```
LCDWriteChar char
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **LCDWriteChar** command will show the specified character on the LCD, at the current cursor position.

char is the ASCII value of the character to show. On most LCDs, characters 0 through 7 are user defined, and can be set using the **LCDCreateChar** command.

Example :

```

'This program draws a smiling face character

'Create an array to store the character until it is copied
Dim CharArray(8)

'Set the array to hold the character
CharArray(1) = b'00011011'
CharArray(2) = b'00011011'
CharArray(3) = b'00000000'
CharArray(4) = b'00000100'
CharArray(5) = b'00000000'
CharArray(6) = b'00010001'
CharArray(7) = b'00010001'
CharArray(8) = b'00001110'

'Copy the character from the array to the LCD
LCDCreateChar 0, CharArray()

'Draw the custom character
LCDWriteChar 0

```

For more help, see [LCDCreateChar](#), [LCD Overview](#)

Supported in <LCD.H>

Locate

Syntax:

```
Locate Line, Column
```

Command Availability:

Available on all microcontrollers.

Explanation:

The Locate command is used to move the cursor on the LCD to the given location.

Line is line number on the LCD display. A byte value from 0 to 255.

Column is column number on the LCD display. A byte value from 0 to 255.

Example :

```
'A Hello World program for GCBASIC.  
'Uses Locate to show "World" on the second line  
  
'General hardware configuration  
#chip 16F877A, 20  
  
'LCD connection settings  
#define LCD_IO 8  
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the  
default width  
#define LCD_DATA_PORT PORTC  
#define LCD_RS PORTD.0  
#define LCD_RW PORTD.1  
#define LCD_Enable PORTD.2  
  
'Main routine  
Print "Hello"  
Locate 1, 5  
Print "World"
```

For more help, see [LCD Overview](#)

Supported in <LCD.H>

Print

Syntax:

```
Print string  
Print byte  
Print word  
Print long  
Print integer
```

Command Availability:

Available on all microcontrollers.

Explanation:

The Print command will show the contents of a variable on the LCD. It can display string, word, byte, long or integer variables.

Example:

```

'A Light Meter program.

'General hardware configuration
#chip 16F877A, 20
#define LightSensor AN0

'LCD connection settings
#define LCD_IO 8
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DATA_PORT PORTC
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

CLS
Print "Light Meter"
Locate 1,2
Print "A GCBASIC Demo"
Wait 2 s

Do
    CLS
    Print "Light Level: "
    Print ReadAD(LightSensor)
    Wait 250 ms
Loop

```

For more help, see [LCD Overview](#)

Supported in <LCD.H>

Put

Syntax:

```
Put Line, Column, Character
```

Command Availability:

Available on all microcontrollers.

Explanation:

The Put command writes the given ASCII character code to the current location on the LCD.

Line is line number on the LCD display. A byte value from 0 to 255.

Column is column number on the LCD display. A byte value from 0 to 255.

Character is the required ASCII code. A byte value from 0 to 255.

Example :

```
'A scrolling star for GCBASIC

'Misc Settings
#define SCROLL_DELAY 250 ms

'General hardware configuration
#chip 16F877A, 20

'LCD connection settings
#define LCD_IO 8
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DATA_PORT PORTC
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

'Main routine
For StarPos = 0 To 16
  If StarPos = 0 Then
    Put 0, 16, 32
    Put 0, 0, 42
  Else
    Put 0, StarPos - 1, 32
    Put 0, StarPos, 42
  End If
  Wait SCROLL_DELAY
Next
```

For more help, see [LCD Overview](#)

Supported in <LCD.H>

Examples

LCD_IO 2 Example

This a connection mode 2 Serial Driver to demonstrate LCD features. This for the 16F877A, but, it can easily be adapted for other microcontrollers.

A 2 by 16 LCD is assumed.

Based on the works by Thomas Henry and then revised Evan R. Venn

```
#chip 16F877A,20

#define LCD_IO 2
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DB portb.2
#define LCD_CB portb.0
#define LCD_NO_RW
    ;Here are various LCD commands which can be used.
    ;These are the LCD commands for the HD44780 controller

#define clrHome = 1           ;clear the display, home the cursor
#define home     = 2           ;home the cursor only
#define RtoL     = 4           ;print characters right to left
#define insR      = 5           ;insert characters to right
#define LtoR      = 6           ;print characters left to right
#define insL      = 7           ;insert characters to left
#define lcdOff    = 8           ;LCD screen off
#define lcdOn     = 12          ;LCD screen on, no cursor
#define curOff    = 12          ;an alias for the above
#define block     = 13          ;LCD screen on, block cursor
#define under     = 14          ;LCD screen on, underline cursor
#define undblk    = 15          ;LCD screen on, blinking and underline cursor
#define CLeft     = 16          ;cursor left
#define CRight    = 20          ;cursor right
#define panR      = 24          ;pan viewing window right
#define panL      = 28          ;pan viewing window left
#define bus4      = 32          ;4-bit data bus mode
#define bus8      = 48          ;8-bit data bus mode
#define mode1     = 32          ;one-line mode (alias)
#define mode2     = 40          ;two-line mode
#define line1     = 128         ;go to start of line 1
#define line2     = 192         ;go to start of line 2
;----- Variables
dim char, msn, lsn, index, ii as byte
;----- Main Program
```

```

LoadEeprom          ;load the EEprom with strings

do forever
    printMsg(0)      ;print first message
    wait 3 S         ;pause 3 seconds
    printMsg(2)      ;print next message
    wait 3 S         ;pause 3 seconds
    repeat 5         ;blink it five times
        LCDCmd(lcdOff) ;display off
        wait 500 mS   ;pause
        LCDCmd(lcdOn) ;display on
        wait 500 mS   ;pause
    end repeat
    wait 1 S         ;pause before next demo
    ;demonstrate panning
    printMsg(4)      ;print next message
    wait 3 S         ;pause 3 seconds
    repeat 16
        LCDCmd(panL) ;pan left a step at a time
        wait 300 mS  ;slow down to avoid blur
    end repeat
    repeat 16
        LCDCmd(panR) ;then pan right
        wait 300 mS
    end repeat
    wait 1 S         ;pause before next demo
    ;demonstrate moving the cursor
    printMsg(6)      ;print next message
    wait 3 S         ;pause 3 seconds
    doHome           ;home cursor
    LCDCmd(under)    ;choose underline cursor
    for ii = 0 to 15 ;move cursor across first line
        LCDCmd(line1+i)
        wait 200 mS
    next i
    for ii = 0 to 15 ;move cursor across second line
        LCDCmd(line2+i)
        wait 200 mS
    next i
    for ii = 15 to 0 step -1 ;move cursor back over second line
        LCDCmd(line2+i)
        wait 200 mS
    next i
    for ii = 15 to 0 step -1 ;move cursor back over first line
        LCDCmd(line1+i)
        wait 200 mS
    next i
    wait 3 S

```



```

;demonstrate blinking block cursor
printMsg(8)           ;print next message
doHome                ;home the cursor
LCDCmd(block)         ;choose blinking block cursor
wait 4 S              ;pause 4 seconds
LCDCmd(mode1)         ;change to one long line mode
doHome                ;home the cursor again
LCDCmd(curOff)        ;and disable it

;demonstrate scrolling a lengthy one-line marquee
for ii = 0xd0 to 0xff ;print next message - the remaining EEPROM
  ERead ii, char      ;fetch directly from eeprom
  print chr(char)
next i
wait 1 S
doHome                ;home cursor once more
repeat 141            ;scroll message twice
  LCDCmd(panR)
  wait 250 mS
end repeat
wait 2 S
LCDCmd(mode2)         ;change back to two line mode
doClr                 ;clear the screen
;demonstrate all of the characters
printMsg(11)          ;print next message
for ii = 33 to 127    ;print first batch of ASCII characters
  LCDCmd(line1+12)    ;overwrite each character displayed
  print chr(ii)        ;this is the ASCII code
  wait 500 mS
next i
for ii = 161 to 255   ;print next batch of ASCII characters
  LCDCmd(line1+12)
  print chr(ii)
  wait 500 mS
next i
;say good-bye
LCDCmd(line2)
printMsg(11)          ;print next message
doHome                ;home the cursor
loop

end

;----- Clear the screen
sub doClr
  LCDCmd(clrHome)
  wait 5 mS           ;this command takes extra time

```

```
end sub
```

```
;----- Home the cursor
```

```
sub doHome
```

```
    LCDCmd(home)
```

```
    wait 5 mS                ;and so does this one
```

```
end sub
```

```
;----- Print a message to the LCD
```

```
;The parameter 'row' points to the start of the string.
```

```
sub printMsg(in row as byte, in Optional StringLength As Byte = 15)
```

```
    LCDCmd(line1)            ;get set for first line
```

```
    for ii = 0 to StringLength
```

```
        index = row*16+ii
```

```
        EPread index, char    ;fetch next character and
```

```
        print chr(char)      ;transmit to the LCD
```

```
    next
```

```
    LCDCmd(line2)            ;get set for second line
```

```
    for ii = 0 to StringLength
```

```
        index = (row+1)*16+ii
```

```
        EPread index, char    ;fetch next character and
```

```
        print chr(char)      ;transmit to the LCD
```

```
    next
```

```
end sub
```

```
sub loadEeprom
```

```
    ' Strings for EEPROM, Strings should be limited to 16 characters for the first 13  
    sstrings, then a long string to fill eeprom
```

```
    location = 0
```

```
    WriteEeprom "First we'll show"
```

```
    WriteEeprom "this message.  "
```

```
    WriteEeprom "Then we'll blink"
```

```
    WriteEeprom "five times.  "
```

```
    WriteEeprom "Now lets pan  "
```

```
    WriteEeprom "left and right. "
```

```
    WriteEeprom "Watch the line  "
```

```
    WriteEeprom "cursor move.  "
```

```
    WriteEeprom "A block cursor  "
```

```
    WriteEeprom "is available.  "
```

```
    WriteEeprom "Characters:    "
```

```
    WriteEeprom "Bye!           "
```

```
    WriteEeprom "in one line mode"
```

```
    WriteEeprom "Next well scroll this long message as a marquee"
```

```
end sub
```

```

; Write to the device eeprom
sub WriteEeprom ( in Estring() ) as string * 64

    for ee = 1 to len ( Estring )
        HSersend Estring(ee)
        epwrite location, Estring(ee)
        location++
    next

end sub

```

LCD_IO 4 Example

This is a connection mode 4 Driver to demonstrate LCD features. This for the 16F877A, but, it can easily be adapted for other microcontrollers.

A 2 by 16 LCD is assumed.

```

#chip 16F877A,20

'Use LCD in 4 pin mode and define LCD pins
#define LCD_IO 4
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width

;----- Main Program

do forever

    Print "GCBASIC 2021"
    wait 3 s
    CLS

loop
end

```

LCD_IO 8 Example

This is an connection mode 8 Driver to demonstrate LCD features. This for the 16F877A, but, it can

easily be adapted for other microcontrollers.

A 2 by 16 LCD is assumed.

Based on the works by Thomas Henry and then revised Evan R. Venn

```
#chip 16F877A,20

'Use LCD in 8 pin mode and define LCD pins
#define LCD_IO 8
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_Data_Port PORTD
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width

;Here are various LCD commands which can be used.
;These are the LCD commands for the HD44780 controller
#define clrHome = 1 ;clear the display, home the cursor
#define home = 2 ;home the cursor only
#define RtoL = 4 ;print characters right to left
#define insR = 5 ;insert characters to right
#define LtoR = 6 ;print characters left to right
#define insL = 7 ;insert characters to left
#define lcdOff = 8 ;LCD screen off
#define lcdOn = 12 ;LCD screen on, no cursor
#define curOff = 12 ;an alias for the above
#define block = 13 ;LCD screen on, block cursor
#define under = 14 ;LCD screen on, underline cursor
#define undblk = 15 ;LCD screen on, blinking and underline cursor
#define CLeft = 16 ;cursor left
#define CRight = 20 ;cursor right
#define panR = 24 ;pan viewing window right
#define panL = 28 ;pan viewing window left
#define bus4 = 32 ;4-bit data bus mode
#define bus8 = 48 ;8-bit data bus mode
#define mode1 = 32 ;one-line mode (alias)
#define mode2 = 40 ;two-line mode
#define line1 = 128 ;go to start of line 1
#define line2 = 192 ;go to start of line 2
;----- Variables
dim char, msn, lsn, index, ii as byte
;----- Main Program
LoadEeprom ;load the EEprom with strings
```

```

do forever
    printMsg(0)                ;print first message
    wait 3 S                   ;pause 3 seconds
    printMsg(2)                ;print next message
    wait 3 S                   ;pause 3 seconds
    repeat 5                   ;blink it five times
        LCDCmd(lcdOff)        ;display off
        wait 500 mS           ;pause
        LCDCmd(lcdOn)         ;display on
        wait 500 mS           ;pause
    end repeat
    wait 1 S                   ;pause before next demo
    ;demonstrate panning
    printMsg(4)                ;print next message
    wait 3 S                   ;pause 3 seconds
    repeat 16
        LCDCmd(panL)          ;pan left a step at a time
        wait 300 mS           ;slow down to avoid blur
    end repeat
    repeat 16
        LCDCmd(panR)          ;then pan right
        wait 300 mS
    end repeat
    wait 1 S                   ;pause before next demo
    ;demonstrate moving the cursor
    printMsg(6)                ;print next message
    wait 3 S                   ;pause 3 seconds
    doHome                     ;home cursor
    LCDCmd(under)              ;choose underline cursor
    for ii = 0 to 15            ;move cursor across first line
        LCDCmd(line1+i)
        wait 200 mS
    next i
    for ii = 0 to 15            ;move cursor across second line
        LCDCmd(line2+i)
        wait 200 mS
    next i
    for ii = 15 to 0 step -1 ;move cursor back over second line
        LCDCmd(line2+i)
        wait 200 mS
    next i
    for ii = 15 to 0 step -1 ;move cursor back over first line
        LCDCmd(line1+i)
        wait 200 mS
    next i
    wait 3 S
    ;demonstrate blinking block cursor
    printMsg(8)                ;print next message

```

```

doHome           ;home the cursor
LCDCmd(block)    ;choose blinking block cursor
wait 4 S         ;pause 4 seconds
LCDCmd(mode1)    ;change to one long line mode
doHome           ;home the cursor again
LCDCmd(curOff)   ;and disable it

;demonstrate scrolling a lengthy one-line marquee
for ii = 0xd0 to 0xff ;print next message - the remaining EEPROM
  ERead ii, char      ;fetch directly from eeprom
  print chr(char)
next i
wait 1 S
doHome               ;home cursor once more
repeat 141           ;scroll message twice
  LCDCmd(panR)
  wait 250 mS
end repeat
wait 2 S
LCDCmd(mode2)        ;change back to two line mode
doClr                ;clear the screen
;demonstrate all of the characters
printMsg(11)         ;print next message
for ii = 33 to 127    ;print first batch of ASCII characters
  LCDCmd(line1+12)    ;overwrite each character displayed
  print chr(ii)        ;this is the ASCII code
  wait 500 mS
next i
for ii = 161 to 255   ;print next batch of ASCII characters
  LCDCmd(line1+12)
  print chr(ii)
  wait 500 mS
next i
;say good-bye
LCDCmd(line2)
printMsg(11)         ;print next message
doHome               ;home the cursor
loop
end

;----- Clear the screen
sub doClr
  LCDCmd(clrHome)
  wait 5 mS           ;this command takes extra time
end sub

;----- Home the cursor

```

```

sub doHome
  LCDCmd(home)
  wait 5 mS                ;and so does this one
end sub

```

```

;----- Print a message to the LCD
;The parameter 'row' points to the start of the string.
sub printMsg(in row as byte, in Optional StringLength As Byte = 15)
  LCDCmd(line1)            ;get set for first line

  for ii = 0 to StringLength
    index = row*16+ii
    EPread index, char      ;fetch next character and
    print chr(char)         ;transmit to the LCD
  next
  LCDCmd(line2)            ;get set for second line
  for ii = 0 to StringLength
    index = (row+1)*16+ii
    EPread index, char      ;fetch next character and
    print chr(char)         ;transmit to the LCD
  next
end sub

```

```

sub loadEeprom

```

' Strings for EEPROM, Strings should be limited to 16 characters for the first 13 sstrings, then a long string to fill eeprom

```

  location = 0
  WriteEeprom "First we'll show"
  WriteEeprom "this message.  "
  WriteEeprom "Then we'll blink"
  WriteEeprom "five times.    "
  WriteEeprom "Now lets pan   "
  WriteEeprom "left and right. "
  WriteEeprom "Watch the line  "
  WriteEeprom "cursor move.    "
  WriteEeprom "A block cursor  "
  WriteEeprom "is available.   "
  WriteEeprom "Characters:     "
  WriteEeprom "Bye!           "
  WriteEeprom "in one line mode"
  WriteEeprom "Next well scroll this long message as a marquee"
end sub

```

```

; Write to the device eeprom
sub WriteEeprom ( in Estring() ) as string * 64
  for ee = 1 to len ( Estring )

```

```
        HSersend Estring(ee)
        epwrite location, Estring(ee)
        location++
    next
end sub
```

LCD_IO 10 Example

LCD_IO 10 provides support for character LCD modules that use an I2C/TWI interface based on the **PCF8574 or PCF8574A I/O expander**. This for the 16F877A, but, it can easily be adapted for any other microcontrollers.

A 2 by 16 LCD is assumed with the LCD being driven using an LCD I2C adapter. Two types are supported the YwRobot LCD1602 IIC V1 / a Sainsmart LCD_PIC I2C adapter or the Ywmjkdz I2C adapter with pot bent over top of chip.

Example - Joy-IT/Raspberry Pi LCD PCB:

This demonstrates using Joy-IT/Raspberry Pi LCD PCB by changing the I2C_LCD_* constants that control this mode of LCD_IO operations.


```

#chip AVR128DA28, 24
#option explicit

// Define Hardware I2C settings
HI2CMode Master
#define HI2C_DATA PORTA.2
#define HI2C_CLOCK PORTA.3

#define LCD_IO 10
// LCD_I2C_Address_1 is not required because default address
// #define LCD_I2C_ADDRESS_1 0X4E

// This settings are suited for a joy-it adapter. Its made for the Raspberry Pi.
// This changes the default expander port mappings.
#define I2C_LCD_E I2C_LCD_BYTE.7
#define I2C_LCD_RW I2C_LCD_BYTE.5
#define I2C_LCD_RS I2C_LCD_BYTE.4
#define I2C_LCD_BL I2C_LCD_BYTE.6
#define I2C_LCD_D4 I2C_LCD_BYTE.0
#define I2C_LCD_D5 I2C_LCD_BYTE.1
#define I2C_LCD_D6 I2C_LCD_BYTE.2
#define I2C_LCD_D7 I2C_LCD_BYTE.3

// ----- Main body of program commences here.

Print "Hello World"

```

Example - Multiple LCDs:

This demonstrates reading a DS18B20 and showing the results on multiple LCDs.

```

#chip mega328p, 16
#include <DS18B20.h>

; ----- Define Hardware settings
' Define I2C settings - CHANGE PORTS
#define I2C_MODE Master
#define I2C_DATA PORTC.4
#define I2C_CLOCK PORTC.5
#define I2C_DISABLE_INTERRUPTS ON

'''Set up LCD
#define LCD_IO 10
#define LCD_I2C_ADDRESS_1 0x4E ; LCD 1
#define LCD_I2C_ADDRESS_2 0x4C ; LCD 2

```

```

; ----- Constants
' DS18B20 port settings - this is required
  #DEFINE DQ PortC.3

; ----- Quick Command Reference:

'''Set LCD_10 to 10 for the YwRobot LCD1602 IIC V1 or the Sainsmart LCD_PIC I2C
adapter
'''Set LCD_10 to 12 for the Ywmjkdz I2C adapter with pot bent over top of chip

; ----- Variables
dim TempC_100 as word    ' a variable to handle the temperature calculations
dim DSdataRaw as Integer

; ----- Main body of program commences here.

'Change to the correct LCD by setting    LCD_I2C_ADDRESS_Current to the correct
address then write to LCD.
LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_1: DisplayInformation ( 1 )
LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_2: DisplayInformation ( 1 )
wait 4 s
LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_1: CLS
LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_2: CLS

ccount = 0
Do forever

    ' The function readtemp12 returns the raw value of the sensor.
    ' The sensor is read as a 12 bit value therefore each unit equates to 0.0625 of a
degree
    DSdataRaw = readtemp12    ; save to this variable to prevent the delay bewtween
screen up dates
    ' The function readtemp returns the integer value of the sensor
    DSdata = readtemp

    LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_1: DisplayInformation ( 2 ) ; update
LCD1
    LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_2: DisplayInformation ( 2 ) ; update
LCD2
    DSdata = DSdataRaw ; Set the data
    LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_1: DisplayInformation ( 3 ) ; update
LCD1
    DSdata= DSdataRaw ; Set the data
    LCD_I2C_ADDRESS_Current = LCD_I2C_ADDRESS_2: DisplayInformation ( 3 ) ; update
LCD2

```

```

ccount++

wait 1 s

loop
End

Sub DisplayInformation ( LCDCommand )

    Select case LCDCommand

    Case 1
        CLS
        print "GCBASIC 2021"
        locate 1,0
        print "DS18B20 Demo"

    Case 2
        ' Display the integer value of the sensor on the LCD
        locate 0,0
        print hex(ccount)
        print " Ceil"
        locate 0,8
        print DSdata
        print chr(223)+"C"+" "

    Case 3

        ' Display the integer and decimal value of the sensor on the LCD

        SignBit = DSdata / 256 / 128
        If SignBit = 0 Then goto Positive
        ' its negative!
        DSdata = ( DSdata # 0xffff ) + 1 ' take twos comp

    Positive:

        ' Convert value * 0.0625. Multitple value by 6 then add result to
multiplication of the value with 25 then divide result by 100.
        TempC_100 = DSdata * 6
        DSdata = ( DSdata * 25 ) / 100
        TempC_100 = TempC_100 + DSdata

        Whole = TempC_100 / 100
        Fract = TempC_100 % 100
        If SignBit = 0 Then goto DisplayTemp
        Print "-"

```

```
DisplayTemp:
  locate 1,0
  print hex(ccount)
  print " Real"
  locate 1,8
  print str(Whole)
  print "."
  ' To ensure the decimal part is two digits
  Dig = Fract / 10
  print Dig
  Dig = Fract % 10
  print Dig
  print chr(223)
  print "C"+" " "
```

```
End Select
```

```
end sub
```

Pulse width modulation

This is the Pulse width modulation section of the Help file. Please refer the sub-sections for details using the contents/folder view for the MicroChip PIC PWM capabilities and the ATMEL AVR PWM capabilities.

Microchip PIC PWM Overview

Introduction:

The methods described in this section allow the generation of Pulse Width Modulation (PWM) signals. PWM signals enables the microcontroller to control items like the speed of a motor, or the brightness of a LED or lamp.

The methods can also be used to generate the appropriate frequency signal to drive an infrared LED for remote control applications.

GCBASIC support the four different method shown below:

- Two methods use the microcontroller CCP module
- One method uses the microcontroller PWM module, and
- One method is a software emulation of PWM.

Hardware PWM using a CCP module

Using PWM with the CCP module: This option requires a CCP module within the microcontroller.

Hardware PWM is only available through the "CCP" or "CCPx" pin. This is a hardware limitation of Microchip PIC microcontrollers.

Microcontrollers with PPS can change the pin - use the PPS tool to set the desired output pin.

This method uses three parameters to setup the PWM.

```
'HPWM channel, frequency, duty cycle  
HPWM 1, 76, 80
```

Hardware PWM using a PWM module

Using microcontroller PWM module. This option requires a PWM module within the microcontroller. Microcontrollers with PPS can change the pin - use the PPS tool to set the desired output pin.

This method uses four parameters to setup the PWM.

```
'HPWM channel, frequency, duty cycle, timer  
HPWM 5, 76, 80, 2
```

Hardware PWM using the CCP1 in fixed mode

Using Hardware PWM on fixed mode PWM requires a CCP1 module.

The fixed mode can use CCP1 only, and, the parameters of the PWM cannot be dynamically changed in the user program. The parameters are fixed by the definition of two constants.

```
#define PWM_Freq 76      'Set frequency in KHz
#define PWM_Duty 80      'Set duty cycle to 80 %

HPWMOn

wait 5 s

HPWMOff
```

Software PWM

Using Software PWM on requires no specific modules with the microcontroller.

The PWM parameters for duty and the number of pulses can be changed dynamically in the user program.

The PWM is **only** operational for the number of cycles stated in the calling method.

```
'A call to use the software PWM on the specific port, with a duty of 127 for 100
cycles

; ----- Constants
'PWM constant. This is a required constant.
#define PWM_Out1 portb.0

; ----- Define Hardware settings
'PWM port out. This is not required but a good practice.
dir PWM_Out1 out

'Pulse the PWM
PWMOut 1, 127, 100
```

Relevant Constants:

A number of constants are used to control settings for the PWM hardware module of the microcontroller. To set them, place a line in the main program file that uses #define to assign a value to the particular constant.

PWM Software Mode

Syntax:

```
PWMOut channel, duty cycle, cycles
```

Command Availability:

Available on all microcontrollers. This method does NOT require a PWM module within the microcontroller.

This command uses a software PWM routine within GCBASIC to produce a PWM signal on the selected port of the chip.

The method **PWMOut** does not make use of any special hardware within the microcontroller. The PWM signal is generated only while the **PWMOut** command is executing - therefore, when the **PWMOut** is not executing by moving onto the next command, the PWM signal will stop.

For more help, see [PWMOut](#)

PWMOut

Syntax:

```
PWMOut channel, duty cycle, cycles
```

Command Availability:

Available on all microcontrollers. This method does NOT require a PWM module within the microcontroller.

This command uses a software PWM routine within GCBASIC to produce a PWM signal on the selected port of the chip.

The method **PWMOut** does not make use of any special hardware within the microcontroller. The PWM signal is generated only while the **PWMOut** command is executing - therefore, when the **PWMOut** is not executing by moving onto the next command, the PWM signal will stop.

Explanation :

channel sets the channel that the PWM is to be generated on. This must have been defined previously by setting the constants **PWM_Out1**

PWM_Out2, **PWM_Out3** or **PWM_Out4**. The maximum number of channels available is 4.

duty cycle specifies the PWM duty cycle, and ranges from 0 to 255. 255 corresponds to 100%, 127 to

50%, 63 to 25%, and so on.

`cycles` is used to set the amount of PWM pulses to supply. This is useful for situations in which a pulse of a specific length is required.

The formula for calculating the time taken for one cycle is:

$$T_{\text{CYCLE}} = (28 + 10C)T_{\text{OSC}} + (255 * \text{PWM_Delay})$$

where:

- C is the number of channels used
- T_{OSC} is the length of time taken to execute 1 instruction on the chip (0.2 us on a 20 MHz chip, 1 us on a 4 Mhz chip)
- PWM_Delay is a length of time specified using the `PWM_DeLay` constant

Example 1 :


```

'This program controls the brightness of an LED on PORTB.0
'using the software PWM routine and a potentiometer.
#chip 16f877a, 20

; ----- Constants
'PWM constant. This is a required constant.
#define PWM_Out1 portb.0

; ---- Optional Constant to add an delay after PWM pulse
''#Define PWM_Delay 1 us

; ----- Define Hardware settings
'PWM port out. This is not required but good practice.
dir PWM_Out1 out

'A potentiometer is attached to AN0

; ----- Variables
' No Variables specified in this example.

; ----- Main body of program commences here.
do
    '100 cycles is a purely arbitrary value as the loop will maintain a relatively
constant PWM
    PWMOut 1, ReadAD(AN0), 100
loop
end

```

Example 2 :

```

'This program controls the brightness of an LED on gpio.1
'using the software PWM routine and a potentiometer.
#chip 12f675, 4

; ----- Constants
'PWM constant. This is a required constant.
#define PWM_Out1 gpio.1

; ----- Define Hardware settings
'PWM port out. This is not required but good practice.
dir PWM_Out1 out

'A potentiometer is attached to AN0

; ----- Variables
' No Variables specified in this example.

; ----- Main body of program commences here.
do
    '100 cycles is a purely arbitrary value
    PWMOut 1, ReadAD(AN0), 100
loop
end

```

HPWM CCP

Syntax:

```
HPWM channel, frequency, duty cycle
```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture/Compare/PWM (CCP) module.

This command supports the CCP 8 bit support with **Timer 2 only**.

For CCP/PWM support for timers 2, 4 and 6, if the specific devices supports alternative CCP/PWM clock sources - see here [HPWM_CCPTimerN](#)

For PWM 10 Bit support with other timers - see here [HPWM 10 Bit](#)

Explanation:

This command sets up the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform of the given frequency and duty cycle.

If you only need one particular frequency and duty cycle, you should use `PWMOn` and the constants `PWM_Freq` and `PWM_Duty` instead.

`channel` is 1, 2, 3, 4 or 5, and corresponds to the pins CCP1, CCP2, CCP3, CCP4 and CCP5 respectively. On chips with only one CCP port, pin CCP or CCP1 is always used, and `channel` is ignored. (It should be set to 1 anyway to allow for future upgrades to more powerful microcontrollers.)

`frequency` sets the frequency of the PWM output. It is measured in KHz. The maximum value allowed is 255 KHz. The minimum value varies depending on the clock speed. 1 KHz is the minimum on chips 16 MHz or under and 2 KHz is the lowest possible on 20 MHz chips. In situations that do not require a specific PWM frequency, the PWM frequency should equal approximately 1 five-hundredth the clock speed of the microcontroller (ie 40 KHz on a 20 MHz chip, 16 KHz on an 8 MHz chip). This gives the best duty cycle resolution possible.

`duty cycle` specifies the desired duty cycle of the PWM signal, and ranges from 0 to 255 where 255 is 100% duty cycle.

To stop the PWM signal use the `HPWMOff` method with the parameter of the channel.

```
'Stop the CCP/PWM signal  
HPWMOff ( 1 )
```

The optional constant `HPWM_FAST` can be defined to enable the recalculation of the timer prescaler when needed. This will provide faster operation, but uses extra byte of RAM and may cause problems if `HPWM` and `PWMOn` are used together in a program. This will not cause any issue when using `HPWM` and `PWMOff` in the same program with `HPWM_FAST`.

The optional constant `DisableCCPFixedModePWM` can be defined to prevent Timer2 from being enabled. This constant may be required when you need to use Timer2 for non-CCP/PWM support.

Example:

```
'This program will alter the brightness of an LED using  
'CCP/PWM.
```

```
'Select chip model and speed  
#chip 16F877A, 20
```

```
'Set the CCP1 pin to output mode  
DIR PORTC.2 out
```

```
'Main code  
do  
    'Turn up brightness over the range  
    For Bright = 1 to 255  
        HPWM 1, 40, Bright  
        wait 10 ms  
    next  
    'Turn down brightness over the range  
    For Bright = 255 to 1 Step -1  
        HPWM 1, 40, Bright  
        wait 10 ms  
    next  
loop
```

HPWMUpdate for CCP/PWM Modules(s)

Syntax:

```
HPWMUpdate ( channel, duty_cycle )
```

Command Availability:

Available on Microchip PIC microcontrollers with the CCP module.

Explanation:

This command updates the **duty cycle only**.

- You **MUST** have previously called the HPWM CCP command using the full command to set the channel specific settings for frequency and timer source. See the example below for the usage.
- You **MUST** specify the constant `#define HPWM_FAST` to support HPWMUpdate when using CCP module.

This command only supports the previously called HPWM CCP command, or, if you have set more than one HPWM CCP channel then to use the command you must have set the channel to the same frequency.

The command only supports the CCP module of the Microchip PIC microcontroller to generate a PWM waveform at the previously defined frequency and timer source.

channel is 1, 2, 3, 4 or 5. These corresponds to the CCP1 through to CCP5 respectively. The channel **MUST** be supported by the microcontroller. Check the microcontroller specific datasheet for the available channel.

duty cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 255 where 255 is 100% duty cycle.

Example for CCP PWM:

```
'This program will alter the brightness of an LED using
'hardware PWM.

#chip 16F1938
#option Explicit

'Set the direction of the CCP/PWM port
DIR portc.2 Out

#define HPWM_FAST          'Required to support HPWMUpdate when using CCP module
HPWM 1, 40, dutyvalue

do
    'use for-loop to show the duty changing a 8bit value
    dim dutyvalue as byte
    for dutyvalue = 0 to 255
        HPWMUpdate 1, dutyvalue
        wait 10 ms
    next
    for dutyvalue = 254 to 1
        HPWMUpdate 1, dutyvalue
        wait 10 ms
    next
loop
```

For more help, see [PWMOFF](#)

HPWMOFF

Syntax:

```
HPWMOff ( channel ) 'selectively turn off the CCP channel
```

```
HPWMOff 'turn off CCP channel 1 only
```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture&Compare/PWM (CCP) modules.

Explanation:

This command will disable the output of the CCP1/PWM module on the Microchip PIC chip.

Example:

```
'Select chip model and speed
#chip 16F877A, 20

'Set the CCP1 pin to output mode
DIR PORTC.2 out

'Main code
do
  'Turn up brightness over 2.5 seconds
  For Bright = 1 to 255
    HPWM 1, 40, Bright
    wait 10 ms
  next

  wait 1 s
  HPWMOff ( 1 ) ' turn off the PWM channel

loop
```

For more help, see [HPWMOff](#)

HPWM_CCPTimerN**Syntax:**

```
HPWM_CCPTimerN channel, frequency, duty cycle [, timer 2, 4 or 6 ]
```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture/Compare/PWM (CCP) module.

This command supports the CCP 8 bit support with selectable **Timer 2, 4 or 6 only** for CCP/PWM only.

For CCP/PWM support for timers 2 only see [HPWM CCPTimer](#)

Explanation:

This command sets up the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform of the given frequency and duty cycle.

If you only need one particular frequency and duty cycle, you should use PWMOn and the constants PWM_Freq and PWM_Duty instead.

channel is 1, 2, 3, 4 or 5, and corresponds to the pins CCP1, CCP2, CCP3, CCP4 and CCP5 respectively. On chips with only one CCP port, pin CCP or CCP1 is always used, and **channel** is ignored. (It should be set to 1 anyway to allow for future upgrades to more powerful microcontrollers.)

frequency sets the frequency of the PWM output. It is measured in KHz. The maximum value allowed is 255 KHz. The minimum value varies depending on the clock speed. 1 KHz is the minimum on chips 16 MHz or under and 2 KHz is the lowest possible on 20 MHz chips. In situations that do not require a specific PWM frequency, the PWM frequency should equal approximately 1 five-hundredth the clock speed of the microcontroller (ie 40 KHz on a 20 MHz chip, 16 KHz on an 8 MHz chip). This gives the best duty cycle resolution possible.

duty cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 255 where 255 is 100% duty cycle.

timer specifies the timer source. Timers 2, 4 and 6 are supported.

To stop the PWM signal use the **HPWMOff** method with the parameter of the channel.

```
'Stop the CCP/PWM signal  
HPWMOff ( 1 )
```

Example:

```

#chip 16F1825, 4

DIR portc Out
DIR porta Out

initialisation:

'Command as follows:
' HPWM_CCPTimerN  CCP_Channel, Frequency, Duty, Timer Source.  Timer source defaults
to timer 2, so, the timersource is optional.

HPWM_CCPTimerN  3, 30, 77 , 4      'CCP/PWM module 3 using timer 4
HPWM_CCPTimerN  4, 40, 102, 6      'CCP/PWM module 4 using timer 6
HPWM  1, 10, 26                      'CCP/PWM module 1 with no parameter therefore
timer 2

do
loop

```

HPWMOff

Syntax:

```

HPWMOff ( channel )  'selectively turn off the CCP channel

HPWMOff              'turn off CCP channel 1 only

```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture&Compare/PWM (CCP) modules.

Explanation:

This command will disable the output of the CCP1/PWM module on the Microchip PIC chip.

Example:


```

'Select chip model and speed
#chip 16F877A, 20

'Set the CCP1 pin to output mode
DIR PORTC.2 out

'Main code
do
    'Turn up brightness over 2.5 seconds
    For Bright = 1 to 255
        HPWM 1, 40, Bright
        wait 10 ms
    next

    wait 1 s
    HPWMOff ( 1 )' turn off the PWM channel

loop

```

For more help, see [HPWMOff](#)

HPWM 10 Bit

Syntax:

```
HPWM channel, frequency, duty cycle, timer [, resolution]
```

Command Availability:

Only available on Microchip PIC microcontrollers with the 10-bit PWM module.

For the Capture/Compare/PWM (CCP) module, see here [HPWM CCP](#)

Explanation:

This command sets up the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform of the given frequency and duty cycle. Once this command is called, the PWM will be emitted until PWMOff is called.

channel is 1, 2, 3, 4, 5, 6, 7 or 8. These corresponds to the HPWM1 through to HPWM8 respectively. The 10-bit PWM channel **MUST** be supported by the microcontroller. Check the microcontroller specific datasheet for the available channel.

frequency sets the frequency of the PWM output. It is measured in KHz. The maximum value allowed is 255 KHz. The minimum value varies depending on the clock speed. 1 KHz is the minimum on chips 16

MHz or under and 2 KHz is the lowest possible on 20 MHz chips. In situations that do not require a specific PWM frequency, the PWM frequency should equal approximately 1 five-hundredth the clock speed of the microcontroller (ie 40 KHz on a 20 MHz chip, 16 KHz on an 8 MHz chip). This gives the best duty cycle resolution possible.

duty_cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 1023 where 1023 is 100% duty cycle. This should be a WORD value. Note: Byte values are supported as a Byte value is factorised to a Word value. To use a Byte value and to ensure the 10-bit resolution you should cast the parameter as a Word, [WORD]byte_value or [WORD]constant_value

timer specifies the desired timer to be used. These can be timer 2, 4 or 6.

Optional **resolution** specifies the desired resolution to be used. These can be either 255 or 1023. The rationale of this optional parameter is to support the duty cycle with a BYTE or a WORD range. If you call the method with a WORD the resolution will be set to 1023.

Notes:

PWM channels 1 and 2 are disabled by default. You must enable using the constants USE_HPWMn where n is the PWM channel you want to enable. You can disable any PWM channel by setting the appropriate change to FALSE.

On some microcontrollers you may need to set the port.pin as an output for PWM to operate as desired.

```
#define USE_HPWM1 TRUE
#define USE_HPWM2 TRUE
```

Example 1:

```

'This program will alter the brightness of an LED using
'hardware PWM.

'Select chip model and speed
#chip 16F18855, 32

'Generated by PIC PPS Tool for GCBASIC
,
'Template comment at the start of the config file
,
#startup InitPPS, 85

Sub InitPPS

    'Module: PWM6
    RA2PPS = 0x000E    'PWM6OUT > RA2

End Sub
'Template comment at the end of the config file

'Set the PWM pin to output mode
DIR PORTA.2 out

dim Bright as word

'Main code
do
    'Turn up brightness over the range
    For Bright = 0 to 1023
        HPWM 6, 40, Bright, 2
        wait 10 ms
    next
    'Turn down brightness over the range
    For Bright = 1023 to 0 Step -1
        HPWM 6, 40, Bright, 2
        wait 10 ms
    next
loop

```

Example 2:

```

'This program will alter the brightness of an LED using
'hardware PWM.

'Select chip model and speed
#chip 16F1705, 32

'Generated by PIC PPS Tool for GCBASIC
,
'Template comment at the start of the config file
,
#startup InitPPS, 85

Sub InitPPS

    'Module: PWM3
    RA2PPS = 0x000E    'PWM3OUT > RA2

End Sub
'Template comment at the end of the config file

'Set the PWM pin to output mode
DIR PORTA.2 out

dim Bright as word

'Main code
do
    'Turn up brightness over the range
    For Bright = 0 to 1023
        HPWM 3, 40, Bright, 2
        wait 10 ms
    next
    'Turn down brightness over the range
    For Bright = 1023 to 0 Step -1
        HPWM 3, 40, Bright, 2
        wait 10 ms
    next
loop

```

For more help, see [PWMOff](#)

HPWMUpdate for PWM Module(s)

Syntax:

```
HPWMUpdate ( channel, duty_cycle )
```

Command Availability:

Available on Microchip PIC microcontrollers with the PWM module.

Explanation:

This command updates the **duty cycle only**.

- You **MUST** have previously called the HPWM 10 Bit command using the full command (see [HPWM 10 Bit](#)) to set the channel specific settings for frequency and timer source. See the example below for the usage.
- You **MUST** have previously called the HPWM 10 Bit command with the same type of variable, or, use casting to ensure the variable type is the same type.

This command only supports the previously called HPWM 10 Bit command, or, if you have set more than one HPWM 10 Bit PWM channel then to use the command you must have set the channel to the same frequency.

The command only supports the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform at the previously defined frequency and timer source.

channel is 1, 2, 3, 4, 5, 6, 7 or 8. These corresponds to the HPWM1 through to HPWM8 respectively. The channel **MUST** be supported by the microcontroller. Check the microcontroller specific datasheet for the available channel.

duty cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 1023 where 1023 is 100% duty cycle.

Example for Hardware PWM:

```

'This program will alter the brightness of an LED using
'hardware PWM.

'Select chip model and speed
#chip 16F18855, 32

'Generated by PIC PPS Tool for GCBASIC
,
'Template comment at the start of the config file
,
#startup InitPPS, 85

Sub InitPPS

    'Module: PWM6
    RA2PPS = 0x000E    'PWM6OUT > RA2

End Sub
'Template comment at the end of the config file

'Set the PWM pin to output mode
DIR PORTA.2 out

'Setup PWM - this is mandated as this specifies the frequency and the clock source.
'Uses casting [word] to ensure the intialisation value of Zero (0) is a treated as a
word. The variable type MUST match the HPWMUpdate variable type.
HPWM 6, 40, [word]0, 2
'Main code
do
    'Turn up brightness over 2.5 seconds
    For Bright = 0 to 1023
        HPWMUpdate 6, Bright
        wait 10 ms
    next
    'Turn down brightness over 2.5 seconds
    For Bright = 1023 to 0 Step -1
        HPWMUpdate 6, Bright
        wait 10 ms
    next
loop

```

For more help, see [PWMOFF](#), [HPWM 10 Bit](#)

HPWMOff

Syntax:

```
HPWMOff ( channel, PWMHardware )
```

Command Availability:

Only available on Microchip PIC microcontrollers with PWM modules.

Explanation:

This command will disable the output of the PWM module on the Microchip PIC chip.

PWMHardware is a GCBASIC defined constant not a user variable.

Example:

```

'This program will alter the brightness of an LED using
'hardware PWM.

'Select chip model and speed
#chip 16F18855, 32

'Generated by PIC PPS Tool for GCBASIC
,
'Template comment at the start of the config file
,
#startup InitPPS, 85

Sub InitPPS

    'Module: PWM6
    RA2PPS = 0x000E    'PWM6OUT > RA2

End Sub
'Template comment at the end of the config file

'Set the PWM pin to output mode
DIR PORTA.2 out

'Main code
For ForLoop = 1 to 4
    'Turn up brightness over 2.5 seconds
    For Bright = 1 to 255
        HPWM 6, 40, Bright, 2
        wait 10 ms
    next
    'Turn down brightness over 2.5 seconds
    For Bright = 255 to 1 Step -1
        HPWM 6, 40, Bright, 2
        wait 10 ms
    next
next

HPWMOff 6, PWMHardware    'where PWMHardware is the defined constant or you can use
TRUE

```

HPWM 16 Bit

Syntax:

HPWM16 channel, frequency, duty cycle	'Enable a 16-bit PWM channel'
HPWM16On channel parameters set by the HPWM16 method'	'Enable a specific PWM channel using
HPWM16Off channel	'Disable a specific PWM channel'

Command Availability:

Only available on Microchip PIC microcontrollers with the 16-bit PWM module. 16-bit PWM support includes both dynamic mode and fixed mode operations. See the examples below for usage.

The PIC microcontroller chip specific DAT file must contain **CHIPPWM16TYPE = 1**. If the chip specific DAT does not contain **CHIPPWM16TYPE = 1** and the microcontroller does support PWM 16 Bit please report the omission to GCBASIC the support forum.

For the Capture/Compare/PWM (CCP) module or the 10-bit PWM module, see the other sections of the Help.

Explanation:

This command sets up the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform of the given frequency and duty cycle. Once this command is called, the PWM will be emitted until HPWM16Off method is called.

channel is 1, 2, 3.. 12. These corresponds to the 16-bit PWM channel respectively.

The 16-bit PWM channel **MUST** be supported by the microcontroller. Check the microcontroller specific datasheet for the available channel.

frequency sets the frequency of the PWM output. It is measured in KHz. The maximum value allowed is 0xFFFF. The minimum value varies depending on the clock speed. 1 KHz is the minimum on chips 16 MHz or under and 2 KHz is the lowest possible on 20 MHz chips. In situations that do not require a specific PWM frequency, the PWM frequency should equal approximately 1 five-hundredth the clock speed of the microcontroller (ie 40 KHz on a 20 MHz chip, 16 KHz on an 8 MHz chip). This gives the best duty cycle resolution possible.

duty cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 0xFFFF where 0xFFFF is 100% duty cycle. This should be a WORD value.

Example 1:

```

' This program will enable dynamic mode PWM signals
,
' All the 12 PWM16 channels can configured at separate dynamic frequencies  dynamic
duty, the syntax is:
,
' HPWM16(xx, frequency, duty )
,
' xx can be 1 through 12, for this specific microcontroller there are three PWM16
channels.
,
' To set the parameters of GCBASIC PWM fixed mode for the channels use the commands
shown below::

#chip 12F1572, 32
#config mclr=on

Dir PORTA Out

HPWM16(1, 30, 16384)  '30 kHz, 25% duty cycle (16384/65535)
HPWM16(2, 30, 16384)  '30 kHz, 25% duty cycle (16384/65535)
HPWM16(3, 30, 16384)  '30 kHz, 25% duty cycle (16384/65535)

do Forever
loop

#define USE_HPWM16_1 TRUE
#define USE_HPWM16_2 TRUE
#define USE_HPWM16_3 TRUE
#define USE_HPWM16_4 FALSE
#define USE_HPWM16_5 FALSE
#define USE_HPWM16_6 FALSE
#define USE_HPWM16_7 FALSE
#define USE_HPWM16_8 FALSE
#define USE_HPWM16_9 FALSE
#define USE_HPWM16_10 FALSE
#define USE_HPWM16_11 FALSE
#define USE_HPWM16_12 FALSE

```

The 16-bit library also supports fixed mode PWM operations. The following two examples show the constants and the commands to control 16-bit PWM Fixed Mode operations.

Example 2:

```

' This program will enable fix mode PWM signals
,
' All the 12 PWM16 channels can configured at separate fixed frequencies and fixed
duty, the syntax is:
,
' #define HPWM16_xx_Freq 38      'Set frequency in KHz on channel xx
' #define HPWM16_xx_Duty 50      'Set duty cycle to 50% on channel xx
,
' xx can be 1 through 12
,
' To set the parameters of GCBASIC PWM fixed mode on channel 1 use the following:
,
' #define HPWM16_1_Freq 0.1 to > 1000      'Set the frequency, but, the clock
speed must be low for low PWM frequency
' #define HPWM16_1_Duty 0.1 to 100          'Set duty cycle as percentage 0-
100%, just change the number
,

#chip 12F1572, 32
#config mclr=on

Dir PORTA Out

#define HPWM16_1_Freq 400      '800Hz to greater than 1mhz... greater than
1mhz at a clock speed of 32hz provides a clipped square wave.
#define HPWM16_1_Duty 50
HPWM16On ( 1 )

do Forever
loop

```

Example 3:

```

' This program will enable fix mode PWM signals
,
' All the 12 PWM16 channels can configured at separate fixed frequencies and fixed
duty, the syntax is:
,
' #define HPWM16_xx_Freq 38      'Set frequency in KHz on channel xx
' #define HPWM16_xx_Duty 50      'Set duty cycle to 50%  on channel xx
,
' xx can be 1 through 12, for this specific microcontroller there are three PWM16
channels.
,
' To set the parameters of GCBASIC PWM fixed mode for the three channels use the
following:

    #chip 12F1572, 32
    #config mclr=on

    Dir PORTA Out

    #define HPWM16_1_Freq 100      '100khz
    #define HPWM16_1_Duty 40      '40% duty
    HPWM16On ( 1 )

    #define HPWM16_2_Freq 200      '200khz
    #define HPWM16_2_Duty 50      '50% duty
    HPWM16On ( 2 )

    #define HPWM16_3_Freq 300      '300khz
    #define HPWM16_3_Duty 60      '60% duty
    HPWM16On ( 3 )

    do Forever
    loop

```

For more help, see [PWMOff](#)

HPWM Fixed Mode

Syntax:

```

PWMOn          'only applies to CCP/PWM channel 1
'or
PWMOff

PWMOn( channel )      'where the parameter can be any valid CCP/PWM channel, 1,
2, 3, 4 or 5
'or
PWMOff( channel )

PWMOn( module_number , PWMModule)      'where the parameter can be any valid
PWM channel 1 .. 9
'or
PWMOff( module_number , PWMModule)

```

Command Availability:

Only available on Microchip PIC microcontrollers with a CCP/PWM or PWM module.

See here [HPWM CCP](#) for the method to change PWM parameters dynamically or to use other CCP channels - this method support CCP1/PWM, CCP2/PWM, CCP3/PWM, CCP4/PWM and CCP5/PWM.

Explanation:

This command sets up the hardware PWM module of the Microchip PIC microcontroller to generate a PWM waveform of the given frequency and duty cycle. Once this command is called, the PWM will be emitted until PWMOff is called.

These constants are required to set the parameters for the PWM. The frequency and the duty applies to all channels when using the method(s) or macro(s) shown above.

Constant Name	Controls	Default Value
<code>PWM_Freq</code>	Specifies the output frequency of the PWM module in KHz.	38
<code>PWM_Duty</code>	Sets the duty cycle of the PWM module output. Given as percentage.	50

For CCP/PWM modules are also supported using a call to a method or a macro, as follows:

Method/Macro	Controls	Default Value
PWMOn	No parameter enables CCP1/PWM only	No parameter
PWMOff	Disables CCP1/PWM only	

Method/Macro	Controls	Default Value
PWMOn(channel)	Where the parameter is any valid CCP/PWM channel	channel can be 1, 2, 3, 4 or 5
PWMOff(channel)	Where the parameter is any valid CCP/PWM channel	channel can be 1, 2, 3, 4 or 5
PWMOn(module , PWMMODULE)	Where the parameter is any valid PWM module	module can be 1..9 See the example below for the constants to control fixed mode PWM using PWM modules.
PWMOff(channel , PWMMODULE)	Where the parameter is any valid CCP/PWM module	module can be 1..9

Fixed Mode PWM for PWM Modules.

To set the Fixed Mode PWM for PWM Modules you need to set a timer frequency, a PWM module cycle and the PWM model source clock.

The options for source clock are shown below. These are the PWM timers supported by the PWM modules, where **nn** is the frequency.

```
PWM_Timer2_Freq `nn` or
PWM_Timer4_Freq `nn` or
PWM_Timer6_Freq `nn`.
```

The PWM module duty is set using PWM_`yy`_Duty **xx** where **yy** is between 1 and 9 and is a valid PWM module, and, **xx** is the Duty cycle for specific channels

```
#define PWM_yy_Duty xx
```

The PMW module clock source us PWM_`zz`_Clock_Source **tt**. Where **zz** is channel and **tt** is the PWM clock source.

```
#define PMW_zz_Clock_Source tt
```

You do not need to define all the timers and or all the channels, just define the constants you need.

The minimum is A timer with a frequency A PWM channel with a duty A PWM channel clock source

Example: For PWM channel 6 with a frequency of 38Khz with a duty of 50% with a clock source of timer 2, use

```
#define PWM_Timer2_Freq 38
#define PWM_7_Duty 50
#define PMW_7_Clock_Source 6
```

Details of the constants with example parameters.

```
#define PWM_Timer2_Freq 20      'Set frequency in KHz, just change the number
#define PWM_Timer4_Freq 40      'Set frequency in KHz, just change the number
#define PWM_Timer6_Freq 60      'Set frequency in KHz, just change the number
```

Supported PWM modules, with example parameters.

```
#define PWM_1_Duty 10           'Set duty cycle as percentage 0-100%, just change the
number
#define PMW_1_Clock_Source 2
```

```
#define PWM_2_Duty 20
#define PMW_2_Clock_Source 4
```

```
#define PWM_3_Duty 30
#define PMW_3_Clock_Source 6
```

```
#define PWM_4_Duty 40
#define PMW_4_Clock_Source 2
```

```
#define PWM_5_Duty 50
#define PMW_5_Clock_Source 6
```

```
#define PWM_6_Duty 60
#define PMW_6_Clock_Source 6
```

```
#define PWM_7_Duty 70
#define PMW_7_Clock_Source 4
```

```
#define PWM_8_Duty 80
#define PMW_8_Clock_Source 4
```

```
#define PWM_9_Duty 90
#define PMW_9_Clock_Source 6
```

Example #1:

Enable CCP1/PWM channel only. This is the legacy command.

```
#chip 16f877a,20

'Set the PWM pin to output mode
DIR PORTC.2 out

'Main code

#define PWM_Freq 38      'Frequency of PWM in KHz
#define PWM_Duty 50      'Duty cycle of PWM (%)

PWMOn    'Will enable CCP1/PWM Only

wait 10 s          'Wait 10 s

PWMOff    'Will disable CCP1/PWM Only

do
loop
```

Example #2:

Enable any CCP/PWM channel using a call to a method.


```

#chip 16f877a,20

'Set the PWM pin to output mode
DIR PORTC.2 out

'Main code

#define PWM_Freq 38      'Frequency of PWM in KHz
#define PWM_Duty 50      'Duty cycle of PWM (%)

PWMOn (2)      'Will enable any valid CCP/PWM channel

wait 10 s      'Wait 10 s

PWMOff (2)     'Will disable any valid CCP/PWM channel

do
loop

```

Example #3:*

Enable any PWM module using a PWM specific method.

```

'A real simple and easy PWM setup for 8 and 10 bit PWM channels
#chip 18f25k42, 16

#startup InitPPS, 85

Sub InitPPS

    'Module: PWM5
    RA0PPS = 0x000D      'PWM5 > RA0
    'Module: PWM6
    RA1PPS = 0x000E      'PWM6 > RA1
    'Module: PWM7
    RA2PPS = 0x000F      'PWM7 > RA2
    'Module: PWM8
    RA3PPS = 0x0010      'PWM8 > RA3

End Sub

'Template comment at the end of the config file
dir porta Out
dir portb Out
dir portc Out

```

'This is the setup section for fixed mode PWM

'The only options are PWM_Timer2_Freq nn|PWM_Timer4_Freq nn|PWM_Timer6_Freq nn.
These are the PWM timers

'The PWM_yy_Duty xx' where yy is between 1 and 9 and is a valid PWM module, and,
xx is the Duty cycle for specific channels

'The PWM_zz_Clock_Source tt. Where zz is channel and tt is the PWM clock source.

'You do not need to define all the timers and channels, just define the constants
you need.

'The minimum is

' A timer with a frequency

' A PWM channel with a duty

' A PWM channel clock source

' For PWM channel 2 with a frequency of 38Khz with a duty of 50% with a clock
source of timer 2, use

' #define PWM_Timer2_Freq 38

' #define PWM_7_Duty 50

' #define PWM_7_Clock_Source 2

#define PWM_Timer2_Freq 20 'Set frequency in KHz, just change the number

#define PWM_Timer4_Freq 40 'Set frequency in KHz, just change the number

#define PWM_Timer6_Freq 60 'Set frequency in KHz, just change the number

' Supported PWM module but not by this specific microcontroller

,

' #define PWM_1_Duty 10 'Set duty cycle as percentage 0-100%, just
change the number

' #define PWM_1_Clock_Source 2

,

' #define PWM_2_Duty 20

' #define PWM_2_Clock_Source 4

,

' #define PWM_3_Duty 30

' #define PWM_3_Clock_Source 6

,

' #define PWM_4_Duty 40

' #define PWM_4_Clock_Source 2

#define PWM_5_Duty 50

#define PWM_5_Clock_Source 6

#define PWM_6_Duty 60

#define PWM_6_Clock_Source 6

#define PWM_7_Duty 70

#define PWM_7_Clock_Source 4

```

#define PWM_8_Duty 80
#define PWM_8_Clock_Source 4

'    Supported PWM module but not by this specific microcontroller
'

'    #define PWM_9_Duty 90
'    #define PWM_9_Clock_Source 6


'    Enable module 7
HPWMOn ( 7, PWMModule )
wait 2 s
'    Disable channel 7
HPWMOff ( 7, PWMModule)
'    wait 2 s


'    Enable others module
HPWMOn ( 5, PWMModule )
HPWMOn ( 6, PWMModule )
HPWMOn ( 7, PWMModule )
HPWMOn ( 8, PWMModule )


'    Enable CCP/PWM channel 1 - uses constants FREQ and DUTY
PWMOn

'    Enable CCP/PWM channel 2
PWMOn ( 2 )
do
loop

End

```

For more help, see [PWMOn](#) and [PWMOff](#) or, for AVR see [Fixed Mode PWM for AVR](#)

PWMOn

Syntax:

```
PWMOn
```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture/Compare/PWM module CCP1.

This command does not operate on any other CCP channel.

Explanation:

Example 1:

This command will enable the output of the CCP1/PWM module on the Microchip PIC microcontroller.

```
'This program will enable a 76 Khz PWM signal, with a duty cycle  
'of 80%. It will emit the signal for 10 seconds, then stop.  
#define PWM_Freq 76      'Set frequency in KHz  
#define PWM_Duty 80      'Set duty cycle to 80 %  
PWMOn                    'Turn on the PWM  
WAIT 10 s                'Wait 10 seconds  
PWMOff                   'Turn off the PWM
```

Example 2:

This command will enable the output of the CCP1/PWM module on the Microchip PIC microcontroller.

Note the chip frequency.

```
'This program will enable a 62Hz PWM signal, with a duty cycle  
'of 50%.  
  
#Chip 12F1840, 1  
  
dir porta.2 out  
#define PWM_Freq .0625    'Set frequency in Hz equates to 62Hz  
#define PWM_Duty 50       'Set duty cycle to 80 %  
PWMON  
  
Do  
loop
```

Example 3:

This command will enable the output of the CCP1/PWM module on the Microchip PIC microcontroller.

Note the chip frequency.

```
'This program will enable a 7.7Hz PWM signal, with a duty cycle  
'of 50%.
```

```
#Chip 12F1840, 0.125
```

```
dir porta.2 out
```

```
#define PWM_Freq .0077      'Set frequency in Hz equates to 7.7Hz
```

```
#define PWM_Duty 50         'Set duty cycle to 50 %
```

```
PWMON
```

```
Do
```

```
loop
```

For more help, also see [PWMOff](#)

PWMOff

Syntax:

```
PWMOff
```

Command Availability:

Only available on Microchip PIC microcontrollers with Capture/Compare/PWM module CCP1.

This command does not operate on any other CCP channel.

Explanation:

This command will disable the output of the CCP1/PWM module on the Microchip PIC chip.

Example:

```
'This program will enable a 76 Khz PWM signal, with a duty cycle
'of 80%. It will emit the signal for 10 seconds, then stop.
#define PWM_Freq 76      'Set frequency in KHz
#define PWM_Duty 80      'Set duty cycle to 80 %
PWMon                    'Turn on the PWM
WAIT 10 s                'Wait 10 seconds
PWMOff                   'Turn off the PWM
```

For more help, also see [PWMon](#)

Hardware PWM Code Optimisation

About Hardware PWM Code Optimisation

For compatibility all channels are supported by default. This maintains backward compatibility.

To minimise the code, use the following to disable support for a specific Capture/Compare/PWM (CCP) module, timers or the PWM module.

Setting a constant to *FALSE* will remove the support of the capability from the method and therefore will reduce the program size.

```
#define USE_HPWMCCP1 FALSE
#define USE_HPWMCCP2 FALSE
#define USE_HPWMCCP3 FALSE
#define USE_HPWMCCP4 FALSE
```

To further minimise the code, use the following to disable support for a specific PWM channels. Only PWM channels 5, 6 and 7 are supported.

```
#define USE_HPWM3 FALSE
#define USE_HPWM4 FALSE
#define USE_HPWM5 FALSE
#define USE_HPWM6 FALSE
#define USE_HPWM7 FALSE
```

To further minimise the code, use the following to disable support for a specific timers.

```
#define USE_HPWM_TIMER2 TRUE
#define USE_HPWM_TIMER4 TRUE
#define USE_HPWM_TIMER6 TRUE
```

Example

This will save 335 bytes of program memory by removing support for CCP1, CCP2 and CCP4.

```
#chip 16f18855,32
#Config MCLRE_ON

UNLOCKPPS
    RC2PPS = 0x0A      'RC2->CCP2:CCP2;
LOCKPPS

#define USE_HPWMCCP1 FALSE      ' This is not used so optimise
#define USE_HPWMCCP2 TRUE       ' This is used so include in the compiled code
#define USE_HPWMCCP3 FALSE      ' This is not used so optimise
#define USE_HPWMCCP4 FALSE      ' This is not used so optimise

'Setting the port an output is VERY important... LED will not work if you do not set
as an output.
dir portC.2 out      ; CCP2

do forever
    For Bright = 1 to 255
        HPWM 2, 40, Bright
        wait 10 ms
    next

loop
```

ATMEL AVR PWM Overview

Introduction:

The methods described in this section allow the generation of Pulse Width Modulation (PWM) signals. PWM signals enables the microcontroller to control items like the speed of a motor, or the brightness of a LED or lamp.

The methods can also be used to generate the appropriate frequency signal to drive an infrared LED for remote control applications.

GCBASIC support the methods described in this section.

Hardware PWM using a Timer/Counter with a OCRnx module

The AVR devices use a Timer/Counter and OCRnx module that has a variable period register. The Hardware PWM is available through the OCnx pin.

The method uses three parameters to setup the HPWM.

```
'HPWM channel, frequency, duty cycle  
HPWM 2, 100, 50
```

Relevant Constants:

A number of constants are used to control settings for PWM hardware module of the microcontroller. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

See [HPWM AVR OCRnx](#)

HPWM AVR OCRnx

Syntax:

```
HPWM channel, frequency, duty cycle
```

Command Availability:

The HPWM command is available on Atmel AVR microcontrollers with an OCnx pin, and is compatible with the PIC HPWM command method. Due to the unique way of AVR PWM implementation, and code efficiency, there are some notable differences in the HPWM initialization and its use.

This command supports the Fast PWM Mode and period registers for their respective devices. Typically Timer0 and Timer2 do not have a period register and the "A" channel is sacrificed to provide that function. Therefore, channel 1 and channel 6 will not be available, but are documented for possible future use. Some device Timers do not have an adjustable period register, so this command is not feasible (consult the datasheet).

Explanation:

The HPWM command sets up the hardware PWM module of the Atmel AVR microcontrollers to generate a PWM waveform of the given frequency and duty cycle. Once this command is called, the PWM will be emitted until the duty cycle parameter is written to zero.

If the need is just one particular frequency and duty cycle, one should use PWMOn and the constants PWM_Freq and PWM_Duty instead. PWMOn for the AVR is uniquely assigned to the OC0B pin, or channel 2. PWMOFF will only shutdown the AVR HPWM channel 2.

channel described as 1, 2, 3,...16 correspond to the pins OCR0A, OCR0B....OCR5C as detailed in the *channel* constant table. Channel 1 and channel 6 are not available.

frequency sets the frequency of the PWM output measured in Khz. The maximum value allowed is 255 KHz. In situations that do not require a specific PWM frequency, the PWM frequency should equal approximately 4 times the clock speed (GCB chipMHz) of the microcontroller (ie 63 KHz on a 16 MHz chip, 32 KHz on 8 MHz, 4 Khz on 1 MHz). This gives the best duty cycle resolution possible. Alternate frequencies with good duty cycle resolution are 1Khz, and 4Khz with chipMhz values of 16 and 8 respectively.

duty cycle specifies the desired duty cycle of the PWM signal, and ranges from 0 to 255 where 255 is 100% duty cycle. The AVR fast PWM mode has a small spike at the extreme setting of 0x00, on most devices, with each period register rollover. By using the HPWM command, and writing 0x00 to the duty cycle parameter, the PWM signal will shutdown completely and avoid the spike. The PWM signal can then be restarted again with a new HPWM command.

Note: Due to the AVR having a timer prescaler of just 1, 8, and 64; the AVR frequency and duty cycle resolution will be different from the PIC frequency and duty cycle resolution. The AVR HPWM parameters will likely need adjusting ,when substituted into an existing PIC program, and where accuracy is required.

HPWM Constants:

The AVR HPWM timer constants for channel number control are shown in the table below. Each timer constant needs to be defined for any one of the channels it controls.

Timer Constants	Controls	Options
AVRTC0	Specifies AVR TC0 associated with <i>channel</i> 1, and 2	Must be defined
AVRTC1	Specifies AVR TC1 associated with <i>channel</i> 3, 4 and 5 Channel 5 present on larger pinout devices	Must be defined
AVRTC2	Specifies AVR TC2 associated with <i>channel</i> 6, and 7	Must be defined
AVRTC3	Specifies AVR TC3 associated with <i>channel</i> 8, 9, and 10	Must be defined
AVRTC4	Specifies AVR TC4 associated with <i>channel</i> 11,12, and 13	Must be defined
AVRTC5	Specifies AVR TC5 associated with <i>channel</i> 14, 15, and 16	Must be defined

The GCBASIC HPWM channel constants for output pin control are shown in the table below. Each HPWM channel used needs to be defined. The Port pin associated with each OCnx must be set to

output.

Channel Constants	Controls	Options
AVRCHAN1	Specifies AVR HPWM <i>channel</i> 1 to the associated output pin OC0A OCR0A is used as period register and thus not available	N/A
AVRCHAN2	Specifies AVR HPWM <i>channel</i> 2 to the associated output pin OC0B	Must be defined
AVRCHAN3	Specifies AVR HPWM <i>channel</i> 3 to the associated output pin OC1A MUX'd with OC0A pin on some ATTiny's	Must be defined
AVRCHAN4	Specifies AVR HPWM <i>channel</i> 4 to the associated output pin OC1B	Must be defined
AVRCHAN5	Specifies AVR HPWM <i>channel</i> 5 to the associated output pin OC1C On some larger pinout devices and MUX'd with OC0A pin	Must be defined
AVRCHAN6	Specifies AVR HPWM <i>channel</i> 6 to the associated output pin OC2A OCR2A is used as a period register and thus not available	N/A
AVRCHAN7	Specifies AVR HPWM <i>channel</i> 7 to the associated output pin OC2B	Must be defined
AVRCHAN8	Specifies AVR HPWM <i>channel</i> 8 to the associated output pin OC3A	Must be defined
AVRCHAN9	Specifies AVR HPWM <i>channel</i> 9 to the associated output pin OC3B	Must be defined
AVRCHAN10	Specifies AVR HPWM <i>channel</i> 9 to the associated output pin OC3C	Must be defined
AVRCHAN11	Specifies AVR HPWM <i>channel</i> 11 to the associated output pin OC4A	Must be defined
AVRCHAN12	Specifies AVR HPWM <i>channel</i> 12 to the associated output pin OC4B	Must be defined
AVRCHAN13	Specifies AVR HPWM <i>channel</i> 13 to the associated output pin OC4C	Must be defined
AVRCHAN14	Specifies AVR HPWM <i>channel</i> 14 to the associated output pin OC5A	Must be defined
AVRCHAN15	Specifies AVR HPWM <i>channel</i> 15 to the associated output pin OC5B	Must be defined
AVRCHAN16	Specifies AVR HPWM <i>channel</i> 16 to the associated output pin OC5C	Must be defined

Example:

```

    'Using HPWM command to alternate ramping leds with the UNO board
#chip mega328,16

'*****pwm*****
'Must define AVRTCx, AVRCHANx, and set OCnX pin dir to out

#define AVRTC0      'Timer0
#define AVRCHAN2
dir PortD.5 Out    'OC0B, UNO pin 5

#define AVRTC1      'Timer1
#define AVRCHAN3
#define AVRCHAN4
dir PortB.1 out    'OC1A, UNO pin 9
dir PortB.2 Out    'OC1B, UNO pin 10

#define AVRTC2      'Timer2
#define AVRCHAN7
dir PortD.3 Out    'OC2B, UNO pin 3

do

'63khz works good with 16MHz
'32khz with 8MHz intosc
'4KHz with 8MHz intosc and ckDiv8 fuse
freq = 63
  For PWMled1 = 0 to 255
    HPWM 2,freq,PWMled1
    PWMled2 = NOT PWMled1
    HPWM 3,freq,PWMled2
    HPWM 4,freq,PWMled2
    HPWM 7,freq,PWMled1
    wait 5 ms
  Next

  For PWMled1 = 255 to 0
    HPWM 2,freq,PWMled1
    PWMled2 = NOT PWMled1
    HPWM 3,freq,PWMled2
    HPWM 4,freq,PWMled2
    HPWM 7,freq,PWMled1
    wait 5 ms
  Next

loop

```

HPWM Fixed Mode for AVR

Syntax:

```
PWMOn  
  
'or  
  
PWMOff
```

Command Availability:

This command is only available on the Atmel AVR microcontrollers with a Timer/Counter0 OC0B register.

Explanation:

The PWMOn command will only enable the output of the OC0B/PWM module of the Atmel AVR microcontroller.

This command is not available for any other OCnx/PWM modules.

This command sets up the hardware PWM module of the Atmel AVR microcontroller to generate a PWM waveform of the given frequency and duty cycle. Once PWMON method is called, the PWM will be emitted until PWMOff is called.

These constants are required for PWMOn.

Constant Name	Controls	Default Value
PWM_Freq	Specifies the output frequency of the PWM module in KHz.	38
PWM_Duty	Sets the duty cycle of the PWM module output. Given as percentage.	50

Example:

```
'This program demonstrates the PWMOn and PWMOff commands  
'of the fixed mode HPWM on OC0B pin.
```

```
#chip mega328p,16
```

```
'activate appropriate PWM output pins  
dir PortD.5 Out      'OC0B
```

```
'define PWM_Freq in kHz  
'define PWM_Duty in %
```

```
#define PWM_Freq 40  
#define PWM_Duty 50
```

```
do
```

```
    'turn on/off single channel 40 KHz PWM on OC0B pin  
    PWMON  
    wait 5 s  
    PWMOFF  
    wait 5 s
```

```
loop
```

For more help, see [PWMOn](#) and [PWMOff](#) or, for Microchip microcontrollers see [Fixed Mode PWM for Microchip](#)

PWMOn for AVR

Syntax:

```
PWMOn
```

Command Availability:

This command is only available on the Atmel AVR microcontrollers with a Timer/Counter0 OC0B register.

Explanation:

The PWMOn command will only enable the output of the OC0B/PWM module of the Atmel AVR microcontroller.

This command is not available for any other OCnx/PWM modules.

Example:

```
'This program demonstrates the PWMOn and PWMOff commands
'of the fixed mode HPWM on OC0B pin.

#chip mega328p,16

'activate appropriate PWM output pins
dir PortD.5 Out      'OC0B

'define PWM_Freq in kHz
'define PWM_Duty in %

#define PWM_Freq 40
#define PWM_Duty 50

do

    'turn on/off single channel 40 KHz PWM on OC0B pin
    PWMON
    wait 5 s
    PWMOFF
    wait 5 s

loop
```

For more help, see [PWMOff](#)

PWMOff for AVR

Syntax:

```
PWMOff
```

Command Availability:

This command is only available on the Atmel AVR microcontrollers with a Timer/Counter0 OC0B register.

Explanation:

The PWMOff command will only disable the output of the OC0B/PWM module of the Atmel AVR microcontrollers.

This command is not available for any other OCnx/PWM modules.

Example:

```
'This program demonstrates the PWMOn and PWMOff commands
'of the fixed mode HPWM on OC0B pin.

#chip mega328p,16

'activate appropriate PWM output pins
dir PortD.5 Out      'OC0B

'define PWM_Freq in kHz
'define PWM_Duty in %

#define PWM_Freq 40
#define PWM_Duty 50

do

    'turn on/off single channel 40 KHz PWM on OC0B pin
    PWMON
    wait 5 s
    PWMOFF
    wait 5 s

loop
```

For more help, see [PWMOn](#)

Random Numbers

This is the Random Numbers section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Overview

Introduction:

These routines allow GCBASIC to generate pseudo-random numbers.

The generator uses a 16 bit linear feedback shift register to produce pseudo-random numbers. The most significant 8 bits of the LFSR are used to provide an 8 bit random number.

When compiling a program, GCBASIC will generate an initial seed for the generator. However, this seed will be the same every time the program runs, so the sequence of numbers produced by a given program will always be the same. To work around this, there is a Randomize subroutine. It can be provided with a new seed for the generator (which will cause the generator to move to a different point in the sequence). Alternatively, Randomize can be set to obtain a seed from some other source such as a timer every time it is run.

Relevant Constants:

These constants are used to control settings for the random number generation. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

Constant Name	Controls	Default Value
RANDOMIZE_SEED	Source of the random seed if Randomize is called without a parameter	Timer0

Example:

```
#define RANDOMIZE_SEED Timer2
```

Random

Syntax:

```
var = Random
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Random** function will generate a pseudo-random number between 0 and 255 inclusive.

The numbers generated by **Random** will follow the same sequence every time, until **Randomize** is used.

Example:

```
'Set chip model
#chip tiny2313, 1

'Use randomize, with the value on PORTD as the seed
Randomize PORTD

'Generate random numbers, and output on PORTB
Do
    PORTB = Random
    Wait 1 s
Loop
```

Randomize**Syntax:**

```
Randomize
Randomize seed
```

Command Availability:

Available on all microcontrollers

Explanation:

Randomize is used to seed the pseudo random number generator, so that it will produce a different sequence of numbers each time it is used.

The random number generator in GCBASIC is a 16 bit linear feedback shift register, which is explained here: http://en.wikipedia.org/wiki/Linear_feedback_shift_register

Generally, you will get the same sequence every time it is used. However, you can seed it so that it will start at a different point at the sequence using the Randomize command.

If you wanted to use an analog reading to seed the generator, this would work:

Randomize ReadAD10(AN0)

If no *seed* is specified, then the RANDOMIZE_SEED constant will be used as the seed. If *seed* is specified, then it will be used to seed the generator.

It is important that the seed is different every time that Randomize is used. If the seed is always the same, then the sequence of numbers will always be the same. It is best to use a running timer, an input port, or the analog to digital converter as the source of the seed, since these will normally provide a different value each time the program runs.

Example:

```
'Set chip model
#chip tiny2313, 1

'Use randomize, with the value on PORTD as the seed
Randomize PORTD

'Generate random numbers, and output on PORTB
Do
    PORTB = Random
    Wait 1 s
Loop
```

7-Segment Displays

This is the 7-Segment Displays section of the Help file. Please refer the sub-sections for details using the contents/folder view.

7 Segment Displays Overview

Introduction

The 7 Segment Displays module provide a cheap red, green, blue or white bright LED Display. The Ebay modules can be had for \$1 to \$4 per piece, sometimes less. They only need 2 pins to control: CLK and DIO for control 4 digit 7 segment LED Display. They often have a colon LED

The GCBASIC 7 segment display methods make it easier for GCBASIC programs to display numbers and letters on 7 segment LED displays.

There are two ways that the 7 segment display routines can be set up.

- A pre 2020 method [7 segment legacy method](#)
- A revised method for [TM1637 4 LEDs](#), or, [TM1637 6 LEDs](#)

7 Segment Displays - Legacy

Introduction

The GCBASIC 7 segment display methods make it easier for GCBASIC programs to display numbers and letters on 7 segment LED displays.

The GCBASIC methods support up to four digit 7 segment display devices, common anode/cathode and inversion of the port logic to support driving the device(s) via a transistor.

There are three ways that the 7 segment display routines can be set up.

Metho d	Description
1	Connect the microcontroller to the 7 segment display (via suitable resistors) using any eight output bits. Use DISP_SEG_x and DISP_SEL_x constants to specify the output ports and the select port(s) to be used.

Metho d	Description
2	Connect the microcontroller to the 7 segment display (via suitable resistors) using whole port (8 bits) of the microcontroller. This implies the connections are consecutive in terms of the 8 output bits of the port. Use the <code>DISPLAYPORTn</code> and <code>DISPSELECTn</code> constants to specify the whole port and the select port(s) to be used. This method will generate the most efficient code.
3	Connect the microcontroller to the 7 segment display (via suitable resistors) using whole port (8 bits) of the microcontroller. This implies the connections are consecutive in terms of the 8 output bits of the port. Use the <code>DISPLAYPORTn</code> and <code>DISP_SEL_n</code> constants to specify the whole port and the select port(s) to be used.

The GCBASIC methods assume the 7 segment display(s) is to be connected to a common parallel bus with a Common Cathode. See the sections [Common Cathode](#) and [Common Anode](#) for examples of using GCBASIC code to control these different configurations

Shown below are the differing constants that must be set for the three connectivity options.

Ind ex	Method	Descr iption	Constants	Default Value
1	<code>DISP_SEG_x</code> & <code>DISP_SEL_x</code>			
		<code>DISP_SEG_x</code>	Specifies the output pin (output bit) used to control a specific segment of the 7 segment display. There are seven constants that must be specified. <code>DISP_SEG_A</code> through <code>DISP_SEG_G</code> . One must be set for each segment. Typical commands are: <code>#define DISP_SEG_A portA.0</code> <code>#define DISP_SEG_B portA.1</code> <code>#define DISP_SEG_C portA.2</code> <code>#define DISP_SEG_D portA.3</code> <code>#define DISP_SEG_E portA.4</code> <code>#define DISP_SEG_F portA.5</code> <code>#define DISP_SEG_G portA.6</code>	Must be specified to use this connectivity option.
		<code>DISP_SEG_DOT</code>	Specifies the output pin (output bit) used to control the decimal point on the 7 segment display. Typical commands are: <code>#define DISP_SEG_DOT portA.7</code>	Optional.

Index	Method	Description	Constants	Default Value
		DISP_SEL_x	Specifies the output pin (output bit) used to control a specific 7 segment display. These constants are used to control the specific 7 segment display being addresses. Typical commands are: <code>#define DISP_SEL_1 portA.0</code> <code>#define DISP_SEL_2 portA.1</code>	A valid output pin (output bit) must be specified. Must be specified to use this connectivity option.
2	DISPLAYPORT_n & DISPSELECT_n			
		DISPLAYPORT_n	Specifies the output port used to control the 7 segment display. Port.bit >> Segment port.0 >> A port.1 >> B port.2 >> C port.3 >> D port.4 >> E port.5 >> F port.6 >> G	Can be DISPLAYPORTA and/or DISPLAYPORTB and/or DISPLAYPORTC and/or DISPLAYPORTD Where DISPLAYPORT_n can be A, B, C or D which corresponding to displays 1, 2, 3 and 4, respectively. Must be specified to use this connectivity option.
		DISPSELECT_n	Specifies the output command used to select a specific 7 segment display addressed by DISPLAYPORT_n . Used to control output pin (output bit) when several displays are multiplexed. Typical commands are: <code>#define DispSelectA Set portA.0 on</code> <code>#define DispSelectB Set portA.1 on</code>	Can be DISPSELECTA and/or DISPSELECTB and/or DISPSELECTC and/or DISPSELECTD Must be specified to use this connectivity option.
		DISPDESELECT_n	An optional command to specify the output command used to deselect a specific 7 segment display addressed by DISPLAYPORT_n . Used to control output pin (output bit) when several displays are multiplexed. Typical commands are: <code>#define DispDeSelectA Set portA.0 off</code> <code>#define DispDeSelectB Set portA.1 off</code>	Can be DISPDESELECTA and/or DISPDESELECTB and/or DISPDESELECTC and/or DISPDESELECTD
3	DISPLAYPORT_n & DISP_SEL_n			

Index	Method	Description	Constants	Default Value
		DISPLAYPORT_n	Specifies the output port used to control the 7 segment display. Port.bit >> Segment port.0 >> A port.1 >> B port.2 >> C port.3 >> D port.4 >> E port.5 >> F port.6 >> G	Can be DISPLAYPORTA and/or DISPLAYPORTB and/or DISPLAYPORTC and/or DISPLAYPORTD Where DISPLAYPORT_n can be A, B, C or D which corresponding to displays 1, 2, 3 and 4, respectively. Must be specified to use this connectivity option.
		DISP_SEL_n	Specifies the output command used to select a specific 7 segment display addressed by DISPLAYPORT_n . Typical commands are: #define DISP_SEL_1 portA.0 #define DISP_SEL_2 portA.1	Must be specified to use this connectivity option. Can be specified as DISP_SEL_1 and/or DISP_SEL_2 and/or DISP_SEL_3 and/or DISP_SEL_4

Example 1:

```
'A Common Cathode 7 Segment display 2 digit example
#chip 16F886, 8

'support for Common Anode
'#DEFINE 7SEG_COMMONANODE

'support for pfet or pnp high side drivers
'#DEFINE 7SEG_HIGHSIDE

' ----- Constants
'You need to specify the port settings
'by one of the following three methods
'The Directions of the ports are automaically set according to the defines

'METHOD 1
'Define individual port pins for segments and selects

#define DISP_SEG_A PORTB.0
#define DISP_SEG_B PORTB.1
#define DISP_SEG_C PORTB.2
#define DISP_SEG_D PORTB.3
#define DISP_SEG_E PORTB.4
#define DISP_SEG_F PORTB.5
#define DISP_SEG_G PORTB.6
#define DISP_SEG_DOT PORTB.7 '' available on some displays as dp or colon

#define DISP_SEL_1 PORTC.5
#define DISP_SEL_2 PORTC.4
```

```

'METHOD 2 Define DISPLAYPORTA (B,C,D) for up to 4 digit display segments
'Define DISPSELECTA (B,C,D) for up to 4 digit display selects

'#DEFINE DISPLAYPORTA PORTB ' same port name can be assigned
'#DEFINE DISPLAYPORTB PORTB

'#DEFINE DispSelectA Set portC.5 off
'#DEFINE DispSelectB Set portC.4 off
'#DEFINE DispDeSelectA Set portC.5 on
'#DEFINE DispDeSelectB Set portC.4 on

'METHOD 3 Define DISPLAYPORTA (B,C,D) for up to 4 digit display segments
'Define port pins for the digit display selects

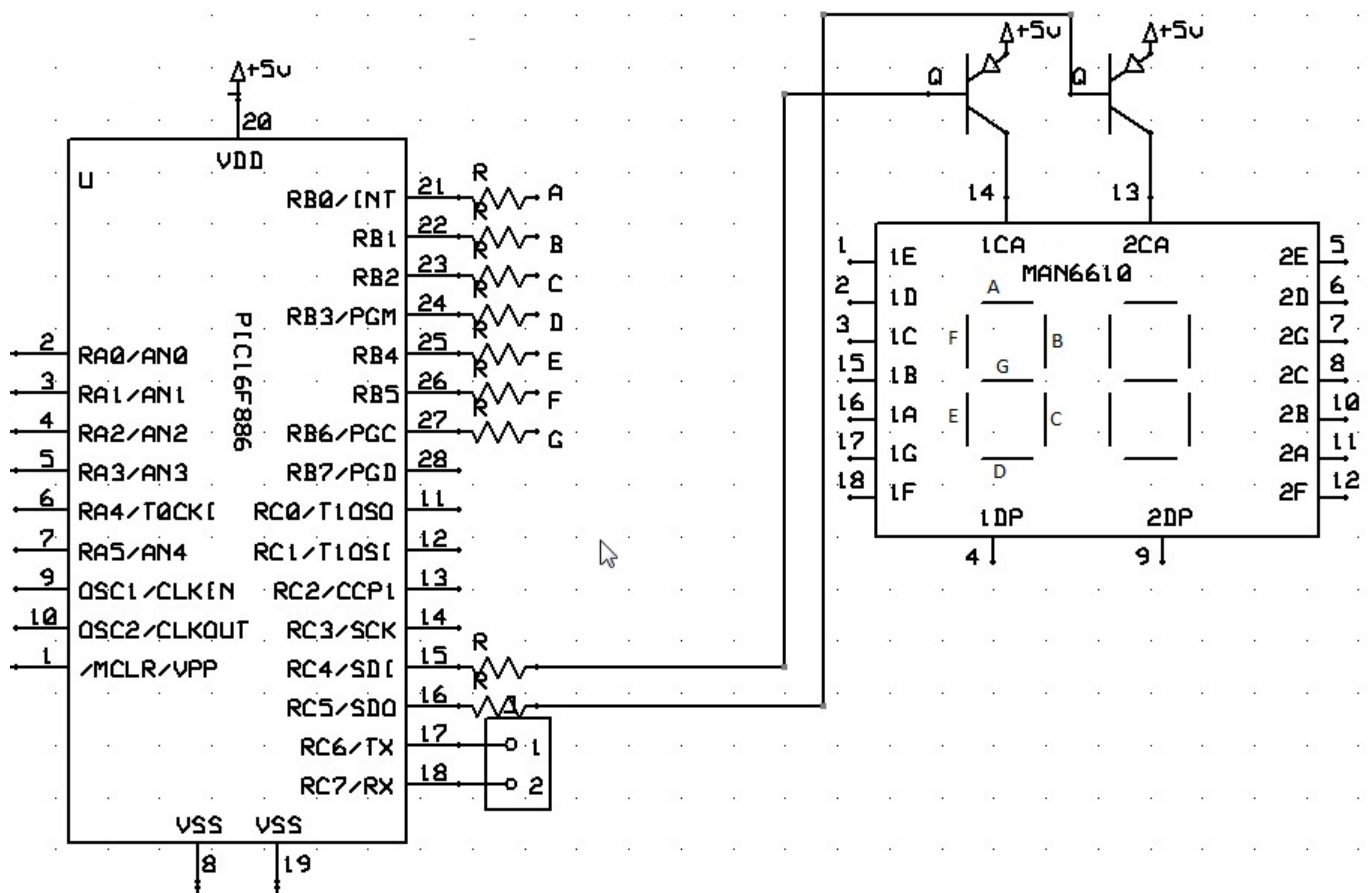
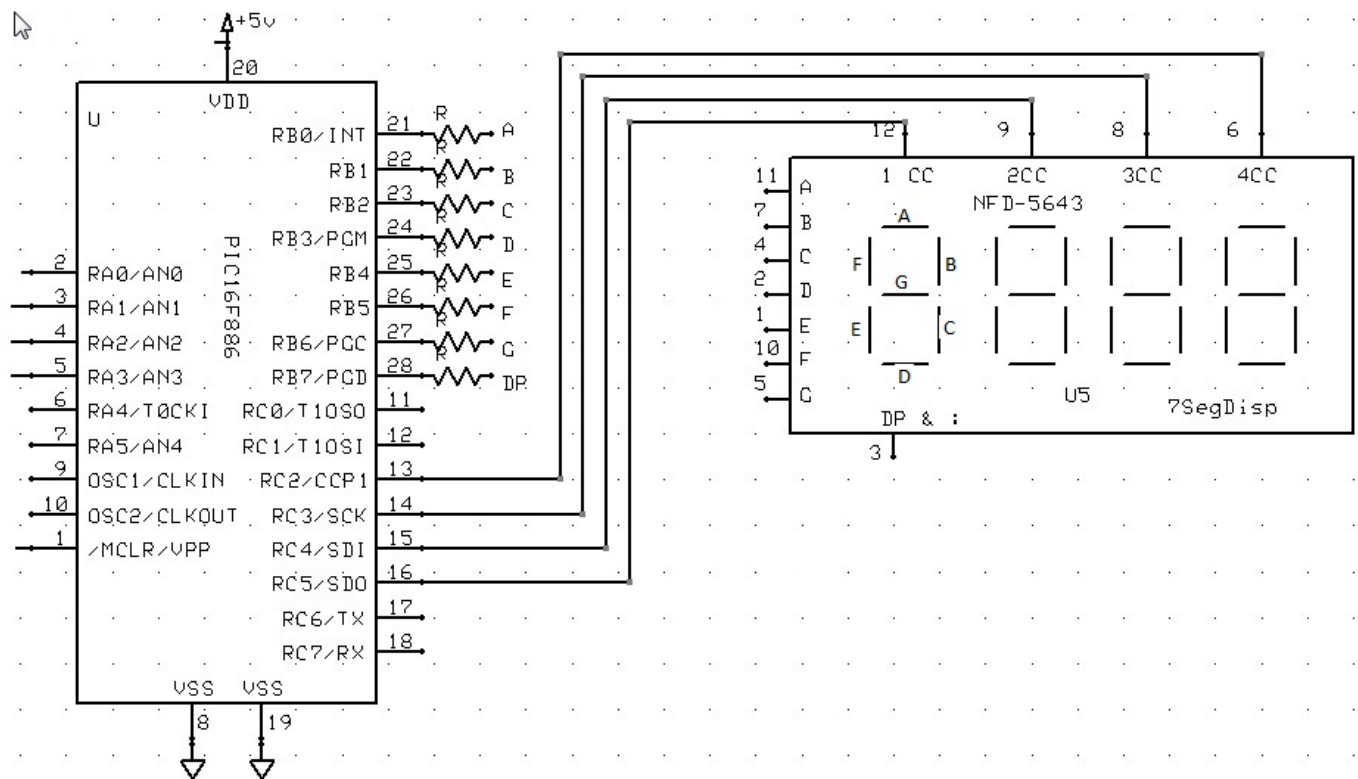
'#DEFINE DISPLAYPORTA PORTB
'#DEFINE DISPLAYPORTB PORTB

'#DEFINE DISP_SEL_1 PORTC.5
'#DEFINE DISP_SEL_2 PORTC.4

Dim Message As String
Message = " HAPPY HOLIDAYS "
Do
    For Counter = 1 to len(Message)-1
        Repeat 50
            Displaychar 1, Message(Counter)
            wait 3 ms
            DisplayChar 2, Message(Counter+1)
            wait 3 ms

        End Repeat
        Wait 100 ms
    Next
Loop

```



Also, see [DisplayChar](#), [DisplayValue](#)

Common Cathode

This is a Common Cathode 7 Segment display example.

No additional configuration is required when using Common Cathode.

Constant Name	Controls	Comment
7Seg_Common Anode	Inverts controls for Common Anode displays	Required for Common Cathode displays
7Seg_HighSide	Support PFET or PNP high side driving of the display	Inverts Common Cathode addressing pin logic for multiplexed displays

This is a Common Cathode 7 Segment display example.

Example:

```
'Chip model
#chip 16f1783,8

'Output ports for the 7-segment device
#define DISP_SEG_A PORTC.0
#define DISP_SEG_B PORTC.1
#define DISP_SEG_C PORTC.2
#define DISP_SEG_D PORTC.3
#define DISP_SEG_E PORTC.4
#define DISP_SEG_F PORTC.5
#define DISP_SEG_G PORTC.6

' This is the usage of the SEG_DOT for decimal point support
' An optional third parameter of '1' will turn on the decimal point
' of that digit when using DisplayValue command
#define DISP_SEG_DOT PortC.7

'Select ports for the 7-segment device
#define Disp_Sel_1 PortA.1
#define Disp_Sel_2 PortA.2
#define Disp_Sel_3 PortA.3

dim count as word
dim number as word

Do Forever
    For count = 0 to 999
        number = count
        Num2 = 0
```

```

        Num3 = 0
        If number >= 100 Then
            Num3 = number / 100
            'SysCalcTempX is the remainder after a division has been completed
            number = SysCalcTempX
        End if
        If number >= 10 Then
            Num2 = number / 10
            number = SysCalcTempX
        end if
        Num1 = number
        Repeat 10
            DisplayValue 1, Num1,1 'Optional third parameter turns on the dp dot on
that digit
            wait 5 ms
            DisplayValue 2, Num2
            wait 5 ms
            DisplayValue 3, Num3
            wait 5 ms

        end Repeat
    Next
Loop

```

Also, see [7 Degment Display Overview](#), [DisplayChar](#), [DisplayValue](#)

Common Anode

This is a Common Anode 7 Segment display example.

Additional configuration is required when using Common Anode.

When setting up the 7 segment Common Anode display you **MUST** use the `7Seg_CommonAnode` constant. You can optionally use the `7Seg_HighSide` constant to support PFET or PNP high side driving of the Common Anode displays as follows:

Constant Name	Controls	Comment
<code>7Seg_CommonAnode</code>	Inverts controls for Common Anode displays	Required for Common Cathode displays
<code>7Seg_HighSide</code>	Support PFET or PNP high side driving of the display	Inverts Common Cathode addressing pin logic for multiplexed displays

Example:

```
'A Common Anode 7 Segment display example using bs250p pfets
```

```

'Chip model
#chip 16f1783,8

'support for Common Cathode
#define 7Seg_CommonAnode

'support for pfet or pnp high side drivers
#define 7Seg_HighSide

#define DISP_SEG_A PORTC.0
#define DISP_SEG_B PORTC.1
#define DISP_SEG_C PORTC.2
#define DISP_SEG_D PORTC.3
#define DISP_SEG_E PORTC.4
#define DISP_SEG_F PORTC.5
#define DISP_SEG_G PORTC.6
#define DISP_SEG_DOT PORTC.7

#define Disp_Sel_1 PortA.1
#define Disp_Sel_2 PortA.2
#define Disp_Sel_3 PortA.3

dim count as word
dim number as word

Do Forever
  For count = 0 to 999
    number = count
    Num2 = 0
    Num3 = 0
    If number >= 100 Then
      Num3 = number / 100
      'SysCalcTempX is the remainder after a division has been completed
      number = SysCalcTempX
    End if
    If number >= 10 Then
      Num2 = number / 10
      number = SysCalcTempX
    end if
    Num1 = number
    Repeat 10
      DisplayValue 1, Num1, 1 'Optional third parameter turns on the dp on that
digit
      wait 5 ms
      DisplayValue 2, Num2
      wait 5 ms
      DisplayValue 3, Num3
      wait 5 ms
    End Repeat
  Next count
End Do

```

```
end Repeat
Next
Loop
```

Also, see [7 Degment Display Overview](#), [DisplayChar](#), [DisplayValue](#)

DisplayValue

Syntax:

```
DisplayValue (display, data, dot)
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will display the given value on a seven segment LED display.

`display` is the number of the display to use. Up to 4 digits.

`data` is the value between 0 and F to be shown.

`dot` is an optional parameter. When it is 1 then the decimal point for that digit is turned on.

The command also support HEX characters in the range between 0x00 and 0x0F (0 to 15). See example two below for usage.

Example 1:

```

    'This program will count from 0 to 99 on two LED displays
    #chip 16F819, 8

    'See 7 Segment Display Overview for alternate ways of defining Ports
    #define DISP_SEG_A PORTB.0
    #define DISP_SEG_B PORTB.1
    #define DISP_SEG_C PORTB.2
    #define DISP_SEG_D PORTB.3
    #define DISP_SEG_E PORTB.4
    #define DISP_SEG_F PORTB.5
    #define DISP_SEG_G PORTB.6
    '#define DISP_SEG_DOT PORTB.7 ' Optional DP

    #define DISP_SEL_1 PORTA.0
    #define DISP_SEL_2 PORTA.1

    Do
        For Counter = 0 To 99

            'Get the 2 digits
            Number = Counter
            Num1 = 0
            If Number >= 10 Then
                Num1 = Number / 10
                'SysCalcTempX stores remainder after division
                Number = SysCalcTempX
            End If
            Num2 = Number

            'Show the digits
            'Each DisplayValue will erase the other (multiplexing)
            'So they must be called often enough that the flickering
            'cannot be seen.
            Repeat 500
                DisplayValue 1, Num1
                Wait 1 ms
                DisplayValue 2, Num2
                Wait 1 ms
            End Repeat
        Next
    Loop

```

Example 2:

```

'This program will count from 0 to 0xff on two LED displays
#chip 16F819, 8

#define DISP_SEG_A PORTB.0
#define DISP_SEG_B PORTB.1
#define DISP_SEG_C PORTB.2
#define DISP_SEG_D PORTB.3
#define DISP_SEG_E PORTB.4
#define DISP_SEG_F PORTB.5
#define DISP_SEG_G PORTB.6

#define DISP_SEL_1 PORTA.0
#define DISP_SEL_2 PORTA.1
#define DISP_SEL_4 PORTA.2
#define DISP_SEL_3 PORTA.3

Do
    For Counter = 0 To 0xff

        'Get the 2 digits
        Number = Counter
        Num1 = 0
        If Number >= 0x10 Then
            Num1 = Number / 0x10
            'SysCalcTempX stores remainder after division
            Number = SysCalcTempX
        End If
        Num2 = Number

        'Show the digits
        'Each DisplayValue will erase the other (multiplexing)
        'So they must be called often enough that the flickering
        'cannot be seen.
        Repeat 500
            DisplayValue 1, Num1
            Wait 1 ms
            DisplayValue 2, Num2
            Wait 1 ms
        End Repeat
    Next
Loop

```

Also, see [7 Segment Display Overview](#), [DisplayChar](#), [DisplaySegment](#)

DisplayChar

Syntax:

```
DisplayChar (display, character, dot)
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will display the given ASCII character on a seven segment LED display.

`display` is the number of the display to use. Up to 4 digits.

`character` is the ASCII character to be shown.

`dot` is an optional parameter. When it is 1 then the decimal point for that digit is turned on.

This example below is a Common Cathode configuration.

Example 1:

```
'This program will show " Hello " on a LED display
'The display should be connected to PORTB and the Enable on PORTA.0

#chip 16F877A, 20

#define DISPLAYPORTA PORTB
#define DISP_SEL_1 PORTA.0

Dim Message As String
Message = " Hello "
Do
  For Counter = 1 to len(Message)
    DisplayChar 1, Message(Counter)
    Wait 250 ms
  Next
Loop
```

This is a Common Anode example There are three different methods for port specification Note the ports are specified bit by bit in this case but could be specified like Example 1 See Overview for further explanation.

Example 2:

```
'This program will show a message on a LED display
'This is a Dual digit Common anode with driver transistors example
#chip 16F886, 8

'support for Common Anode
#define 7Seg_CommonAnode

'support for pfet or pnp high side drivers
#define 7Seg_HighSide

' Constants
' You need to specify the port settings
#define DISP_SEG_A PORTB.0
#define DISP_SEG_B PORTB.1
#define DISP_SEG_C PORTB.2
#define DISP_SEG_D PORTB.3
#define DISP_SEG_E PORTB.4
#define DISP_SEG_F PORTB.5
#define DISP_SEG_G PORTB.6

#define DISP_SEL_1 PORTC.5
#define DISP_SEL_2 PORTC.4

Dim Message As String
Message = " Happy Holidays  "
Do
For Counter = 1 to len(Message)-2
    Repeat 50
        Displaychar 1, Message(Counter)
        wait 3 ms
        DisplayChar 2, Message(Counter+1)
        wait 3 ms
    end Repeat
    Wait 100 ms
Next
Loop
```

Also, see [7 Degment Display Overview](#), [DisplayValue](#), [DisplaySegment](#)

DisplaySegment

Syntax:

DisplayValue (display, data)

Command Availability:

Available on all microcontrollers.

Explanation:

This command will display the given value on a seven segment LED display.

display is the number of the display to use. Up to 4 digits.

data is the value between 0 and 255. Where *data* is the representation of the segments to be set.

Example

```
'This program will count from 10 to 0 then fire the rocket!
'The method DisplaySegment 1, smallTCharacter. Sets the 7 segment to the value of
120, see the constant, 120 equates to a small t.
; ----- Configuration

#chip 16F690, 4

; ----- Define Hardware settings
Dir PORTC Out
DIR PORTA.5 out
DIR PORTA.4 out
DIR PORTA.0 out
DIR PORTA.1 out
DIR PORTA.2 in
DIR PORTB.7 out
; ----- Constants
; You need to specify the port settings
#define DISP_SEG_A PORTC.0
#define DISP_SEG_B PORTC.1
#define DISP_SEG_C PORTC.2
#define DISP_SEG_D PORTC.3
#define DISP_SEG_E PORTC.4
#define DISP_SEG_F PORTC.5
#define DISP_SEG_G PORTC.6
#define DECPNT PORTC.7
#define DISP_SEL_1 PORTA.5
#define DISP_SEL_2 PORTA.4
#define DISP_SEL_3 PORTA.1
#define DISP_SEL_4 PORTA.0
```

```

#define smallTCharacter 120 'raw character for 't' on 7 segment.

#define sw1 PORTA.2

#define firingPort PORTB.7

; ----- Variables
CountDownValue = 10

; ----- Main body of program commences here.
DECPNT = 1 'Decimal Point off

Main:
    ' Push number to 7 Segment Display
    if sw1 = 0 then goto Countdown

    num2 = 1
    num3 = 0
    cnt = 5
    gosub display

goto main

Countdown:

    num2 = CountDownValue/10
    num3 = CountDownValue%10
    cnt = 60

    gosub display

    If sw1 = 0 then goto hld

    if CountDownValue = 0 then
        firingPort = 1
        cnt = 200
        gosub dispfire
        firingPort = 0
        CountDownValue = 10
        goto main
    end if

    CountDownValue = CountDownValue - 1

goto Countdown

```

```

display:
  Repeat cnt
    DisplaySegment 1, smallTCharacter
    wait 5 ms
    Displaychar 2, "-"
    DisplayValue 3, Num2
    wait 5 ms
    DisplayValue 4, Num3
    wait 5 ms
  end Repeat

  return

hld:
  if sw1 = 0 then goto hld
  cnt = 5
  gosub Display
  if sw1 = 1 then goto hld
  goto countdown

DispFire:
  Repeat cnt

    Displaychar 1, "F"
    wait 5 ms
    Displaychar 2, "i"
    wait 5 ms
    Displaychar 3, "r"
    wait 5 ms
    Displaychar 4, "E"
    wait 5 ms

  End Repeat
  return

end

```

Also, see [7 Segment Display Overview](#), [DisplayChar](#)

7 Segment Displays - TM1637 4 Digits

Introduction

The TM1637 display module is used for displaying numbers on a keyboard matrix. The matrix of LEDs consists of four 7-segment displays working together.

The TM1637 specification is

- Two wire interface

- Eight adjustable luminance levels
- 3.3V/5V interface
- Supports Four alpha-numeric digits
- Operating current consumption: 80mA

Why to use TM1637 Display Module?

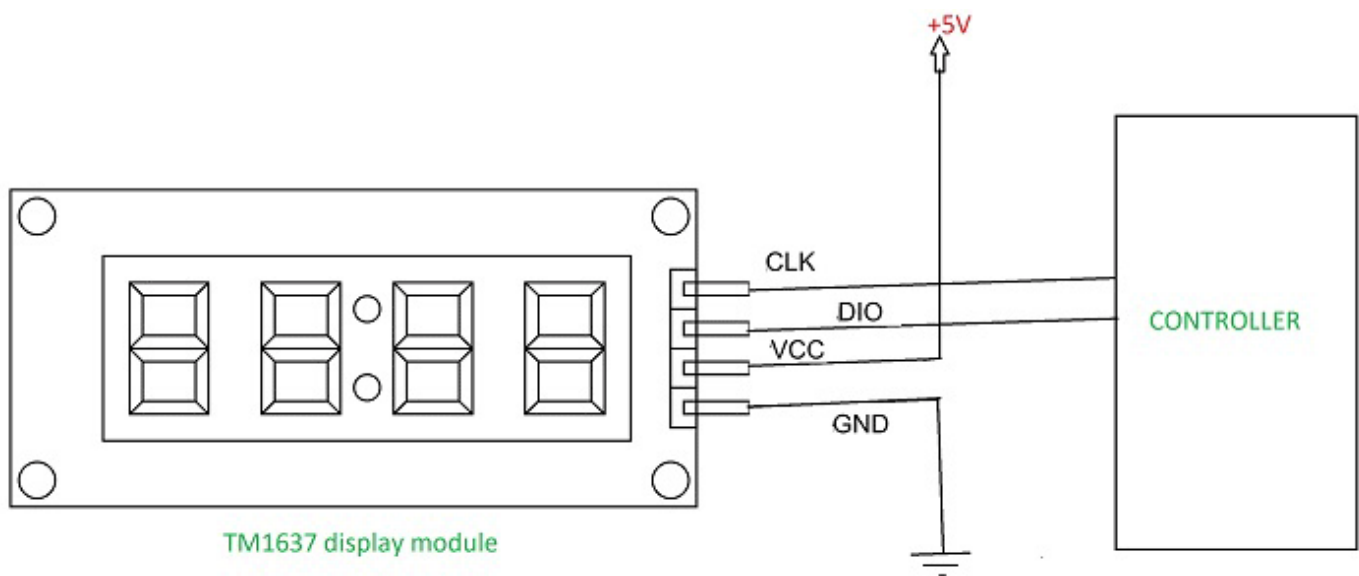
The TM1637 can be interfaced to any system using only two ports. This is the main reason the module is preferred over other module.

Another reason TM1637 display is preferred is because of its low cost. Although there are other display modules present in the market they cost more.

The module design is robust so it can sustain in tough environments and still can perform its function for a long time. The module consumes low power and can be installed in mobile applications.

How to use TM1637 Display Module?

As mentioned earlier the module communication can only be done using the two pins DIO and CLK respectively. The data is sent to the module or received from the module though these two pins. So the characters to be displayed are sent in the form of serial data through this interface. A typical circuit diagram of display module interface to a controller is shown below.



The module can work on +5V regulated power and any higher voltage may lead to permanent damage. The interface is established as shown in figure above. All you need to do is connect DIO and CLK to any of GPIO (General Purpose Input Output) pins of controller and establish serial data exchange through programming.

GCBASIC Support

The GCBASIC 7 segment display methods make it easier for GCBASIC programs to display numbers and letters on 7 segment LED displays.

The GCBASIC methods support up to four digit 7 segment display devices, common anode/cathode and inversion of the port logic to support driving the device(s) via a transistor.

Brightness can be set: 8 is display on minimum bright , 15 is display on max bright. Less than 8 is display off.

The TM1637 chip supports the reading of the keyboard matrix however that is not supported in the library.

DataSheets

The datasheets can found here:

English - [here](#).

Chinese - [here](#).

Usage

The following will set the display.

Constant	Description
TM1637_CLK	Must be a bi-directional port. The direction/port setting is managed by the library.
TM1637_DIO	Must be a bi-directional port. The direction/port setting is managed by the library.

Example program

```
#chip mega328p,16
#include <TM1637a.h>

#define TM1637_CLK PortD.2      ' Arduino Digital_2
#define TM1637_DIO PortD.3      ' Arduino Digital_3

'---- main program -----

TMWrite4Dig (17, 16, 17, 16, 0) 'clear display
    wait 2 s
TMWrite4Dig (17, 16, 17, 16, 10,0) '- -
    wait 2 s
TMchar_Bright = 10
```

TMWrite4Dig

Syntax:

```
TMWrite4Dig (dig1, dig2, dig3, dig4 [, Brightness ], Colon ] ] )
```

Command Availability:

Available on all microcontrollers.

Explanation:

Command defines each digit (left to right) as 0 to 9 OR 0x00 to 0x0F (15). Additionally 0x10 (16) is a blank, 0x11 (17) is a minus sign, 0x12 (18) is a degree sign, 0x13 (19) is a bracket and 0x14 (20) is a question mark.

Brightness set the brightness (8-15). *Colon* turns the colon (only on digit 2) to off (0) or on (1).

TM_Bright

Syntax:

```
TM_Bright = Brightness
```

Command Availability:

Available on all microcontrollers.

Explanation:

Brightness sets the brightness for the display with a range of 8 to 15. Default to 15.

TMDec

Syntax:

```
TMDec Value [, Options ]
```

Command Availability:

Available on all microcontrollers.

Explanation:

Value is a word value. Only values from 0 to 9999 can be displayed, values greater than 9999 will be displayed as ----.

Options as follows:

- 0 or omitted, only decimal value will be displayed;
- 1 decimal value with the leading zeros;
- 2 decimal number with the colon on digit 2;
- 3 decimal number with the colon on digit 2 and the leading zeros.

TMHex

Syntax:

```
TMHex Value
```

Command Availability:

Available on all microcontrollers.

Explanation:

Value is a word value. Only values from 0x0000 to 0xFFFF can be displayed. Non-hex values will be displayed as greater than 9999 will be displayed ??.

TMWriteChar

Syntax:

```
TMWriteChar ( TMaddr, TMchar )
```

Command Availability:

Available on all microcontrollers.

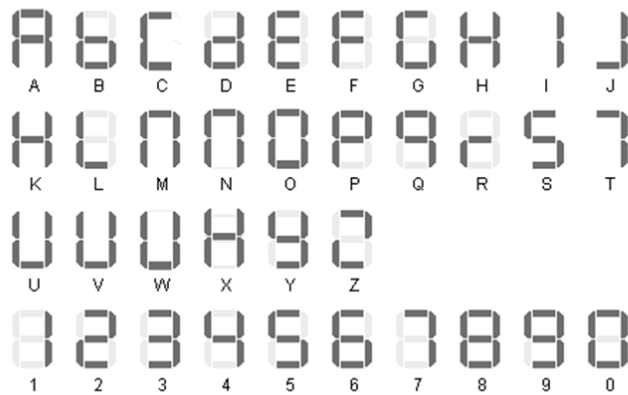
Explanation:

TMaddr is 0 , 1 , 2 , 3 (display left to right) *TMchar* is a letter from A to Z (default alphabet) or from **a** to **z** Siekoo alphabet by Alexander Fakoo, more info at: <http://en.fakoo.de/siekoo.html>. You can insert the special characters (blank, -,) and/or ?).

Character map:

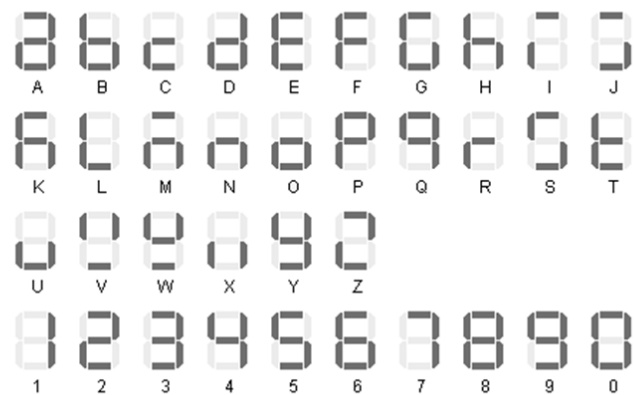
Default 7 - Segment Alphabet

by Mike Otte 2018



7-Segment Alphabet Siekoo

by Alexander Fakoó 2012



(for a confusion-free alphanumeric 7-segment display)

7 Segment Displays - TM1637 6 Digits

Introduction

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The TM1637 specification is

- Two wire interface
- Eight adjustable luminance levels
- 3.3V/5V interface
- Supports six alpha-numeric digits
- Operating current consumption: 80mA

Using the TM1637 Display Module

[graphic] | *TM1637_6d.gif*

Why to use TM1637 Display Module?

The TM1637 can be interfaced to any system using only two ports. This is the main reason the module is preferred over other module.

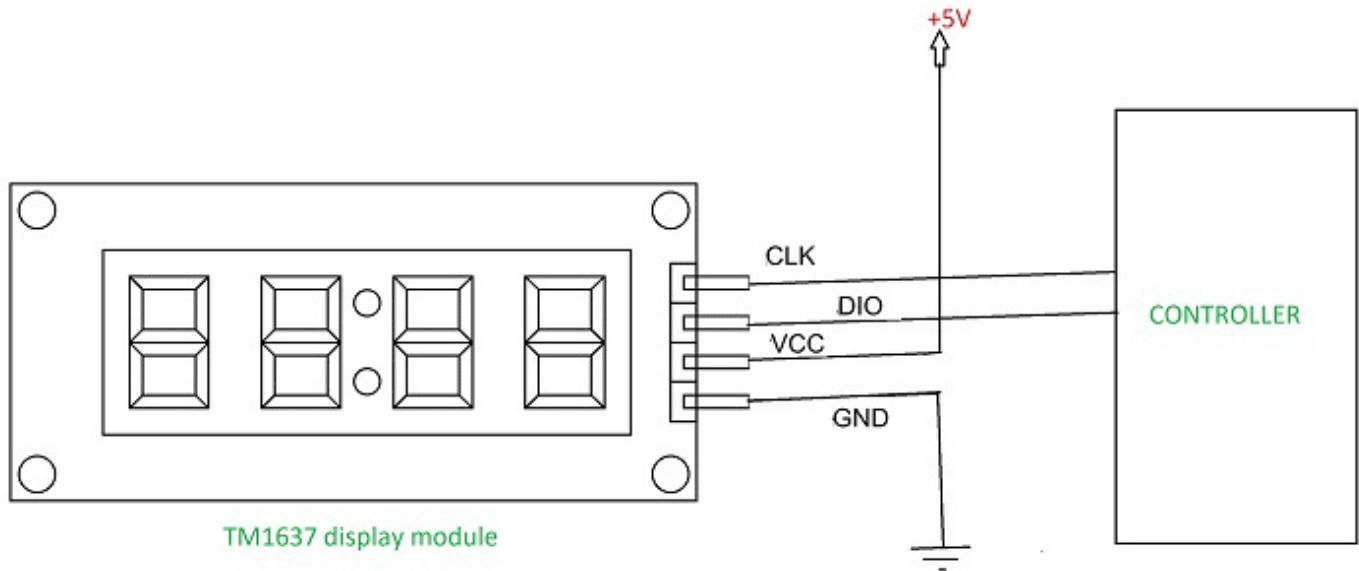
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display modules present in the market they cost more.

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The module can work on +5V regulated power and any higher voltage may lead to permanent damage. The interface is established as shown in figure above. All you need to do is connect DIO and CLK to any of GPIO (General Purpose Input Output) pins of controller and establish serial data exchange through programming.

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The GCBASIC 7 segment display methods make it easier for GCBASIC programs to display numbers and letters on 7 segment LED displays.

The GCBASIC methods supports six 7 segment display devices, common anode/cathode and inversion of the port logic to support driving the device(s) via a transistor.

Brightness can be set: 8 is display on minimum bright , 15 is display on max bright. Less than 8 is display off.

The TM1637 chip supports the reading of the keyboard matrix however that is not supported in the library.

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The following will set the display.

Constant	Description
TM1637_CLK	Must be a bi-directional port. The direction/port setting is managed by the library.
TM1637_DIO	Must be a bi-directional port. The direction/port setting is managed by the library.

Example program

```
#chip mega328p,16
#include <TM1637a.h>

#define TM1637_CLK PortD.2      ' Arduino Digital_2
#define TM1637_DIO PortD.3      ' Arduino Digital_3

'---- main program -----

  TMWrite6Dig (17, 16, 17, 16, 0) 'clear display
    wait 2 s
  TMWrite6Dig (17, 16, 17, 16, 10,0) '- -
    wait 2 s
  TMchar_Bright = 10
```

TMWrite6Dig

Syntax:

```
TMWrite6Dig (dig1, dig2, dig3, dig4, dig5, dig6, Brightness, Point)
```

Command Availability:

Available on all microcontrollers.

Explanation:

Command defines each digit (left to right) as 0 to 9 or 0x00 to 0x0F (15). Additionally 0x10 (16) is a blank, 0x11 (17) is a minus sign, 0x12 (18) is a degree sign, 0x13 (19) is a bracket and 0x14 (20) is a question mark.

Brightness set the brightness (8-15). *Colon* turns the colon (only on digit 2) to off (0) or on (1).

TM_Bright

Syntax:

```
TM_Bright = Brightness
```

Command Availability:

Available on all microcontrollers.

Explanation:

Brightness sets the brightness for the display with a range of 8 to 15. Default to 15.

TM_Bright must be defined before the first use the commands: TMDec, TMHex or TMWriteChar, to set the brightness of the characters (8-15), without this, the display will be blank.

TMDec

Syntax:

```
TMDec Value [, Options ]
```

Command Availability:

Available on all microcontrollers.

Explanation:

Value is a word value. Only values from 0 to 9999 can be displayed, values greater than 9999 will be displayed as ----.

Options as follows:

- 0 or omitted, only decimal value will be displayed;
- 1 decimal value with the leading zeros;
- 2 decimal number with the colon on digit 2;
- 3 decimal number with the colon on digit 2 and the leading zeros.

TMHex

Syntax:

```
TMHex Value
```

Command Availability:

Available on all microcontrollers.

Explanation:

Value is a word value. Only values from 0x0000 to 0xFFFF can be displayed. Non-hex values will be displayed as greater than 9999 will be displayed ??.

TMWriteChar

Syntax:

```
TMWriteChar ( TMaddr, TMchar )
```

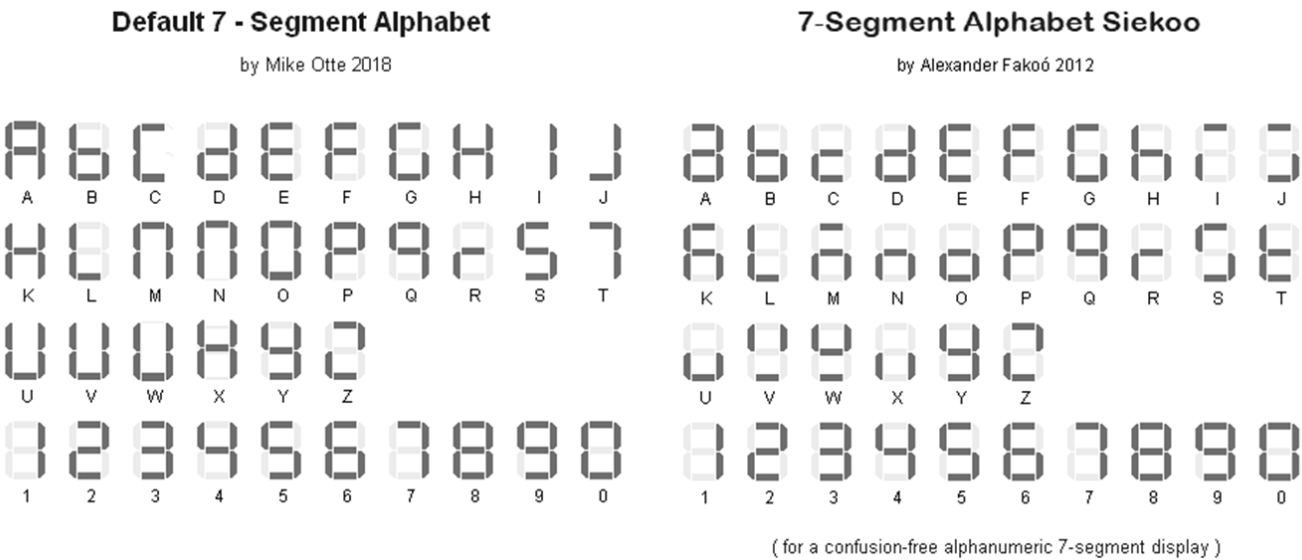
Command Availability:

Available on all microcontrollers.

Explanation:

TMaddr is 0 , 1 , 2 , 3 4, 5 (display left to right) TMchar is a letter from A to Z (default alphabet) or from **a** to **z** Siekoo alphabet by Alexander Fakoo, more info at: <http://en.fakoo.de/siekoo.html>. You can insert the special characters (blank, -,) and/or ?).

Character map:



TM_Point

Syntax:

```
TM_Point = (Point)
```

Command Availability:

Available on all microcontrollers.

Explanation:

Must be defined before use the command TMDec to set the decimal point(s)

Rules for decimal points

You can use the TM_Point and TMWrite6dig commands to turn on one or more decimal points. This is achieved with an 8-bit binary number, with the leftmost bit (MSB) representing the 1st decimal point, the next the 2nd, and so on. The state of the last two bits is ignored because it is only 6 digits.

Examples:

- binary number 0B01010000 (decimal 80) switch on decimal point on digits 2 and 4.
- number 0 switch off all digital points
- 255 (0B11111111) switch all on.

One Wire Devices

This is the One Wire Devices section of the Help file. Please refer the sub-sections for details using the contents/folder view.

DS18B20

The DS18B20 is a 1-Wire digital temperature sensor from Maxim IC.

The sensor reports degrees C with 9 to 12-bit precision from -55C to 125C (+/- 0.5C).

Each sensor has a unique 64-Bit Serial number etched into it. This allows for a number of sensors to be used on one data bus. This sensor is used in many data-logging and temperature control projects.

Reading the temperature from a DS18B20 takes up to 750ms(max).

To use the DS18B20 driver the following is required to added to the GCBASIC source file.

```
#include <DS18B20.h>
```

Note the GCBASIC commands do not work with the older DS1820 or DS18S20 as they have a different internal resolution.

These commands are not designed to be used with parasitically powered DS18B20 sensors.

Command	Usage	Returns
ReadDigitalTemp	Returns two global variables. As follows: DSint the integer value read from the sensors DSdec the string value read from the sensors	Byte variables: DSint String variable: DSdec
ReadTemp	ReadTemp is a function that returns the raw value of the sensor. The temperature is read back in whole degree steps, and the sensor operates from -55 to + 125 degrees Celsius.; Note that bit 7 is 0 for positive temperature values and 1 for negative values (ie negative values will appear as 128 + numeric value).	Word variable via the ReadTemp() function
ReadTemp12	ReadTemp is a function that returns the raw 12bit value of the sensor. The temperature is read back as the raw 12 bit data into a word variable (0.0625 degree resolution).; The user must interpret the data through mathematical manipulation. See the DS18B20 datasheet for more information on the 12 bit temperature/data information construct.	Word variable via the ReadTemp12() function

For more help, see [ReadDigitalTemp](#), [ReadTemp](#) or [ReadTemp12](#)

ReadDigitalTemp

Syntax:

```
ReadDigitalTemp
```

Command Availability:

Available on all microcontrollers.

Explanation:

Return the value of the sensor in two global variables. The following two lines must be included in the GCBASIC source file.

```
#include <DS18B20.h>  
#define DQ PortC.3 ; change port configuration as required
```

This method returns whole part of the sensor value in the byte variable **DSint**, the method also returns decimal part of the sensor value in the byte variable **DSdec**.

Example:

```

'Chip Settings. Assumes the development board with with a 16F877A
#chip 16F877A,1

*#include <DS18B20.h>*

'Use LCD in 4 pin mode and define LCD pins
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7

' DS18B20 port settings
#define DQ PortC.3

do forever

    ReadDigitalTemp

    ' Display the integer value of the sensor on the LCD
    cls
    print "Temp"
    locate 0,8
    print DSInt
    print "."
    print DSdec
    print chr(223)+"C"
    wait 2 s

loop

```

ReadTemp

Syntax:

```
byte_var = ReadTemp
```

Command Availability:

Available on all microcontrollers.

Explanation:

ReadTemp is a function that returns the raw value of the sensor. The following two lines must be included in the GCBASIC source file.

```
#include <DS18B20.h>
#define DQ PortC.3 ; change port configuration as required
```

ReadTemp reads the sensor and stores in output variable. The conversion takes up to 750ms. Readtemp carries out a full 12 bit conversion and then rounds the result to the nearest full degree Celsius.

To read the full 12 bit value of the sensor use the **readtemp12** command.

The temperature is read back in whole degree steps, and the sensor operates from -55 to + 125 degrees Celsius. Note that bit 7 is 0 for positive temperature values and 1 for negative values (ie negative values will appear as 128 + numeric value).

Note the **Readtemp** command does not work with the older DS1820 or DS18S20 as they have a different internal resolution. This command is not designed to be used with parasitically powered DS18B20 sensors, the 5V pin of the sensor must be connected.

Example:

```

'Chip Settings. Assumes the development board with with a 16F877A
#chip 16F877A,1

#include <DS18B20.h>

'Use LCD in 4 pin mode and define LCD pins
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7

' DS18B20 port settings
#define DQ PortC.3

    ccount = 0
    CLS

do forever
    ' The function readtemp returns the integer value of the sensor
    DSdata = readtemp

    ' Display the integer value of the sensor on the LCD
    locate 0,0
    print hex(ccount)
    print " Ceil"
    locate 0,8
    print DSdata
    print chr(223)+"C"

    wait 2 s
    ccount++

loop

```

ReadTemp12

Syntax:

```
byte_var = ReadTemp12
```

Command Availability:

Available on all microcontrollers.

Explanation:

ReadTemp12 is a function that returns the raw value of the sensor. The following two lines must be included in the GCBasic source file.

```
#include <DS18B20.h>
#define DQ PortC.3 ; change port configuration as required
```

Reads sensor and stores in output variable. The conversion takes up to 750ms. **Readtemp12** carries out a full 12 bit conversion.

This command is for advanced users only. For standard 'whole degree' data use the **Readtemp** command.

The temperature is read back as the raw 12 bit data into a word variable (0.0625 degree resolution). The user must interpret the data through mathematical manipulation. See the DS18B20 datasheet for more information on the 12 bit temperature/data information construct.

The function **readtemp12** does not work with the older DS1820 or DS18S20 as they have a different internal resolution. This command is not designed to be used with parasitically powered DS18B20 sensors, the 5V pin of the sensor must be connected.

Example:

```
'Chip Settings. Assumes the development board with with a 16F877A
#chip 16F877A,1

#include <DS18B20.h>

'Use LCD in 4 pin mode and define LCD pins
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RW PORTE.1
#define LCD_RS PORTE.0
#define LCD_Enable PORTE.2
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7

; ----- DS18B20 port settings
#define DQ PortC.3
```

```

; ----- Variables
Dim TempC_100 as word    ' a variable to handle the temperature calculations
Dim CCOUNT, SIGNBIT, WHOLE, FRACT, DIG as Byte
Dim TempC_100 as word    ' a variable to handle the temperature calculations

ccount = 0
CLS

do forever

    'Display the integer and decimal value of the sensor on the LCD

    ' The function readtemp12 returns the raw value of the sensor.
    ' The sensor is read as a 12 bit value. Each unit equates to 0.0625 of a degree
    DSdata = readtemp12
    SignBit = DSdata / 256 / 128
    If SignBit = 0 Then goto Positive
    ' its negative!
    DSdata = ( DSdata # 0xffff ) + 1 ' take twos comp

Positive:

    ' Convert value * 0.0625. Multiple value by 6 then add result to multiplication of
    the value with 25 then divide result by 100.
    TempC_100 = DSdata * 6
    DSdata = ( DSdata * 25 ) / 100
    TempC_100 = TempC_100 + DSdata

    Whole = TempC_100 / 100
    Fract = TempC_100 % 100
    If SignBit = 0 Then goto DisplayTemp
    Print "-"

DisplayTemp:
    locate 1,0
    print hex(ccount)
    print " Real"
    locate 1,8
    print str(Whole)
    print "."
    ' To ensure the decimal part is two digits
    Dig = Fract / 10
    print Dig
    Dig = Fract % 10
    print Dig
    print chr(223)
    print "C"

```

```
        wait 2 s
        ccount++

    loop
```

DS18B20SetResolution

Syntax:

For Single Channel/Device only. The method assumes a single DS18B20 device on the OneWire bus.

```
DS18B20SetResolution ( [DS18B20SetResolution_CONSTANT] )
```

Command Availability:

Available on all microcontrollers.

Explanation:

Set the DS18B20 operating resolution. The configuration register of the DS18B20 allows the user to set the resolution of the temperature-to-digital conversion to 9, 10, 11, or 12 bits. This method set the operating resolution to either 9, 10, 11, or 12 bits.

Calling the method with no parameter will set the operating resolution of the DS18B20 to 12 bits. See example 3 below.

Constants

CONSTANT	Operating resolution	Temperature resolution
DS18B20_TEMP_9_BIT	9 bits	0.5c
DS18B20_TEMP_10_BIT	10 bits	0.25c
DS18B20_TEMP_11_BIT	11 bits	0.125c
DS18B20_TEMP_12_BIT	12 bits	0.0625c

Example Usage 1

The follow example sets the operating resolution of the DS18B20 to 12 bits.

```
#include <DS18B20.h>
#define DQ PortC.3 ; change port configuration as required
DS18B20SetResolution ( DS18B20_TEMP_12_BIT )
```

Example Usage 2

The follow example sets the operating resolution of the DS18B20 to 9 bits.

```
#include <DS18B20.h>
#define DQ PortC.3 ; change port configuration as required
DS18B20SetResolution ( DS18B20_TEMP_9_BIT )
```

Example Usage 3

The follow example sets the operating resolution of the DS18B20 to the default value of 12 bits.

```
#include <DS18B20.h>
#define DQ PortC.3 ; change port configuration as required
DS18B20SetResolution ( )
```

Working Example Program

The following program will display the temperature on a serial attached LCD. Change the `DS18B20SetResolution ( )` method to set the resolution of a specific setting.

You may need to change the chip, edit/remove PPS, and/or the change LCD settings to make this program work with your configuration.

```
#chip 16f18313
#config MCLR=ON
#option Explicit
#include <ds18b20.h>

'Generated by PIC PPS Tool for GCBASIC
'PPS Tool version: 0.0.6.1
'PinManager data: v1.79.0
'Generated for 16f18313
,
'Template comment at the start of the config file
,

#startup InitPPS, 85
#define PPSToolPart 16f18313

Sub InitPPS
```



```

        'Module: EUSART
        RA5PPS = 0x0014      'TX > RA5

End Sub
'Template comment at the end of the config file

'USART settings for USART1
#define USART_BAUD_RATE 115200
#define USART_TX_BLOCKING
#define USART_DELAY OFF

#define LCD_IO 107      'K107
#define LCD_WIDTH 20      ;specified lcd width for clarity only. 20 is the
default width

; ----- Constants
' DS18B20 port settings
#define DQ RA4

; ----- Variables
dim TempC_100 as LONG      ' a variabler to handle the temperature calculations
Dim DSdata,WHOLE, FRACT, DIG as word
Dim CCOUNT, SIGNBIT as Byte

; ----- Main body of program commences here.

ccount = 0
CLS
print "GCBasic 2021"
locate 1,0
print "DS18B20 Demo"
wait 2 s
CLS

DS18B20SetResolution ( DS18B20_TEMP_12_BIT )

do forever
    ' The function readtemp returns the integer value of the sensor
    DSdata = readtemp

    ' Display the integer value of the sensor on the LCD
    locate 0,0
    print hex(ccount)
    print " Ceil"
    locate 0,8

```

```

print DSdata
print chr(223)+"C"

' Display the integer and decimal value of the sensor on the LCD

' The function readtemp12 returns the raw value of the sensor.
' The sensor is read as a 12 bit value therefore each unit equates to 0.0625 of a
degree
DSdata = readtemp12

SignBit = DSdata / 256 / 128
If SignBit = 0 Then goto Positive
' its negative!
DSdata = ( DSdata # 0xffff ) + 1 ' take twos comp

Positive:
' Convert value * 0.0625 by factorisation
TempC_100 = DSdata * 625
Whole = TempC_100 / 10000
Fract = TempC_100 % 10000

If SignBit = 0 Then goto DisplayTemp
Print "-"

DisplayTemp:
Locate 3,0
Print Whole
Print "."
Print leftpad( str(Fract),4,"0")

wait 2 s
ccount++

loop

```

Serial Communications

This is the Serial Communications section of the Help file. Please refer the sub-sections for details using the contents/folder view.

RS232 (software)

This is the Software Serial Communications section of the Help file. Please refer the sub-sections for details using the contents/folder view.

RS232 Software Overview

Introduction:

These routines allow the microcontroller to send and receive RS232 data.

All functions are implemented using software, so no special hardware is required on the microcontroller. However, if the microcontroller has a hardware serial module (usually referred to as UART or USART), and the serial data lines are connected to the appropriate pins, the hardware routines should be used for smaller code, improved reliability and higher baud rates.

Relevant Constants:

These constants are used to control settings for the RS232 serial communication routines. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

Constant Name/s	Controls	Default Value
<code>SendALow</code> , <code>SendBLow</code> , <code>SendCLow</code>	These are used to define the commands used to send a low (0) bit on serial channels A, B and C respectively.	No Default Must be defined
<code>SendAHigh</code> , <code>SendBHigh</code> , <code>SendCHigh</code>	These are used to define the commands used to send a high (1) bit on serial channels A, B and C respectively.	No Default Must be defined
<code>RecALow</code> , <code>RecBLow</code> , <code>RecCLow</code>	The condition that is true when a low bit is being received	<code>Sys232Temp.0 OFF</code> Must be defined
<code>RecAHigh</code> , <code>RecBHigh</code> , <code>RecCHigh</code>	The condition that is true when a high bit is being received	<code>Sys232Temp.0 ON</code> Must be defined

InitSer

Syntax:

```
InitSer channel, rate, start, data, stop, parity, invert
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will set up the serial communications. The parameters are as follows:

`channel` is 1, 2 or 3, and refers to the I/O ports that are used for communication.

`rate` is the bit rate, which is given by the letter r and then the desired rate in bps. Acceptable units are r300, r600, r1200, r2400, r4800, r9600 and r19200.

`start` gives the number of start bits, which is usually 1. To make the microcontroller wait for the start bit before proceeding with the receive, add 128 to `start`. (Note: it may be desirable to use the `WaitForStart` constant here.)

`data` tells the program how many data bits are to be sent or received. In most situations this is 8, but it can range between 1 and 8, inclusive.

`stop` is the number of stop bits. If `start` bit 7 is on, then this number will be ignored.

`parity` refers to a system of error checking used by many devices. It can be odd (in which there must always be an odd number of high bits), even (where the number of high bits must always be even), or none (for systems that do not use parity).

`invert` can be either "normal" or "invert". If it is "invert", then high bits will be changed to low, and low to high.

Example:

Please refer to [SerSend](#) for an example of `InitSer`

For more help, see [RS232 Software Overview](#)

SerSend

Syntax:

```
SerSend channel, data
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will send a byte given by `data` using the RS232 channel referred to as `channel` according to the rules set using `InitSer`.

Example:

'This program will send a byte using PORTB.2, the value of which depends on whether a button is pressed. This can be used with the example for SerReceive.

```
#chip 16F819, 8

#define RS232Out PORTB.2
#define RS232In  PORTB.1

Dir RS232Out Out
Dir RS232In In

'Config Software-UART
#define SendAHigh Set RS232Out ON
#define SendALow Set RS232Out OFF
#define RecAHigh Set RS232In ON
#define RecALow Set RS232In OFF

Dir Button In

InitSer 1, r9600, 1+WaitForStart, 8, 1, none, normal
Do
  If Button = On Then Temp = 100
  If Button = Off Then Temp = 0
  SerSend 1, Temp
  Wait 100 ms
Loop
```

For more help, see [RS232 Software Overview](#), [InitSer](#)

SerReceive**Syntax:**

```
SerReceive channel, output
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will read a byte from the RS232 channel given by `channel` according to the rules set using `InitSer`, and store the received byte in the variable `output`.

Example:

```
'This program will read a byte from PORTB.2, and set the LED on if  
'the byte is more than 50. This can be used with the SerSend  
'example program.
```

```
#chip 16F88, 8
```

```
#define RecAHigh PORTB.2 ON
```

```
#define RecALow PORTB.2 OFF
```

```
#define LED PORTB.0
```

```
Dir PORTB.0 Out
```

```
Dir PORTB.2 In
```

```
InitSer 1, r9600, 1 + WaitForStart, 8, 1, none, normal
```

```
Do
```

```
  SerReceive 1, Temp
```

```
  If Temp <= 50 Then Set LED Off
```

```
  If Temp > 50 Then Set LED On
```

```
Loop
```

For more help, see [RS232 Software Overview](#), [InitSer](#)

SerPrint

Syntax:

```
SerPrint channel, value
```

Command Availability:

Available on all microcontrollers.

Explanation:

SerPrint is used to send a value over the serial connection. **value** can be a string, integer, long, word or byte.

channel is the serial connection to send data through (1 | 2 | 3).

SerPrint will not send any new line characters. If the chip is sending to a terminal, these commands should follow **SerPrint**.

```
SerSend channel, 13
SerSend channel, 10
```

Example:

```
'This program will display any values received over the serial
'connection. If "pot" is received, the value of the analog sensor
'will be sent.

'Chip settings
#chip 18F2525, 8

'LCD settings
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PORTC.7
#define LCD_RW PORTC.6
#define LCD_Enable PORTC.5
#define LCD_DB4 PORTC.4
#define LCD_DB5 PORTC.3
#define LCD_DB6 PORTC.2
#define LCD_DB7 PORTC.1

'Serial settings
#define RS232Out PORTB.0
#define RS232In  PORTB.1

'Set pin direction
Dir RS232Out Out
Dir RS232In In

'Config Software-UART
#define SendAHigh Set RS232Out ON
#define SendALow Set RS232Out OFF
#define RecAHigh Set RS232In ON
#define RecALow Set RS232In OFF
set RS232Out On

Do
  'Potentiometer
  #define POT_PORT PORTA.0
  #define POT_AN AN0

  'Set pin direction
  Dir POT_PORT In
```

```

'Create buffer variables to store received messages
Dim Buffer As String
Dim OldBuffer As String
BufferSize = 0

'Set up serial connection
InitSer 1, r9600, 1 + WaitForStart, 8, 1, none, invert

'Show test messages
Print "Serial Tester"
Wait 1 s
SerPrint 1, "GCBASIC RS232 Test"
SerSend 1, 13
SerSend 1, 10
Wait 1 s

'Main loop
'Get a byte from the terminal
SerReceive 1, Temp

'If Enter key was pressed, deal with buffer contents
If Temp = 13 Then
    Buffer(0) = BufferSize

    'Try to execute commands in buffer
    If Not ExecCommand (Buffer) Then
        'Show message on bottom line, last message on top.
        CLS
        Print OldBuffer
        Locate 1, 0
        Print Buffer
        'Store the message for next time
        OldBuffer = Buffer
    End If

    BufferSize = 0
End If
'Backspace code, delete last character in buffer
If Temp = 8 Then
    If BufferSize > 0 Then BufferSize -= 1
End If
'Received ASCII code between 32 and 127, add to buffer
If Temp >= 32 And Temp <= 127 Then
    BufferSize += 1
    Buffer(BufferSize) = Temp
End If
Loop

```



```

'Takes a sensor reading and sends it to terminal
Sub SendSensorReading
    SerPrint 1, "Sensor Reading: "
    SerPrint 1, ReadAD10(POT_AN)
    SerSend 1, 13
    SerSend 1, 10
End Sub

'Will check the buffer for a command
'If command found, run it and return true
'If not, return false
Function ExecCommand (CmdIn As String)
    ExecCommand = False
    'If received command is "pot", show potentiometer value
    If CmdIn = "pot" Then
        SendSensorReading
        ExecCommand = True
    End If
End Function

```

For more help, see [RS232 Software Overview](#)

RS232 (software optimised)

This is the Software Serial Communications section of the Help file. Please refer the sub-sections for details using the contents/folder view.

RS232 Software Overview - Optimised

Introduction:

These routines allow the microcontroller to send and receive RS232 data.

SoftSerial is a library for the GCBASIC compiler and works on AVR and PIC microcontrollers. These routines allow the microcontroller to send and receive RS232 data. All functions are implemented using software, so no special hardware is required on the microcontroller. SoftSerial uses ASM routines for minimal overhead. If the microcontroller has a hardware serial module (usually referred to as UART or USART) the hardware routines can be used too.

Features

- 3 independent channels Ser1... , Ser2... , Ser3...
- I/O pins user configurable
- polarity can be inverted
- freely adjustable baud rate
- maximum baudrate depends on MCU speed
 - PIC@ 1Mhz 9600 baud
 - PIC@ 4Mhz 38400 baud
 - PIC@ 8Mhz 64000 baud
 - PIC@16Mhz 128000 baud
 - AVR@ 1Mhz 28800 baud
 - AVR@ 8Mhz 115200 baud
 - AVR@16Mhz 460800 baud
- 5 - 8 data bits
- 1 or 2 stop bits
- parity bit not supported

Relevant Constants:

These constants are used to control settings for the RS232 serial communication routines. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

Constant Name/s	Controls	Valid Values	Default value
SER1_TXPORT, SER2_TXPORT, SER3_TXPORT	These are used to define the port for sending on serial channels 1, 2 and 3 respectively. Note, that we also have to define a PortPin (see next line). It is not necessary to define this, if we want to receive only. Sample: #define SER1_TXPORT PortB	PORTA - PORTx	No default defined. An appropriate constant must be defined.
SER1_TXPIN, SER2_TXPIN, SER3_TXPIN	These are used to define the pin (the corresponding bit) for sending on serial channels 1, 2 and 3 respectively. Sample: #define SER1_TXPIN 0	0 - 7	No default defined. An appropriate constant must be defined to enable the TX port.
SER1_RXPORT, SER2_RXPORT, SER3_RXPORT	These are used to define the port for receiving on serial channels 1, 2 and 3 respectively. Note, that we also have to define a PortPin (see next line). It is not necessary to define this, if we want to receive only. Sample: #define SER1_RXPORT PortA	PORTA - PORTx	No default defined. An appropriate constant must be defined to enable the TX port.
SER1_RXPIN, SER2_RXPIN, SER3_RXPIN	These are used to define the pin (the corresponding bit) for receiving on serial channels 1, 2 and 3 respectively. It is not necessary to define this, if we want to send only. Sample: #define SER1_RXPIN 5	0 - 7	No default defined. An appropriate constant must be defined to enable the RX port.
SER1_BAUD, SER2_BAUD, SER3_BAUD	These are used to define the baudrate for sending and receiving on serial channels 1, 2 and 3 respectively. It is not necessary to define this, if we want to send only. Sample: #define SER1_BAUD 19200	75 - 51200 0	No default defined. An appropriate constant must be defined to enable the RX port.
SER1_DATABITS, SER2_DATABITS, SER3_DATABITS	These are used to define the databits for sending and receiving on serial channels 1, 2 and 3 respectively. Sample: #define SER1_DATABITS 7	5 - 8	Optional Default = 8
SER1_STOPBITS, SER2_STOPBITS, SER3_STOPBITS	These are used to define the stopbits for sending and receiving on serial channels 1, 2 and 3 respectively. Sample: #define SER1_STOPBITS 2	1, 2	Optional Default = 1

Constant Name/s	Controls	Valid Values	Default value
SER1_INVERT, SER2_INVERT, SER3_INVERT	These are used to define the polarity for sending and receiving on serial channels 1, 2 and 3 respectively. If it is "On", then high bits will be changed to low, and low to high. This is useful for connection to a PCs native serial port or USB-serial converters with MAX232. Sample: #define SER1_INVERT On	On, Off	Optional Default = Off
SER1_RXNOWAIT, SER2_RXNOWAIT, SER3_RXNOWAIT	These are used to define, if SerNReceive waits for the startbits when receiving on serial channels 1, 2 and 3 respectively. If it is "On", then SerNReceive does not wait for the startbits edge, but directly reads the serial data. Also the time for delaying the startbit is shortened. This is useful when calling SerNReceive from an Interrupt-Service-Routine. Sample: #define SER1_RXNOWAIT On	On, Off	Optional Default = Off
SER1_TXDELAY, SER2_TXDELAY, SER3_TXDELAY	These are used to define, if SerNSend waits for the defined milliseconds after sending a byte of serial data. This is useful when using SerNPrint or SerNSend to a serial device that needs processing time between bytes.	1..255	Optional Default =0
SER1_TXDELAYms, SER2_TXDELAYms, SER3_TXDELAYms	These are used to define, if SerNSend waits for the defined milliseconds after sending a byte of serial data. This is useful when using SerNPrint or SerNSend to a serial device that needs processing time between bytes. Same functionality as SERn_TXDELAY	1..255	Optional Default =0
SER1_TXDELAYus, SER2_TXDELAYus, SER3_TXDELAYus	These are used to define, if SerNSend waits for the defined nanoseconds after sending a byte of serial data. This is useful when using SerNPrint or SerNSend to a serial device that needs processing time between bytes.	1..255	Optional Default =0

SerNSend

Ser1Send, Ser2Send, Ser3Send

Syntax:

```
Ser1Send data  
Ser2Send data  
Ser3Send data
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will send a byte given by data using the channel referred to as Ser1.. , Ser2... , Ser3... according to the rules set by the related defines.

Example:

```
'This program will send one byte using PORTA.5  
  
; ----- Configuration  
#chip 12F1501, 1  
  
; ----- Include library  
#include <SoftSerial.h>  
  
; ----- Config Serial UART for sending:  
#define SER1_BAUD 9600      ; baudrate must be defined  
#define SER1_TXPORT PORTA  ; I/O port (without .bit) must be defined  
#define SER1_TXPIN 5        ; portbit must be defined  
  
; ----- Main body of program commences here.  
Ser1Send 88  'send one byte (88 = X)
```

Exposed in SoftSerial.h authored by Frank Steinberg

SerNPrint**Ser1Print, Ser2Print, Ser3Print****Syntax:**

```
Ser1Print value  
Ser2Print value  
Ser3Print value
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will send a value using the channel referred to as Ser1.. , Ser2... , Ser3... according to the rules set by the related defines. value can be a string, integer, long, word or byte.

Example:

```
'This program will send text and an incrementing value using PORTB.1

; ----- Configuration
#chip 16F886, 16
#option Explicit

; ----- Include library
#include <SoftSerial.h>

; ----- Config Serial UART :
#define SER1_BAUD 115200 ; baudrate must be defined
; Config I/O ports for transmitting:
#define SER1_TXPORT PORTB ; I/O port (without .bit) must be defined
#define SER1_TXPIN 1 ; portbit must be defined

; ----- Variables
Dim xx As Word
xx = 1000

; ----- Main body of program commences here.
Do Forever
  Wait 1 s ; 'time to enjoy the result
  Ser1Send 13 ; 'new line in Terminal
  Ser1Send 10
  Ser1Print "Software-UART: " ; 'send a text
  Ser1Print xx ; 'send the value of xx
  xx += 1
Loop
```

Exposed in SoftSerial.h authored by Frank Steinberg

SerNReceive

Ser1Receive, Ser2Receive, Ser3Receive

Syntax:

```
bytevar = Ser1Receive  
bytevar = Ser2Receive  
bytevar = Ser3Receive
```

Command Availability:

Available on all microcontrollers.

Explanation:

This function will read a byte using the channel referred to as Ser1.. , Ser2... , Ser3... according to the rules set by the related defines. The received byte is stored in the variable bytevar. By default the function waits for the startbit impulse edge before executing the following commands. See the sample files how to realize timeout-functionality or interrupt-driven receiving.

Example:

```

'This program will receive bytes on PORTB.0 and send back using PORTB.1

; ----- Configuration
#chip 16F886, 16
#option Explicit

; ----- Include library
#include <SoftSerial.h>

; ----- Config Serial UART :
#define SER1_BAUD 115200 ; baudrate must be defined
#define SER1_DATABITS 7 ; databits optional (default = 8)
#define SER1_STOPBITS 2 ; stopbits optional (default = 1)
#define SER1_INVERT Off ; inverted polarity optional (default = Off)
; Config I/O ports for transmitting:
#define SER1_TXPORT PORTB ; I/O port (without .bit) must be defined
#define SER1_TXPIN 1 ; portbit must be defined
; Config I/O ports for receiving:
#define SER1_RXPORT PORTB ; I/O port (without .bit) must be defined
#define SER1_RXPIN 0 ; portbit must be defined
#define SER1_RXNOWAIT Off ; don't wait for stopbit optional (default = Off)

; ----- Variables
Dim RecByte As Byte

; ----- Main body of program commences here.
Wait 1 Ms 'delay to prevent garbage if sending too quick after init
Ser1Send 10 'new line in Terminal
Ser1Send 13 '
Ser1Print "Please send a byte!"

Do Forever
  RecByte = Ser1Receive 'receive one byte - wait until detecting startbit
  Ser1Send 13 'newline in Terminal
  Ser1Send 10 '
  Ser1Print "You sent: " 'send a text
  Ser1Send RecByte 'send the sign representing the byte
Loop

```

Exposed in SoftSerial.h authored by Frank Steinberg

RS232 (hardware)

This is the RS232 (hardware) section of the Help file. Please refer the sub-sections for details using the contents/folder view.

RS232 Hardware Overview

Introduction

GCBASIC support programs to communicate easily using RS232.

GCBASIC included microcontroller hardware-based serial routines are intended for use on microcontrollers with built in serial communications modules - normally referred to in datasheets as USART or UART modules. Check the microcontroller data sheet for the defined transmit and receive (TX/Rx) pins. Make sure your program sets the Tx pin direction to Out and the Rx pin direction to In respectively. If the RS232 lines are connected elsewhere, or the microcontroller has no USART module, then the GCBASIC software based RS232 routines must be used.

Initialization of the USART module is handled automatically from your program by defining the chip, speed, and the baudrate. The baudrate generator values are calculated and set, usart is set to asynchronous, usart is enabled , the receive and transmit are enabled. See the table below.

Example:

```
#chip mega328p, 16
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
```

Command Availability:

Available on all microcontrollers with a USART or UART module.
Microchip PIC supports USART1 and 2.
Atmel AVR supports USART 1,2,3 and 4.

The following table explains the methods that can be implemented when using the GCBASIC serial routines.

Commands:

Command	Parameters	Example
Serially print numbers (byte, word, long) or strings.		

Command	Parameters	Example
HSerPrint	Number_constant or number_variable or string [,optional usart address] The optional usart address is microcontroller specific buy can be 1, 2, 3 or 4.	This subroutine prints a variable value to usart 1. No additional parameter for the usart number is used. HSerprint (mynum) To print a variable value to usart 2. Note the additional parameter for the usart address. HSerprint (mynum, 2)
Serially receive ascii number characters and assign to a word variable.		
HSerGetNum	Number_variable [,optional usart address] The optional usart address is microcontroller specific buy can be 1, 2, 3 or 4.	This subroutine ensures that the characters received are numbers. When a carriage return (CR or ASCII code 13) is received this signifies the end of the character stream. Defaults to usart1. To receive number characters use. HSerGetNum (mynum) To receive number characters via usart2 use. HSerGetNum (mynum, 2)
Serially receive characters as a string.		
HSerGetString	User_string_variable [,optional usart address] The optional usart address is microcontroller specific buy can be 1, 2, 3 or 4.	This subroutine ensures that the characters treated as a string. When a carriage return (CR or ASCII code 13) is received this signifies the end of the character stream. GCBASIC will determine the default buffering size for strings. See here for more help on string sizes. Defaults to usart1. To receive a string use. HserGetString (mystring) To a string via usart2 use. HserGetString (mystring, 2)
Serially receive a character using a subroutine.		

Command	Parameters	Example
HSerReceive	byte_variable	This subroutine handles the incoming characters as raw ASCII values. The subroutine receives a single byte value in the range of 0 to 255. The subroutine can receive a byte from usart 1, 2, 3 or 4. The public variable <code>comport</code> can be set before the use of this method to select the desired usart address. If <code>#define USART_BLOCKING</code> is defined then this methods will wait until it a byte is received. If <code>#define USART_BLOCKING</code> is NOT defined then the method will returns ASCII value received or the method will return the value of 255 to indicate not ASCII data was received. You can change the value returned by setting redefining <code>#define DefaultUsartReturnValue = [0-255]</code>. When <code>#define USART_BLOCKING</code> is NOT defined this method becomes a non- blocking method which allows for the testing and handling of incoming ASCII data within the user program. To receive an ASCII byte value in blocking mode use. Defaults to usart1 <pre>#define USART_BLOCKING HSerReceive (user_byte_variable)</pre> To receive an ASCII byte value via usart 3 using blocking mode use <pre>#define USART_BLOCKING Comport = 3 HSerReceive (user_byte_variable)</pre> To receive an ASCII byte value use in non-blocking mode use. Ensure <code>#define USART_BLOCKING</code> is NOT defined. This method fefaults to usart1 <pre>HSerReceive</pre>

Command	Parameters	Example
Serially receive a character using a function specifically via usart1.		
HSerReceive1	none	This function handles the incoming characters as raw ASCII values. The function receives a single byte value in the range of 0 to 255. The function can return only a byte value from usart 1. The blocking and non-blocking mode and the methods are the same as shown in the previous method. To receive an ASCII byte value via usart 1 using blocking mode use <pre>#define USART_BLOCKING user_number_variable =</pre> HSerReceive1 To receive an ASCII byte value use in non-blocking mode use. Ensure #define USART_BLOCKING is NOT defined. user_number_variable = HSerReceive1
Serially receive a character using a function specifically via usart2		

Command	Parameters	Example
<code>HSerReceive2</code>	none	This function handles the incoming characters as raw ASCII values. The function receives a single byte value in the range of 0 to 255. The function can receive only a byte value from usart 2. The blocking and non-blocking mode and the methods are the same as shown in the previous method. To receive an ASCII byte value via usart 2 using blocking mode use <pre>#define USART_BLOCKING user_byte_variable = HSerReceive2</pre> To receive an ASCII byte value use in non-blocking mode use. Ensure #define USART_BLOCKING is NOT defined. user_byte_variable = <code>HSerReceive2</code>
Serially receive a character using a function from either usart ports using a parameter to select the usart.		

Command	Parameters	Example
HSerReceiveFrom	Usart_number, Default is 1	This function handles the incoming characters as raw ASCII values. The function return a single byte value in the range of 0 to 255. The function can receive only a byte value from usart 1 and usart 2 The blocking and non-blocking mode and the methods are the same as shown in the previous method. To receive an ASCII byte value via usart 1 using blocking mode use #define USART_BLOCKING user_byte_variable = HSerReceiveFrom To receive an ASCII byte value use in non-blocking mode use. Ensure #define USART_BLOCKING is NOT defined. 'Chosen_usart = 2 user_byte_variable = HSerReceiveFrom (2)
Serially send a byte using any of the usart ports.		
HSerSend	Byte or byte_variable [optional usart address] + The optional usart address is microcontroller specific buy can be 1, 2, 3 or 4.	This subroutine sends a byte value to usart 1. No additional parameter for the usart number is used. HSerSend (user_byte) To print a variable value to usart 2. Note the additional parameter for the usart address. HSerSend (user_byte, 2)
Serially send a byte and a CR&LF using any of the usart ports		
HSerPrintByteCRLF	Byte or byte_variable + [optional usart address] The optional usart address is microcontroller specific buy can be 1, 2, 3 or 4.	This subroutine sends a byte value to usart 1. HSerPrintCRLF users_byte,2
Serially send CR&LF (can be multiple) using any of the usart ports		

Command	Parameters	Example
HserPrintCRLF	Number of CR&LF to be sent + [optional usart address] The optional usart address is microcontroller specific but can be 1, 2, 3 or 4.	This subroutine sends a CR&LF to port 2. HserPrintCRLF 1,2 ' Will send a CR & LF out of comport 2 to the terminal

Constants These constants affect the operation of the hardware RS232 routines:

Constant Name	Controls	Default Value
USART_BAUD_RATE	Baud rate (in bps) for the routines to operate at.	No default, user must enter a baud. Doesn't have to be a standard baud.
USART_BLOCKING	If defined, this constant will cause the USART routines to wait until data can be sent or received.	No parameter needed. Use “#defining” it implement the action.
USART_TX_BLOCKING	If defined, this constant will cause the Transmit USART routines to wait until Transmit register is empty before writing the next byte which prevents over running the register and losing data.	No parameter needed. Use “#defining” it implement the action.
USART2_BAUD_RATE	Baud rate (in bps) for the routines to operate at.	No default, user must enter a baud. Doesn't have to be a standard baud.
USART2_BLOCKING	If defined, this constant will cause the USART routines to wait until data can be sent or received.	No parameter needed. Use “#defining” it implement the action.
USART2_TX_BLOCKING	If defined, this constant will cause the Transmit USART routines to wait until Transmit register is empty before writing the next byte which prevents over running the register and losing data.	No parameter needed. Use “#defining” it implement the action.
USART3_BAUD_RATE	Baud rate (in bps) for the routines to operate at.	No default, user must enter a baud. Doesn't have to be a standard baud.
USART3_BLOCKING	If defined, this constant will cause the USART routines to wait until data can be sent or received.	No parameter needed. Use “#defining” it implement the action.
USART3_TX_BLOCKING	If defined, this constant will cause the Transmit USART routines to wait until Transmit register is empty before writing the next byte which prevents over running the register and losing data.	No parameter needed. Use “#defining” it implement the action.

Constant Name	Controls	Default Value
USART4_BAUD_RATE	Baud rate (in bps) for the routines to operate at.	No default, user must enter a baud. Doesn't have to be a standard baud.
USART4_BLOCKING	If defined, this constant will cause the USART routines to wait until data can be sent or received.	No parameter needed. Use “#defining” it implement the action.
USART4_TX_BLOCKING	If defined, this constant will cause the Transmit USART routines to wait until Transmit register is empty before writing the next byte which prevents over running the register and losing data.	No parameter needed. Use “#defining” it implement the action.
USART_DELAY	This is the delay between characters.	1 ms To disable this delay between characters ... Use #define USART_DELAY 0 MS, or, To disable this delay between characters ... Use #define USART_DELAY OFF
USART_BLOCKING_TIMEOUT	If defined, this constant will cause the RX USART routines (all USARTs) cease waiting after a specific timeout. PIC only.	#DEFINE USART_BLOCKING_TIMEOUT 125 (where 125 is the time in ms) when using USART_BLOCKING; the timeout is only operation when the constant USART_BLOCKING_TIMEOUT is defined. The objective is to have a very responsive USART received loop by just timing the loop instructions without using timed waits; this sacrifices timing precision but provides excellent communication performance.
CHECK_USART_BAUD_RATE	Instruct the compiler to show the real BPS to be used	Not the default operation
ISSUE_CHECK_USART_BAUD_RATE_WARNING	Instruct the compiler to show BPS calculation errors	Not the default operation
SerPrintCR	Causes a Carriage return to be sent after every HserPrint automatically.	No parameter needed. User “#defining” it implements the action
SerPrintLF	Causes a LineFeed to be sent after every HserPrint. Some communications require both CR and LF	No parameter needed. User “#defining” it implements the action

HSerGetNum

Syntax:


```
HSerGetNum myNum    'Gets a multi digit number  from USART 1
HSerGetNum myNum,1  'Get a multi digit number from USART 1
HSerGetNum myNum,2  'Get a multi digit number from USART 2
```

When the variable type is a word the number range is 0 to 65535
When the variable type is a long the number range is 0 to 99999

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART n_BAUD_RATE 9600` where `n`` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

This command will read a multi digit number received as ascii chars followed by a CR from an external serial source using a hardware serial module. The command checks that only numbers are input disregarding other characters while waiting for the ending <CR>. It can be used only as a subroutine.

Example:

```

    'This program receives a number and CR from a PC terminal and sends it back on both
usarts
    #chip 18f26k22, 16

    'Set the pin directions
    #define USART_BAUD_RATE 9600
    #define USART_BLOCKING
    #define USART2_BAUD_RATE 9600
    #define USART2_BLOCKING

    'Init pins
    #define SerInPort PORTc.7      'usart1 in
    #define SerOutPort PORTc.6    'usart2 out
    'Set pin directions
    Dir SerOutPort Out
    Dir SerInPort In
    Dir PORTB.6 Out               'USART2 out
    Dir PORTB.7 In               'USArt2 in
    Dir PORTB.0 Out              'leds for testing
    Dir PORTB.1 Out              'leds for testing
    Wait 100 Ms

    'Variables
    Dim myNum as Word
    'Main body of program commences here.
    'Message after reset
    HSerPrint "18F26k22"
    HSerPrintCRLF

    'Main routine
    Do forever
        'wait for char from UART
        'HSerReceive InChar
        HSerGetNum myNum,2        'from usart 2
        HSerPrint myNum,1         ' send out usart 1
        HSerPrint myNum,2         'send out usart 2
        HSerPrintCRLF 1,2         'send one CRLF out usart 2
        HserPrintCRLF 1,1         'send one CRLF out usart 1
    loop

```

Example: This program receives number on serial port 1 and displays. This example shows using a Long as the input variable.

Therefore, the result is in the range of 0-99999. The example also shows how to detect a buffer overrun by testing the HSerInByte variable.

```

#chip mega328p, 16

#define USART_BAUD_RATE 9600
#define USART_BLOCKING

Dim myNum as Long ' range 0 to 99999
HSerPrint "Restarted"
HSerPrintCRLF

Do
    HSerGetNum myNum
    HSerPrint myNum

    if HSerInByte <> 13 then
        HSerSend 9
        HSerPrint "Error buffer overrun" 'You should handle error appropriately
    End if
    HSerPrintCRLF
loop
End

```

See also [HSerReceive](#) and [HSerGetString](#)

HSerGetString

Syntax:

```

HSerGetString myString      'Get a multi char string from USART 1
HSerGetString myString,1    'Get a multi char string from USART 1
HSerGetString myString,2    'Get a multi char string from USART 2

```

Variable type is string and the routine checks for numbers, letters, and punctuation.

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART`

`n_BAUD_RATE 9600` where `n`` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

This command will read a multi character string received as ascii input to the hardware serial module followed by a CR from an external serial source. It can be used only as a subroutine. Variable type is string and the routine checks for numbers, letters, and punctuation.

Example:

'This program receives char string and CR from a PC terminal, sends back the string on the serial port, and turns Led's on off by command

```
#chip 18f26k22, 16
```

```
'Set the pin directions
#define USART_BAUD_RATE 9600
#define USART_BLOCKING
#define USART2_BAUD_RATE 9600
#define USART2_BLOCKING
```

```
'InitUSART
#define SerInPort PORTc.7    'USART 1 Rx Pin
#define SerOutPort PORTc.6   'USART 1 Tx Pin
```

```
'Set pin directions
Dir SerOutPort Out
Dir SerInPort In
```

```
Dir PORTB.6 Out    'second USART Tx Pin
Dir PORTB.7 In     'second USART Rx Pin
```

```
Dir PORTB.0 Out    ' LED hooked up for testing
Dir PORTB.1 Out    ' LED hooked up for testing
```

```
Wait 100 Ms
```

```
; ----- Variables
' All byte variables are defined upon use.
Dim myNum as Word
```

```

Dim MyString as String

; ----- Main body of program commences here.
'Message after reset
HSerPrint "18F26k22"
HSerPrintCRLF

'Main routine

Do Forever

    HSerGetString MyString
    HSerPrint MyString
    HSerSend(13)
    If MyString = "LED1 ON" Then
        Set PORTB.0 Off
    End If
    If MyString = "LED1 OFF" Then
        Set PORTB.0 On
    End If
    If MyString = "LED2 ON" Then
        Set PORTB.1 Off
    End If
    If MyString = "LED2 OFF" Then
        Set PORTB.1 On
    End If

Loop

```

See also [HSerReceive](#) and [HSerGetNum](#)

HSerPrint

Syntax:

```

HSerPrint user_value [,1|2|3|4] 'Choose comport with optional parameter
                                'Default comport is 1

'Send a series of ASCII characters using the buffer called SerialPacket
Dim SerialPacket as Alloc
SerialPacket = 66, 105, 108, 108, 38, 69, 118, 97, 110, 13, 10
HserPrint ( SerialPacket, 1 ) 'explicit to comport 1
SerialPacket = 66,44,73,44,82,13,10
HserPrint ( SerialPacket ) 'defaults to comport 1

```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART_n_BAUD_RATE 9600` where `n`` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

`HSerPrint` is used to send a value over the serial connection. `user_value` can be a string, integer, long, word or byte. `HSerPrint` is very similar to `Print`. The data will be sent out the hardware serial module.

`HSerPrint` will not send any new line characters. If the chip is sending to a terminal, these commands should follow every `HSerPrint` :

```
HSerPrint 13
HSerPrint 10
```

Example:

```
'This program will display any values received over the serial
'connection. If "pot" is received, the value of the analog sensor
'will be sent.
'Note: This has been adapted from the SerPrint example.

'Chip settings
#chip 18F2525, 8

'LCD settings
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PORTC.7
#define LCD_RW PORTC.6
#define LCD_Enable PORTC.5
#define LCD_DB4 PORTC.4
```

```

#define LCD_DB5 PORTC.3
#define LCD_DB6 PORTC.2
#define LCD_DB7 PORTC.1

'USART settings
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF

'Potentiometer
#define POT_PORT PORTA.0
#define POT_AN AN0

'Set pin directions
Dir POT_PORT In

'Create buffer variables to store received messages
Dim Buffer As String
Dim OldBuffer As String
BufferSize = 0

'Show test messages
Print "Serial Tester"
Wait 1 s
HSerPrint "GCBASIC RS232 Test"
HSerSend 13
HSerSend 10
Wait 1 s

'Main loop
Do
    'Get a byte from the terminal
    HSerReceive Temp

    'If Enter key was pressed, deal with buffer contents
    If Temp = 13 Then
        Buffer(0) = BufferSize

        'Try to execute commands in buffer
        If Not ExecCommand (Buffer) Then
            'Show message on bottom line, last message on top.
            CLS
            Print OldBuffer
            Locate 1, 0
            Print Buffer
            'Store the message for next time
            OldBuffer = Buffer
        End If
    End If

```

```

    BufferSize = 0
End If
'Backspace code, delete last character in buffer
If Temp = 8 Then
    If BufferSize > 0 Then BufferSize -= 1
End If
'Received ASCII code between 32 and 127, add to buffer
If Temp >= 32 And Temp <= 127 Then
    BufferSize += 1
    Buffer(BufferSize) = Temp
End If
Loop

'Takes a sensor reading and sends it to terminal
Sub SendSensorReading
    HSerPrint "Sensor Reading: "
    HSerPrint ReadAD10(POT_AN)
    HSerSend 13
    HSerSend 10
End Sub

'Will check the buffer for a command
'If command found, run it and return true
'If not, return false
Function ExecCommand (CmdIn As String)
    ExecCommand = False
    'If received command is "pot", show potentiometer value
    If CmdIn = "pot" Then
        SendSensorReading
        ExecCommand = True
    End If
End Function

```

For more help, see also [HSerPrintByteCRLF](#), [HSerPrintStringCRLF](#) and [HSerPrintCRLF](#)

HSerPrintStringCRLF

Syntax:

```

HSerPrintStringCRLF user_string [,1|2|3|4] 'Choose comport with optional parameter
                                           'Default comport is 1

```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.
Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART_n_BAUD_RATE 9600` where `n` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

`HSerPrintStringCRLF` is used to send a string over the serial connection. The parameter can only be a string. `HSerPrintStringCRLF` is very similar to `HSerPrint` but `HSerPrint` can handle all types of variables.

The data will be sent out the hardware serial module.

`HSerPrintStringCRLF` will send new line characters:

Example:

```
'This program will display string over the serial connection.

'Chip settings
#chip 18F2525, 8

'USART settings
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Show string message
HSerPrintStringCRLF "GCBASIC RS232 Test"
Wait 1 s
```

For more help, see also [HSerPrint](#), [HSerPrintByteCRLF](#) and [HSerPrintCRLF](#)

HSerReceive

Syntax:

Used as subroutine:

```
HSerReceive (user_byte_variable)
```

or, if other multiple comports are in use, set the comport before using HSerReceive.

```
comport = 1    '(1|2|3|4|5)Not needed unless using multiple comports in use
HSerReceive (_user_byte_variable_)
```

or, used as function.

```
user_byte_variable = HSerReceive 'Supports only USART1
user_byte_variable = HSerReceive1 'Supports only USART1
user_byte_variable = HSerReceive2 'Supports only USART2
```

or, used to support assigning of received byte to word (or other multi-byte variables). Note the use of casting to ensure the HSerReceive uses byte addressing.

```
Dim dbAdr as Word

HSerReceive [byte]dbAdr_H
HSerReceive [byte]dbAdr
```

For other comports use Function HSerReceiveFrom

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1,2,3,4 and 5.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the USART_BAUD_RATE and to meet your needs. For addition USART ports use #define USART n_BAUD_RATE 9600 where n` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600      'Set the baud rate
#define USART_TX_BLOCKING         'Ensure the transmit buffer is empty
#define USART_BLOCKING            'Ensures a data byte is in the receive buffer
#define USART_DELAY OFF           'Disables USART delays
```

Explanation:

This command will read a byte from the hardware RS232 module. It can be used either as a subroutine or as a function. If used as a subroutine, a variable must be supplied to store the received value in. If used as a function, it will return the received value.

The subroutine HSerReceive can get a byte from any comport but must set the comport number immediately before the call. If "#define USART_BLOCKING" is defined then the HserReceive waits in a loop until it receives a byte. If " #define USART_BLOCKING" is NOT defined then HserReceive returns the new byte that was received OR returns 255 because of "DefaultUsartReturnValue = 255" was defined. This is good because it don't hold up your program from executing other commands and your can check it for new data priodically.

Example:

```
'This program will read a value from the USART, and send it to PORTB.
```

```
#chip 16F877A, 20
```

```
'USART settings
```

```
#define USART_BAUD_RATE 9600 'sets up comport 1 for 9600 baud
```

```
'Set PORTB to output
```

```
Dir PORTB Out
```

```
'Set USART receive pin to input
```

```
Dir PORTC.7 In
```

```
'Main loop
```

```
Do
```

```
  'Get serial data and output value to PortB as 8 bit binary
```

```
  HSerReceive(InChar) 'Receive data as Subroutine from comport 1
```

```
  'InChar = HSerReceive 'Could also be written as Function
```

```
  If InChar <> 255 Then 'If value is 255 then it is old data
```

```
    PortB = InChar 'If new data then it goes to PortB
```

```
  End If
```

```
Loop
```

Example 2:

```

'If you choose no "Blocking" and comment both of them out.
'USART settings
#define USART_BAUD_RATE 9600
'#define USART_BLOCKING      ' just none OR one of the blocking
'#define USART_TX_BLOCKING   ' statements should be defined

'Main loop
Do
  'Get and display value
  'If there is no new data, HSerReceive will return default value.
  comport = 1
  HSerReceive tempvalue
  If tempvalue <> 255    Then    ' don't change PortB if it is default
    PortB = tempvalue
  End If

Loop

```

Example 3:

```

'If you choose no "Blocking" and comment both of them out.
#chip mega328p, 16

#define USART_BAUD_RATE 9600
'#define USART_BLOCKING
'#define USART_TX_BLOCKING

'Don't forget to Set usart pin directions
Dir PortD.1 Out    'com1    USART0
Dir PortD.0 In

Wait 1 s

'Message after reset
HSerPrint "ATmega328P  com test"
HSerPrintCRLF

'Main routine  hook up FTDI232 usb to serial and use terminal program to check
Start:
    comport = 1
    HSerReceive(InChar)    'Subroutine needs the comport set
    'InChar = HSerReceive    ' This function will get from comport 1
    If InChar <> 255 Then    ' check if for received byte
        'return 255 if old data
        HSerSend InChar    'send back char to UART
    End If
    Goto Start

```

See also [RS232 Hardware Overview](#)

HSerReceiveFrom

Syntax:

```

user_byte = HSerReceiveFrom [,1 | 2 | 3 | 4]
user_byte = HSerReceiveFrom          'Defaults to USART1

'other Receive functions
user_byte = HSerReceive1    'from USART1
user_byte = HSerReceive2    'from USART2

```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.
Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART n_BAUD_RATE 9600` where `n` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

This command will read a byte from the hardware RS232 module. It can be only be used as a function. It will return the received value.

Example:

```
'This program will read a value from the USART, and display it on PORTB.

#chip 16F877A, 20

'USART settings
#define USART_BAUD_RATE 9600
#define USART_BLOCKING
#define USART_TX_BLOCKING

'Set PORTB to input
Dir PORTB Out
'Set USART receive pin to input
Dir PORTC.7 In

'Main loop
Do
  'Get byte value
  bytein = HSerReceiveFrom (2)
  'do something useful
Loop
```

See also [HSerReceive](#)

HSerSend

Syntax:

```
HSerSend user_byte [,1|2|3|4] 'Choose comport with optional parameter  
                                'Default comport is 1
```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART n_BAUD_RATE 9600` where `n`` is the required port number.

```
'USART settings for USART1  
#define USART_BAUD_RATE 9600  
#define USART_TX_BLOCKING  
#define USART_DELAY OFF
```

Explanation:

This command will send a byte given by `user_byte` using the hardware RS232 module.

Example:

```

'This program will send the status of PORTB through the hardware
'serial module.

#chip 16F877A, 20

'USART settings
#define USART_BAUD_RATE 9600 'Initializes USART port with 9600 baud
'#define USART_BLOCKING ' Both of these blocking statements will
#define USART_TX_BLOCKING ' wait for tx register to be empty
    ' use only one of the two constants
#define USART_DELAY OFF

'Set PORTB to input
Dir PORTB In
'Set USART transmit pin to output
Dir PORTC.6 Out

'Main loop
Do
    'Send PORTB value through USART
    HSerSend PORTB
    HSerSend(13) ' sends a CR
    'Short delay for receiver to process message
    Wait 10 ms 'probably not necessary with blocking statement
Loop

```

HserPrintByteCRLF

Syntax:

```
HserPrintByteCRLF user_data [, 1 | 2 | 3 | 4 ]
```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.

Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART n_BAUD_RATE 9600` where `n`` is the required port number.


```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

This command will send a byte given by *user_data* using the hardware USART module and then send the ASCII codes 13 and 10. ASCII codes 13 and 10 equate to a carriage return and line feed.

Example:

```
'This program will send the status of PORTB through the hardware serial module.

HserPrintByteCRLF 65      ' Will print a single A on the terminal
HserPrintByteCRLF "A"    ' Will print a single A on the terminal
```

See also [HserPrintCRLF](#)

HserPrintCRLF

Syntax:

```
HserPrintCRLF [optional BYTE] [, 1 | 2 | 3 | 4 ]
```

Command Availability:

Available on all microcontrollers with a USART or UART module.

Microchip PIC supports USART1 and 2.+ Atmel AVR supports USART 1,2,3 and 4.

Enabling Constants:

To enable the use of the USART these are the enabling constants. These constants are required. You can change the `USART_BAUD_RATE` and to meet your needs. For addition USART ports use `#define USART n_BAUD_RATE 9600` where `n`` is the required port number.

```
'USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY OFF
```

Explanation:

This command will send ASCII codes 13 and 10 only using the hardware RS232 module. ASCII codes 13 and 10 equate to a carriage return and line feed.

Optionally, you can add a parameter. The number will determine the number of ASCII codes 13 and 10 set to the hardware RS232 module.

Also you can choose the comport with second optional parameter if it is not the default comport 1. If there is no first optional parameter then you must have atleast a comma before it to indicate this is the second parameter.

Examples:

```
'This Line will send 1 CR and LF
HserPrintCRLF    ' Will send a CR & LF to the terminal

'This Line will send 2 times (CR and LF)
HserPrintCRLF 2   ' Will send 2 times (CR & LF) to the terminal
                  'out of comport 1

'This Line will send 1 CR and LF
HserPrintCRLF 1,2 ' Will send a CR & LF out of
                  'comport 2 to the terminal
```

See also [HserPrintByteCRLF](#)

PS/2

This is the PS/2 section of the Help file. Please refer the sub-sections for details using the contents/folder view.

PS/2 Overview

PS2 Overview

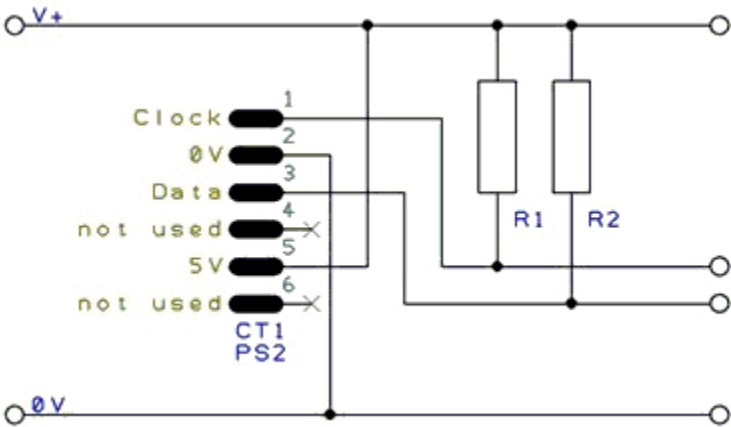
These routines make it easier to communicate with a PS/2 device, particularly an external keyboard.

Relevant Constants

These constants affect the operation of the PS/2 routines:

Constant Name	Controls	Default Value
PS2Data	Pin connected to PS/2 data line	Must be specified
PS2Clock	Pin connected to PS/2 clock line.	Must be specified
PS2_DELAY	This constant can be set to a delay, such as 10 ms. If set, a delay will be added at the end of every byte sent or received.	Not set

Connections between the Keyboard and the Microcontroller The following diagram show a typical connection between the keyboard and the microcontroller. The value of R1 and R2 is typically 4.7k for a 5v system.



InKey

Syntax:

```
output = InKey
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **InKey** function will read the last pressed key from a PS/2 keyboard, and return an ASCII value corresponding to the key. If no key is pressed, then **InKey** will return 0.

It will also monitor Caps Lock, Num Lock and Scroll Lock keys, and update the status LEDs as appropriate.

Example:

```
'A program to accept messages from a standard PS/2 keyboard
'Any keys pressed will be shown on an LCD screen.

'Hardware settings
#chip 18F4620, 20

'LCD connection settings
#define LCD_IO 4
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DB4 PORTD.4
#define LCD_DB5 PORTD.5
#define LCD_DB6 PORTD.6
#define LCD_DB7 PORTD.7
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

'PS/2 connection settings
#define PS2Clock PORTC.1
#define PS2Data PORTC.0
#define PS2_DELAY 10 ms

'Set up key log
Dim KeyLog(32)
DataCount = 0
KeyLog(1) = 32

Main:
'Read the last pressed key
```

```

KeyIn = INKEY
'If no key pressed, try reading again
If KeyIn = 0 Then Goto Main

'Escape pressed - clear message
If KeyIn = 27 Then
    DataCount = 0
    For DataPos = 1 to 32
        KeyLog(DataPos) = 32
    Next
    Goto DisplayData
End If

'Backspace pressed - delete last character
If KeyIn = 8 Then
    If DataCount = 0 Then Goto Main
    KeyLog(DataCount) = 32
    DataCount = DataCount - 1
    Goto DisplayData
End If

'Otherwise, add the character to the buffer
If KeyIn >= 31 And KeyIn <= 127 Then
    DataCount = DataCount + 1
    KeyLog(DataCount) = KeyIn
End If

DisplayData:
'Display key buffer
'LCDWriteChar is used instead of Print for greater control
CLS
For DataPos = 1 to DataCount
    If DataPos = 17 then Locate 1, 0
    LCDWriteChar KeyLog(DataPos)
Next

Goto Main

```

PS2SetKBLEDs

Syntax:

```
PS2SetKBLEDs (LedStatus)
```

Command Availability:

Available on all microcontrollers.

Explanation:

This routine will turn the status LEDs on a keyboard on or off. `LedStatus` is a variable, of which the lower 3 bits correspond to the 3 LEDs. Bit 0 is for Scroll Lock, bit 1 controls Num Lock and bit 2 controls Caps Lock.

Note that this routine does not alter the status variables within the INKEY routine - so even if the Caps Lock LED is turned on, Caps Lock will stay off.

Example:

```
'A spinning LED program for a keyboard
'Will flash Num Lock, then Caps Lock, then Scroll Lock.

'Hardware settings
#chip 16F88, 8

#define PS2Clock PORTB.2
#define PS2Data PORTB.3
#define PS2_DELAY 10 ms

'Main Loop
Do

    'Turn on only Num Lock (bit 1)
    PS2SetKBLeds b'00000010'
    Wait 250 ms

    'Turn on only Caps Lock (bit 2)
    PS2SetKBLeds b'00000100'
    Wait 250 ms

    'Turn on only Scroll Lock (bit 0)
    PS2SetKBLeds b'00000001'
    Wait 250 ms

Loop
```

PS2ReadByte

Syntax:

```
output = PS2ReadByte
```

Command Availability:

Available on all microcontrollers.

Explanation:

PS2ReadByte will read a byte from the PS/2 bus. It will return the byte, or 0 if no data was returned by the PS/2 device.

The PS/2 bus will normally be held in the inhibit state. **PS2ReadByte** will uninhibit the bus for 25 ms. If a response is received, it will be read. Then, the bus will be placed back in the inhibit state.

Example:

For an example, please refer to the **InKey** function. [PS2 Inkey](#)

PS2WriteByte**Syntax:**

```
PS2WriteByte user_data
```

Command Availability:

Available on all microcontrollers.

Explanation:

PS2WriteByte will send a byte to a PS/2 device. Once the byte has been written, the PS/2 bus will be placed in the inhibit state.

Example:

For an example, please refer to the **PS2SetKBLeds** function.

[PS2 Set Keyboard Leds](#)

SPI

This is the Serial Peripheral Interface section of the Help file. Please refer the sub-sections for details using the contents/folder view.

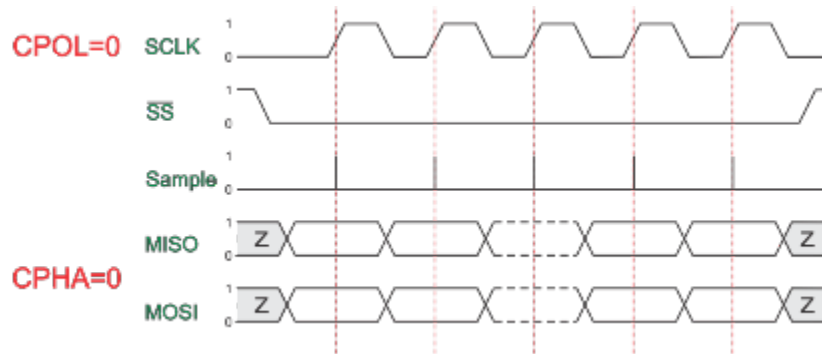
SPI Overview

Syntax:

The SPI interface allows for the transmission and reception of data simultaneously on two lines (MOSI and MISO).

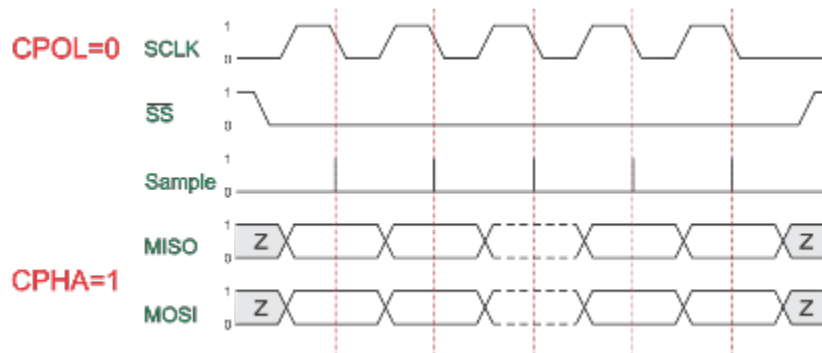
The Clock polarity (CPOL) and clock phase (CPHA) are the main parameters that define a clock format to be used by the SPI bus. Depending on CPOL parameter, SPI clock may be inverted or non-inverted. CPHA parameter is used to shift the sampling phase. If CPHA=0 the data are sampled on the leading (first) clock edge. If CPHA=1 the data are sampled on the trailing (second) clock edge, regardless of whether that clock edge is rising or falling.

CPOL=0, CPHA=0



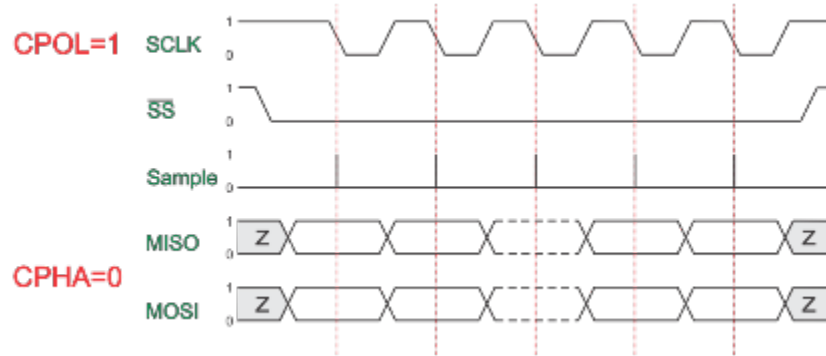
The data must be available before the first clock signal rising. The clock idle state is zero. The data on MISO and MOSI lines must be stable while the clock is high and can be changed when the clock is low. The data is captured on the clock's low-to-high transition and propagated on high-to-low clock transition.

CPOL=0, CPHA=1



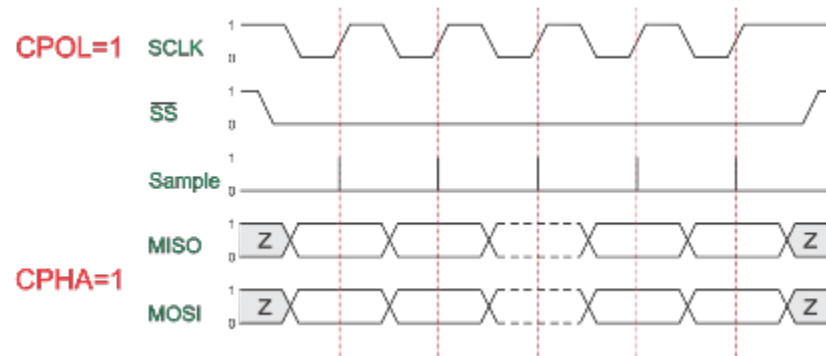
The first clock signal rising can be used to prepare the data. The clock idle state is zero. The data on MISO and MOSI lines must be stable while the clock is low and can be changed when the clock is high. The data is captured on the clock's high-to-low transition and propagated on low-to-high clock transition.

CPOL=1, CPHA=0



The data must be available before the first clock signal falling. The clock idle state is one. The data on MISO and MOSI lines must be stable while the clock is low and can be changed when the clock is high. The data is captured on the clock's high-to-low transition and propagated on low-to-high clock transition.

CPOL=1, CPHA=1



The first clock signal falling can be used to prepare the data. The clock idle state is one. The data on MISO and MOSI lines must be stable while the clock is high and can be changed when the clock is low. The data is captured on the clock's low-to-high transition and propagated on high-to-low clock transition.

Key Commands

```

// Set the mode
SPIMode ( _Mode_ [, SPIClockMode]) //Legacy SPI
SPIMode ( _Mode_ , SPIClockMode) //18FxxQxx, 18FxxK42 and 18xxFK83
microcontrollers

// Send byte and receive data byte
SPITransfer ( _OutByte_, _InByte_ )

// Send data byte
FastHWSPITransfer( _OutByte_ )

// Use MASTERSLOW | MASTER | MASTERFAST | MASTERULTRAFast for specific AVR's only |
MasterSSPADMode for specific PIC's SSPADD support

// The system constant 'HWSPIMode' defaults to MASTERFAST when microcontroller
frequency less or equal to 32 mhz
// Defaults to MASTER when microcontroller frequency more than 32 mhz.
// To change use the following method
#define HWSPIMode MASTERULTRAFast

```

The GCBASIC used the microcontrollers hardware module for SPI. The example below shows an implementation of Hardware and Software SPI. Software SPI allows for a greater choice of ports to be used to control the SPI operations.

To use a second SPI hardware module use the suffix 2, as follows:

```

// Set the mode
SPIMode2 ( _Mode_ [, SPIClockMode]) //Legacy SPI
SPIMode2 ( _Mode_ , SPIClockMode) //18FxxQxx, 18FxxK42 and 18xxFK83
microcontrollers

// Send byte and receive data byte
SPITransfer2 ( _OutByte_, _InByte_ )

// Send data byte
FastHWSPITransfer2( _OutByte_ )

```

Example

This example demonstrates the SPI capabilities for the mega328p. The process is similar of any microcontroller..

This example show using the hardware SPI option and a software SPI option.

Using hardware SPI mode - make sure the `#define SPI_HardwareSPI` is not commented out. Using

software SPI mode - comment out `#define SPI_HardwareSPI`. The example code will then use software SPI.

Using multiple SPI devices

There will be use cases where you need to use more than one SPI target device at a time. In such cases the device defined for SPI must be inserted in your program for each device.

As an example using e-Paper and SRAM at the same time, with an hardware SPI would require `#define SPISRAM_HARDWARESPI` and `#define EPD_HardwareSPI`.

Obviously, when all SPI devices use the same SPI lines, you must select one device at a time by setting SPI Chip Select line to **OFF** for the specific target SPI device, and you must set **ON** the SPI Chip Select line for any other SPI device - this is a normal convention of SPI usage. This is not specific to GCBASIC..

Code overview

For more code examples see the demonstrations and the SPI MODE Help page.

In this example `InitSPIMode` calls `SPIMode`. If needed, when hardware mode, and set the port directions.

The sub `SendByteviaSPI` is called to handle whether to call the Hardware or use Software (bit-banging) SPI.

```
#chip mega328p, 16
#option explicit
#include <UNO_mega328p.h >

#define SPI_HardwareSPI 'comment this out to make into Software SPI but, you may
have to change clock lines

'Pin mappings for SPI - this SPI driver supports Hardware SPI
#define SPI_DC          DIGITAL_8          ' Data command line
#define SPI_CS          DIGITAL_4          ' Chip select line
#define SPI_RESET       DIGITAL_9          ' Reset line

#define SPI_DI          DIGITAL_12         ' Data in | MISO
#define SPI_DO          DIGITAL_11         ' Data out | MOSI
#define SPI_SCK          DIGITAL_13        ' Clock Line

dir SPI_DC    out
dir SPI_CS    out
```

```
dir SPI_RESET out
dir SPI_DO Out
dir SPI_DI In
dir SPI_SCK Out
```

'If DIGITAL_10 is NOT used as the SPI_CS (sometimes called SS) the port must and output or set as input/pulled high with a 10k resistor.

'As follows:

'If CS is configured as an input, it must be held high to ensure Master SPI operation.

'If the CS pin is driven low by peripheral circuitry when the SPI is configured as a Master with the SS pin defined as an input, the

'SPI system interprets this as another master selecting the SPI as a slave and starting to send data to it!

'If CS is an output SPI communications will commence with no flow control.

```
dir DIGITAL_10 Out
```

```
DIM byte1 As byte
```

```
DIM byte2 As byte
```

```
DIM byte3 As byte
```

```
byte1 = 100 ' temp values (will come from potentiometer later)
```

```
byte2 = 150
```

```
byte3 = 200
```

```
InitSPIMode
```

```
do forever
```

```
  set SPI_CS OFF;
```

```
  set SPI_DC OFF;
```

```
  SendByteviaSPI (byte1)
```

```
  set SPI_CS ON;
```

```
  set SPI_DC ON
```

```
  set SPI_CS OFF;
```

```
  set SPI_DC OFF;
```

```
  SendByteviaSPI (byte2)
```

```
  set SPI_CS ON;
```

```
  set SPI_DC ON
```

```
  set SPI_CS OFF;
```

```
  set SPI_DC OFF;
```

```
  SendByteviaSPI (byte3)
```

```
  set SPI_CS ON;
```

```
  set SPI_DC ON
```

```
  wait 10 ms
```

```

loop

sub InitSPIMode

    #ifdef SPI_HardwareSPI
        SPIMode ( MasterFast, SPI_CPOL_0 + SPI_CPHA_0 )
    #endif

    set SPI_DO OFF;
    set SPI_CS ON;    therefore CPOL=0
    set SPI_DC ON;    deselect

End sub

sub SendByteviaSPI( in SPISendByte as byte )

    set SPI_CS OFF
    set SPI_DC OFF;

    #ifdef SPI_HardwareSPI
        FastHWSPITransfer SPISendByte
        set SPI_CS ON;
        exit sub
    #endif

    #ifndef SPI_HardwareSPI
        repeat 8

            if SPISendByte.7 = ON then
                set SPI_DO ON;
            else
                set SPI_DO OFF;
            end if
            SET SPI_SCK On;           ; therefore CPOL=0 ==ON, and, where CPOL=1==ON
            rotate SPISendByte left
            set SPI_SCK Off;          ; therefore CPOL=0 =OFF, and, where CPOL=1==OFF

        end repeat
        set SPI_CS ON;
        set SPI_DO OFF;
    #endif

end Sub

```

See also [SPIMode](#), [SPITransfer](#), [FastHWSPITransfer](#)

SPIMode

Syntax:

Legacy SPI Operations

```
SPIMode ( _Mode_ [, _SPIClockMode_])

// Specfic the hardware SPI operating mode, can be MasterFast, Master, MasterSlow
#define HWSPIMode    MasterFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#define HWSPIClockMode  SPI_CPOL_0 + SPI_CPHA_0
```

AVRDX, 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers

For HWSPI channel 0

```
SPIMode ( _Mode_ , _SPIClockMode_ )

// Specfic the hardware SPI operating mode, can be MasterUltraFast, MasterFast,
Master, MasterSlow
#define HWSPIMode    MasterUltraFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#define HWSPIClockMode  SPI_SS_0 + SPI_CPOL_0 + SPI_CPHA_0

// Optionally change the SPI BAUD RATE from 4000
#define SPI_BAUD_RATE 8000

// Optionally update the SPI baud rate register with an explicit value
// typical use is to entry a specific calculated value
#define SPI_BAUD_RATE_REGISTER  55
```

Command Availability:

Available on Microchip PIC and AVR microcontrollers with Hardware SPI modules.

Explanation:

Mode sets the mode of the SPI module within the microcontroller. These are the possible SPI Modes:

Mode Name	Description
Legacy SPI Operations	

Mode Name	Description
MasterSlow	Master mode, SPI clock is 1/64 of the frequency of the microcontroller.
Master	Master mode, SPI clock is 1/16 of the frequency of the microcontroller.
MasterFast	Master mode, SPI clock is 1/4 of the frequency of the microcontroller.
AVRDX, 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	
MasterSlow	SPI clock baud rate is calculated $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI_BAUD_RATE}) / 16 * 1000) + 1$. Where SPI_BAUD_RATE defaults to 4000. Also, see SPI_BAUD_RATE and SPI_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
Master	SPI clock baud rate is calculated as $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI_BAUD_RATE}) / 4 * 1000) + 1$. Where SPI_BAUD_RATE defaults to 4000. Also, see SPI_BAUD_RATE and SPI_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
MasterFast	SPI clock baud rate is calculated as $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI_BAUD_RATE}) / 2 * 1000) + 1$. Where SPI_BAUD_RATE defaults to 4000. Also, see SPI_BAUD_RATE and SPI_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
MasterUltraFast	SPI1BAUD is set to 0 and therefore the SPI clock baud rate to maximum
Slave Operations	
Slave	Slave mode
SlaveSS	Slave mode, with the Slave Select pin enabled.

For **Legacy microcontrollers SPI operations** *SPIClockMode* is an optional parameter to set the mode of the SPI clock mode. This optional parameter sets both the clock polarity and clock edge.

For **Specific PICs microcontrollers SPI operations** *SPIClockMode* is a mandated parameter to set the mode of the SPI clock mode and the clock polarity bit. This parameter sets both the clock polarity and clock edge. There is no verification by the compiler if you do use the *_SPIClockMode* for the 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers - the compiler uses the default value of [SPI_SS = 0 & SPI_CPOL = 0 & SPI_CPHA = 0](#) The use of SPI_SS_n requires the PPS to be set. If PPS is not set then the SPI_SS will use the default value specified in the specific GCBASIC library.

For the *_SPIClockMode_range*, see the tables below:

SPIClockMode	Description
<i>Legacy SPI operations</i>	
0	SPI_CPOL = 0 & SPI_CPHA = 0
1	SPI_CPOL = 0 & SPI_CPHA = 1

SPIClockMode	Description
2	SPI_CPOL = 1 & SPI_CPHA = 0
3	SPI_CPOL = 1 & SPI_CPHA = 1
18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	
0	SPI_SS = 0 & SPI_CPOL = 0 & SPI_CPHA = 0
2	SPI_SS = 0 & SPI_CPOL = 1 & SPI_CPHA = 0
5	SPI_SS = 1 & SPI_CPOL = 0 & SPI_CPHA = 1
7	SPI_SS = 1 & SPI_CPOL = 1 & SPI_CPHA = 1

You can use a constant value or alternatively you can use constants to set the SPIClockMode as follows:

```
_Legacy SPI microcontrollers_
SPIMode ( MasterFast, SPI_CPOL_n + SPI_CPHA_n )

_18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers_
SPIMode ( MasterFast, SPI_SS_n + SPI_CPOL_n + SPI_CPHA_n )
```

Where the following parameters can be used as a calculation to set the SPIClockMode.

Mode Name	Description
<i>Legacy SPI operations and AVR</i> s	
SPI_CPOL_0	CPOL = 0
SPI_CPOL_1	CPOL = 1
SPI_CPHA_0	CPHA = 0
SPI_CPHA_1	CPHA = 1
18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	
SPI_SS_0	SS = 0 Clear polarity bit
SPI_SS_1	SS = 1 Set polarity bit

Explicitly changing the SPI baud rate on 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers

You can explicitly change the SPI baud rate by defining the **SPI_BAUD_RATE** constant as follows. This will change the default SPI baud from 4000 to the specified numeric value.

```
#DEFINE SPI_BAUD_RATE 8000
```


You can explicitly set the SPI baud rate register by defining the `SPI_BAUD_RATE_REGISTER` constant as follows. This will write the explicit numeric value to the SPI baud register. This overwrites any compiler calculated value.

```
#DEFINE SPI_BAUD_RATE_REGISTER 55
```

Legacy SPI Summary:

When using SPI setting the clock frequency is completed using `SPIMode`, and the master must also configure the clock polarity and phase with respect to the data. Using the two options as `CPOL` and `CPHA`.

The timing diagram is shown below. The timing is further described and applies to both the master and the slave device.

When `CPOL=0` the base value of the clock is zero, i.e. the active state is 1 and idle state is 0.

- For `CPHA=0`, data are captured on the clock's rising edge (low → high transition) and data is output on a falling edge (high → low clock transition).
- For `CPHA=1`, data are captured on the clock's falling edge and data is output on a rising edge.

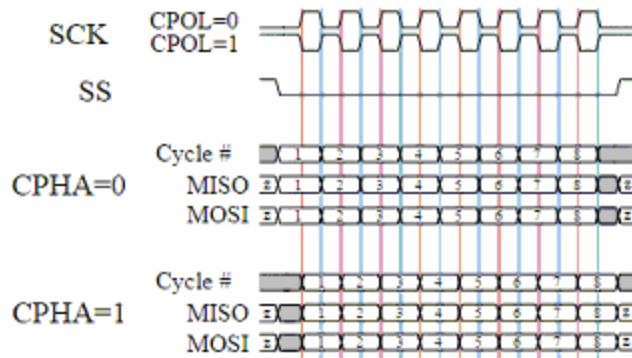
When `CPOL=1` the base value of the clock is one (inversion of `CPOL=0`), i.e. the active state is 0 and idle state is 1.

- For `CPHA=0`, data are captured on clock's falling edge and data is output on a rising edge.
- For `CPHA=1`, data are captured on clock's rising edge and data is output on a falling edge.

When `CPHA=0` means sampling on the first clock edge and , while `CPHA=1` means sampling on the second clock edge, regardless of whether that clock edge is rising or falling. Note that with `CPHA=0`, the data must be stable for a half cycle before the first clock cycle.

In other words, `CPHA=0` means transmitting data on the active to idle state and `CPHA=1` means that data is transmitted on the idle to active state edge. Note that if transmission happens on a particular edge, then capturing will happen on the opposite edge(i.e. if transmission happens on falling, then reception happens on rising and vice versa). The `MOSI` and `MISO` signals are usually stable (at their reception points) for the half cycle until the next clock transition. SPI master and slave devices may well sample data at different points in that half cycle.

This adds more flexibility to the communication channel between the master and slave.



Legacy Example:

This example demonstrates the SPI capabilities for the mega328p. The process is similar of any microcontroller..

You must set the data line as inputs and outputs.

```
#chip mega328p, 16
#option explicit
#include <UNO_mega328p.h >

#define SPI_HardwareSPI 'comment this out to make into Software SPI but, you may
have to change clock lines

'Pin mappings for SPI - this SPI driver supports Hardware SPI
#define SPI_DC      DIGITAL_8      ' Data command line
#define SPI_CS      DIGITAL_4      ' Chip select line
#define SPI_RESET   DIGITAL_9      ' Reset line

#define SPI_DI      DIGITAL_12     ' Data in | MISO
#define SPI_DO      DIGITAL_11     ' Data out | MOSI
#define SPI_SCK     DIGITAL_13     ' Clock Line

dir SPI_DC      out
dir SPI_CS      out
dir SPI_RESET   out
dir SPI_DO      Out
dir SPI_DI      In
dir SPI_SCK     Out

'If DIGITAL_10 is NOT used as the SPI_CS (sometimes called SS) the port must and
output or set as input/pulled high with a 10k resistor.
'As follows:
'If CS is configured as an input, it must be held high to ensure Master SPI
operation.
'If the CS pin is driven low by peripheral circuitry when the SPI is configured
as a Master with the SS pin defined as an input, the
```

```

'SPI system interprets this as another master selecting the SPI as a slave and
starting to send data to it!
'If CS is an output SPI communications will commence with no flow control.
dir DIGITAL_10 Out

dim outbyte, inbyte as byte

#define HWSPILOCKMODE SPI_CPOL_0 + SPI_CPHA_0
SPIMode ( MasterFast, HWSPILOCKMODE )

do
  set SPI_CS OFF//  Select line
  set SPI_DC OFF//  Send Data if off, or, Data if On
  SPITransfer ( outbyte, inbyte )
  set SPI_CS ON//   Deselect Line
  set SPI_DC ON
  wait 10 ms
loop

```

18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers SPI Summary:

When using SPI setting the clock frequency is completed using SPIMode, and the master must also configure the clock polarity and phase with respect to the data. Using the three options as CPOL, CPHA and SS.

The timing diagram is as shown in the previous section that impacts CPOL and CPHA.

If you have set the PPS for SPI1SSPPS then control of the SPI SS (also know as CS / ChipSelect) is automatically controlled by the SPI transmission.

- Example:*

```

#CHIP 18F16Q41,64

#STARTUP InitPPS, 85
#DEFINE PPSToolPart 18F16Q41

// Use PPS to assign SPI capabilities to specific ports
SUB InitPPS
    SPI1SDIPPS = 0x000C
    RB6PPS = 0x001B
    SPI1SCKPPS = 0x000E
    RB5PPS = 0x001C
    RC6PPS = 0x001D
    SPI1SSPPS = 0x0016
END SUB

// Optionally change the SPI BAUD RATE from 4000
// #DEFINE SPI_BAUD_RATE 8000

// Optionally update the SPI baud rate register with an explicit value
// typical use is to entry a specific calculated value
// #DEFINE SPI_BAUD_RATE_REGISTER 1

// Specfic the hardware SPI operating model
// Can be MasterUltraFast, MasterFast, Master, MasterSlow
#DEFINE HWSPIMode MasterUltraFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#DEFINE HWSPIClockMode SPI_SS_0 + SPI_CPOL_0 + SPI_CPHA_0

// Call SPIMode
SPIMode (HWSPIMode, HWSPIClockMode )

// Define the GCBASIC required SPI port constants.
// Must match any PPS defined.
#DEFINE SPI_SCK PORTB.6
#DEFINE SPI_DO PORTB.5
#DEFINE SPI_DI PORTB.4
#DEFINE SPI_DC PortC.1
#DEFINE SPI_CS PortC.6
#DEFINE SPI_RESET PortC.2

DO
    // Send 0x75 via SPI over and over again...
    FastHWSPITransfer 0x75
LOOP

```

See also [SPITransfer](#), [FastHWSPITransfer](#)

SPITransfer

Syntax:

```
SPITransfer tx, rx
```

Command Availability:

Available on Microchip PIC microcontrollers with Hardware SPI modules.

Explanation:

This command simultaneously sends and receives a byte of data using the SPI protocol. It behaves differently depending on whether the microcontroller has been set to act as a master or a slave. When operating as a master, **SPITransfer** will initiate a transfer. The data in **tx** will be sent to the slave, whilst the byte that is buffered in the slave will be read into **rx**. In slave mode, the **SPITransfer** command will pause the program until a transfer is initiated by the master. At this point, it will send the data in **tx** whilst reading the transmission from the master into the **rx** variable.

Example:

There are two example programs for this command - one to run on the slave microcontroller , and one on the master. A reading is taken from a sensor on the slave, and sent across to the master which shows the data on its LCD screen.

Slave Program:

```
'Select chip model and configuration
#chip 16F88, 20
#config MCLR_OFF

'Set directions of SPI pins
dir PORTB.2 out
dir PORTB.1 in
dir PORTB.4 in
'Set direction of analogue pin
dir PORTA.0 in

'Set SPI mode to slave
SPIMode Slave

'Allow other microcontroller to initialise LCD
Wait 1 sec

'Main loop - takes a reading, and then waits to send it across.
do
'Note that rx is 0 - this is because no data is to be received.
SPITransfer ReadAD(AN0), 0
loop
```

Master Program:

```

'General hardware configuration
#chip 16F877A, 20

'LCD connection settings
#define LCD_IO 8
#define LCD_WIDTH 20                ;specified lcd width for clarity only. 20 is the
default width
#define LCD_DATA_PORT PORTC
#define LCD_RS PORTD.0
#define LCD_RW PORTD.1
#define LCD_Enable PORTD.2

'Set SPI pin directions
dir PORTC.5 out
dir PORTC.4 in
dir PORTC.3 out

'Set SPI Mode to master, with fast clock
SPIMode MasterFast

'Main Loop
do
'Read a byte from the slave
'No data to send, so tx is 0
SPITransfer 0, Temp

'Display data
if Temp > 0 then
  CLS
  Print "Light: "
  LCDInt Temp
  Temp = 0
end if

'Wait to allow time for the LCD to show the given value
wait 100 ms
loop

```

See also [SPIMode](#), [FastHWSPITransfer](#)

FastHWSPITransfer

Syntax:

```
FastHWSPITransfer tx
```

Command Availability:

Available on Microchip PIC microcontrollers with Hardware SPI modules.

Explanation:

This command only sends a byte of data using the SPI protocol. This command only supports master mode.

As a master, **FastHWSPITransfer** will initiate a transfer. The data in **tx** will be sent to the slave.

Example:

This is an example for this command.

Master Program:

```
'General hardware configuration
#chip 16F877A, 20

'Set SPI pin directions
dir PORTC.5 out
dir PORTC.4 in
dir PORTC.3 out

'Set SPI Mode to master, with fast clock
SPIMode MasterFast

'Main Loop
do

    'Send the value of 0x55
    FastHWSPITransfer 0x55

loop
```

See also [SPITransfer](#), [SPIMode](#)

SPI2Mode

Syntax:

Legacy SPI Operations


```

SPI2Mode ( _Mode_ [, _SPIClockMode_]

// Specfic the hardware SPI operating mode, can be MasterFast, Master, MasterSlow
#define HWSPI2Mode    MasterFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#define HWSPI2ClockMode SPI_CPOL_0 + SPI_CPHA_0

```

AVRDX, 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers

```

SPI2Mode ( _Mode_ , _SPIClockMode_ )

// Specfic the hardware SPI operating mode, can be MasterUltraFast, MasterFast,
Master, MasterSlow
#define HWSPI2Mode    MasterUltraFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#define HWSPI2ClockMode SPI_SS_0 + SPI_CPOL_0 + SPI_CPHA_0

// Optionally change the SPI BAUD RATE from 4000
#define SPI2_BAUD_RATE 8000

// Optionally update the SPI baud rate register with an explicit value
// typical use is to entry a specific calculated value
#define SPI2_BAUD_RATE_REGISTER 55

```

Command Availability:

Available on Microchip PIC and AVR microcontrollers with Hardware SPI modules.

Explanation:

Mode sets the mode of the SPI module within the microcontroller. These are the possible SPI Modes:

Mode Name	Description
Legacy SPI Operations	
MasterSlow	Master mode, SPI clock is 1/64 of the frequency of the microcontroller.
Master	Master mode, SPI clock is 1/16 of the frequency of the microcontroller.
MasterFast	Master mode, SPI clock is 1/4 of the frequency of the microcontroller.
AVRDX, 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	

Mode Name	Description
MasterSlow	SPI clock baud rate is calculated $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI2_BAUD_RATE}) / 16 * 1000) + 1$. Where SPI2_BAUD_RATE defaults to 4000. Also, see SPI2_BAUD_RATE and SPI2_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
Master	SPI clock baud rate is calculated as $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI2_BAUD_RATE}) / 4 * 1000) + 1$. Where SPI2_BAUD_RATE defaults to 4000. Also, see SPI2_BAUD_RATE and SPI2_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
MasterFast	SPI clock baud rate is calculated as $\text{INT}(\text{ChipMHz} / \text{INT}(\text{SPI2_BAUD_RATE}) / 2 * 1000) + 1$. Where SPI2_BAUD_RATE defaults to 4000. Also, see SPI2_BAUD_RATE and SPI2_BAUD_RATE_REGISTER for changing SPI Baud Rate and setting the SPI Baud Rate register with an explicit value
MasterUltraFast	SPI1BAUD is set to 0 and therefore the SPI clock baud rate to maximum
Slave Operations	
Slave	Slave mode
SlaveSS	Slave mode, with the Slave Select pin enabled.

For **Legacy microcontrollers SPI operations** *SPIClockMode* is an optional parameter to set the mode of the SPI clock mode. This optional parameter sets both the clock polarity and clock edge.

For **Specific PICs microcontrollers SPI operations** *SPIClockMode* is a mandated parameter to set the mode of the SPI clock mode and the clock polarity bit. This parameter sets both the clock polarity and clock edge. There is no verification by the compiler if you do use the *_SPIClockMode* for the 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers - the compiler uses the default value of [SPI_SS = 0 & SPI_CPOL = 0 & SPI_CPHA = 0](#). The use of SPI_SS_n requires the PPS to be set. If PPS is not set then the SPI_SS will use the default value specified in the specific GCBASIC library.

For the *_SPIClockMode_range*, see the tables below:

SPIClockMode	Description
<i>Legacy SPI operations</i>	
0	SPI_CPOL = 0 & SPI_CPHA = 0
1	SPI_CPOL = 0 & SPI_CPHA = 1
2	SPI_CPOL = 1 & SPI_CPHA = 0
3	SPI_CPOL = 1 & SPI_CPHA = 1
18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	
0	SPI_SS = 0 & SPI_CPOL = 0 & SPI_CPHA = 0

SPIClockMode	Description
2	SPI_SS = 0 & SPI_CPOL = 1 & SPI_CPHA = 0
5	SPI_SS = 1 & SPI_CPOL = 0 & SPI_CPHA = 1
7	SPI_SS = 1 & SPI_CPOL = 1 & SPI_CPHA = 1

You can use a constant value or alternatively you can use constants to set the SPIClockMode as follows:

```
_Legacy SPI microcontrollers_
SPIMode ( MasterFast, SPI_CPOL_n + SPI_CPHA_n )

_18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers_
SPIMode ( MasterFast, SPI_SS_n + SPI_CPOL_n + SPI_CPHA_n )
```

Where the following parameters can be used as a calculation to set the SPIClockMode.

Mode Name	Description
<i>Legacy SPI operations and AVR's</i>	
SPI_CPOL_0	CPOL = 0
SPI_CPOL_1	CPOL = 1
SPI_CPHA_0	CPHA = 0
SPI_CPHA_1	CPHA = 1
18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers	
SPI_SS_0	SS = 0 Clear polarity bit
SPI_SS_1	SS = 1 Set polarity bit

Explicitly changing the SPI baud rate on 18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers

You can explicitly change the SPI baud rate by defining the **SPI2_BAUD_RATE** constant as follows. This will change the default SPI baud from 4000 to the specified numeric value.

```
#DEFINE SPI2_BAUD_RATE 8000
```

You can explicitly set the SPI baud rate register by defining the **SPI2_BAUD_RATE_REGISTER** constant as follows. This will write the explicit numeric value to the SPI baud register. This overwrites any compiler calculated value.

```
#DEFINE SPI2_BAUD_RATE_REGISTER 55
```

Legacy SPI Summary:

When using SPI setting the clock frequency is completed using SPIMode, and the master must also configure the clock polarity and phase with respect to the data. Using the two options as CPOL and CPHA.

The timing diagram is shown below. The timing is further described and applies to both the master and the slave device.

When CPOL=0 the base value of the clock is zero, i.e. the active state is 1 and idle state is 0.

- For CPHA=0, data are captured on the clock's rising edge (low → high transition) and data is output on a falling edge (high → low clock transition).
- For CPHA=1, data are captured on the clock's falling edge and data is output on a rising edge.

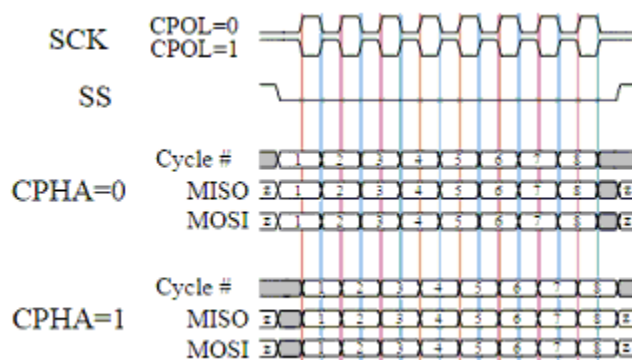
When CPOL=1 the base value of the clock is one (inversion of CPOL=0), i.e. the active state is 0 and idle state is 1.

- For CPHA=0, data are captured on clock's falling edge and data is output on a rising edge.
- For CPHA=1, data are captured on clock's rising edge and data is output on a falling edge.

When CPHA=0 means sampling on the first clock edge and , while CPHA=1 means sampling on the second clock edge, regardless of whether that clock edge is rising or falling. Note that with CPHA=0, the data must be stable for a half cycle before the first clock cycle.

In other words, CPHA=0 means transmitting data on the active to idle state and CPHA=1 means that data is transmitted on the idle to active state edge. Note that if transmission happens on a particular edge, then capturing will happen on the opposite edge(i.e. if transmission happens on falling, then reception happens on rising and vice versa). The MOSI and MISO signals are usually stable (at their reception points) for the half cycle until the next clock transition. SPI master and slave devices may well sample data at different points in that half cycle.

This adds more flexibility to the communication channel between the master and slave.



Legacy Example:

This example demonstrates the SPI capabilities for the mega328p. The process is similar of any

microcontroller..

You must set the data line as inputs and outputs.

```
#chip mega328p, 16
#option explicit
#include <UNO_mega328p.h >

#define SPI2_HardwareSPI 'comment this out to make into Software SPI but, you
may have to change clock lines

'Pin mappings for SPI - this SPI driver supports Hardware SPI
#define SPI2_DC      DIGITAL_8      ' Data command line
#define SPI2_CS      DIGITAL_4      ' Chip select line
#define SPI2_RESET    DIGITAL_9      ' Reset line

#define SPI2_DI      DIGITAL_12     ' Data in | MISO
#define SPI2_DO      DIGITAL_11     ' Data out | MOSI
#define SPI2_SCK      DIGITAL_13     ' Clock Line

dir SPI_DC    out
dir SPI_CS    out
dir SPI_RESET out
dir SPI_DO    Out
dir SPI_DI    In
dir SPI_SCK   Out

'If DIGITAL_10 is NOT used as the SPI2_CS (sometimes called SS) the port must and
output or set as input/pulled high with a 10k resistor.
'As follows:
'If CS is configured as an input, it must be held high to ensure Master SPI
operation.
'If the CS pin is driven low by peripheral circuitry when the SPI is configured
as a Master with the SS pin defined as an input, the
'SPI system interprets this as another master selecting the SPI as a slave and
starting to send data to it!
'If CS is an output SPI communications will commence with no flow control.
dir DIGITAL_10 Out

dim outbyte, inbyte as byte

#define HWSPI_CLOCKMODE SPI_CPOL_0 + SPI_CPHA_0
SPIMode ( MasterFast, HWSPI_CLOCKMODE )
```

```
do
  set SPI_CS OFF//  Select line
  set SPI_DC OFF//  Send Data if off, or, Data if On
  SPITransfer ( outbyte, inbyte )
  set SPI_CS ON//   Deselect Line
  set SPI_DC ON
  wait 10 ms
loop
```

18FxxQxx, 18FxxK42 and 18xxFK83 microcontrollers SPI Summary:

When using SPI setting the clock frequency is completed using SPIMode, and the master must also configure the clock polarity and phase with respect to the data. Using the three options as CPOL, CPHA and SS.

The timing diagram is as shown in the previous section that impacts CPOL and CPHA.

If you have set the PPS for SPI1SSPPS then control of the SPI SS (also know as CS / ChipSelect) is automatically controlled by the SPI transmission.

- Example:*

```

#CHIP 18F16Q41,64

#STARTUP InitPPS, 85
#DEFINE PPSToolPart 18F16Q41

// Use PPS to assign SPI capabilities to specific ports
SUB InitPPS
    SPI1SDIPPS = 0x000C
    RB6PPS = 0x001B
    SPI1SCKPPS = 0x000E
    RB5PPS = 0x001C
    RC6PPS = 0x001D
    SPI1SSPPS = 0x0016
END SUB

// Optionally change the SPI BAUD RATE from 4000
// #DEFINE SPI2_BAUD_RATE 8000

// Optionally update the SPI baud rate register with an explicit value
// typical use is to entry a specific calculated value
// #DEFINE SPI2_BAUD_RATE_REGISTER 1

// Specfic the hardware SPI operating model
// Can be MasterUltraFast, MasterFast, Master, MasterSlow
#DEFINE HWSPIMode MasterUltraFast

// You can use a shared constant to set a consant with the desired SPIClockMode
#DEFINE HWSPIClockMode SPI_SS_0 + SPI_CPOL_0 + SPI_CPHA_0

// Call SPIMode
SPIMode (HWSPIMode, HWSPIClockMode )

// Define the GCBASIC required SPI port constants.
// Must match any PPS defined.
#DEFINE SPI2_SCK PORTB.6
#DEFINE SPI2_DO PORTB.5
#DEFINE SPI2_DI PORTB.4
#DEFINE SPI2_DC PortC.1
#DEFINE SPI2_CS PortC.6
#DEFINE SPI2_RESET PortC.2

DO
    // Send 0x75 via SPI over and over again...
    FastHWSPITransfer 0x75
LOOP

```

See also [SPITransfer](#), [FastHWSPITransfer](#)

SPI2Transfer

Syntax:

```
SPI2Transfer tx, rx
```

Command Availability:

Available on Microchip PIC microcontrollers with Hardware SPI modules.

Explanation:

This command simultaneously sends and receives a byte of data using the SPI protocol. It behaves differently depending on whether the microcontroller has been set to act as a master or a slave. When operating as a master, **SPI2Transfer** will initiate a transfer. The data in **tx** will be sent to the slave, whilst the byte that is buffered in the slave will be read into **rx**. In slave mode, the **SPI2Transfer** command will pause the program until a transfer is initiated by the master. At this point, it will send the data in **tx** whilst reading the transmission from the master into the **rx** variable.

See also [SPIMode](#), [FastHWSPI2Transfer](#)

FastHWSPI2Transfer

Syntax:

```
FastHWSPI2Transfer tx
```

Command Availability:

Available on Microchip PIC microcontrollers with Hardware SPI modules.

Explanation:

This command only sends a byte of data using the SPI protocol. This command only supports master mode.

As a master, **FastHWSPI2Transfer** will initiate a transfer. The data in **tx** will be sent to the slave.

See also [SPITransfer](#), [SPIMode](#)

I2C Software

This is the I2C Software section of the Help file. Please refer the sub-sections for details using the contents/folder view.

I2C Overview

Introduction:

These software routines allow GCBASIC programs to send and receive I2C messages. They can be configured to act as master or slave, and the speed can also be altered.

No hardware I2C module is required for these routines - all communication is handled in software. However, these routines will not work on 12-bit instruction Microchip PIC microcontrollers (10F, 12F5xx and 16F5xx chips).

Relevant Constants:

These constants control the setup of the software I2C routines:

Constant	Controls	Default Value
I2C_MODE	Mode of I2C routines (Master or Slave)	Master
I2C_DATA	Pin on microcontroller connected to I2C data	N/A
I2C_CLOCK	Pin on microcontroller connected to I2C clock	N/A
I2C_BIT_DELAY	Time for a bit (used for acknowledge detection)	2 us
I2C_CLOCK_DELAY	Time for clock pulse to remain high	1 us
I2C_END_DELAY	Time between clock pulses	1 us
I2C_USE_TIMEOUT	Set to true if the I2C routines should stop waiting for the I2c bus - the routine will exit if a timeout occurs. Should be used when you need to prevent system lockups on the I2C bus. Supports both software I2C master and slave mode. Will return the variable I2CAck = FALSE when a timeout has occurred.	Not Set
I2C_DISABLE_INTERRUPTS	Disable interrupts during I2C routines. Important when an i2C clock is part of your solution	Not defined.

Example: This example examines the IC2 devices and displays on a terminal. This code will require adaption but the code shows an approach to discover the IC2 devices.

```
' I2C Overview - using the ChipIno board, see here for information
#chip 16F886, 8
#config MCLRE_ON
```

```

' Define I2C settings
#define I2C_MODE Master
#define I2C_DATA PORTC.4
#define I2C_CLOCK PORTC.3
#define I2C_DISABLE_INTERRUPTS ON

'USART/SERIAL PORT via a MAX232 TO PC Terminal
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dir PORTc.6 Out
#define USART_DELAY 0 ms

HSerPrintCRLF 2
HSerPrint "I2C Discover using the ChipIno"
HSerPrintCRLF 2

HSerPrint "Started: "
HSerPrint "Searching I2C address space: v0.85"
HSerPrintCRLF

wait 100 ms
dim DeviceID as byte
for DeviceID = 0 to 255
    I2CStart
    I2CSend ( deviceID )
    I2CSend ( 0 )
    I2CSend ( 0 )
    i2cstop

    if I2CSendState = True then

        HSerPrint    "__"
        HSerPrint    "ID: 0x"
        HSerPrint    hex(deviceID)
        HSerPrint    " (d"
        HSerPrint    Str(deviceID)
        HSerPrint    ")"
        HSerPrintCRLF
    end if
next
HSerPrint    "End of Device Search": HSerPrintCRLF 2
End

```

Supported in <I2C.H>

I2CAckPollState

Syntax:

```
<test condition> I2CAckPollState
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

Should only be used when I2C routines are operating in Master mode, this command will return the last state of the acknowledge response from a specific I2C device on the I2C bus.

I2CACKPOLL sets the state of variable **I2CAckPollState**. **I2CAckPollState** can only read - it cannot be set.

Example:

```
...
' ACK polling removes the need to for the 24xxxxx device to have a 5ms
write time
I2CACKPOLL( eeprom_device )
' You check the exit state,
' Use I2CAckPollState to check the state of a target device
...
```

Supported in <I2C.H>

I2CAckpoll

Syntax:

```
I2CAckpoll ( I2C_device_address )
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

Should only be used when I2C routines are operating in Master mode, this command will look for a

specific I2C device on the I2C bus.

This sets a global variable `I2CAckPollState` that can be inspected in your calling routine.

Example:

```
...
' ACK polling removes the need to for the 24xxxxx device to have a 5ms write time
I2CACKPOLL( eeprom_device )
' You check the exit state, use I2CAckPollState to check the state of
' the acknowledge from the target device
...
```

Supported in <I2C.H>

I2CReceive

Syntax:

```
I2CReceive data
I2CReceive data, ack
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

The I2CReceive command will send `data` through the I2C connection. If `ack` is TRUE, or no value is given for `ack`, then `I2CReceive` will send an ack.

If in master mode, `I2CReceive` will read the data immediately.

If in slave mode, `I2CReceive` will wait for the master to send the data before reading. When the method `I2CReceive` is used in Slave mode the global variable `I2CMatch` will be set to true when the received value is equal to the constant `I2C_ADDRESS`.

Example 1 - Master Mode:

' I2C Receive - using the ChipIno board, see here for information. ' This program reads an I2C register and LED is set to on if the value is over 100.

' This program will read from address 83, register 1.

```
#chip 16F886, 8
#config MCLRE_ON
```

```
'I2C settings
#define I2C_MODE Master
#define I2C_DATA PORTC.4
#define I2C_CLOCK PORTC.3
```

```
'Misc settings
#define LED PORTB.5
dir LED Out
```

```
'Main loop
```

```
Do
```

```
  'Send start
  I2CStart
```

```
  'Request value
  I2CSend 83
  I2CSend 1
```

```
  'Read value
  I2CReceive ValueIn
```

```
  'Send stop
  I2CStop
```

```
  'Turn on LED if received value > 100
  Set LED Off
  If ValueIn > 100 Then Set LED On
```

```
  'Delay
  Wait 20 ms
```

```
Loop
```

Example 2 - Slave Mode:

See the [I2C Overview](#) for the Master mode device to control this Slave mode device.

' I2CReceive_Slave.gcb - using a 16F88.

' This program receives commands from a GCB Master. This Slave has three LEDs attached.

```

;----- Configuration

#chip 16F88, 8
#config MCLR_OFF

#define I2C_MODE    Slave    ;this is a slave device now
#define I2C_CLOCK    portb.4    ;SCL on pin 10
#define I2C_DATA    portb.1    ;SDA on pin 7
#define I2C_ADDRESS 0x60      ;address of the slave device

;----- Variables

dim addr, reg, value as byte

;----- Program
#define LED0    porta.2        ;pin 1
#define LED1    porta.3        ;pin 2
#define LED2    porta.4        ;pin 3

dir LED0 out                ;0, 1 and 2 are outputs (LEDs)
dir LED1 out                ;0, 1 and 2 are outputs (LEDs)
dir LED2 out                ;0, 1 and 2 are outputs (LEDs)

do
    I2CStart                ;wait for Start signal
    I2CReceive( addr )      ;then wait for an address

    if I2CMatch = true then    ;if it matches, proceed

        I2CReceive(regval, ACK) ;get the register number
        I2CReceive(value, ACK)  ;and its value
        I2CStop                ;release the bus from this end

        select case regval      ;now turn proper LED on or off
            case 0:
                if value then
                    set LED0 on
                else
                    set LED0 off
                end if

            case 1:
                if value then
                    set LED1 on
                else
                    set LED1 off
            end if
        end select
    end do

```

```

        end if

        case 2:
        if value then
            set LED2 on
        else
            set LED2 off
        end if
        case else
            ;other register numbers are ignored
        end select
    else
        I2CStop          ;release bus in any event
    end if

loop

```

Supported in <I2C.H>

I2CReset

Syntax:

```
I2CReset
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

This will attempt a reset of the I2C by changing the state of the I2C bus.

Example:

```

...
    I2CReset
...

```

Supported in <I2C.H>

I2CRestart

Syntax:

I2CRestart

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

If the I2C routines are operating in Master mode, this command will send a start and restart condition in a single command.

Example:

```
...  
  I2CRESTART  
....
```

Supported in <I2C.H>

I2CSend

Syntax:

```
I2CSend data  
I2CSend data, ack
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

The I2CSend command will send *data* through the I2C connection. If *ack* is TRUE, or no value is given for *ack*, then **I2CSend** will wait for an Ack from the receiver before continuing. If in master mode, **I2CSend** will send the data immediately. If in slave mode, **I2CSend** will wait for the master to request the data before sending.

Example 1:

```
' I2CSend - using the ChipIno board, see here for information.  
' This program send commands to a GCB Slave with three LEDs attached.
```



```

#chip 16F886, 8
#config MCLRE_ON

'I2C settings
#define I2C_MODE Master
#define I2C_DATA PORTC.4
#define I2C_CLOCK PORTC.3
#define I2C_BIT_DELAY 20 us
#define I2C_CLOCK_DELAY 30 us

#define I2C_ADDRESS 0x60      ;address of the slave device
;----- Variables

dim reg as byte

;----- Program

do

    for reg = 0 to 2          ;three LEDs to control
        I2CStart              ;take control of the bus
        I2CSend I2C_ADDRESS    ;address the device
        if I2CSendState = ACK then
            I2CSend reg         ;address the particular register
            I2CSend ON          ;command to turn on LED
        end if
        I2CStop                ;relinquish the bus
        wait 100 ms
    next reg
    wait 1 S                   ;pause to show results

    for reg = 0 to 2          ;similarly, turn them off
        I2CStart              ;take control of the bus
        I2CSend I2C_ADDRESS    ;address the device
        if I2CSendState = ACK then
            I2CSend reg         ;address the particular register
            I2CSend OFF         ;command to turn off LED
        end if
        I2CStop                ;relinquish the bus
        wait 100 ms
    next reg
    wait 1 S                   ;pause to show results

loop

```

Example 2:

```
'This program will act as an I2C analog to digital converter
'When data is requested from address 83, registers 0 through
'3, it will return the value of AN0 through AN3.
```

```
'Chip model
#chip 16F88, 8
```

```
'I2C settings
#define I2C_MODE Slave
#define I2C_CLOCK PORTB.0
#define I2C_DATA PORTB.1
```

```
#define I2C_DISABLE_INTERRUPTS ON
```

```
'Main loop
```

```
Do
```

```
    'Wait for start condition
    I2CStart
```

```
    'Get address
```

```
    I2CReceive Address
```

```
    If Address = 83 Then
```

```
        'If address was this device's address, respond
```

```
        I2CReceive Register
```

```
        OutValue = ReadAD(Register)
```

```
        I2CSend OutValue
```

```
    End If
```

```
    I2CStop
```

```
    Wait 5 ms
```

```
Loop
```

Specific control of I2CSend

The I2CSend method can be controller with command(s) the change the behaviour of method. The behaviour can be changed as a Prefix or Suffix therefore the start or end of the method.

The two macros (defined constants) are I2CPreSendMacro and I2CPostSendMacro. The macros must be a single line, with colon delimiters are permitted.

Examples

The following defined macros change the start and end behaviour.

```

#define I2CPreSendMacro if LabI2CState <> True then exit Sub 'I2CPreSendMacro to
ensure GLCD operations only operate within specific lab
#define I2CPostSendMacro if LabI2CState = True then MSSP =1 'I2CPostSendMacro
to ensure GLCD operations only operate within specific lab setting a specific variable.

```

The following defined macro changes- the start behaviour to call an alternative I2CSend method.

```

#define I2CPreSendMacro      myI2CSend: exit sub

sub myI2CSend
    // your i2C handler
end sub

```

The following defined macros changes- the start behaviour to call an alternative I2CSend method, then jump to the I2CPostSendMacroLabel which is at the end of I2CSend method.

```

#define I2CPreSendMacro      myI2CSend: goto I2CPostSendMacroLabel
#define I2CPostSendMacro    NOP

sub myI2CSend
    // your i2C handler
end sub

```

This will generate the following ASM. The I2CPreSendMacro calls the MYI2CSEND() method, then BRANCHes to the label I2CPOSTSENDMACROLABEL as the end of the method.

```

;Source: i2c.h (339)
I2CSEND
;I2CPreSendMacro
    rcall    MYI2CSEND
    bra I2CPOSTSENDMACROLABEL
;I2C_CLOCK_LOW          'begin with SCL=0
    bcf TRISC,3,ACCESS
    bcf LATC,3,ACCESS
...
lots of ASM
...
;wait I2C_BIT_DELAY      'wait the usual bit length
    nop
    nop
I2CPOSTSENDMACROLABEL
;I2CPostSendMacro
    nop
    return

```

Supported in <I2C.H>

I2CStart

Syntax:

```
I2CStart
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

If the I2C routines are operating in Master mode, this command will send a start condition. If routines are in Slave mode, it will pause the program until a start condition is sent by the master. It should be placed at the start of every I2C transmission.

If interrupt handling is enabled, this command will disable it.

Example:

Please see [I2CSend](#) and [I2CReceive](#) for an example.

Supported in <I2C.H>

I2CStartoccurred

Syntax:

```
I2CStartoccurred
```

Command Availability:

Available on all microcontrollers except 12 bit instruction Microchip PIC microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

If the I2C routine IS operating in Slave mode, this function will check if a start condition has occurred since the last run of this function. 'Only used in slave mode

Example:

Please see [I2CSend](#) and [I2CReceive](#) for an example.

Supported in <I2C.H>

I2CStop

Syntax:

```
I2CStop
```

Command Availability:

Available on all microcontrollers except 12 bit instruction microcontrollers (10F, 12F5xx, 16F5xx chips)

Explanation:

When in Master mode, this command will send an I2C stop condition, and re-enable interrupts if [I2CStart](#) disabled them. In Slave mode, it will re- enable interrupts.

[I2CStop](#) should be called at the end of every I2C transmission.

Example:

Please see [I2CSend](#) and [I2CReceive](#) for an example.

Supported in <I2C.H>

I2C/TWI Hardware Module

This is the I2C/TWI section of the Help file. Please refer the sub-sections for details using the contents/folder view.

HI2C Overview

Introduction:

These methods allow GCBASIC programs to send and receive Inter- Integrated Circuit (I2C™) messages via:

- Master Synchronous Serial Port (MSSP) module of the microcontroller for the Microchip PIC architecture, or
- ATMEL 2-wire Serial Interface (TWI) for the Atmel AVR microcontroller architecture.

These methods are serial interfaces that are useful for communicating with other peripheral or microcontroller devices. These peripheral devices may be serial EEPROMs, shift registers, display drivers, A/D converters, etc.

This method can operate in one of two operational modes:

- Master Mode, or
- Slave mode (with general address call)

These methods fully implement all the I2C master and slave functions (including general call support) and supports interrupts on start and stop bits in hardware to determine a free bus (multi-master function).

These methods implement the standard mode specifications as well as 7-bit and 10-bit addressing. A “glitch” filter is built into the SCL and SDA pins when the pin is an input. This filter operates in both the 100 KHz and 400 KHz modes. In the 100 KHz mode, when these pins are an output, there is a slew rate control of the pin that is independent of device frequency.

A hardware I2C/TWI module within the microcontroller is required for these methods.

The driver supports two hardware I2C ports. The second port is addressed by the suffix HI2C2. All HI2C commands are applicable to the second HI2C2 port.

For the Microchip I2C modules Specific for the 18F class including the K42, K83 and Q10, see the later section regarding clock sources and I2C frequencies.

The method supports the following frequencies:

Frequency	Description	Support
Up to 400 kbit/s	I2C/TWI fast mode : Defined as transfer rates up to 400 kbit/s.	Supported
Up to 100 kbit/s.	I2C/TWI standard mode : Defined as transfer rates up to 100 kbit/s.	Supported
Up to 1 Mbit/s.	I2C fast-mode plus : Allowing up to 1 Mbit/s.	Supported on I2C Module Only Requires alternative clock source to be set.
Up to 3.4 Mbit/s.	I2C high speed : Allowing up to 3.4 Mbit/s.	Supported on I2C Module Only Requires alternative clock source to be set.

Relevant Constants:

These constants control the setup of the hardware I2C methods:

Constant	Controls	Usage
Master	Operational mode of the microcontroller	<code>HI2CMode (Master)</code>
Slave	Operational mode of the microcontroller	<code>HI2CMode (Slave)</code>
HI2C_BAUD_RATE	Operational speed of the microcontroller. Defaults to 100 kbit/s	For Microchip SSP or MSSP modules and AVR microcontrollers: <code>#define HI2C_BAUD_RATE 400</code> or <code>#define HI2C_BAUD_RATE 100</code> . Where <code>#define HI2C_BAUD_RATE 100</code> is the default value and therefore does need to be specified. For Microchip I2C module: 'define HI2C_BAUD_RATE 125' is the default KHz. You can override this value if you set up an alternative clock source. To change use:
HI2CITSLWaitPeriod	Sets the TSCL period to Zero as the Stop condition must be held for TSCL after Stop transition. Default to 70, some solutions can use this set to 0. The clock source and clock method must be reviewed before changing this setting. To change use: <code>#define HI2CITSLWaitPeriod = 70</code>	<code>HI2CITSLWaitPeriod = 70</code>
HIC2Q2XBUFFERSIZE	The 18FxxQ20 and 18FxxQ24 I2C buffer size. These specific microcontrollers require an I2C buffer to support I2C TX and RX operations. To change use: <code>#define HIC2Q2XBUFFERSIZE = 16</code>	<code>HIC2Q2XBUFFERSIZE = 128</code>

Port Settings:

The settings of the pin direction is critical to the operation of these methods.

For the Microchip SSP/MSSP modules both ports **must** be set as **input**.

For the Microchip I2C module both ports **must** be set as **output**. And, configure the pins as open-drain and set the I2C levels - see example below for usage.

In all case the data and clock line **must ** be pulled up with an appropriate resistor (typically 4.k @ 5.0v for 100Mkz transmissions).

Constant	Controls	Default Value
HI2C_DATA	Pin on microcontroller connected to I2C data	Must be defined
HI2C_CLOCK	Pin on microcontroller connected to I2C clock)	Must be defined

Microchip I2C modules Specific Support - 18F class including the K42, k47, K83, Q43, Q40/Q41, Q83/Q84, and Q71

Clock Sources: The Microchip I2C can select one of ten clocks sources as shown in the table below. I2C1Clock_MFINTOSC is the default which supports 125KHz.

It is important to change the clock source from the default of 125KHz if you want faster I2C communications. Change the following constant to change the clock source. Obviously, you setup the clock source correctly for I2C to operate:

```
#define I2C1CLOCKSOURCE I2C1CLOCK_MFINTOSC
```

Clock Constants that can be I2C clock sources are

Constant	Clock Source	Default Value
I2C1CLOCK_SMT1	SMT	0x09+ You MUST setup the SMT clock source.
I2C1CLOCK_TIMER6 PSO	Timer 6 Postscaler	0x08+ You MUST setup the timer6 clock source.
I2C1CLOCK_TIMER4 PSO	Timer 4 Postscaler	0x07+ You MUST setup the timer4 clock source.
I2C1CLOCK_TIMER2 PSO	Timer 2 Postscaler	0x06+ You MUST setup the timer3 clock source.
I2C1CLOCK_TIMER0 OVERFLOW	Timer 0 Overflow	0x05+ You MUST setup the timer0 clock source.
I2C1CLOCK_REFERE NCEOUT	Reference clock out	0x04+ You MUST ensure the clock source generates a within specification clock source. Check the datasheet for more details.

Constant	Clock Source	Default Value
I2C1CLOCK_MFINTOSC	MFINTOSC	0x03 (default)+ This is the default and will set the I2C clock to 125KHz
I2C1CLOCK_HFINTOSC	HFINTOSC	0x02+ You MUST ensure the clock source generates a within specification clock source. Check the datasheet for more details.
I2C1CLOCK_FOSC	FOSC	0x01+ You MUST ensure the clock source generates a within specification clock source. Check the datasheet for more details.
I2C1CLOCK_FOSC4	FOSC/4	0x00+ You MUST ensure the clock source generates a within specification clock source. Check the datasheet for more details.

This an example of using a Clock Source. This example uses the SMTClock source as the clock source, the following methods implement the SMT as the clock source. The defintion of the constant, the include, setting of the SMT period, initialisation and starting of the clock source are ALL required.

```
'Set the clock source constant
#define I2C1CLOCKSOURCE I2C1CLOCK_SMT1

'include the SMT capability
#include <SMT_Timers.h>

'Setup the SMT
'400 KHz @ 64MHZ
Setsmt1Period ( 39 )
' 100 KHz @ 64MHZ
' Setsmt1Period ( 158 )
'Initialise and start the SMT
InitSMT1(SMT_FOSC,SMPres_1)
StartSMT1
```

For other clock sources refer to the appropriate datasheet.

Error Codes: This module has extensive error reporting. For the standard error report refer to the appropriate datasheet. GCBASIC also exposes the following error messages to enable the user code to handle the errors appropriately. These are exposed via the variable **HI2C1lastError** - the bits of the **HI2C1lastError** are set as in the table shown below.

Constant	Error Value/Bit
I2C1_GOOD	0
I2C1_FAIL_TIMEOUT	1

Constant	Error Value/Bit
I2C1_TXBE_TIMEOUT	2
I2C1_START_TIMEOUT	4
I2C1_RESTART_TIMEOUT	8
I2C1_RXBF_TIMEOUT	16
I2C1_ACK_TIMEOUT	32
I2C1_MDR_TIMEOUT	64
I2C1_STOP_TIMEOUT	128

Shown below are two examples of using Hardware I2C with GCBASIC.

Example 1:

This example examines the IC2 modules using the Microchip SSP/MSSP module and the AVR microcontrollers. This will display the result on a serial terminal. This code will require adaption but the code shows an approach to discover the IC2 devices.

```
#chip mega328p, 16
#config MCLRE_ON

' Define I2C settings
#define HI2C_BAUD_RATE 400
#define HI2C_DATA PORTC.5
#define HI2C_CLOCK PORTC.4
'I2C pins need to be input for SSP module when used on Microchip PIC device
Dir HI2C_DATA in
Dir HI2C_CLOCK in

'MASTER MODE
HI2CMode Master

'USART/SERIAL PORT WORKS WITH max232 THEN TO PC Terminal
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
Dir PORTc.6 Out
#define USART_DELAY 0 ms

HSerPrintCRLF 2
HSerPrint "Hardware I2C Discover using the "
HSerPrint CHipNameStr
HSerPrintCRLF 2
```

```

for deviceID = 0 to 255
  HI2CStart
  HI2CSend ( deviceID )

  if HI2CAckPollState = false then

    if (( deviceID & 1 ) = 0 ) then
      HSerPrint "W"
    else
      HSerPrint "R"
    end if
    HSerSend 9
    HSerPrint  "ID: 0x"
    HSerPrint  hex(deviceID)
    HSerSend 9
    HSerPrint "(d)" + str(deviceID)
    HSerPrintCRLF
    HI2CSend ( 0 )

  end if

  HI2CStop
next
HSerPrintCRLF
HSerPrint  "End of Device Search"
HSerPrintCRLF 2

```

This example examines the IC2 devices and displays on a serial terminal for the I2C module only. This code will require adaption but the code shows an approach to discover the IC2 devices. This code will only operate on the Microchip I2C module.

```

#chip 18f25k42, 16
#option Explicit
#config MCLRE_ON

#startup InitPPS, 85

Sub InitPPS

  RC4PPS =      0x22    'RC4->I2C1:SDA1
  RC3PPS =      0x21    'RC3->I2C1:SCL1
  I2C1SCLPPS =  0x13    'RC3->I2C1:SCL1
  I2C1SDAPPS =  0x14    'RC4->I2C1:SDA1

```

```

'Module: UART1
RC6PPS = 0x0013      'TX1 > RC6
U1RXPPS = 0x0017      'RC7 > RX1

```

```
End Sub
```

```
'Template comment at the end of the config file
```

```

'Setup Serial port
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

```

```

' Define I2C settings
#define HI2C_BAUD_RATE 125
#define HI2C_DATA PORTC.4
#define HI2C_CLOCK PORTC.3
'Initialise I2C - note for the I2C module the ports need to be set to Output.
Dir HI2C_DATA out
Dir HI2C_CLOCK out
RC3I2C.TH0=1  'Port specific controls may be required - see the datasheet
RC4I2C.TH0=1  'Port specific controls may be required - see the datasheet

```

```

'For this solution we can set the TSCL period to Zero as the Stop condition must be
held for TSCL after Stop transition
#define HI2CITSCLWaitPeriod 0

```

```

'*****
*****

```

```
'Main program commences here.. everything before this is setup for the board.
```

```

dim DeviceID as byte
Dim DISPLAYNEWLINE as Byte

```

```
do
```

```

HSerPrintCRLF
HSerPrint "Hardware I2C "
HSerPrintCRLF 2

```

```

' Now assumes Serial Terminal is operational
HSerPrintCRLF
HSerPrint "  "
'Create a horizontal row of numbers
for DeviceID = 0 to 15
  HSerPrint hex(deviceID)

```

```

        HSerPrint " "
    next

    'Create a vertical column of numbers
    for DeviceID = 0 to 255
        DisplayNewLine = DeviceID % 16
        if DisplayNewLine = 0 Then
            HSerPrintCRLF
            HserPrint hex(DeviceID)
            if DisplayNewLine > 0 then
                HSerPrint " "
            end if
        end if
        HSerPrint " "

        'Do an initial Start
        HI2CStart
        if HI2CWaitMSSPTimeout <> True then

            'Send to address to device
            HI2CSend ( deviceID )

            'Did device fail to respond?
            if HI2CAckPollState = false then
                HI2CSend ( 0 )
                HSerPrint hex(deviceID)
            Else
                HSerPrint "--"
            end if
            'Do a stop.
            HI2CStop

        Else
            HSerPrint "! "
        end if

    next

    HSerPrintCRLF 2
    HSerPrint "End of Search"
    HSerPrintCRLF 2
    wait 1 s
    wait while SwitchIn = On
loop

```

Supported in <HI2C.H>

HI2CAckPollState

Syntax:

```
<test condition[s]> HI2CAckPollState
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

Should only be used when I2C routines are operating in Master mode, this command will return the last state of the acknowledge response from a specific I2C device on the I2C bus.

HI2CSend sets the state of variable **HI2CAckPollState**.

HI2CAckPollState can only read - it cannot be set.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
<test condition[s]> HI2C2AckPollState
```

Example:

This example code would display the devices on the I2C bus.

```
...
for deviceID = 0 to 255
  HI2CStart
  HI2CSend ( deviceID )

  if HI2CAckPollState = false then
    HSerPrint "ID: 0x"
    HSerPrint hex(deviceID)
    HSerSend 9
  end if
next
...
```

Supported in <HI2C.H>

HI2CReceive

Syntax:

```
HI2CReceive data  
  
HI2CReceive data, ACK  
HI2CReceive data, NACK
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

The HI2CReceive command will send *data* through the I2C connection. If *ack* is TRUE, or no value is given for *ack*, then **HI2CReceive** will send an ack to the I2C bus.

If in master mode, **HI2CReceive** will read the data immediately. If in slave mode, **HI2CReceive** will wait for the master to send the data before reading.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2Receive _data_  
  
HI2C2Receive _data_, ACK  
HI2C2Receive _data_, NACK
```

Example 1:

```
'This program reads an I2C register and sets an LED if it is over 100.  
  
'It will read from I2C device with an address of 83, register 1.  
' Change the processor  
#chip 16F1937, 32  
#config MCLRE_ON  
  
' Define I2C settings  
#define HI2C_BAUD_RATE 400  
  
#define HI2C_DATA PORTC.4  
#define HI2C_CLOCK PORTC.3
```

```

'I2C pins need to be input for SSP module
Dir HI2C_DATA in
Dir HI2C_CLOCK in

'MASTER I2C Device
HI2CMode Master

'Misc settings
#define LED_PORTB.0

'Main loop
Do
  'Send start
  HI2CStart

  'Request value
  HI2CSend 83
  HI2CSend 1

  'Read value
  HI2CReceive ValueIn

  'Send stop
  HI2CStop

  'Turn on LED if received value > 100
  Set LED Off
  If ValueIn > 100 Then Set LED On

  'Delay
  Wait 20 ms

Loop

```

Example 2:

See the [I2C Overview](#) for the Master mode device to control this Slave mode device.

```

' I2CHardwareReceive_Slave.gcb - using a 16F88.
' This program receives commands from a GCB Master. This Slave has three LEDs
attached.

; This Slave device responds to address 0x60 and may only be written to.
; Within it, there are three registers, 0,1 and 2 corresponding to the three LEDs.
Writing a zero
; turns the respective LED off. Writing anything else turns it on.

```



```

#chip 16F88, 4
#config MCLR_Off

#define I2C_MODE    Slave    ;this is a slave device now
#define I2C_CLOCK    portb.4    ;SCL on pin 10
#define I2C_DATA    portb.1    ;SDA on pin 7
#define I2C_ADDRESS 0x60      ;address of the slave device

#define I2C_BIT_DELAY 20 us
#define I2C_CLOCK_DELAY 10 us
#define I2C_END_DELAY 10 us

'Serial settings
#define SerInPort PORTB.6
#define SerOutPort PORTB.7

#define SendAHigh Set SerOutPort OFF
#define SendALow Set SerOutPort On
'Set pin directions
Dir SerOutPort Out
Dir SerInPort In

'Set up serial connection
InitSer 1, r2400, 1 + WaitForStart, 8, 1, none, INVERT
wait 1 s

#define LED0    porta.2    ;pin 1
#define LED1    porta.3    ;pin 2
#define LED2    porta.4    ;pin 3

;----- Variables

dim addr, reg, value,location as byte
addr = 255
reg = 255
value = 255
location = 0
mempointer = 255

;----- Program

dir LED0 out    ;0, 1 and 2 are outputs (LEDs)
dir LED1 out    ;0, 1 and 2 are outputs (LEDs)
dir LED2 out    ;0, 1 and 2 are outputs (LEDs)

```

```

set LED0 off
set LED1 off
set LED2 off

#define SerialControlPort portb.3
dir SerialControlPort in

'Set up interrupt to process I2C

    dir I2C_CLOCK in          ; required to input for MSSP module
    dir I2C_DATA in           ; required to input for MSSP module
    SSPADD=I2C_ADDRESS        ; Slave address
    SSPSTAT=b'00000000'       ; configuration
    SSPCON=b'00110110'        ; configuration
    PIE1.SSPIE=1              ; enable interrupt

repeat 3                      ;flash LEDs
    set LED0 on
    set LED1 on
    set LED2 on
    wait 50 ms
    set LED0 off
    set LED1 off
    set LED2 off
    wait 100 ms
end Repeat

oldvalue = 255                ; old value, set up value only
oldreg = 255                  ; old value, set up value only

UpdateLEDS                    ; call method to set LEDs
                             ; set up interrupt
On Interrupt SSP1Ready call I2C_Interrupt

do forever
    if reg <> oldreg then      ; only process when the reg is a new value
        oldreg = reg          ; retain old value
        show = 1              ; its time to show the LEDS!
        if value <> oldvalue then ; logic for tracking old values. You only want to
update terminal once per change
            oldvalue = value
            show = 1
        end if
    end if

    UpdateLEDS                ; Update date LEDs

```

```

        ; update serial terminal
if show = 1 and SerialControlPort = 1 then

    SerPrint 1, "0x"+hex(addr)
    SerSend 1,9

    SerPrint 1, STR(reg)
    SerSend 1,9

    SerPrint 1, STR(value)
    SerSend 1,10
    SerSend 1,13

    show = 0
end if
loop

Sub I2C_Interrupt
' handle interrupt
IF SSPIF=1 THEN                ; its a valid interrupt

    IF SSPSTAT.D_A=0 THEN      ; its an address coming in!
        addr=SSPBUF
        IF addr=I2C_ADDRESS THEN ; its our address

            mempointer = 0      ; set the memory pointer. This code emulates an
EEPROM!

        end if
        IF addr = ( I2C_ADDRESS | 1 ) THEN ; its our write address
            CKP = 0              ; acknowledge command
                                ; If the SDA line was low (ACK), the transmit data must be
loaded into

                                ; the SSPBUF register which also loads the SSPSR
                                ; register. Then, pin RB4/SCK/SCL should be enabled
                                ; by setting bit CKP.

            mempointer = 10      ; set a pointer to track incoming write
requests

            if I2C_DATA = 0 then
                SSPBUF = 0x22
                CKP = 1
                readpointer = 0x55
            end if
        end if

    else

```

```

sequence
    if SSPSTAT.P = 1 then
        ' Stop bit has been detected - out of
        ' handle event
    end if

    IF SSPSTAT.S = 1 THEN
        ' Start bit has been detected - out of
        ' handle event
    END IF

    IF SSPSTAT.R_W = 0 THEN
        ' Write operations requested

        SELECT CASE mempointer
            CASE 0
                reg = SSPBUF
                mempointer++
                ' incoming value
                ' increment our counter
            CASE 1
                value = SSPBUF
                mempointer++
                ' incoming value
                ' increment our counter
            CASE ELSE
                dummy = SSPBUF
                ' incoming value
        END SELECT

    ELSE
        ' Read operations
        SSPBUF = readpointer
        readpointer++
        ' incoming value
        ' increment our counter

    END IF
END IF
CKP = 1
SSPOV = 0
' acknowledge command
' acknowledge command
END IF
SSPIF=0
END SUB

```

sub UpdateLEDS

```

select case reg
case 0
    if value = 1 then
        set LED0 on
    else
        set LED0 off
    end if
;now turn proper LED on or off

```

```
case 1
  if value = 1 then
    set LED1 on
  else
    set LED1 off
  end if

case 2
  if value = 1 then
    set LED2 on
  else
    set LED2 off
  end if

end select

End Sub
```

Supported in <HI2C.H>

HI2CRestart

Syntax:

```
HI2CRestart
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

If the HI2C routines are operating in Master mode, this command will send a start and restart condition in a single command.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2Restart
```

Example:

```

do
    HI2CReStart                ;generate a start signal
    HI2CSend(eepDev)            ;inidcate a write
loop While HI2CAckPollState

HI2CSend(eepAddr_H)            ;as two bytes
HI2CSend(eepAddr)
HI2CReStart
HI2CSend(eepDev + 1)           ;indicate a read

eep_i = 0                      ;loop consecutively
do while (eep_i < eepLen)      ;these many bytes
    eep_j = eep_i + 1          ;arrays begin at 1 not 0
    if (eep_i < (eepLen - 1)) then
        HI2CReceive(eepArray(eep_j), ACK) ;more data to get
    else
        HI2CReceive(eepArray(eep_j), NACK ) ;send NACK on last byte
    end if
    eep_i++                    ;get set for next
loop
HI2CStop

```

Supported in <HI2C.H>

```
==== HI2CSend
```

Syntax:

```
HI2CSend data
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

The HI2CSend command will send **data** through the I2C connection. If in master mode, HI2CSend will send the data immediately. If in slave mode, HI2CSend will wait for the master to request the data before sending.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2Send  data
```

Example:

This example code retrieves multiple bytes from an EEPROM memory device.

```
do
    HI2CReStart                ;generate a start signal
    HI2CSend(eepDev)           ;indicate a write
loop While HI2CAckPollState

HI2CSend(eepAddr_H)           ;as two bytes
HI2CSend(eepAddr)
HI2CReStart
HI2CSend(eepDev + 1)          ;indicate a read

eep_i = 0                     ;loop consecutively
do while (eep_i < eeplen)     ;these many bytes
    eep_j = eep_i + 1         ;arrays begin at 1 not 0
    if (eep_i < (eeplen - 1)) then
        HI2CReceive(eepArray(eep_j), ACK) ;more data to get
    else
        HI2CReceive(eepArray(eep_j), NACK ) ;send NACK on last byte
    end if
    eep_i++                   ;get set for next
loop
HI2CStop
```

Supported in <HI2C.H>

HI2CStart

Syntax:

```
HI2CStart
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

If the HI2C routines are operating in Master mode, this command will send a start condition. If routines are in Slave mode, it will pause the program until a start condition is sent by the master. It

should be placed at the start of every I2C transmission.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2Start
```

Example:

Please see [HI2CSend](#) and [HI2CReceive](#) for examples.

Supported in <HI2C.H>

HI2CStartOccurred

Syntax:

```
HI2CStartOccurred
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

Check if a start condition has occurred since the last run of this function

Only used in slave mode.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2StartOccurred
```

Supported in <HI2C.H>

HI2CMode

Syntax:

```
HI2CMode Master | Slave
```


Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

Sets the microcontroller to either a Master device or a Slave device.

Only used in slave mode

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2Mode Master | Slave
```

Supported in <HI2C.H>

HI2CSetAddress**Syntax:**

```
HI2CSetAddress address_number
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

Sets the microcontroller address number in Slave mode.

Only used in slave mode.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2SetAddress address_number
```

Supported in <HI2C.H>

HI2CStop**Syntax:**

HI2CStop

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

HI2CStop should be called at the end of every I2C transmission.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

HI2C2Stop

Example:

Please see [HI2CSend](#) and [HI2CReceive](#) for an example.

Supported in <HI2C.H>

HI2CStopped

Syntax:

HI2CStopped

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

In Slave mode only. Check if start condition received since last used of HI2CStopped.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

HI2C2Stopped

Supported in <HI2C.H>

HI2CWaitMSSP

Syntax:

```
HI2CWaitMSSP
```

Command Availability:

Only available for microcontrollers with the hardware I2C or TWI module.

Explanation:

The methods sets the global byte variable *HI2CWaitMSSPTimeout* to 255 (or True) if the MSSP module has timeout during operations.

HI2CWaitMSSPTimeout can tested for the status of the I2C bus.

Note:

This command is also available on microcontrollers with a second hardware I2C port.

```
HI2C2WaitMSSP
```

Supported in <HI2C.H>

Sound

This is the Sound section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Sound Overview

Introduction:

These GCBASIC methods generate tones of a given frequency and duration.

Method	Controls
Tone	Generate a specified tone for a specified duration in terms of a frequency of a specified Mhz and units of 10ms
ShortTone	Generate a specified tone for a specified duration in terms of a frequency of a 10Mhz and units of 1ms
Play	Play a tune string. The format of the string is the QBASIC play command.
PlayRTTTL	Play a tune string. The format of the string is the Nokia cell phone RTTTL format.

Relevant Constants:

These constants are used to control settings for the tone generation routines. To set them, place a line in the main program file that uses `#define` to assign a value to the particular constant.

Constant Name	Controls	Default Value
<code>SoundOut</code>	The output pin to produce sound on	N/A - <i>Must be defined</i>

Note: If an exact frequency is required, or a smaller program is needed, these routines should not be used. Instead, you should use code like this:

```
Repeat count
PulseOut SoundOut, period us
Wait period us
End Repeat
```

Set `count` and `period` to the appropriate values as follows:

`period` to $1000000 / \text{desired frequency} / 2$

`count` to $\text{desired duration} / \text{period}$.

Tone

Syntax:

```
Tone Frequency, Duration
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will produce the specified tone for the specified duration. **Frequency** is measured in Hz, and **Duration** is in 10 ms units.

Please note that this command may not produce the exact frequency specified. While it is accurate enough for error beeps and small pieces of monophonic music, it should not be used for anything that requires a highly precise frequency.

Example:

```
'Sample program to produce a constant A note (440 Hz)
'on PORTB bit 1.
#chip 16F877A, 20
#define SoundOut PORTB.1

Do
    Tone 440, 1000
Loop
```

For more help, see [Sound Overview](#)

ShortTone

Syntax:

```
ShortTone Frequency, Duration
```

Command Availability:

Available on all microcontrollers.

Explanation:

This command will produce the specified tone for the specified duration. Frequency is measured in

units of 10 Hz, and Duration is in 1 ms units. Please note that this command may not produce the exact frequency specified. While it is accurate enough for error beeps and small pieces of monophonic music, it should not be used for anything that requires a highly precise frequency.

Example:

```
'Sample program to produce a tone on PORTB bit 1, based on the
'reading of an LDR on AN0 (usually PORTA bit 0).

#chip 16F88, 20
#define SoundOut PORTB.1

Dir PORTA.0 In

Do
    ShortTone ReadAD(AN0), 100
Loop
```

For more help, see [Sound Overview](#)

Play

Syntax:

```
Play SoundPlayDataString
```

You must specify the following include and the port of the sound device.

```
#include <songplay.h>
#define SOUNDOUT PORTN.N
```

Command Availability:: Available on all microcontrollers.

Explanation: This command will plays a QBASIC sequence of notes. The SoundPlayDataString is a string representing a musical note or notes to play where Notes are A to G.

Command	Description
A - G	May be followed by length: 2 = half note, 4 = quarter, also may be followed by # or + (sharp) or - (flat).
On	Sets current octave. n is octave from 0 to 6
Pn	Pause playing. n is length of rest

Command	Description
Ln:	Set default note length. n = 1 to 8.
< or >	Change down or up an octave
Tn:	Sets tempo in L4s/minute. n = 32 to 255, default 120.
Nn	Play note n. n = 0 to 84, 0 = rest.

Unsupported QBASIC commands are

Command	Description
M	Play mode
.	Changes note length

For more information on the QBASIC PLAY command set, see <https://en.wikibooks.org/wiki/QBasic/Appendix>

Example:

```
'Sample program to play a string
'on PORTB bit 1.
#chip 16F877A, 20
#include <songplay.h>
#define SoundOut PORTB.1

play "C C# C C#"
```

For more help, see [Sound Overview](#)

Play RTTTL

Syntax:

```
PlayRTTTL SoundPlayRTTTLDataString
```

You must specify the following include and the port of the sound device.

```
#include <songplay.h>
#define SOUNDOUT PORTN.N
```

Command Availability:: Available on all microcontrollers.

Explanation: This command will play a sequence of notes in the Nokia RTTTL string format.

The `SoundPlayRTTTLDataString` is a string representing a musical note or notes to play where Notes are A to G. This format and information below is credited to WikiPedia, see [here](#). To be recognized by ringtone programs, an RTTTL/Nokring format ringtone must contain three specific elements: name, settings, and notes. For example, here is the RTTTL ringtone for Haunted House:

HauntHouse: d=4,o=5,b=108: 2a4, 2e, 2d#, 2b4, 2a4, 2c, 2d, 2a#4, 2e., e, 1f4, 1a4, 1d#, 2e., d, 2c., b4, 1a4, 1p, 2a4, 2e, 2d#, 2b4, 2a4, 2c, 2d, 2a#4, 2e., e, 1f4, 1a4, 1d#, 2e., d, 2c., b4, 1a4

The three parts are separated by a colon.

- Part 1: name of the ringtone (here: "HauntHouse"), a string of characters represents the name of the ringtone
- Part 2: settings (here: d=4,o=5,b=108), where "d=" is the default duration of a note. In this case, the "4" means that each note with no duration specifier (see below) is by default considered a quarter note. "8" would mean an eighth note, and so on. Accordingly, "o=" is the default octave. There are four octaves in the Nokring/RTTTL format. And "b=" is the tempo, in "beats per minute".
- Part 3: the notes. Each note is separated by a comma and includes, in sequence: a duration specifier, a standard music note, either a, b, c, d, e, f or g, and an octave specifier. If no duration or octave specifier are present, the default applies.

Example 1:

```
#chip 16f877a
#include <songplay.h>

#define SOUNDOUT PORTA.4
PlayRTTTL "HauntHouse: d=4,o=5,b=108: 2a4, 2e, 2d#, 2b4, 2a4, 2c, 2d, 2a#4, 2e., e,
1f4, 1a4, 1d#, 2e., d, 2c., b4, 1a4, 1p, 2a4, 2e, 2d#, 2b4, 2a4, 2c, 2d, 2a#4, 2e., e,
1f4, 1a4, 1d#, 2e., d, 2c., b4, 1a4"
```

Example 2:


```

    #chip 16f877a
    #include <songplay.h>

    'Defines
    #define SoundOut PORTC.0

    Dir SoundOut Out
    Dim SoundPlayRTTTLDataString as String

    wait 1 s
    SoundPlayRTTTLDataString =
    "Thegood,:d=4,o=6,b=63:32c,32f,32c,32f,c,8g_5,8a_5,f5,8p,32c,32f,32c,32f,c,8g_5,8a_5,d_"
    PlayRTTTL(SoundPlayRTTTLDataString)

    wait 1 s
    SoundPlayRTTTLDataString
    ="LedZeppel:d=4,o=6,b=80:8g,16p,8f_,16p,8f,16p,8e,16p,8d,8a5,8c,16p,8b5,16p,a_5,8a5,16f5,
    16e5,16d5,8p,16p,16a_5,16a_5,16a_5,8p,16p,16b5,16b5,16b5,8p,16p,16b5,16b5,16b5,8p,16p,16c
    ,16c,16c,8p,16p,16c,16c,16c"
    PlayRTTTL(SoundPlayRTTTLDataString)

    Do Forever
    Loop
    End

```

For more help, see [Sound Overview](#)

Timers

This is the Timers section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Timer Overview

GCBASIC supports methods to set, clear, read, start and stop the microcontroller timers.

GCBASIC supports the following timers.

```
Timer 0
Timer 1
Timer 2
Timer 3
Timer 4
Timer 5
Timer 6
Timer 7
Timer 8
Timer 10
Timer 12
```

Not all of these timers available on all microcontrollers. For example, if a microcontroller has three timers, then typically only **Timer0**, **Timer1** and **Timer2** will be available.

Please refer to the datasheet for your microcontroller to determine the supported timers and if a specific timer is 8-bit or 16-bit.

Calculating a Timer Prescaler:

To initialise and change the timers you may have to change the Prescaler.

A Prescaler is an electronic counting circuit used to reduce a high frequency electrical signal to a lower frequency by integer division. The prescaler takes the basic timer clock frequency and divides it by some value before being processed by the timer, according to how the Prescaler register(s) are configured. The prescaler values that may be configured might be limited to a few fixed values, see the timer specific page in this Help file or refer to the datasheet.

To use a Prescaler some simple integer maths is required, however, when calculating the Prescaler there is often be a tradeoff between resolution, where a high resolution requires a high clock frequency and range where a high clock frequency willl cause the timer to overflow more quickly. For example, achieving 1 us resolution and a 1 sec maximum period using a 16-bit timer may require some clever thinking when using 8-bit timers. Please ask for advice via the GCBASIC forum, or, search for some of the many great resources on the internet to calculate a Prescaler value.

Common Language:

Using timers has the following terms /common language. This following paragraph is intended to explain the common language.

The Oscillator (OSC) is the system clock, this can be sourced from an internal or external source, OSC is same the as microcontroller Mhz. This is called the the Frequency of the OSCillator (FOSC) or the System Clock.

On a Microchip PIC microcontroller, one machine code instruction is executed for every four system clock pulses.

This means that instructions are executed at a frequency of $FOSC/4$.

The Microchip PIC datasheets call this $FOSC/4$ or $FOSC4$.

All Microchip PIC timer prescales are based on the $FOSC/4$, not the FOSC or the System Clock.

As Prescale are based upon $FOSC/4$, you must use $FOSC/4$ in your timer calculations to get the results you expect.

All Prescale and Postscale values are integer numbers.

On Atmel AVR microcontroller, most machine code instructions will execute in a single clock pulse.

Timer differences between Microchip PIC and Atmel AVR microcontrollers:

Initialising a timer for a Microchip PIC microcontroller may not operate as expected when using the same code for an Atmel AVR microcontroller by simply changing the `#chip` definition. You **must** recalculate the Prescaler of a timer when moving timer parameters between Microchip PIC and Atmel AVR microcontrollers. And, of course, the same when moving timer parameters between Atmel AVR and Microchip PIC microcontrollers.

Timer Best Practices:

Initialising microcontrollers with very limited RAM using GCBASIC needs carefull consideration. RAM may be need to be optimised by using ASM to control the timers. You can use GCBASIC to create the timer related GCBASIC ASM code then manually edit the GCBASIC ASM to optmise RAM usage. Add your revised and optimised ASM back into your program and then remove the no longer required calls the the GCBASIC methods. If you need advice on this subject please ask for advice via the GCBASIC forum.

Using Timers 2/4/6/8 on Microchip PIC microcontrollers.

A Microchip PIC microcontroller can have one of two types of 8-bit timer 2/4/6/8.

The first type has only one clock source and that clock is the $FOSC/4$ source.

The second type is much more flexible and can have many different clock sources and supports more prescale values.

The timer type for a Microchip PIC microcontroller can be determined by checking for the existence of a `T2CLKCON` register, either in the Datasheet or in the GCBASIC "dat file" for the specific

microcontroller.

If the microcontroller DOES NOT have a T2CLKCON register then ALL Timer 2/4/6/8 timers on that chip are the first type, and are configured using:

```
_InitTimer2 (PreScale, PostScale)_ 'Timer2 is example for timer 2/4/6 or 8
```

If the microcontroller DOES have a T2CLKCON register then ALL Timer 2/4/6/8 timers on that chip are the second type and are configured using:

```
_InitTimer2 (Source, PreScale, PostScale)_ 'Timer2 is example for timer 2/4/6 or 8
```

The possible *Source*, *PreScale* and *PostScale* constants for each type are shown in the GCBASIC Help file. See each timer for the constants.

The "Period" of these timers is determined by the system clock speed, the prescale value and 8-bit value in the respective timer period register. The timer period registers are PR2, PR4, PR6 or PR8 for timer2, timer4, timer6 and timer8 respectively. These registers are also called PRx and TMRx where the **x** refers to specific timer number.

When a specific timer is enabled/started the TMRx timer register will increment until the TMRx register matches the value in the PRx register. At this time the TMRx register is cleared to 0 and the timer continues to increment until the next match of the PRx register, and so on until the timer is stopped. The lower the value of the PRx register, the shorter the timer period will be. The default value for the PRX register at power up is 255.

The timer interrupt flag (TMRxIF) is set based upon the number of match conditions as determine by the postscaler. The postscaler does not actually change the timer period, it changes the time between interrupt conditions.

ClearTimer

Syntax:

```
ClearTimer TimerNo
```

Command Availability:

Available on all Microchip PIC and Atmel AVR microcontrollers with built in timer modules.

Explanation:

`ClearTimer` is used to clear the specified timer to a value of 0.

`Cleartimer` can be used on-the-fly if desired, so there is no requirement to stop the timer first.

Example:

```
.....  
'Clear timer 1  
ClearTimer 1  
.....
```

See also, [InitTimer1](#) article for an example.

InitTimer0

Syntax:

```
InitTimer0 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 0 module.

See also see: [InitTimer0 8bit/16bit](#) for support for microcontrollers with a 8 bit/16 bit Timer 0 module.

Explanation:

`InitTimer0` will set up timer 0.

Parameters are required as detailed in the table below:

Parameter	Description
<code>source</code>	The clock source for this specific timer. Can be either <code>Osc</code> or <code>Ext</code> where <code>Osc</code> is an internal oscillator and <code>Ext</code> is an external oscillator. <code>Osc</code> - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). <code>Ext</code> - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits.

Parameter	Description
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values for Microchip PIC or the Atmel AVR microcontrollers.

When the timer overflows from 255 to 0, a `Timer0Overflow` interrupt will be generated. This can be used in conjunction with `On Interrupt` to run a section of code when the overflow occurs.

Microchip PIC microcontrollers:

On Microchip PIC microcontrollers where the `prescaler` rate select bits are in the range of 2 to 256 you should use one of the following constants. If the `prescaler` rate select bits are in the range of 1 to 32768 then see the subsequent table.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:2	<code>PS0_2</code>	0
1:4	<code>PS0_4</code>	1
1:8	<code>PS0_8</code>	2
1:16	<code>PS0_16</code>	3
1:32	<code>PS0_32</code>	4
1:64	<code>PS0_64</code>	5
1:128	<code>PS0_128</code>	6
1:256	<code>PS0_256</code>	7

These correspond to a prescaler of between 1:2 and 1:256 of the oscillator speed where the oscillator speed is (FOSC/4). The prescaler applies to both the internal oscillator or the external clock.

Atmel AVR microcontrollers:

On Atmel AVR microcontrollers `prescaler` must be one of the following constants:

The prescaler will only apply when the timer is driven from the `Osc` the internal oscillator - the prescaler has no effect when the external clock source is specified.

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:1	PS_1	PS_0_1	1
1:8	PS_8	PS_0_8	2
1:64	PS_64	PS_0_64	3
1:256	PS_256	PS_0_256	4
1:1024	PS_1024	PS_0_1024	5

Example 1 for 8-bit timer 0:

This code uses Timer 0 and On Interrupt to generate a Pulse Width Modulation signal, that will allow the speed of a motor to be easily controlled.

```
#chip 16F88, 8

#define MOTOR PORTB.0

'Call the initialisation routine
InitMotorControl

'Main routine
Do
    'Increase speed to full over 2.5 seconds
    For Speed = 0 to 100
        MotorSpeed = Speed
        Wait 25 ms
    Next
    'Hold speed
    Wait 1 s
    'Decrease speed to zero over 2.5 seconds
    For Speed = 100 to 0
        MotorSpeed = Speed
        Wait 25 ms
    Next
    'Hold speed
    Wait 1 s
Loop

'Setup routine
Sub InitMotorControl
    'Clear variables
```

```

    MotorSpeed = 0
    PWMCounter = 0

    'Add a handler for the interrupt
    On Interrupt Timer0Overflow Call PWMHandler

    'Set up the timer using the internal oscillator with a prescaler of 1/2 (Equates
to 0)
    'Timer 0 starts automatically on a Microchip PIC microcontroller, therefore,
    StartTimer is not required.
    InitTimer0 Osc, PS0_2

End Sub

'PWM sub
'This will be called when Timer 0 overflows
Sub PWMHandler
    If MotorSpeed > PWMCounter Then
        Set MOTOR On
    Else
        Set MOTOR Off
    End If
    PWMCounter += 1
    If PWMCounter = 100 Then PWMCounter = 0
End Sub

```

Example 1 for 18-bit timer 0 operating an 8-bit timer:

The same example for a 16-bit timer 0 operating as an 8-bit timer.

```

#chip 16f18855,32
#option Explicit
'timer test Program

dim speed, MotorSpeed, PWMCounter as byte

#define MOTOR PORTb.0
dir MOTOR out

'Call the initialisation routine
InitMotorControl

'Main routine
Do
    'Increase speed to full over 2.5 seconds
    For Speed = 0 to 100
        MotorSpeed = Speed
    
```



```

        Wait 25 ms
    Next
    'Hold speed
    Wait 1 s
    'Decrease speed to zero over 2.5 seconds
    For Speed = 100 to 0
        MotorSpeed = Speed
        Wait 25 ms
    Next
    'Hold speed
    Wait 1 s
Loop

'Setup routine
Sub InitMotorControl
    'Clear variables
    MotorSpeed = 0
    PWMCounter = 0

    'Add a handler for the interrupt
    On Interrupt Timer0Overflow Call PWMHandler

    InitTimer0(Osc, TMR0_FOSC4 + PRE0_1 , POST0_1)
    StartTimer 0
End Sub

'PWM sub
'This will be called when Timer 0 overflows
Sub PWMHandler

    If MotorSpeed > PWMCounter Then
        Set MOTOR On
    Else
        Set MOTOR Off
    End If
    PWMCounter += 1
    If PWMCounter = 100 Then PWMCounter = 0
End Sub

```

Supported in <TIMER.H>

InitTimer0 8bit/16bit

Syntax:

InitTimer0 source, prescaler + clocksource, postscaler

Timers are useful as timers can generate interrupts. Timers can be used in conjunction with **On Interrupt** to run a section of code when a specific timer event occurs. Example events are when the timer matches a specific value, or, the timer resets to a zero value. For more details on timer events see **On Interrupt**.

Command Availability:

Available on microcontrollers with a Timer 0 where the timer has the capability of operating as an 8 bit or 16 bit timer. This type of Timer 0 can be found on Microchip PIC 18(L)F, as well as small number of 18C and 16(L)F microcontrollers. These timers can be configured for either 8-bit or 16-bit operation.

You may need to refer to the datasheet for your microcontroller to determine if it supports both 8-bit and 16-bit operations.

Explanation for 8-bit timer:

The default operation is as an 8-bit timer. **InitTimer0** will set up timer 0.

Parameters are required as shown in the table below:

Parameter	Description
source	The clock source for this specific timer. Can be either Osc or Ext where `Osc` is an internal oscillator and Ext is an external oscillator. Osc - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). Ext - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits.
prescaler	The value of the prescaler for this specific timer, and , the clocksource See the tables below for permitted values for Microchip PIC or the Atmel AVR microcontrollers.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values for Microchip PIC or the Atmel AVR microcontrollers.

8 bit Example:

The example show in the **osc** as an internal source, a **prescaler** value of 256 witht the **HFINTOSC clocksource** and a **postscaler** value of 2

```
InitTimer0 Osc, PRE0_256 + TMR0_HFINTOSC , POST0_2
'also, note when in 8-bit mode you MUST set the 8bit timer value to the upper byte of
a WORD, when setting the `SetTimer`
SetTimer 0, 0x5800 'Setting the HIGH byte!!!
```

16 bit Example:

To use the 16 bit timer you need to add the constant `#define TMR0_16bit`.

The example show in the `osc` as an internal source, a `prescaler` value of 256 witht the HFINTOSC `clocksource` and a `postscaler` value of 2

```
#define TMR0_16bit
InitTimer0 Osc, PRE0_256 + TMR0_HFINTOSC , POST0_2
```

Differences in Timer0 Operations

The section refers to chips with a 8/16-bit Timer0.

When these chips are operating in 8-bit mode, Timer0 behaves much like Timers2/4/6. In 8-bit mode the TMR0H register does not increment. It instead becomes the Period or Match register and is aliased as "PR0" (Period Register 0).

In 8-bit mode, Timer0 does not technically overflow. Instead when TMR0L increments and matches the value in the PR0 register, TMR0L is reset to 0. The interrupt flag bit (TMR0IF) bit is then set (based upon Postscaler).

The default value in the PR0 "match register" is 255. This value can be set/changed in the user program to set/change the timer period. This can be used to fine tune the timer period.

Timer 0 mandated constants:

Paramet er	Description
<code>source</code>	The clock source for this specific timer. Can be either <code>Osc</code> or <code>Ext</code> where <code>Osc</code> is an internal oscillator and <code>Ext</code> is an external oscillator.

Parameter	Description
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values for Microchip PIC or the Atmel AVR microcontrollers. You may also be required to specify one of the following clock sources. TMR0_CLC1 TMR0_SOSC TMR0_LFINTOSC TMR0_HFINTOSC TMR0_FOSC4 TMR0_T0CKIPPS_Inverted TMR0_T0CKIPPS_True You should use a simple addition to concatenate the prescaler with a specific clock source. For example. PRE0_16 + TMR0_HFINTOSC
postscaler	See the tables below for permitted values for Microchip. Also, refer to the specific datasheet postcaler values.

Microchip PIC microcontrollers where the prescaler rate select bits are in the range of 1 to 32768 you should use one of the following constants.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PRE0_1	0
1:2	PRE0_2	1
1:4	PRE0_4	2
1:8	PRE0_8	3
1:16	PRE0_16	4
1:32	PRE0_32	5
1:64	PRE0_64	6
1:128	PRE0_128	7
1:256	PRE0_256	8
1:512	PRE0_512	9
1:1024	PRE0_1024	10

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:2048	PRE0_2048	11
1:4096	PRE0_4096	12
1:8192	PRE0_8192	13
1:16384	PRE0_16384	14
1:32768	PRE0_32768	15

These correspond to a prescaler of between 1:1 and 1:32768 of the oscillator speed where the oscillator speed is (FOSC/4). The prescaler applies to both the internal oscillator or the external clock.

On Microchip PIC microcontrollers where the **prescaler** rate select bits are in the range of 2 to 256 you should use one of the following constants. If the **prescaler** rate select bits are in the range of 1 to 32768 then see the subsequent table.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:2	PS0_2	0
1:4	PS0_4	1
1:8	PS0_8	2
1:16	PS0_16	3
1:32	PS0_32	4
1:64	PS0_64	5
1:128	PS0_128	6
1:256	PS0_256	7

These correspond to a prescaler of between 1:2 and 1:256 of the oscillator speed where the oscillator speed is (FOSC/4). The prescaler applies to both the internal oscillator or the external clock.

On Microchip PIC microcontroller that require **postscaler** is can be one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postcaler Value	Primary GCB Constant	Use Numeric Constant
1:1	POST0_1	0

Postcaler Value	Primary GCB Constant	Use Numeric Constant
1:2	POST0_2	1
1:3	POST0_3	2
1:4	POST0_4	3
1:5	POST0_5	4
1:6	POST0_6	5
1:7	POST0_7	6
1:8	POST0_8	7
1:9	POST0_9	8
1:10	POST0_10	9
1:11	POST0_11	10
1:12	POST0_12	11
1:13	POST0_13	12
1:14	POST0_14	13
1:15	POST0_15	14
1:16	POST0_16	15

Example:

This code uses Timer 0 and On Interrupt to flash an LED.

```

/*

Remember four things to setup a timer.

1. InitTimer0 source, prescaler + clocksource, postscaler

2. SetTimer (byte_value, value ), or
   SetTimer (word_value [where the High byte sets the timer], value )

3. StartTimer 0

   and, optionally use

4. ClearTimer 0

*/

'Chip Settings.
#CHIP 16f18313, 32

Dir porta.1 Out

'Setup the timer.
'          Source, Prescaler + Clock Source      , Postscaler
InitTimer0 Osc,    PRE0_16384 + TMR0_HFINTOSC    , POST0_11

' Set the Timer start value. Use the HIGH byte of the word when using an 8/16bit
timer in 8 bit mode
SetTimer ( 0, 0x5800 )

' Start the Timer by writing to TMR0ON bit
StartTimer 0

Do
    Wait While TMR0IF = 0
    ' Clearing timer flag
    TMR0IF = 0
    porta.1 = ! porta.1

Loop

```

Supported in <TIMER.H>

See also see: [InitTimer0](#) for microcontroller with only an 8 bit Timer 0 module.

InitTimer1

Syntax:

```
InitTimer1 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 1 module.

Explanation:

InitTimer1 will set up timer 1.

Parameters are required as detailed in the table below:

Parameter	Description
source	The clock source for this specific timer. Can be either Osc, Ext or ExtOsc where: Osc is an internal oscillator. Ext is an external oscillator. Osc - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). Ext - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits. ExtOsc is an external oscillator and only available on a Microchip PIC microcontroller. Enhanced Microchip PIC microcontrollers with a dedicated TMRxCLK register support additional clock sources. This includes, but limited to, the following devices: 16F153xx, 16F16xx, 16F188xx and 18FxxK40 Microchip PIC microcontroller series. On these devices the clock source can be one of the following: Osc is an internal oscillator which is the same source as FOSC4. Ext is an external oscillator which is the same source as TxXKIPPS. ExtOsc is an external oscillator which is the same source as SOSC. FOSC is an internal oscillator which is the Frequency of the OSCillator. FOSC4 is an internal oscillator which is the Frequency of the OSCillator divided by 4. SOSC is an external oscillator which is the same source as SOSC. MFINTOSC is an internal 500KHz internal clock oscillator. LFINTOSC is an internal 31Khz internal clock oscillator. HFINTOSC is an oscillator as specified within the datasheet for each specific microcontroller. TxCKIPPS is an oscillator input on TxCKIPPS Pin.

Parameter	Description
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted vales for Microchip PIC or the Atmel AVR microcontrollers.

When the timer overflows an interrupt event will be generated. This interrupt event can be used in conjunction with **On Interrupt** to run a section of code when the interrupt event occurs.

Microchip PIC microcontrollers:

On Microchip PIC microcontrollers **prescaler** must be one of the following constants:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS1_1	0
1:2	PS1_2	16
1:4	PS1_4	32
1:8	PS1_8	48

These correspond to a prescaler of between 1:2 and 1:8 of the oscillator (FOSC/4) speed. The prescaler will apply to either the oscillator or the external clock input.

Atmel AVR microcontrollers:

On the majority of Atmel AVR microcontrollers **prescaler** must be one of the following constants:

The prescaler will only apply when the timer is driven from the **Osc** the internal oscillator - the prescaler has no effect when the external clock source is specified.

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:0	PS_0	PS_1_0	0
1:1	PS_1	PS_1_1	1
1:8	PS_8	PS_1_8	2
1:64	PS_64	PS_1_64	3
1:256	PS_256	PS1_256	4

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:1024	PS_1024	PS_1_1024	5

On Atmel AVR ATtiny15/25/45/85/216/461/861 microcontrollers **prescaler** must be one of the following constants:

The prescaler will only apply when the timer is driven from the **Osc** the internal oscillator - the prescaler has no effect when the external clock source is specified.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:0	PS_1_0	0
1:1	PS_1_1	1
1:2	PS_1_2	2
1:4	PS_1_4	3
1:8	PS_1_8	4
1:16	PS_1_16	5
1:32	PS_1_32	6
1:64	PS_1_64	7
1:128	PS_1_128	8
1:256	PS_1_256	9
1:512	PS_1_512	10
1:1024	PS_1_1024	11
1:2048	PS_1_2048	12
1:4096	PS_1_4096	13
1:8192	PS_1_8192	14
1:16384	PS_1_16384	15

Example 1 (Microchip):

This example will measure that time that a switch is depressed (or on) and will write the results to the EEPROM.

```
#chip 16F819, 20
#define Switch PORTA.0

Dir Switch In
DataCount = 0

'Initilise Timer 1
InitTimer1 Osc, PS1_8

Dim TimerValue As Word

Do
    ClearTimer 1
    Wait Until Switch = On
    StartTimer 1
    Wait Until Switch = Off
    StopTimer 1

    'Read the timer
    TimerValue = Timer1

    'Log the timer value
    EPWrite(DataCount, TimerValue_H)
    EPWrite(DataCount + 1, TimerValue)
    DataCount += 2
Loop
```

Example 2 (Atmel AVR):

This example will flash the yellow LED on an Arduino Uno (R3) once every second.

```
#Chip mega328p, 16 'Using Arduino Uno R3
```

```
#define LED PORTB.5
Dir LED OUT
```

```
Inittimer1 OSC, PS_256
Starttimer 1
Settimer 1, 3200 ;Preload Timer
```

```
On Interrupt Timer10verflow Call Flash_LED
```

```
Do
    'Wait for interrupt
loop
```

```
Sub Flash_LED
    Settimer 1, 3200 'Preload timer
    pulseout LED, 100 ms
End Sub
```

Supported in <TIMER.H>

InitTimer2

Syntax: (MicroChip PIC)

```
InitTimer2 prescaler, postscaler
```

or, where you required to state the clock source, use the following

```
InitTimer2 clocksource, prescaler, postscaler
```

Syntax: (Atmel AVR)

```
InitTimer2 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 2 module. As shown above a Microchip microcontroller can potentially support two types of methods for initialisation.

The first method is:

```
InitTimer2 prescaler, postscaler
```

This is the most common method to initialise a Microchip microcontroller timer. With this method the timer has only one possible clock source, this is mandated by the microcontrollers architecture, and that clock source is the System Clock/4 also known as FOSC/4.

The second method is much more flexible in terms of the clock source. Microcontrollers that support this second method enable you to select different clock sources and to select more prescale values. The method is shown below:

```
InitTimer2 clocksource, prescaler, postscaler
```

How do you determine which method to use for your specific Microchip microcontroller ?

The timer type for a Microchip microcontroller can be determined by checking for the existence of a T2CLKCON register, either in the Datasheet or in the GCBASIC "dat file" for the specific device.

If the Microchip microcontroller **DOES NOT** have a T2CLKCON register then timers 2/4/6/8 for that specific microcontroller chip use the first method, and are configured using:

```
InitTimer2 (PreScale, PostScale)
```

If the microcontroller **DOES** have a T2CLKCON register then ALL timers 2/4/6/8 for that specific microcontroller chip use the second method, and are configured using:

```
InitTimer2 (Source,PreScale,PostScale)
```

The possible Source, Prescale and Postscale constants for each type are shown in the tables below. These tables are summary tables from the Microchip datasheets.

Period of the Timers

The Period of the timer is determined by the system clock speed, the prescale value and 8-bit value in the respective timer period register. The timer period for timer 2 is held in register PR2.

When the timer is enabled, by starting the timer, it will increment until the TMR2 register matches the value in the PR2 register. At this time the TMR2 register is cleared to 0 and the timer continues to increment until the next match, and so on.

The lower the value of the PR2 register, the shorter the timer period will be. The default value for the

PR2 register at power up is 255.

The timer interrupt flag (TMR2IF) is set based upon the number of match conditions as determined by the postscaler. The postscaler does not actually change the timer period, it changes the time between interrupt conditions.

Timer constants for the MicroChip microcontrollers

Parameters for this timer are detailed in the tables below:

Parameter	Description
clocksource	This is an optional parameter. Please review the datasheet for specific usage. Source can be one of the following numeric values: 1 equates to OSC (FOSC/4). The default clock source 6 equates to EXTOSC same as SOSC 5 equates to MFINTOSC 4 equates to LFINTOSC 3 equates to HFINTOSC 2 equates to FOSC 1 equates to FOSC/4 same as OSC 0 equates to TxCIPPS same as EXTOSC and EXT (T1CKIPPS) Other sources may be available but can vary from microcontroller to microcontroller and these can be included manually per the specific microcontrollers datasheet.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values.

Table 1 shown above

prescaler can be one of the following settings, if your MicroChip microcontroller has the T2CKPS4 bit then refer to table 3:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS2_1	0
1:4	PS2_4	1
1:16	PS2_16	2

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:64	PS2_64	3

Table 2 shown above

Note that a 1:64 prescale is only available on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS2_1	0
1:2	PS2_2	1
1:4	PS2_4	2
1:8	PS2_8	3
1:16	PS2_16	4
1:32	PS2_32	5
1:64	PS2_64	6
1:128	PS2_128	7

Table 3 shown above

postscaler slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postcaler Value	GCB Constant	Eqautes to
1:1 Postscaler	POST_1	0
1:2 Postscaler	POST_2	1
1:3 Postscaler	POST_3	2
1:4 Postscaler	POST_4	3
1:5 Postscaler	POST_5	4
1:6 Postscaler	POST_6	5

Postcaler Value	GCB Constant	Eqautes to
1:7 Postscaler	POST_7	6
1:8 Postscaler	POST_8	7
1:9 Postscaler	POST_9	8
1:10 Postscaler	POST_10	9
1:11 Postscaler	POST_11	10
1:12 Postscaler	POST_12	11
1:13 Postscaler	POST_13	12
1:14 Postscaler	POST_14	13
1:15 Postscaler	POST_15	14
1:16 Postscaler	POST_16	15

Table 4 shown above

Explanation:(Atmel AVR)

`InitTimer2` will set up timer 2, according to the settings given.

`source` can be one of the following settings: Parameters for this timer are detailed in the table below:

Parameter	Description
<code>source</code>	The clock source for this specific timer. Can be either <code>Osc</code> or <code>Ext</code> where <code>Osc</code> is an internal oscillator and <code>Ext</code> is an external oscillator.

Table 5 shown above

`prescaler` for Atmel AVR Timer 2 is chip specific and can be selected from one of the two tables shown below. Please refer to the datasheet determine which table to use and which prescales within that table are supported by a specific Atmel AVR microcontroller.

Table1: Prescaler Rate Select bits are in the range of 1 to 1024

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:0	PS_0	PS_2_0	1
1:1	PS_1	PS_2_1	1
1:8	PS_8	PS_2_8	2
1:64	PS_64	PS_2_64	3
1:256	PS_256	PS2_256	4
1:1024	PS_1024	PS_2_1024	5

Table 6 shown above

Prescaler Rate Select bits are in the range of 1 to 16384

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:1	PS_2_1	none	1
1:2	PS_2_2	none	2
1:4	PS_2_4	none	3
1:8	PS_2_8	none	4
1:16	PS_2_16	none	5
1:32	PS_2_32	none	6
1:64	PS_2_64	none	7
1:128	PS_2_128	none	8
1:256	PS_2_256	none	9
1:512	PS_2_512	none	10
1:1024	PS_2_1024	none	11
1:2048	PS_2_2048	none	12
1:4096	PS_2_4096	none	13
1:8192	PS_2_8192	none	14
1:16384	PS_2_16384	none	15

Table 7 shown above

Example:

This code uses Timer 2 and On Interrupt to flash an LED every 200 timer ticks.

```
#chip 16F1788, 8

#DEFINE LED PORTA.1
DIR LED OUT

#Define Match_Val PR2 'PR2 is the timer 2 match register
Match_Val = 200      'Interrupt after 200 timer ticks

On interrupt timer2Match call FlashLED 'Interrupt on match
Inittimer2 PS2_64, 15 'Prescale 1:64 /Postscale 1:16 (15)
Starttimer 2

Do
  ' Waiting for interrupt on match val of 100
Loop

'This sub will be called when Timer 2 matches "Match_Val" (PR2)
SUB FlashLED
  pulseout LED, 5 ms
END SUB
```

InitTimer3**Syntax:**

```
InitTimer3 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 3 module.

Explanation:

InitTimer3 will set up timer 3.

Parameters are required as detailed in the table below:

Parameter	Description
source	The clock source for this specific timer. Can be either Osc, Ext or ExtOsc where: Osc is an internal oscillator. Ext is an external oscillator. Osc - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). Ext - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits. ExtOsc is an external oscillator and only available on a Microchip PIC microcontroller. Enhanced Microchip PIC microcontrollers with a dedicated TMRxCLK register support additional clock sources. This includes, but limited to, the following devices: 16F153xx, 16F16xx, 16F188xx and 18FxxK40 Microchip PIC microcontroller series On these devices the clock source can be one of the following: Osc is an internal oscillator which is the same source as FOSC4. Ext is an external oscillator which is the same source as TxCKIPPS. ExtOsc is an external oscillator which is the same source as SOSC. FOSC is an internal oscillator which is the Frequency of the OSCillator. FOSC4 is an internal oscillator which is the Frequency of the OSCillator divided by 4. SOSC is an external oscillator which is the same source as SOSC. MFINTOSC is an internal 500KHz internal clock oscillator. LFINTOSC is an internal 31Khz internal clock oscillator. HFINTOSC is an oscillator as specified within the datasheet for each specific microcontroller. TxCKIPPS is an oscillator input on TxCKIPPS Pin.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted vales for Microchip PIC or the Atmel AVR microcontrollers.

When the timer overflows an interrupt event will be generated. This interrupt event can be used in conjunction with **On Interrupt** to run a section of code when the interrupt event occurs.

Microchip PIC microcontrollers:

On Microchip PIC microcontrollers **prescaler** must be one of the following constants:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS3_1	0
1:2	PS3_2	16

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:4	PS3_4	32
1:8	PS3_8	48

These correspond to a prescaler of between 1:2 and 1:8 of the oscillator (FOSC/4) speed. The prescaler will apply to either the oscillator or the external clock input.

Atmel AVR microcontrollers:

On the majority of Atmel AVR microcontrollers `prescaler` must be one of the following constants:

The prescaler will only apply when the timer is driven from the `Osc` the internal oscillator - the prescaler has no effect when the external clock source is specified.

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:0	PS_0	PS_3_0	1
1:1	PS_1	PS_3_1	1
1:8	PS_8	PS_3_8	2
1:64	PS_64	PS_3_64	3
1:256	PS_256	PS_3_256	4
1:1024	PS_1024	PS_3_1024	5

Supported in <TIMER.H>

InitTimer4

Syntax: (MicroChip PIC)

```
InitTimer4 prescaler, postscaler
```

or, where you required to state the clock source, use the following

```
InitTimer4 clocksource, prescaler, postscaler
```

Syntax: (Atmel AVR)

```
InitTimer4 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 4 module. As shown above a Microchip microcontroller can potentially support two types of methods for initialisation.

The first method is:

```
InitTimer4 prescaler, postscaler
```

This is the most common method to initialise a Microchip microcontroller timer. With this method the timer has only one possible clock source, this is mandated by the microcontrollers architecture, and that clock source is the System Clock/4 also known as FOSC/4.

The second method is much more flexible in terms of the clock source. Microcontrollers that support this second method enable you to select different clock sources and to select more prescale values. The method is shown below:

```
InitTimer4 clocksource, prescaler, postscaler
```

How do you determine which method to use for your specific Microchip microcontroller ?

The timer type for a Microchip microcontroller can be determined by checking for the existence of a T2CLKCON register, either in the Datasheet or in the GCBASIC "dat file" for the specific device.

If the Microchip microcontroller **DOES NOT** have a T4CLKCON register then timers 2/4/6/8 for that specific microcontroller chip use the first method, and are configured using:

```
InitTimer4 (PreScale, PostScale)
```

If the microcontroller **DOES** have a T2CLKCON register then ALL timers 2/4/6/8 for that specific microcontroller chip use the second method, and are configured using:

```
InitTimer4 (Source,PreScale,PostScale)
```

The possible Source, Prescale and Postscale constants for each type are shown in the tables below. These tables are summary tables from the Microchip datasheets.

Period of the Timers

The Period of the timer is determined by the system clock speed, the prescale value and 8-bit value in the respective timer period register. The timer period for timer 4 is held in register PR4.

When the timer is enabled, by starting the timer, it will increment until the TMR4 register matches the value in the PR4 register. At this time the TMR4 register is cleared to 0 and the timer continues to increment until the next match, and so on.

The lower the value of the PR4 register, the shorter the timer period will be. The default value for the PR4 register at power up is 255.

The timer interrupt flag (TMR4IF) is set based upon the number of match conditions as determine by the postscaler. The postscaler does not actually change the timer period, it changes the time between interrupt conditions.

Timer constants for the MicroChip microcontrollers

Parameters for this timer are detailed in the tables below:

Parameter	Description
clocksource	If required by method select. Source can be one of the following numeric values: 1 equates to OSC (FOSC/4). The default clock source 6 equates to EXTOSC same as SOSC 5 equates to MFINTOSC 4 equates to LFINTOSC 3 equates to HFINTOSC 2 equates to FOSC 1 equates to FOSC/4 same as OSC 0 equates to TxCKIPPS same as EXTOSC and EXT (T1CKIPPS) Other sources may be available but can vary from microcontroller to microcontroller and these can be included manually per the specific microcontrollers datasheet.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values.

Table 1 shown above

prescaler can be one of the following settings, if you MicroChip microcontroller has the T4CKPS4 bit then refer to table 2:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS4_1	0
1:4	PS4_4	1
1:16	PS4_16	2
1:64	PS4_64	3

Table 2

Note that a 1:64 prescale is only avaialable on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS4_1	0
1:2	PS4_2	1
1:4	PS4_4	2
1:8	PS4_8	3
1:16	PS4_16	4
1:32	PS4_32	5
1:64	PS4_64	6
1:128	PS4_128	7

Table 3

postscaler slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postcaler Value	Use Numeric Constant
1:1 Postscaler	0

Postcaler Value	Use Numeric Constant
1:2 Postscaler	1
1:3 Postscaler	2
1:4 Postscaler	3
1:5 Postscaler	4
1:6 Postscaler	5
1:7 Postscaler	6
1:8 Postscaler	7
1:9 Postscaler	8
1:10 Postscaler	9
1:11 Postscaler	10
1:12 Postscaler	11
1:13 Postscaler	12
1:14 Postscaler	13
1:15 Postscaler	14
1:16 Postscaler	15

Explanation:(Atmel AVR)

`InitTimer4` will set up timer 4, according to the settings given.

`source` can be one of the following settings: Parameters for this timer are detailed in the table below:

Parameter	Description
<code>source</code>	The clock source for this specific timer. Can be either <code>Osc</code> or <code>Ext</code> where `Osc` is an internal oscillator and <code>Ext</code> is an external oscillator.

`prescaler` for Atmel AVR Timer 4 can be selected from the table below.

Prescaler Rate Select bits are in the range of 1 to 1024

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:0	PS_0	PS_4_0	1
1:1	PS_1	PS_4_1	1
1:8	PS_8	PS_4_8	2
1:64	PS_64	PS_4_64	3
1:256	PS_256	PS4_256	4
1:1024	PS_1024	PS_4_1024	5

Example:

This code uses Timer 4 and On Interrupt to generate a 1ms pulse 20 ms.

```
#chip 18F25K80, 8

#DEFINE PIN3 PORTA.1
DIR PIN3 OUT

#Define Match_Val PR4 'PR4 is the timer 2 match register
Match_Val = 154      'Interrupt afer 154 Timer ticks (~20ms)

On interrupt timer4Match call PulsePin3 'Interrupt on match
Inittimer4 PS4_16, 15 'Prescale 1:64 /Postscale 1:16 (15)
Starttimer 4

Do
    'Waiting for interrupt on match val of 154
Loop

Sub PulsePin3
    pulseout Pin3, 1 ms
End Sub
```

InitTimer5

Syntax:

Command Availability:

Available on all microcontrollers with a Timer 5 module.

Explanation:

InitTimer5 will set up timer 5.

Parameters are required as detailed in the table below:

Parameter	Description
source	The clock source for this specific timer. Can be either Osc, Ext or ExtOsc where: Osc is an internal oscillator. Ext is an external oscillator. Osc - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). Ext - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits. ExtOsc is an external oscillator and only available on a Microchip PIC microcontroller. Enhanced Microchip PIC microcontrollers with a dedicated TMRxCLK register support additional clock sources. This includes, but limited to, the following devices: 16F153xx, 16F16xx, 16F188xx and 18FxxK40 Microchip PIC microcontroller series. On these devices the clock source can be one of the following: Osc is an internal oscillator which is the same source as FOSC4. Ext is an external oscillator which is the same source as TxCKIPPS. ExtOsc is an external oscillator which is the same source as SOSC. FOSC is an internal oscillator which is the Frequency of the OSCillator. FOSC4 is an internal oscillator which is the Frequency of the OSCillator divided by 4. SOSC is an external oscillator which is the same source as SOSC. MFINTOSC is an internal 500KHz internal clock oscillator. LFINTOSC is an internal 31KHz internal clock oscillator. HFINTOSC is an oscillator as specified within the datasheet for each specific microcontroller. TxCKIPPS is an oscillator input on TxCKIPPS Pin.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values for Microchip PIC or the Atmel AVR microcontrollers.

When the timer overflows an interrupt event will be generated. This interrupt event can be used in conjunction with **On Interrupt** to run a section of code when the interrupt event occurs.

Microchip PIC microcontrollers:

On Microchip PIC microcontrollers **prescaler** must be one of the following constants:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS5_1	0
1:2	PS5_2	16
1:4	PS5_4	32
1:8	PS5_8	48

These correspond to a prescaler of between 1:2 and 1:8 of the oscillator (FOSC/4) speed. The prescaler will apply to either the oscillator or the external clock input.

Atmel AVR microcontrollers:

On the majority of Atmel AVR microcontrollers **prescaler** must be one of the following constants:

The prescaler will only apply when the timer is driven from the **Osc** the internal oscillator - the prescaler has no effect when the external clock source is specified.

Prescaler Value	Primary GCB Constant	Secondary GCB Constant	Constant Equates to value
1:0	PS_0	PS_5_0	0
1:1	PS_1	PS_5_1	1
1:8	PS_8	PS_5_8	2
1:64	PS_64	PS_5_64	3
1:256	PS_256	PS_5_256	4
1:1024	PS_1024	PS_5_1024	5

Supported in <TIMER.H>

InitTimer6

Syntax: (MicroChip PIC)

```
InitTimer6 prescaler, postscaler
```

or, where you required to state the clock source, use the following

```
InitTimer6 clocksource, prescaler, postscaler
```

Syntax: (Atmel AVR)

```
InitTimer6 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 6 module. As shown above a Microchip microcontroller can potentially support two types of methods for initialisation.

The first method is:

```
InitTimer6 prescaler, postscaler
```

This the most common method to initialise a Microchip microcontroller timer. With this method the timer has only one possible clock source, this mandated by the microcontrollers architecture, and that clock source is the System Clock/4 also known as FOSC/4.

The second method is much more flexible in term of the clock source. Microcontrollers that support this second method enable you to select different clock sources and to select more prescale values. The method is shown below:

```
InitTimer6 clocksource, prescaler, postscaler
```

How do you determine which method to use for your specific Microchip microcontroller ?

The timer type for a Microchip microcontroller can be determined by checking for the existence of a T2CLKCON register, either in the Datasheet or in the GCBASIC "dat file" for the specific device.

If the Microchip microcontroller **DOES NOT** have a T2CLKCON register then timers 2/4/6/8 for that specific microcontroller chip use the first method, and are configured using:

```
InitTimer6 (PreScale, PostScale)
```

If the microcontroller **DOES** have a T2CLKCON register then ALL timers 2/4/6/8 for that specific microcontroller chip use the second method, and are configured using:

```
InitTimer6 (Source,PreScale,PostScale)
```

The possible Source, Prescale and Postscale constants for each type are shown in the tables below. These table are summary tables from the Microchip datasheets.

Period of the Timers

The Period of the timer is determined by the system clock speed, the prescale value and 8-bit value in the respective timer period register. The timer period for timer 6 is held in register PR6.

When the timer is enabled, by starting the timer, it will increment until the TMR6 register matches the value in the PR6 register. At this time the TMR6 register is cleared to 0 and the timer continues to increment until the next match, and so on.

The lower the value of the PR6 register, the shorter the timer period will be. The default value for the PR6 register at power up is 255.

The timer interrupt flag (TMR6IF) is set based upon the number of match conditions as determine by the postscaler. The postscaler does not actually change the timer period, it changes the time between interrupt conditions.

Timer constants for the MicroChip microcontrollers

Parameters for this timer are detailed in the tables below:

Parameter	Description
clocksource	This is an optional parameter. Please review the datasheet for specific usage. Source can be one of the following numeric values: 1 equates to OSC (FOSC/4). The default clock source 6 equates to EXTOSC same as SOSC 5 equates to MFINTOSC 4 equates to LFINTOSC 3 equates to HFINTOSC 2 equates to FOSC 1 equates to FOSC/4 same as OSC 0 equates to TxCKIPPS same as EXTOSC and EXT (T1CKIPPS) Other sources may be available but can vary from microcontroller to microcontroller and these can be included manually per the specific microcontrollers datasheet.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values.

Table 1 shown above

prescaler can be one of the following settings, if you MicroChip microcontroller has the T6CKPS4 bit then refer to table 3:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS6_1	0
1:4	PS6_4	1
1:16	PS6_16	2
1:64	PS6_64	3

Table 2

Note that a 1:64 prescale is only available on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS6_1	0
1:2	PS6_2	1
1:4	PS6_4	2
1:8	PS6_8	3
1:16	PS6_16	4
1:32	PS6_32	5
1:64	PS6_64	6
1:128	PS6_128	7

Table 3

`postscaler` slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postcaler Value	Use Numeric Constant
1:1 Postscaler	0
1:2 Postscaler	1
1:3 Postscaler	2
1:4 Postscaler	3
1:5 Postscaler	4
1:6 Postscaler	5
1:7 Postscaler	6
1:8 Postscaler	7
1:9 Postscaler	8
1:10 Postscaler	9
1:11 Postscaler	10
1:12 Postscaler	11
1:13 Postscaler	12

Postcaler Value	Use Numeric Constant
1:14 Postscaler	13
1:15 Postscaler	14
1:16 Postscaler	15

InitTimer7

Syntax:

```
InitTimer7 source, prescaler
```

Command Availability:

Available on Microchip microcontrollers with a Timer 7 module.

Explanation:

`InitTimer7` will set up timer 7.

Parameters are required as detailed in the table below:

Parameter	Description
source	The clock source for this specific timer. Can be either Osc, Ext or ExtOsc where: Osc is an internal oscillator. Ext is an external oscillator. Osc - Selects the clock source in use, as set by the microcontroller specific configuration (fuses or #config). This could be an internal clock or an external clock source (external clock sources are typically attached to the XTAL pins). Ext - Selects the clock source attached to a specific external interrupt input port. This allows a different clock frequency than the main clock to be used, such as 32.768 kHz crystals commonly used for real time circuits. ExtOsc is an external oscillator and only available on a Microchip PIC microcontroller. Enhanced Microchip PIC microcontrollers with a dedicated TMRxCLK register support additional clock sources. This includes, but limited to, the following devices: 16F153xx, 16F16xx, 16F188xx and 18FxxK40 Microchip PIC microcontroller series. On these devices the clock source can be one of the following: Osc is an internal oscillator which is the same source as FOSC4. Ext is an external oscillator which is the same source as TxCKIPPS. ExtOsc is an external oscillator which is the same source as SOSC. FOSC is an internal oscillator which is the Frequency of the OSCillator. FOSC4 is an internal oscillator which is the Frequency of the OSCillator divided by 4. SOSC is an external oscillator which is the same source as SOSC. MFINTOSC is an internal 500KHz internal clock oscillator. LFINTOSC is an internal 31Khz internal clock oscillator. HFINTOSC is an oscillator as specified within the datasheet for each specific microcontroller. TxCKIPPS is an oscillator input on TxCKIPPS Pin.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted vales for Microchip PIC or the Atmel AVR microcontrollers.

When the timer overflows an interrupt event will be generated. This interrupt event can be used in conjunction with **On Interrupt** to run a section of code when the interrupt event occurs.

Microchip PIC microcontrollers:

On Microchip PIC microcontrollers **prescaler** must be one of the following constants:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS7_1	0
1:2	PS7_2	16

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:4	PS7_4	32
1:8	PS7_8	48

These correspond to a prescaler of between 1:2 and 1:8 of the oscillator (FOSC/4) speed. The prescaler will apply to either the oscillator or the external clock input.

Supported in <TIMER.H>

InitTimer8

Syntax: (MicroChip PIC)

```
InitTimer8 prescaler, postscaler
```

or, where you required to state the clock source, use the following

```
InitTimer8 clocksource, prescaler, postscaler
```

Syntax: (Atmel AVR)

```
InitTimer8 source, prescaler
```

Command Availability:

Available on all microcontrollers with a Timer 8 module. As shown above a Microchip microcontroller can potentially support two types of methods for initialisation.

The first method is:

```
InitTimer8 prescaler, postscaler
```

This the most common method to initialise a Microchip microcontroller timer. With this method the timer has only one possible clock source, this mandated by the microcontrollers architecture, and that clock source is the System Clock/4 also known as FOSC/4.

The second method is much more flexible in term of the clock source. Microcontrollers that support

this second method enable you to select different clock sources and to select more prescale values. The method is shown below:

```
InitTimer8 clocksource, prescaler, postscaler
```

How do you determine which method to use for your specific Microchip microcontroller ?

The timer type for a Microchip microcontroller can be determined by checking for the existence of a T2CLKCON register, either in the Datasheet or in the GCBASIC "dat file" for the specific device.

If the Microchip microcontroller **DOES NOT** have a T2CLKCON register then timers 2/4/6/8 for that specific microcontroller chip use the first method, and are configured using:

```
InitTimer8 (PreScale, PostScale)
```

If the microcontroller **DOES** have a T2CLKCON register then ALL timers 2/4/6/8 for that specific microcontroller chip use the second method, and are configured using:

```
InitTimer8 (Source,PreScale,PostScale)
```

The possible Source, Prescale and Postscale constants for each type are shown in the tables below. These table are summary tables from the Microchip datasheets.

Period of the Timers

The Period of the timer is determined by the system clock speed, the prescale value and 8-bit value in the respective timer period register. The timer period for timer 8 is held in register PR8.

When the timer is enabled, by starting the timer, it will increment until the TMR8 register matches the value in the PR8 register. At this time the TMR8 register is cleared to 0 and the timer continues to increment until the next match, and so on.

The lower the value of the PR8 register, the shorter the timer period will be. The default value for the PR8 register at power up is 255.

The timer interrupt flag (TMR8IF) is set based upon the number of match conditions as determine by the postscaler. The postscaler does not actually change the timer period, it changes the time between interrupt conditions.

Timer constants for the MicroChip microcontrollers

Parameters for this timer are detailed in the tables below:

Parameter	Description
clocksource	This is an optional parameter. Please review the datasheet for specific usage. Source can be one of the following numeric values: 1 equates to OSC (FOSC/4). The default clock source 6 equates to EXTOSC same as SOSC 5 equates to MFINTOSC 4 equates to LFINTOSC 3 equates to HFINTOSC 2 equates to FOSC 1 equates to FOSC/4 same as OSC 0 equates to TxCKIPPS same as EXTOSC and EXT (T1CKIPPS) Other sources may be available but can vary from microcontroller to microcontroller and these can be included manually per the specific microcontrollers datasheet.
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values.

Table 1 shown above

prescaler can be one of the following settings:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS8_1	0
1:4	PS8_4	1
1:16	PS8_16	2
1:64	PS8_64	3

Note that a 1:64 prescale is only available on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

postscaler slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postscaler Value	Use Numeric Constant
1:1 Postscaler	0
1:2 Postscaler	1
1:3 Postscaler	2
1:4 Postscaler	3
1:5 Postscaler	4
1:6 Postscaler	5
1:7 Postscaler	6
1:8 Postscaler	7
1:9 Postscaler	8
1:10 Postscaler	9
1:11 Postscaler	10
1:12 Postscaler	11
1:13 Postscaler	12
1:14 Postscaler	13
1:15 Postscaler	14
1:16 Postscaler	15

InitTimer10

Syntax:

```
InitTimer10 prescaler, postscaler
```

Command Availability:

Available on Microchip microcontrollers with a Timer 10 module.

Explanation:

Parameters for this timer are detailed in the table below:

Parameter	Description
prescaler	The value of the prescaler for this specific timer. See the tables below for permitted values.
postscaler	The value of the postscaler for this specific timer. See the tables below for permitted values.

prescaler can be one of the following settings:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS10_1	0
1:4	PS10_4	1
1:16	PS10_16	2
1:64	PS10_64	3

Note that a 1:64 prescale is only available on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

postscaler slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postcaler Value	Use Numeric Constant
1:1 Postscaler	0
1:2 Postscaler	1
1:3 Postscaler	2
1:4 Postscaler	3
1:5 Postscaler	4
1:6 Postscaler	5
1:7 Postscaler	6

Postcaler Value	Use Numeric Constant
1:8 Postscaler	7
1:9 Postscaler	8
1:10 Postscaler	9
1:11 Postscaler	10
1:12 Postscaler	11
1:13 Postscaler	12
1:14 Postscaler	13
1:15 Postscaler	14
1:16 Postscaler	15

InitTimer12

Syntax:

```
InitTimer12 prescaler, postscaler
```

Command Availability:

Available on Microchip microcontrollers with a Timer 12 module.

Explanation:

Parameters for this timer are detailed in the table below:

Parameter	Description
<code>prescaler</code>	The value of the prescaler for this specific timer. See the tables below for permitted values.
<code>postscaler</code>	The value of the postscaler for this specific timer. See the tables below for permitted values.

`prescaler` can be one of the following settings:

Prescaler Value	Primary GCB Constant	Constant Equates to value
1:1	PS12_1	0
1:4	PS12_4	1
1:16	PS12_16	2
1:64	PS12_64	3

Note that a 1:64 prescale is only available on certain midrange microcontrollers. Please refer to the datasheet to determine if a 1:64 prescale is supported by a specific microcontroller.

postscaler slows the rate of the interrupt generation (or WDT reset) from a counter/timer by dividing it down.

On Microchip PIC microcontroller one of the following constants where the Postscaler Rate Select bits are in the range of 1 to 16.

Postscaler Value	Use Numeric Constant
1:1 Postscaler	0
1:2 Postscaler	1
1:3 Postscaler	2
1:4 Postscaler	3
1:5 Postscaler	4
1:6 Postscaler	5
1:7 Postscaler	6
1:8 Postscaler	7
1:9 Postscaler	8
1:10 Postscaler	9
1:11 Postscaler	10
1:12 Postscaler	11
1:13 Postscaler	12
1:14 Postscaler	13
1:15 Postscaler	14

Postcaler Value	Use Numeric Constant
1:16 Postscaler	15

Settimer

Syntax:

```
Settimer timernumber, byte_value
```

```
Settimer timernumber, word_value
```

Command Availability:

Available on all microcontrollers with a Timer modules.

Explanation:

Settimer will set the value of the specified timer with either byte value or a word value. 8-bit timers use a byte value. 16-bit timers use a word value.

Settimer can be used on-the-fly, so there is no requirement to stop the timer first.

Refer to the datasheet for timer specific information.

Example:

This example shows the operation of setting two timers - is not intended as a meaningful solution.

```

#chip 16f877a, 4
On Interrupt Timer1Overflow call Overflowed
Set PORTB.0 On

InitTimer1 Osc, PS1_8
SetTimer 1, 1
StartTimer 1

InitTimer2 PS2_16, PS2_16
SetTimer 2, 255
StartTimer 2

'Manually set Timer2Overflow to create a second event
'this will event will be handled by the Interrupt sub routine
TMR2IE = 1
end

Sub Interrupt
    Set PORTB.2 On
    TMR2IF = 0
End Sub

Sub Overflowed
    Set PORTB.1 On
    TMR1IF = 0
End Sub

```

Supported in <TIMER.H>

StartTimer

Syntax:

```
StartTimer TimerNo
```

Command Availability:

Available on all microcontrollers with a Timer module.

Explanation:

StartTimer is used to start the specified timer.

Timer 0:

Please refer to the datasheet to determine if Timer 0 on specific Microchip PIC microcontroller can be

started and stopped with `starttimer` and `stoptimer`. If the Microchip PIC microcontroller has a register named "T0CON" then it supports `stoptimer` and `starttimer`.

On Microchip PIC 18(L)Fxxx microcontrollers Timer 0 can be started with `starttimer`.

On Microchip PIC baseline and midrange microcontrollers `starttimer` (and `stoptimer`) has no effect upon Timer 0.

Example:

This example will measure that time that a switch is depressed (or on) and will write the results to the EEPROM.

```
#chip 16F819, 20
#define Switch PORTA.0

Dir Switch In
DataCount = 0

'Initilise Timer 1
InitTimer1 Osc, PS1_8

Dim TimerValue As Word

Do
    ClearTimer 1
    Wait Until Switch = On
    StartTimer 1
    Wait Until Switch = Off
    StopTimer 1

    'Read the timer
    TimerValue = Timer1

    'Log the timer value
    EPWrite(DataCount, TimerValue_H)
    EPWrite(DataCount + 1, TimerValue)
    DataCount += 2
Loop
```

Supported in <TIMER.H>

StopTimer

Syntax:

```
StopTimer TimerNo
```

Command Availability:

Available on all microcontrollers with a Timer modules. **Explanation:**

On the Microchip PIC 18(L)Fxxx microcontrollers Timer 0 can be stopped with `stopttimer`.

With respect to Timer 0 on the Microchip PIC baseline and midrange range of microcontrollers `stoptimer` (and `starttimer`) has no effect as Timer 0.

Example:

This example will measure that time that a switch is depressed (or on) and will write the results to the EEPROM.

The example shows how to stop a timer when not in use.

```

#chip 16F819, 20
#define Switch PORTA.0

Dir Switch In
DataCount = 0

'Initilise Timer 1
InitTimer1 Osc, PS1_8

Dim TimerValue As Word

Do
    ClearTimer 1
    Wait Until Switch = On
    StartTimer 1
    Wait Until Switch = Off
    StopTimer 1

    'Read the timer
    TimerValue = Timer1

    'Log the timer value
    EPWrite(DataCount, TimerValue_H)
    EPWrite(DataCount + 1, TimerValue)
    DataCount += 2
Loop

```

Supported in <TIMER.H>

Reading Timers

GCBASIC has the following functions to read the current timer value. They are:

```

Timer0()
Timer1()
Timer2()
Timer3()
Timer4()
Timer5()
Timer6()
Timer7()
Timer8()
Timer10()
Timer12()

```

Note that these functions should only be used to read the timer value. To write the timer value, `settimer` should be used.

Not all of these functions are available on all microcontrollers. For example, if a microcontroller has three timers, then typically only `Timer0`, `Timer1` and `Timer2` will be available.

Please refer to the datasheet for your microcontroller to determine the supported timer numbers, and if a specific timer is 8-bit or 16-bit.

SMT Timers

The Signal Measurement Timer (SMT) capability is a 24-bit counter with advanced clocking and gating logic, which can be configured for measuring a variety of digital signal parameters such as pulse width, frequency and duty cycle, and the time difference between edges on two signals.

Syntax:

```
SETSMT1PERIOD ( 4045000 )      ' 1.000s period
                                ' a perfect internal clock would be 4000000

SETSMT2PERIOD ( 9322401 )      ' 4.600s period

InitSMT1(SMT_FOSC,SMPres_1)
InitSMT2(SMT_FOSC4,SMPres_8)

On Interrupt SMT1Overflow Call yourSMT1InterruptHandler
On interrupt SMT2Overflow Call yourSMT1InterruptHandler

StartSMT1
StartSMT2
```

Command Availability:

Available on Microchip microcontrollers with the SMT timer module.

This command set supports the use of the SMT as a 24-bit timer only.

Microchip PIC Microcontrollers have either 1 or 2 Signal Measurement Timers (SMT). A 24-bit timer allows for very long timer periods/high resolution and can be quite useful for certain applications.

SMT timers support multiple clock sources and prescales. Interrupt on overflow/match is also supported.

SMT timers will "overflow" when the 24-bit timer value "matches" the 24-bit period registers.

The timer period can be precisely adjusted/set by writing a period value to the respective period register for each timer.

The maximum period is achieved by a period register value of 16,777,215. 16,777,215 is the default value at POR. The timer period is also affected by the ChipMhz, TimerSource and Timer Prescale.

The library supports "normal" timer operation of SMT1/SMT2. The library does not support the advanced signal measurement features.

Explanation:

Commands are detailed in the table below:

Command	Description	Example
InitSMT1(Source,Presscaler)	Source can be one of the below: SMT_AT1_perclk equates to 6 SMT_MFINTOSC equates to 5 (500KHz) SMT_MFINTOSC_16 equates to 4 (500Khz / 16) SMT_LFINTOSC equates to 3 (32Khz) SMT_HFINTOSC equates to 2 SMT_FOSC4 equates to 1 (FOSC/4) SMT_FOSC equates to 0 Prescaler can be one of the following: SMTPres_1 equates to 1:1 SMTPres_2 equates to 1:2 SMTPres_4 equates to 1:4 SMTPres_8 equates to 1:8	InitSMT1(SMT_FOSC, SMTPres_1)
InitSMT2(Source,Presscaler)	Source can be one of the below: SMT_AT1_perclk equates to 6 SMT_MFINTOSC equates to 5 (500KHz) SMT_MFINTOSC_16 equates to 4 (500Khz / 16) SMT_LFINTOSC equates to 3 (32Khz) SMT_HFINTOSC equates to 2 SMT_FOSC4 equates to 1 (FOSC/4) SMT_FOSC equates to 0 Prescaler can be one of the following: SMTPres_1 equates to 1:1 SMTPres_2 equates to 1:2 SMTPres_4 equates to 1:4 SMTPres_8 equates to 1:8	InitSMT2(SMT_FOSC4 ,SMTPres_8)
ClearSMT1	Clears the timer. No parameter required.	ClearSMT1
ClearSMT	Clears the timer. No parameter required.	ClearSMT2
SetSMT1(TimerValue)	Sets the timer to the specific value. The value can be 1 to 16777215	SETSMT1(4045000)

Command	Description	Example
SetSMT2(TimerValue)	Sets the timer to the specific value. The value can be 1 to 16777215	SETSMT2(4045000)
StopSMT1	Stops the timer. No parameter required.	StopSMT2
StopSMT2	Stops the timer. No parameter required.	
StartSMT1	Starts the timer. No parameter required.	StartSMT1
StartSMT2	Starts the timer. No parameter required.	StartSMT2
SetSMT1Period (PeriodValue)	Sets the timer period to the specific value. The value can be 1 to 16777215	SETSMT1PERIOD(4045000)
SetSMT2Period (PeriodValue)	Sets the timer period to the specific value. The value can be 1 to 16777215	SETSMT1PERIOD(9322401)

Example 1 (Microchip Only):

This example will ..

```
#Chip 16F18855, 32

#option explicit
#include <SMT_Timers.h>
#config CLKOUTEN_ON

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D1 |-
'' -----
''

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
#define Potentiometer PORTA.4

Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT
DIR    Potentiometer In
```



```

    SETSMT1PERIOD ( 4045000 )      ' 1.000s period with the parameters of SMT_FOSC and
SMTPres_1 within the clock variance of the interclock
                                   ' a perfect internal clock would be 4000000

    SETSMT2PERIOD ( 9322401 )      ' 4.600s period with the parameters of SMT_FOSC4
and SMTPres_8

    InitSMT1(SMT_FOSC,SMTPres_1)
    InitSMT2(SMT_FOSC4,SMTPres_8)


    On Interrupt SMT1Overflow Call BlinkLEDD2
    On interrupt SMT2Overflow Call BlinkLEDD3


    StartSMT1
    StartSMT2


    Do
        '// Waiting for interrupts

    LOOP


    Sub BlinkLEDD2
        LEDD2 = !LEDD2
    End SUB


    Sub BlinkLEDD3
        LEDD3 = !LEDD3
    End SUB

```

Supported in <SMT_Timers.h>

Variables Operations

This is the Variables Operations section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Using Variables

Explanation

Using and accessing bytes within word and long numbers etc may be required when you are creating your solution. This can be done with some ease.

Example 1:

You can access the bytes within word and longs variables using the following as a guide using the Suffixes **_H**, **_U** and **_E**

```
Dim workvariable as word
workvariable = 21845
Dim lowb as byte
Dim highb as byte
Dim upperb as byte
Dim lastb as byte

lowb = workvariable
highb = workvariable_H
upperb = workvariable_U
lastb = workvariable_E
```

To further explain, where

```
Dim rB as Byte
Dim sW as Word
```

To extract the bytes from a WORD of 16 bits use the Suffix **_H**

```
'To use the bits 7-0 [lower byte] in the Word variable sW
rB = sW

'For bits 15-8 [upper byte] in the Word variable sW, use sw_H
rB = sW_H
```

To extract the bytes from a LONG of 32 bits use the Suffixes **_H**, **_U** and **_E**, where

```
Dim rB as Byte
Dim tL as Long

' For bits 7-0 [lowest byte #0] in Long variable tL
rB = tL

' For bits 15-8 [lower middle byte #1] in Long variable tL
rB = tL_H

' For bits 23-16 [upper middle byte #2] in Long variable tL
rB = tL_U

' For bits 31-24 [highest byte #3] in Long variable tL
rB = tL_E
```

To extract nibbles from the variable **rB**

```
lower_nibble = rB & 0x0F
upper_nibble = (rB & 0xF0) / 16
```

Example 2:

Assigning values to Word and Long variables via the the Byte variable (the Least Significant Byte [**lsb**]) of the same Word and Long variable.

Because a Long (or Word) variable and the Least Significant Byte, of the variable, have the same variable assignments to specific byte elements (**_e**, **_u** and **_h**) assignment must be appropriate to the element.

The code below uses a Long variable but the same principle is used for a Word.

Assigning two values, a byte and a word constant value, to the variable **tL** to compare resulting impact on Long variable.

```

Dim tL as Long

tL = 255 'All bits of the value 255 will reside in the lowest byte of the Long
variable tL
tL = 286 'This assignment will flow into tL_H where tL_H =1 and tL=30.

```

Assigning values to specific elements of a Long variable.

```

'Assign value to specific elements
tL_E = 0xF7
tL_U = 0xC5
tL_H = 0xE3

'is same as the following assignment, we show the use of casting for clarification
only.
[Long] tL = 0xF7C5E300 The lower byte (tL) is empty (zero).

'or, treat the Long as a byte and assign a byte.
[byte]tL = [byte]0xA4

```

Assigning values to the byte element of a long variable.

```

'This will assign the lowest byte with 0xA4 but this assignment will also clear the
upper 3 byte elements of the long variable.
tL = 0xA4

'To assign the lowest byte
tL = ( tL and 0xffffffff00 ) + 0xA4 'Wwill preserve the upper bytes and ensure the
lowest byte is assigned correctly.

```

A method to check a variable is assigned as expected is to use HserPrint and HserPrint hex(), as follows:

```

' HserPrint hex() only prints one byte so we need to handle the four elements
HserPrint " Print tL_E, tL_U, tL_H & tL as hex"
HserPrint hex (tL_E)
HserPrint hex (tL_U)
HserPrint hex (tL_H)
HserPrint hex (tL)
HserPrintCRLF
HserPrint "Variable tL = "
HserPrint tL

```

The user code above will result in an output as follows:

```
Print tL _E, tL_U, tL_H & tL as hexF7C5E3A4  
Variable tL = 4156941220
```

More on setting Variables and Constants

Explanation

Within GCBASIC you can use regular variable assignments. But, you can also use C like maths assignments.

The following methods are also supported.

```
GLCDPrintLoc += 6  
CharCode -= 15  
CharCode++  
CharCode---
```

Within GCBASIC you can define binary, hexadecimal and decimal constants, see [Constants](#). Please note what is and what is not support with respect to assigning numbers to constants. An example program examines what is supported.

```

#chip 16F88, 4
#config Osc = MCLRE_OFF

' All these work
#define Test1 0b11111111
#define Test2 0B11111111
#define Test3 255
#define Test4 0xFF
#define Test5 0xff
#define Test6 0Xff

# Proof - select each option one in turn
dir porta Out

porta = test1
porta = test2
porta = test3
porta = test4
porta = test5
porta = test6

```

You can assigned values/numbers with all the methods shown above (for constants and variables) but please be aware that you must Use '0' not '00'. One zero equates to zero and two zeros will give you an unassigned variable.

Constants:

A few critical constants are defined within GCBASIC , you can re-use these constants. They include:

```

#define ON 1      ' These are defined in System.h
#define OFF 0
#define TRUE 255
#define FALSE 0

#define OSC = 1   ' These are defined in TIMER.H
#define EXT = 2   ' and, are used by InitTimer0 command
#define EXTOSC = 3

```

Setting Variables

Syntax:

```
Variable = data
```

Explanation:

Variable will be set to **data**.

data can be either a fixed value (such as 157), another variable, or a sum.

All unknown byte variables are assigned Zero. A variable with the name of **Forever** is not defined by GCBASIC and therefore defaults to the value of zero.

If **data** is a fixed value, it must be an integer between 0 and 255 inclusive.

If **data** is a calculation, then it may have any of the following operands:

```
+ (add)
- (subtract, or negate if there is no value before it)
* (multiply)
/ (divide)
% (modulo)
& (and)
| (or)
# (xor)
! (not)
= (equal)
<> (not equal)
< (less than)
> (greater than)
<= (less than or equal)
>= (more than or equal)
```

The final six operands are for checking conditions. They will return FALSE (0) if the condition is false, or TRUE (255) if the condition is true.

The **And**, **Or**, **Xor** and **Not** operators function both as bitwise and logical operators.

GCBASIC understands order of operations. If multiple operands are present, they will be processed in this order:

```
Brackets
Unary operations (not and negate)
Multiply/Divide/Modulo
Add/Subtract
Conditional operators
And/Or/Xor
```

There are several modes in which variables can be set. GCBASIC will automatically use a different mode for each calculation, depending on the type of variable being set. If a byte variable is being set, byte mode will be used; if a word variable is being set, word mode will be used. If a byte is being set but the calculation involves numbers larger than 255, word mode can be used by adding [WORD] to the start of one of the values in the calculation. This is known as casting - refer to the Variables article for more information.

And with other operations

The order of operations, comparison operations have a higher precedence than boolean operations. GCBASIC behaves the same way as most other languages. Source code like this (randomly taken from `glcd_ili9326.h`) works.

```
if GLCDFntDefaultSize = 2 and CurrCharRow = 7 then
```

It is an easy mistake to compare values and get the precedent incorrect. Generally, if you can use an individual bit check, that is generally the best way to go. These are a lot simpler for the compiler to deal with and result in much nicer assembly.

This works using the correct order of precedence.

```
if (H_Byte & 0x10) = 0x10 Then ...  
  
'or, using the individual bit check to do the same  
if H_Byte.4 Then
```

This will fail as the order of precedence as shown below.

```
if H_Byte & 0x10 = 0x10 Then ...  
  
'the code above equates. This is not achieve the testing of the H_byte.4  
if H_Byte & ( 0x10 = 0x10 ) Then ...
```

Divide or division

GCBASIC support division.

When using division you will get accurate results, within the limitations of integer numbers, by completing any multiplication first and the division last. But, you may have issues with variables overflowing - ensure your variable type are correct for the calculation type.

If you that calculation a division, the compiler will use the long division routine, if the value may

overflow, and then fit the result into a word. This code provides the correct result, again within the limitations of integer numbers:

```
dim L1s as word
dim L1p as word
L1s = 6547200 / L1p
```

Division also sets the global system variable SysCalcTempX to the remainder of the division. However the following simple rules apply.

- If both of the parameters of the division are constants, the compiler will do the calculation itself and use the result rather than making the microcontroller work out the same thing every time. So, if there are two constants used, the microcontroller division routine does not get used, and SysCalcTempX does not get set.
- If either of the parameters of the division are variables, the compiler will ensure the microcontroller does the calculation as the result could be different every time. So, in this case the microcontroller division routine does get used, and SysCalcTempX is set.

If you prefer, you can add **Let** to the start of the line. It will not alter the execution of the program, but is included for those who are used to including it in other BASIC dialects.

Example:

```
'This program is to illustrate the setting of variables.
Chipmunk = 46      'Sets the variable Chipmunk to 46
Animal = Chipmunk  'Sets the variable Animal to the value of the variable Chipmunk
Bear = 2 + 3 * 5    'Sets the variable Bear to the result of 2 + 3 * 5, 17.
Sheep = (2 + 3) * 5 'Sets the variable Sheep to the result of (2 + 3) * 5, 25.
Animal = 2 * Bear   'Sets the variable Animal to twice the value of Bear.

LargeVar = 321      'LargeVar must be set as a word - see DIM.
Temp = LargeVar / [WORD]5 'Note the use of [WORD] to ensure that the calculation is
performed correctly
```

Setting Explicit Bits of a Variable/Register:

GCBASIC supports the method setting a specific bit of a variable or register. Use the following method:

```
'variable.bit method
myByteVariable.0 = 1    'will set bit 0 to 1
myByteVariable.1 = 0    'will set bit 1 to 0
myByteVariable.2 = 1    'will set bit 2 to 1
```

To set more than one bit in one command GCBASIC supports the bits method.

GCBASIC also supports setting specific bits of a variable or register. Use the following method:

```
'variable.bitS method
SPLLEN, IRCF3, IRCF2, IRCF1, IRCF0 = b'01111'
' would generate ASM [for your specific microcontroller like the following.
' bcf OSCUNE,PLLEN,ACCESS
' bsf OSCCON,IRCF2,ACCESS
' bsf OSCCON,IRCF1,ACCESS
' bsf OSCCON,IRCF0,ACCESS
```

This method is limited to literal values. You cannot use value from another variable as the setting value (at v0.98.00).

Setting Explicit Bits of a Variable/Register with Error Handling

To set more than one bit in one command GCBASIC supports the bits method.

GCBASIC also supports setting specific bits of a variable or register with error handling. Use this method to prevent errors when a specified bit does not exist.

The `[canskip]` prefix will handle the error condition when a specific bit or specific bits do not exist. The following example shows the usage.

```
[canskip] SPLLEN, IRCF3, IRCF2, IRCF1, IRCF0 = b'01111'
```

This method is limited to literal values. You cannot use value from another variable as the setting value (at v0.98.00).

This example shows how the error handler compares to not have the `[canskip]` prefix

```
' Of these two lines, only the first compiles:
[canskip] SPLLEN, IRCF3, IRCF2, IRCF1, IRCF0 = b'01111'      'first line with error
handler
      SPLLEN, IRCF3, IRCF2, IRCF1, IRCF0 = b'01111'          'second line with no
error handler

'Second line produces this message:
'samevar.gcb (16): Error: Bit IRCF3 is not a valid bit and cannot be set
```

Setting a String - set a string with Escape characters

An example showing how to set a string to an escape sequence for an ANSI terminal. You can
`Dim`ension a string and then assign the elements similar to setting array elements.

```
dim line2 as string
line2 = 27, "[", "2", "H", 27, "[", "K"
HSerPrint line2
```

Will send the following to the terminal. <esc>[2H<esc>[K

For more help, see: [Variables](#)

Variable Lifecycle

Explanation

Within GCBASIC you can use variables. This section details the Variable Lifecycle when using variables.

Variable rules - with #Option Explicit

As shown below in the rule without #Option Explicit but ALL variables MUST be defined including bytes variables.

Variable rules - without #Option Explicit

Scope - every variable is global from an addressing/usage point of view.

Once a variable is defined, and then the variable it is used the variable persists.

Aliasing - You can reduce memory usage by Aliasing. Remember all variables are global so you must be careful.

If there are two variables with the same name, they will be placed in the same memory location. You can reuse the same variable name in two subs/functions, and you can make the variables different types, but writing to the variable in one sub will overwrite the value from the other sub, see the example below.

As a general guide define any shared variables near the start of the program for easier readability.

All variables should be initialised with a desired initialisation value. Do not assume the initialisation value is Zero.

Variables local to particular subroutines are not implemented.

Specific rules to specific variable types

All variables are global. Bit variables defined in subs/function are global.

Byte variables do not need to be defined using the DIM statement. See #Option Explicit above. Just to clarify byte is default type, this means:

```
Dim MainVar As Byte is unnecessary.  
MainVar = 128    automatic defines the MainVar variable
```

Bit, Word, Longs, Integers and Strings variables must be defined.

All variables are global, but, if they are defined inside a particular subroutine then their type is not, see the example below:

Example code:

```

Dim MainVar As Byte
Dim OtherVar As Word

MainVar = 128
OtherVar = 514

DemoSub
'At this point:
'MainVar is a byte, value 128
'OtherVar is a word, value 514
'Counter is a byte, value 2
'(Byte is default type, but location shared with that of Counter in DemoSub. High
byte ignored)

Sub DemoSub
    Dim Counter As Word
    Counter = 2050
    'At this point:
    'MainVar and OtherVar as byte and word, as in main routine
    'Counter is a word, value 2050
End Sub

```

In DemoSub, Counter is a word. But anywhere else in the program it is a byte unless otherwise specified. If the variable is used/read in the main routine, it will be treated as a byte, and only the low 8 bits will be returned. In this example the low 8 bits of 2050 are 2.

The main reason for keeping the type inside the subroutine was for the following scenario: A subroutine uses a temporary variable of type byte, and relies on it overflowing.

Another subroutine uses a temporary variable of the same name, but of word type.

If the first subroutine is already in the program, and then the second one is added, the behaviour of the first one will not change at all due to the addition of the second one.

The handling of variable types using this method minimises the size of the generated assembly code.

For more help, see [Option Explicit](#)

Dim

Syntax:

For Variables:

```
Dim variable // will define a Byte variable
Dim variable [As type]
Dim variable [As type] [= initialvalue]
Dim variable[, variable2 [, variable3]] [As type] [Alias othervar [, othervar2]] [= initialvalue]
Dim variable[, variable2 [, variable3]] [As type] [At location] [= initialvalue]
```

For Arrays:

```
Dim array(size) [As type] [At location]
```

For Strings:

```
Dim string [* size] [At location]
```

Explanation:

The **Dim** command is used to declare variables, arrays, and strings. It can also create aliases for existing variables or place variables at specific memory locations.

GCBASIC supports **optional initialisation** at the point of declaration:

```
Dim byte_var As Byte = 1
```

This sets the variable to the specified value at program start. The traditional form remains fully valid:

```
Dim byte_var As Byte
```

If no initial value is provided, the variable is initialised to zero unless the compiler determines otherwise.

Command Availability: Available on all microcontrollers.

Variable Declaration

Dim can declare one or more variables on a single line. When multiple variables are declared together, the type at the end of the line applies to all variables:

```
Dim A, B, C As Word = 100
```

If no type is specified, the variable defaults to **Byte**.

Initial Values

You may optionally assign an initial value using `= value`:

```
Dim Counter As Byte = 10
Dim Temperature As Integer = -5
Dim Flag As Bit = 1
```

This improves readability and reduces the need for separate initialisation code.

Initialisation is supported for:

- Byte, Integer, Word, Long, Float
- Bit variables
- Aliased variables (if the alias target is valid)
- Arrays (initial value applies to all elements if supported by the compiler)

Example:

```
Dim Buffer(8) As Byte = 0
```

Alias

Alias creates a variable that shares the same memory location as another variable:

```
Dim ADResult As Word Alias ADRESH, ADRESL
```

Alias and **At** cannot be used together on the same declaration line.

Alias does not support **Bit** variables. For bit-level aliasing, use constants or **#Define** macros.

At (Fixed Memory Location)

At places a variable at a specific RAM location:

```
Dim SerialBuffer As Byte At 0x20
```

If the location is already used or invalid, a warning is generated.

At and **Alias** are mutually exclusive.

Strings

A fixed-length string is declared as:

```
Dim Name * 16
```

This allocates a 16-byte string (plus one extra byte internally). Fixed strings are not null-terminated and do not resize automatically.

New Example 1: Basic Variables with Initial Values

```
' Demonstrates variable declarations with and without initial values

Dim Count As Byte = 5
Dim Threshold As Word = 300
Dim Mode As Byte
Dim Ready As Bit = 1

Mode = 2
```

New Example 2: Arrays and Initialisation

```
' Create an array of 10 bytes, all initialised to zero
Dim DataList(10) As Byte = 0

' Create a word array with no initial values
Dim WordList(4) As Word

DataList(1) = 15
WordList(0) = 1024
```


New Example 3: Bit-Level Access Using Constants

```
#Option Explicit
#Chip 16F1825, 32

Dim SerialByte As Byte = 0

#Define StatusReady  SerialByte.0
#Define StatusError  SerialByte.1
#Define StatusMotor   SerialByte.2

Do
    SerialByte = SerialByte + 1

    If StatusReady = 1 Then
        StatusError = 0
    End If

    If StatusError = 1 Then
        StatusMotor = 0
    End If
Loop
```

For more help, see: [SerPrint](#) for string handling examples and [Variables Types](#) for more information on variable types.

Alloc

About Alloc

Alloc creates a special type of variable - an array variant. This array variant can store values. The values stored in this array variant must be of the same type.

Essentially, ALLOCate will reserve a memory range as described by the given layout that can be used as a RAM buffer or as an array variant.

Layout:

```
Dim variable_name as ALLOC * memory_size at memory_location
```

The allocated block of memory will not be initialized.

Example Usage:

```
Dim my256bytebuffer as alloc * 256 at 0x2400
```

There is a pointer to allocated memory. Use @variable_name.

Example Pointer

```
HSerPrint @my256bytebuffer
```

Extents

This method can be unsafe because undefined behaviour can result if the caller does not ensure that buffer extents are not maintained. Buffer extents are 0 (zero) to the memory_size - 1

Example Extents:

```
my256bytebuffer(0)    = some_variable. Will address location 0x2400
my256bytebuffer(255)  = some_variable. Will address location 0x24FF ' the 256th byte
of the allocated memory
```

Implementers of ALLOC must ensure memory constraints remain true.

Safety

This method is unsafe because undefined behaviour can result if the caller does not ensure that buffer extents are not maintained. If buffer extents are exceeded the program may address areas of memory that have adverse impact on the operation of the microcontroller.

Examples of unsafe usage:

```
my256bytebuffer(256) = some_variable. Will address location 0x2500 ' this is the
first byte of BUFFER RAM on the 18FxxQ43 chips... bad things may happen
my256bytebuffer(65535) = some_variable. Will address location 0x123FF ' this is the
beyond the memory limit and the operation will write an SFR.
```

Example Program

The following example program shows the ALLOCation of a 256 byte buffer at a specific address. The array variant is then populated with data and then shown on a serial terminal.

```
' Chip Settings and preamble
#CHIP 18F27Q43
#OPTION EXPLICIT
```

```
'Generated by PIC PPS Tool for GCBASIC - this explicit to a specific chip
#startup InitPPS, 85
#define PPSToolPart 18f27q43
```

```
Sub InitPPS
```

```
    'Module: UART pin directions
    Dir PORTC.6 Out    ' Make TX1 pin an output
    'Module: UART1
    RC6PPS = 0x0020    'TX1 > RC6
```

```
End Sub
```

```
'Template comment at the end of the config file
```

```
' USART settings for USART1
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING
#define USART_DELAY 0
```

```
'-----
```

```
' Main Program
```

```
#DEFINE BUFFERSIZE 256 ' gives range of 0 to 255
```

```
'DIMension an ArrayVariant using ALLOC to create an ArrayVariant with the size of
BUFFERSIZE.
```

```
'This array is created at memory location 0x2400.
```

```
'This memory location is specific to this chip ( you must ensure other
microcontrollers address are valid).
```

```
Dim mybuffer1 as ALLOC * BUFFERSIZE at 0x2400
```

```
'A data table
```

```
Table myDataTable
```

```
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
    0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
```

```

0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
0,1,2,3,4,5,6,7,8,9,0x0A,0x0B,0x0C,0x0D,0x0E,0x0F
End Table

```

```

Dim iLoop, tableDataValue, memoryDataValue as byte

```

```

'These variables are ONLY used to demonstrate the showing of the allocated memory
address

```

```

Dim mybuffer1startaddress, mybuffer1endaddress as word

```

```

mybuffer1startaddress = @mybuffer1
mybuffer1endaddress = mybuffer1startaddress + BUFFERSIZE - 1

```

```

HSerPrintCRLF 2
HSerPrint "Buffer test - 256 bytes "
HSerPrint " at address: 0x"
HSerPrint hex( mybuffer1startaddress_h )
HSerPrint hex( mybuffer1startaddress )
HSerPrint " to 0x"
HSerPrint hex( mybuffer1endaddress_h )
HSerPrint hex( mybuffer1endaddress )
HSerPrintCRLF 2

```

```

'Load buffer with table data
for iLoop = 0 to 255
    ReadTable myDataTable, [word]iLoop+1, tableDataValue
    mybuffer1( iLoop ) = tableDataValue
next

```

```

wait 100 ms

```

```

HserPrint "Print dataDump array to serial terminal"
HSerPrintCRLF
for iLoop = 0 to 255
    HSerPrint leftpad(str( myBuffer1(iLoop)),3)
    If iLoop % 16 = 15 Then HSerPrintCRLF
next

```

```

Wait 100 ms
HSerPrintCRLF
HserPrint "Print memory to serial terminal using PEEK to get the memory location
byte value"
HSerPrintCRLF
for iLoop = 0 to 255
    memoryDataValue = PEEK ( @myBuffer1+iLoop )
    HSerPrint leftpad(str( memoryDataValue ) ,3)

```

```
If iLoop % 16 = 15 Then HSerPrintCRLF
next
HSerPrintCRLF
Wait 100 ms
```

For more help, see [Declaring arrays with DIM](#)

BcdToDec_GCB

Syntax:

```
BcdToDec_GCB ( ByteVariable )
```

Command Availability:

Available on all microcontrollers.

Support Bytes only.

Explanation:

Converts numbers from Binary Coded Decimal format to decimal.

You can add this function. Just add this to your GCBASIC program and then call it when you need it.

Example:

```
Function BcdToDec(va) as byte
    BcdToDec=(va/16)*10+va%16
End Function
```

Also see [DecToBcd_GCB](#)

DecToBcd_GCB

Syntax:

```
DectoBcd( ByteVariable )
```

Command Availability:

Available on all microcontrollers.

Explanation:

Converts numbers from Decimal to Binary Coded Decimal format. Support Bytes only.

You can add this function. Just add this to your GCBASIC program and then call it when you need it.

Example:

```
Function DecToBcd(va) as Byte
    DecToBcd=(va/10)*16+va%10
End Function
```

Also see [BcdToDec_GCB](#)

Rotate

Syntax:

```
Rotate variable {Left | Right} [Simple]
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Rotate** command will rotate *variable* one bit in a specified direction. The bit shifted will be placed in the Carry bit of the Status register (**STATUS.C**). **STATUS.C** acts as a ninth bit of the variable that is being rotated.

variable supports Bytes, Word and Long variables.

When a variable is **rotated right**, the bit in the **STATUS.C** location is placed into the MSB of the variable being rotated, and the LSB of the variable is placed into **STATUS.C** location.

When **rotated left** the opposite occurs. The MSB of the variable is shifted to the **STATUS.C** bit and the LSB of the variable will contain what was previously in the **STATUS.C** bit location.

This table shows the operation of the **Rotate Left** command

Command	<i>variable</i>	STATUS.C
Values at start:	b'01110011'	0
Rotate Left	b'11100110'	0
Rotate Left again	b'11001100'	1
Rotate Left third time	b'10011001'	1

As you may notice the STATUS.C bit added a 0 to the rotation. So this will take 9 shifts left to get back to the original value.

Simple option

Many times you want to rotate the variable around like the STATUS.C bit wasn't there so the MSB of the variable fills the LSB of the variable on Rotate Left or the LSB fills the MSB on Rotate Right. That is where the SIMPLE option comes in. It adds a hidden step that shifts the STATUS.C bit twice so the bit moves from one end of the variable to the other.

Command	<i>variable</i>	STATUS.C
Values at start:	b'01110011'	0
Rotate Left	b'11100110'	0
Rotate Left again	b'11001101'	1
Rotate Left third time	b'10011011'	1

Notes: The carry is also called SREG bit C, or simply C flag on AVR.

In some cases the Status.C or C flag may already be set because of prior operations in your program. Therefore, it may be necessary to clear the C flag before using **Rotate**. Use **Set C Off** before using the **Rotate** command to clear the flag.

Example:

```
'This program will use Rotate to show a chasing LED.
'8 LEDs should be connected to PORTB, one on each pin.

#chip 16F819, 8

'Set port direction
Dir PORTB Out

'Set initial state of port (bits 0 and 4 on)
PORTB = b'00010001'

'Chase
C = 0
Do
    Rotate PORTB Right Simple
    Wait 250 ms
Loop
```

Set

Syntax:

```
Set variable.bit {On | Off}
```

Command Availability:

Available on all microcontrollers.

Explanation:

The purpose of the Set command is to turn individuals bits on and off.

The Set command is most useful for controlling output ports, but can also be used to set variables.

Often when controlling output ports, Set is used in conjunction with constants. This makes it easier to adapt the program for a new circuit later.

Example:

```
'Blink LED sample program for GCBASIC
'Controls an LED on PORTB bit 0.

'Set chip model and config options
#chip 16F84A, 20

'Set a constant to represent the output port
#define LED PORTB.0

'Set pin direction
Dir LED Out

'Main routine
Do
    Set LED On
    Wait 1 sec
    Set LED OFF
    Wait 1 sec
Loop
```

SWAP4

Syntax:


```
SWAP4( VariableA)
```

Command Availability:

Available on all microcontrollers.

Support Bytes only.

Explanation:

A function that swaps (or exchanges) nibbles (or the 8 bits of a byte in nibbles).

Example:

```
dim ByteVariable as Byte

' Set variable to 0x12
ByteVariable = 0x12

ByteVariable = Swap4( ByteVariable )

HSerPrint hex(ByteVariable)

' Would return 0x21
```

SWAP

Syntax:

```
SWAP( VariableA, VariableB)
```

Command Availability:

Available on all microcontrollers.

Support Bytes and Words only.

Explanation:

A function that swaps (or exchanges) one byte or word for another. SWAP support the use of byte and word variables.

String Manipulation

This is the String Manipulation section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Asc

Syntax:

```
bytevar= ASC(string, [position] )
```

Command Availability:

Available on all microcontrollers

Explanation:

Returns the character code of the character at the specified position in a string.

ASC returns the character code of a particular character in the string. If the string is an ANSI string, the returned value will be in the range of 0 to 255. This function DOES NOT support UNICODE.

The optional position parameter determines which character is to be checked. The first character is one, the second two, etc. If the position parameter is missing, the first character is presumed.

CHR is the natural complement of **ASC**. **CHR** produces a one-character string corresponding to its ASCII.

Note:

If the string passed is null (zero-length) or the position is zero or greater than the length of the string the returned value will be 0.

Example:

```
charpos = ASC( "ABCD" )      ' Returns 65
```

```
charpos = ASC( "ABCD", 2 )  ' Returns 66
```

For more help, see [Chr](#)

ByteToBin

Syntax:

```
stringvar = ByteToBin(bytevar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **ByteToBin** function creates a string of a ANSI (8-byte) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Note: Supports BYTE variables only. For WORD variables use **WordToBin**

Example:

```
string = ByteToBin( 1 ) ' Returns "00000001"
```

```
string = ByteToBin( 254 ) ' Returns "11111110"
```

For more help, see [WordToBin](#)

ByteToHex

Syntax:

```
stringvar = ByteToHex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Hex** function will convert a byte number into hexadecimal format. The input **number** should be a byte variable, or a fixed number between 0 and 255 inclusive. After running the function, the string variable **stringvar** will contain a 2 digit hexadecimal number.

Example:

```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 255
    'Send location
    HSerPrint ByteToHex(CurrentLocation)
    HSerPrint ":"
    'Read byte and send
    EPRead CurrentLocation, CurrByte
    HSerPrint Hex(CurrByte)
    'Send carriage return/line feed
    HSerPrintCRLF
Next

```

See Also [WordToHex](#), [LongToHex](#), [IntegerToHex](#), [SingleToHex](#)

ByteToString

Syntax:

```
stringvar = ByteToString(byte_variable)    'supports byte.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **ByteToString** function will convert a number into a string. **number** can be any byte variable, or a fixed number constant between 0 and 255 inclusive. For Word number use **WordToString()**, Long numbers use **LongToString()**, for Integer numbers use **IntegerToString()** and for Single numbers use **SingleToString()**

The string variable **stringvar** will contain the same number, represented as a string. The length of the string returned is 5 characters.

This function is especially useful if a number needs to be added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Example1:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dim SensorReading as Byte

'Take an A/D reading
SensorReading = ReadAD(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = ByteToString(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF
```

When using the functions ByteToString() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```
' use this, note this is no space between the ByteToString() and the left brace!
ByteToString(number_variable)
' do not use, note the space!
ByteToString (number_variable)
```

Example2:

```
'''
'''
'''
'''
'''*****
```

```

'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board
''' Date: June 2021
'''

' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config CLRE_ON
#option Explicit

; ----- Define Hardware settings

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim bytevar as Byte

; ----- Main body of program commences here.
bytevar = 0xff

do
    wait 100 ms

    HSerPrint ByteToString( bytevar )
    HSerPrintCRLF
    wait 1 s
loop
end

; ----- Support methods. Subroutines and Functions

```

See Also [WordToString](#), [LongToString](#), [IntegerToString](#), [SingleToString](#), [ByteToHex](#)

Chr

Syntax:

```
stringvar = CHR(bytevar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **CHR** function creates a string of a ANSI (1-byte) character.

ASC is the natural complement of **CHR**.

Example:

```
string = CHR( 65 ) ' Returns "A"
```

```
string = CHR( 66 ) ' Returns "B"
```

For more help, see [Asc](#)

Fill

Syntax:

```
stringvar = Fill ( byte_value_of_the_new_length , pad_character )
```

Command Availability:

Available on all microcontrollers

Explanation:

The Fill function is used to create string to a specific length that is of a specific character.

The length of the string is specified by the first parameter. The character used to pad the string is specified by the second parameter, this parameter is optional as the " "(space) character is assumed.

A typical use is to fill a string to be displayed on an LCD or serial terminal.

Example:

```
'Set chip model
#chip 16F886

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

HserPrint Fill ( 16, "" ) ;will print a string of '*'
HserPrintCRLF
```

For more help, see [Asc](#)

IntegerToBin

Syntax:

```
stringvar = IntegerToBin(integervar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **IntegerToBin** function creates a string of a ANSI (signed 15 digit string) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Note: Supports Integer variables only. For BYTE variables use **VarToBin**, for Word variables use **WordToBin** and for LONG variables use **LongToBin**

Example:

```
string = IntegerToBin( 1 )    ' Returns "+000000000000001"
string = IntegerToBin( -1 )  ' Returns "-000000000000001"
```

For more help, see [ByteToBin](#), [WordToBin](#), [LongToBin](#)

IntegerToHex

Syntax:


```
stringvar = IntegerToHex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Hex** function will convert a Integer number into hexadecimal format. The input **number** should be a Integer variable, or a fixed number between -32767 and 32767 inclusive. After running the function, the string variable **stringvar** will contain an 4 digit hexadecimal number.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 65535
    'Send location
    HSerPrint IntegerToHex(CurrentLocation)
    HSerPrint ":"
    'Read Integer and send
    EPRead CurrentLocation, CurrInteger
    HSerPrint Hex(CurrInteger)
    'Send carriage return/line feed
    HSerPrintCRLF
Next
```

See Also [ByteToHex](#), [WordToHex](#), [LongToHex](#), [SingleToHex](#)

IntegerToString

Syntax:

```
stringvar = IntegerToString(Integer_variable)    'supports Integer.
```

Command Availability:

Available on all microcontrollers

Explanation:

The `IntegerToString` function will convert a number into a string. `number` can be any Integer variable, or a fixed number constant between 0 and 4294967295 inclusive. For Byte number use `ByteToString()`, Word numbers use `WordToString()`, for Integer numbers use `IntegerToString()` and for Single numbers use `SingleToString()`

The string variable `stringvar` will contain the same number, represented as a string. The length of the string returned is 9 characters.

This function is especially useful if a number needs to be added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Example1:

```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dim SensorReading as Integer

'Take an A/D reading
SensorReading = ReadAD10(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = IntegerToString(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF

```

When using the functions IntegerToString() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```

' use this, note this is no space between the IntegerToString() and the left brace!
IntegerToString(number_variable)
' do not use, note the space!
IntegerToString (number_variable)

```

Example2:

```

'''
'''
'''
'''
'''*****
'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board

```

```

''' Date: June 2021
'''
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config CLRE_ON
#option Explicit

; ----- Define Hardware settings

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim IntegerVar as Integer

; ----- Main body of program commences here.
IntegerVar = -10

do
    wait 100 ms

    HSerPrint IntegerToString( IntegerVar )
    HSerPrintCRLF
    wait 1 s
loop
end

; ----- Support methods. Subroutines and Functions

```

See Also [ByteToString](#), [WordToString](#), [LongToString](#), [SingleToString](#), [ByteToHex](#)

Instr

Syntax:

```
location = Instr(source, find)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Instr** function will search one string to find the location of another string within it. **source** is the string to search inside, and **find** is the string to find. The function will return the location of **find** within **source**, or 0 if **source** does not contain **find**.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "Hello, world!"

'Display the location of "world" within the string
'Will return 8, because "w" in world is the 8th character
'of "Hello, world!"
HSerPrint Instr(TestData, "world")
HSerPrintCRLF

'Display the location of "planet" within the string
'Will display 0, because "planet" does not occur inside
'the string "Hello, world!"
HSerPrint Instr(TestData, "planet")
HSerPrintCRLF
```

LCase

Syntax:

```
output = LCase(source)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **LCase** function will convert all of the letters in the string **source** to lower case, and return the result.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "Hello, world!"

'Display the string in lower case
'Will display "hello, world!"
HSerPrint LCase(TestData)
HSerPrintCRLF
```

See Also [UCase](#)

Left**Syntax:**

```
output = Left(source, count)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Left** function will extract the leftmost **count** characters from the input string **source**, and return them in a new string.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "Hello, world!"

'Display the leftmost 5 characters
'Will display "Hello"
HSerPrint Left(TestData, 5)
HSerPrintCRLF
```

See Also [Mid](#), [Right](#)

LeftPad

Syntax:

```
LeftPad(string_variable,byte_value_of_the_new_length,pad_character)
```

Command Availability:

Available on all microcontrollers

Explanation:

The LeftPad function is used to create string to a specific length that is extended with a specific character to the left hand side of the string.

The length of the string is specified by the second parameter.

The character used to pad the string is specified by the third parameter.

A typical use is to pad a string to be displayed on a serial terminal or LCD.

Example:

```
'Set chip model
'Set chip model
#chip 16f877a
```

```

DIR PORTA 0x03

' make port C as output
Dir PortC 0x0

'Defines (Constants)
#define LCD_SPEED slow
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_NO_RW
#define LCD_Enable PORTc.0
#define LCD_RS PORTc.1
#define LCD_DB4 PORTa.5
#define LCD_DB5 PORTa.4
#define LCD_DB6 PORTa.3
#define LCD_DB7 PORTa.2
'''-----
'''-----End of board-specific settings-----
'''-----

'''DEMO for padding strings left with
'''1st character of a given string.
'''if no string is given, blanks are used

; ---- variables
DIM inString as string * 5
DIM outString1 as String
DIM outString2 as String

; ---- main body of program begins here

inString = "12345"

outString1 = leftpad(inString, 9, "*")
outString2 = leftpad(inString, 9)

'show results on LCD-Display
cls

print instring
print " "

```



```
print outstring1
locate 1,0
print instring
print " "
print outstring2

end
```

Len

Syntax:

```
output= Len( string )
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Len** function returns an byte value which is the length of a phrase or a sentence, including the empty spaces. The format is:

```
target_byte_variable = Len("Phrase")
```

or another example. This code will loop through the for-next loop 12 times as determined by the length of the string:

```
' create a test string of 12 characters
dim teststring as string * 12

teststring = "0123456789AB"
for loopthrustring = 1 to len(teststring)
    hserprint mid(teststring, loopthrustring , 1)
next
```

LongToBin

Syntax:

```
stringvar = LongToBin(longvar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **LongToBin** function creates a string of a ANSI (32) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Note: Supports LONG variables only. For BYTE variables use **VarToBin**, for WORD variables use **VarWToBin** and for INTEGER variables use **VarIntegerToBin**

Example:

```
string = LongToBin( 1 ) ' Returns "0000000000000001"
string = LongToBin( 254 ) ' Returns "0000000011111110"
```

For more help, see [VarToBin](#), [VarWToBin](#), [VarIntegerToBin](#)

LongToHex

Syntax:

```
stringvar = LongToHex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Hex** function will convert a Long number into hexadecimal format. The input **number** should be a Long variable, or a fixed number between 0 and 4294967295 inclusive. After running the function, the string variable **stringvar** will contain an 8 digit hexadecimal number.

Example:

```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 65535
    'Send location
    HSerPrint LongToHex(CurrentLocation)
    HSerPrint ":"
    'Read Long and send
    EPRead CurrentLocation, CurrLong
    HSerPrint Hex(CurrLong)
    'Send carriage return/line feed
    HSerPrintCRLF
Next

```

See Also [ByteToHex](#), [WordToHex](#), [IntegerToHex](#), [SingleToHex](#)

LongToString

Syntax:

```
stringvar = LongToString(Long_variable)    'supports Long.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **LongToString** function will convert a number into a string. **number** can be any Long variable, or a fixed number constant between 0 and 4294967295 inclusive. For Byte number use **ByteToString()**, Word numbers use **WordToString()**, for Integer numbers use **IntegerToString()** and for Single numbers use **SingleToString()**

The string variable **stringvar** will contain the same number, represented as a string. The length of the string returned is 10 characters.

This function is especially useful if a number needs to be added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Example1:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dim SensorReading as Long

'Take an A/D reading
SensorReading = ReadAD12(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = LongToString(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF
```

When using the functions LongToString() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```
' use this, note this is no space between the LongToString() and the left brace!
LongToString(number_variable)
' do not use, note the space!
LongToString (number_variable)
```

Example2:

```
'''
'''
'''
'''
'''*****
```

```

'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board
''' Date: June 2021
'''

' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config CLRE_ON
#option Explicit

; ----- Define Hardware settings

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim Longvar as Long

; ----- Main body of program commences here.
Longvar = 0xffffffff

do
    wait 100 ms

    HSerPrint LongToString( Longvar )
    HSerPrintCRLF
    wait 1 s
loop
end

; ----- Support methods. Subroutines and Functions

```

See Also [ByteToString](#), [WordToString](#), [IntegerToString](#), [SingleToString](#), [ByteToHex](#)

Ltrim

Syntax:

```
stringvar = LTRIM(stringvar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Ltrim** function will trim the 7-bit ASCII space character (value 32) from the LEFT hand side of a string.

Use **Ltrim** on text that you have received from another source that may have irregular spacing at the left hand end of the string.

See Also [Trim](#), [Rtrim](#)

Mid

Syntax:

```
output = Mid(source, start[, count])
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Mid** function returns a string variable containing a specified number of characters from a source string.

source is the variable to extract from. If **source** is a zero length string - a zero length string is returned equating to "".

start is the position of the first character to extract. If **start** is greater than the number of characters in string, Mid returns a zero-length string equating to "".

count is the number of characters to extract. If **count** is not specified, all characters from **start** to the end of the source string will be returned.

Example:

```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "The cat sat on the mat"

'Extract "cat". The c is at position 5, and 3 letters are needed
HSerPrint "The animal is a "
HSerPrint Mid(TestData, 5, 3)

'Extract the action. "sat" starts at position 9.
HSerPrint "The animal "
HSerPrint Mid(TestData, 9)
HSerPrintCRLF

```

See Also [Left](#), [Right](#)

Pad

Syntax:

```
out_string = Pad( string_variable, byte_value_of_the_new_length, pad_character)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Pad** function is used to create string to a specific length that is extended with a specific character.

The length of the string is specified by the second parameter. The character used to pad the string is specified by the third parameter.

A typical use is to pad a string to be displayed on a LCD display.

Example:

```

'Set chip model
  #chip 16f877a

DIR PORTA 0x03

' make port C as output
Dir PortC 0x0

'Defines (Constants)
#define LCD_SPEED slow
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_NO_RW
#define LCD_Enable PORTc.0
#define LCD_RS PORTc.1
#define LCD_DB4 PORTa.5
#define LCD_DB5 PORTa.4
#define LCD_DB6 PORTa.3
#define LCD_DB7 PORTa.2
'''-----
'''-----End of board-specific settings-----
'''-----

'''DEMO for pad strings to a length
'''1st character of a given string.
'''if no string is given, blanks are used

; ---- variables
'Define the string
Dim TestData As String * 16
TestData = "Location"

'show results on LCD-Display
cls
Print Pad ( TestData, 16, "*" )
Locate 1,0
Print Pad ( TestData, 16, )

end

```


Right

Syntax:

```
output = Right(source, count)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Right** function will extract the rightmost **count** characters from the input string **source**, and return them in a new string.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "Hello, world!"

'Display the rightmost 6 characters
'Will display "world!"
HSerPrint Right(TestData, 6)
HSerPrintCRLF
```

Rtrim

Syntax:

```
stringvar = Rtrim(stringvar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Rtrim** function will trim the 7-bit ASCII space character (value 32) from the RIGHT hand side of a string.

Use **Rtrim** on text that you have received from another source that may have irregular spacing at the right hand end of the string.

See Also [Trim](#), [Ltrim](#)

Trim

Syntax:

```
stringvar = Trim(stringvar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Trim** function will trim the 7-bit ASCII space character (value 32) from text.

Trim removes all spaces from text except for single spaces between words. Use **Trim** on text that you have received from another source that may have irregular spacing at the left or right hand ends of the string.

See Also [Ltrim](#), [Rtrim](#)

UCase

Syntax:

```
output = UCase(source)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **UCase** function will convert all of the letters in the string **source** to upper case, and return the result.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Fill a string with a message
Dim TestData As String
TestData = "Hello, world!"

'Display the string in upper case
'Will display "HELLO, WORLD!"
HSerPrint UCase(TestData)
HSerPrintCRLF
```

See Also [LCase](#)

SingleToBin

Syntax:

```
stringvar = SingleToBin(Singlevar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **SingleToBin** function creates a string of a ANSI (32) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Note: Supports Single variables only. For BYTE variables use **VarToBin**, for WORD variables use **VarWToBin** and for INTEGER variables use **VarIntegerToBin**

Example:

```
string = SingleToBin( 1 ) ' Returns "00000000000000000000000000000001"

string = SingleToBin( 254 ) ' Returns "000000000000000000000000011111110"
```

For more help, see [ByteToBin](#), [WordToBin](#), [IntegerToBin](#)

SingleToHex

Syntax:

```
stringvar = SingleToHex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Hex** function will convert a Single number into hexadecimal format. The input **number** should be a Single variable, or a fixed number between -3.4×10^{38} and $+3.4 \times 10^{38}$ inclusive. After running the function, the string variable **stringvar** will contain an 4 digit hexadecimal number.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 9999999
    'Send location
    HSerPrint SingleToHex(CurrentLocation)
    HSerPrint ":"
    'Read Single and send
    EPRead CurrentLocation, CurrSingle
    HSerPrint Hex(CurrSingle)
    'Send carriage return/line feed
    HSerPrintCRLF
Next
```

See Also [ByteToHex](#), [WordToHex](#), [LongToHex](#), [IntegerToHex](#)

SingleToString

Syntax:

```
stringvar = SingleToString(Single_variable)    'supports Single.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **SingleToString** function will convert a number into a string. **number** can be any Single variable. For Byte numbers use **ByteToString()**, Word numbers use **WordToString()**, for Integer numbers use **IntegerToString()** and for Long numbers use **LongToString()**

The string variable **stringvar** will contain the ACSII representation of the input number. The length of the string is variable length dependent on the input variable value.

This function is especially useful if a number needs to added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Operational Returned Controls

The function returns either the number string or the message "Error ". The reasons for "Error " are:

- Very small number that actaully compute as 0.0
- The input values is too large
- Too many chars-out of range

There is a public variable available after using this function. ``SysByte_STS_Err`` - this variable returns the following:

`SysByte_STS_Err` where 1 or 9 equates to no error.

1 equates to a properly formatted number string.

8 equateq to a properly formatted integer number string.

Bitwise returned details

```
SysByte_STS_Err.0 : 1 = good, or, 0 = bad
SysByte_STS_Err.1 : 1 = decimals places to many chars, or, 0 = ok
SysByte_STS_Err.2 : 1 = integer places to many chars-out of range, or, 0 = ok
SysByte_STS_Err.3 : 1 = no decimal point, info only
SysByte_STS_Err.4 : 1 = non numeric chars found
```

Example Usage 1:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dim SensorReading as Single

'Take an A/D reading
SensorReading = ReadAD(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = SingleToString(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF
```

When using the functions SingleToString() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```
' use this, note this is no space between the SingleToString() and the left brace!
SingleToString(number_variable)
' do not use, note the space!
SingleToString (number_variable)
```

```
Do
Loop
```

```
End
```

Example Usage 2:

```

'''
'''
'''
'''
'''*****
'''
'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board
''' Date: June 2021
'''
' ----- Configuration
' Chip Settings.
' #chip 16f18855,32
' #Config CLRE_ON
' #option Explicit

; ----- Define Hardware settings

' Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim Singlevar as Single

; ----- Main body of program commences here.
Singlevar = -10

do
    wait 100 ms

    HSerPrint SingleToString( Singlevar )
    HSerPrintCRLF
    wait 1 s
loop

end

```

See Also [ByteToString](#), [WordToString](#), [LongToString](#), [SingleToString](#), [ByteToHex](#)

StringToByte

Syntax:

```
var = StringToByte(string)    'Supports decimal byte range strings only.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **StringToByte** function will extract a number from a string variable, and store it in a byte variable. One potential use is reading numbers that are sent in ASCII format over a serial connection.

The **StringToByte** function will not extract a StringToByte from a hexadecimal string.

Example1:

```
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config MCLRE_ON

; ----- Define Hardware settings

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D2 |-
'' -----
''

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING
```



```

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT


#define Potentiometer    PORTA.4
DIR    Potentiometer In


#define SWITCH_DOWN      0
#define SWITCH_UP        1
#define SWITCH            PORTA.5
Dir SWITCH              In


; ----- Variables
dim bytevar as Byte
dim wordvar as Word
dim longvar as long


bytevar = 0
wordvar = 0
longvar = 0


; ----- Main body of program commences here.


#option Explicit


do
    wait 100 ms


    bytevar = StringToByte( "255" )
    HSerPrint bytevar
    HSerPrintCRLF


    wait 1 s
loop
end


; ----- Support methods.  Subroutines and Functions

```

StringToLong

Syntax:

```
var = StringToLong(string)    'Supports decimal Long range strings only.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **StringToLong** function will extract a number from a string variable, and store it in a Long variable. One potential use is reading numbers that are sent in ASCII format over a serial connection.

The **StringToLong** function will not extract a StringToLong from a hexadecimal string.

Example1:

```
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config MCLRE_ON

; ----- Define Hardware settings

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D2 |-
'' -----
''

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
```

```

#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT

#define Potentiometer    PORTA.4
DIR    Potentiometer In

#define SWITCH_DOWN      0
#define SWITCH_UP        1
#define SWITCH            PORTA.5
Dir SWITCH                In

; ----- Variables
dim bytevar as Byte
dim wordvar as Word
dim longvar as long

bytevar = 0
wordvar = 0
longvar = 0

; ----- Main body of program commences here.

#option Explicit

do
    wait 100 ms

    Longvar = StringToLong( "255" )
    HSerPrint Longvar
    HSerPrintCRLF

    wait 1 s
loop
end

; ----- Support methods.  Subroutines and Functions

```

StringToSingle

Syntax:

```
var = StringToSingle(string)    'Supports decimal Single range strings only.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **StringToSingle** function will extract a number from a string variable, and store it in a Single variable. One potential use is formatting a serial number received via a serial connection.

The **StringToSingle** function will not extract a StringToSingle from a hexadecimal string.

The function supports two messages to support usage.

```
' SysByte_STS_Err = 0 if no error
' SysByte_STS_Err.0 = 1 good - 0 - bad
' SysByte_STS_Err.1 = 1 decimals places to many chars, 0 = ok
' SysByte_STS_Err.2 = 1 integer places to many chars-out of range, 0 = ok
' SysByte_STS_Err.3 = 1 no decimal point, info only
' SysByte_STS_Err.4 = non numeric chars found
```

Example Usage 1:

```

' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config MCLRE_ON

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim bytevar as Byte
dim wordvar as Word
dim Singlevar as Single

bytevar = 0
wordvar = 0
Singlevar = 0

; ----- Main body of program commences here.

#option Explicit

do
    wait 100 ms

    Singlevar = StringToSingle( "255" )
    HSerPrint SingltoString(Singlevar)
    HSerPrintCRLF

    wait 1 s
loop
end

; ----- Support methods.  Subroutines and Functions

```

StringToWord

Syntax:

```
var = StringToWord(string)    'Supports decimal Word range strings only.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **StringToWord** function will extract a number from a string variable, and store it in a Word variable. One potential use is reading numbers that are sent in ASCII format over a serial connection.

The **StringToWord** function will not extract a StringToWord from a hexadecimal string.

Example1:

```
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config MCLRE_ON

; ----- Define Hardware settings

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D2 |-
'' -----
''

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
```

```

Dir      LEDD5 OUT

#define Potentiometer      PORTA.4
DIR      Potentiometer In

#define SWITCH_DOWN        0
#define SWITCH_UP          1
#define SWITCH              PORTA.5
Dir SWITCH                  In

; ----- Variables
dim bytevar as Byte
dim wordvar as Word
dim longvar as long

bytevar = 0
wordvar = 0
longvar = 0

; ----- Main body of program commences here.

#option Explicit

do
    wait 100 ms

    Wordvar = StringToWord( "65535" )
    HSerPrint WordVar
    HSerPrintCRLF

    wait 1 s
loop
end

; ----- Support methods.  Subroutines and Functions

```

ULongIntToBin

Syntax:

```
stringvar = ULongIntToBin(ULongIntvar)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **ULongIntToBin** function creates a string of a ANSI (32) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Note: Supports ULongInt variables only. For BYTE variables use **VarToBin**, for WORD variables use **VarWToBin** and for INTEGER variables use **VarIntegerToBin**

Example:

```
string = ULongIntToBin( 1 ) ' Returns "00000000000000000000000000000001"
string = ULongIntToBin( 254 ) ' Returns "000000000000000000000000000011111110"
```

For more help, see [ByteToBin](#), [WordToBin](#), [IntegerToBin](#)

WordToBin

Syntax:

```
stringvar = WordToBin(bytevar)
```

Command Availability: Available on all microcontrollers

Explanation:

The **WordToBin** function creates a string of a ANSI (16-byte) characters. The function converts a number to a string consisting of ones and zeros that represents the binary value.

Example:

```
string = WordToBin(1) ' Returns "0000000000000001"
string = WordToBin(37654) ' Returns "1001001100010110"
```

For more help, see [ByteToBin](#)

WordToHex

Syntax:


```
stringvar = WordToHex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Hex** function will convert a Word number into hexadecimal format. The input **number** should be a Word variable, or a fixed number between 0 and 65525 inclusive. After running the function, the string variable **stringvar** will contain a 4 digit hexadecimal number.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 65535
  'Send location
  HSerPrint WordToHex(CurrentLocation)
  HSerPrint ":"
  'Read Word and send
  EPRead CurrentLocation, CurrWord
  HSerPrint Hex(CurrWord)
  'Send carriage return/line feed
  HSerPrintCRLF
Next
```

See Also [ByteToHex](#), [LongToHex](#), [IntegerToHex](#), [SingleToHex](#)

WordToString**Syntax:**

```
stringvar = WordToString(Word_variable)    'supports Word.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **WordToString** function will convert a number into a string. **number** can be any Word variable, or a fixed number constant between 0 and 65535 inclusive. For Word number use **WordToString()**, Long numbers use **LongToString()**, for Integer numbers use **IntegerToString()** and for Single numbers use **SingleToString()**

The string variable **stringvar** will contain the same number, represented as a string. The length of the string returned is 5 characters.

This function is especially useful if a number needs to be added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Example1:

```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

Dim SensorReading as Word

'Take an A/D reading
SensorReading = ReadAD10(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = WordToString(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF

```

When using the functions WordToString() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```

' use this, note this is no space between the WordToString() and the left brace!
WordToString(number_variable)
' do not use, note the space!
WordToString (number_variable)

```

Example2:

```

'''
'''
'''
'''
'''*****
'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board

```

```

''' Date: June 2021
'''
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config CLRE_ON
#option Explicit

; ----- Define Hardware settings

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

; ----- Variables
dim Wordvar as Word

; ----- Main body of program commences here.
Wordvar = 0xffff

do
    wait 100 ms

    HSerPrint WordToString( Wordvar )
    HSerPrintCRLF
    wait 1 s
loop
end

; ----- Support methods. Subroutines and Functions

```

See Also [ByteToString](#), [LongToString](#), [IntegerToString](#), [SingleToString](#), [ByteToHex](#)

Concatenation

Syntax:

```
stringvar = variable1 + variable2
```

Command Availability:

Available on all microcontrollers

Explanation:

The method joins two variables into another variable.

This method does not change the existing strings, but returns a **new** string containing the text of the joined variables, see Concatenated String Constraint below.

Concatenation joins the elements of a specified values using the specified separator between each variable.

WARNING

Using concatenation as a parameter with commands like HSerPrint or Print the compiler will create a system string variable. An examples of concatenating two strings constants like HSerPrint ("123"+"456") may yield incorrect results. Use the constant `SYSDEFAULTCONCATSTRING` to resolve. Without using `SYSDEFAULTCONCATSTRING` there is a risk that the compiler does not allocate sufficient RAM to hold the concatenated string. The resulting string may be corrupted as the size of the system string variable is not sufficient. Use `SYSDEFAULTCONCATSTRING` within the source program to resolve.

Set a specific size of compiler created system string variable

Use the following to set the size of the system string variable used during concatenation.

The compiler will create system string variables when you concatenate on a commands line like `HSerPrint`, `Print` and many others commands. Using concatenate with a command is bad practice, using a lot of RAM and may create a number of system string variables. It is recommended to define a string (of a known length), concatenate using an assignment then use the string.

To control the size of system string variable use the following. Also, use this constant to set the size when the compiler does not create a system string variable.

```
'Define the constant to control the size of system created string variables called  
SYSSTRINGPARAM1, SYSSTRINGPARAM2 etc.
```

```
Use #DEFINE SYSDEFAULTCONCATSTRING 4
```

```
'Then, use
```

```
HSerPrint "A"+"123" 'will print A123. Without the SYSDEFAULTCONCATSTRING constant  
some microcontrollers may corrupt the result of the concatenation.
```

This concatenation constraint does not apply using concatenation as an assignment.

Example 1:

```
timevariable = 999
stringvar = "Time = " + str(timevariable) ' Convert the timevariable to a String.
This operation returns Time = 999
```

Example 2:

An example showing how to set a string to an escape sequence for an ANSI terminal. You can `Dim`ension a string and then assign the elements like an array. {empty} + {empty} +

```
dim line2 as string
line2 = 27, "[", "2", "H", 27, "[", "K"
HSerPrint line2
```

Will send the following to the terminal. <esc>[2H<esc>[K

Example 3: Assigning concatenated string to same string

For reliable coding you must not assign a string concatenation to same source variable. You must assign the result of string concatenation to another string. To resolve see below:

```
Dim outstring, tmpstring as string * 16
Dim outnumber as byte

outnumber = 24
outstring = "Result = "
'This concatenation may yield an incorrect string on 10f, 12f or 16f chips
outstring = outstring + str(outnumber)
HserPrintCRLF 2
HserPrint outstring
HserPrintCRLF 2

outstring = "Result = "
'This concatenation will yield an the correct string. With tmpstring1 containing the
correct concatenated string
tmpstring = outstring +str(outnumber)
HserPrint tmpstring
HserPrintCRLF 2
end
```

To resolve the constraint simply assign the source string to another string.

Deprecated string functions.

Use the alternative updated functions.

Hex

Syntax:

```
stringvar = Hex(number)
```

Command Availability:

Available on all microcontrollers

Explanation:

The `Hex` function will convert a number into hexadecimal format. The input `number` should be a byte variable, or a fixed number between 0 and 255 inclusive. After running the function, the string variable `stringvar` will contain a 2 digit hexadecimal number.

Example:

```
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Send EEPROM data over serial connection
'Uses Hex to display as hexadecimal
For CurrentLocation = 0 to 255
    'Send location
    HSerPrint Hex(CurrentLocation)
    HSerPrint ":"
    'Read byte and send
    EPRead CurrentLocation, CurrByte
    HSerPrint Hex(CurrByte)
    'Send carriage return/line feed
    HSerPrintCRLF
Next
```

See Also [Str](#), [Val](#)

Str

Syntax: Deprecated use ByteToString()

```
stringvar = Str(number)      'supports decimal byte and word strings only.  
  
'Use the following to support decimal long number strings.  
stringvar = Str32(long number)  'supports decimal long number strings.  
  
'Use the following to support decimal integer number strings.  
stringvar = StrInteger(integer number)  ' decimal integer number strings.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Str** function will convert a number into a string. **number** can be any byte or word variable, or a fixed number between 0 and 65535 inclusive. For Long numbers use **Str32** and for Integer numbers use **StrInteger**.

The string variable **stringvar** will contain the same number, represented as a string. The length of the string returned is 5, 10 or 6 characters for Byte & Word, Long and Integer respectively.

This function is especially useful if a number needs to be added to the end of a string, or if a custom data sending routine has been created but only supports the output of string variables.

These methods will not support conversion of hexadecimal number strings.

Example1:


```

'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Take an A/D reading
SensorReading = ReadAD(AN0)

'Create a string variable
Dim OutVar As String

'Fill string with sensor reading
OutVar = Str(SensorReading)

'Send
HSerPrint OutVar
HSerPrintCRLF

```

When using the functions STR() do not leave space between the function call and the left brace. You will get a compiler error that is meaningless.

```

' use this, note this is no space between the STR and the left brace!
STR(number_variable)
' do not use, note the space!
STR (number_variable)

```

Example2:

```

'''
'''
'''
'''
'''
'''*****
'''
''' PIC: 16F18855
''' Compiler: GCB
''' IDE: GCode
'''
''' Board: Xpress Evaluation Board
''' Date: June 2021

```

```

''
' ----- Configuration
' Chip Settings.
' #chip 16f18855,32
' #Config CLRE_ON
' #option Explicit

; ----- Define Hardware settings

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D2 |-
'' -----
''

' Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010    'RC0->EUSART:TX;
    RXPPS  = 0x0011    'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT

#define Potentiometer    PORTA.4
DIR    Potentiometer In

#define SWITCH_DOWN      0
#define SWITCH_UP        1
#define SWITCH           PORTA.5
Dir SWITCH              In

; ----- Variables
dim bytevar as Byte
dim wordvar as Word
dim longvar as long
dim integervarP, integervarN, integervar as Integer

```

```

; ----- Main body of program commences here.
bytevar = 0xff
wordvar = 0xffff
longvar = 0xffffffff
integervarP = 127
integervarN = -127
integervar = 0

do
    wait 100 ms

    HSerPrint str( bytevar )
    HSerPrintCRLF
    HSerPrint str( wordvar )
    HSerPrintCRLF
    HSerPrint str32( longvar )
    HSerPrintCRLF
    HSerPrint StrInteger( integervarP )
    HSerPrintCRLF
    HSerPrint StrInteger( integervarN )
    HSerPrintCRLF
    HSerPrint StrInteger( integervar )
    HSerPrintCRLF
    wait 100 ms
    HSerPrintCRLF

    wait 1 s
loop
end

; ----- Support methods.  Subroutines and Functions

```

See Also [Hex](#), [Val](#)

Val

Syntax:

```
var = Val(string)    'Supports decimal byte and word strings only.
```

'use the following for strings that represent Long numbers

```
var = Val32(string)  'Supports decimal long number strings only.
```

Command Availability:

Available on all microcontrollers

Explanation:

The **Val** function will extract a number from a string variable, and store it in a word variable. One potential use is reading numbers that are sent in ASCII format over a serial connection.

The **Val32** function will extract a long number from a string variable, and store it in a long variable.

The **Val** function will not extract a value from a hexadecimal string.

Example1:

```

'Program for an RS232 controlled dimmer
'Set chip model
#chip 16F1936

'Set up hardware serial connection
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

'Set pin directions for USART and PWM

'Variable for output level
Dim OutputLevel As Word

'Variables for received bytes
Dim DataIn As String
DataInCount = 0

'Main Loop
Do
    'Get serial byte
    Wait Until USARHasData
    HSerReceive InByte

    'Process latest byte
    'Enter key?
    If InByte = 13 Then
        'Convert output level to numeric variable
        OutputLevel = Val(DataIn)

        'Output
        HPWM 1, 32, OutputLevel

        'Clear output buffer for next command
        DataIn = ""
        DataInCount = 0
    End If

    'Number?
    If InByte >= 48 and InByte <= 57 Then
        'Add to end of DataIn string
        DataInCount += 1
        DataIn(DataInCount) = InByte
        DataIn(0) = DataInCount
    End If
Loop

```

Example2:

```
' ----- Configuration
'Chip Settings.
#chip 16f18855,32
#Config MCLRE_ON

; ----- Define Hardware settings

'' -----LATA-----
'' Bit#:  -7---6---5---4---3---2---1---0---
'' LED:   -----|D5 |D4 |D3 |D2 |-
'' -----
''

'Set the PPS of the RS232 ports.
UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;
LOCKPPS

; ----- Constants
#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

#define LEDD2 PORTA.0
#define LEDD3 PORTA.1
#define LEDD4 PORTA.2
#define LEDD5 PORTA.3
Dir    LEDD2 OUT
Dir    LEDD3 OUT
Dir    LEDD4 OUT
Dir    LEDD5 OUT

#define Potentiometer      PORTA.4
DIR    Potentiometer In

#define SWITCH_DOWN        0
#define SWITCH_UP          1
#define SWITCH             PORTA.5
Dir SWITCH                In

; ----- Variables
```

```

dim bytevar as Byte
dim wordvar as Word
dim longvar as long

bytevar = 0
wordvar = 0
longvar = 0

; ----- Main body of program commences here.

#option Explicit

do
    wait 100 ms

    bytevar = Val( "255" )
    HSerPrint bytevar
    HSerPrintCRLF

    wordvar = Val( "65535" )
    HSerPrint wordvar
    HSerPrintCRLF

    longvar = Val32( "65536" )
    HSerPrint longvar
    HSerPrintCRLF 2

    wait 1 s
loop
end

; ----- Support methods.  Subroutines and Functions

```

See Also [Hex](#), [Str](#)

Miscellaneous Commands

This is the Miscellaneous Commands section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Dir

Syntax:

```
Dir port.bit {In | Out}           (Individual Form)  
Dir port {In | Out | DirectionByte} (Entire Port Form)
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Dir** command is used to set the direction of the ports of the microcontroller chip. The individual form sets the direction of one pin at a time, whereas the entire port form will set all bits in a port.

In the individual form, specify the port and bit (ie. **PORTB.4**), then the direction, which is either In or Out.

The entire port form is similar to the **TRIS** instruction offered by some Microchip PIC microcontrollers. To use it, give the name of the port (i.e. **PORTA**), and then a byte is to be written into the **TRIS** variable. This form of the command is for those who are familiar with the Microchip PIC microcontrollers internal architecture.

Note: Entire port form will work differently on Atmel AVR microcontrollers when a value other than IN or OUT is used. Atmel AVR microcontrollers use 0 to indicate in and 1 to indicate out, whereas Microchip PIC microcontrollers use 0 for out and 1 for in. When IN and OUT are used there are no compatibility issues.

Example:


```

'This program sets PORTA bits 0 and 1 to in, and the rest to out.
'It also sets all of PORTB to output, except for B1.
'Individual form is used for PORTA:
DIR PORTA.0 IN
DIR PORTA.1 IN
DIR PORTA.2 OUT
DIR PORTA.3 OUT
DIR PORTA.4 OUT
DIR PORTA.5 OUT
DIR PORTA.6 OUT
DIR PORTA.7 OUT
'Entire port form used for B:
DIR PORTB b'00000010'

'Entire port form used for C:
DIR PORTC IN

```

Automatic DIRection setting by the compiler

The compiler will set the automatic pin DIRection using the following logic.

Any time that the user program reads a pin or port, the compiler records that. Any time that the user program writes to a pin or entire port, the compiler also records that.

Once all input code has been compiled, the compiler examines the list of reads and writes.

If a pin is only ever written to, the compiler makes it an output.

If a pin is only ever read, the compiler does not know if the intent is to read the latch or an input value, so it sets that pin to be an input.

If the compiler sees a pin being read and written to, the compiler does not know if you are using a pin for some sort of bidirectional communication, or if you are just reading the latch. To avoid making incorrect assumptions, the compiler will expect you to set the pin direction manually.

If you use "portA.2 = 1", you've only written to the pin, so the compiler knows it must be an output.

If you use "portA.2 = not portA.2", the compiler sees that you are reading and writing to the pin, and will expect the user program set the direction instead of trying to guess what you are doing.

The compiler also records any use of the Dir command, and will not do any automatic direction setting on a pin if Dir has been used on that pin anywhere in the user program..

GetUserID

Syntax:

Command Availability:

Available on all Microchip microcontrollers that support UserIDs.

Explanation:

Reads the memory location and returns the ID for a specific microcontroller.

If the microcontroller does not support GetUserID then the following message will be issued during compilation **Warning: GetUserID not supported by this microcontroller.**

The method reads the memory location $0x8000 + \text{Index}$ and returns it as a Word value, where the Index $0x00$ to $0x0B$ as follows:

Address	Function	Read	Write
8000h-8003h	User IDs	Yes	Yes
8006h/8005h	Device ID/Revision ID	Yes	No
8007h-800Bh	Configuration Words 1 through 5	Yes	No

Refer to your particular Device Datasheet to confirm the address table

Example:

```

#chip 16F1455
#Config MCLRE_ON

#include <GetUserID.h>

#define USART_BAUD_RATE 19200
#define USART_TX_BLOCKING

'Implement ANSI escape code for serial terminal NOT using a LCD!
#define ESC    chr(27)
#define CLS    HSerPrint(ESC+"[2J")
#define HOME   HSerPrint(ESC+"[H")
#define Print  HSerPrint

CLS
HOME

dim UserIDRegister as word

For Index = 0 to 0xF
    UserIDRegister = GetUserID(Index)
    HserPrint "80" + hex(NVIndex)
    HserPrint " : "
    HserPrint hex( UserIDRegister_H )
    HserPrint hex( UserIDRegister )
Next Index

End

```

Pot

Syntax:

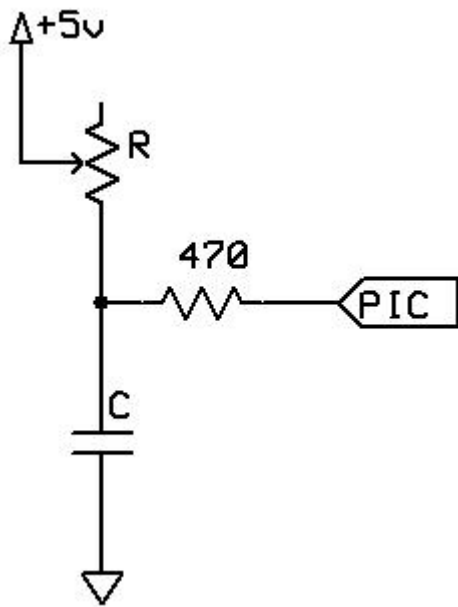
Pot pin, output

Command Availability:

Available on all microcontrollers.

Explanation:

Pot makes it possible to measure an analog resistance with a digital port, with the addition of a small capacitor. This is the required circuit:



The command works by using the microcontroller pin to discharge the capacitor, then measuring the time taken for the capacitor to charge again through the resistor.

The value for the capacitor must be adjusted depending on the size of the variable resistor. The charging time needs to be approximately 2.5 ms when the resistor is at its maximum value. For a typical 50 k potentiometer or LDR, a 50 nf capacitor is required.

This command should be used carefully. Each time it is inserted, 20 words of program memory are used on the chip, which as a rough guide is more than 15 times the size of the Set command.

`pin` is the port connected to the circuit. The direction of the pin will be dealt with by the `Pot` command.

`output` is the name of the variable that will receive the value.

Example 1:

```

'This program will beep whenever a shadow is detected
'A potentiometer is used to adjust the threshold

#chip 16F628A, 4

#define ADJUST PORTB.0
#define LDR PORTB.1
#define SoundOut PORTB.2

Dir SoundOut Out

Do
    Pot ADJUST, Threshold
    Pot LDR, LightLevel
    If LightLevel > Threshold Then
        Tone 1000, 100
    End If
Loop

```

Example 2:

This program is an implementation of the capacitor and resistor principle using the chips internal capacitor and the internal pullup resistor.

The will test the state of the GPIO.3 port by using these internal components, and, after the charge state has been complete the LED PWM will represent the detected value of signal on the GPIO.3 port.

It should be note that GCBASIC will set the DIREction of GPIO.2 and GPIO.3 automatically. And, this solution is specific to the 12F509 and therefore the 12F509 register called **NOT_GPPU** may be different on another chip.

```

#chip 12F509
#option Explicit

;Defines (Constants)
#define PWM_Out1 GPIO.2

;Variables
Dim TimeCount As byte
Dim OPTION_REG as byte

Do Forever

    NOT_GPPU = Off
    Wait 1 ms
    NOT_GPPU = On
    TimeCount = 0

    'Do while held high by the internal capacitance
    Do While GPIO.3 = 1

        TimeCount = TimeCount + 1
        If TimeCount = 255 Then
            Exit Do
        End If

    Loop

    PWMOut 1, TimeCount, 5

Loop

```

See also ladyada.net/library/rccalc.html or cvs1.uklinux.net/cgi-bin/calculators/time_const.cgi for calculating capacitor value. These sites are not associated with GCBASIC.

PulseOut

Syntax:

```
PulseOut pin, time units
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **PulseOut** command will set the specified pin high, wait for the specified amount of time, and then set the pin low again. The pin is specified in the same way as it is for the Set command, and the time is the same as for the **Wait** command.

Example:

```
'This program flashes an LED on GPIO.0 using PulseOut
#chip 12F629, 4

'The DIRection of the port is set to show the command. It is not required to set the
DIRection when using the PulseOut command.
Dir GPIO.0 Out
Do
    PulseOut GPIO.0, 1 sec 'Turn LED on for 1 sec
    Wait 1 sec             'Wait 1 sec with LED off
Loop
```

PulseOutInv

Syntax:

```
PulseOutInv pin, time units
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **PulseOutInv** command will set the specified pin low, wait for the specified amount of time, and then set the pin high. The pin is specified in the same way as it is for the **Set** command, and the time is the same as for the **Wait** command.

Example:

```
'This program flashes an LED on GPIO.0 using PulseOutInv
#chip 12F629, 4

Dir GPIO.0 Out
Do
    PulseOutInv GPIO.0, 1 sec 'Turn LED off for 1 sec
    Wait 1 sec               'Wait 1 sec with LED on
Loop
```

PulseIn

Syntax:

```
PulseIn pin, user_variable, time units
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **PulseIn** command will monitor the specified pin when the pin is high, and then measure the high time. It will store the time in the user variable. The user variable must be a WORD if returned units are expected to be > 255 (Example: Pulse is 500 ms)

PulseIn is not recommended for accurate measurement of microsecond pulses

Example:

```
#chip 12F629, 4

Dir GPIO.0 In
Dim TimeResult as WORD

Do while GPIO.0 = Off      'Wait for next positive edge to start measuring
Loop

Pulsein GPIO.0, TimeResult, ms
```

PulseInInv

Syntax:

```
PulseInInv pin, user_variable, time units
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **PulseIn** command will monitor the specified pin when the pin is low, and then measure the low time. It will store the time in the user variable. The user variable must be a WORD if returned units are

expected to be > 255 (Example: Pulse is 500 ms)

PulseInInv is not recommended for accurate measurement of microsecond pulses.

Example:

```
#chip 12F629, 4

Dir GPIO.0 In
Dim TimeResult as WORD

Do while GPIO.0 = On      'Wait for next negative edge to start measuring
Loop

PulseInInv GPIO.0, TimeResult, ms
```

Peek

Syntax:

```
OutputVariable = Peek (location)
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Peek** function is used to read information from the on-chip RAM of the microcontroller.

location is a word variable that gives the address to read. The exact range of valid values varies from chip to chip.

This command should not normally be used, as it will make the porting of code to another chip very difficult.

Example #1 :

```
'This program will read and check a value from PORTA
'This specific peek will only work on some microcontrollers
Temp = Peek(5)
IF Temp.2 ON THEN SET green ON
IF Temp.2 OFF THEN SET red ON
```

Example #2

```

' This subroutine will toggle the pin state.
' You must change the parameters for your specific chip.
' Usage as show in examples below.
'
'     Toggle @PORTE, 2 ' equates to RE1.
'     Wait 100  ms
'     Toggle @PORTE, 2
'     Wait 100 ms

' Port , Pin address in Binary
' Pin0 = 1
' Pin1 = 2
' Pin2 = 4
' Pin3 = 8
'
' You can toggle any number of pins.
' Toggle @PORTE, 0x55
Sub Toggle ( In DestPort As word, In DestBit )
    Poke DestPort, Peek(DestPort) xor DestBit
End sub

```

See Also [Poke](#)

Poke

Syntax:

```
Poke(location, value)
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Poke** command is used to write information to the on-chip RAM of the microcontroller.

location is a word variable that gives the address to write. The exact range of valid values varies from chip to chip. **value** is the data to write to the location.

This command should not normally be used, as it will make the porting of code to another chip very difficult.

Example 1:

```
'This program will set all of the PORTB pins high
POKE (6, 255)
```

Example 2:

```
;Chip Settings
#chip 16F88

Dir PORTB out

Do Forever
    FlashPin @PORTB, 8
    Wait 1 s
Loop

Sub FlashPin (In DestVar As word, In DestBit)
    Poke DestVar, Peek(DestVar) Or DestBit
    Wait 1 s
    Poke DestVar, Peek(DestVar) And Not DestBit
End Sub
```

Using @ before the name of a variable (including a special function register) will give you the address of that variable, which can then be stored in a word variable and used by **Peek** and **Poke** to indirectly access the location.

See Also [Peek](#)

Weak Pullups

Weak pullups provide a method within many microcontrollers such as the Atmel AVR and Microchip PIC microcontrollers to support internal/selectable pull-ups for convenience and reduced parts count.

If you require **Weak pullups** these internal pullups can provide a simple solution. For example, you can use them to ground input pins with a switch closure - with the pullup enabled, the pin is held in a high state until the input line pulls it to ground. Be aware of possible EMI interference and also make sure to use a debounce routine.

If you need your weak pullups to exactly control current (rare for most microcontroller applications), then you should consider 10k resistors ($5V/10K = 500\mu A$) Why? If you review in the microcontroller data sheet, there is no resistance given for the weak pullups. That is because they are not weak pull-resistors they are weak pullups consisting of what appear to be high-resistance channel pFETs. Their channel resistance will vary with temperature and between parts; not easy to characterize.

The data sheet gives a current range for the internals as 50-400 μA (at 5V).

PORTs can have an individually controlled weak internal pullup. When set, each bit of the appropriate Microchip PIC register enables the corresponding pin pullup. There is a master bit within a specific register bit that enables pullups on all pins which also have their corresponding weak pull bit set. Also when set, there is a weak pull register bit to disable all weak pullups.

The weak pullup is automatically turned off when the port pin is configured as an output. The pullups are disabled on a Power-on Reset.

Each specific microcontroller has different registers/bits for this functionality.

You should review the datasheet for the method for a specific microcontroller.

The following code demonstrates how to set the weak pullups available on port B of an 18F25K20.

Example:

```
'A program to show the use of weak pullups on portb.
'Set chip model
#chip 18F25k20,16 'at 16 MHz
#config MCLR = Off

Set RBPU = 0 'enabling Port B pullups in general.
SET WPUB1 = 1 'portb.1 pulled up
Set WPUB2 = 1 'portb.2
Set WPUB3 = 1 'portb.3
Set WPUB4 = 1 'portb.4

Dir Portb in
Dir Portc out

do
    portc.1 = portb.1 'in pin 22, out pin 12
    portc.2 = portb.2 'in pin 23, out pin 13
    portc.3 = portb.3 'in pin 24, out pin 14
    portc.4 = portb.4 'in pin 25, out pin 15

loop 'jump back to the start of the program

'main line ends here
end
```

Also, see I2C Slave Hardware for an example using a 16F microcontroller.

Maths

This is the Maths section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Abs

Syntax:

```
integer_variable = Abs( integer_variable )
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Abs** function will compute the absolute value of a integer number therefore in the range of -32767 to +32767.

Example:

```
absolute_value = Abs( -127 ) ' Will return 127  
absolute_value = Abs( 127 )  ' Will return 127 also. :-)
```

Average

Syntax:

```
integer_variable = Average(byte_variable1 , byte_variable2)
```

Command Availability:

Available on all microcontrollers.

Explanation:

A function that returns the average of two numbers. This only supports byte variables.

Provides a very fast way to calculate the average of two 8 bit numbers.

Example:

```
average_value = Average(8,4) ' Will return 6
```

Difference

Syntax:

```
Difference ( word_variable1 , word_variable2 ) or  
Difference ( byte_variable1 , byte_variable2 )
```

Command Availability:

Available on all microcontrollers.

Explanation:

A function that returns the difference between of two numbers. This only supports byte or word variables.

Example:

```
Difference( 8 ,4 ) ' Will return 4  
Difference( 0xff01 , 0xffffa ) ' Will return 0xf9 or 249d
```

Int

Syntax:

```
integer_variable = Int( single_variable )
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Int** function will compute the integer value of an integer number therefore will return the range of single variable.

Example:

```
integer_value = Int( _singlevariable_ ) ' Will return the range of single  
variable.
```

Logarithms

Explanation:

GCBASIC support logarithmic functions through the include file <maths.h>.

These functions compute base 2, base e and base 10 logarithms accurate to 2 decimal places, +/- 0.01.

The values returned are fixed-point numbers, with two decimal places assumed on the right. Or if you prefer, think of the values as being scaled up by 100.

The input arguments are word-sized integers, 1 to 65535. Remember, logarithms are not defined for non-positive numbers. It is the calling program's responsibility to avoid these. Output values are also word-sized.

Local variables consume 9 bytes, while the function parameters consume another 4 bytes, for a grand total of 13 bytes of RAM used. The lookup table takes 35 words of program memory.

For more help, see [Log10](#), [Log2](#), [Loge](#)

Supported in <MATHS.H>

Log2

Syntax:

```
returned_word_variable = Log2 ( word_value )
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Log2** command will return the base-2 logarithm, to 2 decimal places.

The values returned are fixed-point numbers, with two decimal places assumed on the right. or if you prefer, think of the values as being scaled up by 100.

Example:

```
dim log_value as word
log_value = log2 ( 10 ) 'return 3321 equate to 3.321
```

Supported in <MATHS.H>

Loge

Syntax:

```
returned_word_variable = Loge ( word_value )
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Loge** command will return the base-e logarithm, to 2 decimal places.

The values returned are fixed-point numbers, with two decimal places assumed on the right. or if you prefer, think of the values as being scaled up by 100.

Example:

```
dim log_value as word  
log_value = loge ( 10 )
```

Supported in <MATHS.H>

Log10

Syntax:

```
returned_word_variable = Log10 (word_value)
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **Log10** command will return the base-10 logarithm, to 2 decimal places.

The values returned are fixed-point numbers, with two decimal places assumed on the right. or if you prefer, think of the values as being scaled up by 100.

Example:


```
dim log_value as word
log_value = log10 ( 10 )      'return 230 equate to 2.30
```

Supported in <MATHS.H>

Power

Syntax:

```
power( base, exponent )
```

Explanation:

This function raises a base to an exponent, i.e, `power(base,exponent)`. Calculation powers will become large, in terms of long numbers, you must ensure the program manage numbers remain within range of the defined variables.

The `base` and `exponent` are Byte sized numbers in this method.

The returned result is a Long.

Non-negative numbers are assumed throughout.

Note: 0 raised to 0 is meaningless and should be avoided, but, any other non-zero base raised to 0 is handled correctly.

Example:

```

;Thomas Henry -- 5/2/2014

;----- Configuration

#chip 16F88, 8           ;PIC16F88 running at 8 MHz
#config mclr=off         ;reset handled internally

#include <maths.h>        ;required maths.h

;----- Constants

#define LCD_IO 4          ;4-bit mode
#define LCD_WIDTH 20      ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS PortB.2    ;pin 8 is LCD Register Select
#define LCD_Enable PortB.3 ;pin 9 is LCD Enable
#define LCD_DB4 PortB.4   ;DB4 on pin 10
#define LCD_DB5 PortB.5   ;DB5 on pin 11
#define LCD_DB6 PortB.6   ;DB6 on pin 12
#define LCD_DB7 PortB.7   ;DB7 on pin 13
#define LCD_NO_RW 1       ;Ground the RW line on LCD

;----- Variables

dim i, j as byte

;----- Program

dir PortB out             ;all outputs to the LCD
for i = 1 to 10           ;do all the way from
  for j = 0 to 9          ;1^0 on up to 10^9
    cls
    print i
    print "^"
    print j
    print "="
    locate 1,0
    print power(i,j)      ;here's the invocation
    wait 1 S
  next j
next i

```

Supported in <MATHS.H>

RoundSingle

Syntax:

```
rounded_single_value = RoundSingle( single_variable )
```

Command Availability:

Available on all microcontrollers.

Explanation:

The **RoundSingle** function will return a floating point number that is a rounded version of the specified number.

This operates the same as Microsoft floor().

Example:

```
rounded_single_value = RoundSingle( _singlevariable_ ) ' Will return the range of  
single variable.
```

Scale

Syntax:

```
integer_variable = Scale (value_word , fromLow_integer , fromHigh_integer ,  
toLow_integer , toHigh_integer [, calibration_integer] )
```

Command Availability:

Available on all microcontrollers. The parameters are:

value: the number to scale. A value between 0 and 0xFFFFF - all values passed will be treated as Word variables.

fromLow: the lower bound of the value's current range. An Integer value between -32767 and 32767.

fromHigh: the upper bound of the value's current range. An Integer value between -32767 and 32767.

toLow: the lower bound of the value's target range. An Integer value between -32767 and 32767.

toHigh: the upper bound of the value's target range. An Integer value between -32767 and 32767.

calibration: optional calibration offset value. An Integer value between -32767 and 32767.

This is also an overloaded method. You can also use word variables to provide a returned result of 0-65535.

```
word_variable = Scale (value_word , fromLow_word , fromHigh_word , toLow_wordr ,  
toHigh_word [, calibration_integer] )
```

Available on all microcontrollers. The parameters are:

value: the number to scale. A value between 0 and 0xFFFFF - all values passed will be treated as Word variables.

fromLow: the lower bound of the value's current range. A word value.

fromHigh: the upper bound of the value's current range. A word value.

toLow: the lower bound of the value's target range. A word value.

toHigh: the upper bound of the value's target range. A word value.

calibration: optional calibration offset value. An Integer value between -32767 and 32767.

Explanation:

Scales, re-maps, a number from one range to another. That is, a value of fromLow would get scaled to toLow, a value of fromHigh to toHigh, values in-between to values in-between, etc.

The method does not constrain values to within the integer range returned, because out-of-range values are sometimes intended and useful.

Note that the "lower bounds" of either range may be larger or smaller than the "upper bounds" so the scale() method may be used to reverse a range of numbers, for example:

```
my_newvalue = scale ( ReadAD10(An0) , 0, 1023, 135, 270)
```

The method also handles negative integer numbers well, so that this example:

```
my_newvalue = scale(ReadAD(An0), 0, 255, 50, -100);
```

This method is similar to the Arduino Map() function.

Sqrt

Syntax:

```
word_variable = sqrt ( word )
```

Explanation:

A square root routine for GCBASIC. The function only involves bit shifting, addition and subtraction, which makes it fast and efficient.

This method required a word variable as the input and a word variable as the output. The method will handle arguments of up to 4294.

Command Availability:

Available on all microcontrollers, required MATHS.H include file.

Example:

```
;Demo: Show the first 100 square roots to 2 decimal places.
;This uses the maths.h include file.

;----- Configuration

#chip 16F88, 8                ;PIC16F88 running at 8 MHz
#config mclr=off              ;reset handled internally

#include <maths.h>             ;required maths.h

;----- Constants

#define LCD_IO      4          ;4-bit mode
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS      PortB.2    ;pin 8 is LCD Register Select
#define LCD_Enable  PortB.3    ;pin 9 is LCD Enable
#define LCD_DB4     PortB.4    ;DB4 on pin 10
#define LCD_DB5     PortB.5    ;DB5 on pin 11
#define LCD_DB6     PortB.6    ;DB6 on pin 12
#define LCD_DB7     PortB.7    ;DB7 on pin 13
#define LCD_NO_RW   1          ;ground the RW line on LCD

;----- Variables

dim length as byte
dim i as word
dim valStr, outStr as string

;----- Program
```

```

dir PortB out      ;all outputs to the LCD

for i = 0 to 100   ;print first 100 square roots
  cls
  print "sqrt("
  print i
  print ")="

  valStr = str(sqrt(i))      ;format decimal nicely
  length = len(valStr)

  select case length
    case 1:
      outStr = "0.00"        ;zero case
    case 3:
      outStr = left(valStr,1)+ "."+right(valStr,2)
    case 4:
      outStr = left(valStr,2)+ "."+right(valStr,2)
    case 5:
      outStr = left(valStr,3)+ "."+right(valStr,2)
  end select

  print outStr            ;display results
  wait 2 S
next i

```

Supported in <MATHS.H>

Trigonometry Sine, Cosine and Tangent

Syntax:

```

integer_variable = sin( integer_variable )

integer_variable = cos( integer_variable )

integer_variable = tan( integer_variable )

```

Explanation:

GCBASIC supports Three Primary Trigonometric Functions

GCBASIC supports the following functions, sin(x), cos(x), tan(x), where x is a signed integer representing an angle measured in a whole number of degrees. The output values are also integers, represented as fixed point decimal fractions.

Details:

The sine, cosine and tangent functions are available for your programs simply by including the header file offering the precision you need.

```
#INCLUDE <TRIG2PLACES.H> gives two decimal places
#INCLUDE <TRIG3PLACES.H> gives three decimal places
#INCLUDE <TRIG4PLACES.H> gives four decimal places
```

In fixed point representation, the decimal point is assumed. For example, with two places of accuracy, $\sin(60)$ returns 87, which you would interpret as 0.87. With three places, 866 is returned, to be interpreted as 0.866, and so on. Another way of thinking of this is to consider the two-place values as scaled up by 100, the three-place values scaled up by 1000 and the four-place values scaled up by 10,000.

Sine and Cosine are always defined, but remember that tangent fails to exist at 90 degrees, 270 degrees and all their coterminal angles. It is the responsibility of the calling program to avoid these special values.

Note that the tangent function is not available to four decimal places, since its value grows so rapidly, exceeding what the Integer data type can represent.

These routines are completely general. The input argument may be positive, negative or zero, with no restriction on the size. Further observe that lookup tables are used, so the routines are very fast, efficient and accurate.

Example: Show the trigonometric values to three decimal places.

```
;----- Configuration
#CHIP 16F88, 8                ;PIC16F88 RUNNING AT 8 MHZ
#CONFIG MCLR=OFF              ;RESET HANDLED INTERNALLY

#INCLUDE <TRIG3PLACES.H>

;----- Constants

#define LCD_IO      4          ;4-bit mode
#define LCD_WIDTH 20           ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RS      PortB.2    ;pin 8 is LCD Register Select
#define LCD_Enable  PortB.3    ;pin 9 is LCD Enable
#define LCD_DB4     PortB.4    ;DB4 on pin 10
#define LCD_DB5     PortB.5    ;DB5 on pin 11
#define LCD_DB6     PortB.6    ;DB6a on pin 12
#define LCD_DB7     PortB.7    ;DB7 on pin 13
#define LCD_NO_RW   1          ;ground the RW line on LCD
```

```

;----- Variables

dim ii as integer
dim outStr, valStr as string

;----- Program

dir PortB out                ;all outputs to the LCD

for ii = -720 to 720          ;arguments from -720 to 720
cls
print "sin("                  ;print the label
print ii                      ;and the argument
print ")="                    ;and closing parenthesis
locate 1,0
printTrig(sin(ii))            ;print value of the sine
wait 500 mS                   ;pause to view

cls                            ;do likewise for cosine
print "cos("
print ii
print ")="
locate 1,0
printTrig(cos(ii))
wait 500 mS                   ;pause to view
cls                            ;do likewise for tangent
print "tan("
print ii
print ")="
locate 1,0
printTrig(tan(ii))
wait 500 mS                   ;pause to view
next i

sub printTrig(in value as integer)
    ;print decently formatted trig results

    outStr = ""                ;assume positive (no sign)

    if value < 0 then           ;handle negatives
        outStr = "-"           ;prefix a minus sign
        value = -1 * value     ;but work with positives
    end if

    valStr = str(value)
    length = len(valStr)
    select case length

```



```

        case 1:
        outStr = outStr + "0.00" + valStr
        case 2:
        outStr = outStr + "0.0" + valStr
        case 3:
        outStr = outStr + "0." + valStr
        case 4:
        outStr = outStr + left(valStr,1) + "." + right(valStr,3)
        case 5:
        outStr = outStr + left(valStr,2) + "." + right(valStr,3)
    end select
    print outStr
end sub

```

Trigonometry ATAN

Syntax:

```

#include <maths.h>

integer_variable = ATan (_x_vector_,_y_vector_)

```

Explanation:

GCBASIC supports the trigonometric function for ATan.

Details:

GCBASIC supports the following functions ATan(x, y) where x and y are the vectors. The function returns an Integer result representing the angle measured in a whole number of degrees.

The function also returns a global byte variable NegFlag with returns the quadrant of the angle.

```

Quadrant 1 = 0 to 89
Quadrant 2 = 90 to 179
Quadrant 3 = 180 to 269
Quadrant 4 = 270 to 359

```

This ATan function is a fast XY vector to integer degree algorithm developed in Jan 2011, see [www.RomanBlack.com](http://www.romanblack.com/integer_degree.htm) and see http://www.romanblack.com/integer_degree.htm

The function converts any XY vectors including 0 to a degree value that should be within +/- 1 degree of the accurate value without needing large slow trig functions like ArcTan() or ArcCos().

At least one of the X or Y values must be non-zero. This is the full version, for all 4 quadrants and will

generate the angle in integer degrees from 0-360. Any values of X and Y are usable including negative values provided they are between -1456 and 1456 so the 16bit multiply does not overflow.

Peripheral Pin Select

This is the Peripheral Pin Select section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Peripheral Pin Select for Microchip microcontrollers.

Introduction:

Peripheral Pin Select (PPS) enables the digital peripheral I/O pins to be changed to support mapping of external pins to different pins.

In older 8-bit Microchip devices, a peripheral was hard-wired to a specific pin (example: PWM1 output on pin RC5).

PPS allows you to choose from a number of output and input pins to connect to the digital peripheral.

This can be extremely useful for routing circuit boards.

There are cases where a change of I/O position can make a circuit board easier to route. Sometimes mistakes are found too late to fix so having the option to change a pinout mapping in software rather than creating a new printed circuit board can be very helpful.

You **must** use the command **UnLockPPS** to enable setting of the PPS if the PPS have been previously locked, and, you can, optionally, use **LockPPS** to prevent unintentional change to PPS settings.

GCBASIC includes these two macros to ensure this process is handled correctly.

Also, see <http://microchip.wikidot.com/8bit:pps> for more information.

Example:

'Please check configuration before using on an alternative microcontroller.

```
#chip 16f18855,32
```

```
#option explicit
```

'Set the PPS of the I2C and the RS232 ports.

```
#startup InitPPS, 85
```

```
Sub InitPPS
```

```
LOCKPPS
```

```
RC0PPS = 0x0010      'RC0->EUSART:TX;
```

```
RXPPS  = 0x0011      'RC1->EUSART:RX;
```

```
SSP1CLKPPS = 0x14    'RC3->MSSP1:SCL1;
```

```
SSP1DATPPS = 0x13    'RC4->MSSP1:SDA1;
```

```
RC3PPS = 0x15        'RC3->MSSP1:SCL1;
```

```
RC4PPS = 0x14        'RC4->MSSP1:SDA1;
```

```
UnLockPPS
```

```
End Sub
```

For more help, see: [UnlockPPS](#) and [LockPPS](#).

UnLockPPS

Syntax:

```
UNLOCKPPS
```

Explanation:

Peripheral Pin Select (PPS) has an operation mode in which all input and output selections can be prevented to stop inadvertent changes.

PPS selections are unlocked by setting by the use of the [UnLockPPS](#) command.

Using this command will ensure the special sequence of Microchip assembler is handled correctly.

Command Availability:

Available on all Microchip microcontrollers only.

```

#chip 16f18855,32
#option explicit

'Set the PPS of the I2C and the RS232 ports.
#startup InitPPS, 85
Sub InitPPS
  UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;

    SSP1CLKPPS = 0x14    'RC3->MSSP1:SCL1;
    SSP1DATPPS = 0x13    'RC4->MSSP1:SDA1;
    RC3PPS = 0x15        'RC3->MSSP1:SCL1;
    RC4PPS = 0x14        'RC4->MSSP1:SDA1;
  LockPPS
End Sub

```

For more help, see: [LockPPS](#).

LockPPS

Syntax:

```
LOCKPSS
```

Explanation:

Peripheral Pin Select (PPS) has an operation mode in which all input and output selections can be prevented to stop inadvertent changes.

PPS selections are locked by setting by the use of the [LockPPS](#) command.

Using this command will ensure the special sequence of Microchip assembler is handled correctly.

Command Availability:

Available on all Microchip microcontrollers only.

```

#chip 16f18855,32
#option explicit

'Set the PPS of the I2C and the RS232 ports.
#startup InitPPS, 85
Sub InitPPS
  UNLOCKPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;

    SSP1CLKPPS = 0x14    'RC3->MSSP1:SCL1;
    SSP1DATPPS = 0x13    'RC4->MSSP1:SDA1;
    RC3PPS = 0x15        'RC3->MSSP1:SCL1;
    RC4PPS = 0x14        'RC4->MSSP1:SDA1;
  LOCKPPS
End Sub

```

For more help, see: [UnlockPPS](#).

Compiler Directives

This is the Compiler Directives section of the Help file. Please refer the sub-sections for details using the contents/folder view.

#asmraw

Syntax:

```
#asmraw [label]  
#asmraw [Mnemonics | Directives | Macros] [Operands] ['comments']
```

for ASM blocks use

```
#asmraw[  
    [label]  
    [Mnemonics | Directives | Macros] [Operands] ['comments']  
#asmraw]
```

Explanation:

The **#asmraw** directive is used to specify the assembly that GCBASIC will use.

Anything following this directive will be inserted into ASM source file with no changes other than trimming spaces - no replacement of constants.

Assembly is a programming language you may use to develop the source code for your application. The directive must conform to the following basic guidelines. Each line of the source file may contain up to four types of information:

- Labels
- Mnemonics, Directives and Macros
- Operands
- Comments

The order and position of these are important. For ease of debugging, it is recommended that labels start in column one and mnemonics start in column two or beyond. Operands follow the mnemonic.

Comments may follow the operands, mnemonics or labels, and can start in any column. The maximum column width is 255 characters.

White space or a colon must separate the label and the mnemonic, and white space must separate the mnemonic and the operand(s). Multiple operands must be separated by commas.

White space is one or more spaces or tabs. White space is used to separate pieces of a source line. White space should be used to make your code easier for people to read.

Example 1

```
#asmraw lds SysValueCopy,TCCR0B
#asmraw andi SysValueCopy, 0xf8
#asmraw inc SysValueCopy
#asmraw sts TCCR0B, SysValueCopy
```

Example 2

```
#asmraw[
    lds SysValueCopy,TCCR0B
    andi SysValueCopy, 0xf8
    inc SysValueCopy
    sts TCCR0B, SysValueCopy
#asmraw]
```

This example will generate the following in the ASM source file.

```
lds SysValueCopy,TCCR0B
andi SysValueCopy, 0xf8
inc SysValueCopy
sts TCCR0B, SysValueCopy
```

#chip

Syntax:

```
#chip model, frequency

#chip model, frequency / numeric constant
```


Explanation:

The `#chip` directive is used to specify the chip model and frequency that GCBASIC will use.

The `model` is the specific microcontroller - examples are "16F819".

The `frequency` is the frequency of the chip in MHz, and is required for the delay and PWM routines. The following constants simplify setting specific frequencies. `31k`, `32.768K`, `125k`, `250k` or `500k`. Any of these constant can be used. As shown in the example below.

If `frequency` is not present the compiler will select a frequency default frequency that should work for the microcontroller.

If `numeric constant` is specified then the compiler will complete a simple math calculation to determine the frequency. The only supported math instruction is divide.

1. If the chip has an internal oscillator, the compiler will use that and pick the highest frequency it supports.
2. If the chip does not have an internal oscillator, then GCBASIC will assume that the chip is being run at its maximum possible clock frequency using an external crystal.
3. If you are using an external crystal then you must specify a chip frequency.

When using an AVR:

1. There is no need to specify "AT" before the name.
2. Only AVRDX chips support setting the internal frequency using the `frequency` statement.
3. megaAVR assumes an external oscillator, and, therefore the `frequency` must match the external oscillator frequency. For Arduino products this is typically 16Mhz

When using a LGT:

1. Only LGT support setting the internal frequency using the `frequency` statement.

Examples:

```
#chip 12F509, 4
#chip 18F4550, 48
#chip 16F88, 0.125
#chip tiny2313, 1
#chip mega8, 16
#chip 12f1840, 31k
#chip 12f1840, 500k
#chip 12f1840, 250k
#chip 12f1840, 125k

#chip lgt8x328p, 4
#chip tiny3127, 16 / 48
```

'Select the internal low frequency oscillator. The microcontroller must have a low frequency oscillator option. The internal oscillator is automatically selected on PIC.
#chip 16f18326, 31k

'Select the external SOSC clock source.
#chip 16f18855, 32.768k
#config osc=SOSC

Setting Other Clock frequencies: Some alternative compilers allow value of the clock frequency to be set with the numerical value in Hertz (*i.e.* 24576000). This can be useful when using the clock frequencies other than standard frequencies.

GCBASIC requires clock frequencies to be specified in MHz, but will accept decimal points. For example, if you wanted to run a 16F1827 at 24576000 Hz, you would write the following:

```
#chip 16F1827, 24.576
```

GCBASIC support for microcontrollers:

Each microcontroller has a microcontroller data file. This file is located in \GCBasic\chipdata\ folder when installed.

An example is 12F1840.dat

There are two sections in the microcontroller data file that control the "chip frequency", they are:

```
*[ChipData]* and *[ConfigOps]*
```

ChipData section

The ChipData section for 12F1840 microcontroller. The 12F1840 is used as an example

```
[ChipData]
Prog=4096
EEPROM=256
RAM=256
I/O=6
ADC=4
MaxMHz=32
IntOsc=32, 16, 8, 4, 2, 1, 0.5, 0.25, 0.125
31kSupport=INTOSC,OSCCON,2
Pins=8
Family=15
ConfigWords=2
PSP=0
MaxAddress=4095
```

The IntOsc line specifies the supported internal clock frequencies - The 12F1840 microcontroller supports nine internal frequencies (ChipMHz). #Chip is used as follows: The 31kSupport line specifies the chip supports 31k for internal clock frequency.

```
#chip 12F1840, 32
```

A ChipMHz of 32 does two things.

1. When using the internal oscillator, it tells the compiler to set the chip clock frequency (FOSC) to 32MHz
2. It tells the compiler to calculate all delays (wait times) based upon FOSC of 32 MHz. Unlike Picaxe Basic (and other compilers) GCBASIC delays ("wait") are correct regardless of the setting of FOSC. If you set the internal oscillator to 4 MHz a "wait 1 ms" will still be 1 ms.

If you set chipMHz to something other than the valid options in the [ChipData] IntOsc section of the microcontroller specific dat file, then, the compiler assumes that you are using an external oscillator and will calculate the delays according to the value you use. The wait times will be incorrect if you are not using an external oscillator at the same frequency as ChipMhz.

```
Example: #chip 12F1840, 12
```

Since "12" is not a valid internal osc frequency, the microcontroller FOSC will default to 8 MHz because there is no external crystal installed. However, the wait times will be incorrect as they will be calculated by the compiler based upon a 12 Mhz clock.

ConfigOps section

The [ConfigOps] section of 12F1840.dat is towards the end of the chip data file. For the 12F1840 it looks

like this

```
[ConfigOps]
OSC=LP,XT,HS,EXTRC,INTOSC,ECL,ECM,ECH
WDTE=OFF,SWDTEN,NSLEEP,ON
PWRTE=ON,OFF
MCLRE=OFF,ON
CP=ON,OFF
CPD=ON,OFF
BOREN=OFF,SBODEN,NSLEEP,ON
CLKOUTEN=ON,OFF
IESO=OFF,ON
FCMEN=OFF,ON
WRT=ALL,HALF,BOOT,OFF
Pllen=OFF,ON
STVREN=OFF,ON
BORV=HI,LO,19
LVP=OFF,ON
```

OSC specifies which oscillator options are available for the specific microcontroller. **INTOSC** is the internal oscillator. All others are some form of external clock source. **Pllen** sets the internal Phase Lock Loop either on or off. With this chip the default clock frequency is 8 Mhz. The PLL multiplies this by 4. So to get 32 Mhz the basic internal oscillator will be 8 Mhz then multiplied by 4. For 16 MHz it will be 4 multiplied by 2.

GCB sets the PLL automatically, so this option should generally be left alone. If Pllen is set to ON, then GCB may not be able to set the correct frequency of the internal oscillator. Only set PLL = ON if you know what you are doing.

It is a good practice to set the oscillator source in #config at the beginning of your code when you are not using the internal oscillator. This prevents potential errors. Example:

```
#Chip 12F1840, 16
#Config OSC = INTOSC 'This is normally not required as the internal oscillator is
the default oscillator.
```

In this example above, GCBASIC will automatically set the necessary OSC bits for the microcontroller. Frequency bits will be set to 4 MHz and the PLL will be turned on and wait times will be calculated on an FOSC of 16.

You can set the clock to other frequencies but you have to put the PIC into **EC** or **External Clock** mode and then supply that specific clock frequency to the OSC1 pin.

There are three EC modes on the PIC12F1840:

```
ECL - 0 MHz - 0.5 MHz  
ECM - 0.5 MHz - 4 MHz  
ECH = 4 MHz - 32MHz
```

Example: For a 2.1 MHz clock you would need to set the #config and the clock frequency, and, provide the OSC1 pin with a 2.1 MHz signal.

```
#chip 12f1840,2.1  
#config OSC = ECM
```

Notes

When "#config osc=" is not specified in the source code, most microcontrollers will default to an external oscillator source. This means at runtime the chip is expecting an external clock signal. If the external clock signal is not present, the chip detects a "failure" of the external clock and will "falls back" to the default internal oscillator setting.

The PLLEN bit defaults to OFF. The PLL is enabled depending upon the ChipMhz in #Chip xxxxxx, ChipMhz.

The GCBASIC defaults - This is how the bits are set if there is no #config in the source code, GCBASIC does set certain bits. To examine what bits are set on a particular chip you can omit #config in the source code, thenm compile the code and then use "Open ASM" in the GCBASIC IDE. The bits that are set will be in the config section. All other bits (those not specifically set) with #Config will be at the POR setting as described below, The **POR** settings are shown in the datasheet for each microcontroller.

Currently GCBASIC sets the **LVP** bit **OFF** by default on many chips. This does not affect normal HV programming like a with a PicKit3. The default of LVP = OFF will prevent the microcontroller from being programmed with Low Voltage Programmer. This means that if a PIC microcontroller has previously been programmed with with "LVP = OFF", then it must be erased or reprogrammed with LVP = ON using a HVP programmer prior to using certain programming devices e.g. Curiosity development boards, or "NS" programmers as these required that LVP = ON.

When LVP = ON, the MCLR pin is automatically set to EXTERNAL MCLR. This means that the MCLRE pin CANNOT be sue for general purpose I/O functions.

The native **POR** (Power On Reset) defaults. This is the state of the config bits after Power on if the ASM code has no configuration entries or on a blank factory chip. The only way to power up in this state with GCB code is to use " #option NoConfig" in the GCBASIC source code.

#config

Syntax:

```
#config option1, option2, ... , optionN
```

Explanation:

The `#config` directive is used to specify configuration options for the chip. There is a detailed explanation of `#config` in the Configuration section of help.

See Also [Configuration](#)

#DEFINE

Syntax:

```
#DEFINE NAME [String]

or

#DEFINE NAME [ = String]
```

Explanation:

`#DEFINE` allows to declare string-based preprocessor constants.

This directive defines a text substitution string. Wherever `name` is encountered in the program or assembly code, `string` will be substituted.

The expansion is done recursively, until there is nothing more to expand and the compiler can continue analyzing the resulting code.

The use of an existing constant is supported. The order of constants when using an existing constant is strict. The constant must exist prior to use.

To ensure evaluation of a calculation. A ``=`` must be used. This is strict.

`#UNDEFINE` can be used to make the compiler delete an existing constant.

Using the directive with no `string` causes a definition of `name` to be noted internally and may be tested for using the `ifdef` directive/conditional processing. See the examples below.

Constants defined with this method are available for viewing in the CDF file. Creation of the CDF file is controlled with the Programmer Preferences Utility.

GCBasic does not support creation of SUBroutine or FUNCTION with this directive.

Examples:

This progress shows the creation of constants, the processing and showing of constants within a script, creation of specific constants within a script and use of constants within a program.

```
#chip MEGA4809
#option Explicit

// Numeric constants
#define LENGTH 20
#define CONTROL 0x19,7
#define SINGLEPI = 22/7 // evaluated
numeric string
#define INTPI = INT(22/7) // evaluated
numeric string
#define FACTOREDPI = INT((22/7 - INT(22/7))*1000) // evaluated
numeric string
#define LENGTHSQUARED = LENGTH * LENGTH // evaluated
numeric string
#define PI 3.142
#define RADIUS 10
#define CIRCUMFERENCECALC = PI * RADIUS // evaluated
numeric string, with constant substitution
#define FACTORISEDCIRCUMFERENCE = INT(CIRCUMFERENCECALC*100) // evaluated
numeric string, with constant substitution

//String(s) required double quotes and NO '=' assignment
#define MYSTRING "This is a string"
// A string assignment is not required
#define ACONSTANTTHATEXISTS

// Macros are not supported.. just define the sub!
// #define BADPOSITION(XX,YY,ZZ) (YY-(2 * ZZ +XX))

// Unused Constants that are invalid may not report an error until try to use within
your program- as in this example
// String assignment with an equal sign will fail. Do not use '='
#define BADMYSTRING = "This is a string"

#script
//Scripts can modify and manage CONSTANTS

// Warning can be used to show values during compilation
WARNING LENGTH
WARNING CONTROL
WARNING SINGLEPI
```

```

WARNING INTPI
WARNING FACTOREDPI
WARNING MYSTRING

If DEF(ACONSTANTTHATEXISTS) Then
    WARNING "The constant ACONSTANTTHATEXISTS exists"
End If

// Test for a value in the [ChipData] section of the DAT file. Always has the
prefix CHIP
If DEF(CHIPAVRDX) Then
    WARNING "This is an AVRDX chip"
End If

// Good practice constant testing, see the code below for BAD and GOOD practice

// Set the constant to 0 we can use this to test for validity
// Use prefix of SCRIPT as this is clear in the program
SCRIPTAN1CONSTANT = 0
If DEF(AVRDX) Then
    // Is this an AVRdx chip
    If DEF(AIN1) Then
        SCRIPTAN1CONSTANT = AIN1
    End If
End If
If NODEF(AVRDX) Then
    // This is NOT an Avr dx chip
    If DEF(AVR) Then
        // This is an AVR
        SCRIPTAN1CONSTANT = AN1
    End If
    If DEF(PIC) Then
        // This is an AVR
        SCRIPTAN1CONSTANT = ANA1
    End If
End If
// Now test for validate result
If SCRIPTAN1CONSTANT = 0 Then
    WARNING Script has determined that no valid ADC port exists, or, some
other message
End If

#endscript

// Some conditional examples

#IF DEF( CONSTANTTHATEXISTS )
    //! Cause a compiler error - as the constant exists. Remove comment to test

```



```

#ELSE
    //! Cause a compiler error - as the constant does not exists. Remove comment to
test
#ENDIF

#IFDEF
Oneof(CHIP_18F24K40,CHIP_18F25K40,CHIP_18F26K40,CHIP_18F27K40,CHIP_18F45K40,CHIP_18F46K40
,CHIP_18F47K40,CHIP_18F65K40,CHIP_18F66K40,CHIP_18LF24K40, CHIP_18LF25K40,
CHIP_18LF26K40, CHIP_18LF27K40, CHIP_18LF45K40, CHIP_18LF46K40, CHIP_18LF47K40,
CHIP_18F65K40, CHIP_18LF65K40, CHIP_18F66K40, CHIP_18LF66K40, CHIP_18F67K40,
CHIP_18LF67K40 )
    //~ Do something
#ENDIF

dim myStringVar as String
myStringVar = MYSTRING

// BAD PRACTICE = code is hard to understand
// Use constant test to determine correct ADC to read. Bad practice, see the
#SCRIPT section
dim mybyteVar as Byte
#IF DEF(AIN1)
    mybyteVar = readAD( AIN1 )
#ELSE
    #IF DEF(ANA1)
        mybyteVar = readAD( ANA1 )
    #ELSE
        mybyteVar = readAD( AN1 )
    #ENDIF
#ENDIF

// GOOD PRACTICE
dim mybyteVar as Byte
mybyteVar = readAD( SCRIPTAN1CONSTANT )

dim myArray(2)
myArray = CONTROL

```

See Also [DEFINES](#)

#UNDEFINE

Syntax:

```
#UNDEFINE existing-symbol
```

Explanation:

#UNDEFINE undefines a symbol previously defined with **#DEFINE**.

Can be used to ensure that a symbol has a limited lifespan and does not conflict with a similar macro definition that may be defined later in the source code.

(Note: **#UNDEFINE** should not be used to undefine variable or function names used in the current program. The names are needed internally by the compiler and removing them can cause strange and unexpected results.)

See Also [Defines](#)

#if

Syntax:

```
#if Condition
...
[#else]
...
#endif
```

Explanation:

The **#if** directive is used to prevent a section of code from compiling unless **Condition** is true.

Condition has the same syntax as the condition in a normal GCBASIC if command. The only difference is that it uses constants instead of variables and does not use "then".

Example:

```

'This program will pulse an adjustable number of pins on PORTB
'The number of pins is controlled by the FlashPins constant
#chip 16F88, 8

'The number of pins to flash
#define FlashPins 2

'Initialise
Dir PORTB Out

'Main loop
Do
    #if FlashPins >= 1
        PulseOut PORTB.0, 250 ms
    #endif
    #if FlashPins >= 2
        PulseOut PORTB.1, 250 ms
    #endif
    #if FlashPins >= 3
        PulseOut PORTB.2, 250 ms
    #endif
    #if FlashPins >= 4
        PulseOut PORTB.3, 250 ms
    #endif
Loop

```

#ifnot

Syntax:

```

#ifnot Condition
    ...
[#else]
    ...
#endif

```

Explanation:

The **#ifnot** directive is used to prevent a section of code from compiling unless **Condition** is false.

Condition has the same syntax as the condition in a normal GCBASIC if command. The only difference is that it uses constants instead of variables and does not use "then".

Example:

```
'This program will set the constant to true only if NOT a PIC family
#chip 16F88, 8

#ifnot ChipFamily = 14

    #define myConstant True

#endif
```

#ifdef

Syntax:

```
#ifdef Constant | Constant Value | Var(VariableName)
...
[#else]
...
#endif
```

Explanation:

The **#ifdef** directive is used to selectively enable sections of code.

There are several ways in which it can be used:

- Checking if a constant is defined
- Checking if a constant is defined and has a particular value
- Checking if a system variable exists
- Checking if a system bit has been defined

The advantage of using **#ifdef** rather than an equivalent series of **IF** statements is the amount of code that is downloaded to the chip. **#ifdef** controls what code is compiled and downloaded, **IF** controls what is run once on the chip. **#ifdef** should be used whenever the value of a constant is to be checked.

GCBASIC also supports the **#ifndef** directive - this is the opposite of the **#ifdef** directive - it will remove code that **#ifdef** leaves, and vice versa.

Note: The code in the following sections will not compile, as it is missing **#chip** directives and **Dir** commands. It is intended to act as an example only.

Example 1: *Enabling code if a constant is defined*

```

#define Blink1

#ifdef Blink1
    PulseOut PORTB.0, 1 sec
    Wait 1 sec
#endif
#ifdef Blink2
    PulseOut PORTB.1, 1 sec
    Wait 1 sec
#endif

```

This code will pulse `PORTB.0`, but not `PORTB.1`. This is because `Blink1` has been defined, but `Blink2` has not. If the line was added at the start of the program, then both pins would be pulsed.

```

#define Blink2

```

The value of the constant defined is not important and can be left off of the `#define`.

Example 2: *Enabling code if a constant is defined and has a given value*

```

#define PinsToFlash 2

#ifdef PinsToFlash 1,2,3
    PulseOut PORTB.0, 1 sec
#endif
#ifdef PinsToFlash 2,3
    PulseOut PORTB.1, 1 sec
#endif
#ifdef PinsToFlash 3
    PulseOut PORTB.2, 1 sec
#endif

```

This program uses a constant called `PinsToFlash` that controls how many lights are pulsed. `PORTB.0` is pulsed when `PinsToFlash` is equal to 1, 2 or 3, `PORTB.1` is pulsed when `PinsToFlash` equals 2 or 3, and `PORTB.2` is flashed when `PinsToFlash` is 3.

Example 3: *Enabling code if a system variable is defined*

```

#ifdef NoVar(ANSEL)
    SET ADCON1.PCFG3 OFF
    SET ADCON1.PCFG2 ON
    SET ADCON1.PCFG1 ON
    SET ADCON1.PCFG0 OFF
#endif
#ifdef Var(ANSEL)
    ANSEL = 0
#endif

```

The above section of code has been copied directly from a-d.h. It is used to disable the A/D function of pins, so that they can be used as standard digital I/O ports. If **ANSEL** is not declared as a system variable for a particular chip, then the program uses **ADCON1** to control the port modes. If **ANSEL** is defined, then the chip is newer and its ports can be set to digital by clearing **ANSEL**.

Example 4: *Enabling code if a system bit is defined*

Similar to above, except with **Bit** and **NoBit** in the place of **Var** and **NoVar** respectively.

See Also [Defines](#), [#define](#)

#ifndef

Syntax:

```

#ifndef Constant | Constant Value | Var(VariableName)
...
[#else]
...
#endif

```

Explanation:

The **#ifndef** directive is used to selectively enable sections of code. It is the opposite of the **#ifdef** directive - it will delete code in cases where **#ifdef** would leave it, and will leave code where **#ifdef** would delete it.

See the [#ifdef](#) article for more information.

#include

Syntax:

```
#include filename
```

Explanation:

#include tells GCBASIC to open up another file, read all of the subroutines and constants from it, and then copy them into the current program.

There are two forms of include; absolute and relative.

Absolute is used to refer to files in the `..\GCBASIC\include` directory. The name of the file is specified in between < and > symbols. For instance, to include the file `srf04.h`, the directive is:

```
#include <srf04.h>
```

Relative is used to read files in the same folder as the currently selected program. Filenames are given enclosed in quotation marks, such as where `mycode.h` is the name of the file that is to be read.

```
#include "mycode.h"
```

NOTES: It is not essential that the include file name ends in `.h` - the important thing is that the name given to GCBASIC is the exact name of the file to be included.

Those who are familiar with **#include** in assembly or C should bear in mind that **#include** in GCBASIC works differently to **#include** in most other languages - code is not inserted at the location of the **#include**, but rather at the end of the current program.

#ignorecompile

Syntax:

```
#ignorecompile
```

Explanation:

The **#ignorecompile** directive is used to control whether a source file can be compiled.

This can be used to prevent libraries and other source files from being processed as a source file for compilation.

This therefore will prevent creation of all the GCBASIC outputs files in the folder.

#insert

Syntax:

```
#insert filename
```

Explanation:

#insert tells GCBASIC to open up another file, read all of the subroutines and constants from it, and then copy them into the current program at the specific line where the **#insert** directive is located.

There are two forms of include; absolute and relative.

Absolute is used to refer to files in the `..\GCBASIC\include` directory. The name of the file is specified in between < and > symbols. For instance, to include the file `toolchain.il`, the directive is:

```
#insert <"toolchain.il">
```

Relative is used to read files in the same folder as the currently selected program. Filenames are given enclosed in quotation marks, such as where `mycode.h` is the name of the file that is to be read.

```
#insert "toolchain.il"
```

Difference from #include:

This is very different from **#include**. With **#include** you can organize constant, method and macro definitions and then use **#include** directive to add them to any source file. Include files are also useful for incorporating declarations of external variables and complex data types. The types may be defined and named only once in an include file created for that purpose. The compiler will optimise the include files to determine the best order/location in your program.

Using **#insert** you are determining the location of the code segment. It will be inserted exactly where you specify. The optimisation will only be applied to any methods that you insert but the rest of the code essentially exits at the point of insertion.

#Insert does not support Conversion:

There is no conversion of the inserted file. For conversion use **#Include**.

If you need to convert a file from an external source then see the Converters section of the Help.

Usage Notes:

The file must exist. An error message is issued if not found. When an error is encountered in the

inserted file the error line number is in the format of xxxxyyyy. Where xxxx is the code line number in the user program and the yyyy is the the line number in the inserted file.

An example error message. Where the source insert instruction is on line 6 and the error in the inserted file is on line 4.

```
An error has been found:
insertexample.gcb (60004): Error: Syntax Error
The message has been logged to the file Errors.txt.
```

#script

Syntax:

```
#script
[scriptcommand1]
[scriptcommand2]
...
[scriptcommandn]
#endscript
```

Explanation:

The **#script** block is used to create small sections of code which GCBASIC runs during compilation. A detail explanation and example are included in the Scripts article.

See Also [Scripts](#)

#startup

Syntax:

```
#startup SubName [priority]
```

Explanation:

#startup is used in include files to automatically insert initialization routines. If a define or subroutine from the file is used in the program, then the specified subroutine will be called.

The **priority** to #startup support the setting of the priority of the subroutines for all the libraries in a project.

Subroutines will be called in order from smallest to largest priority number.

InitSys has priority 80, lowlevel communication routines have the priority of 90
All other subroutines defaults to 100.

Notes: Limitations on this directive are:

startup may only occur once within a source file.

No parameters can be passed the the subroutine that is specificed.

Example 1:

This example from the hardware I2C library set the subroutine with the priority of 90.

```
#startup HIC2Init, 90
```

Example 2:

This example from would be included in user code to ensure the PPS setting are set prior to use of the MSSP or USART.

```
#chip 16f18855,32
#option explicit

'Set the PPS of the I2C and the RS232 ports.
#startup InitPPS, 85
Sub InitPPS
    RC0PPS = 0x0010      'RC0->EUSART:TX;
    RXPPS  = 0x0011      'RC1->EUSART:RX;

    SSP1CLKPPS = 0x14    'RC3->MSSP1:SCL1;
    SSP1DATPPS = 0x13    'RC4->MSSP1:SDA1;
    RC3PPS = 0x15        'RC3->MSSP1:SCL1;
    RC4PPS = 0x14        'RC4->MSSP1:SDA1;
End Sub
```

#mem

This directive is obsolete.

GCBASIC determines the amount of memory on a chip automatically, and will ignore the **#mem** directive.

It is recommended that this directive is removed from all programs.

Enum

Overview

Enums in GCBASIC provide a convenient way to define named constants, improving code readability and maintainability. Instead of using raw numeric values, users can define meaningful names for states, modes, or categories.

Syntax:

The **Enum** keyword allows users to declare an enumeration with automatic assignment of values:

```
Enum ModbusState [Reset]
    MODBUS_IDLE      ' Waiting for a new message
    MODBUS_SYNC      ' Sync with Modbus device address
    MODBUS_FUNCTION_CODE ' Identify function code
    MODBUS_PROCESS_DATA ' Read incoming data bytes
    MODBUS_COMPLETE   ' Packet processing finished
End Enum
```

The optional **Reset** parameter will start this specific enum at zero. If **Reset** is not specified then every enum element will be sequentially/uniquely numbered.

Enhanced Compatibility with External Systems

Many protocols, such as Modbus, require specific numeric values for states or function codes. By letting users define their own values, they can align their enums with standardised protocols. The **Enum** keyword also allows users to declare an enumeration with named constants:

```
Enum ModbusState
    MODBUS_IDLE = 1
    MODBUS_SYNC = 0x02
    MODBUS_FUNCTION_CODE = 0x03
    MODBUS_PROCESS_DATA = MODBUS_FUNCTION_CODE + 1
    MODBUS_COMPLETE = INT((30/2)+1)
End Enum
```

Users can manually assign enumeration values using constants and expressions. **Calculations must use the equals sign (=), following #DEFINE rules.**

Usage

Users can declare variables of an enum type and assign values directly:

```
Dim currentState As Byte
currentState = MODBUS_IDLE
```

Enums can be used in conditions for more readable logic:

```
If currentState = MODBUS_FUNCTION_CODE Then
    ' Handle function code processing
End If
```

Users can also define additional Enum values using constants:

```
#DEFINE CUSTOM_STATE = MODBUS_PROCESS_DATA + 2
If currentState = CUSTOM_STATE Then
    Print "Custom processing step reached."
End If
```

Conversion Support

Enums in GCBASIC can be used in mathematical operations and references to other constants. This allows developers to integrate enums seamlessly into calculations and logical expressions.

Math Operations

Users can use enums in arithmetic expressions since they resolve to numeric values:

```
Dim nextState As Byte
nextState = MODBUS_SYNC + 1 ' Evaluates to MODBUS_FUNCTION_CODE
```

Enums can also be used in expressions involving comparisons:

```
If currentState < MODBUS_COMPLETE Then
    ' Keep processing data
End If
```

Reference to Other Constants

Since enum values are numeric, users can reference them in calculations that involve predefined constants:

```
#DEFINE MAX_STATE = MODBUS_COMPLETE
If currentState = MAX_STATE Then
    Print "Processing complete."
End If
```

Summary

- Users can use automatic enumeration values.
- Users can manually assign enumeration values using constants and calculations.
- Calculations must use the equals sign (=), following GCBASIC's **#DEFINE** rules.
- Enums allow users to define named constants.
- Values start at 0 and increment automatically if not manually assigned.
- Enums improve readability and eliminate the need for magic numbers.
- Enums can be used in mathematical operations and references to constants.

Other directives

The built-in **#defines** are used to support the **#IFDEF** command set are as follows. The table also shows which **#defines** are supported as string in HSerPrint, SerPrint and other string related commands.

Constants	Type	Usage	Decription
CHIPADC	Constant	Conditional compilation or output commands	The number of A/D inputs on the current chip
CHIPASSEMBLER	Constant	Conditional compilation or output commands	The select assemble GCASM/MPASM/PICAS etc
CHIPLEEPROM	Constant	Conditional compilation or output commands	The number of Bytes in EEPROM memmory
CHIPIO	Constant	Conditional compilation or output commands	The number of general purpose IO pins
CHIPMHZ	Constant	Conditional compilation or output commands	The microcontroller clock speed
CHIPNAME	Constant	Conditional compilation only	The microcontroller type
CHIPNAMESTR	Constant	Conditional compilation or output commands	The microcontroller name
CHIPPINS	Constant	Conditional compilation or output commands	The number of microcontroller pins.

Constants	Type	Usage	Decription
CHIPRESERVE HIGHPROG	Constant	Scripts, Conditional compilation, and output commands	The value of the words reserved
CHIPOSC	Constant	Scripts, Conditional compilation, and output commands	The frequency selected
CHIPUSINGIN TOSC	Constant	Scripts, Conditional compilation, and output commands	The constant exists if the compiler has determened the program is using the internal oscillator
CHIPPROGRAM MERNAMESTR	String constant	Name of the chip type to be used by a programmer	The psuedo microcontroller type
CHIPRAM	Constant	Conditional compilation or output commands	The RAM size
CHIPSHARED RAM	Constant	Conditional compilation or output commands	The first RAM location
CHIPFAMILY	Constant	Conditional compilation or output commands	See the table below
CHIPWORDS	Constant	Conditional compilation or output commands	The number of WORDS in Flash memmory
SOURCEFILE	Constant string	Conditional compilation or output commands	The name of the source GCB file
Function	Type	Usage	Description
Var()	Function	Conditional compilation only	True if a register is declared (or false if not declared) in the currently specficed microcontroller's .dat file. Var(register_name)
NoVar()	Function	Conditional compilation only	True if a register is NOT declared (or false if declared) in the currently specficed microcontroller's .dat file. NoVar(register_name)
Bit()	Function	Conditional compilation only	True if a bit is declared (or false if not declared) in the currently specficed microcontroller's .dat file. Bit(bit_name)
NoBit()	Function	Conditional compilation only	True if a bit is NOT declared (or false if declared) in the currently specficed microcontroller's .dat file. NoBit(bit_name)
Allof()	Function	Conditional compilation only	True if all defines are declared: Allof(define1, define2, ...)

Constants	Type	Usage	Decription
<code>OneOf()</code>	Function	Conditional compilation only	True if one of the defines is declared: <code>OneOf(define1, define2, ...)</code>

The table below shows two special directives that support the mapping for one variable or bit to another variable or bit. This is useful when creating portable code or libraries to ensure GCBASIC

Directive	Explanation	Usage
<code>#samebit</code>	The compiler checks each item in the list to see which ones are implemented on the current microcontroller. If any of the bits do not exist, the compiler will create a constant mapping to the name of the first parameter in the list of parameters that does exist. + If none of the bits exist the no constant is created.	<code>#samebit PLEN, SPLLEN, SPLLMULT</code> Set SPLLEN On
<code>#samevar</code>	The compiler checks each item in the list to see which ones are implemented on the current microcontroller. If any of the variables do not exist, the compiler will create a constant mapping to the name of the first parameter in the list of parameters that does exist. + If none of the variables exist the no constant is created.	<code>#samevar CMCON, CMCON0, CMCONbob</code> <code>#ifdef Var(CMCONbob)</code> <code>CMCONbob = 7</code> <code>#endif</code> Compiles to: <code>;CMCONbob = 7</code> <code>movlw 7</code> <code>movwf CMCON,ACCESS</code>

This table shows the ChipFamily constants mapped to the microcontroller architecture.

ChipFamily Value	
AVR	Microcontroller Characteristics
100	AVR core version V0E class microcontrollers
110	AVR core version V1E class microcontrollers
120	AVR core version V2E class microcontrollers

ChipFamily Value	
-120 Subtype: 121	AVR core version AVR8L, also called AVRrc, reduced core class microcontrollers. ATtiny4-5-9-10 and ATtiny102-104 with only 16 GPR's from r16-r31 and only 54 instructions.
-120 Subtype: 122	LGT microcontrollers.
-120 Subtype: 123	AVR core version V2E class microcontrollers with one USART like the mega32u4, mega16u4 - they have different registers for the usart.
121	Tiny4-5-9-10 and tiny102-104. Only 16 GPR's from r16-r31 and only 54 instructions.
130	AVR core version V3E class microcontrollers but essentially the mega32u6 only
140	AVRDX microcontrollers. Series 0, series1, series2, DA series and DB series.
PIC	Microcontroller Characteristics
12	Baseline devices. 12 Bit instruction set
15	Mid-range core devices. 14 Bit instruction set with enhanced instruction set class
15 plus familyVariant=1	Mid-range core devices. 14 Bit instruction set with enhanced instruction set and with large memory capability class
16	High end core devices. 16 Bit instruction set, memory addressing architecture and an extended instruction set. Chip family 16 also have a sub chip family Constant. These constants are shown below: ChipFamily18FxxQ10 = 16100 ChipFamily18FxxQ43 = 16101 ChipFamily18FxxQ41 = 16102 ChipFamily18FxxK42 = 16103 ChipFamily18FxxK40 = 16104 ChipFamily18FxxQ40 = 16105 ChipFamily18FxxQ84 = 16106 ChipFamily18FxxK83 = 16107 ChipFamily18FxxQ83 = 16108 ChipFamily18FxxQ71 = 16109 ChipFamily18FxxQ20 = 16110 ChipFamily18FxxQ24 = 16111

Compiler Options

This is the Compiler Options section of the Help file. Please refer the sub-sections for details using the contents/folder view.

#Option Explicit

Syntax:

```
#option explicit
```

This option ensures that all variables are dimensioned in the user program. The scope is the user code only and no other code space like .h or include files.

#option explicit requires all variables, including bytes, in the user program to be defined.

Variables can be defined and not used within your user program. Unused variables will not allocate memory.

Introduction:

Example:

```
'Set chip model
#chip 16f877a

'Example command
#option explicit

dim myuserflag as byte

myuserflag = true
```

For more help, see [Variable Lifecycle](#)

#Option NoConfig

Syntax:

```
#option NoConfig
```

This option will prevent the generated assembler from generating _Config items.

`#option NoConfig` is used when using a bootloader.

Introduction:

Example:

```
'Set chip model
#chip 16f877a

'Example command
#option NoConfig

'User Code.....
```

#Option Bootloader

Syntax:

```
#option bootloader address
```

Explanation:

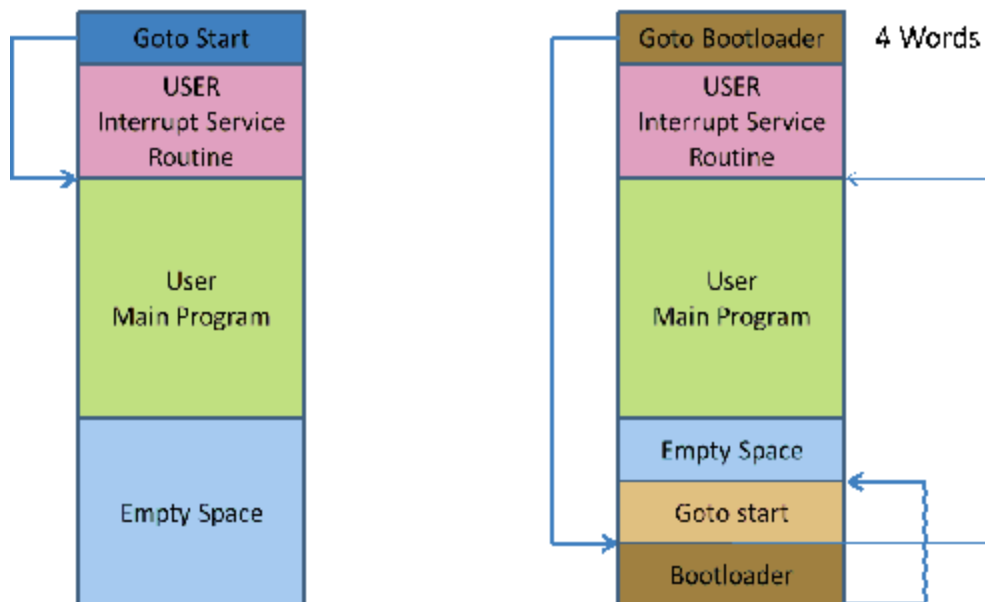
`#option bootloader` prevents the overwriting of any pre-loaded bootloader code, vectors, etc. below the specified address. The GCBASIC code will start at specified `address`.

A bootloader is a program that stays in the microcontroller and communicates with the PC, typically through the serial interface. The bootloader receives a user program from the PC and writes it in the flash memory, then launches this program in execution. Bootloaders can only be used with those microcontrollers that can write their flash memory through software.

The bootloader itself must be written into the flash memory with an external programmer.

In order for the bootloader to be launched after each reset, a `goto bootloader` instruction must exist somewhere in the first 4 instructions; There are two types of bootloaders, some that require that the user reallocate the code and others that by themselves reallocate the first 4 instructions of the user program to another location and execute them when the bootloader exits.

The diagram below shows the architecture of a bootloader. The left hand is the operation of the instructions without a bootloader. The right hand shows the initial instruction of goto the bootoader, then, when the bootloader has initialised the execution of the start code.



See [example bootloader software](#).

Example:

```
#option bootloader 0x800
```

#Option NoContextSave

Syntax:

```
#option NoContextSave
```

Explanation:

Interrupts can occur at almost any time, and may interrupt another command as it runs. To ensure that the interrupted command can continue properly after the interrupt, some temporary variables (the context) must be saved. Normally GCBASIC will do this automatically, but in some cases it may be necessary to prevent this. If porting some existing assembly code to GCBASIC, or creating a bootloader using GCBASIC that will call another program,

NoContextSave can be used to prevent the context saving code from being added automatically.

Be very careful using this option - it is very easy to cause random corruption of variables. If creating your own context saving code, you may need to save several variables. These are:

1. For Microchip PIC microcontrollers 12F/16F: W, STATUS, PCLATH
2. For Microchip PIC microcontrollers 12F1/16F1/18F: W, STATUS, PCLATH, PCLATU, BSR

For Atmel AVR microcontrollers: All 32 registers

Other variables may also need to be saved, depending on what commands are used inside the interrupt handler. Everything that is saved will also need to be restored manually when the interrupt handler finishes.

Example:

```
' This shows an example that could be used by a bootloader to call some application
code.

' The application code must deal with context save and restore
' Suppose that application code starts at location 0x100, with interrupt vector at
0x108

'Chip model
#chip 18F2620

'Do not save context automatically
#option NoContextSave

'Main bootloader routine
Set PORTB.0 On
'Do other stuff to make this an actual bootloader and not a trivial example
'Transfer control to application code
goto 0x100

'Interrupt routine - this will be placed at the interrupt vector
Sub Interrupt
    'If any interrupt occurs, jump straight to application interrupt vector
    goto 0x108
End Sub
```

#Option NoLatch

Syntax:

```
#option nolatch
```

This option disables PORTx to LATx redirection.

Introduction:

The GCBASIC compiler will redirect all I/O pin writes from PORTx to LATx registers on 16F1/18F Microchip PIC microcontrollers.

The Microchip PIC mid-range microcontrollers use a sequence known as **Read-Modify-Write** (RMW) when changing an output state (1 or 0) on a pin. This can cause unexpected behavior under certain circumstances.

When your program changes the state on a specific pin, for example RB0 in PORTB, the microcontroller first **READS** all 8 bits of the PORTB register which represents the states of all 8 pins in PORTB (RB7-RB0).

The microcontroller then stores this data in the MCU. The bit associated with RB that you've commanded to **MODIFY** is changed, and then the microcontrollers **WRITES** all 8 bits (RB7- RB0) back to the PORTB register.

During the first reading of the PORT register, you will be reading the actual state of the physical pin. The problem arises when an output pin is loaded in such a way that its logic state is affected by the load. Instances of such loads are LEDs without current-limiting resistors or loads with high capacitance or inductance.

For example, if a capacitor is attached between pin and ground, it will take a short while to charge when the pin is set to 1. On the other hand, if the capacitor is discharged, it acts like a short circuit, forcing the pin to '0' state, and, therefore, a read of the PORT register will return 0, even though we wrote a 1 to it.

GCBASIC resolves this issue using the LATx register when writing to ports, rather than using PORTx registers. Writing to a LATx register is equivalent to writing to a PORTx register, but readings from LATx registers return the data value held in the port latch, regardless of the state of the actual pin. So, for reading use PORTx.

Note:

You can use the `#option nolatch` if problems occur with compiler redirection.

#Option Required

Syntax:

```
#option REQUIRED PIC|AVR CONSTANT %message.dat entry%  
#option REQUIRED PIC|AVR CONSTANT "Message string"  
  
or  
  
#option REQUIRED DISABLE
```

This option ensure that the specific CONSTANT exists within a library to ensure a specific capability is available with the microcontroller.

Introduction:

This will cause the compiler check the `CONSTANT` is a non zero value. If the `CONSTANT` does not exist it will be treated as a zero value.

Example:

This example tests the `CONSTANT CHIPUSART` for both the PIC and AVR microcontrollers. If the `CONSTANT` is zero or does not exist then the string will be displayed as an error message.

```
#option REQUIRED PIC CHIPUSART "Hardware Serial operations. Remove USART commands to  
resolve errors."  
#option REQUIRED AVR CHIPUSART "Hardware Serial operations. Remove USART commands to  
resolve errors."
```

Disabling:

To disable checking capability, add the following directive.

```
#option REQUIRED DISABLE
```

#Option Shadowregister

Syntax:

```
#option Shadowregister GPIO|PORTA|PORTB
```

This option enables RMW-safe output operations for the specified port by using a shadow latch to track intended output states, preventing unintended changes to input pin states.

Introduction:

Without this option, instructions like `bsf GPIO,2` on chips without `LATx` registers (e.g., PIC12F675) trigger a read-modify-write (RMW) operation on the port register. If any pins are configured as inputs, their bits reflect external voltage levels during the read, which can corrupt output latches when writing back. This can cause unexpected output levels, corrupted latch states during interrupts, or hard-to-trace bugs in mixed input/output setups.

When `#option Shadowregister port` is enabled, the compiler uses a shadow latch variable to track output states. All pin changes update the shadow latch, and writes to the port are done using full-byte transfers from the shadow latch, avoiding RMW issues.

Example:

This example enables the shadow register for **GPIO** on a PIC12F675, ensuring RMW-safe output for **GP2 = 1**.

```
#chip 12F675, 4
#option Shadowregister GPIO

Dir GP0 In
GP2 = 1
```

This compiles to RMW-safe code:

```
; Set GP2 high without affecting GP0
bsf    LATA, 2      ; Update shadow latch
movf   LATA, W
movwf  GPIO         ; Safe write to port
```

#Option Volatile

Syntax:

```
#option volatile `bit`
```

This option ensure port setting are glitch-free.

Introduction:

#option volatile bit where bit is an IO bit, like PORTB.0 appended.

This will cause the compiler to set the bit without any glitches when copying a value from another variable, but will increase code size slightly.

Example:

```
'Set chip model
#chip 16f877a

'Example command
#option volatile portb.0

dir portb.0 out

do forever

    portb.0 = !portb.0

loop
```

#Option ReserveHighProg

Syntax:

```
#option ReserveHighProg [words]
```

This option reserves program memory to be kept free at the top end of memory. This useful for HEF/SAF or bootloaders.

The option provided a reservation for the memory region that is normally assumed to be available to the compiler for the application code storage. In order to avoid any possible conflict (overlapping code and data usage), it is important to reserve the devices pecific memory range by using the compiler option (shown above) in the project configuration.

Using the `#option ReserveHighProg [words]` exposes the constant `'ChipReserveHighProg` in the user program.

Defined constants

The compiler has constants that can be used as an alternative to the parameter `[words]`.

The compiler constants are: `OPTIBOOT`, `OPTIBOOTUSB`, `ARDUINONANO`, `ARDUINOMEGA2560` or `TINYBOOTLOADER`.

Where these constants equate to:


```
OPTIBOOT      = 1024
OPTIBOOTUSB   = 2048
ARDUINONANO   = 1024
ARDUINOMEGA2560 = 1024
TINYBOOTLOADER = 128
TINYBOOTLOADER128 = 128
TINYBOOTLOADER125 = 256
```

Examples 1

In the example below the region 0x1F80 to 0x1FFF (flash block for a PIC16F1509 microcontroller) has been removed from the default space available for code storage using the compiler option.

```
'Set chip model
#chip 16F1509

'Directive
#option ReserveHighProg 128
```

Examples 2

In the example below the bootloader area of Program Memory is protected.

This will ensure the program size does not overwrite the OptiBoot bootloader.

```
'Set chip model
#chip MEGA328P

'Directive
#option ReserveHighProg OPTIBOOT
```

Using Assembler

This is the Using Assembler section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Assembler Overview

Introduction:

You can use microcontroller assembler code within your GCBASIC code.

You can put the assembler code inline in with your source code. The assembler code will be passed through to the assembly file associated with your project.

GCBASIC should recognise all of the commands in the microcontroller datasheet.

The commands should be in lower case, this is good practice, and have a space or tab in front of the command.

Even if the mnemonics are not being formatted properly, `gputils/MPASM` should still be capable of assembling the source code.

Format commands as follows:

Example:

```
btfsc STATUS,Z  
bsf PORTB,1
```

Macros

This is the Macros section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Macros Overview

Introduction:

You can use macros within your GCBASIC code.

Macros are similar to subroutines. But during compilation, everything is inserted inline. This may increase the code size slightly, but it also reduces stack usage.

Parameters are handled in a similar way to how constants are handled, so there is a lot more freedom when passing things in to a macro. (Unlike subs or functions, where everything must be stored in a variable.)

For example, for **PulseOut** one parameter is a pin, and the other is a time length like "500 ms". Neither of those parameters could be stored in a variable, but passing them in as macro parameters is possible.

Demonstration Program:

```
'PulseOut Macro
macro Pulseout (Pin, Time)
    Set Pin On
    Wait Time
    Set Pin Off
end macro
```

Example Macros

This is the Example Macros section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Measuring a Pulse Width

Introduction

The demonstration shows how a macro can be used to optimised code by compiling code inline.

When the measurement of a pulse width to sub-microsecond resolution is required for instance measuring the high or low pulse width of an incoming analog signal a comparator can be combined with a timer to provide the pulse width.

Microchip PIC has published a "Compiled Tips 'N Tricks Guide" that explains how to do certain tasks with Microchip PIC 8-bit microcontrollers.

This guide provides the steps that need to be taken to perform the task of measuring a pulse width. The guide provides guidance on measuring a pulse width using Timer 1 and the CCP module. This guidance was used as the basis for the GCBASIC port the shown below. The guidance was generic and in this example polling the CCP flag bit was more convenient than using an interrupt.

In this demonstration shown below, a 16F1829 microcontroller operating at 32 Mhz uses the internal oscillator. The demonstration code is based on a macro that uses Timer1 and CCP4. However, any of the four CCP modules could be used, the 16F1829 microcontroller has four CCP module.

The timer resolution of this method uses a timer Prescaler of 1:8 and a microcontroller frequency of 32 MHz giving a pulse width resolution is 1ms. With the timer Prescaler of 1:2 and the microcontroller frequency of 32MHz the resolution is 250 ns.

The accuracy is dependent upon the accuracy of the system clock, but oscilloscope measurements have show an accuracy of +- 1us from 3us to 1000us.

In this demonstration the following was implemented

- Using GCBASIC a macro to ensure the generated assembler is inline to ensure the timing is consistent and no sub routines are called.
- Another microcontroller was used to generate the pulses to be measured
- A TEK THS730A oscilloscope was used to measure/verify pulse widths
- A 4x20 LDC module with an I2C Backpack was used to display the results. However, as an alternative, a serial output to a terminal program to view the data could be used

This demonstration could be improved by adding code to poll the TIMER1 overflow flag. If the timer

overflows, then either no pulse was detected or the pulse was longer than allowed by the prescaler/OSC settings. In this case, return a value of zero for pulse width.

Usage:

To get positive pulse width use:

```
PULSE_IN
```

PULSE_IN returns a global word variable Pulse_Width

Demonstration Program:

```
#Chip 16F1829, 32
#CONFIG MCLRE = OFF

'Setup Software I2C
#define I2C_MODE Master
#define I2C_DATA PORTA.2
#define I2C_CLOCK PORTC.0
#define I2C_DISABLE_INTERRUPTS ON

'Set up LCD
#define LCD_IO 10
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_SPEED FAST
#define LCD_Backlight_On_State 1
#define LCD_Backlight_Off_State 0

'Note: This example can be improved by adding code to poll the 'TIMER1 overflow flag.
IF the timer overflows, then either no 'pulse was detected or the pulse was longer than
allowed by the 'prescaler/OSC settings. In this case, return a value of zero 'for pulse
width.

CLS
PRINT "Pulse Width Test"
DIM PULSE_WIDTH AS WORD
DIR PORTC.6 IN

'Setup timer
'Set timer1 using PS1_2 gives 250ns resolution
InitTimer1 OSC, PS1_8
wait 1 s
CLS
```

```

'MAIN PROGRAM LOOP
DO
  PULSE_IN    'Call the Macro to get positive pulse width.
  Locate 0,0
  PRINT Pulse_Width
  PRINT "    "
  wait 1 s
Loop

MACRO PULSE_IN 'Measure Pulse Width
  'Configure CCP4 to Capture rising edge
  CCP4CON = 5 'Set to 00000101
  StartTimer 1
  CCP4IF = 0

  do while CCP4IF = 0 'Wait for rising edge
  loop

  TMR1H = 0: TMR1L = 0 'Clear timer to zero
  CCP4IF = 0           'Clear flag

  'Configure CCP4 to Capture Falling Edge
  CCP4CON = 4 '00000100'

  do while CCP4IF = 0 'Wait for falling edge
  loop

  StopTimer 1 'Stop the time
  Pulse_Width = TIMER1 'Save the timer value
  CCP4IF = 0 'Clear the CCP4 flag
End MACRO

```

Also see [Macros Overview](#)

Implementing a method with a Pin name as a parameter

Introduction

A constant such as a Pin name cannot be passed to a sub routine or a function. This is a constraint of GCBASIC.

A macro can be used to implement a method of passing a constant to reusable code section.

The example shown below implements a button press routine and takes an input port constant and prints the result on an LCD display.

Note: A macro will use more program memory as the macro will be compiled as inline code. Therefore, for every use of the macro will use additional program memory - the same amount of program memory for each call to the macro.

Demonstration Program:

```
#chip 16F877a, 16
#define Button PORTC.1      ' Switch on PIN 14 via 10K pullup resistor
DIR Button In
wait 1 sec

'USART settings
#define USART_BAUD_RATE 9600
#define USART_TX_BLOCKING

;===== MAIN PROGRAM LOOP =====
HSerPrint "Button Test"
HSerPrintCRLF 2
Do
    Test_button ( button )
Loop
;=====

Macro Test_button (Button)
    if Button = ON then
        wait 10 ms          'debounce
        ButtonCount = 0

        Do While Button = On
            Wait 10 ms
            ButtonCount += 1
        Loop

        if ButtonCount > 5  then
            if ButtonCount > 50 then    'Long push
                hserprint "Long push"
            else                        'Short push
                hserprint "Short push"
            end if
            HSerPrintCRLF
        end if
        wait 1 s
    end if
End Macro
```

Also see [Macros Overview](#)

Example Programs

Flashing LEDs and an Interrupt

Explanation:

This code implements four flashing LEDs. This is based on the Microchip PIC Low Pin Count Demo Board.

The example program will blink the four red lights in succession. Press the Push Button Switch, labeled **SW1**, and the sequence of the lights will reverse. Rotate the potentiometer, labeled **RP1**, and the light sequence will blink at a different rate.

This implements an interrupt for the switch press, reads the analog port and set the LEDs.

Demonstration program:

```
#chip 18F14K22, 32
#config MCLRE_OFF

'Works with the low count demo board

'Set the input pin direction
  #define SwitchIn1 PORTa.3
  Dir SwitchIn1 In

#define LedPort PORTc
  DIR PORTC OUT

'Setup the ADC pin direction
  Dir PORTA.0 In
  dim ADCreading as word

'Setup the input pin direction
  #define IntPortA PORTA.1
  Dir IntPortA In

'Variable and constants
  #define intstate as byte
  intstate = 0
  #define minwait 1

  dim ccount as byte
  dim leddir as byte
```

```

ccount = 8
leddir = 0

SET PORTC = 15
WAIT 1 S

SET PORTC = 0

'Setup the Interrupt
  Set IOCA.3 on
  Dir porta.3 in
  On Interrupt PORTABCHANGE Call Setir

'Set initial LED direction
  setLedDirection

DO FOREVER

  INTON
  ADCreading = ReadAD10(AN0)
  if ADCreading < minwait then ADCreading = minwait

  'Set LEDs
  Set PortC = ccount
  wait ADCreading ms

  if leddir = 0 then
    rotate ccount left simple
    'Restart LED position
    if ccount = 16 then
      ccount = 128
    end if
  end if

  if leddir = 1 then
    rotate ccount Right simple
    'Restart LED position
    if ccount = 128 then
      ccount = 8
    end if
  end if

  'Reset interrupt - this may be been reset so set to zero so interrupt can
operate.
  intstate = 0

```

Loop

```
'Interrupt routine.
sub Setir

    if IntPortA = 0 and intstate = 0 then
        intstate = 1
        wait while SwitchIn1 = 0
        setLedDirection
    end if

end sub

sub setLedDirection

    'Set LED values
    select case leddir

        case 0
            leddir = 1
            ccount = 8

        case 1
            leddir = 0
            ccount = 1

    end select

End Sub
```

See Also [Interrupts](#), [ReadAD10](#)

Flashing LED with timing parameters

Explanation:

This is an example of how to define a subroutine.

When called, this subroutine will blink an LED for the number of times and duration as determined by the input parameters.

The syntax of the subroutine is:

```
' Flash_LED (numtimes, OnTime, (optional) OffTime)
' Where numtimes is from 1 - 255 and OnTime/OffTime is
' from 0 - 65535 ms. If OffTime is not entered, then
' OffTime = OnTime.

Sub Flash_LED (in numtimes, in OnTime as WORD, Optional OffTime as WORD = OnTime)
    repeat numtimes
        set LED on
        wait OnTime ms
        set LED OFF
        wait OffTime ms
    end repeat
End Sub
```

Shown below is a working example program using a Microchip PIC 18F25K22.

Change Settings/PORTS as needed for other Chips.

Connect an LED to the LED pin via a 1K series resistor.

Demonstration program:

```

#chip 18F25K22, 16
#define LED PORTC.1      'Led on PIN 14 via 1K resistor
DIR LED OUT
wait 1 sec

;===== MAIN PROGRAM LOOP =====
Do
    Flash_LED ( 3,250 )      '3 Flashes 250 ms equal on/off time
    Wait 2 Sec
    Flash_LED ( 5,250,500 )  '5 flashes On 250 ms / off 500 ms
    Wait 2 Sec
    Flash_LED ( 10,100 )     '10 rapid flashes
    Wait 2 Sec
Loop
;=====

Sub Flash_LED (in numtimes, in OnTime as WORD, optional OffTime as word = OnTime)
    repeat numtimes
        set LED on
        wait OnTime ms
        set LED OFF
        wait OffTime ms
    end repeat
End Sub

```

Generate Accurate Pulses

Explanation:

The **PulseOut** Command is a reliable method for generating pulses if accuracy is not critical, the **PulseOut** command uses a calculation of the clock to speed for the timing .

If you need better accuracy and resolution then an alternative approach is required.

To generate pulses in the 100 us to 2500 us range with an accuracy of +- 1us over this range is practical using the approach shown in this example.

This example code works on a midrange PIC16F690 operating at 8Mhz. However, it should work on any Microchip PIC microcontroller, but may need some minor modifications.

Usage:

```
Pulse_Out_us ( word_value )
```

How It Works:

Timer1 is loaded with a preset value based upon the variable passed to the sub routine. The timer (**Timer1**) is started and the pulse pin (the output pin) is set high. When **Timer1** overflows the Timer1 interrupt flag bit (**TMR1IF**) is set. This causes the program to exit a polling loop and set the pulse Pin off. Then, **Timer1** is stopped and **TMRIF** flag is cleared and the sub routine exits.

This method supports delays between 5 us and 65535 us and uses Timer1.

Test Results:

These tests were completed using a Saleae Logic Analyzer.

Pulse setting	Time Results
Pulse_Out_us (2500)	2501.375 us
Pulse_Out_us (1000)	1000.750 us
Pulse_Out_us (100)	100. 125 us
Pulse_Out_us (10)	10.125 us
Pulse_Out_us with less then 4	Unreliable results

Demonstration program:

```

;*****
; Code:  Output an accurate pulse
; Author: William Roth 03/13/2021
;*****

#chip 16F690,8

; ---- Define Hardware settings
; ---- Define I2C settings - CHANGE PORTS AS REQUIRED
#define I2C_MODE Master
#define I2C_DATA PORTB.4
#define I2C_CLOCK PORTB.6
#define I2C_DISABLE_INTERRUPTS ON

; ---- Set up LCD - Using I2C LCD Backpack
#define LCD_IO 10
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_I2C_Address_1 0x4e ; default to 0x4E
; ---- May need to use SLOW or MEDIUM if your LCD is a slower device.
#define LCD_SPEED Medium
#define LCD_Backlight_On_State 1
#define LCD_Backlight_Off_State 0

CLS
```

```

; ---- USART settings
#define USART_BAUD_RATE 38400
#define USART_TX_BLOCKING
DIR PORTB.7 OUT

; ---- Setup Pulse parameters
#define PulsePin PORTC.4
Dim Time_us As WORD
Dir PulsePin Out      'Pulsout pin
Set PulsePin off

; ---- Setup Timer
InitTimer1 Osc, PS1_2 'For 8Mhz Chip
'InitTimer1 Osc, PS1_4, 'For 16 Mhz Chip
TMR1H = 0: TMR1L = 0   'Clear timer1
TMR1IF = 0 'Clear timer1 int flag
TMR1IE = on 'Enable timer1 Interrupt (Flag only)

' **** This is the MAIN loop ****
Do
    PULSE_OUT_US (2500) 'Measured as 2501.375 us
    wait 19 ms
    Pulse_Out_US (1000) 'Measured as 1000.750 us
    wait 19 ms
    Pulse_Out_US (100)   'Measured as 100.125 us
    wait 19 ms
    Pulse_Out_US (10)    'Measured as 10.125 us
    Wait 19 ms
Loop

SUB PULSE_OUT_US (IN Variable as WORD)
TMR1H = 65535 - Variable_H      'Timer 1 Preset High
TMR1L = (65535 - Variable) + 4 'Timer 1 Preset Low
Set TMR1ON ON                   'Start timer1
Set PulsePin ON                 'Set Pin high
Do While TMR1IF = 0             'Wait for Timer1 overflow
    Loop
Set PulsePin off                ' Pin Low
Set TMR1ON OFF                  ' Stop timer 1
TMR1IF = 0                      'Clear the Int flag
END SUB

```

Also see [PulseOut](#)

Graphical LCD Demonstration

Explanation:

This demonstration code shows the set of commands supported by GCBASIC.

Demonstration program:

```
;Chip Settings
#chip 16F877a,16

#include <glcd.h>

'Setup the GLCD
#Define glcd_rw PORTD.3      'RW pin on LCD
#Define glcd_reset PORTD.4   'Reset pin on LCD
#Define glcd_cs1 PORTD.1     'CS1, CS2 can be reversed
#Define glcd_cs2 PORTD.2     'CS1, CS2 are be reversed
#Define glcd_rs PORTD.5      'D/I pin on LCD
#Define glcd_enable PORTD.4  'E pin on LCD
#Define glcd_db0 PORTB.0     'D0
#Define glcd_db1 PORTB.1     'D1
#Define glcd_db2 PORTB.2     'D2
#Define glcd_db3 PORTB.3     'D3
#Define glcd_db4 PORTB.4     'D4
#Define glcd_db5 PORTB.5     'D5
#Define glcd_db6 PORTB.6     'D6
#Define glcd_db7 PORTB.7     'D7 on LCD

'Specify the type of GLCD
#define GLCD_TYPE GLCD_TYPE_KS0108
#define GLCD_WIDTH 128
#define GLCD_HEIGHT 64
#define GLCD_PROTECTOVERRUN

wait 1 s
GLCDCLS
GLCDPrint 0, 1, "GCBASIC "
wait 1 s
GLCDCLS

rrun = 0
dim msg1 as string * 16

do forever

    GLCDCLS
    Box 18,30,28,40
    Line 0,0,127,63
```



```
Line 0,63,127,0
```

```
wait 1 s
```

```
FilledBox 18,30,28,40
```

```
wait 1 s
```

```
GLCDCLS
```

```
GLCDDrawString 30,0,"ChipMhz@"
```

```
GLCDDrawString 78,0, str(ChipMhz)
```

```
Circle(10,10,10,1) 'upper left
```

```
Circle(117,10,10,1) 'upper right
```

```
Circle(63,31,10,1) 'center
```

```
Circle(63,31,20,1) 'center
```

```
Circle(10,53,10,1) 'lower left
```

```
Circle(117,53,10,1) 'lower right
```

```
wait 1 s
```

```
GLCDDrawString 30,0,"ChipMhz@"
```

```
GLCDDrawString 78,0, str(ChipMhz)
```

```
FilledCircle(10,10,10,1) 'upper left
```

```
FilledCircle(117,10,10,1) 'upper right
```

```
FilledCircle(63,31,10,1) 'center
```

```
FilledCircle(63,31,20,1) 'center
```

```
FilledCircle(10,53,10,1) 'lower left
```

```
FilledCircle(117,53,10,1) 'lower right
```

```
wait 1 s
```

```
GLCDCLS
```

```
GLCDDrawString 30,0,"ChipMhz@"
```

```
GLCDDrawString 78,0, str(ChipMhz)
```

```
Circle(10,0,10,1) 'upper left
```

```
Circle(117,0,10,1) 'upper right
```

```
Circle(63,31,10,1) 'center
```

```
Circle(63,31,20,1) 'center
```

```
Circle(10,63,10,1) 'lower left
```

```
Circle(117,63,10,1) 'lower right
```

```
wait 1 s
```

```
GLCDCLS
```

```
GLCDDrawString 0,10,"Hello" 'Print Hello
```

```
wait 1 s
```

```
GLCDDrawString 0,10, "ASCII #:" 'Print ASCII #:
```

```
Box 18,30,28,40 'Draw Box Around ASCII Character
```

```
for char = 0x30 to 0x39 'Print 0 through 9
```

```
GLCDDrawString 16, 20 , Str(char)+" "
```

```
GLCDdrawCHAR 20, 30, char
```

```
wait 250 ms
```

```

next
line 0,50,127,50           'Draw Line using line command
for xvar = 0 to 80         'Draw line using Pset command
    pset xvar,63,on
next
FilledBox 18,30,28,40      'Draw Box Around ASCII Character '
wait 1 s
GLCDCLS
GLCDDrawString 0,10,"End  "
wait 1 s
GLCDCLS

workingGLCDDrawChar:
dim gtext as string
gtext = "KS0108"

    for xchar = 1 to gtext(0)      'Print 0 through 9
        xxpos = (1+(xchar*6)-6)
        GLCDDrawChar xxpos , 0 , gtext(xchar)
    next

GLCDDrawString 1, 9, "GCBASIC @2021"
GLCDDrawString 1, 18,"GLCD 128*64"
GLCDDrawString 1, 27,"Using GLCD.H from GCB"
GLCDDrawString 1, 37,"Using GLCD.H GCB@2021"
GLCDDrawString 1, 45,"GLCDDrawChar method"
GLCDDrawString 1, 54,"Test Routines"

wait 1 s
GLCDCLS

msg1 = "Run = " +str(rrun)
rrun++
GLCDPrint 0, 3, msg1
wait 1 s
GLCDCLS

```

loop

For more help, see [Graphical LCD Demonstration](#), [GLCDCLS](#), [GLCDDrawChar](#), [GLCDPrint](#), [GLCDReadByte](#), [GLCDWriteByte](#), [Pset](#)

InfraRed Remote

Explanation:

GCBASIC support interfacing with IR remote controls. The header file contains explanations, for both

hardware and software.

This has been tested on many different IR sensors, and different remote controls.

Demonstration program:

The example is expected to work with most any IR sensor running at a 38 kHz carrier frequency.

```
;This demo prints the device number and key number sent by  
;a Sony compatible IR remote control unit to an LCD
```

```
;Thomas Henry --- 4/23/2014
```

```
#chip 16F88, 8           ;PIC16F88 running at 8 MHz  
#config mclr=off         ;reset handled internally  
#include <SonyRemote.h>  ;include the header file
```

```
;----- Constants
```

```
#define LCD_IO      4      ;4-bit mode  
#define LCD_WIDTH 20      ;specified lcd width for clarity only. 20 is the  
default width  
#define LCD_RS      PortB.2 ;pin 8 is Register Select  
#define LCD_Enable  PortB.3 ;pin 9 is Enable  
#define LCD_DB4     PortB.4 ;DB4 on pin 10  
#define LCD_DB5     PortB.5 ;DB5 on pin 11  
#define LCD_DB6     PortB.6 ;DB6 on pin 12  
#define LCD_DB7     PortB.7 ;DB7 on pin 13  
#define LCD_NO_RW   1      ;ground RW line on LCD  
  
#define IR_DATA_PIN PortA.0 ;sensor on pin 17
```

```
;----- Variables
```

```
dim device, button as byte
```

```
;----- Program
```

```
dir PortA in           ;A.0 is IR input  
dir PortB out          ;B.2 - B.6 for LCD  
  
do  
  readIR_Remote(device, button) ;wait for button press  
  
  cls                  ;show device code  
  print "Device: "  
  print device  
  
  locate 1,0  
  print "Button: "  
  print button          ;show button code  
  
  wait 10 mS           ;ignore any repeats  
loop                   ;repeat forever
```

See also [SonyRemote.h](#).

SonyRemote.h

Explanation: Sony IR Remote Control Library for GCBASIC

This include file will let you easily read and use the infrared signals from a Sony compatible television remote control. In particular, the remote control transmits a pulse modulated signal, the sensor detects this, and the subroutine in this header file decodes the signal, returning two numbers: one representing the device (television, VCR, DVD, tuner, etc.), while the other returns the key which has been depressed (VOL+, MUTE, channel numbers 0 through 9, etc.).

This has been tested and confirmed with a fixed remote control purchased surplus for \$2.00 from All Electronics, as well as an universal remote control, set to Sony mode.

Moreover it has also been tested with a Panasonic IR sensor and a Vishay sensor, both purchased surplus for about fifty cents.

Every combination performed well, and it is probably the case that most any garden variety 38 kHz IR sensor will work. The only tricky bit is making sure you get the pinout for your sensor correct, search out the datasheet for whichever device you use.

There are only three pins: Ground Vcc Data

It is essential to filter the power applied to the Vcc pin. Do this by connecting a 100 ohm resistor from the +5V power supply to the Vcc pin, and bridge the pin to ground with a 4.7uF electrolytic capacitor.

The Data pin requires a 4.7k pullup resistor.

There is only one constant required of the calling program. It indicates which port line the IR sensor is connected to. For example,

```
#DEFINE IR_DATA_PIN PORTA.0
```

There is one subroutine:

```
readIR_Remote(IR_rem_dev, IR_rem_key)
```

The values returned are, respectively, the device number mentioned earlier and the key that is currently pressed. Both are byte values.

Seventeen local bytes are consumed, and two bytes are used for the output parameters. That's a grand total of nineteen bytes required when invoking this subroutine.

Header File

```

sub readIR_Remote(out IR_rem_dev as byte, out IR_rem_key as byte)
  dim IR_rem_count, IR_rem_i as byte
  dim IR_rem_width(12) as byte          ;pulse width array

  do
    IR_rem_count = 0                    ;wait for start bit
    do while IR_DATA_PIN = 0            ;measure width (active low)
      wait 100 uS                        ;24 X 100 uS = 2.4 mS
      IR_rem_count++
    loop
  loop while IR_rem_count < 20           ;less than this so wait

  for IR_rem_i = 1 to 12                ;read/store the 12 pulses
    do
      IR_rem_count = 0
      do while IR_DATA_PIN = 0          ;zero = 6 units = 0.6 mS
        wait 100 uS                    ;one = 12 units = 1.2 mS
        IR_rem_count++
      loop
      loop while IR_rem_count < 4        ;too small to be legit

      IR_rem_width(IR_rem_i) = IR_rem_count ;else store pulse width
    next IR_rem_i

    IR_rem_key = 0                      ;command built up here
    for IR_rem_i = 1 to 7               ;1st 7 bits are the key
      IR_rem_key = IR_rem_key / 2        ;shift into place
      if IR_rem_width(IR_rem_i) > 10 then ;longer than 10 mS
        IR_rem_key = IR_rem_key + 64    ;so call it a one
      end if
    next

    IR_rem_dev = 0                      ;device number built up here
    for IR_rem_i = 8 to 12              ;next 5 bits are device number
      IR_rem_dev = IR_rem_dev / 2
      if IR_rem_width( IR_rem_i ) > 10 then
        IR_rem_dev = IR_rem_dev + 16
      end if
    next
  end sub

```

Midpoint Circle Algorithm

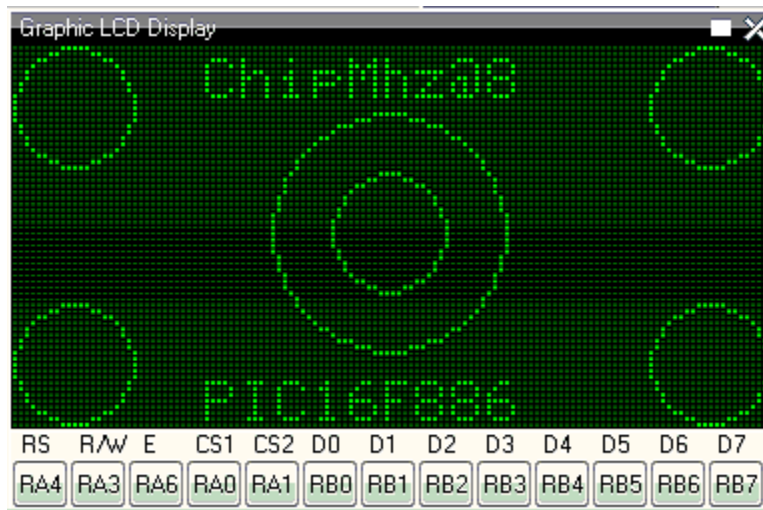
Explanation:

GCBASIC can draw circles using the midpoint circle algorithm. The midpoint circle algorithm

determines the points needed for drawing a circle. The algorithm is a variant of [Bresenham's line algorithm](#), and is thus sometimes known as Bresenham's circle algorithm, although not actually invented by [Jack E. Bresenham](#).

The example program below show the midpoint circle algorithm within GCBASIC.

Example Output on GLCD Device:



```
'Midpoint Circle algorithm
'Chip model
#chip 16F886, 8           ;PIC16F88 running at 8 MHz
#config mclr=off         ;reset handled internally

#include <glcd.h>

;----- Constants

;Pinout is shown for the LCM12864H-FSB-FBW
;graphical LCD available from Amazon.

;      +5V                ;LCD pin 1
;      ground             ;LCD pin 2
;      Vo = wiper of pot   ;LCD pin 3
#define GLCD_DB0 PORTB.0    ;LCD pin 4
#define GLCD_DB1 PORTB.1    ;LCD pin 5
#define GLCD_DB2 PORTB.2    ;LCD pin 6
#define GLCD_DB3 PORTB.3    ;LCD pin 7
#define GLCD_DB4 PORTB.4    ;LCD pin 8
#define GLCD_DB5 PORTB.5    ;LCD pin 9
#define GLCD_DB6 PORTB.6    ;LCD pin 10
#define GLCD_DB7 PORTB.7    ;LCD pin 11
#define GLCD_CS2 PORTA.0     ;LCD pin 12
#define GLCD_CS1 PORTA.1     ;LCD pin 13
```

```

#define GLCD_RESET PORTA.2 ;LCD pin 14
#define GLCD_RW PORTA.3 ;LCD pin 15
#define GLCD_RS PORTA.4 ;LCD pin 16
#define GLCD_ENABLE PORTA.6 ;LCD pin 17
; Vee = pot low side ;LCD pin 18
; backlight anode ;LCD pin 19
; backlight cathode ;LCD pin 20

#define GLCD_TYPE GLCD_TYPE_KS0108
#define GLCD_WIDTH 128
#define GLCD_HEIGHT 64

;----- Program

Do forever

    GLCDDrawString 30,0,"ChipMhz@"
    GLCDDrawString 78,0, str(ChipMhz)
    Circle(10,10,10,0) ;upper left
    Circle(117,10,10,0) ;upper right
    Circle(63,31,10,0) ;center
    Circle(63,31,20,0) ;center
    Circle(10,53,10,0) ;lower left
    Circle(117,53,10,0) ;lower right
    GLCDDrawString 30,54,"PIC16F886"

loop

```

I2C Master Hardware

Explanation:

This program demonstrates how to read and write data from an EEPROM device using the serial protocol called I2C.

This program uses the hardware I2C module within the microcontroller. If your microcontroller does not have a hardware I2C module then please use the software I2C GCBASIC library.

This program has three sections.

1. Read a single byte from the EEPROM
2. Write and read a page of 64 bytes to and from the EEPROM, and
3. Finally display the contents of the EEPROM.

This program has an interrupt driven serial handler to capture and manage input from a serial terminal.

Demonstration program:

```
'Change the microcontroller, frequency and config to suit your needs.
#chip 16F1937, 32
#config MCLRE_ON

'Required Library to read and write to an EEPROM
#include <I2CEEPROM.h>

' Define I2C settings - CHANGE PORTS
#define HI2C_BAUD_RATE 400
#define HI2C_DATA PORTC.4
#define HI2C_CLOCK PORTC.3
'I2C pins need to be input for SSP module for Microchip PIC devices.
Dir HI2C_DATA in
Dir HI2C_CLOCK in
'I2C MASTER MODE
HI2CMode Master

' THIS CONFIG OF THE SERIAL PORT WORKS WITH max232 THEN TO PC
' USART settings
#define USART_BAUD_RATE 9600
Dir PORTc.6 Out
Dir PORTc.7 In
#define USART_DELAY OFF
#define USART_TX_BLOCKING
wait 500 ms

'Create a Serial Interrupt Handler
On Interrupt UsartRX1Ready Call readUSART

' Constants etc required for the serial Buffer Ring
#define BUFFER_SIZE 32
#define bkbhit (next_in <> next_out)
' Required Variables for the serial Buffer Ring
Dim buffer(BUFFER_SIZE)
Dim next_in as byte: next_in = 1
Dim next_out as byte: next_out = 1

Dim syncbyte as Byte
wait 125 ms

' Read ONE byte from the EEPROM
dim DeviceID as byte
dim EepromAddress, syscounter as word
#define EEpromDevice 0xA0
```

```

'Master Main Loop
location = 0
'Define our array
dim outarray(64), inarray(64)

do
    HSerPrintCRLF 2
    HSerPrint "Commence Array Write and Read"
    'Populate the array
    for tt = 1 to 64
        outarray(tt) = tt
    next

    'Library write call is: eeprom_wr_array(device_number, page_size, address,
array_name, number_of_bytes)
    eeprom_wr_array(EEpromDevice, 64, location, outarray, 64)

    'Library read call is: eeprom_rd_array(device_number, address, array_name,
number_of_bytes)
    eeprom_rd_array(EEpromDevice, location, inarray, 64)

    'Show results of the read of the I2C EEPROM
    HSerPrintCRLF 2
    for tt = 1 to 64

        if outarray(tt) <> inarray(tt) then
            Hserprint "!"
            HSerPrint inarray(tt)
        else
            HSerPrint inarray(tt)
        end if
        HSerPrint ","
    next

    HSerPrintCRLF 2
    HSerPrint "Commence Write and Read a single byte":HSerPrintCRLF
    HSerPrint "Read value should be "
    HSerPrint str(location):HSerPrintCRLF
    HSerPrint "Read = "
    'Use library to write and read from the I2C EEPROM
    eeprom_wr_byte (EEpromDevice, location, location)
    eeprom_rd_byte (EEpromDevice, location, bbyte )

    HSerPrint bbyte
    location++
    HSerPrintCRLF 2

```

```

    'Show the connected I2C devices on the Serial terminal.
    HI2CDeviceSearch
    HSerPrint "Commence Dump of the EEPROM"
    validateEEPROM
Loop
End

```

```

'Show the attached I2C devices
sub HI2CDeviceSearch
    'Assumes serial is operational
    HSerPrintCRLF
    HSerPrint "I2C Device Search"
    HSerPrintCRLF 2
    for deviceID = 0 to 255
        HI2CStart
        HI2CSend ( deviceID )
        if HI2CAckPollState = false then
            HSerPrint "ID: 0x"
            HSerPrint hex(deviceID)
            HSerSend 9
            testid = deviceID | 1
            select case testid
                case 49
                    Hserprint "DS2482_1Channel_1Wire Master"
                case 65
                    Hserprint "Serial_Expander_Device"
                Case 73
                    Hserprint "Serial_Expander_Device"
                case 161
                    Hserprint "EEProm_Device_Device"
                case 163
                    Hserprint "EEProm_Device_Device"
                case 165
                    Hserprint "EEProm_Device_Device"
                case 167
                    Hserprint "EEProm_Device_Device"
                case 169
                    Hserprint "EEProm_Device_Device"
                case 171
                    Hserprint "EEProm_Device_Device"
                case 173
                    Hserprint "EEProm_Device_Device"
                case 175
                    Hserprint "EEProm_Device_Device"
                case 209
                    Hserprint "DS1307_RTC_Device"
            end select
        end if
    next deviceID
end sub

```

```

        case 249
            Hserprint "FRAM_Device"
        case else
            Hserprint "Unknown_Device"
        end select
        HI2CSend ( 0 )
        HSerPrintCRLF
    end if
    HI2CStop
next
    HSerPrint  "End of Device Search"
    HSerPrintCRLF 2
end sub

'Validation EEPROM code
sub validateEEPROM
    EepromAddress = 0
    HSerPrintCRLF 2
    HSerPrint "Hx"
    HSerPrint hex(EepromAddress_h)
    HSerPrint hex(EepromAddress)
    HSerPrint " "

    for EepromAddress = 0 to 0xffff
        'Read from EEPROM using a library function
        eeprom_rd_byte EEPromDevice, EepromAddress, objType

        HSerPrint hex(objType)+" "
        if ((EepromAddress+1) % 8 ) = 0 then
            HSerPrintCRLF
            HSerPrint "Hx"
            syscounter = EepromAddress + 1
            HSerPrint hex(syscounter_h)
            HSerPrint hex(syscounter)
            HSerPrint " "
        end if
        'Has serial data been received
        if bkbhit then
            syschar = bgetc
            select case syschar
                case 32
                    do while bgetc = 32
                        loop
                    case else
                        HSerPrintCRLF
                        HSerPrint "Done"
                        exit sub
            end select
        end if
    next
end sub

```

```

        end if
    next
    HSerPrintCRLF
    HSerPrint "Done"
end Sub

' Start of Serial Support functions
' Required to read the serial port
' Assumes serial port has been initialised
Sub readUSART
    buffer(next_in) = HSerReceive
    tempnt = next_in
    next_in = ( next_in + 1 ) % BUFFER_SIZE
    if ( next_in = next_out ) then ' buffer is full!!
        next_in = tempnt
    end if
End Sub

' Serial Support functions
' Get characters from the serial port
function bgetc
    wait while !(bkbhit)
    bgetc = buffer(next_out)
    next_out=(next_out+1) % BUFFER_SIZE
end Function

```

I2C Slave Hardware

Explanation:

This program demonstrates how to control and display using a LCD the code for the keypad.

This program can be adapted. This program uses the hardware I2C module within the microcontroller. If your microcontroller does not have a hardware I2C module then please use the software I2C GCBASIC library for most microcontrollers.

This program also has an interrupt driven I2C handler to manage the I2C from the Start event.

Demonstration program:

```

'Code for the keypad and LCD Microchip PIC microcontroller on the Microlab board v2
'microcontroller is responsible for:
' - Reading keypad
' - Displaying data on LCD
' - communication with main Microchip PIC microcontroller via I2C

```

```

' - providing 5 keypad lines to main Microchip PIC microcontroller (for
compatibility)
' - receiving remote control signals for button and keypad

'This code has support for two keypad layouts. This is one possible layout:
'0123
'4567
'89AB
'CDEF
'And this is the other possible layout:
'123A
'456E
'789D
'A0BC
'Select the keypad layout by uncommenting one of these lines:
#define KEYPAD_KEYMAP KeypadMap0123
#define KEYPAD_KEYMAP KeypadMap123F

'Chip and config
#chip 16F882, 8

'Ports connected to keypad
'Column ports need pullups, hence columns are on PORTB for built in weak pullups
#define KEYPAD_COL_1 PORTB.4
#define KEYPAD_COL_2 PORTB.5
#define KEYPAD_COL_3 PORTB.6
#define KEYPAD_COL_4 PORTB.7
#define KEYPAD_ROW_1 PORTA.4
#define KEYPAD_ROW_2 PORTA.3
#define KEYPAD_ROW_3 PORTA.2
#define KEYPAD_ROW_4 PORTA.1

'Ports connected to LCD
#define LCD_IO 4
#define LCD_WIDTH 20 ;specified lcd width for clarity only. 20 is the
default width
#define LCD_RW PORTA.7
#define LCD_RS PORTA.6
#define LCD_Enable PORTA.5
#define LCD_DB4 PORTA.4
#define LCD_DB5 PORTA.3
#define LCD_DB6 PORTA.2
#define LCD_DB7 PORTA.1
#define BACKLIGHT PORTA.0

'Button port (for remote control)
#define BUTTON PORTB.0

```

```

'Keypad ports connected to main Microchip PIC microcontroller
'These are disabled when KeyoutDisabled = true
#define KEYOUT_A PORTC.5
#define KEYOUT_B PORTC.2
#define KEYOUT_C PORTC.1
#define KEYOUT_D PORTC.0
#define KEYOUT_DA PORTB.1

'I2C ports
#define I2C_DATA PORTC.4
#define I2C_CLOCK PORTC.3

'RS232/USART settings
'To do if/when remote support needed

'Initialise
Dir KEYOUT_A Out
Dir KEYOUT_B Out
Dir KEYOUT_C Out
Dir KEYOUT_D Out
Dir KEYOUT_DA Out

Dir BACKLIGHT Out
Dir BUTTON In 'Is an output, turn off by switching pin to Hi-Z

'Initialise I2C Slave
'I2C pins need to be input for SSP module
Dir I2C_DATA In
Dir I2C_CLOCK In
HI2CMode Slave
HI2CSetAddress 128

'Buffer for incoming I2C messages
'Each message takes 4 bytes
Dim I2CBuffer(10)
BufferSize = 0
OldBufferSize = 0

'Set up interrupt to process I2C
On Interrupt SSP1Ready Call I2CHandler

'Enable weak pullups on B4-7 (keypad col pins)
NOT_RBPU = 0
WPUB = b'11110000'

'Key buffers
'255 indicates no key, other value gives currently pressed key
RemoteKey = 255

```

```

OutKey = 255
KeyoutDisabled = False 'False if KEYOUT lines used to send key

'Main loop
Do

    'Read keypad, send value
    CheckPressedKeys
    SendKeys

    'Check for I2C messages
    ProcessI2C

Loop

'This keymap table is for this arrangement:
'0123
'4567
'89AB
'CDEF
Table KeypadMap0123
    3
    7
    11
    15
    2
    6
    10
    14
    1
    5
    9
    13
    0
    4
    8
    12
End Table

'This keymap table is for this arrangement:
'123F
'456E
'789D
'A0BC
Table KeypadMap123F
    15
    14
    13

```



```

12
3
6
9
11
2
5
8
0
1
4
7
10
End Table

Sub CheckPressedKeys
    'Subroutine to:
    ' - Read keypad
    ' - Check remote keypress
    ' - Decide which key to output

    'Read keypad
    If RemoteKey <> 255 Then
        OutKey = RemoteKey
    Else
        EnableKeypad
        OutKey = KeypadData

    End If

End Sub

Sub EnableKeypad
    'Disable LCD so that keypad can be activated
    Set LCD_RW Off 'Write mode, don't let LCD write

    'Re-init keypad
    InitKeypad

End Sub

Sub I2CHandler
    'Handle I2C interrupt
    'SSPIF doesn't trigger for stop condition, only start!

    'If buffer full flag set, read

    Do While HI2CHasData

```

```

HI2CReceive DataIn

'Sending code
If BufferSize = 0 Then
    LastI2CWasRead = False
    'Detect read address
    If DataIn = 129 Then
        LastI2CWasRead = True

        HI2CSend OutKey

        KeyoutDisabled = True
        Dir KEYOUT_A In
        Dir KEYOUT_B In
        Dir KEYOUT_C In
        Dir KEYOUT_D In
        Dir KEYOUT_DA In

        Exit Sub
    End If
End If

If BufferSize < 10 Then I2CBuffer(BufferSize) = DataIn
BufferSize += 1
Loop

End Sub

Sub SendKeys

'Don't run if not using KEYOUT lines
If KeyoutDisabled Then Exit Sub

'Send pressed keys
If OutKey <> 255 Then
    'If there is a key pressed, set output lines
    If OutKey.0 Then
        KEYOUT_A = 1
    Else
        KEYOUT_A = 0
    End If
    If OutKey.1 Then
        KEYOUT_B = 1
    Else
        KEYOUT_B = 0
    End If
    If OutKey.2 Then
        KEYOUT_C = 1

```

```

Else
    KEYOUT_C = 0
End If
If OutKey.3 Then
    KEYOUT_D = 1
Else
    KEYOUT_D = 0
End If

KEYOUT_DA = 1
Else
    'If no key pressed, clear data available line to main Microchip PIC
microcontroller
    KEYOUT_DA = 0
End If

End Sub

Sub ProcessI2C

    If HI2CStopped Then
        IntOff

        If LastI2CWasRead Then BufferSize = 0

        If BufferSize <> 0 Then
            OldBufferSize = BufferSize
            BufferSize = 0
        End If
        IntOn
    End If

    If OldBufferSize <> 0 Then

        CmdControl = I2CBuffer(1)

        'Set backlight
        If CmdControl.6 = On Then
            Set BACKLIGHT On
        Else
            Set BACKLIGHT Off
        End If

        'Set R/S bit
        LCD_RS = CmdControl.4

        'Send bytes to LCD
        LCDDataBytes = CmdControl And 0x0F
    
```

```

        If LCDDDataBytes > 0 Then
            For CurrSendByte = 1 To LCDDDataBytes
                LCDWriteByte I2CBuffer(LCDDDataBytes + 1)
            Next
        End If
        'LCDWriteByte I2CBuffer(2)

        OldBufferSize = 0
    End If

End Sub

```

RGB LED Control

Explanation:

This program demonstrates how to drive an RGB LED to create 4096 different colors. Each of the three elements (red, green and blue) responds to 16 different levels. A value of 0 means the element never turns on, while a value of 15 means the element never shuts off. Values in between these two extremes vary the pulse width.

This program is an interrupt driven three channel PWM implementation.

The basic carrier frequency depends upon the microcontroller clock speed. For example, with an 8 MHz clock, the LED elements are modulated at about 260 Hz. The interrupts are generated by Timer 0. With an 8 MHz clock they occur about every 256 μ S. The interrupt routine consumes about 20 μ S.

Do not forget the current limiting resistors to the LED elements. A value of around 470 ohms is typical, but you may want to adjust the individual values, to balance the color response.

In this demonstration, three potentiometers are used to set the color levels using the slalom technique.

Demonstration program:

```

;----- Configuration
#chip 16F88, 8                ;PIC16F88 running at 8 MHz
#config mclr=off              ;reset handled internally

;----- Constants
#define LED_R PortB.0         ;pin to red element
#define LED_G PortB.1         ;pin to green element
#define LED_B PortB.2         ;pin to blue element
;----- Variables
dim redValue, greenValue, blueValue, ticks as byte
;----- Program
dir PortA in                  ;three pots for inputs
dir PortB out                 ;the LED outputs
on interrupt Timer0Overflow call update
initTimer0 Osc, PS0_1/2
do
    redValue = readAD(AN0)/16 ;red -- 0 to 15
    greenValue = readAD(AN1)/16 ;green -- 0 to 15
    blueValue = readAD(AN2)/16 ;blue -- 0 to 15
loop

Sub update                    ;interrupt routine
    ticks++                  ;increment master timekeeper
    if ticks = 15 then       ;start of the count
        ticks = 0
        if redValue <> 0 then ;only turn on if nonzero
            set LED_R on
        end if
        if greenValue <> 0 then
            set LED_G on
        end if
        if blueValue <> 0 then
            set LED_B on
        end if
    end if
    if ticks = redValue then ;time to turn off red?
        set LED_R off
    end if
    if ticks = greenValue then ;time to turn off green?
        set LED_G off
    end if
    if ticks = blueValue then ;time to turn off blue?
        set LED_B off
    end if
end sub

```

Serial/RS232 Buffer Ring

Explanation:

This program demonstrates how to create a serial input buffer ring.

The program receives a character into the buffer and sends back. Try sending large volumes of characters.....

This program uses an interrupt event to capture the incoming byte value and place in the buffer ring. When a byte is received the buffer ring is incremented to ensure the next byte is handled correctly.

Testing `bkbhit` will set to True when a byte has been received. Reading the function `bgetc` will return the last byte received.

Demonstration program:

This demonstration program will support up to 256 bytes. For a larger buffer change the variables to words.

```
#chip 16F1937
// #chip mega4809
// #chip mega328p, 16

// Add PPS if appropriate for your chip
// [change to your config] This is the config for a serial terminal

// assumes USART1 ( or USART0 on AVRdx ), if you select USART1/2/3/4 then you MUST
add the comport parameter to all HSerxxxxx functions.
// turn on the RS232 and terminal port.
// Define the USART settings
#define USART_BAUD_RATE 9600

// This assumes you are using an ANSI compatible terminal. Use PUTTY.EXE it is very
easy to use.

// Main program

// Wait for terminal to settle
wait 3 s

// Create the supporting variables
Dim next_in As Byte
Dim next_out As Byte
Dim syncbyte As Byte
```

```

Dim tempnt As Byte

// Constants etc required for Buffer Ring
#define BUFFER_SIZE 8
#define bkbhit (next_in <> next_out)

// Define the Buffer
Dim buffer( BUFFER_SIZE - 1 ) // we will use element 0 in the array as part of out
buffer

// Call init the buffer
InitBufferRing

HSerSend 10
HSerSend 13
HSerSend 10
HSerSend 13
HSerPrint "Started: Serial between two devices"
HSerSend 10
HSerSend 13

// Get character(s) and send back
// Get character(s) and send back
Do

    // Do we have data in the buffer?
    if bkbhit then

        // Send the next character in the buffer to the terminal, exposed via the
function `bgetc` back the terminal
        HSerSend bgetc

    end if
Loop

// Supporting subroutines

Sub readUSART

    buffer(next_in) = HSerReceive
    tempnt = next_in
    next_in = ( next_in + 1 ) % BUFFER_SIZE
    If ( next_in = next_out ) Then // buffer is full!!
        next_in = tempnt
    End If

```

End Sub

Function bgetc

Dim local_next_out as Byte // maintain a local copy of the next_out variable
to ensure it does not change when an Interrupt happens

local_next_out = next_out

bgetc = buffer(local_next_out)

local_next_out=(local_next_out+1) % BUFFER_SIZE

INTOFF

next_out = local_next_out

INTON

End Function

Sub InitBufferRing

// Set the buffer to the first address

next_in = 0

next_out = 0

// Interrupt Handler - some have RCIE and some have U1RXIE, so handle

//

// You would need to change the Interrupt if you use USART1,2,3, or 4

//

#IFDEF BIT(RCIE)

On Interrupt UsartRX1Ready Call readUSART

#ENDIF

#IFDEF BIT(U1RXIE)

On Interrupt UART1ReceiveInterrupt Call readUSART

#ENDIF

#IFDEF AVR

#IFDEF CHIPAVRDX

//~ Support for legacy AVR

On Interrupt UsartRX1Ready Call readUSART

#ELSE

//~ Support for AVRDX - AVRDX chips are USART0,USART1,USART2,USART3 and
USART4 not USART.

On Interrupt Usart0RXReady Call readUSART

#ENDIF

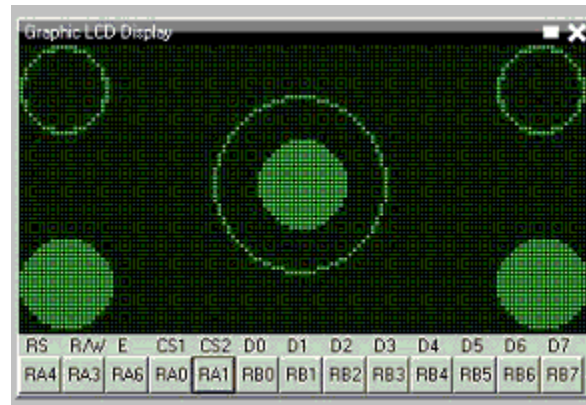
#ENDIF

End Sub

Trigonometry Circle

Explanation:

GCBASIC can draw circles on a Graphical LCD device using GCBASIC library functions.



Demonstration program:

```
;Circle and filled circle commands on a graphic LCD.
;This uses the 2-place trigonometric routines found in the include file.

;----- Configuration
#CHIP 16F88, 8           ;PIC16F88 RUNNING AT 8 MHZ
#CONFIG MCLR=OFF        ;RESET HANDLED INTERNALLY
#OPTION EXPLICIT
#DEFINE USELEGACYFORNEXT ;WILL ENSURE THE OLD FOR-NEXT Loop is used just to save
some memory as this is a very simple FOR-NEXT loop

#include <GLCD.H>
#include <TRIG2PLACES.H>

;----- Constants

;Pinout is shown for the LCM12864H-FSB-FBW
;graphical LCD available from Amazon.

;      +5V           ;LCD pin 1
;      ground       ;LCD pin 2
;      Vo = wiper of pot ;LCD pin 3
#define GLCD_DB0 PORTB.0 ;LCD pin 4
#define GLCD_DB1 PORTB.1 ;LCD pin 5
#define GLCD_DB2 PORTB.2 ;LCD pin 6
#define GLCD_DB3 PORTB.3 ;LCD pin 7
#define GLCD_DB4 PORTB.4 ;LCD pin 8
#define GLCD_DB5 PORTB.5 ;LCD pin 9
#define GLCD_DB6 PORTB.6 ;LCD pin 10
```

```

#define GLCD_DB7 PORTB.7      ;LCD pin 11
#define GLCD_CS2 PORTA.0      ;LCD pin 12
#define GLCD_CS1 PORTA.1      ;LCD pin 13
#define GLCD_RESET PORTA.2    ;LCD pin 14
#define GLCD_RW PORTA.3       ;LCD pin 15
#define GLCD_RS PORTA.4       ;LCD pin 16
#define GLCD_ENABLE PORTA.6   ;LCD pin 17
;      Vee = pot low side    ;LCD pin 18
;      backlight anode       ;LCD pin 19
;      backlight cathode     ;LCD pin 20

#define GLCD_TYPE GLCD_TYPE_KS0108
#define GLCD_WIDTH 128
#define GLCD_HEIGHT 64

;----- Variables

dim cx, cy, edge, jj as byte
dim ii as word

;----- Program

myCircle(10,10,10)
;upper left
myCircle(117,10,10)           ;upper right
myCircleFilled(63,31,10)      ;center
myCircle(63,31,20)            ;center
myCircleFilled(10,53,10)      ;lower left
myCircleFilled(117,53,10)     ;lower right

;----- Subroutines

sub myCircle(cenX, cenY, rad)
    ;Center of circle = (cenX,cenY), radius = rad

    for ii = 0 to 358 step 2          ;every two degrees
        cx = cenX -((10*rad*cos(ii))/100+5)/10 ;properly rounded x value
        cy = cenY -((10*rad*sin(ii))/100+5)/10 ;properly rounded y value

        ;the following ignores the pixel if off the screen
        if (cx>=0 and cx<=GLCD_WIDTH and cy>=0 and cy<=GLCD_HEIGHT) then
            Pset(cx, cy, on)
        end if
    next ii
end sub

sub myCircleFilled(cenX, cenY, rad)

```

```

;Center of circle = (cenX,cenY), radius = rad

for ii = 0 to 358 step 2
  cx = cenX -((10*rad*cos(ii))/100+5)/10
  cy = cenY -((10*rad*sin(ii))/100+5)/10
  edge = 2 * cenX - cx                      ;compute right edge

  for jj = cx to edge                      ;fill entire line, uses legacy for
next permitting CX to be less than edge
  if (jj>=0 and jj<=GLCD_WIDTH and cy>=0 and cy<=GLCD_HEIGHT) then
    Pset(jj,cy,on)
  end if
next jj
next ii
end sub

```

See also [Trigonometry](#), [Circle](#), [FilledCircle](#),

Graphical GCBASIC

This is the Graphical GCBASIC section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Code Documentation

Documenting GCBASIC is key for ease of use. This section is intended for developers only.

Documenting is the ability to read some extra information from comments in libraries.

Some comments that start with `'''` have a special meaning, and will be displayed as tooltips or as information to the user. These tooltips helps inexperienced users to use extra libraries.

1. GCBASIC uses `;` (a semicolon) to show comments that it has placed automatically, and `//` or `'` to indicate ones that the user has placed. When loading a program, it will not load any that start with a `;` (semi-colon). The use of comments do not impact the users but it worthy of note.
2. As for code documentation comments, GCBASIC will load information about subroutines/functions and any hardware settings that need to be set.
3. For subroutines, a line before the Sub or Function line that starts with `'''` will be used as a tooltip when the user hovers over the icon. A line that starts with `'''@` will be interpreted differently, depending on what comes after the `@`. `'''@param ParamName Parameter Description` will add a description for the parameter. For a subroutine, this will show in the Icon Settings panel under the parameter when the user has selected that icon.
4. For functions, it will show at the appropriate time in the Parameter Editor wizard. `'''@return` Returned value applies to functions only. It will be displayed in the Parameter Editor wizard when the user is asked to choose a function. An example of all this is given in `srf04.h`:

```
'''Read the distance to the nearest object
'''@param US_Sensor Sensor to read (1 to 4)
'''@return Distance in cm
Function USDistance(US_Sensor) As Word
```

5. If a subroutine or command is used internally in the library, but GCBASIC users should not see it, it can be hidden by placing `'''@hide` before the Sub or Function line. Another example from `srf04.h`:

```
'''@hide
Sub InitUSSensor
```

These should hopefully be pretty easy to add. It is also possible to add Hardware Settings. A particular setting can be defined anywhere in the file, using this syntax:

```
'''@hardware condition, display name, constant, value type
```

These comments informs GCBASIC when to show the setting. Normally, this is All, but sometimes it can include a constant, a space and then a comma separated list of values. display name is a friendly name for the setting to display. constant is the constant that must be set, and valuetype is the type that will be accepted for that constant's value. To allow for multiple value types, enter a list of types with a | between them.

6. Allowed types are:

```
free - Allows anything
label - Allows any label
condition - Allows a condition
table - Allows a data table
bit - Allows any bit from variable, or bit variable
io_pin - Allows an IO pin
io_port - Allows an entire IO port
number - Allows any fixed number or variable
rangex:y - Allows any number between x and y
var - Allows any variable
var_byte - Allows any byte variable
var_word - Allows any word variable
var_integer - Allows any integer variable
var_string - Allows any string variable
const - Allows any fixed number
const_byte - Allows any byte sized fixed number
const_word - Allows any word sized fixed number
const_integer - Allows any integer sized fixed number
const_string - Allows any fixed string
byte - Allows any byte (fixed number or variable)
word - Allows any word
integer - Allows any integer
string - Allows any string
array - Allows any array
```

7. When the library is added the program, GCBASIC will show a new device with the name of the library file on the Hardware Settings window. The user can then set the relevant constants without necessarily needing to see any code. Adding a GCBASIC library to GCBASIC will not result in any changes to the library. GCBASIC uses the information it reads to help edit the user's program, but then the user's program is passed to the compiler along with the unchanged library.
8. Hardware Settings are a bit more involved to add, but hopefully the bit of extra documentation for subroutines will be straight forward.

Windows .NET Support

From Graphical GCBASIC version 0.941 supports use on newer Windows versions without having the pre-requisite of .NET 3.5.

GCBASIC for Linux

This is the GCBASIC for Linux section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Overview - Linux Operating System

Introduction: GCBASIC can be used when using the Linux Operating System.

This instructions are not distribution specific, but are for Linux only (not Windows).

Instructions: Complete the following steps to compile and install GCBASIC for Linux:

1. Install FreeBasic from your distributions repository or <http://www.freebasic.net/wiki/CompilerInstalling>
2. Download the "GCBASIC - Linux Distribution" from SourceForge at <https://sourceforge.net/projects/gcbasic/files/>
3. Unrar/unpack GCBASIC.rar to a location of your choice within your home directory (eg. within Downloads) with either a file manager or from a console.
4. From a console, change to the GCBASIC Sources in the unpacked directory:

```
eg. cd ~/Downloads/sources/linuxbuild/
```

5. Make sure that `install.sh` is set as executable (ie. `chmod +x install.sh`), and then execute: `./install.sh build`
6. You will need root privileges for this step. You can switch user (su) to root, or optionally use `sudo`.

```
Execute: [sudo] ./install.sh install
```

7. If you su'd to root, use `exit` to drop back to your normal user. Then, be sure to follow the instructions given by the script for updating your path.
8. Confirm proper execution, and the version, of GCBASIC by executing: `gcbasic /version`

Now you can create and compile GCBASIC source files.

Programming microcontrollers:

To program your microcontroller with your GCBASIC created hex file, you will need additional programming and programmer software.

For Microchip PIC microcontroller programming, you might find what you need at:

For Atmel AVR microcontroller programming, you will need **avrdude**. It should be available in your distributions repository. If not, check here: <http://www.nongnu.org/avrdude/>

Make ASM, Make HEX and Programming Operations using the provided Linux scripts

The scripts provided are intended to assist in the creation of the ASM file (from a GCBASIC source file), creation of the HEX file (also from a GCBASIC source file) and to support programming operations (often called `FLASH`ing the microcontroller).

Script	Usage	Example
makeasm.sh	To compile the GCBASIC source program to create the ASM.	makeasm.sh sourcefile.gcb
makehex.sh	To compile and asseble the GCBASIC source program to create the ASM and a microcontroller specific HEX file.	makehex.sh sourcefile.gcb
flash.sh	To compile,asseble the GCBASIC source program to create the ASM and a microcontroller specific HEX file, and, then to program the microcontroller	flash.sh sourcefile.gcb

Examples

There are multiple constructs to run multiple programs on a single command line. The most common are ';' and '&&'

To run another command immadiately after running makehex.sh. Use the following:

```
makehex.sh sourcefile.gcb; anothercommand
```

To run another command only if makehex.sh does not exit with an error such as a compiler error. Use the following:

```
makehex.sh sourcefile.gcb && anothercommand
```


GCBASIC for ARM & Pi

This is the GCBASIC for Pi section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Overview - Raspberry Pi

Introduction:

GCBASIC can be used when using the Raspberry Pi.

You can install the command-line version of GCBASIC on a Raspberry Pi (and similar single-board computers) and use it to compile your GCBASIC programs.

You can also program most PICs and AVR using only the Pi's GPIO pins (see [\[Programming\]](#)), as well as communicate with your device over the Pi's serial port. This makes it easy to program, modify, and communicate with a PIC or AVR using just a Pi and an SSH connection.

GCBASIC is not published for ARM-based computers, there is currently no pre-compiled version for ARM-based computers, so you will have to compile it from source. The GCBASIC compiler is written in [FreeBASIC](#) (an open-source version of BASIC), so you will need to first install the FreeBASIC compiler on your Pi, then use it to compile the GCBASIC compiler from its source code. This is relatively simple.

FreeBASIC is not included in any of the major Linux repositories, but there is a customized version for ARMv7 boards like the Raspberry Pi on their [web site](#).

The following procedure will work with any ARMv7 single-board computer running a Debian derivative like [Raspberry Pi OS](#) or [Armbian](#). This includes the Raspberry Pi 2 and 3, and all single board computers with an Allwinner H2+ or H3 microprocessor (Orange Pi PC, Orange Pi Zero, Nano Pi R1, etc). It has not been tested with a Raspberry Pi 4.

Instructions:

All commands should be performed on your Pi board, either through a remote SSH terminal or using a keyboard and monitor connected to your Pi.

Installing FreeBASIC

1. Install FreeBASIC dependencies

```
$ sudo apt-get install libx11-dev libxext-dev libxpm-dev libxrandr-dev libncurses5  
libncurses5-dev
```

2. Download the latest version FreeBASIC for ARMv7 from <https://users.freebasic-portal.de/stw/builds/linux-armv7a-hf-debian/> :

```
$ cd ~
$ wget https://users.freebasic-portal.de/stw/builds/linux-armv7a-hf-
debian/fbc_linux_armv7a_hf_debian_0376_2020-09-17.zip
$ unzip fbc_linux_armv7a_hf_debian_0376_2020-09-17.zip
```

3. Install FreeBASIC

```
$ cd fbc_linux_armv7a_hf_debian/
$ chmod +x install.sh
$ sudo ./install.sh -i
```

Installing GCBASIC

1. Download and extract the GCBASIC sources:

```
$ wget "https://downloads.sourceforge.net/project/gcbasic/GCBasic%20-
%20Linux%20Distribution/GCB%40Syn.rar"
$ sudo apt install unar
# the password when requested in the next step is "GCB"
$ unar GCBASIC.rar
$ cd GCBASIC/sources/linuxbuild/
```

2. Build and install the compiler:

```
$ chmod +x install.sh
$ ./install.sh build
$ sudo ./install.sh install
```

3. Verify the compiler is properly installed and view the full list of compiler options

```
$ gcbasic
```

Now you can create and compile GCBASIC source files. For example, to compile a program you created named **myprogram.bas** into **myprogram.hex**, you could run:

```
$ gcbasic -A:GCASM -R:none -K:A -WX -V myprogram.bas
```

This will:

- use GCBASIC's internal assembler,

- turn off report creation,
- preserve all code in the assembly file output (useful for debugging)
- treat warnings as errors, and
- print compiler messages in verbose mode

Programming

To transfer your compiled .hex program files from your Pi to your microcontroller, you will need additional software.

For most PIC microcontrollers, you should use [PICkitPlus for Pi](#). PICKitPlus support the widest range of PICs including the latest PICs. It is fully supported application.

For AVR microcontrollers, you will need [avrdude](#). It should be available in your distribution's repository. If not, check here: <http://www.nongnu.org/avrdude/> . Instructions on how to use it can be found [here](#).

GCBASIC for Apple macOS

This is the GCBASIC for Apple macOS section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Overview - Apple macOS GCBASIC

Introduction

The GCBASIC compiler can be used with the Apple macOS operating system. It should run on any version from Snow Leopard 10.6 and above. It is guaranteed to run on both High Sierra 10.13 and Mojave 10.14 which have been extensively tested.

You have a choice to make. You can either:

1. download a macOS installer which will install a precompiled 64 bit binary for the GCBASIC compiler along with support files and some optional components; or
2. download, compile and install the GCBASIC compiler from the source files.

There are instructions below for both choices. If I was you, I would use the macOS GCBASIC Installer and save valuable programming time :-)

Instructions for using the GCBASIC Installer

1. Download the GCBASIC - macOS Installer disk image (.dmg) file from <https://sourceforge.net/projects/gcbasic/files/GCBasic-macOS-Installer.dmg/download>
2. Double click the .dmg file to mount it on your Desktop and a window will open which contains the Installer.
3. Double click the REAME_FIRST.txt file and read it for any important information you may need before proceeding.
4. Double click the GCBASIC icon and follow the installer prompts.

Instructions for building your own GCBASIC binary

Complete the following steps to compile and install the GCBASIC compiler:

1. Download the FreeBASIC 1.06 macOS binary compilation from: <http://tmc.castleparadox.com/temp/fbc-1.06-darwin-wip20160505.tar.bz2>
2. Download the GCBASIC UNIX Source Distribution from SourceForge at <http://gcbasic.sourceforge.net/Typesetter/index.php/Download>
3. Note: the following instructions assume the distribution file is named GCBASIC-UNIX-v0_98_05.rar however the version number (v0_98_05) may change before these instructions are updated, so you may have to adjust the filename to match the version you have downloaded.

4. Unfortunately Apple replaced the gcc compiler with the clang compiler and FreeBASIC needs the real gcc due to a certain use of goto... so, you can compile your own version of gcc following the instructions at <https://solarianprogrammer.com/2017/05/21/compiling-gcc-macos/> or you can take the low road and just download the pre-compiled binary version from <https://github.com/sol-prog/macos-gcc-binary/releases/download/v8.3/gcc-8.3.macos.tar.bz2>
5. Open a Terminal window (Terminal can be found in Applications > Utilities).
6. Move gcc-8.3.tar.bz2 from your Downloads directory to your Home directory by typing the following command in your Terminal window:

```
mv ~/Downloads/gcc-8.3.tar.bz2 ~/
```

6. Unpack the gcc-8.3.tar.bz2 compressed tar file by typing the following command into your Terminal window:

```
gzcat gcc-8.3.tar.bz2 | tar xvf -
```

This will produce a new directory called gcc-8.3.

7. You now need to link the binary gcc-8.3 to just gcc by typing the following commands into your Terminal window:

```
cd gcc-8.3
ln -s gcc-8.3 gcc
cd ..
```

8. Move the gcc-8.3 directory to the /usr/local/ directory by typing the following commands into your Terminal window:

```
sudo mv gcc-8.3 /usr/local
```

Note: You will be asked for your password to execute the above command.

9. Ensure that the Apple Developer Xcode app is installed. Xcode can be downloaded and installed from the App Store for free.
10. Ensure that the Xcode command line tools are installed by typing the following command in your Terminal window:

```
xcode-select --install.
```

11. Move the FreeBASIC compressed tar file from your Downloads directory to your home directory by typing the following command in your Terminal window:

```
mv ~/Downloads/fbc-1.06-darwin-wip20160505.tar.bz2 ~/
```

12. Unpack the FreeBASIC compressed tar file by typing these commands in your Terminal window:

```
gzcat fbc-1.06-darwin-wip20160505.tar.bz2 | tar xvf -
```

This will produce a new directory called fbc-1.06.

13. Move the GCBASIC compressed tar file from your Downloads directory to your home directory by typing the following command in your Terminal window:

```
mv ~/Downloads/GCBASIC-UNIX-v0_98_05.rar ~/
```

14. Unpack the GCBASIC compressed tar file by typing these commands in your Terminal window:

```
unrar x GCBASIC-UNIX-v0_98_05.rar
```

This will produce a new directory called GCBASIC. **Note:** If you do not currently have the unrar program which can extract rar file archives you can download and install it for free from the App Store.

15. Change to the GCBASIC/Sources directory by typing this command in your Terminal window:

```
cd ~/GCBASIC/Sources
```

16. Compile the GCBASIC binary (gcbasic) by typing the following command into your Terminal window:

```
sh DarwinBuild/build.sh
```

Note 1: If you did not install the various files with the same names as in the instructions above into your Home directory, you will need to edit the build.sh script file and change the file paths and filenames to the appropriate values.

Note 2: You may need to alter the library and include paths in the build.sh script depending on your version of macOS (it is currently setup for the Xcode High Sierra 10.13 and Mojave 10.14 versions of macOS).

17. Confirm the proper execution, and the version, of GCBASIC by typing the following command in the Terminal window:

```
gcbasic
```

Now you should be able create GCB source files with your favourite editor and compile those files with the gcbasic compiler.

Programming microcontrollers

To program your microcontroller with your GCBASIC-created hex file, you will need additional hardware and software.

1. For Microchip PIC microcontroller programming, you might find what you need at: <https://www.microchip.com/DevelopmentTools/ProductDetails.aspx?PartNO=pg164120> and the macOS version of the `pk2cmd` v1.2 command line programming software.
2. For Atmel AVR microcontroller programming, you will need the `avrdude` programming software. Check here: <http://www.nongnu.org/avrdude/> for it.

Alternatively, if you use Virtual Machine software such as *Parallels* or *VMWare Fusion* to run Windows programs, you can use Windows GUI programming software.

- For Microchip, the PICKit 2 and PICKit 3 standalone GUI software or even better the PICKitPlus software (<https://sourceforge.net/projects/pickit3plus/>) for both the PICKit 2 (<https://www.microchip.com/DevelopmentTools/ProductDetails.aspx?PartNO=pg164120>) and PICKit 3 (<https://www.microchip.com/Developmenttools/ProductDetails/PG164130>) which has fixed various bugs in those programs and been updated to program the latest Microchip 8 bit microcontrollers.

Help

GCBASIC Help documentation is installed in the Documentation subdirectory in your GCBASIC directory.

If at any time you encounter an issue and need help, you will find it over at the friendly GCBASIC discussion forums at <https://sourceforge.net/p/gcbasic/discussion/>

GCBASIC for FreeBSD

This is the GCBASIC for FreeBSD OS section of the Help file. Please refer the sub-sections for details using the contents/folder view.

Overview - FreeBSD GCBASIC

Introduction

The GCBASIC compiler can be used with the FreeBSD operating system.

Instructions for using the GCBASIC install.sh script

Complete the following steps to compile and install the GCBASIC compiler for FreeBSD:

1. Download one of the nightly builds of FreeBASIC 1.06 for the FreeBSD 32 bit or 64 bit binary compilation from: <http://users.freebasic-portal.de/stw/builds/freebsd32/> (32 bit) or <http://users.freebasic-portal.de/stw/builds/freebsd64/> (64 bit) The filenames are in the format fbc_freebsd[32|64]_[BuildNumber]_[Date].zip.
2. Download the GCBASIC UNIX Source Distribution from SourceForge at <https://gcbasic.sourceforge.net/Typesetter/index.php/Download>
3. Move the FreeBASIC ZIP file from your download directory to your home directory.
4. Unzip the FreeBASIC ZIP file which will produce a new directory called `fbcb_freebsd[32|64]`. The FreeBASIC compiler `fbcb` is in the `bin` subdirectory. You should add the path to `fbcb` to your path so that the installation script can find it.
5. Move the GCBASIC compressed tar file from your download directory to your home directory.
6. Unpack the GCBASIC compressed tar file by typing the command below. **Note:** the version number (v0_98_05) in the filename may change before these instructions are updated - adjust depending on the version number of the file you downloaded.

```
unrar x GCBASIC-UNIX-v0_98_05.rar
```

This will produce a new directory called GCBASIC. **Note:** If you do not already have the unrar program installed you can either compile it from the ports collection or use the pkg command to install the binary and any required dependancies.

7. Change to the `GCBASIC/Sources` directory.
8. Execute the FreeBSDBuild/install.sh shell script from the Sources directory.

```
sh FreeBSDBuild/install.sh [all | build | install]
```

The build script arguments are:

- *all* - will compile **and** install the GCBASIC compiler and its support files.
- *build* - will just compile the binary for the GCBASIC compiler.
- *install* - will install the GCBASIC compiler and its support files.

When choosing *all* or *install* you will be prompted for an installation directory. The default is `/usr/local/gcb-[version]` for which you will need to run the installation script as root. Alternatively, you can choose to install in your home directory (eg `~/bin/gcb`). The installation script will automatically append the GCBASIC version so that directory would become `~/bin/gcb-[version]`

9. Add the directory where you installed `gcbasic` to your path, or use the full path to the `gcbasic` installation directory and confirm the proper execution, and the version, of GCBASIC by executing `gcbasic`.

Now you should be able create GCB source files with your favourite editor and compile those files with the GCBASIC compiler.

Programming microcontrollers

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1. For Microchip PIC microcontroller programming, you might find what you need at: <https://www.microchip.com/DevelopmentTools/ProductDetails.aspx?PartNO=pg164120> and the FreeBSD version of the `pk2cmd` v1.2 command line programming software.
2. For Atmel AVR microcontroller programming, you will need the `avrdude` programming software. `avrdude` can be compiled and installed from the FreeBSD ports directory or the precompiled binary and any missing dependancies can be installed using `pkg install avrdude`.

Alternatively, if you use Virtual Machine software such as *Virtual Box* to run Windows programs, you may be able to use Windows GUI programming software.

- For Microchip, the PICKit 2 and PICKit 3 standalone GUI software or even better the PICKitPlus software (<https://sourceforge.net/projects/pickit3plus/>) for both the PICKit 2 (<https://www.microchip.com/DevelopmentTools/ProductDetails.aspx?PartNO=pg164120>) and PICKit 3 (<https://www.microchip.com/Developmenttools/ProductDetails/PG164130>) which has fixed various bugs in those programs and been updated to program the latest Microchip 8 bit microcontrollers.

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GCBASIC Maintenance and Development

This is the GCBASIC maintenance section of the Help file. Please refer the sub-sections for details using the contents/folder view.

GCBASIC Maintenance

Introduction: GCBASIC maintenance covers the key processes that the developers use to maintain and build the solution.

These insights are not distribution specific.

Solution Architecture: These components are key for a complete solution:

1. GCBASIC installer
2. GCBASIC chip specific .DAT files
3. GCBASIC Help
4. GCBASIC IDE

GCBASIC installers:

The Windows GCBASIC installer uses the *InnoSoft* installer with packaging completed using R2Build.

The process uses a Gold build structure. The R2Build software creates four packages for Windows and one package for the Linux distribution. The process is automated with automatic versioning and configuration.

The macOS GCBASIC installer uses the *Packages* installer (<http://s.sudre.free.fr/Software/Packages/about.html>) with packaging completed using the Bourne shell script `pkg2dmg.sh` to create a compressed disk image file containing the installer.

GCBASIC chip specific .DAT files:

What are the .DAT files?

The DAT files are the GCBASIC representation of the capabilities of a specific microcontroller. The DAT is based upon a number of vendor sources and corrections/omissions added by the GCBASIC development team. The DAT file is exposed to the user program as a set of registers and register bits that can be used to configure the program in terms of the microcontroller specifics.

The process to create the .DAT file for microcontrollers is as follows:

Step	Description
1	Obtain the MPASM *.INC or the AVR *.XML files to be used. These files determine the scope of registers and register bits.
2	For Microchip only. Place the source INC files in the ..DAT\incfiles\OrgFiles. Process the file using 'Preprocess.bat'. This is an AWK text processor - you will need AWK.EXE in the executing folder. This preprocessing will examine all the INC files in the ..DAT\incfiles\OrgFiles folder. The resulting files will be placed in the ..DAT\incfiles folder. The resulting files will have the 'BITS' section sorted in port priority - this priority ensures the bits are assigned in the target DAT in the same order every time.
3	Update the database of support microcontrollers. This database contains the microcontroller configuration that GCBASIC requires as the core information for the DAT files. The database is called chipdata.csv or avr_chipdata.csv for Microchip and AVR respectively. - These files are comma delimited. The first row of data specifies the field name - these field names control the chip conversion program, see later notes. The database fields are controlled by the GCBASIC development team and the specification of the database may change between releases to support new capabilities. Database fields will have the suffix of Variant such as PWMTimerVariant and SMTClockSourceVariant and are microcontroller specific configuration settings to support the various microcontroller settings. The values of these variants are determined by the examination of the microcontroller datasheets as this information is NOT specified in the source files. Variants are exposed in the user program with the prefix of Chip. If the variant is called PWMTimerVariant a constant called ChipPWMTimerVariant will be exposed in the user program.
4	Update the CriticalChanges.txt file, if required. The CriticalChanges.txt file contains changes to the INC file during processing that are corrections or additions to the source files. The format of each line is the filename , Append (new line) or Replace, find, replace . Each line is comma delimited and spaces are NOT critical. Essentially, the processing with find a partial line and replace or append the whole line. example: p16lf1615.inc,R,OSCFIE EQU H'0007',OSFIE EQU H'0007' - so, when processing p16lf1615.inc find the line OSCFIE EQU H'0007' and replace with OSFIE EQU H'0007' .
5	If required. This is not normally edited. Update the 18FDefaultASMConfig.txt to set the 18f configuration defaults.
6	Maintain the conversion program. The conversion program may require maintenance. The programs are written in FreeBASIC and therefore require compilation. An example of maintenance is a new variant field is required. The source program will need to be updated to support the new variant - simply edit the source, compile and publish. Another example is the addition of a new Interrupt - follow the same process to edit, compile and publish.

Step	Description
7	Execute the program to convert the source files to the DAT files for Microchip or AVR. There are two programs for each architecture. Executing the conversion program without a parameter will process ALL the entries in the database (the csv file), passing a single parameter to the conversion program will only convert the single microcontroller. The conversion program will process as follows: a) Read the database for the chip specifics b) If a .DEV file or .INFO file is not present a routine called GuessDefaultConfig is invoked. This method sets the bit(s). In all cases the default mask is sometimes specified for a particular config option and that is used for ASMConfig See the section below for the processing of a .DEV file. c) For all microcontrollers read the CriticalChanges.txt file and process. d) For 18f microcontrollers read the 18FDefaultASMConfig.txt . This simply overwrites all options stated in 18FDefaultASMConfig.TXT and marks this in the output DAT file. e) Create the output DAT file.
8	Test and publish the DAT file(s) to the distribution as required.

An example the processing of a .DEV.

This is the 18F25K20 example. For this microcontroller **Disabled** is default:

```
<SWITCH_INFO_TYPE><FCMEN><Fail-Safe Clock Monitor><40><2>
<SETTING_VALUE_TYPE><OFF><Disabled><0>
<SETTING_VALUE_TYPE><ON><Enabled><40>
```

Where the default is selected from the Info_Type.

Prog = . An explanation of the parameter. The Prog value is measured in words. It is the same in the device specific.dat files.

Microchip in the past have used words, but then they started using bytes on the website instead to make their chips appear to have larger capacity.

An example: If a device has 8192 words, which is $8192 * 14 = 114688$ bits, or 14336 bytes. It is an odd measurement because dividing 14336 by 14/8 to see how many instructions you can use is extra maths work within the compiler.

The GCBASIC PROGram memory analysis are in words.

Development Guide

There are lots of ways to contribute to the GCBASIC project: coding, testing, improving the build

process and tools, or contributing to the documentation. This guide provides information that will not only help you get started as a GCBASIC contributor, but that you will find useful to refer to EVEN if you are already an experienced contributor.

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We know EVEN before you start contributing that getting set up to work on GCBASIC and finding a bug that is a good fit for your skills can be a challenge. We are always looking for ways to improve this process: making GCBASIC more open, accessible, and easier to participate with. If you are having any trouble following this documentation, or hit a barrier you cannot get around, please contact us via the discussion forum. We will solve hurdles for new contributors and make GCBASIC better.

This section addresses developing libraries but this guide is appropriate to any GCBASIC development.

The section covers the recommended programming style, Constants, Variables, Script syntax (gotchas) and tab usage.

PROGRAMMING STYLES

Indenting is standardized.

All scripts within a specific library should be the first major section the library. Scripts within methods should not be used.

Some #defines may need to be placed before the script to provide clarity to the structure of the library.

```
#startup  startupsub

#Define I2C_ADDRESS_1  0x4E    'The default address if user does not specify in
the user program

#SCRIPT
    ... code script
    ... code script
    ... code script
#ENDSCRIPT
```

Scripts support structures like IF <CONDITION> THEN <ACTION> END IF. Scripts supports the <condition> argument that must generate a TRUE result, meaning that at a literal level, your conditional formatting rule is an If - THEN statement along the lines of "If this condition is TRUE, THEN process the <ACTION>". the condition must use logical "AND" and "OR" to test two conditions. Using

"AND" or "OR" reduces the script size, however, it is essential the the conditional test(s) are valid. If a test fails then you may not get the results you expect.

```
IF .. THEN

END IF
```

CONSTANTS

A constant is a value that cannot be altered by the program during normal execution. Within GCBASIC there are two ways to create constants. 1. with the **#DEFINE** instruction, or, 2. via ``#SCRIPT .. #ENDSCRIPT'`. ; Within a script constants can be created and changed. A script is process that is executed prior to the GCBASIC source program is processing the main user program.

An example of using **#DEFINE** is

```
#DEFINE TIME_DELAY_VALUE    10
```

The script construct is

```
#SCRIPT
    'Create a constant
    TIME_REPEAT_VALUE = 10
#ENDSCRIPT
```

Guide for constants

The following rules are recommended.

- 1 - All CONSTANTS are capitalized
- 2 - Do not define a constant in a library unless required
- 3 - Create all library constants within a script (see example below **Constrain a Constant Example** on how to constrain a constant)
- 2 - Underscores are permitted in constant names within Scripts **
- 3 - No prefix is required when a CONSTANT is PUBLIC. A PUBLIC constant is one that the user sets or the user can use.
- 4 - Prefix CONSTANTS with SCRIPT_ when the CONSTANT is used outside of the library specific script section AND ARE NOT EXPOSED AS PUBLIC Constants.

5 - Prefix CONSTANTS with __ (two underscores) when the CONSTANT is ONLY used inside the library specific script section

6 - For PUBLIC prefix CONSTANTS with the capability name, _ (one underscore) and then a meaningful title, as follows GLCD_HEIGHT SPISRAM_TYPE

7 - All scripts within a specific library should be the first major section the library. Scripts within methods (Sub, Functions) should not be used.

8 - All scripts within a specific library should be the first major section the library. Scripts within methods (Sub, Functions) should not be used.

9 - Other naming recommendations. Do not use underscores in subroutine, function or variable names

Example script within a library

```
#startup  startupsub
#DEFINE I2C_ADDRESS_1  0x4E    'Default address if user omits
#SCRIPT
    'script code
    'script code
    'script code
#ENDSCRIPT
```

Simple Example

```
#SCRIPT  'Calculate Delay Time
    __LCD_DELAY  = ( __LCD_TIMEPERIOD - __LCD_DELAYS) - (INT((4/ChipMHZ) *
__LCD_INSTRUCTIONS))
    SCRIPT_LCD_POSTWRITDELAY = __LCD_DELAY
    SCRIPT_LCD_CHECKBUSYFLAG = TRUE
#ENDSCRIPT

'usage within user code or code outside of script
#IF SCRIPT_LCD_CHECKBUSYFLAG = TRUE
    WaitForReady    'Call subroutine to poll busy flag
    set LCD_RW OFF  'Write mode
#ENDIF
WAIT SCRIPT_LCDPOSTWRITDELAY us
```

Create Constant Example

Background: All constants are always processed, regardless of where they are placed in the user code or library. This includes any constant defined anywhere in user code or any library - the constant will

be processed and the constant will be defined. The only method to constrain a constant is via a script.

The following code segment will not constrain the constant. The constant **MYCONSTANT** will be created. The **#IFDEF PIC** will not constrain even if an AVR or LGT chip.

```
#IFDEF PIC
    #DEFINE MYCONSTANT 255
#ENDIF
```

The recommended method follows. The constant will only be created when a PIC.

```
#SCRIPT
  IF PIC then
    MYCONSTANT = 255
  End IF
#ENDSCRIPT
```

Constrain a Constant Example

An example to constrain a constant is to test if a user constant is define in the user source program. In this example the constant **SENDALOW** is defined in user source program.

- If yes, then define the library specific constants.
- If no, then do not define the library specific constants.

Using the method below defines constants only when the user requires the constants assuming they have defined **SENDALOW** in the user source program.


```

#SCRIPT
  IF SENDALOW then
    NONE = 0 : ODD = 1 : EVEN = 2 : NORMAL = 0 : INVERT = 1
    WAITFORSTART = 128 : SERIALINITDELAY = 5
  END IF

  IF SENDALOW then
    NONE = 0 : ODD = 1 : EVEN = 2 : NORMAL = 0 : INVERT = 1
    WAITFORSTART = 128 : SERIALINITDELAY = 5
  END IF

  IF SENDALOW then
    NONE = 0 : ODD = 1 : EVEN = 2 : NORMAL = 0 : INVERT = 1
    WAITFORSTART = 128 : SERIALINITDELAY = 5
  END IF
#ENDSCRIPT

```

SCRIPTS VARIABLES

Scripting has the concept of variable that can be used within the script. The variables are NOT available as variables to a user program or a library beyond the scope of the script. The variables are available to a user program as constants. The variables will be integer values, if accessed in a user program.

SCRIPT SYNTAX

Scripting support the preprocessing of the program to create specific constants. Scripting has a basic syntax and this is detailed in the HELP. However, this guide is intended to provide insights into the gotchas and best practices.

Script Insights

Scripting handles the creation of specific constants that can be used within the library. Many libraries have script to create constants to support PWM, Serial, HEFSAF etc.

You can use the limited script language to complete calculations using real numbers but you MUST ensure the resulting constant is an integer value. Use the IN() method to ensure an integer is assigned.

You can use IF-THEN-ENDIF but if your IF conditional test uses a chip register or a user define constant then you must ensure the register or constant exists. If you do not check the register or constant exists the script will fail to operate as expected.

There is limited syntax checking. You must ensure the quality of the script by extensive testing.

```
int( register +1s)) 'Will not create an error, but, simple give an unexpected result.
```

TAB USAGE AND INDENTING

Four spaces are to be used. A tab is not permitted

Example follows where the indents are all four spaces.

```
sub ExampleSub (In VariableName)
  select case VariableName
    case 1
      Do This
    case 2
      Do That
  end select
end sub
```

Not like this:

```
SUB ExampleSub (In VariableName)
  Select Case VariableName
    Case 1
      Do This
    Case 2
      Do That
  End Select
End SUB
```

and, not like this

```
Sub ExampleSub (In VariableName)
Select Case VariableName
Case 1
Do This
Case 2
Do That
End Select
End Sub
```

OPTION REQUIRED

#Option Required supports ensuring the microcontroller has the mandated capabilities, such as EEPROM, HEF, SAF, USART.

Syntax:

```
#option REQUIRED PIC|AVR CONSTANT %message.dat entry%
#option REQUIRED PIC|AVR CONSTANT "Message string"
```

This option ensure that the specific CONSTANT exists within a library to ensure a specific capability is available with the microcontroller.

This will cause the compiler check the CONSTANT is a non zero value. If the CONSTANT does not exist it will be treated as a zero value.

Example:

This example tests the CONSTANT CHIPUSART for both the PIC and AVR microcontrollers. If the CONSTANT is zero or does not exist then the string will be displayed as an error message.

```
#option REQUIRED PIC CHIPUSART "Hardware Serial operations. Remove USART commands to
resolve errors."
#option REQUIRED AVR CHIPUSART "Hardware Serial operations. Remove USART commands to
resolve errors."
```

RAISING COMPILER ERROR CONDITIONS

From build 1131 the compiler now supports raising a compiler error message.

The method uses the ``RaiseCompilerError "<string>"|<string>"`` method to pass an error message to the compilation process.

An example from USART.H/INITUSART subroutine is shown below. This example tests for the existence

of one of the three supported baud rate constants. If none of the constants exist and the constant (in this example) `STOPCOMPILERERRORHANDLER` does not exist the `RaiseCompilerError` with the string will be passed to the assembler for error processing. This permits the inspect of the user program with appropriate messages to inform the user.

```
....
#ifdef ONEOF(USART_BAUD_RATE,USART1_BAUD_RATE,USART2_BAUD_RATE) THEN
  'Look for one of the baud rates CONSTANTS
  #ifndef STOPCOMPILERERRORHANDLER
    'Use one of the following - the string MUST be start and end with a double quote

    ' Use the message.dat file
    ' RaiseCompilerError "%USART_NO_BAUD_RATE%"

    ' Use hard code text
    ' RaiseCompilerError "USART not setup correctly. No baud rate specified - please
correct USART setup"

    RaiseCompilerError "%USART_NO_BAUD_RATE%"

  #endif
#endif
....
```

The `RaiseCompilerError` handler can be stopped using the constant `STOPCOMPILERERRORHANDLER` as shown above.

LCD ERROR HANDLING

The setup of an LCD is inspected and an appropriate error message is displayed. The Compiler now controls error messages when LCD is not setup up correctly. This the text displayed is held in the `messages.dat` file - `LCD_Not_Setup` entry.

Development Guide for GCBASIC.EXE compiler

There are lots of ways to contribute to the GCBASIC project: coding, testing, improving the build process and tools, or contributing to the documentation. This guide provides information that will not only help you get started as a GCBASIC contributor, but that you will inform you as an experienced contributor wanting to help.

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This addresses the changes and updates to the GCBASIC compiler.

BACKGROUND

The compiler was created by Dr. Hugh Considine when he was 12 years old. That was in 2005. Hugh came up with the idea for a new compiler - of the then available compilers - they were hard to use and not free. And, he had some spare time.

Hugh believes that GCBASIC should be free to all - forever.

The original software was called Great Cow BASIC, but, he had some resistance in getting high schools in Australia to use and agree to the use of text based programming. Graphical GCBASIC was created to address the need for a graphical user interface. Graphical GCBASIC acts like a set of training wheels. The concept of Graphical GCBASIC is that the icons make it less scary, and since they all share names with the BASIC commands it is then easy to remember what command corresponds to each icon.. Using Graphical GCBASIC users can then switch to text mode whenever they want to, go backwards and forwards a few times if they want, and finally end up using just the text programming. It is a journey from a graphical user interface to text based programming.

Those who already have programming experience can go straight to GCBASIC, while those who would prefer a lighter learning curve can take the Graphical GCBASIC option. The two approaches targets two different sets of users who ultimately want to do the same thing.

As for the **name**, it was the fourth name Hugh tried. First name was BASPIC, but it did not seem memorable enough. Then, he considered some animal names - first thought was Chipmunk BASIC, but someone already used that! Then, Bear BASIC, but decided against it on finding out the slang meaning of bear. Final name was GCBASIC, which is named after something his sister and he came up with (when aged 12 and 15!!). No-one else had that name, it had no meanings that could offend, and it was something odd enough to be memorable, so Great CoW CBASIC it was.

In 2013 Evan Venn joined the team as a compiler developer, with others joining in Bernd Dau, Trevor Roydhouse, Pete Everett, Theo Loermans, Giuseppe D'Elia, Derek Gore, Ian Smith, Bernd Dau, Theo Loermans, Urs Hopp, Kent Schafer, and Frank Steinberg. Some those that joined in drove changes to

the compiler, some changed the source code, some built tools and some built libraries. They all had one thing in common - improvements to the GCBASIC compiler.

In 2021 we are still having new developers join the project like ToniG adding a new capability for handling Tables.

In 2023 we renamed to GCBASIC. The Cow is now deadbeef ... a hex number .. :-).

THE COMPILER

The compiler executable is called GCBASIC.EXE. The compiler source is written in FreeBASIC. FreeBASIC is a multiplatform, free/open source (GPL) BASIC programming language and a compiler for Microsoft Windows, protected-mode MS-DOS (DOS extender), Linux and FreeBSD.

The official website is <https://www.freebasic.net/>

FreeBASIC provides syntax compatibility with programs originally written in Microsoft QuickBASIC (QB). FreeBASIC is a command line only compiler, unless users manually install an external integrated development environment (IDE) of their choice. IDEs specifically made for FreeBASIC include FBide and FbEdit, while more graphical options include WinFBE Suite and VisualFBEditor.

The source code is Open Source. And, has a GNU GENERAL PUBLIC LICENSE.

The source code for the compiler can be found on [SourceForge](#)

Use SVN to UPDATE and COMMIT code changes. You require developer access to SourceForge but if you have got this far then you already know this. You are therefore required to use SVN for source code management.

When COMMITting you MUST update the change log, then, when you commit an update use the change log with the SourceForge commit number. Then, add the new change at the end of the change log. The COMMIT message should be the same as the description in the change log. Add the [COMMIT NUMBER] to the description in the change log to show the COMMIT number.

You will find the changelog [here](#). The change log is an EXCEL spreadsheet.

COMPILER ARCHITECTURE

The compiler is relatively simple in terms of the architecture. There is a main source program with a set of header files that contain other methods or declarations. The GCBASIC header files are the following:

1. `preprocessor.bi` - methods, statements, defines, declarations, prototypes, constants, enumerations, or similar types of statements
2. `utils.bi` - methods that are shared across the architecture
3. `variables.bi` - methods that control the creation and management of variables
4. `assembly.bi` - methods specific to the generation of GCAssembler (GCASM)
5. `file.bi` - the FreeBASIC files library
6. `string.bi` - the FreeBASIC string library

The supporting files are:

1. `messages.dat` - the English messages source file. All user messages from the compiler are sourced from this file.
2. `reservedwords.dat` - the list of system reserved words

The compiler process is simple. The process, shown below, generates the ASM source and the HEX file from the user source program.

1. Create the indexes
2. Declare the methods, arrays and variables
3. Process the user source programs using PreProcessor method. This includes
 - i. Loading of all source files including including files
 - ii. Translate files, if needed
 - iii. Examine source for comments, tables, `asm`, `rawasm`, `functions`; `subs`; `macros`, set origin of valid code


```
Origin = ";?F" + Str(RF) + "L" + Str(LC) + "S" + Str(SBC) + "?"
```

```
RF = File number
```

```
L = Line number in source file
```

```
S = Sub Routine number
```
 - iv. Find compiler directives, except `SCRIPT`, `ENDSCRIPT`, `IFDEF` and `ENDIF` - including all the `#DEFINEs` outside of conditional statements
 - v. If `GLCD_TYPE` in user source program is found, then, determine the library and load that library with all dependent libraries. This method improves compiler performance by only loading the required libraries
 - vi. `ReadChipData`
 - vii. `CheckClockSpeed`
 - viii. `ReadOptions`
 - ix. `PreparePageData`
 - x. `PrepareBuiltIn`. Initialise built-in data, and prepare built-in subs.
 - xi. `RunScripts`
 - xii. `BuildMemoryMap`
 - xiii. `Process samevar` and `samebit`
 - xiv. `RemIfDefs`. Remove any `#IFDEFs` that do not apply to the program.

- xv. Prepare programmer, need to know chip model and need to do this before checking config
- xvi. Replace Constants
- xvii. Replace table value. Replace constants and calculations in tables with actual values
- 4. Compile the program using the CompileProgram method
 - i. Compile calls to other subroutines, insert macros
 - ii. Compile DIMs again, in case any come through from macros
 - iii. Compile FOR commands
 - iv. Process arrays
 - v. Add system variable(s) and bit(s)
 - vi. Compile Tables
 - vii. Compile Pot
 - viii. Compile Do
 - ix. Compile Dir
 - x. Compile Wait
 - xi. Compile On Interrupt
 - xii. Compile Set(s)
 - xiii. Compile Rotate
 - xiv. Compile Repeat
 - xv. Compile Select
 - xvi. Compile Return
 - xvii. Compile If(s)
 - xviii. Compile Exit Sub
 - xix. Compile Goto(s)
- 5. Allocate RAM using the AllocateRAM method
- 6. Optimise the generated code using the TidyProgram method
- 7. Combine and locate the subroutines and functions for the selected chip using the MergeSubroutines method
- 8. Complete the final optimisation using the FinalOptimise method
- 9. Write the assembly using the WriteAssembly method
- 10. Assemble and generate the hex file using GCASM, MPASM, PICAS or some other define Assembler
- 11. Optionally, pass programming operations to the programmer
- 12. Write compilation report using the WriteCompilationReport method
- 13. If needed, write the error and warning log using the WriteErrorLog method
- 14. Exit, setting the ERRORLEVEL

Note #1: Constants are can be created in many places and the order is critical when trying to understand the process.

Step 3.iv; Step 3.xi, 3.xiv and xvi. These are Find compiler directives; Runscripts, process IFDEFs and replace Constants values respectively. This means constants that are not created by the Find compiler directives step are clearly not available in the RunScripts step, and the same applies to the process IFDEFs step. So, please consider the order of constant creation in terms of these steps. Always think about the precedence of constant creation.

Note #2: When using IFDEFs Conditional statements you should #UNDEFINE all constants prior to #DEFINE. Whilst there will be cases where the constant does not exist, or where the Preprocessor can determine the outcome of the Conditional statements there will be cases, specifically nested IFDEFs Conditional statements, where you will be required to use #UNDEFINE to remove all warnings.

Note #3: Good practice is NOT to create constants in a library where the user can overwrite the value of the same constant. You must determine if the user has created the constant and then create a default value if the user has not defined a value. An example:

```
IF NOTDEF(AD_DELAY) THEN
    'Acquisition time. Can be reduced in some circumstances - see PIC manual for details
    AD_DELAY = 2 10US
END IF
```

This will create the constant AD_DELAY only when the user program does not define a value.

FreeBASIC COMPILATION OF GCBASIC SOURCE CODE

The compiler is relatively simple in terms of the compilation.

Use the following versions of the FreeBASIC compiler to compile the GCBASIC source code.

For Windows 32 bit

```
FreeBASIC Compiler - Version 1.07.1 (2019-09-27), built for win32 (32bit)
Copyright (C) 2004-2019 The FreeBASIC development team.
```

For Windows 64 bit

```
FreeBASIC Compiler - Version 1.07.1 (2019-09-27), built for win64 (64bit)
Copyright (C) 2004-2019 The FreeBASIC development team.
```

Using other version of Windows FREEBASIC compiler are NOT tested and may fail. Use the specific versions shown above.

The compile use the following command lines. Where "%ProgramFiles%" is the root location of the FreeBASIC installation, and \$SF is the location of the source files and the destination of the compiled executable.

For Windows 32 bit

```
"%ProgramFiles%\FreeBASIC\win32\fbcc.exe" $SF\gcbasic.bas -exx -arch 586 -x
$SF\gcbasic32.exe
```

For Windows 64 bit

```
"%ProgramFiles%\FreeBASIC\win64\fbcbasic.exe" $SF\gcbasic.bas -x $SF\gcbasic64.exe -ex
```

Linux, FreeBSD and Pi OS are also supported. Please see [Online Help](#) and search for the specific operating system.

FreeBASIC COMPILER TOOLCHAIN

To simplify the establishment of development environment download a complete installation from [here](#). This includes the correct version of FreeBASIC and the libraries - all ready for use. Simply unzip the ZIP to a folder and the toolchain is ready for use. For an IDE please see the information above.

BUILDING THE GCBASIC EXECUTABLE USING THE FBEDIT IDE

To build GCBASIC from the source files. The list shows the installation of the FBEdit IDE.

Complete the following:

1. Download and install FreeBASIC from url shown above.
2. Download and install fbedit from <https://sourceforge.net/projects/fbedit/?source=dlp>
3. Download the GCBASIC source using SVN into a gcbasic source folder.
4. Run fbedit (installed at step #2). Load project GCBASIC.fbp from GBASIC source folder.
5. Hit <f5> to compile.

CODING STYLES

Remember, Hugh was 12 when he started this project. You must forgive him for being a genius, but, he did not implement many programming styles and conventions that are common place today.

There is a general lack of documentation. We are adding documentation as we progress. This can make the source frustrating initially but can find the code segments as they are clearly within method blocks.

The following rules are recommended.

1. All CONSTANTS are capitalized
2. Do not use TAB - use two spaces
3. You can rename a variable to a meaningful name. Hugh used a lot of single character variables many years ago. This should be avoided in new code.
4. Document as you progress.
5. Ask for help.

COMPILER SOURCE INSIGHTS

There are many very useful methods, a lot of methods, look at existing code before adding any new method. The compiler is mature from a functionality standpoint. Just immature in terms of documentation.

COMPILER DEBUGGING

To debug or isolate a specific issue use lots of messages using PRINT or HSERPRINT Both of these methods are easy to setup and use.

Specific to #SCRIPT you can use WARNING messages to display results of calculations or assignments.

Specific to CONDITIONAL Compilation use `conditionaldebugfile` (see above) to display conditional statement debug for the specified file. Options are any valid source file or nothing. Nested conditions are evaluated sequentially, therefore the first, second, third etc etc. The compiler does not at this point rationalise the hierarchy of nested conditions. It simply finds a condition and then matches to an #ENDIF. So, the compiler walks through the nested conditions as the outer nested, then the next nest, the next nest etc. etc. This compiler is completing the following actions:

1. If the conditional is not valid. Remove the code segment include the #IF and the #ENDIF
2. If the conditional is valid. Remove the just the #IF and the #ENDIF

So, in this context the compiler walks the code many times (as these are lists not arrays this is blindly fast) removing code segments.

The following program shows the impact of nested conditions.. Each nest is evaluated until all conditions have been assessed.. See the comment section of the listing to see the output from the debugging.

```
#CHIP 18F16Q41
#OPTION EXPLICIT

; ----- Add the following line to USE.ini -----
```

```

;
; conditionaldebugfile = IFDEF_TEST.gcb
;
; -----

#IFDEF PIC
  #IFDEF ONEOF(CHIP_18F15Q41, CHIP_18F16Q41)
    #IF CHIPRAM = 2048 'TRUE
      #IF CHIPWORDS = 32768 ' TRUE
        #IFDEF VAR(NVMLOCK) 'TRUE
          #IFDEF VAR(OSCCON2) 'TRUE
            #IFDEF VAR(NVMCON0) 'TRUE    set var1 to 1
              DIM _VAR1
              _VAR1 = 1
            #ENDIF
          #ENDIF
        #ENDIF
      #ENDIF
    #ENDIF
  #ENDIF

  #IF CHIPRAM = 4096 'TRUE
    #IF CHIPWORDS = 32768 ' TRUE
      #IFDEF VAR(NVMLOCK) 'TRUE
        #IFDEF VAR(OSCCON2) 'TRUE
          #IFDEF VAR(NVMCON0) 'TRUE    = set var1 to 0
            DIM _VAR1
            _VAR1 = 0
          #ENDIF
        #ENDIF
      #ENDIF
    #ENDIF
  #ENDIF
#ENDIF

Do
Loop

// =====
// *** Below is debugger output for this file ***
// =====

// GCBASIC (0.99.02 2022-07-21 (Windows 32 bit) : Build 1143)

// Compiling c:\Users\admin\Downloads\IFDEF_TEST.gcb

//          13: #IFDEF PIC
//          15: #IFDEF ONEOF(CHIP_18F15Q41, CHIP_18F16Q41)

```

```

//      17: #IF CHIPRAM = 2048
//      19: #IF CHIPWORDS = 32768
//      21: #IFDEF VAR(NVMLOCK)
//      23: #IFDEF VAR(OSCCON2)
//      25: #IFDEF VAR(NVMCON0)
//      ;DIM _VAR1
//      27: DIM _VAR1
//      ;_VAR1 = 1
//      28: _VAR1 = 1

//      15: #IFDEF ONEOF(CHIP_18F15Q41, CHIP_18F16Q41)
//      17: #IF CHIPRAM = 2048
//      19: #IF CHIPWORDS = 32768
//      21: #IFDEF VAR(NVMLOCK)
//      23: #IFDEF VAR(OSCCON2)
//      25: #IFDEF VAR(NVMCON0)
//      ;DIM _VAR1
//      27: DIM _VAR1
//      ;_VAR1 = 1
//      28: _VAR1 = 1

//      39: #IF CHIPRAM = 4096
//      41: #IF CHIPWORDS = 32768
//      43: #IFDEF VAR(NVMLOCK)
//      45: #IFDEF VAR(OSCCON2)
//      47: #IFDEF VAR(NVMCON0)
//      ;DIM _VAR1
//      49: DIM _VAR1
//      ;_VAR1 = 0
//      50: _VAR1 = 0

//      41: #IF CHIPWORDS = 32768
//      43: #IFDEF VAR(NVMLOCK)
//      45: #IFDEF VAR(OSCCON2)
//      47: #IFDEF VAR(NVMCON0)
//      ;DIM _VAR1
//      49: DIM _VAR1
//      ;_VAR1 = 0
//      50: _VAR1 = 0

//      43: #IFDEF VAR(NVMLOCK)
//      45: #IFDEF VAR(OSCCON2)
//      47: #IFDEF VAR(NVMCON0)
//      ;DIM _VAR1
//      49: DIM _VAR1
//      ;_VAR1 = 0
//      50: _VAR1 = 0

```

```

//          45: #IFDEF VAR(OSCCON2)
//          47: #IFDEF VAR(NVMCON0)
//          ;DIM _VAR1
//          49: DIM _VAR1
//          ;_VAR1 = 0
//          50: _VAR1 = 0

//          47: #IFDEF VAR(NVMCON0)
//          ;DIM _VAR1
//          49: DIM _VAR1
//          ;_VAR1 = 0
//          50: _VAR1 = 0

// Program compiled successfully (Compile time: 1 seconds)

// Assembling program using GCASM
// Program assembled successfully (Assembly time: 0.125 seconds)
// Done

```

The resulting ASM from the about code is as expected. The assignment of **VAR1 = 0**.

```

;DIM _VAR1
;_VAR1 = 0
    clrf    _VAR1,ACCESS
;Do
SysDoLoop_S1
;Loop
    bra SysDoLoop_S1
SysDoLoop_E1

```

COMPILERDEBUG

The COMPILERDEBUG setting in the USE.INI file for GCBASIC is used to enable or disable debugging features for the compiler. When the bits of the setting COMPILERDEBUG are set to 1, it activates additional debug information during compilation, which can be helpful for developers to diagnose and fix issues.

To see the permissible bits for COMPILERDEBUG first opening and closing the PREFERENCES editor (this does imply that the PREFERENCES EDITOR is maintained to show the Header in the USE.INI), and then edit USE.INI. The help section will display the following:

```
'Preferences file for GCBASIC Preferences 3.14

... lots of help, then

'   compilerdebug = 0  - 1 = COMPILECALCADD
'                       - 2 = VAR SET
'                       - 4 = CALCOPS
'                       - 8 = COMPILECALCMULT
'                       - 16 = AUTOPINDIR
'                       - 32 = ADRDX
'                       - 64 = GCASM
'                       - 128 = COMPILESUBCALLS
'                       - 256 = COMPILEUPDATESUBMAP
```

To see the debug add or edit the [gcbasic] section of USE.INI.

```
[gcbasic]
'change to a bitwise value
compilerdebug = 0
```

As previously stated. This setting can be helpful for developers to diagnose and fix issues within the compiler.

ABOUT GLCD Library Support

The GLCD capability supports over 40 GLCDs. GCBASIC loads automatically the specific library required. The loading of the specific library(ies) by the compiler improves performance and significantly reduces the compilation time.

The GLCD libraries, that are automatically loaded, are controlled by the use of an incude statement in the use program `#include <glcd.h>` with a constant to define the specific GLCD driver to be loaded `#define GLCD_TYPE GLCD_TYPE_SSD1289`.

The process the compiler uses the file `include\glcd.dat`. The `glcd.dat` has a strict format where the row index has parameters as shown below.

Format. This is strict.

Use a comma delimiter; single quote to a comment line; ampersand to group libraries. No other format controls permitted.

For each row.

```
Index, glcd type, library[&library]&library]
```

Index = the reference number from GLCD.H. There is a unique reference number per type of glcd.

Type = the type. per type of glcd. must match those definition in GLCD.H
this is use as the search/match in the constant 'GLCD_TYPE' in user source program

this is not case sensitive

Library = library to be loaded, or, group of dependent libraries
delimiter must be '&'

Development Guide for GCBASIC Preferences Editor

This section deals with the GCBASIC Preferences Editor (Pref Editor). The Prefs Editor is the software enables the user to select programmers, select the options when compiling, select the assembler and other settings. The Prefs Editor uses an ini to read and store the compiler settings. The INI structure is explained the first section, then, the Prefs Editor in detail.

ABOUT THE INI FILES

You can provide the compiler an INI file with a number of settings and programmers.

The following section provide details of the specifics within an example INI file. The comments are NOT part of an INI file.

The settings are in the INI section called [gcbasic].

```
[gcbasic]
programmer = arduinouno, pickitpluscmd1, lgt8f328p-1, xpress, pickit2cmdline, nsprog
- the currently selected available programmers
showprogresscounters = n
- show percent values as compiler runs. requires Verbose = y
verbose = y
- show verbose compiler information
```



```

    preserve = n
- preservice source program in ASM
    warningsaserrors = n
- treat Warnings from scripts as errors. Errors will cause the compiler to cease on an
Error(s)
    pauseaftercompile = n
- pause after compiler. Do not do this with IDEs
    flashonly = n
- Flash the chip is source older than hex file
    assembler = GCASM
- currently selected Assembler
    hexappendgcbmessage = n
- appends a message in the HEX file
    laxsyntax = n
- use lax syntax when Y, the compiler will not check that reserved words are being used
    mutebanners = n
- mutes the post compilation messages
    evbs = n
- show extra verbose compiler information, requires Verbose = y
    nosummary = n
- mutes almost all messages post compilation
    extendedverbosemessages = n
- show even more verbose compiler information, requires Verbose = y
    conditionaldebugfile =
- creates CDF file
    columnwidth = 180
- ASM width before wrapping
    picasdebug = n
- adds PIC-AS preprocessor message to .S file
    datfileinspection = y
- inspects DAT for memory validation
    methodstructureddebug = n
- show method structure start & end for validation
    floatcapability = 1
- 1 = singles

- 2 = doubles

- 4 = longint

- 8 = uLongINT
compilerdebug = 0
- 1 = COMPILECALCADD

- 2 = VAR SET

- 4 = CALCOPS

```

- 8 = COMPILECALCMULT
- 16 = AUTOPINDIR
- 32 = ADRDX
- 64 = GCASM
- 128 = COMPILESUBCALLS
- 256 = COMPILEUPDATESUBMAP

The section shows an example [tool] assembler section.

```
[tool=pic-as]
'An assembler
type = assembler
'Location of the assembler using a parameter substitution.
command = %picaslocation%\pic-as.exe
'Parameters
params = -mcpu=%ChipModel% "%Fn_NoExt%.S" -msummary=-mem,+psect,-class,-hex,-file,
-sha1,-sha256,-xml,-xmlfull -WL -mcallgraph=std -mno-download-hex -o"%Fn_NoExt%.hex"
-WL,-Map="%Fn_NoExt%.map" -Wa,-a

[tool=mpasm]
'An assembler
type = assembler
'Location of the assembler using a parameter substitution.
command = %mpasmlocation%\mpasmx.exe
'Parameters
params = /c- /o- /q+ /l+ /x- /w1 "%FileName%"
```

The section shows an example [patch] section.

This section shows and explicit set of patches applied to PIC-AS assembler.

```
[patch=asm2picas]
desc = PICAS correction entries. Format is STRICT as follows: Must have quotes and
the equal sign as the delimiter. PartName +COLON+"BadConfig"="GoodConfig" Where
BadConfig is from .s file and GoodConfig is from .cfgmap file
16f88x:"intoscio = "="FOSC=INTRC_NOCLKOUT"
16f8x:"intrc = IO"="FOSC=INTOSCIO"
12f67x:"intrc = OSC_NOCLKOUT"="FOSC=INTRCIO"
```

The section shows an example [programmer] section.

```
[tool = pk4_pic_ipecmd_program_release_from_reset]
'Description
desc = MPLAB-IPE PK4 CLI for PIC 5v0
'A programmer
type = programmer
'Command line using a parameter substitution.
command = %mplabxipedirectory%\ipecmd.exe
'Parameters using a parameter substitution.
params = -TPPK4 -P%chipmodel% -F"%filename%" -M -E -OL -W5
'Working directory using a parameter substitution.
workingdir = %mplabxipedirectory%
'Use if constraints - this shows none
useif =
'Mandated programming config constraints - this shows none
progconfig =
```

ABOUT THE PREFERENCES EDITOR

This is a utility for editing GCBASIC ini files. It is derived from the Graphical GCBASIC utilities, and requires some files from Graphical GCBASIC to compile.

The software is developed using Sharp Develop v.3.2.1 (not Visual Studio).

The latest source is in Sourceforge.

COMPILING

Ensure that the "Programmer Editor" folder is in the same folder as a "Graphical GCBASIC" folder. The "Graphical GCBASIC" folder must contain the following files from GCGB: - Preferences.vb - PreferencesWindow.vb - ProgrammerEditor.vb - Translator.vb - ProgrammerEditor.resources

Once these files are in place, it should be possible to compile the Programmer Editor using SharpDevelop 3.2 (or similar).

USING PREFS EDITOR

If run without any parameters, this program will create an ini file in whatever directory it is located in. If it is given the name of an ini file as a command line parameter, it will use that file.

As well as the ini file it is told to load, this program will also read any files that are included from that file.. This makes it possible to keep the settings file in the Application Data folder if GCBASIC is installed in the Program Files directory.. To put the settings file into the Application Data folder, create

a small ini file containing the following 3 lines and place it in the same directory as this program:

```
include %appdata%\gcgb.ini
[gcgb]
useappdata = true
```

The include line tells the program (and GCBASIC) to read from the Application Data folder. The useappdata=true line in the [gcgb] section will cause this program to write any output to a file in Application Data called gcgb.ini. The hard coding of GCGB is required this program is based on GCGB. It will result in programmer definitions being shared between GCGB and any other environment using this editor, which may be a positive side effect.

BUILDING THE PROGRAMMER EDITOR EXECUTABLE USING SHARP DEVELOP

To build Prefs Editor from the source files. The list shows the installation of the Sharp Develop IDE.

Complete the following:

1. Download and install Sharp Develop from [https://sourceforge.net/projects/sharpdevelop/files/SharpDevelop%203.x/3.2/\[SourceForge\]](https://sourceforge.net/projects/sharpdevelop/files/SharpDevelop%203.x/3.2/[SourceForge])
2. Download the Prefs Editor source using SVN into a source folder. This is the folder `..\utils\Programmer Editor`
4. Run Sharp Develop (installed at step #1). Load project "Programmer Editor.sln" from source source folder.
5. Hit <f8> to compile.

GCBASIC with the AVRISP or MKII Programmer

This is the GCBASIC section of the Help file that explains how to use an AVRISP MKII or USBtinyISP for ATTINY10 chip under Windows 10

Setup an AVRISP MKII or USBtinyISP for ATTINY10 chip under Windows

AVRISP MKII Windows Driver Validation/Changing

Windows 10 and 11 tested.

When using the AVRDUDE, which is used as part of the GCBASIC toolchain, you need to ensure these programmers are operating with the correct Windows device driver. The Windows driver must be **libusbK** not **libusb** or **libusb-win32** Windows driver.

You must ensure the Windows driver is **libusbK**. Using the utility Zadig enables changing the Windows device driver to **libusbK**.

- Download and install Zadig Driver Utility software <https://zadig.akeo.ie/>
- Connect the USB cable from the AVRISP MKII to your PC.
- Open Zadig and from the menu select Options / List All Devices.
- From the device list select AVRISP MKII.
- Select the target driver libusbK and click (Install / Replace Driver) button.

After the installation is completed, open Windows Device Manager and verify the driver installation.

Close Zadig

[Setup] | 1%20%20-%20AVRISP%20MKII%20Zadig%20Setup.png

NOTE

If you use ATMEL Studio you must ensure the Windows device driver is **libusb** or **libusb-win32**.

NOTE

If you use AVRDUDE then you must ensure the Windows device driver is **libusbK**.

Next in the process is to upgrade the firmware of the AVRISP MKII.

Option 1:

This process shows the installation of Atmel Studio 7, however, you may have to use Atmel Studio 6 because of operation system constraint.

Download and Install Atmel Studio 7 from the GCBASIC file store [here](#)

After the installation open ATMEL STUDIO 7

- Select Tools / Device Programming
- Make sure the AVRISP MKII is selected

- At this point ATMEL STUDIO will notified to upgrade the firmware to version 1.8
- Click the Upgrade button
- Once the firmware is upgraded, close ATMEL STUDIO 7

[Studio] | *2%20%20-%20ATMEL%20STUDIO.png*

Option 2:

To update the firmware please, follow the steps listed below.

- Connect the programmer to the USB and with a sharp object (needle or pin) press the upgrade pin - it is in a small hole at the back of the board (this will start the bootloader and will turn off the LED, it will also probably show a new unrecognized device in the device manager for which we will install drivers in step 3)
- Download and install the latest version of "Atmel Flip" software (it can be downloaded from the Atmel's web-site, or from the GCBASIC file store [here](#))
- Open its install folder and update the software of the unrecognized device (usually under the "Other devices" tab) with the drivers from folder named "usb"; the device should now be recognized as AT90USB162 under "libusb-win32" tab
- Start "Atmel FLIP" and click "Select a target device" → choose AT90USB162
- Click "Select a Communication Medium" and then USB medium
- Download the firware and unpack [Olimex website](#) or the GCBASIC files store [here](#)
- From "File → load HEX file" choose the latest HEX and click "RUN" in the "Operations Flow" section
- Disconnect the AVR-ISP-MK2 from the USB and connect it again

For more information about AVR-ISP-MK2 see this guide [here](#)

AVRISP MKII to ATTINY10 Connections:

Connect the AVRISP MKII to the ATTINY10 as show in the Diagram.

[Connections] | *3%20%20-%20AVRISP%20MKII%20Connections.png*

GCStudio Programmer Setup:

Open GCStudio and set the Programmers to use as show below

For AVRISP MKII use the AVR ISP XP2 [KANDA] and drag it to the top of the list and click OK.

4%20%20-

%20GCStudio%20AVRISP%20MKII%20use%20the%20AVR%20ISP%20XPII%20%5BKANDA%5D.png

Now you're ready to upload your first program to an ATTINY10.

USBtinyISP Setup:

- From Zadig select USBtinyISP in the device list.
- Select the target driver libusb-win32 and click (Install / Replace Driver) button.
- After the installation, open Windows Device Manager and verify the driver installation.
- Close ZADIG

[5%20%20 %20USBtinyISP%20Zadig%20setup] | *5%20%20-%20USBtinyISP%20Zadig%20setup.png*

USBtinyISP to ATTINY10 Connections:

Connect the USBtinyISP to the ATTINY10 as show in the Diagram.

[6%20%20 %20USBtinyISP%20Copnections] | *6%20%20-%20USBtinyISP%20Copnections.png*

GCStudio Programmer Setup:

Open GCStudio and set the Programmers to use as show below

Select and Drag the Avrdude (USBtinyISP) programmer to the top of the list.

[7%20%20 %20GCStudio%20Avrdude%20%28USBtinyISP%29] | *7%20%20-*

%20GCStudio%20Avrdude%20%28USBtinyISP%29.png

Now you are ready to upload your first program to an ATTINY10

For more information programming review these resources:

A guide

http://gcbasic.sourceforge.net/library/Programming_an_Attiny10_with_AVRISP_mkII_and_AVR_Studio_5.pdf[here]

A blog

http://gcbasic.sourceforge.net/library/Technoblogy_Programming_the_ATTiny10.pdf[here]